# The cosmology of Cosmological Relativity:
## the expansion history from causal reassignment on a de Sitter background, the cosmogenesis branch point, and the scalar perturbation sector

Daryl Janzen
Department of Physics and Engineering Physics
University of Saskatchewan

**Abstract**

Cosmological Relativity (CR) obtains the observed cosmological expansion by a causal reassignment on a de Sitter background at the cosmogenesis branch point, with the Einstein field equations unchanged. This paper develops that cosmology and completes it with its scalar perturbation sector. The reassignment selects the Nariai member [37] of Schwarzschild–de Sitter, whose proper frame is a homogeneous, observationally isotropic, *non-synchronous* cosmology with areal radius $r(\tilde{\tau}) \propto \sinh^{2/3}$—exactly the flat-ΛCDM scale factor—its rate fixed by the cosmological constant alone, with radiation and matter inherited content read off the cosmic clock rather than terms that source the expansion. The cosmic beginning is the branch point $r = 0$—a genuine curvature singularity of the SdS metric read over the $r$-chart [36], and *not* a singularity of the underlying substrate, whose throat is the defining constant $\alpha$; the divergence of the $r$-chart invariants there is the areal coordinate degenerating; the constant-proper-time slices are exactly Euclidean ($\Omega_k = 0$) while the cosmological layers are a closed $S^3$—spatial curvature and dark energy one $\Lambda$ read on two slicings. The acoustic scale follows from that rate rather than calibrating it: with the plasma handed over at the branch point there is no early-universe parameter to carry the angle, so the angle is computed, and it and the baryon-acoustic distances fix $H_0$ together at $\simeq$68.6. The light-element composition is produced on the same footing: the infalling matter heats above the deuterium bottleneck and the expanding leg cools back through it, so that leg is a standard big-bang nucleosynthesis run from free nucleons, fixing the composition with the inherited baryon-to-photon ratio $\eta$—the one boundary datum of that handover. Helium-4 and deuterium sit at their observed values—the freeze-out temperature the data demands being the bottleneck the standard rate delivers, un-tuned; the near-zero metallicity of the oldest systems follows from the handover; and because the run is standard, lithium-7 carries the standard threefold over-prediction, the lithium problem shared with flat ΛCDM rather than dissolved.

On this cosmology the primordial scalar spectrum *factorizes*. The de Sitter substrate *determines* its structure—the geometric coherence of the acoustic phases on the null boundary (not a super-horizon freeze-out), the classical non-vacuum character of the fluctuations (the substrate vacuum lies some $10^{113}$ below the observed $A_s$), the throat's isotropization, a parameter-free low-multipole discreteness floor, and the faithful transmission of the spectral shape—while the progenitor collapse *supplies* its content, the amplitude $A_s$ and tilt $n_s$, inherited as boundary data exactly as flat ΛCDM inherits the baryon-to-photon ratio. *The transmission is supported by a mechanism rather than by the absence of a scale alone* (§3): on the contracting leg $aH$ grows without bound as the branch point is approached, so the comoving horizon shrinks to zero and every mode *exits* it and freezes before the crossing; the imaginary-time segment's Euclidean kernel damps oscillatory content and has nothing to act on in a frozen mode, so amplitude and tilt cross unaltered while the collapse-leg acoustic phase does not—which the observed peak positions independently require, excluding a transmitted leg phase by a factor of 21–41 in the required distance [9].

The decisive result is a *transmission dichotomy*: a non-degenerate horizon's exponential near-horizon approach imprints a scale-invariant spectrum (the inflationary de Sitter-horizon

mechanism), whereas the degenerate Nariai double root's power-law approach carries no scale, and the branch point where the modes cross—a third case, $f$ diverging so that the tortoise coordinate stays finite—accumulates no phase at all; the super-horizon transfer across it is computed and scale-invariant, so the progenitor spectrum is transmitted unaltered and CR has no inflationary scale-invariant attractor, no consistency relation, and no substrate-sourced primordial $B$-modes.

Because the distance slicing is flat, the discrete closed-$S^3$ source projects through the flat transfer to a parameter-free low-multipole deficit ($\ell \lesssim 8$)—mild, and on a genuine Boltzmann transfer a dip whose *minimum falls at* $\ell = 4$ (0.36 of the ΛCDM expectation there, 0.487 and 0.435 at $\ell = 2$ and 3; *the shape and the depth both cross-validated between two independent transfers*, which agree to three per cent at worst once the post-recombination source is carried to the present), and non-discriminating in sum: scored against the measured multipoles with cosmic variance carried, *the quadrupole favours this construction and* $\ell = 3$ *to* 5 *favour the standard model*, the two nearly cancelling at $\Delta(-2\ln L) = +1.6$ over $2 \leq \ell \leq 10$.

Taken together, these results carry a reckoning the paper draws in full: what flat ΛCDM assembles from added, tunable apparatus—inflation for the horizon, flatness, coherence, and near-scale-invariance, a dark-energy component for the late-time acceleration, added early-universe physics for the tensions—this cosmology recovers from the cosmological constant alone, each feature required by structure already present rather than produced by a sector introduced to yield it. By the economy of assumption that distinguishes a required structure from a merely permitted one, that comparison is settled on the theory-choice axis in this cosmology's favour; and on the data axis it is split: the expansion history fits the low-redshift distance ladder on the SDSS DR12 consensus and, at the current state of the art, the DESI DR2 dataset (thirteen measurements across seven tracers) at $\chi^2/\mathrm{dof} \simeq 1$, which fixes $\Omega_m \simeq 0.30$ and $H_0 \simeq 68.6$ and leaves the discrepancy with the local distance ladder where the standard model leaves it; the light-element abundances are confirmed within $1\sigma$ at the microwave-background baryon density (the cosmogenesis paper [19]); and the acoustic peak structure, computed from that same background with nothing fitted to it, reproduces the comb to a part in six hundred and is still rejected on shape—at 2.57 times the control's distance from the data as computed, and 1.57 times once both arms' parameters are refitted; what it leaves is about one multipole at each peak.

The cosmology's exposed edges are named: a signature in the diffusion *scale*, the two rates differing by the radiation term alone so that the plasma's Silk scale and the foliation's distances do not share a rate [27]—a computed effect, non-reabsorbable by the baryon density, whose size is fixed by the handover, read with both lengths carried to the peak of the visibility function, and which the joint fit locates as a Gaussian residual in $\ell$ that no tilt removes, a refit absorbing a Gaussian only window-locally (the low multipoles being a mild, non-discriminating deficit rather than a sharp edge either way); and the acoustic peak structure. *The full-spectrum likelihood is run on the computed spectrum*: scored against Planck 2018 `plik_lite` over the full-range lensed configuration, with one amplitude fitted so that what is compared is shape, this arm sits at 2.57 times the control's distance from the data (§4.7). *That is a spectrum computed on a background fitted to the distances and then confronted*, which is a different statement from a refit of the spectrum's own parameters to the microwave data; the second is the comparison the standard model is usually judged by, and it is reported in §4.7 beside the first.

We mark maturity throughout. *Established*: the cosmological derivation, the transmission dichotomy, the amplitude floor, the throat, the flat slice, the flat-projection transfer, the data confrontation, and the time-reversal driving equality as a Fourier theorem. *Argued*: the coherence mechanism. *Measured*: the peak *heights*, which this arm's own instrument returns within a few per cent of the sky's first two ratios on the polarisation path and above them on the fluid path, so the heights are the path's and the positions are the construction's.

*Run, and returning a partial agreement*: the end-to-end branch-point-to-recombination transfer, the standard line-of-sight integral across a visibility of finite width, validated on its control to 0.23% in the first-to-second height ratio in the configuration this construction's arm is read in, and carried through to that arm, which returns the peak spacing to one part in six hundred and the first three peaks within a grid step of the measured ones, with the fourth not resolved against the sky once the likelihood's own covariance is propagated; the likelihood rejects the arm's spectrum on $TT$ shape nonetheless. *Open*: the *derivation*—as

against the measurement—of $A_s$ and $n_s$. *The cosmology carries no early-universe parameter, and what it does carry should be stated plainly*: the plasma is handed over at the branch point, where every acoustic mode is outside the horizon and the potential sits at its regular value, so the sound-horizon integral has no lower endpoint to fit and the acoustic angle is an output of the rate rather than a calibration of it. The background is fixed by the late-time data alone—the epoch $x_0$ (equivalently $\Omega_m = 2/(x_0^3+2)$) and $H_0$, from the baryon-acoustic distances together with that angle, which agree with one another within a twentieth of a unit at $H_0 \simeq 68.6$. *Matter–radiation equality is then fixed by the same epoch*, $1 + z_{\rm eq} = \omega_m/\omega_r$ with $\omega_r$ measured, so one quantity carries the expansion history and the acoustic driving together and the computed spectrum is a test of the background rather than a further fit to it. The substrate's single $\Lambda$ is shown to govern both the global expansion and, per structure, the local scale at which structure resists it—the Hubble–Eddington radius—one constant read at two ranges.

# 1 Introduction: the cosmological theory and its scalar perturbation sector

Cosmological Relativity (CR) recovers the flat-ΛCDM expansion history by a causal reassignment on a de Sitter background at the cosmogenesis branch point: the expansion rate is fixed by the cosmological constant alone, radiation carries no term in it, and the present matter density is a reading of the cosmic epoch rather than an independent quantity [9, 11]. The cosmic beginning is the branch point $r = 0$—a genuine curvature singularity of the SdS metric read over the $r$-chart, and *not* a singularity of the substrate, whose throat is the defining constant $\alpha$, and the construction makes the constant-proper-time slices exactly flat while the cosmological layers are a closed $S^3$. This paper develops that cosmological theory in the form its observational tests require (§2) and completes it with the scalar perturbation sector the framework paper's outlook named open—"perturbations within the Schwarzschild–de Sitter representation"—the primordial density perturbations and their imprint on the microwave background (§§4–8), closing on the open frontier the construction sets up (§10). Across these sections one comparison recurs and is drawn together at the close (§9): the causal contact, flatness, coherence, isotropy, and near-scale-invariance that the standard picture obtains from an added, tunable inflationary sector are here read off structure already present and fixed by $\Lambda$ alone—an economy-of-assumption difference the paper argues is decidable on the theory-choice axis, where it falls to this cosmology—and on the data axis the expansion history fits the low-redshift distance ladder, which fixes $H_0$ alongside the epoch, and the light-element abundances are confirmed against the data (§2.4), while the exposed edge is the acoustic peak structure—where the transfer, run on that background with nothing fitted to it, reproduces the comb to a part in six hundred and is still rejected on $TT$ shape (§4.7)—and the damping-*scale* signature, a computed, non-reabsorbable effect whose size the handover fixes, read with both lengths carried to the peak of the visibility function, and which the joint fit locates as a Gaussian residual no tilt removes, the low multipoles a mild, non-discriminating deficit rather than a sharp edge either way. The companion dynamics paper develops the perturbation sector's *tensor* half—the propagating graviton as the leaf's shear, a massless de Sitter mode on the cosmic foliation [10]; the scalar half is here.

The result is organized by one thesis, which we state at the outset because it controls everything below.

> *The CR primordial scalar spectrum factorizes: the de Sitter substrate geometry determines its structure, the progenitor collapse supplies its content, and the branch point's null geometry assigns each its job.*

Substrate-determined, and developed here: the coherence of the acoustic phases (§4); the classical, non-vacuum character of the fluctuations (§5); the isotropizing throat (§6); a parameter-

free low-multipole discreteness floor (§7); and the faithful transmission of the spectral shape (§8). Progenitor-supplied, and inherited rather than derived: the amplitude $A_s$ and tilt $n_s$. This is the same division of labour by which flat ΛCDM carries the baryon-to-photon ratio $\eta$ as a measured datum from a baryogenesis it does not model; this cosmology carries that same $\eta$ from the hot handover of the progenitor collapse, and the transmission result of §8 routes $A_s$ and $n_s$ through the *same* handover. There is one boundary-condition supplier, not two.

A word on what is settled before the spectrum. The acoustic *scale* (the peak spacing) is settled in the cosmological development above (§2.3): it is computed from the rate, with the sound speed the ordinary baryon-loaded value of pressureless branch-point matter, and the Hubble constant is fixed alongside it by the baryon-acoustic distances. *The sky's own side of that comparison is measured here rather than quoted. Locating the acoustic peaks of the Planck 2018* `plik_lite` *binned TT spectrum [24] by parabolic fit to each hump above* $\ell = 100$ *returns* $\ell = 220.4$, $537.7$, $817.3$ *and* $1123.9$, *which reproduce the standard quoted values to better than one per cent without having been fitted to them. Fitting* $\ell_n = \ell_A(n + \phi/\pi)$ *to the first three gives the sky* $\phi/\pi = -0.2404$ *against* $-0.2361$ *for a* ΛCDM *control measured by the identical procedure—an offset of* $0.0043$, *so the control stands in for the sky on this quantity to within seven parts in a thousand.* The move has a name and the name is worth stating, because it is what licenses the comparison*: this is* matched-procedure differencing*—the control is passed through the identical extraction, so whatever bias the procedure itself carries appears in both arms and cancels in their difference, and what survives is attributable to the physics rather than to the instrument.* The same discipline governs every control in this sector*: a residual is credited against CR only once the control, run through the same machinery, has been shown not to produce it.* What this construction's own arm returns against that measured comparison is reported in §4.7*, the two-arm transfer being validated on the control and carried to convergence on both arms.* These are settled there and are not reopened in the perturbation sector; in particular, the standard radiation-governed sound horizon laid on this cosmology's rate is a calculation belonging to neither framework, and any "tension" it produces is an artifact, not a CR result. What is genuinely downstream is the *spectrum*—the peak heights, measured below against this arm's own transfer (§4), and the full oscillating-medium transfer, run on both arms in §4.7—which is where this paper's substance lives.

The register is the programme's [14, 9]: the field equations are Einstein's, unchanged; the existing object is the evolving three-dimensional layer; "the substrate determines" and "the branch point transmits" are statements about a real geometry and its real boundary [17, 16], not about a modification of the dynamics.

## 2 The cosmological theory the perturbations complete

This paper develops the cosmology of Cosmological Relativity from first principles and carries it to its empirical confrontation, completing it with the scalar perturbation sector. It takes from the companion framework paper [9] the *general* primitives—the layered ontology, the projection principle, and the representational freedom by which one fixed manifold admits distinct Lorentzian metrics—together with the Null–Boundary Correspondence theorem, the structural result (grounded in the metric-singularity causal structure of gravitational collapse [4]) that a collapse event-of-events and a de Sitter cosmic-entry slice are one ontological layer under two causal assignments. The cosmology proper is built here: the causal reassignment that selects the Nariai member, the proper-frame expansion law, the branch point at which the collapse leg becomes the expansion leg—the beginning, reached a lap on from the horizon the Correspondence anchors—and the computation of the acoustic scale and the confrontation with the distance data; the perturbation sector (§4–§8) is its completion.

## 2.1 The causal reassignment and the Nariai member

The vacuum solution of $R_{\mu\nu} = \Lambda g_{\mu\nu}$ with a localized source is Schwarzschild–de Sitter (SdS),

$$ds^2 = -\frac{r}{\frac{\Lambda}{3}r^3 + \frac{2GM}{c^2} - r}\,dr^2 + \frac{\frac{\Lambda}{3}r^3 + \frac{2GM}{c^2} - r}{r}\,dt^2 + r^2 d\Omega^2, \tag{1}$$

whose areal radius $r$ is timelike in the cosmological regime $\Lambda G^2 M^2/c^4 \geq 1/9$. Building the cosmology, we reassign causal roles on the de Sitter substrate—a move licensed by the representational freedom of the framework, distinct Lorentzian metrics on one fixed manifold sharing its foliation [9]: the de Sitter null rulings whose causal sense matches a collapse horizon's generators are promoted to the timelike fundamental congruence, the at-rest comoving worldlines become the null photon congruence, and the foliation by expanding 3-spheres is preserved. This selects the *Nariai* member $\Lambda G^2 M^2/c^4 = 1/9$, $M = c^2/(3\sqrt{\Lambda}G)$—the unique non-pivoting tilt, fixed by $\Lambda$ alone—at which the two positive horizons merge at $r_N = \alpha/\sqrt{3}$, $\alpha = \sqrt{3/\Lambda}$ the de Sitter 3-sphere size (a scale distinct from the merged-horizon radius $\alpha/\sqrt{3}$ and from the areal-radius amplitude below; we keep the three apart throughout). The selection is geometric—the fixed point of the slicing curve's root-exchange involution, equivalently the limiting tangent orientation a collapse selects—not a fitted mass [5]. That the selection is moreover *unique* is a structural fact about the description groupoid rather than a fitting: the generic vantages fall into two-cycles of the involution, the Nariai member is its one fixed point, and the reassignment promoting a null direction to the fundamental congruence is forbidden at every generic vantage, so it selects Nariai and no other [6].

## 2.2 The proper-frame cosmology, derived

In the proper frame of the fundamental observers the SdS metric is a homogeneous, non-synchronous cosmology. The derivation [15] proceeds from (1) in five steps. (i) A timelike radial geodesic has a conserved energy $E$ (the metric being independent of $t$); eliminating $dt/d\tau$ in the normalization $L = -1$ gives the radial equation $(dr/d\tau)^2 = E^2 - V_{\text{eff}}(r)$, with $V_{\text{eff}} = (r - 2GM/c^2 - \frac{\Lambda}{3}r^3)/r$. (ii) The fundamental congruence is the worldlines whose motion in $r$ is produced purely by the field—those at rest exactly where $V_{\text{eff}} \equiv 1$ and the metric reduces to Minkowski—which is $E = 1$; this fixes the rest frame by the field rather than by stipulation, *and it is worth saying where that locus is*: $V_{\text{eff}} = 1$ is $r^3 = -2GM\alpha^2/c^2$, so the rest frame is defined at $r = -(6GM/c^2\Lambda)^{1/3}$—a *negative* areal radius, on the conjugate branch of the lap, between the progenitor's horizon and the branch point [9]. *An earlier derivation carried that locus as a formal one*—"a negative mathematical abstraction that lies beyond a singularity at $r = 0\ldots$ simply a matter of algebraic consistency" [2, App.]—*and on the lap it is a place.* The same locus fixes the sign, since $t = \tau$ there if and only if $E = +1$. (iii) With $E = 1$, integrating the radial equation gives, along one worldline, $r(\tau) \propto \sinh^{2/3}(\frac{3}{2}\sqrt{\Lambda c^2/3}\,\tau)$. *That this integration closes at all is a second thing the same condition buys.* Writing the radial equation as $(\mathrm{d}r/\mathrm{d}\tau)^2 = 2GM/c^2 r + r^2/\alpha^2 - k$ with $k = 1 - E^2$, the quadrature $\int \mathrm{d}r/\sqrt{\cdot}$ is elementary at $k = 0$—it is $\frac{2\alpha}{3}\,\mathrm{arcsinh}\big(r^{3/2}/\sqrt{2GM\alpha^2/c^2}\big)$, which inverts to the $\sinh^{2/3}$ above—while for $k \neq 0$ the substitution $r = u^2$ carries it to a sextic under the root and it is not elementary. *So $E = 1$ fixes the rest frame by the field and, by the same stroke, is the one non-degenerate energy at which the cosmology integrates in closed form*; the only other elementary cases are the degenerate ones, $\Lambda = 0$ giving the dust cycloid and $M = 0$ giving pure de Sitter[L14]. (iv) Labelling the congruence by a comoving coordinate $\chi$ through the integration constant $r(0)$, the areal radius across the congruence depends on $\tau, \chi$ only through their sum, $r(\tau, \chi) = r(\tau + \chi)$, with $\partial_\chi r = \partial_\tau r$. (v) Fixing the transformation $t(\tau, \chi)$ by requiring the comoving coordinate orthogonal to proper time, $g_{\chi\tau} = 0$, forces the integration function to be $\chi$ itself, and the remaining components follow ($g_{\tau\tau} = -1$, $g_{\chi\chi} = (\partial_\chi r)^2$). The result is the proper-frame line element

$$ds^2 = -d\tau^2 + (\partial_\chi r)^2\,d\chi^2 + r^2 d\Omega^2, \tag{2}$$

with cosmic time $\tilde{\tau} \equiv \tau + \chi$ (by Weyl's principle the congruence issues from a common origin and the constant-cosmic-time slices are those of constant $\tilde{\tau}$), areal radius

$$r(\tilde{\tau}) = \left(\frac{6GM}{\Lambda c^2}\right)^{1/3} \sinh^{2/3}\left(\tfrac{3}{2}\sqrt{\tfrac{\Lambda c^2}{3}}\,\tilde{\tau}\right), \tag{3}$$

and redshift $1 + z = r(\tilde{\tau}_0)/r(\tilde{\tau}_e)$. The constant-$\tilde{\tau}$ slices sit at 45° to the fundamental rest frame: the cosmology is homogeneous and observationally isotropic, yet *non-synchronous*—the constant-$\tau$ slices are exactly Euclidean (§7) but are not the cosmological ones. *Two presentations of this one geometry are in use and it is worth saying plainly how the cosmological layer appears in each, because the second is easy to misread.* In the de Sitter presentation the layer *is* the $S^3$: the at-rest geodesics are the photon bundle, one of the two null bundles on the cosmological horn carries the late-time matter geodesics, and the universe is an $S^3$ in which all matter falls uniformly along null lines at constant velocity in the expanding sphere's Hopf fibration. *Under the reassignment that gives* (2)*, that same null bundle becomes the constant-$\chi$ geodesics and the other becomes the constant-$\tau$ ones*—both straight lines in the $\tau$–$\chi$ plane, with constant $\tilde{\tau}$ the 45° lines. *So in this chart a constant-$\tilde{\tau}$ surface is not presented as a sphere*: it is that line together with an $S^2$ of directionality, the Hopf direction having been traded into $\chi$, and reading its induced metric returns $\mathbb{R} \times S^2$ accordingly. *That is what the reassignment does and not a statement that the layer is other than the $S^3$*—the uniformity of matter across cosmic-time slices is the same fact read either way, since matter stationary within an expanding $S^3$ and space spinning uniformly along the Hopf direction are one description. Of the FLRW kinematic assumptions—(i) a geodesic congruence, (ii) hypersurface orthogonality, (iii) homogeneity, (iv) isotropy—this construction keeps (i) and drops (ii); Lemaître's reading [30] of the Euclidean slices as "space ejected from $r = 0$" is causally incoherent and is not the physics. Equation (3) is moreover a single analytic function of $\tilde{\tau}$ (Fig. 1): the expanding cosmology is its real branch $\tilde{\tau} > 0$, and the cosmogenesis *branch point* at $\tilde{\tau} = 0$, where $r \to 0$, is the branch point of the scale factor rather than a real $r = 0$ boundary—and it is *not* a seam: the seams are the two unit-speed loci of the lap, $r = -2\alpha/\sqrt{3}$ and $r = +\alpha/\sqrt{3}$ [9], and $r = 0$ lies strictly between them, at the close of the lift—the analytic counterpart of the beginning at the branch point, where the substrate's curvature stays finite while the geometry read over the $r$-chart diverges [4], and the cosmological reading of the slicing curve's closed lap—the *cosmogenetic bead* the framework establishes as one closed object [5, 9], of which this scale factor's real branch $\tilde{\tau} > 0$ is the expansion leg and its analytic continuation the prior universe's collapse—on which $r = 0$ is a branch point and not a barrier, the substrate smooth across the locus the chart so labels [5].

### 2.3 Exact recovery of flat ΛCDM, and the amplitude fixed by Λ

*What fixes this rate is worth stating before what it recovers, and the count is smaller than it looks.* The rate this cosmology computes on is set by the substrate's geometry and by nothing else. *It has one parameter*—the substrate curvature radius $\alpha = \sqrt{3/\Lambda}$, an invariant length—*and one epoch.* At the Nariai member the mass is no longer free (5), so the amplitude and the rate of the expansion law are both set by $\Lambda$ and the scale factor is one function of it; what remains is where the present sits on that one curve. *The quantity $x_0$ carries that*: it is the present areal radius in units of the Nariai radius, $x_0 = r(\tilde{\tau}_0)/r_N$, and since $r_N = \alpha/\sqrt{3}$ is fixed by $\Lambda$ the number is a clock reading and not a second geometric datum[P15R14]. *The geometry's own offset is fixed twice over*: it is the double root of the horizon cubic at the Nariai mass, and it is the constant ledger's $2M = \alpha\big((r_0/\alpha) - (r_0/\alpha)^3\big)$ with $M$ at that mass [5, 17]—*so "the offset" names the unit, and $x_0$ the reading taken against it. So the ratio it fixes is a reading of cosmic epoch*, which is what (7) says of the density ratio directly. *Neither the parameter nor the epoch is a*

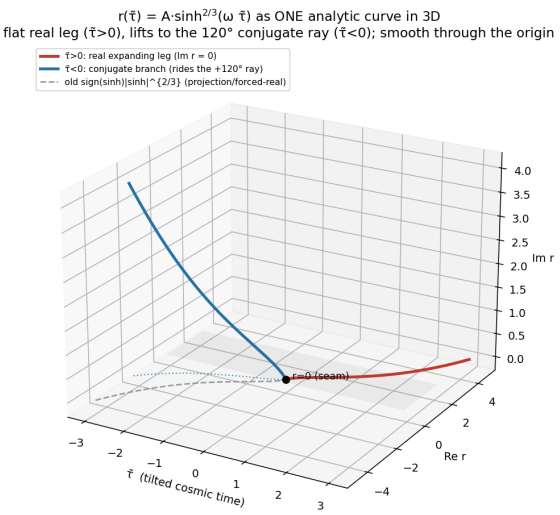



**Figure 1:** The scale factor (3) as one analytic curve $r(\tilde{\tau})$ in $(\tilde{\tau}, \mathrm{Re}\, r, \mathrm{Im}\, r)$. The expanding cosmology is the real leg $\tilde{\tau} > 0$, lying in the real plane; the branch point $\tilde{\tau} = 0$, $r \to 0$—*not* a seam—is where the curve leaves the real plane onto a conjugate branch at constant phase $2\pi/3$ for $\tilde{\tau} < 0$. The beginning is this analytic branch point, not a real $r = 0$ boundary. Forcing $r$ real (dashed) projects the conjugate branch onto the negative axis and manufactures a cusp the smooth curve does not carry.

*content of the universe*:

$$c^2 H^2(z) \;=\; \frac{c^2}{\alpha^2}\Big(1 + \frac{2(1+z)^3}{x_0^3}\Big). \tag{4}$$

The offset is measured directly and calibration-free—$z_{\mathrm{acc}} = x_0 - 1$, (17) below—without a distance ladder, without the microwave background and without any density. *What the rate is a rate of is the stacking of the foliation*, and content does not enter it: radiation is absent from the slicing operator's vacuum kernel, requiring $m'(r) \neq 0$, a bend of the cut; and the second term is the offset itself rather than a density supplied by hand [7, 9].

*The familiar cosmological form is accordingly a coincidence of form, and saying so is the substance of this subsection.* Written in the fitted pair the same rate reads $H^2 = H_0^2\big[\Omega_m(1+z)^3 + \Omega_\Lambda\big]$, with $\Omega_m = 2/(x_0^3+2)$ and $\Omega_\Lambda = x_0^3/(x_0^3+2)$ following identically—*a translation between parameter sets, and not a decomposition into components.* In the standard reading cosmic time is orthogonal to the three-dimensional world and the expansion is driven: by initial conditions, and by whatever densities obtain at each moment, through the field equations. Here it is read off a fixed background whose foliation is *not* orthogonal to the proper time of observers at rest, so that such observers move through the universe over cosmic time in a geometrically fixed manner. *The two readings agree on the function and share nothing beneath it.*

*Where the field equations act is the leaf, and they act there in full.* All stress-energy gravitates, by the ordinary Friedmann readout of the self-gravitating congruence; the perturbations, the plasma's sound horizon and its diffusion length are processes running *in* the content and take that rate, while a comoving separation read across leaves takes the stacking rate (§2.2). *What the layered framework keeps apart is not two physics but one physics and its projection*: a quantity is computed on the level it belongs to, carried on the layer and then projected.

Equation (3) is exactly the flat-ΛCDM scale factor. At Nariai the mass is no longer free, and the amplitude becomes a pure Λ-length,

$$\left(\frac{6GM}{\Lambda c^2}\right)^{1/3} = \frac{2^{1/3}}{\sqrt{\Lambda}}, \tag{5}$$

so both factors of the expansion law are set by $\Lambda^{\mathsf{P15R39}}$: the rate $\frac{1}{2}\sqrt{3\Lambda}\,c$ is the standard late-

time rate and the amplitude is the third Λ-scale (distinct from $\alpha$ and $\alpha/\sqrt{3}$). The Friedmann equation the $\sinh^{2/3}$ law satisfies,

$$H^2 = \frac{\Lambda c^2}{3}\coth^2\Big(\tfrac{1}{2}\sqrt{3\Lambda}\,c\,\tilde{\tau}\Big) = \tfrac{1}{3}\big(8\pi G\rho + \Lambda c^2\big), \tag{6}$$

makes the density ratio a clock[P15R39],

$$\frac{\Omega_m}{\Omega_\Lambda} = \operatorname{csch}^2\Big(\tfrac{1}{2}\sqrt{3\Lambda}\,c\,\tilde{\tau}\Big), \tag{7}$$

so $\Omega_{m,0}$ records the cosmic epoch $\tilde{\tau}_0$ at which we observe rather than an independent density, and the coincidence problem becomes the observation that we exist at a time of order the geometry's single timescale. The densities are bookkeeping for one Λ-set geometry; *radiation, like matter, is inherited content read off that clock, not a term that sources the rate*, and the stacking rate has no radiation-dominated era [11], the leaf, where radiation gravitates, carrying one (§4). The objection that a perturbing radiation fluid "must gravitate and so alter the rate" is a category error against a construction that reads (6) leftward: the rate is set, and the fluid's energy is among the contents the set rate carries. The acoustic *scale* follows from this rate, and $H_0$ is fixed with it by the distance data; this is developed next (§2.4).

### 2.4 The acoustic scale, the expansion rate, and where the ruler comes from

The two frameworks reproduce the late-time expansion identically and part in the early universe, where radiation carries no term in the foliation's own rate (6) while the plasma's scales accumulate on the leaf's, which carries it. *What follows from that one structural fact is not that the standard model's tensions dissolve but that the acoustic ruler has a different provenance*, and the two lengths the observed angle is built from sit on opposite sides of the split.

*The three levels, and which rate each quantity uses.* The reading of every rate-bearing quantity below is fixed by locating it on one of three levels, which the geometry keeps apart and whose conflation is the standing error the layered framework names [9]. **(L1) The foliation stacking rate**: the Λ-set $\sinh^{2/3}$ law (6) read *leftward* in the sense the slicing paper fixes [7]—the cut is primary and $\rho$ is the name of its bend, not its cause, the rate-side of the slicing operator's reading whose content-side is *matter is the bend of the cut* [7]—the layer's own geometric expansion, set by $\alpha$ and the offset $x_0$, and *empirically forced* rather than modelled: the microwave-redshift isotropy floor forces uniform expansion to some three orders of magnitude, so a matter-tracking (radiation-sourced) rate is already excluded by data [11].

This is what the observable cosmology rides at *every* epoch, the branch point included: it is the rate the foliation stacks at, not a regime that begins somewhere. **(L2) The leaf-level local dynamics**: the expansion scalar of a self-gravitating congruence, the ordinary Friedmann readout $\mathcal{H}^2 = \frac{8\pi G}{3}\rho_{\rm tot} + \frac{\Lambda}{3}$ with radiation gravitating normally—the valid phenomenological reading of general relativity on the progenitor's collapse excursion, where nucleosynthesis runs [19]; it is the same single $E{=}1$ geodesic read *inward* as dust collapse that L1 reads outward as cosmology [7, 4]. **(L3) The $E = 1$ projection**: the reassigned shadow the observer reads, through which an L1 or L2 quantity appears as distance and redshift—the Projection Principle of the framework paper, its geometric origin the *second ruling* of the slicing (the synchronous space is the second ruling) [9, 7]; reading this appearance as the existent is precisely the naïve-realist collapse the epistemology forbids [12]. The decision rule is structural, not a choice: a self-gravitating local excursion runs on L2, diffuse content riding the global foliation on L1, and every observable is an L1 or L2 quantity seen through L3; and *there is no L1/L2 boundary in time.* The distinction is kinematic and holds at every epoch: the vacuum kernel of the slicing operator is the SdS family exactly—Λ and the cut's offset $2M$, and no more [7]—so the stacking rate is $c^2H^2_{\rm stack} = (c^2/\alpha^2)(1 + 2(1+z)^3/x_0^3)$, while radiation, requiring $m'(r) \neq 0$, is a *bend* of

the cut and therefore content. The decomposition is exact, $H^2_{\rm leaf} = H^2_{\rm stack} + H_0^2\Omega_r(1+z)^4$: the two rates differ by the radiation term alone. A comoving separation read across leaves takes the stacking rate; the scales the plasma itself accumulates—$r_{\rm s}$ and $r_D$—take the leaf's, together with recombination's microphysics and the perturbations, because they are accumulated by a fluid running *in* the content whose own radiation gravitates in its own expansion. The two must take the same rate, and the peak spacing the construction computes is the leaf accumulation: fitted to the computed peaks, the comb reproduces that integral to two per cent and the stacking one not at all. The assignment looked temporal only because $\Omega_r(1+z)^4$ is negligible below $z \sim 10$ [9, 4]. Each mis-assignment is quantitatively fatal—the geometric L1 rate carried down into the nuclear window is $\sim$300$\times$ too slow, and the radiation-sourced L2 rate carried into the distances would pin them to a radiation-era density the foliation's own rate does not carry [19]. Crucially L1 is the layer's own *ontological* expansion, not the L3 shadow: computing a process against the stacking rate is reading the geometry, not the appearance. Every rate-bearing quantity below carries its level, so the reading is settled by the geometry rather than defaulted to the synchronous machinery.

*The Hubble constant.* The late-time expansion is reproduced exactly, so $H_0$ enters this cosmology as it enters the standard one: through the distances. With the plasma handed over at the branch point there is no early-universe parameter left to carry the acoustic angle, and the angle and the baryon-acoustic distances then fix the rate together, at $H_0 \simeq 68.6$—the angle alone and the distance data agreeing to a twentieth of a unit. *So this construction does not dissolve the discrepancy with the local distance ladder* [25]: it leaves it where the standard model leaves it, as a disagreement between the ladder and the early-universe inference, and what it supplies in its place is stated in §4.7—the acoustic spectrum computed from that background with nothing fitted to it.

*The acoustic scale, computed.* Recombination is observable cosmology from the branch point onward, and the two lengths its angle is built from sit on opposite sides of the rate rule: the comoving distance $D_M$ is a separation read across leaves and rides the stacking rate, while the sound horizon $r_{\rm s}$ is accumulated by the plasma and rides the leaf's. The observed angle $\theta_* = r_{\rm s}/D_M$ is their projection.

The robust ingredients are in hand: $D_M$ is the standard observable, and $H(z)$ is the fixed $\sinh^{2/3}$ law (6) with radiation absent from the rate.

The sound horizon $r_{\rm s} = \int_{z_{\rm rec}}^{\infty} (c_s/H)\, dz$—whose baryon-loaded $c_s$ is ordinary content microphysics and whose expansion is the leaf rate the plasma runs on—carries no early-universe parameter at all once the plasma is handed over at the branch point, where every acoustic mode is outside the horizon and the integral converges at its upper end.

The branch point is a locus at which the areal chart's invariants diverge—$r = 0$ is a genuine infinite-curvature locus of that chart—while the curvature of the manifold the cut is taken on is set by $\alpha$ everywhere, so the divergence is the areal coordinate degenerating and not a scale of the geometry [19, 17]. *And it is the substrate reading of the branch point that makes the integral well posed*: the modes are outside the horizon there and the plasma's own expansion carries them, so the sound horizon converges at its upper end without a floor having to be placed under it by hand. Taking $c_s$ the ordinary baryon-loaded value—the matter the plasma carries is pressureless, and the geometric throat ratio $\sqrt{3}$ does not set it—the acoustic scale is computed rather than met: integrated on the leaf rate from the branch point, it returns an angle whose comb sits at 298.0 against the sky's 298.4 with the background fixed by the distance data alone. *This is where the two rates part, and it is not where the standard model's radiation sits.* There the whole expansion carries radiation and the ruler is pinned by the matter density's physical value; here the foliation's own rate carries none, so the distances are radiation-free while the plasma's scales ride a leaf rate that is not, and the two lengths the angle is built from sit on opposite sides of that split. The beginning is well posed: the branch point—finite in the substrate's curvature, whatever the $r$-chart geometry does there (§2.2)—carries every acoustic

mode outside the horizon and the potential at its regular value, so the sound-travel integral converges at its upper end and needs no lower endpoint placed by hand.

*What the construction fixes, and what it does not.* The empirical forcing of the cosmic foliation bears on the rate, not the matter content—the matter density being the bend of the spatial cut [11, 7, 8]—so the content the plasma carries is measured rather than set by the geometry; the cold seam (Gibbons–Hawking temperature [33] $\sim 10^{-30}\,\mathrm{K}$, some thirty orders below recombination) confirms there is no geometric source to set it instead.

The quantities this cosmology actually uses are the epoch $x_0$ (equivalently $\Omega_m = 2/(x_0^3+2)$; *$x_0$ is the expansion read in units of the geometry's own fixed offset $r_N$, not the offset itself, which the Nariai condition fixes at $\alpha/\sqrt{3}$*), which the rate carries and the baryon-acoustic data fix; $H_0$, fixed by the same data together with the acoustic angle; and $\omega_b$ and $\omega_\gamma$, which the sound speed carries and which are measured directly and framework-independently. *There is no early-universe parameter among them*: with the plasma handed over at the branch point the sound horizon has no lower endpoint to place, so the angle is an output of the rate rather than a calibration of it.

*The physical matter density $\omega_m = \Omega_m h^2$ then does enter, and it enters once, through equality.* Matter–radiation equality is $1 + z_{\mathrm{eq}} = \omega_m/\omega_r$ with $\omega_r$ measured, so the epoch the distance data fix also fixes how long the plasma spends radiation-dominated, and with it the driving that sets the peak heights and the wavenumber at which the envelope turns over. *One quantity carries the expansion history and the acoustic driving together*, which is why the spectrum below is a test of the background rather than a further fit to it.

*One thing about $\Omega_m$ is worth stating plainly, since the construction's matter sector invites the question.* The dust the rate carries exceeds the baryons by about a factor of six, $\Omega_m \simeq 0.31$ against $\omega_b/h^2 \simeq 0.05$, so most of it is non-baryonic. *That is not a component this cosmology introduces, and it is not one it derives.* What the matter sector fixes is the number of families, their chirality, the $S_3$ relating the three walls and the cyclic $\mathbb{Z}_3$ relating the generations, and *nothing about their internal content* [20]; abundances are measured content throughout, the matter density being the bend of the spatial cut. So the non-baryonic fraction stands with $\eta$, $A_s$ and $n_s$: inherited, not set by the geometry. *What the construction does supply is a place rather than an abundance.* Its correspondence with the Standard Model's content carries a right-handed neutrino as the sixteenth Weyl fermion of each generation, and fixes that state's place while fixing none of its couplings [20]; a species that is never thermalized contributes nothing to $N_{\mathrm{eff}}$, so nucleosynthesis is untouched and no prediction is made or needed here. *The point of the paragraph is the division of labour*: a candidate the construction already contains, an abundance it does not claim, and no third thing wedged in between.

The radiation amplitude at a start placed by hand, $\omega_r(1 + z_{\mathrm{start}})/\omega_m$, is that start read in units of a density rather than an inheritance; with the plasma handed over at the branch point there is no such start and no such amplitude. *And the epoch the distance data fix lands where the microwave background's own matter density sits*: the rate's $\Omega_m$ at the $H_0$ the angle and the distances return together gives $\omega_m = \Omega_m h^2$ within a couple of per cent of the standard inference, so equality falls within a couple of per cent of flat ΛCDM's with nothing fitted to a spectrum[P15R25].

What this cosmology does carry from outside is the composition: the baryon-to-photon ratio $\eta$, exactly as flat ΛCDM carries it from a baryogenesis outside the model proper—measured, not derived—and with it the light elements, synthesized from the hot handover of the progenitor collapse, this universe having formed at the event horizon of a black hole in a previous one, equally outside the construction and equally measured.

That the composition is measured rather than derived is no defect: a framework is credited for *requiring* the structure it explains—here the rate, and the acoustic structure that follows from it—not faulted for measuring the boundary data it does not. *One naturalness argument about the handover fails, because it compares a quantity with itself at two epochs nine orders*

*apart.*

Complete collapse does liberate binding energy of order the rest mass ($GM/Rc^2 = \frac{1}{2}$ at the Schwarzschild radius), and it is tempting to read that as making a handover carrying radiation of order the matter density the natural scale—order unity, rather than the small value $\eta$ takes. But $\rho_r = \rho_m$ occurs at $T \simeq 0.8\,\mathrm{eV}$, whereas the handover sits four orders *above* the deuterium bottleneck at 0.07 MeV, since that is what the nucleosynthesis computed on it requires [19]: at the bottleneck the ratio is already $\sim 9 \times 10^4$ and at the handover proper $\sim 9 \times 10^8$. A handover carrying radiation merely of order the matter density would be a handover with $\eta \simeq 0.5$, which leaves neither light elements nor acoustic peaks. *So the naturalness question runs the other way*: what the progenitor must deliver is a plasma with of order a billion photons per baryon, and the binding-energy scale offers no head start toward it. *One route to that derivation is closed from the geometry's side*: the constituent count the matter sector carries is exactly conserved on this history, its parity-odd source vanishing on the spherically symmetric class the cosmogenesis premise fixes, so what such a derivation needs is a chiral *state* on the spatial layer rather than a chiral member[P15R152]—and that state is not available on this layer, the same thermal regularity that fixes the tower's boundary condition carrying no helicity label, so the two helicity towers are populated alike and the parity-odd content cancels between them[P15R154]. *And the amplitude a start placed by hand would carry is not a feature of this construction either*: given $\eta$ and the measured matter-to-baryon ratio, both of which the composition inherits regardless, it follows from standard thermodynamics alone, $[1+\nu](\pi^4/30\zeta_3)\,T/[\eta\,(\omega_m/\omega_b)\,m_N]$. *So what the handover supplies is one composition datum* (receipt: `the_two_data_are_one.py`[P15R26]). $\eta$ may remain a measured boundary condition indefinitely at no cost to what this paper computes. *And the inheritance has a structural reason rather than being a gap left standing* [9, 19]: the infall thermalizes four orders above the deuterium bottleneck, so the progenitor's composition is erased and the abundances are synthesized in the window rather than inherited, while what survives that passage is baryon number, which no dissociation destroys. *So $\eta$ is inherited and the abundances are predicted*, and the asymmetry is worth stating exactly: baryon number protects the numerator, while what protects the *ratio* is the adiabatic invariance of the specific entropy — the peak perturbing it by $5.5 \times 10^{-10}$ of a standing $1.15 \times 10^{10}\,k_B$ per baryon, so that $\eta$ survives the erasure because $\eta$ is small [19]. *And a derivation of $\eta$ is not needed, because there is nothing here for one to be about.* It enters the progenitor from the ambient universe by composition-sharing and leaves the branch point unchanged to one part in $2 \times 10^9$: *transmitted, not determined* — which is the exact dual of the radiation amplitude's closure, that one turning on a quantity with no single value to transmit. *So the neighbouring statement strengthens from may to must: $\eta$ remains a measured boundary condition, at no cost.*

*The synthesized composition.* The same collapse fixes the light-element composition, and the geometry sharpens how. The infalling matter compresses adiabatically—the plasma is optically thick, so the photon bath compresses with it—and heats far above the deuterium bottleneck ($T_\mathrm{D} \simeq 0.07\,\mathrm{MeV}$) for every progenitor: the compression does not stop at horizon crossing but continues to the branch point, and the peak is bounded below by the infall energy at the horizon, of order the rest mass per nucleon, independently of the progenitor's mass (the cosmogenesis paper [19]); and across the branch point the expanding leg cools back through the window. *The window is therefore crossed twice*, and at the branch point the temperature exceeds the binding thresholds by any margin, so nothing the infall built survives to be counted. Because the effective rate through the window is the standard Friedmann rate, the expanding leg is a standard big-bang nucleosynthesis run from free nucleons [19]. On a first sample this is favourable, and on one element advantageous. The helium-4 fraction $Y_p \approx 0.245$ sits at the observed value; the near-zero metallicity of the oldest systems follows from the handover photodissociating the progenitor's processed composition back to light elements; and because the run is a standard nucleosynthesis, the lithium-7 abundance carries its standard threefold over-prediction—the standing lithium problem shared with flat ΛCDM, neither dissolved nor worsened. Deuterium

sits at the observed $D/H \approx 2.5 \times 10^{-5}$: the freeze-out temperature the data demands is the standard bottleneck, and that is exactly the temperature at which the standard rate delivers a freeze-out—target and mechanism coincide un-tuned. The peak calculation that forces this synthesis reading over an inherited one, and the full multi-abundance account, are given in the cosmogenesis paper [19], where the network is integrated explicitly on the cooling leg and returns deuterium, helium-4 and helium-3 at their observed values with lithium-7 at the shared over-prediction, jointly from the single inherited datum—so one progenitor handover does fit the pattern together; what stands as its open edge is the last-percent precision against the measured abundances, a data-confrontation the produced pattern exposes rather than a computation still to be run.

*The load-bearing claim.* The acoustic scale is computed from the rate rather than calibrated against it; the standard $c_s$ it uses presumes the inherited medium to be an ordinary photon–baryon plasma, which the abundance comparison confirms—helium-4 and deuterium at their observed values, lithium-7 the shared standard problem (the cosmogenesis paper). The load-bearing, falsifiable claim remains the prior one—that radiation carries no term in the foliation's own rate—of which the split between the two rates, and with it the whole acoustic reading below, are the consequences. The sound horizon is where a purely geometric expansion history and the standard radiation-governed one are told apart in the data.

*The expansion history, confronted with the late-time data.* The distances and the ruler do not share a common factor: $D_M$ and $D_H = c/H$ carry the stacking rate's $1/H_0$, while the sound horizon is accumulated on the leaf rate, which carries the radiation the stacking rate does not. So the baryon-acoustic observables $D_M/r_\mathrm{s}$ and $D_H/r_\mathrm{s}$ constrain $H_0$ here as they do in flat $\Lambda$CDM, and the fit returns it: against the DESI DR2 dataset the rate fits at $\chi^2/\mathrm{dof} \simeq 1$ with $\Omega_m \simeq 0.30$ and $H_0 \simeq 68.6$, and the acoustic angle alone, from independent data, agrees within a twentieth of a unit. *What the two frameworks still do not share is the ruler's provenance*: there it is pinned by a radiation-era physical density, here it is the plasma's own accumulation on a rate whose geometric part carries no radiation at all. *The two rulers make the constraint differently, and the difference is where the radiation sits.* In flat $\Lambda$CDM the ruler is pinned in physical length by the radiation-era matter density, so the same distance data tie the fit to one $H_0$; here the ruler is the plasma's own accumulation on the leaf rate and the distances ride the stacking rate, so the two enter the fit through different quantities and the constraint is the joint one[L9].

The confrontation carried to the state-of-the-art DESI DR2 dataset [23]—thirteen distance measurements across seven tracers ($0.3 \lesssim z \lesssim 2.3$), with the full per-tracer $D_M/r_\mathrm{s}$–$D_H/r_\mathrm{s}$ correlations—returns $\chi^2/\mathrm{dof} \simeq 1$ at $\Omega_m \simeq 0.30$ and $H_0 \simeq 68.6$[P15R157], and the acoustic angle computed on the same background agrees with that $H_0$ from independent data. *The determination is the leaf ruler's*: on a radiation-free ruler $H_0$ would cancel algebraically from the ratios and the data would not see it, while the leaf rate carries $\omega_r/h^2$, so the same thirteen measurements pin the rate here as they pin it in flat $\Lambda$CDM.

*One thing about $\Omega_m$ is worth stating exactly.* It is fitted from the distance data here rather than taken from the microwave background, and the invariant is the offset $x_0$, read as $\Omega_m$, and not $\omega_m$—holding the physical density fixed instead imports the $\Lambda$CDM reading. The expansion history the fit returns is a property of the whole redshift range rather than of one angle, and it rides the very rate whose branch-point-located scoping keeps the nucleosynthesis window standard [19]. The cosmic chronometers [26], which read $H(z)$ directly, test the shape of that history independently of any ruler.

### 2.5 Where the acoustic modes stand when the plasma is handed over

The cosmological beginning is the branch point $r = 0$ at the close of the lift, where the conjugate branch becomes the matter branch [9, 5]. The curvature invariants of the $r$-chart diverge there because the areal coordinate degenerates, while the throat itself is the substrate's defining

constant $\alpha$ and the bead continues through by analytic continuation [36]. The lap's seams lie elsewhere, at the two unit-speed loci $r = -2\alpha/\sqrt{3}$ and $r = +\alpha/\sqrt{3}$. We record the consequence that organizes the perturbation sector.

**Proposition 1** (Every acoustic mode is outside the horizon at the branch point)**.** *Approaching $r = 0$ on the contracting leg the comoving Hubble scale $aH$ grows without bound, so the comoving horizon shrinks to zero and every mode leaves it before the crossing, on either rate and for every wavenumber.* What is handed to the expanding leg is therefore the frozen adiabatic state—*the effective temperature at an extremum with no velocity and the potential at its regular value—and the collapse leg's contribution to it is an amplitude and not a phase.*

*What the modes then meet on the expanding leg is fixed by the same background.* The leaf congruence carries a radiation term and the perturbations evolve on it, so the wavenumber entering the horizon at equality, $\ell_{\rm eq} \simeq 144$ at the epoch the distance data fix, separates the modes driven through a radiation-dominated entry from those entering later: the first acoustic peak sits above it and is driven, which is what §4.1 computes in closed form and §4.7 measures.

*The two loci a reader may run together are distinct, and the distinction is load-bearing*[P15R79]. The branch point is the locus $r = 0$, which no finite cosmic-time layer reaches and which therefore carries no finite redshift at all; a configuration that instead starts the plasma at a finite redshift on the expanding leg is starting it somewhere else, and there the census is the opposite one—modes inside the horizon, and a state handed over mid-oscillation.

*Argument.* The comoving horizon on the contracting leg, $1/(aH)$, vanishes as $r \to 0$ because $2M/r$ diverges there, so $k/(aH) \to 0$ for every $k$; that census is re-derived from the metric function directly, the comoving horizon being largest at the front seam and falling to zero toward $r = 0$[P15R79]. The equality multipole follows from $1 + z_{\rm eq} = \omega_m/\omega_r$ on the leaf background at the epoch the distance data fix, projected on the stacking rate the distances take: $k_{\rm eq} = a_{\rm eq}H(a_{\rm eq})/c$ evaluated on the background rather than on its matter-dominated closed form, which is a third out at equality[P15R161]. □

*What this settles is a premise and not a mechanism*: it fixes what the plasma is handed and where each mode stands when it starts, and it does not measure how much of the comb the subsequent driving supplies. That measurement is the transfer of §4.7, which computes the driving directly; the census above is what the transfer is carried from.

## 3 What crosses, and what does not

The transmission dichotomy above is an argument from the absence of an imprinted scale. A mechanism is given for it, and it also settles which content the branch point does *not* carry.

The imaginary-time segment preceding the branch point acts on a perturbation as a Euclidean kernel [35], damping a mode of frequency $\omega$ by $e^{-\omega|\Delta\eta|}$. *The segment's conformal length is known in closed form rather than by quadrature*: on the bead's own conformal time $\mathrm{d}\eta = \mathrm{d}\tilde{\tau}/r$ the lap closes as an exact $1 : \sqrt{3} : 2$ right triangle whose imaginary leg is $c_0 B(\frac{1}{6}, \frac{1}{2})/2 = 3.33874$ with $c_0 = 2/(\sqrt{3}\,2^{1/3})$—a pure number, carrying neither mass nor epoch—so $|\Delta\eta| \simeq 1.7 \times 10^4\,$Mpc on this construction's scales[P15R245]. *What the kernel's exponent is, as against its length, is not settled here, and the gap is a factor and not a detail*: the rate applied below takes the mode's frequency to be $\omega = kc_s$, at the plasma's sound speed, while the substrate field whose equation the lift carries has no sound speed in it at all, and the two exponents differ by $\sqrt{3}$ on this one segment. *Which of the two the segment's own field obeys remains open*, and the consequences drawn below are those the sound-speed reading supports. Taken naively this would annihilate everything: at $\ell = 220$ the factor is $10^{-68}$. *But $\omega$ is the frequency of an* oscillating *mode, and at the branch point no mode is oscillating.* The comoving Hubble scale $aH = |\mathrm{d}r/\mathrm{d}\tilde{\tau}| = \sqrt{|1 - f|}$ grows without bound as $r \to 0$—normalised at the seam, where it is exactly unity, it is 1.96 at $|r| = 0.1\,\alpha$ and 19.6 at $10^{-3}\alpha$—so the comoving

horizon $1/aH$ shrinks to zero and *every* mode exits it and freezes before the crossing. A frozen mode has no oscillation for the kernel to damp. *So the amplitude and the tilt cross unaltered, and the argument from the absence of a scale is recovered as a consequence of horizon exit rather than assumed. One limit of that census should be stated where it is used rather than left to be discovered*: it is computed on the branch where $2M/r$ is positive, on which the rate has no turning point at all, while the branch carrying the imaginary segment turns around at $|r| = (2M)^{1/3}$—the areal radius $A$ identically—where the rate is *zero* and every mode is inside the horizon instead. *Which of the two the freezing argument should be run on is the same question as the exponent above*, and the corpus computes a census on only one[P15R244]. *And the two branches are not each other's image, which closes off the identification a reader would reach for first*: the lap's own closure is the *translation* $r \mapsto r + \sqrt{3}\,\alpha$, which is what carries the back unit-speed locus to the front one, $-2\alpha/\sqrt{3} + \sqrt{3}\,\alpha = +\alpha/\sqrt{3}$ exactly, whereas the reflection that $|r|$ implements carries neither locus to a locus at all. *The two agree at one point of the lap and nowhere on the imaginary segment*—at $r = -\sqrt{3}\,\alpha/2$, the antipode of the branch point—so the census's stretch and the segment's are two distinct places on one closed lap rather than one place read twice[P15R248]. *What separates the two is the radiation content itself, which is what makes the question a determinate one rather than a matter of presentation*: written on the areal radius, the propagation equation on the radiation-carrying congruence and on the vacuum curve differ by $A_r$ times the radial Laplacian of the perturbation and by nothing else—wavenumber-independent, exactly linear in the radiation constant, and vanishing only where that constant does. *So the two are not one equation re-timed*: the same equation carried to the areal radius by the proper and the conformal clock agrees on both congruences up to the measure factor, so no monotone re-timing reaches the difference[P15R250]. *And on the construction's own inherited datum that added term is not a correction but the larger of the two*: the ratio of the radiation term to the matter term is $2r_N/r$ exactly, so it is 2 at the seam, $2^{2/3}$ at the turnaround, and falls to unity only at the lap's other unit-speed value. *That last coincidence is forced rather than fitted*—equality sits at $A_r/2M$, which the datum places at $2r_N$, and the two unit-speed radii stand in the exact ratio 2:1 about the branch point[P15R251].

*What does not cross is the oscillatory content itself*, and that is the collapse leg's acoustic phase: a mode that was sub-horizon earlier carries an oscillation, and the kernel annihilates it—an exponent of $-152$ at the first acoustic peak, integrated across the segment's own sound-speed profile rather than at the single value the estimate above uses[P15R80][P15R245]. This is required independently by the observed peak positions, which exclude a transmitted leg phase by a factor of 21–41 in the required distance [9], and it is what leaves the comb to be set on the expansion side by the characteristic data of §4. *The two results are one statement*: the crossing transmits what is frozen and destroys what oscillates, so the progenitor supplies the spectrum's amplitude and shape while supplying none of its phase.

*The exponent has two factors, and the argument above varies only one of them.* It is $e^{-kc_s|\Delta\eta|}$: holding $c_s$ at the plasma's value and varying $k$ reads it as a *scale* criterion, frozen against oscillating, which is the reading just given. Holding $k$ and varying $c_s$ reads the same inequality as a *species* criterion, and that reading supplies something the first does not. A pressureless component has $c_s = 0$ identically—not small, not slowly varying—so its exponent vanishes at every wavenumber; on the segment its mode equation is $\psi'' = 0$, whose constant solution the kernel leaves untouched with no approximation at all. *The matter sector's inheritance is therefore exact rather than adiabatic*: it does not rest on horizon exit and it does not degrade at small scales, where the frozen-mode argument does. The two arguments agree wherever both apply, and only the pressureless one is available everywhere. *Neither reading carries a constraint the other lacks.* On this geometry the exponent at the first acoustic peak is $\simeq 1.6 \times 10^2\, c_s/c$, so a component arriving *oscillating* with a per-cent sound speed would be suppressed by $e^{-1.6}$ and one with ten per cent by $e^{-16}$ (receipt: `the_filter_is_a_sound_speed_filter.py`[P15R80]). *But the exponent has nothing to act on.* The kernel acts on *oscillatory* content—a frozen,

zero-frequency mode is a fixed point of it—and on the progenitor's own interior nothing arrives oscillating: $|aH| = |a'/a|$ diverges as $1/x$ at the branch point while $c_s$ saturates at $1/\sqrt{3}$, so $c_s k/|aH| \to c_s k x \to 0$ for every $k$, and every mode from $\ell \simeq 28$ to $\ell \simeq 2475$ exits the comoving sound horizon strictly before the crunch—the highest with a fifth of a per cent of the collapsing leg still to run [19]. *So the two readings do not divide the work between them: the scale criterion is satisfied by everything, and the species criterion selects nothing. The crossing is lossless for every species rather than for cold ones alone*, and the pressureless argument's positive content—that the inheritance is exact rather than adiabatic—holds of every species and not of dust alone.

*One consequence of that freezing bears on the acoustic sector rather than on the crossing, and it is computed rather than left as an inference.* Every mode is outside the comoving horizon at the branch point (Prop. 1), so each re-enters on the expanding leg at its own time, and what it meets there is fixed by the rate the perturbations run on. *On the leaf rate that re-entry is an event*: the plasma carries radiation, the potential decays on the sound-crossing time, and the modes entering while radiation dominates—everything above $\ell_{\rm eq} \simeq 144$ at the epoch the distance data fix—are driven. *That driving is not an inference about the potential's form but a shape read off the two arms*: subtracting an undriven reference that calibrates at unity, the control's acoustic phase at first turnover is flat in wavenumber to 2.1% while this arm's varies by a factor of 3.9 across the same band[P15R64], and §4.1 carries the same driving in closed form.

*What the rate rule closes is the alternative reading*, on which the modes re-enter under pressureless matter and nothing is imprinted: the potential equation for a pressureless component, $\Phi'' + 3\mathcal{H}(1+w)\Phi' + [2\mathcal{H}' + (1+3w)\mathcal{H}^2]\Phi + wk^2\Phi = 0$, *contains no $k$ at all once $w = 0$*, so on a radiation-free rate the problem is scale-free and a scale-free problem imprints nothing[P15R62]. That is the stacking rate's answer, and the perturbations do not take it: they take the leaf's, which carries the radiation term and therefore the driving.

# 4 What the substrate determines: coherence from the null boundary

The acoustic peaks are sharp because the modes of a given wavenumber share a common phase at last scattering. Inflation produces this coherence dynamically, by freezing modes while superhorizon. CR obtains it instead from the character of the branch point.

The branch point is a *null* surface—the cosmological horizon promoted to the initial layer [9]. The initial-value problem on a null surface is characteristic, not Cauchy: the data that determine the future are one free function per mode on the surface together with regularity along the generators, rather than a field and an independent momentum. A single datum per mode is a single phase per mode; there is no second, independently specifiable quantity to randomize the relative phase.

We therefore argue that coherence is not imposed but is what regular characteristic data on a null surface *is*.

This is borne out quantitatively, and the scope of the demonstration is stated exactly.

Evaluating the tightly-coupled mode function at last scattering with a *common* phase on that surface gives a sharp regular comb, whereas drawing the phase independently per mode—the second, free datum a Cauchy surface would admit—washes it out entirely, leaving flat power (peak-to-trough contrast $\sim 1$ against the coherent comb's many orders of magnitude; receipt: `verify_coherence_comb.py`[P15R18]). *That establishes the mechanism—one characteristic datum per mode is one phase per mode, and coherence follows—and it establishes nothing about where the peaks fall.* The receipt writes the mode function down as $\cos(kr_{\rm s})$ with $r_{\rm s}$ and $D_C$ as literals—and the $r_{\rm s}$ it writes is the *standard* 147.0 Mpc rather than any value this construction derives; no mode is propagated in it, as its own scope note says, so the spacing $\Delta\ell \simeq \pi D_C/r_{\rm s} \approx 296$ is

the arithmetic of an assumed expression on an imported ruler rather than an output. *It is not the result of propagating the modes, and is not offered as one*[P15R61].

*What the peak spacing lacks here is not agreement but a derivation*: this receipt asserts it, and it is the propagated modes of §4.7 that compute it.

The null boundary admits only the first column: one characteristic datum per mode is one phase per mode.

The peak *heights* are then carried by a structural argument rather than a bespoke transfer. The driven acoustic amplitude is the resonant Fourier magnitude of the potential's evolution, $|\tilde{\Psi}(\omega{=}1)|$—exactly invariant under time reversal, $|\widetilde{\Psi(x)}| = |\widetilde{\Psi(-x)}|$, a Fourier theorem confirmed numerically on the radiation-era transfer function and on a random control[P15R20].

The equality is a theorem about Fourier magnitudes and holds whichever side the driving lives on. *Which side it lives on is fixed by the rate rule*: the perturbations run on the leaf congruence, whose matter–radiation equality is fixed by the epoch the distance data fix, so the plasma's early history is radiation-dominated and the modes carrying the first peak enter the horizon while it is. *The driving is therefore on the expanding leg*, and the collapse leg's contribution to the state that meets it is an amplitude and not a phase.

*What the state at the handover is has to be said at one locus, and the construction supplies one.*

*At the branch point every acoustic mode is outside the horizon and the leg's closed-form potential has returned to its regular value*, so the state arriving there is the ordinary frozen adiabatic one: the effective temperature at half the potential with no velocity, the potential unspent, and the collapse leg's contribution surviving as an amplitude alone—the wavenumber-independent $0.4835\,\Psi_i$, within three per cent of the frozen value $\Psi_i/2$ the free oscillator carries at every wavenumber. Specified that way the radiation-era driving is applied exactly once, and the specification carries no free choice, the other three codings each making one. *It is a limit rather than a parameter*: carried back over two decades in starting redshift the peaks hold to the grid step and the height ratios to half a per cent, while a start an order of magnitude later has not converged.

The collapse-side driving that argument relied on carries a qualification of its own receipt: completeness holds for the modes whose entry precedes the horizon maximum at $r_* = 1.53\,r_{\rm seam}$, entry on the rising branch being earlier the higher the wavenumber, so the acoustic band is on the safe side of that boundary by construction and the low-$k$ end is where it would bite[P15R48]; what a Boltzmann reference returns for flat ΛCDM's own heights is $P_1/P_2 \approx 2.2$, $P_2/P_1 \approx 0.45$, $P_3/P_1 \approx 0.44$ (`camb_reference.py`[P15R21])—*those are the reference's values, not this arm's*, and what this construction's own instrument returns is set out above. The *damping* tail ($\ell \gtrsim 1000$) is fixed by the three-level rule (§2.4) rather than left to judgement. Recombination is observable cosmology from the branch point onward, and the photon diffusion length $r_D$—like $r_{\rm s}$—is a scale the plasma itself accumulates, so it takes the leaf's rate: the random walk runs *in* the content, whose own radiation gravitates in its own expansion, its Thomson microphysics ordinary content physics and its expansion the leaf's (§2.4).

This is the same scoping the cosmogenesis paper derives for the nucleosynthesis window [19], and the two windows fall on the *same* side of it rather than on opposite sides: there the self-gravitating excursion sets the leaf rate radiation is included in, and here the plasma accumulating its own scales takes that same rate. *What differs between the two is not the rate but what is read off it*—an abundance there, an angle here—and the angle alone is divided by a distance read across leaves, which is where the rate radiation is excluded from enters.

This is forced by consistency: one may not take the rate geometric for the peak spacing and radiation-included for the diffusion—and *the peak spacing this construction computes is the leaf accumulation*, which is measured rather than assumed (§2.4), so the demand puts both lengths on the leaf and not on the foliation's rate. *The two rates are far apart*: the geometric one is ∼13% below the radiation-included one at recombination ($\rho_r/\rho_m \approx 0.3$) and further below

it earlier, where the inherited radiation is a larger fraction, and the closed-form derivation of §4.2 puts 10.8% of that gap into the diffusion length by itself. *But a rate the two lengths share cancels from their ratio*, and what the sky reads is the ratio, $\theta_D/\theta_* = r_D/r_\mathrm{s}$. *So the rate gap is not what the signature is made of.* What it is made of is what §4.3 derives on the leaf assignment—where the plasma is handed over, and how far the diffusion integral is carried—with the handover fixing the size and the common endpoint at the visibility peak fixing the sign. *One consequence of the signature being that small should be stated rather than left implicit*: the reference the instrument is validated against agrees on $r_\mathrm{s}$ to 0.28% (144.0 against 144.4), which was a comfortable margin against a signature of several per cent and is a third of this one, so the word *validated* here carries a thinner margin than it did and the figure is quoted with that in view[12]. What a stated tolerance carries anywhere in this corpus's computational layer is measured rather than assumed: every `abs(...)<tol` comparison in the receipts rewritten twice, once to displace the compared quantity above its tolerance and once to collapse it below, so that 243 of the 245 receipts that pass are shown to have at least one such comparison that actually gates their verdict, the two exceptions being a display filter and a skip that prints its own reason[12L22].

This is a genuinely CR-specific piece of the height pattern, and *what makes it so is the handover and not a rate*: a rate the two lengths share cancels from their ratio, while the epoch at which the plasma is handed to the expanding leg is fixed by the construction and is not a parameter. So the signature is not an artefact of laying the standard radiation-governed $r_\mathrm{s}$ on a rate it was not accumulated on, the calculation belonging to neither framework this paper warns against (§1).

One thing about this signature is settled and one is not, and it is essential to keep them apart.

Settled: the effect in $r_D$ is genuinely CR-specific and does not dissolve under the density that would ordinarily rescale it—$r_D \propto \omega_b^{-0.31}$, so a percent-level $\omega_b$ (fixed by primordial deuterium and the peak-height ratio) shifts it by well under a percent (receipt: `damping_reabsorption.py`[P15R23]). *And what the signature does to the observed high-$\ell$ power is measured, with a verdict that is neither of the two a reader expects.* Scaling the diffusion length by the instrument's own measured ratio multiplies the damping by a Gaussian in $\ell$, and a tilt is a power law — so over a restricted high-$\ell$ range the one is well approximated by the other. *Refitting amplitude and tilt therefore absorbs the signature almost entirely in shape*, and what it costs is a shift in the inferred tilt. *Both scale with $r^2-1$, and at the adjudicated ratio that factor is small*: with both lengths carried to the visibility peak, $r = 0.9918$, the signature costs 1.70 in $\chi^2$ over the 185 full-range bins—under a hundredth per bin—and a tilt of $+0.0034$ absorbs seven tenths of it. *Terminating the diffusion integral at a sharp recombination cut instead, $r = 1.0231$, gives* 12.6 *and a tilt of* $-0.0093$, and a ratio of 1.082 gives 160 and $-0.034$ on the same base and the same damping scale[P15R168]. *So the hundredfold collapse is the configuration moving and not the fit improving. So the signature does not vanish under a refit — it moves out of the spectrum's shape and into an inferred parameter. What a refit buys in shape it pays for in parameters*, and that direction is the robust part.

*And how local that approximation is can be stated exactly, which settles what a joint fit would actually meet.* The signature's own logarithmic slope is $\mathrm{d}\ln(\text{ratio})/\mathrm{d}\ln\ell = -2(r^2-1)(\ell/\ell_D)^2$ *— not a constant, but growing as $\ell^2$*: across the likelihood's own $\ell = 30$–$2508$ it runs from $-0.0002$ to $-1.31$, a factor of seven thousand, which is exactly $(\ell_\mathrm{max}/\ell_\mathrm{min})^2$[P15R3]. *So a single tilt is a one-parameter fit to a function whose slope varies by that factor over the data it is scored on*, and the displacement quoted above is the signature's true slope at $\ell = 381$ and at no other multipole. *The consequence is the useful part*: a joint fit does not encounter a shifted tilt but a residual whose *shape* is a Gaussian in $\ell$, and no tilt removes a Gaussian — so what such a fit weighs is the shape residual, which is the quantity already bounded from above. *And a displacement in $n_s$ is not by itself a tension here*, the tilt being inherited boundary data rather

than predicted: a different value of an inherited input is a different input. *What could be a tension is a constraint on $n_s$ that does not run through the damping tail* — the low- and mid-$\ell$ shape, the polarisation spectra, lensing — which is what a joint fit means and is where the question properly sits.

*Two bounds travel with that and neither can be left to a reader. The shape residual is an upper bound*: freeing only the named degeneracy is the least absorption a fuller refit could achieve, so a wash is a wash *a fortiori. The tilt displacement is not a bound*, because more free parameters would share it — the baryon density moves the diffusion length itself. *And the magnitude is not robust to a factor of three*: imposing the ratio on the measured spectrum gives the figures above, while applying it inside the instrument's own source — where it also reaches the Doppler term through a derivative — gives 0.92 in $\chi^2$ per bin. *The larger comes from the more faithful route, so the wash is stated at the weaker bound* and closing that factor is named here as work not done.

*The joint fit returns the residual this reading predicts.* Fitting amplitude and tilt together against the signature-carrying spectrum, on `plik_lite`'s own bandpowers and covariance, the residual left after the fit is the analytic Gaussian-minus-power-law shape to a correlation of 1.000000 — *not merely non-zero, but the predicted form. The residual's form is the finding and its size is the configuration's*: at the adjudicated ratio it is under a hundredth in $\chi^2$ per bin, and 1.50 per bin at a ratio of 1.0926 on the same window. *And the displacement is not one number*: fitted over $\ell = 100$–$1760$ and over $\ell = 1400$–$2508$ it differs by a factor of 6.3 across windows of the same data, so a displacement is the value of nothing without the window it was fitted over. *And the tilt takes a fixed share of it rather than a variable one*: seven tenths at the adjudicated ratio and seven tenths at 1.0926, so what the refit can and cannot do is a property of the shapes and not of the size[P15R4].

*Not* settled, and unchanged by that run: whether a displacement of that size in the inferred tilt is a *tension.* The tilt is inherited boundary data here, not predicted, so a different value of it is a different input; what would make it a tension is a constraint on $n_s$ that does not run through the damping tail, which this fit does not supply. The observable is $\theta_D/\theta_* = r_D/r_\mathrm{s}$, and translating it into a spectrum requires the full acoustic transfer, which is run (§4.7) and does not return high-$\ell$ peaks tracking flat ΛCDM's. A leading-order pass that simply *assumes* the high-$\ell$ peaks equal ΛCDM's and applies the extra damping produces a large high-$\ell$ deficit (receipt: `full_transfer_verdict.py`[P15R27])—but that assumption may not be imported for free: the same handover that fixes $r_D$ also governs the high-$\ell$ driving envelope, so CR's high-$\ell$ spectrum rests on short-wavelength collapse-phase driving rather than on the boost an expanding radiation era supplies.

*And the observable ratio is read at one epoch on both lengths, which is what fixes it.* The two-arm transfer measures the raw diffusion length at the visibility peak, and the sound horizon is carried to the same place, because a ratio of integrals taken to different epochs is not a ratio of anything (§4.3). *The endpoint is therefore not a free choice*, and the difference between terminating the two lengths together and terminating them apart is larger than anything the cosmological parameters do to the signature.

*That driving is computed below* (§4.1), and the calculation removes the licence the shortcut lacked: the envelope is derived on the collapse leg rather than imported.

Diffusion on the leaf rate therefore constrains, but does not by itself determine, the high-$\ell$ shape; the net effect is computed in §4.2; whether it is a tension, a wash, or a distinctive but consistent feature turns on a parameter refit and on the early integrated Sachs–Wolfe term. *Two confrontations are to be distinguished and both are run.* The spectrum computed on the background the distances fix is scored as it stands, one amplitude fitted so that what is compared is shape, and sits at 2.57 times the control's distance from the data. *Refitting the spectrum's own parameters to the microwave data*—the comparison the standard model is usually judged by—*leaves it at* 1.57 *times*: freedom closes about a third of the gap and leaves the rest (§4.7). The

control reproduces the sky essentially perfectly, which is what makes the comparison readable at all; and the verdict it returns is that the spectrum derived here is disfavoured against the standard model on this data. We state it as it stands. The effect the fit is run on is separately established: a real, computed, non-reabsorbable CR-specific diffusion-scale signature, under a per cent at the adjudicated endpoint and fixed by the handover rather than by a rate (§4.3), its observable consequence derived rather than entangled (§4.2). *What the fit does not reach is the join*: it weighs the five-parameter spectrum construction, and what carries that construction back to the geometry is the transfer work of this section, which is where the open end lies. The structural point that stands is the *mechanism of phase*: where inflation freezes the phase dynamically, CR fixes it geometrically on the single null layer; how the amplitudes are driven is what the transfer of §4.7 measures.

### 4.1 The collapse-phase driving envelope

The driving named above can be carried out in closed form, and doing so removes the entanglement the previous subsection had to leave standing. On the radiation-dominated collapse leg the potential obeys $\Psi'' + (4/\eta)\Psi' + (k^2/3)\Psi = 0$, whose regular solution is elementary, $\Psi = 3\Psi_i(\sin x - x\cos x)/x^3$ with $x = k\eta/\sqrt{3}$, and which is *even* in $x$[P15R47]—so the contracting leg carries the same closed form pointwise, a sharper statement than the Fourier-magnitude equality of §4 and consistent with it. Writing $\hat{\Theta} = \Theta_0 + \Psi$ removes the source exactly, $\hat{\Theta}'' + (k^2/3)\hat{\Theta} = 0$: the effective temperature oscillates freely, and the $k$-dependence rides entirely in the amplitude at horizon entry and in the decay of $\Psi$.

Horizon entry is fixed by the construction rather than fitted. The comoving Hubble radius is built from the slicing paper's own turnaround function, $(rH)^2 = (1-f) + A/r^2$ [5], and has a single turning point on the whole excursion, at $r_* = 1.53\, r_{\rm seam}$ on the corpus's inherited datum; the *seam* sits just inside that turning point, on the rising branch, so the acoustic modes are already within the comoving horizon there—a property of the turnaround function's geometry rather than a numerical accident of the background, and a statement about the seam that Prop. 1, being about the branch point, does not make[P15R48]. *The locus matters and the two available ones give opposite answers*: the branch point lies at the far end of the same rising branch, where $2M/r \to \infty$ carries $aH$ up without bound and the comoving horizon to zero, so there *every* mode is outside it (§3). It is the seam, not the crossing, that this construction reads—the inversion derived from the metric function rather than asserted[P15R79]. On the leg the radiation term dominates because it is constant while the others vanish, so $\eta = r/\sqrt{A}$ to about two per cent at the entry radius of the lowest mode in question and better the deeper the mode enters—sub-percent from $k \gtrsim 30\, k_s$[P15R49], and entry then occurs at $x \to 1/\sqrt{3}$ *in the deep sub-horizon limit*—the radiation era's own scale invariance, recovered here from the substrate geometry. *The envelope is therefore flat to the accuracy of that limit*: a mode at $k \simeq 10\, k_s$ enters within 2% of $1/\sqrt{3}$ and one at $300\, k_s$ within 0.1%, all of them reaching the *seam*—where the collapse leg ends—with the same driving amplitude $\Psi(1/\sqrt{3})/2$, and the residual potential is below a percent *on average* while oscillating up to some seven per cent of it where $\cos x_{\rm seam}$ is near an extremum[P15R50]. *Both are qualifications rather than identities: the limit is a limit and the average an average.* Its turnover sits at equality, and the equality is the background's own, $\Omega_m/\Omega_r$ on the leaf rate: at the epoch and rate the baryon-acoustic data fix it falls within a couple of per cent of flat ΛCDM's, and it moves with the matter density rather than being assumed[P15R53].

Two corrections that might have been expected do not enter. The baryon loading is proportional to the scale factor and the driving happens where the scale factor is smallest, so $R \lesssim 0.1$ where the driving happens and orders below that at entry; the odd/even asymmetry is imprinted afterwards, on the expansion side at $R \approx 0.6$, which is ordinary content physics on the observable leg. *The sky's own value, located by parabolic fit on the Planck binned spectrum, is* $P_1/P_2 = 2.2564 \pm 0.0772$*, and flat ΛCDM's is* 2.200*;* what this construction returns for the same

quantity is 2.264 on the polarisation path, on the transfer of §4.7 with its wavenumber integral converged, since a peak height is exactly what an unconverged integral misreports[P15R70P15R51]. The structural point stands without it*: the loading is the same in both because it is imprinted on the expansion side, and the argument is at the level of contents rather than of the driving—where the loading ratio alone would give* 3.5. And the free-streaming neutrinos the cooling-leg nucleosynthesis requires do carry anisotropic stress at these temperatures—the branch point is far below their decoupling—but their deep sub-horizon effect is a constant phase shift and a constant suppression, and a $k$-independent correction leaves a flat envelope flat[P15R52]. *The transfer carries the effect, and what it carries is a composite*: removing the anisotropic stress at both of its dynamical sites—the neutrino Euler equation and the shear term in $\Psi$—leaves the sector a perfect fluid at the same background density, and pushes every peak on both arms to larger multipole, the direction the free-streaming pull has, with the amplitude moving separately—its envelope by about a quarter. *But that operation removes the constant phase shift and the change in the potentials' decay together*, so what it measures is not the deep sub-horizon constant the argument above is about, and no value for that constant is read off here[P15R178]. What it does not leave is the closed form for $\Psi$ itself, the equation becoming integro-differential; the closure is a property of the skeleton, and the flatness is what survives.

## 4.2 What the envelope buys, and what remains

With the envelope derived rather than assumed, the diffusion signature can be stated without the entanglement. The diffusion length reduces to a single integral over the scale factor in which every microphysical constant sits outside, $1/k_D^2 \propto \int \mathrm{d}a\, g(R)/(Hx_e)$, so in the ratio the Thomson physics and the ionisation history cancel identically and the whole difference is carried by $H(a)$—which is the division of labour this section has argued for throughout. Both integrals close in elementary form, and on the inherited datum the geometric rate gives a diffusion length 10.8% longer[P15R54]. *That is a statement about the two rates and not about the observable*: the plasma accumulates both lengths on the leaf's rate (§2.4), so a difference common to them cancels from $\theta_D/\theta_*$, and the figure is quoted here because it sizes the rate gap the division of labour turns on rather than because it sizes the signature. *The elementary route and the code-validated one agree on it to about a point and a half*, which is the relationship an elementary derivation should bear to a code-validated one. The sound horizon is the same kind of integral, and it must be taken *from the branch point*: there is no observable expansion below it. Doing so returns $r_s = 146.4$ Mpc against 145.4 on the radiation-included rate—within 0.7% of each other, the exact crossing sitting a little lower at $z \simeq 6.7 \times 10^3$[P15R55]—the acoustic-scale calibration, met from the other end and to that accuracy rather than exactly. *Both are the elementary integral; carried through a full recombination history the radiation-included one comes out at* 144.0 *against a Boltzmann reference's* 144.4*, a per cent lower than the elementary value*[P15R22]. The observable, in which the common distance cancels, is then $\theta_D/\theta_*$ moved by under a per cent, its size fixed by the handover and its sign by the common endpoint both lengths are carried to (§4.3), and $\ell_* = \pi D_M/r_s = 302.2$ against the measured 301.76[P15R44].

The high-$\ell$ consequence follows with no free parameter: $C_\ell^{\rm CR}/C_\ell^{\Lambda\rm CDM} = \exp[-(\ell/\ell_D)^2(r^2 - 1)]$ with $r = \theta_D/\theta_*$—the same ratio the preceding paragraph derives, and nothing else—so with both lengths carried to the peak of the visibility function, $r = 0.992$, it is a rise of about two per cent at $\ell_D$ and four at $1.5\,\ell_D$, where terminating the diffusion integral at a sharp recombination cut instead, $r = 1.023$, turns it into a fall of four and ten[P15R168], the sentence and the ratio it defines being checked against each other rather than corrected separately[12]. *Whether that is a tension is a separate question this paper does not settle*: a confrontation refits the cosmological parameters, which is a likelihood analysis and not a closed-form calculation, and asserting exclusion from these numbers would repeat, at one remove, the error of holding everything else fixed while varying one thing. What has changed is the standing of the signature. The signature is a requirement of the construction, following from the level assignment and the handover and

nothing else, with its sign fixed by the endpoint the diffusion length is carried to.

*The early integrated Sachs–Wolfe contribution is present here, and the step that once denied it is worth setting out because the denial was an inference and not a measurement.* The rate at recombination is matter-dominated to nine orders, the substrate term being utterly negligible there[P15R46]: *that is true, and it is a statement about the rate.* The geometric rate carries no radiation term, radiation being inherited content read off the clock rather than something that sources the rate.

*But the effect is driven by the radiation* ***content*** *through the perturbation source, not by the rate.* The content enters the source under either rate assignment, and *at recombination this cosmology's content is* 21.7% *radiation*, $\rho_r/\rho_m = 0.2765$ at $z = 1088$ — which no statement about the rate's composition bears on. *So the inference from a radiationless rate to an absent term does not hold*, and the term is present.

*Measured, as the integrated* $|e^{-\tau}(\Phi' + \Psi')|$ *after recombination in units of the Sachs–Wolfe amplitude there*: 0.348 on the leaf congruence the rate rule assigns the perturbations to, and 0.229 on the stacking rate. *Non-zero on both. The leaf assignment makes the term half again larger; it does not create it.*

*One scope qualification is owed here, and it narrows the claim without touching the conclusion.* The constancy argument runs on $\Phi'' + 3\mathcal{H}\Phi' = 0$, which is the *super-horizon* equation: it drops the $k^2\Phi$ term. For the acoustic modes—inside the horizon through the observable leg—that term does not vanish, and the potential does decay on the observable leg, by a factor of order two across the first few peaks[P15R45]. The decay is not a radiation effect: zeroing the radiation fractions in the constraint makes it *larger*, and the rate responsible is $k^2/(3\mathcal{H})$, which grows with $k$ and refers to no content at all. So the statement that survives is that *the substrate rate supplies no early integrated Sachs–Wolfe term*—which is what the leftward reading delivers and what the comparison with flat ΛCDM turns on—while the sub-horizon potential decay is an ordinary consequence of the perturbation's own scale, present in any cosmology.

Whatever difference in that term remains between the arms cannot offset the diffusion signature, and the reason is structural rather than numerical. It sits near the first peaks and dies away well before the damping tail begins, so the two live at opposite ends of the spectrum; and the first peak is where the amplitude is anchored, by an $A_s$ this construction inherits rather than predicts. A few per cent there is absorbed by that inherited normalisation, exactly as it would be in a framework carrying $\eta$ from outside. The diffusion signature is not so absorbed: it is a change of *shape* at high multipole, and rescaling the amplitude to restore the first peak would worsen the tail rather than repair it. *What stands between the derivation and a verdict is therefore the likelihood confrontation*—a parameter refit, not a further calculation of this kind, and it is run: the joint fit leaves the Gaussian residual (§4) and the full-spectrum likelihood rejects this arm's spectrum on $TT$ shape (§4.7).

### 4.3 The diffusion scale, and the error budget

The acoustic scale is computed from the rate, and the diffusion scale is computed from the same one. Both lengths are accumulated by the plasma on the leaf rate, so the projection cancels from their ratio and what is left depends on two things only: where the plasma is handed over, and how far the diffusion integral is carried. *Neither is a free parameter, and the first is the larger.* With the handover at the branch point the ratio sits within a couple of per cent of flat ΛCDM's on either endpoint convention; a start placed later on the expanding leg raises it steeply, reaching some thirty-five per cent by the redshift at which the angle would be met.

*The remaining choice is the epoch both integrals terminate at, and it is settled by the same consistency the rate assignment is.* $\theta_D/\theta_*$ is a ratio of two lengths the plasma accumulates, and *a ratio of integrals taken to different epochs is not a ratio of anything*: the observed angle is read at last scattering, which is the peak of the visibility function, so that is where both the sound horizon and the diffusion length are carried to. *Terminating them differently is what produces*

*the spread*: with the sound horizon anchored at the observed angle and the diffusion length taken instead to a sharp recombination cut, the signature reads +2.2% where a common endpoint at the visibility peak gives −0.9% — a swing of three points where the cosmological parameters together move it by two hundredths[P15R162]. *So the size is the handover's and the epoch is not a free choice*, and the ordering of the four handover-and-rate combinations is unchanged on either reading. *What the signature is not* is reabsorbable: the baryon density has about one per cent of freedom against it, and the matter density could only absorb it by moving some thirty-seven per cent, which is a different cosmology rather than a refit.

Two contributions that might have been expected do not enter: the last-scattering surface's thickness scales with the rate in both its width and its conversion to comoving distance, so it cancels from the ratio; and the recombination redshift, integrated on this rate through the Peebles equation [38] rather than inherited, moves by only −0.11%[P15R59]. The same integration returns $z_* = 1091$ on the radiation-included rate, against the 1090 of the standard treatment, which is the check that the ionisation history is being handled correctly.

### 4.4 The integrated Sachs–Wolfe contribution across the first-peak region

The first acoustic peak is a separate matter and it does not move. Its offset from $\ell_*$—220 against 301.7—is largely the phase shift the potential's decay imparts while the modes are driven, and that decay has the same closed form on the collapse leg as on an expanding radiation era (§4.1), so it transfers. What does not transfer is the early integrated Sachs–Wolfe contribution, and that turns out to be a nearly *flat* multiplicative factor across the first-peak region—some ten per cent in amplitude, varying by two parts in a hundred between $\ell = 150$ and $\ell = 300$—so it displaces the maximum by one or two multipoles rather than by tens[P15R57]. *So the early integrated Sachs–Wolfe contribution does not move the first peak*; what sets this arm's first peak, and how its phase stands against the sky's, is taken up at §4.7. *The reabsorption of the damping signature, by contrast, can be bounded here*, and bounding it is worth doing because the reabsorption question is the whole of what a likelihood would decide about this signature. Pinning $\ell_*$ to the measured 301 and minimising the residual in $\theta_D/\theta_*$ over the baryon density and the matter density—the former held to the one per cent primordial deuterium allows, the latter to the five per cent the distance and growth data allow—leaves a residual of several per cent [P15R58]. The baryon density has about one per cent of freedom against it (§4.3); the matter density could absorb it only by moving some thirty-seven per cent, which is not a refit but a different cosmology. *So the shift has a floor, and the floor is the prediction.* We state this as a bound and not as a likelihood: a full comparison fits the whole spectrum with every parameter and its covariances, where this fixes one observable and minimises one derived ratio. It settles that the signature does not dissolve under reabsorption; it does not settle the verdict.

### 4.5 What the instrument's residual is made of

The two-arm comparison's floor is set by components of the instrument rather than by either cosmology. Two candidates for the accuracy it lacks are measured here.

*Of the two, one cannot pay at all*[P15R73L23]. Above $\ell \simeq 40$ the whole effect of reionisation on the temperature spectrum is the constant screening $e^{-2\tau}$—the second visibility bump contributes only below $\ell \simeq 30$, outside this comparison's range—and the comparison fits one amplitude in closed form. *An amplitude fitted as* $A = (m^{\mathsf{T}}Fd)/(m^{\mathsf{T}}Fm)$ *is homogeneous of degree* $-1$ *in the model, so rescaling the model returns* $A/s$ *and leaves the residual unchanged*: the degeneracy is exact rather than approximate, and it is a property of the statistic rather than of the size of $\tau$. Checked at $\tau = 0.054$, 0.10 and 0.30, the change in $\chi^2$ is zero to machine precision at all three. *Lensing is a different matter.* A smoothing in multipole, scanned in width, takes the control from 1320 to 921; and a width that *grows* with $\ell$ beats a constant one by a factor of 1.8 in the improvement, each compared at its own optimum, so the preference is for a shape

and not merely for a size—*which is lensing's own signature, since a deflection of fixed angle subtends a larger fraction of an acoustic wavelength at high $\ell$. We state what that is and is not.* The width is fitted, so the improvement bounds what a derived calculation has to play for rather than measuring a lensing amplitude; and even at its own optimum some 70% of the residual survives it. *The lensed spectrum is therefore what the instrument's remaining floor is made of, and it is computed rather than fitted.* Applying ΛCDM's own lensed-to-unlensed ratio to the arm—a derived operation with no free width—takes the control from 1320 to 989, an improvement of 331, concentrated in the damping tail where a fixed-angle deflection fills the troughs. *That is below the* 400 *the fitted width bounded, as a zero-parameter derivation must be*, and the acoustic peaks do not move under it. The corpus's own first-order kernel overshoots the bound to 508 by breaking down in that same tail—returning a spurious +13% at $\ell = 1900$ where the full operator gives +6.5%—which is why the derived number is the full operator's. *Most of the fitted headroom is recovered by a genuine lensing, and about* 989 *of the* 1320 *survives it*: the verdict of the last paragraph stands from the derived side, lensing real and not the last build[P15R74].

*And what the residual is MADE of was decomposed alongside it, which is what let the size of that gain be anticipated rather than discovered*[P15R76P15R238]. Projecting named templates through the same inverse covariance the comparison uses, the peak *positions* account for a tenth of a per cent of the residual—*which is why the acoustic peaks do not move under the lensing above, and the two findings corroborate each other*—while the peak-trough *contrast* at fixed position accounts for two-fifths of it, this transfer's contrast running some thirteen per cent too high. *Reducing contrast without moving peaks is what lensing does.* A smooth multiplicative envelope accounts for a further sixth, and *about half the residual is in neither*—so a complete lensing calculation was expected to leave a comparable amount behind, and it did.

### 4.6 The shear coefficient, derived

*The coefficient is derived rather than fitted, and the derivation settles it against the fit.* The coefficient is measured by running the same oscillator twice on the same background: once as the perfect tight-coupled fluid, which is undamped by construction, and once as a photon hierarchy carried to $\ell = 24$ with Thomson scattering, the polarisation multipoles evolved alongside and the baryons given their own velocity. *Their ratio is the damping and nothing else*—the slowly varying prefactor, the drift of the sound speed and the baryon-loading dependence of the oscillation are common to the two runs and divide out identically—and reading it locally, between consecutive extrema, lets the residual be extrapolated in $(k/\tau')^2$ rather than smeared over an interval. With the polarisation multipoles suppressed the quadrupole's scattering term reduces to $-\frac{9}{10}\tau' F_2$, which is the unpolarised closure, and the coefficient comes back as 0.8852 against 8/9; with them carried it is 1.0618 against 16/15. *The polarisation correction itself, which is the quantity in dispute, is a factor* 1.1996 *against* 6/5*: three parts in ten thousand.* Both absolute values sit about half a per cent low by an amount proportional to the coefficient, so the residual is a common normalisation and cancels in the ratio[P15R71].

### 4.7 The acoustic comparison

The angular scale $\ell_* = \pi D_M / r_\mathrm{s}$ is built from two lengths that do not ride the same rate here: the distance carries the stacking rate's $1/H_0$ and the sound horizon is accumulated by the plasma on the leaf rate, which carries the radiation the other does not. So the measured angle constrains $H_0$, as it does in flat ΛCDM where the sound horizon is fixed instead by the matter and baryon densities[P15R56]. *So the acoustic scale is computed rather than accommodated*, and it is the comb of the computed spectrum that carries it: fitted to those peaks by the procedure this paper runs on the sky, it is 298.0 against the sky's 298.4, with the background fixed by the distance data alone[P15R163]. *A start placed by hand and fitted to the angle would report a different*

*quantity*: what such a fit sets is the scale reported, not the spacing the spectrum carries, the two differing by four per cent.

*The datum §4 states leaves two things to be fixed, and the handover fixes both.* That datum is one amplitude per mode and a *common* phase; it says that the phase is common without saying which phase it is, and it does not say what the amplitude is flat in wavenumber *at*. At the branch point both follow: every mode is outside the horizon there, so the phase is an extremum with no velocity, and the amplitude is the frozen value the closed-form leg carries, $0.4835\,\Psi_i$.

*How much the comb depends on that choice is measured rather than assumed.* Across the seventeen readings admitted by a four-peak criterion fixed before the numbers, the first peak spans 148 to 228—a factor of 1.541, and across the fifteen numeric-phase readings alone 1.118—with the spacing spanning 276.0 to 330.4, the phase intercept $-0.4044$ to $-0.2161$ and the alternation ratio 0.475 to 1.088. *Within that span position and alternation move against each other*, Spearman $\rho = +0.782$ at $p = 2.1 \times 10^{-4}$ between the first peak and the gap ratio, with every contracting reading sitting at $\ell_1 \leq 212$. *The single-locus reading is not inside that trade*: it returns the first peak at 222 and a contracting second gap together, which is what fixing the state at one locus rather than choosing a reading of it buys.

*The instrument that would measure the rest is built, and its guard and its control are what make it an arbiter.* Both arms run on one set of equations, so no comparison between them is a difference of machinery[P15R63]. *The guard first, because it is what licenses the rest.* With every coupling to the potential removed at every site—the $4\Phi'$ in the photon, neutrino and cold-matter continuity equations and the $k^2\Psi$ in the photon–baryon and neutrino Euler equations—the source comb lands on the integers on *both* arms. *So the sound horizon, the measure, the wavenumber grid and the phase extraction are validated as an ensemble, and every departure from $n\pi/r_\mathrm{s}$ in a driven run is the driving and nothing else.*

*The control is the arm whose answer is known independently, and it reproduces it*[P15R72P15R108]. The photons are carried as a Boltzmann hierarchy with the polarisation multipoles alongside, the baryons given their own velocity and density contrast, and the source carrying $g\Pi/4$ and $(3/4k^2)\,\mathrm{d}^2(g\Pi)/\mathrm{d}\eta^2$. Run on the flat-ΛCDM arm it returns peaks at 220, 540 and 812 against the sky's 220.6, 538.1 and 809.8, and a first-to-second height ratio of 2.1969 against a standard code's 2.200—agreement to 0.14%, and within the sky's own uncertainty.

*Two conditions on how such a transfer is run are what that agreement rests on, and the tighter of them is numerical.* The projection must carry the polarisation source, which a damping envelope cannot supply at any coefficient; and the wavenumber integral must be carried past the highest multipole *reported*, since $\int P(k)\,\Delta_\ell(k)^2\,\mathrm{d}k$ is not converged there. *Each is worth a factor on its own and neither alone reaches the answer*: the height ratio runs from 2.721 at an $\ell$-equivalent cutoff of 900 to 2.393 at 2400, so a cutoff set at the reported multipole is a systematic and not a rounding[P15R108].

*The perturbation sector runs on the leaf congruence, which is what the framework assigns it.* A process running *in* the content—the sound horizon, the diffusion length, recombination, the perturbations—takes the leaf's rate (§2.4), while the comoving separations and the projection keep the stacking rate; the change is an exact chain rule between two monotone conformal times, so every spline and grid is untouched, and on the flat-ΛCDM arm the two rates are identical and the assignment is provably a no-op there—which is its own self-check.

*The comparison is made on the polarisation path, in the configuration the control is calibrated in.* There the control returns a first-to-second height ratio of 2.195 at peaks 220, 536 and 814, against a standard code's 2.200—0.23%: *so the instrument is calibrated where this cosmology's number is taken, and that is the figure quoted for it, the closer* 0.14% *belonging to a different run and not to this one*[P15R160P15R63P15R68]. *Both arms are converged in the wavenumber integral.* The control's own movement between cutoffs is 0.000% and 0.046% on the two height ratios, and this cosmology's arm does not move at all—0.000% and 0.000%—so the arm is converged

at the lower cutoff and the figures below are not a truncation artefact. *The same comparison on the fluid path reads* 0.042% *and* 0.036% *against* 0.000% *and* 0.045%: both paths converge, and the two sets of figures are not interchangeable. *A rung below that was refused outright by the instrument's own truncation guard, on both arms, which is data and not an error.*

*On that path this cosmology's arm returns the following, with the plasma handed over at the branch point and the background fixed by the distance data alone.* The peaks fall at 222, 538, 818 and 1134 against the sky's 220.4, 537.7, 817.3 and 1123.9; the comb fitted to them by the procedure this paper runs on the sky is 298.0 against the sky's 298.4, and the phase intercept $\varphi/\pi = -0.2349$ against $-0.2405$. The height ratios are $P_1/P_2 = 2.264$ and $P_1/P_3 = 2.298$ against the sky's 2.217 and 2.277. *The first three peaks agree with the sky inside the spread the locating itself carries, and the fourth is not resolved against it*: located through one procedure with the likelihood's own covariance propagated, this arm sits 0.9 to 1.3 standard deviations from the measured fourth peak across the admissible window, so *the data cannot tell the two models apart there*[P15R163P15R179]. *That one procedure is a requirement and not a convenience*, because the fitting window is a source of bias rather than of scatter: across the admissible windows the located peak moves by a fraction of a multipole at the first peak and by some six at the fourth, a wider fit on an increasingly asymmetric and damping-suppressed hump pulling its apex down the envelope's slope—*and it moves the sky and both models the same way*, so it cancels from a difference taken through one procedure and does not cancel from anything else. *And what offset there is, is mostly not this construction's*: both models sit high of the located sky, the control by 1.2 standard deviations and this arm by 2.1, with well over half of this arm's offset being the control's own. *Two wavenumber cutoffs agree to every digit, and the positions are converged in the reported multipole grid*: read on a grid of step 2 and refined to step 1, they move by a single multipole at the first peak and not at all above it.

*And the residual is neither of the two instrument faults a reader would reasonably suspect*, because both are removed in this configuration: the wavenumber integral is carried past the reported multipole, and the projection carries the polarisation source rather than a damping envelope[P15R70]. *And the truncation's removal is measured on the arm whose answer is known*: it was 78% of the control's remaining $\chi^2$, and taking it out brings the control to 1.18 per degree of freedom over the full range against a reference 1.01[P15R90]. *The diffusion damping is derived on each arm's own opacity and rate rather than imported or fitted*, and putting it back moves the height ratios by tens of per cent and drops the likelihood's floor by a factor of five without the verdict turning over[P15R69]. *The polarisation source pulls both arms, and* nothing about either operation differs between the arms, which is what makes what survives them attributable to the source rather than to the machinery. *The size of that pull is not quoted here*: the operation is arm-independent by construction, which is the load-bearing point, and the pull's magnitude on the configuration this section reports has not been measured.

*Two likelihood configurations are quoted in this section and they are not interchangeable, so which is which is stated rather than left to be inferred.* The full-range lensed one, on 185 bins, is the better-converged: it is where the control reaches 1.18 per degree of freedom against a reference 1.01, and it is the configuration in which an instrument fault can be told from a physical one. The 133-bin unlensed one is where the shape rejection is quoted, and its control sits at 2.10 per bin because the ceiling cuts the damping tail[P15R236]. *The second is not a worse measurement of the first's quantity*: it is a different configuration, chosen for the comparison it makes, and the two sit beside each other rather than one superseding the other.

*The driving's size on the observable leg is measured rather than inferred, and its wavenumber dependence is known in closed form*—the $k$-dependence living in the potential's gradient rather than in its decay, with the turnover at a fixed multiple of the horizon scale[P15R65P15R66]. *Subtracting the driving is what measures it*: removing every coupling to the potential moves the control's first peak from 220 to 274 and $\ell_1/\ell_A$ by 0.1792, so the modes carrying the first peak are driven and the shift is the driving's size[P15R165P15R77]. *Run on this arm the same subtraction*

*gives* 0.1717*—the potential's grip within four per cent of the control's, and slightly weaker rather than stronger*[P15R165]. *And this one is not path-specific*: the two instrument paths return the same shift to four decimal places, where the heights do not.

*One of the two position statistics is not a projection quantity, and saying which keeps them apart.* The ratio of the first peak to the acoustic scale is $\ell_1/\ell_A = k_1 r_\mathrm{s}/\pi$*—the distance cancels identically*—which is why removing each source term, the damping envelope and the visibility function in turn leaves it untouched. What it compares is the sound horizon each arm *reports* against the phase its modes actually accumulate. *Both arms integrate that horizon from* $a \to 0$, the plasma being handed over at the branch point on this one and the control having no other start, so the comparison does not confound the rate with the start[P15R7]; and here the two agree, the comb at 298.0 against a reported scale of 302.9. *The two sides of that agreement are computed independently*: the reported scale reaches the transfer nowhere—no source term, no projection and no solver reads it—so the comb it is compared against is not a quantity it was used to set[P15R163P15R178].

*And the likelihood is run, both arms scored on the same bins by the same code, with the amplitude fitted so that what is compared is shape. The result is still a rejection of this arm's spectrum and is stated as one.* On the full-range lensed configuration—$\ell = 100$–1996, 185 bins, the lensing carried by the same derived operator on both arms—the control returns $\chi^2 = 214.1$, or 1.16 per bin, and this arm 550.5, or 2.98 per bin: *the arm sits at* 2.57 *times the control's distance from the data*[P15R164P15R236P15R67]. *That is the unfavourable of the two configurations and it is the one quoted*, the same comparison unlensed over 133 bins giving 279.4 against 556.7, a factor 1.98; lensing is worth proportionally more to a spectrum whose peaks are already in place[P15R236]. *The band-by-band comparison is even across the range* — 1.70, 2.22, 4.23 and 9.29 per bin against the control's 0.77, 1.44, 2.62 and 3.73 over the four quarters from $\ell = 100$ to 1300, read on the unlensed configuration — so the disagreement is not one feature carrying the whole of it[P15R163]. *And the spectrum's own parameters are refitted, which is the comparison the standard model is usually judged by.* With the expansion rate, the matter and baryon densities and the tilt free in each arm and the amplitude in closed form, scored on the full range and verified by running the spectra at the minima, the control settles at 1.01 in $\chi^2$ per bin and this arm at 1.58: *the arm at* 1.57 *times the control's distance, where the computed spectrum sits at* 2.56 *times it on the same bins*—so *freedom closes about a third of the gap and leaves the rest*[P15R236P15R279P15R170P15R163P15R167P15R278].

*What the refit does not move is the background.* Given four free parameters this arm's expansion rate moves by 0.028% and its matter density by 0.031% from the values the baryon-acoustic distances and the acoustic angle fix with no spectrum involved: *the background the distances fix is the background the spectrum wants. And the size of that agreement is read against the distance data's own width rather than left as a percentage*: the refit's matter density sits 0.046 of a standard deviation from the distance fit's, at a cost of 0.010 in $\chi^2$ to the distance data—one part in a hundred—where the same fit run on the distances alone, on this construction's own ruler with the distances on its radiation-free rate and the sound horizon on the leaf clock, returns the pair at 0.958 per degree of freedom. *So the spectrum is not pulling the background anywhere*, and what remains is left after the parameters have been given away rather than being a parameter still to be set[P15R278]. *The control's own displacement is larger and is not a comparison of the same kind*: its reference matter density is a microwave fit rather than a distance fit, so what its refit moves is the standing disagreement between those two datasets and not anything this comparison introduces. *And they are at a sharp minimum rather than unconstrained*—one-step excursions cost 2527, 740, 141 and 31 in $\chi^2$ on the four parameters, which is the distinction between a parameter that does not move and one that cannot. *The one parameter that does move is the tilt*: this arm prefers 0.998 against the control's 0.954, bluer by 0.044, with the damping tail's leverage landing there and on the baryon density as the shorter range predicted it would.

*And what the refit leaves is measured rather than quoted as a ratio, because a residual's size says nothing about whether the model could have removed it.* Taking the whitened response of the fit's own directions on each arm—the amplitude and the four parameters at the step sizes the grid was built on—the five are four stiff directions and one soft one *on both arms, with the same degeneracy*: the expansion rate and the matter density respond at $|\cos| = 0.99$ on each, which is the familiar geometric degeneracy and not something this background introduces. *So this arm is not short of freedom; it has the control's, direction for direction.* What separates them is the *reach*. At the exact least-squares minimum of the model linearised about its base—where the residual is orthogonal to all five directions by construction, so nothing reachable is left in it—the control leaves 186.0 in $\chi^2$ against $n - 5 = 180$, within a third of a standard deviation of the noise expectation, *and this arm leaves* 278.8*, at* 5.2: about a hundred in $\chi^2$ lies outside every direction the model has. *And the direction it lies along is nameable.* Decomposing each parameter's response on an amplitude, an acoustic *shift* and an acoustic *contrast* template of the base spectrum, all four parameters move the shift strongly and the contrast barely—the $4 \times 2$ matrix of their (shift, contrast) responses having a singular-value ratio of 0.040 on the control and 0.036 here, so *the contrast is nearly unmoved by any declared parameter on either background*. Freeing one uniform contrast template as a sixth direction then removes 56.4 of this arm's residual, a fifth of it, against 0.8 on the control, at a fitted coefficient of $-0.0636 \pm 0.0085$ against the control's $-0.0075 \pm 0.0083$: *the data ask this arm's acoustic contrast to be* 6.4 *per cent lower at better than seven standard deviations, and ask the control for nothing.* An $\ell$-modulated contrast buys 4.5 and a modulated shift 0.4, so what is wanted is a uniform level and not a drift. *And the leftover is not noise*: in $\ell$ order this arm's residual changes sign 54 times against $88 \pm 10$ for correlated draws of the likelihood's own covariance with the same five directions projected out, while the control's 87 sits inside one standard deviation of it[P15R223]. *Two things follow, and the second is the one that matters.* The 6.4 per cent is the size of this sector's standing arm-to-control contrast excess, measured independently and reported above, so the quantity the fit cannot supply and the quantity the projection manufactures are one quantity. And *no re-parametrisation reaches it*: a residual lying outside the span of every declared direction is closed by changing what *sets* the contrast, which is the assignment of the scales and the distances to their rates (§10), and not by freeing a parameter.

*One thing the ratio does not carry, and it is stated here rather than folded in.* The two arms fit the same four parameters, so no information criterion separates them—penalties of about two per fitted parameter against a difference of some hundred in $\chi^2$—and the asymmetry between the frameworks is not in what was fitted but in what had to be fixed before there was a model to fit. *The standard model reaches flat $\Lambda$CDM by setting a curvature parameter to zero on inflation's authority, and uses the same added epoch to supply the coherent phases*; here the flat slices, the expansion law and the coherence are consequences of the construction (§2.3, §4) and there is no curvature parameter to free. *And it runs the other way too*: both frameworks inherit the composition, the amplitude and the tilt, and neither derives the spectrum it is scored on. *Two axes, and no single number crosses them* [18].

*And the phase intercept is not a residual at all, once the sky's own is located by the same procedure.* Running one locator over the binned likelihood and both arms alike, the sky's intercept is $-0.2378 \pm 0.0099$ on the first three peaks and $-0.2448 \pm 0.0074$ on all four; this arm's is $-0.2278$ and $-0.2445$. *So the arm sits inside the locator's own window-to-window spread on the one fit and on the sky's value in the other*, and the sign of the offset flips between them — *the shape of a difference carried by a single peak rather than a phase error*[P15R172]. *What dissolves it is the sky's own locating spread and not a correction to the arm*: the binned likelihood cannot pin that intercept tighter than a hundredth across the windows the locator admits, and the offset is inside that range. *The spread quoted is the locator's movement across those windows on one realisation and not a noise level*, the covariance-propagating reading of the same quantity having been left unopened.

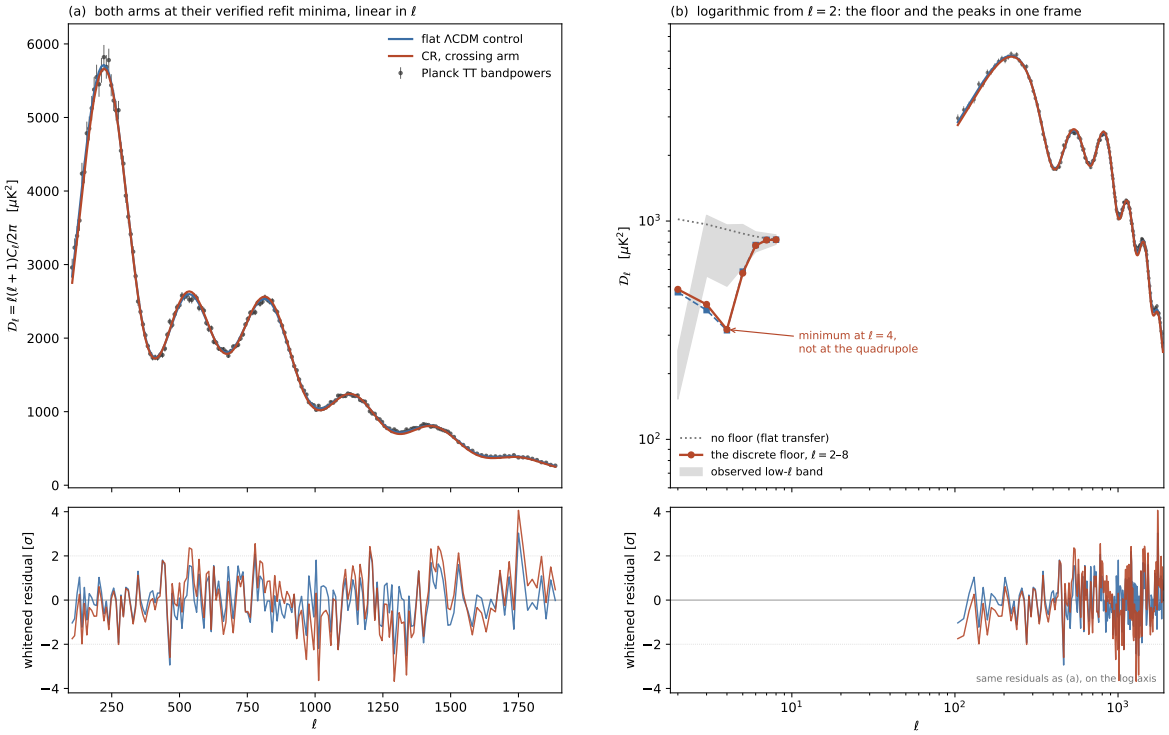



**Figure 2:** Both frameworks refit on the same bins with the same freedom, against the measured temperature spectrum. *Above*, the acoustic range in $D_\ell$, each arm at its verified refit minimum with the amplitude taken on the full bandpower covariance; *below*, the same on a logarithmic multipole axis from $\ell = 2$, so the discrete low-multipole floor and its minimum at $\ell = 4$ sit in one frame with the peaks — that panel's floor being a reconstruction against the control's own low-multipole spectrum, since the sweep carries it as a ratio. *Whitened residuals beneath each.* The peak positions and the spacing are the result to read in the upper panels; *the residuals are where the rejection lives*, and the control's own are drawn beside this arm's so the size of the difference is visible.

*What remains is about one multipole at each peak, and it is not the damping envelope.* Imposing the control's envelope on this arm moves the fourth peak by a seventh of a multipole against an offset of two, and the inverse operation returns the same displacement with the opposite sign, so the test is sound and the answer negative. *And read as the difference between the two arms, located sub-bin by the same procedure on each, the offsets are one near-constant multipole at all four peaks*: +1.61, −0.19, +1.07 and +1.31. *That is the shape of a small common phase and not of a ruler or a driving error*—a constant displacement fits it half again better than a constant fractional one, which would have to grow from half a multipole at the first peak to two and a half at the fourth, and a driving error would grow faster still. *What each arm shares with the other is larger than what separates them*: both sit low of the located sky at the second and third peaks and high at the fourth, and at the fourth the control carries four fifths of the offset[P15R192]. *At the spread the sky's own locating carries, one multipole is not distinguishable from none*, so this is a shape the comparison exhibits rather than a disagreement it records.

*And what the rejection is made of is measured rather than described.* Whitened against the full covariance, the two arms' residuals sit at a cosine of 0.855, and projecting the control's direction out of this arm's leaves 73% of it shared — stable across three bin cuts. *That fraction is not an attribution, and reading it as one would be an error*: both arms are fitted to the same bandpowers, so their residuals differ by the model difference alone and the fraction measures only how alike the two models are — a control deliberately tilted away from its own best fit,

which nobody would defend, scores higher still. *What the decomposition does say runs the other way.* The control already fits these bins at 0.99 per bin, so there is no shared defect for this arm's misfit to be an amplification of; this arm exceeds it by 105 in $\chi^2$ and the model difference alone carries 77 of that, so *the excess is this construction's*, and the factor of 1.3 between the two residuals' sizes is that difference's own size[P15R174]. *Within it, position rather than amplitude, tilt or damping*: admitting one further parameter at a time, a peak-position rescale buys 4.7 in $\chi^2$, the damping 1.4 and the tilt 0.1 — and on the control the position direction buys nothing at all, so *the position part is this construction's.* It is small: the rescale is under a multipole at the fourth peak, and 4.7 of 283 is under two per cent[P15R173]. *Amplitude, tilt, damping and position together do not account for the shape rejection*: they carry 5.7% of the model difference between them. *What does is a contrast difference*, and the sky's noise does not enter it.

*Taking the realisation out is what makes the difference readable.* With $\Delta$ the whitened difference of the two arms at their own minima, $\chi^2$ decomposes exactly as $282.96 = 177.88 + 27.82 + 77.26$: *the model difference carries three quarters of the excess and the cross term a quarter*, so the rejection is not this sky's luck — set the cross term to its expectation and this arm still scores 1.43 per bin against the control's 0.99 on a typical sky. *And $\Delta$ is trough-dominated*: $+0.011$ at the peaks against $-0.760$ at the troughs, so the two arms agree about where the peaks are and differ between them. Split about a smooth envelope, the oscillatory part carries seven tenths of it and is peak-high and trough-low—*antisymmetric about the envelope rather than about each peak, which is a contrast and not a phase and not an envelope.* This arm's acoustic oscillation is 1.040 times the control's about its own envelope, and one direction built from the control alone—raise the oscillation at fixed envelope, with nothing fitted—carries half of $\|\Delta\|^2$[P15R175P15R177].

*Decomposed about its own envelope the excess is an amplitude and not a lopsidedness*: splitting the band variance of the normalised oscillation at its zero gives 1.067 on the peak side against 1.056 on the trough side, and at the located extrema themselves the arm exceeds the control by 5.7 per cent at the maxima against 5.3 at the minima. *So heights and depths carry it equally*—and $\Delta$'s trough dominance is a property of the whitened residual rather than of the contrast, the two not being the same decomposition. *The trough depths are nevertheless where the excess becomes observable, and for a reason that is the sky's rather than the model's*: the arm exceeds the control by the same amount in each, $+0.0106$ in depth against $+0.0103$ in height, while the sky measures depth twice as precisely—so the same excess reads at 1.8 of the sky's own spread in the depths and 0.9 in the heights. *And the depths are the one statistic in this comparison whose residual is not shared*: the control lands on the measured trough depths at five hundredths of a standard deviation while this arm sits 1.8 above them[P15R193]. *That is a shape the data can see and not a detection*, at the spread one statistic carries; and the locating must be anchored to the comb rather than searched for, a free search under the same noise returning a spread six times wider because it latches onto noise minima. *One systematic is stated rather than minimised*: the envelope's abscissa moves the measured depth by about a third of that spread across the admissible acoustic scales.

*And the three features this arm's residual shows and the control's does not are all $\Delta$'s*: the peak-and-trough asymmetry, the $\ell \simeq 1000$ trough where $\Delta$ carries $-1.074$ of the arm's $-1.177$ in fifteen bins, and the growth with multipole—*which $\Delta$ does monotonically where neither residual does*, a residual being $\Delta$ on a noise floor of order one that flattens it[P15R175]. *The channel is identified and the cause is not*: the contrast direction is the balance between the monopole and the Doppler terms of the source, bracketed with opposite signs by removing each and re-projecting, on both instrument paths and with the same signs on both; and *neither deletion returns the two arms to a common contrast*, so what the bracket locates is where the difference enters rather than what produces it[P15R176]. No mechanism for the imbalance is offered here. *And the neutrinos' free-streaming, which the shape invites, is not the difference either*: removing it moves both arms' peaks by the same amount—band by band the two agree to a fifth of a

multipole, and in units of each arm's own acoustic scale to two parts in ten thousand—while its large effect falls on the envelope rather than on the contrast. *A correction the two arms share cannot be what separates them*[P15R178].

*And the trough-filling a contrast excess would be a shortage of is not short on this arm.* Four channels carry that, each measured and each larger here. Deeper troughs are what excess baryon loading produces, through the displacement of the oscillation's zero point—and measured against each arm's own loading, *this arm's monopole swings about a point that falls short of its own* $-R\Psi$, by an effective baryon density of $-2.9$ per cent, which is the wrong sign for that reading. And the visibility is *wider*, 43.6 against 38.0 Mpc, so this arm averages over more acoustic phase and not less. *So the channels that would each shallow the troughs are larger here*, and the same is true of the driving, whose measured grip is weaker.

*And the Doppler term, which fills troughs, is stronger here*: the dipole-to-monopole ratio of *oscillation amplitudes* at the visibility peak sits two per cent above the control's and rises with wavenumber. *Which of two ratios that is matters, and the distinction is the physics rather than a convention*: trough-filling is an oscillating term a quarter period out of phase adding into another's troughs, so *what fills a trough is an oscillating amplitude*, while the smooth part of the same field displaces the envelope and fills nothing. A ratio of band-integrated powers mixes the two and reads the opposite sign on the same spectra—*so the two readings are different quantities and not a disagreement about one*, and the one this comparison needs is the amplitude ratio[P15R198].

*One channel does carry a definite share, and it is the first in this comparison that does.* The source is weighted across the visibility by a function rather than a constant, and that weighting is not the same on the two arms: read in the phase the oscillation actually accumulates, the window's spread is 0.0652 of the acoustic scale on the control against 0.0642 on this arm. *That the two are within a per cent and a half of each other is itself the two-rate structure, and it is a cancellation rather than an absence*—the arm's visibility is fourteen per cent wider in conformal time while its phase accumulates fourteen per cent more slowly per unit of it, and the product of the two comes to within one part in ninety of unity. *Imposing one arm's weighting on the other moves the contrast the way a narrower window should*, and the size can be read two ways that share no machinery: from the instrument's own response to the weighting, which gives between a quarter and a half of the band excess above the damping scale, and from the direct substitution, which closes between a fifth and two fifths of it. *So the weighting is a real channel carrying about a third, measured rather than modelled, and from a quantity nobody chose.*

*And it cannot be the whole of it, for a reason that is independent of every one of those sizes.* The excess is symmetric about the envelope, heights and depths carrying it equally; the weighting is not. Imposed at the arm's own size it delivers four fifths of the peak-height excess and one seventh of the trough-depth excess—*a five-fold asymmetry against a symmetric object*[P15R194]. *A peak-weighted channel cannot carry a symmetric excess alone, whatever its share*, and two further readings agree without being needed: the measured response extrapolates to a non-zero offset at zero wavenumber, where a window's own characteristic function is identically one, and a weighting large enough to deliver the whole excess moves the first peak by eighteen multipoles and the heights by nearly half. *So what this settles is a share and not a mechanism.*

*A second channel carries the other half of the shape, and it is the term mix rather than the window.* The two arms hold different fractions of their window's power in the monopole, and imposing one on the other at a coefficient solved from the profiles themselves—one constant to two per cent across every band—moves the contrast with the signature the long-wavelength offset needs: *the response is independent of wavenumber*, an intercept against a slope worth three per cent of it, where a window's smearing must die at long wavelength. *It is the first channel measured to have that shape. And it is not the remainder either, for the same kind of reason the window was not*: it is peak-weighted where what the window leaves is trough-weighted, it over-delivers the whole excess by between one and three times, and it moves the

first two peaks the wrong way and outside the widths the sky's own locating carries[P15R195].

*And the two together are what the measurement actually settles, because their shares do not add to what is observed but to something larger*: a third from the window and above one and a half from the term mix make more than one and a half times the excess. *How two such channels compose is then a measurement rather than an assumption*, and it is made by imposing both on one control spectrum at the sizes their own comparisons solved, with neither coefficient re-chosen and no third channel admitted.

*The rule is multiplicative.* The joint response lands within a pre-registered tolerance of the product in every band and of the sum in every band, and of a quadrature combination in none of them—*the last by three and a half times that tolerance, which is the one separation this measurement can make and the only one claimed.* That the product and the sum differ here by less than a third of the tolerance was declared before the run, so they are not separated and are not reported as though they were. *The pair is slightly sub-multiplicative and consistently so*, below both the product and the sum in every band and above quadrature in every band. *And the composed pair reproduces the two shares' arithmetic directly*: what adding the separate measurements put at a little over one and a half times the excess reads, composed in one spectrum, within one per cent of that value[P15R196].

*So the remainder left by dividing one response into the excess is a well-defined object, and the over-delivery is evidence about the sizes rather than about the rule. What does not survive is a size reading of it at all, and the reason is a shape.* The joint response is the flattest thing this comparison has measured, and the excess is not flat: the pair over-delivers by nearly four times at the longest wavelengths and by under one and a half at the shortest. *No coefficient on either channel reproduces the excess*—set to match at long wavelength it falls short at short, and set to match at short it is an order too large at long—*which is an obstruction of a different kind from a size discrepancy, because a size discrepancy is what a coefficient absorbs and a shape mismatch is not. And that reading is available only because the rule was measured*: with the rule assumed, the same numbers read as a size a smaller coefficient could take up.

*What the excess's own dependence on wavenumber is has then to be measured before it can be searched for, and the instrument is what has to be measured first.* Six forms injected into the control's own spectrum, two of them carrying a scale, are recovered to about a per cent. *But the estimator manufactures structure between bands*: its bias varies from band to band by about a per cent while the six injected forms agree with each other on that bias seven times more closely, *so a contrast that is constant reads as a wobble, and band ordering at the per-cent level is a property of the statistic and not of the spectrum. One alternative anchoring is excluded by its mechanism rather than by its number*: a median envelope returns a value three times larger, and the tell is that it does not move with its own smoothing window—at short wavelength the running median collapses onto the curve it is meant to be an envelope of, a median of a locally monotone stretch being its central value whatever the window. *So the rise is real and the wobble is not*, and the distinction is established on the instrument before anything is concluded from it[P15R197].

*And the rise decelerates within the statistic as defined, which excludes the form this comparison's own statistics are written in.* A departure growing linearly in the squared wavenumber is rejected—the only one of the six so rejected—and every concave form is admitted with none preferred among them, so *the direction is settled and the functional form is not. No scale is separated from another either*, and the spread that comparison is judged against is the fit's own injection-and-recovery error rather than a noise level. *And two things about that deceleration are stated here because a reader would otherwise assume both.* It is *robust within the statistic*—negative under every variation of the envelope's width, the band edges and the sampling—*and it is carried by the lowest band alone*: with that band set aside the sign is not determined at all, the deceleration being that band lying low rather than the upper bands bending. *And it does not survive a change of envelope*: a running median in place of the arithmetic one, with

everything else held, reverses it. *That is a different statistic and the difference is measured*—the median absorbs about half the highest band's oscillation on both arms, so its ratios are formed on a much smaller residual—*but it is not on its face the worse one, since it leaves a departure of zero mean where the other does not. So the sign of the curvature belongs to the statistic and not yet to the excess*[P15R201]. *And the rise is a preference over flatness rather than an exclusion of it*, which is stated because the difference matters: a constant departure is a worse description, not an impossible one. *The consequence for the numbers above is that a slope read against an intercept is a description and not a model*, and the growth of the departure itself is the scale-free quantity that comparison needs.

*Read with that quantity, the search this comparison can run in the positive direction leaves exactly one candidate standing.* The excess's departure grows by a factor of three and a half across the range with negative curvature, which gives two conditions rather than one, and *the source terms and the three channels with measured shares all fail the first*: they are flat where the excess grows. *The dipole's amplitude ratio does not*, and the range over which that can be said has since been extended by an estimator built for it: holding the oscillation's period and fitting its amplitude and phase, rather than reading an amplitude off a window, reaches well below the window estimator's floor. *Its own first finding is that the period is not the period*—set by construction to one value, it runs about two per cent short of it in the fields and by a *different* amount for the two of them, so the period is measured per field and per arm rather than assumed. *And it is validated before it is used*: the two estimators agree on the mean where both are sound, and where the window estimator does not survive its own width—its ripple running from under one per cent to above ten as the window narrows—the held-period one stays near two and a half per cent at every width. *The one cost it carries is measured rather than noted*, and it is half a per cent of the dependence being measured, for a mechanism: the fit re-fits the phase in every window, so an error in the held period is absorbed there and costs only a common amplitude factor, which cancels in a ratio of ratios.

*On the range that unlocks, the candidate's departure is positive in every band as the target's is and grows faster*, so *it passes both of the conditions that can be applied to it. And it stops where the fields stop rather than where the method does, at a floor that is exact and was named before it was used.* An oscillation's amplitude is identified at a point only if the window holds a turning point on either side of it, because across a span with no turning point the held-period pair is monotone and the baseline already spans it. *The monopole has no turning point below its first*—the first excursion is one-sided—*and the monopole is what binds*, putting the floor well above the lowest band, which lies far below it while every other band lies entirely above. *And the floor is the fields' and not the window's, shown three ways*: the same widths spread four times as far below it as above and change sign there; sweeping the window's edge, the departure is negative for every window that opens past the monopole's first turning point and positive for every one that stays above it, *what lies below being the field's rise from its initial condition rather than an oscillation at all*; and every centre below the floor spreads more than every centre above it, which the floor was not fitted to. *So no estimator of an oscillation amplitude can reach inside the first excursion*—an acoustic oscillation has a first extremum, and before it there is no amplitude because there is no oscillation[P15R200].

*And the condition that would decide cannot be applied, for an exact reason rather than a statistical one.* The excess decelerates across the full range, and *that deceleration is carried entirely by the one band no estimator of the candidate can reach*: removing that band alone reverses the sign of the excess's own curvature, while removing any other single band leaves it where it was. *So the filter is out of conditions rather than the candidate out of chances*—the candidate's own curvature is stable there, and what is missing is the target's sign and not the candidate's number. *That is a statement about the filter, and it was one of the two outcomes written down before the estimator was built*[P15R200].

*And the way that condition is unavailable is worse than a simple absence, which is worth*

*stating plainly.* It needs a value in the band the fields close, and that band is shut to *amplitude* estimators but not to a candidate computed from the projection's kernel—which had one, and which the condition duly excluded. *So the filter's full strength is available exactly for the class that cannot carry the excess, and its third condition is permanently unavailable exactly for the class that can*, because what fills a trough is an oscillating amplitude and that is what closes the band.

*Which leaves the question of what a two-condition filter is, and the answer is that it is what it always was. As a device for excluding it has lost nothing*: each remaining condition is necessary, and every exclusion this comparison has made was made on growth alone. *As a device for confirming it was never one, with three conditions or with two*—all three are conditions on shape, and the surviving candidate matches the excess in sign and in direction of growth while being several times too small at the highest band. *So the question this comparison now carries is the coupling and not the shape*: a quantitative question, living where the candidate is measured, and needing the closed band not at all.

*That coupling is measured, and its sign comes out the same everywhere.* Scaling the Doppler source across three banked spectra gives the contrast's response as a finite difference—*three points, so linearity is checked rather than assumed*—and the response steepens towards the unscaled value, so the slope there is both the applicable one and the one taken over the shortest step. *It is negative at every band, on both envelopes, over every interval*: forty-two readings of the sign and not one positive. *Which is this comparison's own trough-filling physics read forwards*—more of an oscillating term in quadrature fills more trough and *lowers* the contrast.

*And the construction fixes it, shown by what fails to change it.* The control's own ratio of that term's band power to the monopole's runs across a factor of more than six from the lowest band to the highest, *and the coupling's sign is the same at every one. A quantity whose sign is invariant across a factor of six in the very ratio it depends on is not set by a knob*: the knob is the probe and not the setter, so the size is a prediction rather than a fit.

*And it is measured on the arm itself rather than carried across from the control.* The decomposition the instrument already writes resolves each spectrum into its bilinear terms multipole by multipole—ten of them, summing to the spectrum to a part in $10^{15}$—and their scaling in the source amplitude is exact, quadratic on the term's own square and linear on each of its crosses, so the response follows analytically on either arm rather than from a difference of runs. *It runs from* $-0.48$ *to* $-0.86$ *across the bands, the two arms agreeing to a few per cent and each agreeing with the finite difference above to better than a tenth—the same quantity reached two ways, on the arm's own spectrum.*

*The contribution the coupling multiplies is then read by a swap rather than by a product, which is what removes the ambiguity in its input.* Replacing the arm's four Doppler-carrying bilinear terms by the control's, envelope-scaled and on the arm's own grid, gives the share of the excess those terms carry as a ratio of contrast differences—*the same operation the instrument already sanctions for the clock, and with no coupling-times-amplitude product anywhere in it. The answer is a minority share on either envelope*: from a per cent to a quarter of the excess against the arithmetic envelope, and from a tenth to a half against the median one. *So the Doppler term is a real contributor and it is not the carrier.*

*And the share grows with wavenumber while being sign-indefinite at the bottom*—negative at the second band on both envelopes—*so at the bands where the excess carries its own structure it changes sign rather than settling to a share. Which is exactly how far the envelope ambiguity reaches*: the coupling's sign survives both envelopes at every band, so the contribution does not reverse with the statistic, while the share's magnitude moves by about a factor of two between them—*so the share inherits that ambiguity where the sign does not*[P15R203].

*And the excess every candidate here is measured against has one dominant feature, which is a step and not a trend.* Its lowest band is a fraction of the mean of the others and sits below them by nearly five times the spread the upper bands themselves show about their own

trend, while above that band a constant already fits—a line adds a mild slope and halves the residual and no more. *So the bands above the lowest carry very little structure with which to tell a flat candidate from a rising one*, and that stretch is where every shape comparison across wavenumber is made: a flat candidate matches the featureless part and misses the step, a rising one matches the mild rise and misses the step, and across those bands the two are barely distinguishable.

*That the step is the excess's own and not the statistic's is established before anything is read off it, and by a reading that uses no statistic at all.* Three running envelopes differing in *kind*—an arithmetic mean, a median and a local quadratic—and a fourth quantity that needs no window whatever, the peak-to-trough depth of each acoustic cycle, all put the lowest band below the rest, and none of them puts it near parity. *What the window-free family settles is the step's freedom from the statistic and not its size*: in the lowest band it holds a single half-cycle transition, the first locatable pair of extrema lying above that band's lower edge, so what it returns there is consistent with zero and equally with no step at all—*unresolved rather than small.* An extremal envelope built from located extrema alone reaches only a quarter of that band, rests on the same single extremum below the step and interpolates across it. *So the size is read from the statistics that resolve the band, and on every one of them the lowest band runs between a third and two thirds of the rest*—a range because the statistic moves it, and a range that excludes parity throughout. *And the one artefact that could produce a step at the lowest band is excluded in the direction that excludes it*: the band's envelope comes within five multipoles of truncating, and moving the band's lower edge *away* from that truncation makes it read *higher*, where an artefact of truncation would have to make it read lower.

*Located rather than merely noticed, the step sits at the second acoustic peak.* Its half-rise point, taken in sliding windows at three widths, is stable to better than a per cent and lands within about one per cent of the second peak on either arm, where the nearest other scale this construction fixes is seven times further away; and it is sharp, a clean factor of two on all three widths reached within about a tenth of a comb period—*narrower than the narrowest window that resolves it. So the excess in the first acoustic cycle runs between a third and two thirds of its size in every cycle above. What that locates is not a length belonging to this arm*: the peaks sit at the same wavenumber on both arms, so the step says that the first acoustic cycle behaves differently rather than that some new scale has entered.

*Scored against that feature rather than against a trend, every candidate in this comparison is evaluable, and between them they carry a minority of the step.* A candidate's step is its lowest band's departure from the trend its own upper bands set, read on four bases—linear and logarithmic, in the wavenumber and in its square—with a departure counted only where it survives all four. *The window weighting steps the wrong way*, upward by about half, working against the step rather than making it; *the term mix steps the right way* at some two thirds of the excess's own departure; and *the pair as actually composed*—one spectrum with both coefficients at the sizes their own profiles solve—*carries about two fifths of the excess's own step*, exceeding twice that same upper-band spread on three of the four bases; the share is against the excess's own extrapolated size and not against the larger step the arms have in common, the two denominators differing by about a factor of six on the one departure. *The projection width carries none, and the basis family is what shows why*: being linear in wavenumber to a few per cent of its own range, it flips sign across the four, so a logarithmic basis would manufacture a step for it. And the Doppler contribution, read at the lowest band by the swap above, changes sign repeatedly in sliding half-period windows with means on either side of the step equal and both near zero—*scatter at the step's own resolution, and no transition at all. So against that statistic the step has a partial account*: no single channel produces it and two of them together reach a minority of it.

*Scored instead on the differential estimator—where the shared nine tenths never enters and only the part belonging to this cosmology remains—the same channels read differently, and the*

*accounting does not survive the change.* Each knob is read there as it must be, by the width of its own difference from the control, against the arm's. *The window weighting steps, and steps the opposite way to the arm*; *the term mix no longer steps at all* at that resolution, where against the mostly-shared quantity it reached two thirds; and *the realised pair changes sign across the bases and measures zero to within its own scatter*, where against the mostly-shared quantity it reached two fifths. *So a channel that accounts for a substantial share of a quantity nine tenths of which is common to both cosmologies accounts for nothing of the part that is this one's. The weaker word is the supported one*: those two are not resolved on the differential estimator, which is not the same as being excluded.

*And they cannot be resolved by the family of statistics this comparison is built on, which is a statement about that family proved over its whole reach rather than inferred from a failure.* A channel is decided when its own departure at the lowest band exceeds twice the scatter of its upper bands, so the smallest share of the step a channel could carry and still be decided is a computable floor. *Every route this construction can take to lower that floor was taken*: both aggregations of the width, both pairings of the two arms' multipoles, a wider span of bands and a finer division of the same span. *The floor does not move below three fifths of the step, where the two channels carry about a third and about a thirtieth. And the reason it does not move is worth the sentence*: the phase-insensitive aggregation lowers the scatter by the factor predicted and lowers the signal by the same factor at the same time, so *a reduction in scatter is a gain only where the signal does not fall with it. More of the same statistic would therefore decide nothing.*

*Nor would it, for a second and independent reason.* Of the three channel readings, only the one that steps the wrong way holds its sign across every aggregation; *the other two change sign between two legitimate aggregations of the same statistic*, so their departures are not established quantities that a sharper measurement would sharpen. *What is demonstrated is accordingly narrower than that the channels do not produce the step, and narrower is what is earned*: *no statistic of this family could tell whether they do*[P15R208].

*And the family is what the limit belongs to rather than the comparison, because the likelihood these spectra are confronted with does not share it.* Its bins are a thirty-fourth of a comb period where every statistic above averages over seven tenths of one, two dozen of them fall inside the lowest band, and its bin-to-bin correlation is a flat floor rather than a coupling that grows across a period—*so it bins without banding, and it does not inherit the width limit through its covariance either. Read band by band from its own bins, the two arms are distinguishable everywhere, and at the lowest band the step appears in that distinguishability*—standing where its upper bands' trend predicts half again as much, with the same sign as every other reading of it and with the instrument's own noise for an uncertainty rather than a scatter taken across a noiseless theory spectrum. *And the two channels separate from each other there at some twenty-five standard deviations*, which is the measurement the banded family cannot make at all. *The two floors are not one number and neither bounds the other*—one a share of the lowest band's difference under real instrument noise, the other a share of the step under an empirical scatter of a noiseless spectrum—*so what compares is the verdicts*: the banded family cannot separate the channels there and the likelihood separates them[P15R209].

*But what the data can distinguish and what it rejects the arm on are two quantities, and at the lowest band they part company completely.* Of the excess in fit that the covered bands carry between them, the lowest band holds *under one per cent—and that share changes sign* as smooth global shape is absorbed into free coefficients, where every band above it holds to within a few per cent of its value. *So the step is not a distinct feature of the residual and not the shape rejection read locally either, the shape rejection being barely present there at all*: *it lives in the upper half of the range, where four bands carry between them some six sevenths of the total. That location is read off an exact decomposition and not a per-band approximation*: because the two arms' residuals differ by the difference of the spectra, the excess is identically a term in that

difference alone plus a term coupling it to the control's own residual, and the two reproduce the total to a part in $10^{13}$. *And the split is the distinction itself written as algebra*—the first term never asks where the data sits and the second is the only part that does—*which is why a quantity can stand at some twenty-five standard deviations in the one and at nothing in the other. Which puts the comparison's own structure and the data's own verdict in different places*—the excess's one prominent feature across wavenumber sits where the rejection is cheapest, and the rejection sits where no smooth shape across wavenumber finds anything at all.

*But the rejection is not a level there, and what it does carry is acoustic.* Across the individual bins those four bands hold, a constant accounts for none of the scatter, a trend for under a fiftieth of it and a quadratic for under a thirtieth—*and a projection at the comb's own period is reached by none of two thousand draws of the instrument's own covariance carried through the same statistic, nor by any of twenty thousand when they are pooled. The period it sits at is fixed only to the resolution the range allows*: the peak spans periods 0.88 to 1.19, which is the acoustic period to within the resolution of the two and four fifths periods of data the bands cover, and not to a hundredth of it.

*And no banded reading of that stretch can find it, which is not attenuation but a limit of representation.* One number per band at seven tenths of a period spacing has a Nyquist period of twice that, so a modulation at the comb period lies beyond what such numbers can represent at all, and the averaging inside each band passes barely more than a third of it before that. *The likelihood bins thirty-four times finer*, at a Nyquist period a twentieth of the comb's and a transfer indistinguishable from unity at it—*so the instrument that rejects the arm is the one instrument in this comparison that can represent what the rejection carries*, and the limit that closed the lowest band is the same limit, met at the other end of the range[P15R213].

*And the modulation is a fact about the data rather than the difference looking at itself*—the difference is a comb, so the question is which term carries it, and five sixths of it comes from the term that couples the difference to the control's residual against one sixth from the term in the difference alone. *The first clears the same noise null and the second has no noise in it to clear*, being a property of the difference and not of the data[P15R211].

*Scored in the metric the step does live in, every candidate departs the way the arm does.* Read as each channel's own lowest-band departure from the trend its upper bands set, in the likelihood's units, the window weighting delivers about a third more than the arm's, the term mix about six sevenths of it, and the pair as actually composed about half—*the first time this comparison has held candidates that depart the arm's way on an instrument able to carry the question. And the window weighting's verdict differs between instruments*, going the wrong way on the differential estimator and the right way here, *which is reported as a disagreement between instruments rather than resolved into a story. The pair again falls well short of every composition rule*, product, sum and quadrature alike, delivering about a third of the closest. *And what that costs the account belongs in the same breath as the shares*: these are shares of a departure in what the data can distinguish, and that feature carries one part in a hundred of what the data rejects the arm on[P15R210].

*Scored instead in the metric the data's own rejection lives in, the channels sort differently, and one of them sorts the right way.* Read across the four bands that carry the rejection, the term mix delivers about four thirds of the arm's cost *and delivers it where the arm's falls*—the two patterns correlating at seven tenths band by band, on a regression slope of three quarters, with nearly half the arm's cost left unexplained. *That is the first channel this comparison has held whose cost is in the right places at all*, and it is neither the carrier identified nor no carrier found: *four thirds of the cost at that correlation is not the absence of a quantity, and half of it misplaced is not the presence of one. The window weighting is the contrary case, and its small share is the wrong reading of it*: its cost is *anti*-correlated with the arm's at seven tenths, *so it is a different cost and not a small share of the same one*—the reverse of its standing on separating power, where it led. *And the pair as composed matches worse than the term mix alone*, raising

the total while degrading the pattern, *which is the sharpest form in which this comparison can say that a total is not a match*: any spectrum that is not the control costs something, so what separates costing the arm's way from merely also being rejected is the pattern and never the share.

*Whether that channel carries what the cost is made of, as against carrying the cost in the right bands, is a second question, and against the instrument's own noise the two do not come apart.* Projected at the comb period and scored against two thousand draws of the covariance through the same statistic, the channel's own cost is reached by no draw, and neither is the part of the arm's excess the channel does not explain—the latter at the arm's own phase to within a sixth of a radian. The part the channel misplaces is reached by none either, and is not quoted with them. *The statistic quoted is the tail count and not the ratio to the ensemble's largest value*, that maximum being itself a noisy quantity whose spread across seeds is about a tenth, where the tail count is stable and holds when the draws are pooled tenfold. *The channel's own clearance is carried by the term that knows where the data sits*, its other term being one the null holds fixed and therefore one no draw could have made it fail. *So every piece of this decomposition is acoustically modulated, which is what a detection test can establish here and is not what authorship needs—a knob that moves the balance of two oscillating source terms produces an oscillating cost by construction, so clearing a noise floor does not distinguish a carrier from a participant. A null built instead from the same data at neighbouring frequencies cannot be used for the question at all*, its amplitudes carrying the feature they are meant to score: over this range such an ensemble spans only about three independent draws, and its largest values run twice those of the instrument's own noise[P15R214][P15R212][P15R213].

*And the anti-correlated channel is combed too, in antiphase, which locates the reversal without naming it.* Its own cost projects at the comb period above its null with a phase $\pi$ from the arm's to within a 0.021 rad of $\pi$, so it is a structure and not a smooth offset. *Yet the two channels' modulations in the spectra themselves are in phase*, each a quarter to a third of a radian from the arm's, and the smaller one's amplitude predicts the larger's to within about an eighth from that ratio alone—*so the magnitude composes and the sign does not, and the reversal appears only after the likelihood's own weighting is applied. What that is not is one structure read with two signs*: the window's spectrum difference is not a negative multiple of the arm's, the residual of the attempt keeping almost all of the window's power[P15R212].

*The accounting that would settle authorship closes exactly and still cannot attribute, and the reason is an identity rather than a want of precision.* Writing each channel's modulation as a vector in the plane the cosine and sine of the comb span, the channel's contribution and the residue left after it reconstruct the arm's own vector to numerical precision—*so it is an accounting and not an estimate. But the share a least-squares projection returns in such a plane is the cosine of the phase offset and nothing else*: it depends only on the direction the channel points and not at all on how large it is, so *a channel a millionth the size scores the same. And the anti-correlated channel is the proof rather than an illustration*—pointing $\pi$ from the arm it is assigned essentially all of the authorship, at a negative scale that turns its own cost upside down to reach it. *A quantity that credits a channel with the whole of a feature for pointing the opposite way is not measuring authorship. The other available scale, the ratio of the two modulations in the spectra themselves, fails differently and by measurement*: it predicts the realised amplitude for one channel to about an eighth and misses for the other by more than that channel's whole size. *So one scale falls to algebra and the other to measurement, and neither failure is the other one twice.*

*And the second scale has no statistical content to appeal to, which closes the last route.* The difference between the two spectra moves with the data only through two fitted amplitudes, so a share or a phase read at that level is a property of the two theories with a systematic uncertainty and *no statistical floor at all*—the same thing the contrast statistic is, and for the same reason. *At the level of the fit the floors are finite*, about a twentieth of a radian in phase and a few per

cent in ratio, *but it is there that the projection is amplitude-blind. And the degeneracy is exactly dimensional*: two channels span the plane, so any two of them reconstruct the arm exactly and do it as arithmetic. *So what this comparison leaves is a discrepancy whose size, location, period and shape are measured and whose carrier no instrument it holds can attribute*—and what attribution would take is named rather than awaited: *a channel independent of the plane these two span, or an estimator that uses the channels' amplitudes where those amplitudes carry statistical content. Neither is a refinement of the instruments above, which is why the question is recorded as located rather than open*[P15R216P15R217].

*And the step does not compose the way the contrast does, which is a property of the step rather than of the knobs.* The contrast composes multiplicatively, measured band by band. On the step, the two channels' own departures put a product, a sum and a quadrature all at a large *positive* value where the realised pair measures zero to within its scatter on the phase-dependent aggregation, every rule several scatters away—*so on that reading the two very nearly cancel on the step where they multiply on the contrast. The cancellation is a property of the reading and not of the pair, and that is measured rather than left open*: it vanishes within its own scatter on every basis of the phase-dependent aggregation and on none of the phase-insensitive one, and in the metric the data's own rejection lives in *the two channels add*, the realised pair standing at the sum of the two singles to within a fiftieth—*so of the three metrics that can carry the question, two show the cancellation and the third shows the contrary, which is what makes it the reading's property and not the pair's. The residuals are quoted in departure units against that scatter and not as a share*, a fraction of a near-zero measurement being no fraction at all[P15R207P15R211].

*What remains to be explained is accordingly what distinguishes the first acoustic cycle from the rest*[P15R205P15R201].

*And the step is very largely common to the two arms, which is the decomposition that says where it belongs.* Read on each arm's own contrast against the trend its own upper bands set, with no reference to the other arm, *both arms step upward at the lowest band*—on all four bases and on the windowed and window-free contrasts alike—*and this arm's step is always the shallower, by about a tenth and no more. So roughly nine tenths of the step is shared and cancels in the ratio, and the excess's step is the part that fails to cancel*: the feature belongs to the acoustic physics the two arms have in common, and what this cosmology contributes is the small residual—*which is also why the step's location is the same wavenumber on both. That residual is readable because the ratio is smoother than either spectrum*, standing at some seven times its own upper-band spread where each arm's own step stands at four. *And a small residual of two large common features is exactly where a nearly-common systematic would sit*, so the feature is read again on an estimator that is differential by construction rather than a difference of two large numbers[P15R205].

*On that estimator it survives.* Dividing the two spectra multipole by multipole before any width is taken leaves the common oscillation cancelled rather than subtracted, so the quantity read is the width of the *difference* of the two arms' oscillations and the shared step is never formed: the lowest band's departure is negative on all four trend bases, on both pairings of the two arms' multipoles, and on both aggregations of the width—*a band's own dispersion and a held-period amplitude, which are not the same quantity because a band spans less than one full comb period and its dispersion is therefore phase-dependent by construction*—the two differing by about a third in size and neither chosen. *So the part that fails to cancel is not an artefact of forming two large numbers and differencing them—and the same construction shows which step of the two it belongs to*: dividing instead by a smoothed control removes the differencing while keeping the division, and the departure turns positive, back toward each arm's own.

*And the width of a difference is fed by a relative phase as well as a relative amplitude, so which of the two it is has to be settled rather than assumed—and it is settled by an identity.* For two oscillations of the same period the width of their difference is a closed function of the

amplitude ratio and the phase offset alone, and that function reproduces the measured difference pointwise, so the split into an amplitude-only and a phase-only part is exact rather than fitted. *Split that way, the step is the amplitude's*: the amplitude-only term carries almost all of the statistic at the lowest band and steps with the same sign, while the phase-only term steps the *other* way. *So the first acoustic cycle is weakened in this cosmology rather than displaced in it. What the split does not settle is the phase channel's size*: the relative phase is a fraction of a degree, an order of magnitude below the relative amplitude, and its step reverses sign as the reading window widens where the amplitude's does not—*so the phase term sits at this estimator's own resolution and the honest statement is the sign without the share. Read against the comb over several periods rather than locally, the relative phase is determined to a few tenths of a degree, at seven times its stretch-to-stretch standard error—and the lowest band still cannot be reached that way*, being narrower than the period the method needs, so a projection across it leaks the baseline and is both noisier and biased. *Against a control of stretches matched to its own width the lowest band's phase sits inside one and a half scatters, where against the wide-window error bar it would have read five—so the phase step is not resolved and the amplitude reading is the complete account rather than a verdict on the sign alone. That the lowest band is narrower than every period this comparison measures with is now the same limit met in three places*, on the window-free family, on the amplitude-and-phase split, and here[P15R208][P15R207].

*The shared part, for its own sake, does not reduce to one term of the source.* Removing each bilinear term of the decomposition in turn and asking whether the step leaves with it, the fractional losses sum to nearly twice the whole—*so the step is a property of the sum and no term carries it. What the same table does show is that leverage is not flat*: weighted by how much of the step each term accounts for against how much of the oscillation it holds, the two leading terms are both Doppler—one at more than five times its weight and one at nearly four—where the monopole holds seven eighths of the oscillation and carries the step at less than its weight. *The ordering is not simply Doppler against the rest*, one Doppler cross term telling against it and the integrated Sachs–Wolfe autocorrelation lying lower still. *So a step-weighted reading of the source points at a sector where an amplitude-weighted one points at the monopole*—a ranking and not an attribution[P15R206].

*One whole class is excluded by structure rather than by measurement, and needs no run at all.* Every quantity that belongs to the two clocks—the leaf clock, the ruler, the Jacobian between them, the visibility and the optical depth—is a function of conformal time alone, and *a function of conformal time has the same value at every wavenumber.* So it cannot vary across one. *That is neither measured-and-failed nor unmeasured*: it is a statement that the class cannot carry the effect, from the shape of the objects in it.

*And the projection itself holds a width nothing in this construction has ever imposed, which is where the row's own name has pointed since it opened.* Two widths govern the last-scattering window and they are not the same width: *the acoustic phase swept across it* goes with the window's width measured in the leaf sound horizon, while *the projection's own smearing* goes with its width measured in conformal time, that being what the projection's argument carries. *In a one-rate cosmology those are one object up to the sound speed; on two rates they are not*, because the clocks differ. *And the difference between them is one object and not a residue*: the two sound horizons on this arm are related pointwise by the Jacobian between the two rates, checked across the window to a part in a million, and identically unity on any single-rate cosmology. *Measured against the ruler clock the two arms are the same instrument*—their windows agree in width to a few parts in a thousand on either definition, the visibility being laid down by scattering on the physical background, which is the same physics on both. *Measured against the leaf clock the arm's window is some fourteen per cent wider, and the whole of that is the Jacobian*, the arm's own leaf-to-ruler width ratio reproducing the Jacobian's mean to a fraction of a per cent. *Which quantity a width is has to be said before the number means anything*: a root-mean-square width returns half as much, because the window has long tails

that it weights and that neither the projection nor the phase sweep responds to, while the full width at half maximum and the characteristic function's own half-fall agree with each other to better than a hundredth of a per cent.

*So the different ratio is a prediction of the two-rate assignment rather than a property of this instrument—the Jacobian carries no free coefficient and is fixed by the background solution once the arm is specified*—and the ruler clock is what establishes it, the two arms agreeing there being what leaves the leaf-clock difference nowhere else to come from. *And it cannot be imposed as a knob, which follows from that same identity rather than from the instrument's limitations*: giving the control this arm's conformal width at the same leaf width is giving it this arm's Jacobian, and a control carrying the Jacobian is not a control.

*Applied in the projection's own kernel instead—holding the source's phase and stretching only the projection's argument—it is a real channel whose size the statistic does not fix and it is not the carrier. Its share depends on the anchoring by an order of magnitude*: in the band where the excess is largest it delivers about a quarter of what the excess needs on the mean anchoring and about a fiftieth on the peak anchoring, so the quarter is one of the two readings and not the channel's size. *What both readings agree on is the sign of its wavenumber dependence*, and it is the wrong one—*it accelerates across wavenumber where the excess decelerates*, under either anchoring. *So it is excluded as the carrier on its shape rather than on its size, and the size is left as the range the two anchorings give. And the width ratio is a second reading of the Jacobian rather than a restatement of the first*: the comb reads its *cumulative* value at last scattering, which is what sets the peak spacing this paper already matches, while the widths read its *window-local average*—and the two differ by about a third. *So a construction tuned to the comb must still return the right local Jacobian to match the widths*, which is a second and independent demand on the same object, with neither carrying a free coefficient and both identically unity on a single rate. *What it is not is separately observable*: the width the damping reads differs between the arms by under a per cent where the width the projection reads differs by more than ten, and the projection's own contrast effect is bounded above at about two per cent with its sign undetermined[P15R199].

*One thing about the statistic belongs with those numbers, because without it none of them can be read.* The anchored contrast is measured against a running-mean envelope that is recomputed for each spectrum, so a purely symmetric change of amplitude does not read as unity on it: applied to the control's own spectrum, amplitude and envelope operations both read near two, a loading below one and a smearing negative, and the near-two is stable across a fivefold range of sizes. *So the baseline against which a class is identified is that value and not one*, and read against it the anchored statistic and the band-variance split of the excess agree to about a tenth—*the same decomposition reached twice rather than two readings in tension. And the baryon density cannot supply the contrast at any value*: its response to $\omega_b$ saturates, spanning 0.028 over the whole range with increments falling monotonically, so four per cent is beyond its reach rather than away from its fitted point[P15R177].

*And what kind of statement this statistic makes is worth saying once, because every number above inherits it.* The amplitude it fits to the data cancels exactly against the envelope it divides by, *so as defined it compares two computed spectra and reads no data at all*: its floor is numerical, at some six parts in a thousand, and every spread quoted on it above is a spread of computed quantities rather than a noise level. *Its sky counterpart is a different instrument and carries a real floor*, built by running the same statistic on noisy realisations of the control against the control through the likelihood's own covariance: across the whole range that floor is 0.018, against which this arm's +0.047 stands at about two and a half floors. *Read one band at a time it is coarser than the excess it would measure*—between 0.043 and 0.086 per band, where the excess's own per-band values run from +0.022 to +0.076, *so band by band the contrast is a statement about theories and would not be an observation*, the lowest band's value standing at under half of that band's floor. *Which is why the rejection this section closes on is read in*

*the likelihood and not here*: the whole-range contrast is what this instrument supports, and the band-by-band structure is a property of the computed spectra that the sky's own noise would not resolve[P15R215].

*Those four channels all act on the source, and the contrast difference is not in the source.* The same statistic that reads 1.047 on the reported spectrum reads 0.996 on the source at last scattering and 0.992 on the $\eta$-integrated source, over four envelope windows, both envelope definitions, three term subsets and with or without the $k$-measure—the source rungs spanning 0.971 to 1.005 and the multipole rung 1.042 to 1.052, against a floor of 0.6 per cent set by the statistic's own bias on a known injected contrast. *So the difference is made between wavenumber and multipole*, and in wavenumber this arm's relative oscillation is if anything the shallower of the two—which is the direction all four channels point, and is why none of them has anything to overcome. *The stage is located here and the mechanism is named below*: within the projection, this arm retains 1.054 times as much of its own source oscillation as the control does, the excess rises with multipole where the source ratio is flat, and it is neither the visibility width acting before the kernel, nor the lensing, nor the binning, nor the two arms' different samplings of wavenumber—replacing this arm's physical ladder with the control's uniform grid at the same mode count leaves the multipole rung unchanged. *Nor is it the distance*: the two arms' comoving distances to last scattering differ by 0.449 per cent with this arm's the larger, their acoustic angles agreeing to 0.085 per cent, and projecting either arm's own source through the other's distance moves its contrast by under a per cent—in the direction that makes the difference larger rather than smaller[P15R186].

*Nor is it one term of the source, nor the interference between two.* Decomposing the projection bilinearly—one transfer per source term, so that $C_\ell = \sum_{a\le b} w_{ab} \sum_k P\,\Delta^a \Delta^b$ closes on $D_\ell$ to one part in $10^{15}$—*deleting the monopole–Doppler cross term moves the contrast ratio from* 1.0452 *to* 1.0451, its carry running $-0.0013$ to $+0.0008$ across four envelope windows. *Each piece separately already carries the excess instead*: the monopole at 1.034, the Doppler at 1.099 and the polarisation at 1.058, against the full spectrum's 1.045, while every term's source ratio is at or below 0.992—*so the projection raises this arm's contrast on nearly everything it projects, and no single term is the channel.* Of the difference, about two fifths is this arm weighting the terms differently and the rest is each term's own response, two reweightings run in opposite directions agreeing to 0.001. *And the fraction of the source's oscillation the projection retains rises with wavenumber*: its arm-to-control ratio goes from 1.030 in the lowest acoustic band to 1.089 in the highest, with a slope of $+0.0139$ per acoustic period, the twelve envelope settings spanning 0.0021 about it and none of them returning a negative value. *The intercept of that rise covers one across the same twelve settings*, 1.017 with a spread of 0.010, so what is established is the growth with wavenumber and not a constant offset: *whatever the projection does differently here, it does nothing at the longest acoustic wavelengths and more of it at each shorter one*[P15R188]. *That slope's size is not all retention, and the part that is not is measured rather than allowed for.* Fed one comb on both arms—no difference to retain—the statistic still returns a rise at the real comb's phase, 36 to 39 per cent of the reported slope and in its direction, *because the two arms' own source combs sit at periods* 1.0170 *and* 1.0160 *and so drift apart by between* 0.013 *and* 0.042 *of a radian across the range, which a band root-mean-square taken over seven tenths of a period reads as a difference in retention. Replacing the two combs by one removes it*, the null slope falling to a sixth or less, while flattening the envelope leaves most of it—*so the carrier is the source's phase and not the envelope. The rise survives and its size does not*: about two thirds of the quoted slope is retention and the rest is that drift. *The step at the lowest band is the sounder of the two readings*, its false value at the real phase running a quarter to two fifths of the measured step. *And the statistic's phase systematic is what limits both*, carrying three to three and a half independent modes across the seven bands where its noise response carries nearly seven—*the slope's own effective count being between five and nine*, so a wavenumber slope read across these bands rests on fewer independent points

than its regression error assumes[P15R220].

*And it is not in the source at all.* Replacing the source with a pure oscillation on the visibility—no transfer, no terms, no weights, carrying $k^{(1-n_s)/2}$ so that the smooth part of the projected integrand is the same on both arms—and projecting it through each arm's own kernel, visibility and wavenumber grid gives an arm-to-control retained ratio of 1.066 at a slope of +0.023 per acoustic period, *exceeding* what the real source gives; the injection's phase does not change it. *So the projection's geometry accounts for the whole of the difference and the source is irrelevant to it.* And the two factors left in that integral are not separable: the comoving distance the kernel reads is $\eta_0 - \eta$ on both arms, so there is nothing to exchange there, while the visibility's width and the clock the acoustic phase accumulates on close the gap only together—forcing both arms onto one clock moves the comb rather than isolating a factor, and exchanging the widths amplifies the wavenumber dependence eightfold instead of removing it. *What they close on is one ratio.* Across the visibility's full width at half maximum the *sound horizon* this plasma accumulates agrees between the arms to 0.08 per cent, 17.307 against 17.294 Mpc, while the *comoving distance* across that same window differs by 14.6 per cent, 38.04 against 43.59—because the scales accumulate on the leaf rate while conformal time, and so that distance, is the stacking rate's. *The sound speed the kernel sees is therefore* 0.455 *on the control and* 0.397 *here*, and a difference in that ratio is term-independent by construction, grows with wavenumber, and vanishes for a window narrower than one acoustic period—which is the signature above[P15R189P15R225]. *That places the residual on the rate assignment of §2.4 rather than on the projection*: what is open is whether that assignment is right, not whether the transfer implements it.

*And the retention is not merely located at that acceptance; it is a law, and there is nothing in the law to fit.* For a *standing* oscillation the line-of-sight integral factorises with no approximation at all, $\Delta_\ell(k) = \cos(kr_\mathrm{s}^*)\, k^{(1-n_s)/2} G_\ell(k)$ with $G_\ell(k) = \int g\, j_\ell(k\chi)\, \mathrm{d}\eta$, and the factorisation holds to a relative $3\times10^{-15}$ on multipoles spread across the reported range—*so the comb reaches the transfer undamped, and the only thing left that can damp it is the sum over $k$ at fixed $\ell$.* That sum's weight is $W_\ell(k) = P(k)\, k^{1-n_s} G_\ell(k)^2 = G_\ell(k)^2\, \mathrm{d}k/k$, the tilt cancelling identically, and since $C_\ell = \int W_\ell \cos^2(kr_\mathrm{s}^*)\, \mathrm{d}k = \frac{1}{2}\int W_\ell\, \mathrm{d}k + \frac{1}{2}\,\mathrm{Re}\int W_\ell e^{2ikr_\mathrm{s}^*}\, \mathrm{d}k$, the comb's fractional amplitude in $C_\ell$ is

$$A_\ell = \frac{\left|\int W_\ell(k)\, e^{2ikr_\mathrm{s}^*}\, \mathrm{d}k\right|}{\int W_\ell(k)\, \mathrm{d}k}, \tag{8}$$

built from the window the kernel reads and the comb's period, both of them the background's own. *There is no coefficient in it, so it predicts an* absolute *amplitude and one measurement can kill it*—and against the amplitude measured band by band it holds to within ten per cent everywhere, worst deviation 6.7 per cent. *What that comparison is worth turns on the measuring statistic's own error, which is not a single number*: fed a comb of known amplitude the statistic is accurate to 3.74 per cent at one injection phase and to ten or eleven per cent at the worst phase, *so the smaller figure is one phase's error and not a floor. But the phase is measured rather than assumed, and the real comb sits at that phase to* 0.03 *of a radian*, where the statistic is wrong by about four per cent in the lowest band and by under one per cent in the other six. *Divided out, the law holds at every band on both arms to better than four per cent*—worst 3.4 and 3.6—*and the lowest band's apparent six-and-a-half per cent departure is around two thirds the statistic's own. And the two arms are one test of it rather than two*: their deviation patterns correlate at 0.998, as do the statistic's errors at the two arms' phases. *Within an arm the seven bands are close to independent under the instrument's noise response*, 6.9 effective of seven across four hundred draws of the whole instrument, *while the instrument's phase systematic carries only some two and a half independent modes across the same seven*—so the comparison is seven measurements against noise and nearer two against that systematic. *The acceptance the law integrates over is the narrower on this arm*, spanning 4.47 of acoustic phase against the control's 5.14—13.03 per cent narrower, *and by that same amount at every one of twelve*

*numerical settings, the spread across them being five thousandths of a percentage point*, where the sound-speed ratio above independently gives 12.8: *two routes to one quantity, and the agreement is built in nowhere. Read physically the law says one thing*: a wider $\chi$-extent narrows the kernel's acceptance in wavenumber, so less of an acoustic period is averaged over and more of the comb survives; this arm's visibility is the 14.6 per cent wider in $\chi$ while accumulating the same sound horizon to 0.08 per cent, and that is why it retains *more*[P15R218P15R221P15R219]. *Three things the law does not do, and they are stated because it would otherwise be read as doing them. The statistic the comparison is scored on does not move under refinement, and that statistic is the band-RMS ratio rather than the acceptance or the retention*: swept to full coverage over every numerical axis—the wavenumber ceiling and the mode count, the reported multipole grid, the conformal-time resolution, and the visibility's half-width and its split—the ratio sits at 1.0587 on the control and between 1.0659 and 1.0660 on this arm at every refinement of every axis, *the worst last step over all twelve axis readings being* 0.008 *per cent against a floor of* 0.6 *per cent pre-registered before the sweep ran*, with both base points reproducing their banked values exactly. *What that buys is stability and not convergence, and the two are not the same purchase*: by the rule fixed in advance a sequence that has not turned over is not converged whatever its last step, and monotonicity is undefined on two points—so *exactly one reading of the twelve is reportable as converged*, seven having not turned over and four carrying two points only. *The axis count is read from the run log rather than from the design*: five axes moved the ratio on this arm and six on the control, not twelve, because this arm's mode count is inert by construction—its wavenumber ladder never reaching the decimation cap, so those six configurations never ran at all—while the same setting moves the control outright. *And the acceptance's own inert set is a different set, which is why the two are counted separately*: three of the twelve pairs cannot move $A_\ell$, the reported multipole grid because the acceptance is read at the same multipoles by construction and this arm's mode count for the reason just given, the two refinements writing a wavenumber axis, measure, conformal time, kernel argument and visibility identical to the base run to the byte[P15R222]. *What none of that settles is the sum the spectrum is made of*: the retention and the peak heights are measured on spectra rather than on the window and are a different quantity from the ratio swept here, and §10 keeps that as its own open reading—*a stable statistic beside an unconverged one would itself be a finding, which is why the two are not reported together.* The oscillation it is checked against is a known input transferred through the projection, so that the law over-delivers is sufficiency and not identity with the real source. And naming $A_\ell$ does not say why the construction puts its scales and its distances on different rates, which is where the residual sits and is untouched by anything here.

*What the rule does reach, and it reaches the projection directly, is the distance the kernel argument is built from.* The assignment is not a modelling choice on either side: a comoving separation read across leaves—$D_M$, $D_H$, $D_V$, the observable expansion—takes the stacking rate, while a process running *in* the content—the plasma's sound horizon, its diffusion length, recombination, the perturbations—takes the leaf's, and *there is no locus at which the rate switches. So the two rates do not compete for the line-of-sight integral; they divide it.* The integral $\Delta_\ell(k) = \int S\, j_\ell(k(\eta_0 - \eta))\, \mathrm{d}\eta$ runs over one variable, and that variable is the stacking conformal time because $\chi = \eta_0 - \eta$ is a separation read across leaves; the plasma's quantities then enter it as functions of that variable, carried through the Jacobian $\mathrm{d}\eta_{\mathrm{leaf}}/\mathrm{d}\eta_{\mathrm{stack}}$, which on this arm runs 0.789 to 0.913 across $\pm 3$ half-widths of the visibility and is identically one on the control[P15R189P15R224]. *Two rates inside one integral is what the rule requires. One rate throughout is what it forbids*, since the single rate would have to be the leaf's for the plasma's sake and would then carry $D_M$ on it—*reading a stacking quantity on the leaf's rate, which is the reification the construction's own proposition rules out.*

*And the division is carried out rather than asserted, which is checkable and is checked.* The sound horizon the oscillator's phase is reckoned in accumulates on the leaf rate, and the integral runs over the stacking variable, so the two statements of one quantity—$\int c_s\, \mathrm{d}\eta_{\mathrm{leaf}}$ and

$\int c_s \, (\mathrm{d}\eta_{\mathrm{leaf}}/\mathrm{d}\eta_{\mathrm{stack}}) \, \mathrm{d}\eta_{\mathrm{stack}}$—must agree identically. *Across the plasma's whole run on this arm they agree to a part in* $10^8$, while the same integrand carried on the stacking rate throughout stands 75 per cent away. *So the Jacobian is doing the work the rule assigns it*, and the phase the oscillator turns over on is the leaf's even though the variable it is expressed in is not.

*And the size of what the division decides is worth stating, because the distance is a poor witness to it.* One event—recombination at a fixed redshift—carries two conformal times, 485.5 Mpc on the stacking rate and 282.3 on the leaf, and *both readings are correct*: the first is what $\chi$ and $D_M$ are built from and the second is what the plasma's accumulations run on. *What does not distinguish them is* $D_M$: built either way it moves only from 14017 to 13947 Mpc, half a per cent, because the present epoch moves with the recombination epoch. *So the distance cannot be read as a check on the assignment*—the two clocks differ by seventy per cent in a quantity the projection distance records at half-a-per-cent[P15R225][P15R226].

*The same division settles where the plasma's evolution starts, and settles it as a consequence rather than a choice.* A sound horizon's memory of its own lower endpoint scales as a power of the scale factor there, and *the power is the radiation term*—the one thing by which the two rates differ. On the stacking rate, with no radiation era, the integrand runs as $a^{-1/2}$ and the memory is a square root: measured at 0.5008, the integral is still 1.1 per cent short at $z = 10^7$ and 43.1 per cent short at $z \simeq 6.8 \times 10^3$. On the leaf rate the integrand is flat in $a$ and the memory is linear, measured at 0.9983, leaving the accumulation 0.0064 per cent short at $z = 3 \times 10^7$. *So a sound horizon carried on the leaf rate has no lower endpoint to place and the acoustic angle is an output, exactly as the handover requires*; the same quantity carried on the stacking rate would need its start supplied by hand, and the control, whose single rate obeys the leaf law at 0.9985, is why integrating from $a \simeq 0$ was never in question there[P15R224]. *And the start is super-horizon on both rates alike*—the comoving horizon vanishes at the handover whichever rate is read, so the condition the coherence argument needs does not turn on this division at all.

*With every assignment settled, the configuration the construction specifies can be run against the one it rules out, and the two are measured side by side on the same grid with the same steps.* The test was fixed before either was read: a $\chi^2$ that improves while the residual still swings is not the fix, and a residual that stops swinging is, even if $\chi^2$ moves little—a settled residual having *more* sign changes, *shorter* runs of one sign and *smaller* excursions. *The configuration the rate rule specifies moves neither statistic.* The contrast coefficient the declared parameters cannot supply moves by 0.08 of its own error, its magnitude *rising* one per cent to $-0.0642$; the longest run of one sign goes from 18 to 22 bins, the wrong way; $\chi^2$ per bin goes from 3.05 to 3.20. *Carrying the projection's distance on the leaf rate instead—which is what the rule forbids—moves the same coefficient by* 6.7 *of that error to* $-0.0069 \pm 0.0083$, *its magnitude falling* 89 *per cent onto the value the control itself returns*, while the sign changes rise from 30 to 44, the longest run falls from 18 to 12 and $\chi^2$ per bin falls to 1.47. *And the acoustic scale follows it*: $\ell_A$ moves from 302.889 to 301.380 against the control's 301.375. *The control's own coefficient is identical across all three grids*, which is what says the move belongs to this arm and not to the method[P15R228].

*What that comparison can and cannot carry is fixed by where the discriminating feature lives, and it is measured rather than assumed.* Across four reading ceilings the three configurations are *indistinguishable below* $\ell \simeq 800$—every one of them with a longest run of eight or nine bins—and separate only above it, where the specified configuration's run grows 9, 9, 13, 22 and the forbidden one's stays 9, 9, 9, 12. *So the statistic decides nothing when read low, and what it decides when read high is not an artefact of where the grids stop, which is measured rather than conceded.* The projection's reach is a fixed multiple of the inverse distance on every arm—$k_{\mathrm{max}} D_M = 3998$ against $\ell_{\mathrm{max}} = 2000$ on each—so a shorter distance does not carry an arm nearer its ceiling and the coupling that would manufacture the separation cancels by construction. *More than half the separation is already present eight hundred multipoles below the*

*ceiling*: re-fitting all three statistics to data truncated at a succession of ceilings, the specified configuration's excess over the forbidden one runs $-1.0$, $+14.4$, $+51.0$, $+67.9$, $+74.7$, $+84.0$ and $+93.8$ as the truncation moves from $\ell = 850$ to 1996, with the bulk of it, about half, falling between $\ell \simeq 850$ and 1500; and *at every one of those truncations the forbidden configuration sits within about one unit of the control*, which a flatness produced by distance from a ceiling would not do. *What the comparison therefore carries is a rigidity across scales rather than a feature confined to high multipole*: its onset falls between $\ell \simeq 850$ and 1000, it is already decisive by $\ell \simeq 1200$, and it is *not* converged—it grows to the last bin—so the quoted separation is a statement about the range read and not a saturated value. *Below the onset the parameters absorb everything*, even though a third of the separation on the full-range fit sits there, which is what makes the failure a cross-scale one: no single parameter set fits the low and the high multipoles together[P15R234].

*Taken together these fix the open end of this section precisely, and it is narrower than the section has previously been able to state. The implementation now does what the construction says on every object*: the plasma's sound horizon, its diffusion length, its ionisation history and the perturbations on the leaf rate, the optical depth and the projection's distance on the stacking rate, and no free time origin anywhere. *What the sky prefers differs from that in exactly one assignment*—the rate the projection's own comoving distance is built on—*and it prefers it by a large margin on three statistics that do not share a definition. Rebuilt consistently on the assignments the construction makes, the fit's unreachable* $\chi^2$ *moves from* 278.795 *to* 278.788—0.007 *per cent of a gap of* 92.8 *against the control's* 186.007—*and the residual's sign changes and longest run of one sign do not move at all, on a rebuild that moved the spectrum itself by six and a half per cent. Built instead with the projection's distance on the leaf rate it closes the whole of it*: $\chi^2 = 184.989$, below the control's own value; 91 sign changes against the control's 87; a longest run of 8, which is the control's exactly[P15R230P15R235].

*And what that second configuration is, said plainly, is what makes the comparison a statement about this cosmology rather than about its implementation.* The leaf rate and the stacking rate differ by the radiation term alone, and *the control's single rate carries that term*—so building this arm's distance on the leaf rate gives it a rate of the control's own form, radiation present, differing from the control's only in the parameters the distance data fix. *The arm keeps everything else that makes it this cosmology's*: its own handover datum, its own discrete wavenumber ladder, its own sound horizon. *So the configuration the acoustic contrast prefers is the one in which radiation is restored to the rate that builds the distance*—which is the single feature this construction removes, and removes not as a choice but because the vacuum kernel fixes the stacking rate from $\Lambda$ and the cut's offset alone with radiation a bend of the cut and no part of it (§2).

*Stated as the two-sided thing it is*: that same absence is what the distance ladder confirms at $\chi^2/\mathrm{dof} \simeq 1$ and what reproduces the acoustic *scale*, $\pi D_C/r_s$ coming to 302.87 from the background alone against the 302.889 the grid measures. *What disagrees is the acoustic* contrast*, and it disagrees in the one direction the construction cannot move. So this section leaves neither a residual in search of a parameter nor an arithmetic to re-run*: it leaves a localised, quantified disagreement between this cosmology and the microwave sky, with one bounded calculation outstanding that could still bear on it (§10).

*And the rule does reach the object that difference lives in, by the one property that distinguishes the two sides of it.* The visibility $g = \tau' e^{-\tau}$ looks ambiguous because the optical depth counts the plasma's own scatterings while being differentiated per unit conformal time. *The rate it carries is settled by asking what it is an integral* over*, not what it is an integral* of. Thomson scattering has a single rate per unit proper time, so $\tau' = \Gamma a/c$ is the same expression on either reading and the whole question is the accumulation; and by the definition of a conformal time, $a\,\mathrm{d}\eta = c\,\mathrm{d}t$, so the proper length a photon covers between two leaves is $a\,\mathrm{d}\chi = c\,\mathrm{d}t_{\mathrm{stack}}$. *Weighting* $\mathrm{d}\tau$ *by the leaf-to-stacking Jacobian would therefore count the scat-*

*terings of a photon crossing that same proper length in less time*—at 1.130$c$ at recombination and 1.175$c$ by $z = 1500$—*and a measure that moves the photon off the null cone of the path it is integrated along is not a second admissible reading. So the optical depth takes the stacking rate, as a photon-path observable alongside the redshift and the distances, and the content enters it through $n_e$ rather than through the measure. The apparent counter-case is distinguished by its integrand rather than overruled*: the diffusion length's $1/k_D^2$ carries the baryon loading and the polarisation coefficient, so it is the oscillator's own viscous damping and shares the oscillator's clock, while $\tau$ carries nothing but $n_e\sigma_T a$ and shares the kernel's. *Each integral takes the clock of the evolution it runs over, and the Thomson rate appears in both while setting the measure in neither*[P15R227]. *What is left in the visibility's neighbourhood is the ionisation history alone*, which is the content's own chemistry over its own duration and so the leaf's: carried there it moves last scattering by 0.59 per cent in redshift and its width by a tenth of a per cent.

*The family that assignment sits in is measured anyway, because what the sky can see of it is worth knowing even once the derivation has settled it. The peak positions are the only one of the three readings with an external referent*—the retained oscillation's level and its wavenumber slope have none—so they are the only one against which the span can be scored at all, and the measurement below is of what that referent resolves rather than of what it decides. Letting the weighting on the optical depth run continuously from one clock to the other through the one-parameter family $1 + f(\mathrm{d}\eta_{\mathrm{leaf}}/\mathrm{d}\eta_{\mathrm{stack}} - 1)$, every reading is linear in $f$: the first peak runs $\ell_1 = 221.93 + 4.21f$ to three hundredths of a multipole. *So the whole span between the two admissible assignments is four multipoles*—0.01395 in $\ell_1/\ell_A$, which is 4.2 times a one-multipole locating width and 2.1 times a two-multipole one—and the peak positions therefore pin the assignment to about a quarter of that family and no further. *Those two widths are assumed rather than measured here*: the only locating uncertainty this comparison propagates through the sky's own covariance is the fourth peak's, at 2.19 multipoles, *so whether the first three locate to one multipole is not settled by anything above and the span is quoted against a stated assumption.* The retained oscillation's *level* is the shallower reading, at 1.75 of its own band spread across the family, and its wavenumber *slope* reaches 3.80 of its own regression error over that family. *Those two are not the peak positions' statistical equals and are not quoted as such*: the comb's width is a locating width assumed of the sky where both of the others are spreads taken across a family of computed curves, *so what the comparison supports is which reading moves and which does not*—the contrast's spread grows about fivefold across the family while the comb's locating width does not move at all. *And the decisive result is not about the choice between the clocks.* The sky's 0.7312 lies at $f = -0.304$ on that same straight line—outside the family, on the far side of the clock the transfer already uses—so no intermediate weighting fits the peak positions better than the endpoint in use, there is no fitted clock to declare, and *the residual first-peak disagreement cannot be absorbed by the clock assignment at all, because the direction it would need is not admissible.* The height ratio $P_1/P_2$ agrees independently, moving $2.141 \to 2.017$ away from the sky's 2.217. *And the motion the alternative produces is separable into its causes*, the projection of an analytic oscillating source with no plasma dynamics in it isolating them: about a third is the visibility peak relocating, so that the sound horizon at last scattering is read at a slightly different time, a tenth is the visibility's re-weighting of the projection kernel, and three fifths is the plasma's own acoustic phase responding to the re-weighted optical depth[P15R191].

*The comparison between the arms is taken as a distance between the two models rather than as a difference of two numbers each read against the sky*, which is what makes it independent of the configuration's own floor[P15R93].

*The instrument's own accuracy is measured rather than assumed.* On the full range its control sits at 1.16 per bin lensed, within about a seventh of what a standard code returns on the same bins, so the ratio above is a statement about the arm and not about the machinery. *What the comb does not carry is the disagreement*: at 298.0 against 298.4 the spacing is right

to a part in six hundred, on both instrument paths, so what remains is about one multipole at each peak.

*The result is reproduced independently on a second instrument.* A separate implementation—the hierarchy path rather than the line-of-sight one, 185 covered bins rather than 133, its own wavenumber grid—returns peaks at 221.96, 536.11, 815.40 and 1130.53, located sub-bin on a grid refined to unit step and recovered there to a few hundredths of a multipole, against this path's 222, 538, 818 and 1134; height ratios 2.273 and 2.319 against 2.264 and 2.298; and 4.02 per bin in the band from $\ell = 700$ to 1000 against 4.23: *within a multipole on every peak and within one per cent on the heights. The fourth peak's excess reproduces there too*, so it is a property of the spectrum rather than of either implementation—and on that path, where the sub-bin location is available on both arms, it is a property the control carries with it[P15R166][P15R192].

*The heights are the polarisation path's and the positions are not.* Both paths return the same comb to four figures; the height ratios differ between them by more than either differs from the sky — 2.264 and 2.298 on the polarisation path against 2.448 and 2.934 on the fluid one — so every height quoted here is the polarisation path's and is stated as such, while the position, the phase and the alternation are the construction's.

*Where the plasma starts moves the scale and not the peak, which is why the start is not free.* Scanning a hand-placed start over a factor of 3.6 runs $\ell_A$ from 360.6 down to 239.3 while the first peak sits between 204 and 208 throughout[P15R6]: it moves the denominator of $\ell_1/\ell_A$ and leaves the numerator where it is[P15R24], so the spacing such a start reports is not the spacing its spectrum carries.

*What that leaves is the equality, and it is measured rather than supposed.* Holding everything else fixed and moving matter–radiation equality alone carries the third peak and the band it sits in with it: the height ratio $P_1/P_3$ tracks equality on both routes to it — through the radiation density and through the matter density, which move the projection in opposite directions — agreeing to within a per cent across a 12 to 24% change in the matter density, so what the band responds to is the equality and not the route. *And the epoch the distance data fix is what sets it*: equality is $\omega_m/\omega_r$ with the radiation density measured, so the background that meets the baryon-acoustic distances is the same background that drives the peaks.

*One candidate for the residual's carrier is measurable, and it is measured shut.* The visibility's width in comoving distance is 14.3 per cent greater on this arm than on the control while the two accumulate the same sound horizon to 0.08 per cent, and it is what sets the kernel's acceptance in wavenumber—so it is the quantity to reach for. The swap is well posed and both directions are run. Widening the control's visibility to this arm's moves its folded harmonic at $\ell_A$ from $0.3704 \pm 0.0132$ to $0.3639 \pm 0.0139$, *away* from the arm's 0.4003 rather than toward it; narrowing the arm's to the control's leaves it at 0.4007, unmoved to one part in a thousand. All four folds are resolved at 26–30$\sigma$, so a two-per-cent move is one this statistic would have seen, and the arm-to-control difference survives the swap and slightly grows, from +8.1 to +10.1 per cent. *A mechanism that made the residual would unmake it when undone, and this one does neither.* The same candidate fails a second way: a wider visibility is Silk damping, which acts strongly at high multipole and barely at low, so it predicts an arm-to-control ratio that climbs with $\ell$—and the measured ratio is flat to 9.5 per cent of its mean across 100–1900, against the 230 per cent the absolute amplitude moves over the same bands[P15R283][P15R282].

*And the residual's own period is not the comb's, which is where this sector's open work now sits.* The fold is taken at $\ell_A$ by construction and nothing had asked whether that is where the modulation actually lies. Fitted as a free parameter over $240 \le p < 380$, the unfitted arm prefers 312.0, on a one-sigma interval $[308.5, 315.2]$ that excludes $\ell_A = 298$ at $\Delta\chi^2 = 15.0$—an offset of +4.7 per cent; freeing the four parameters pulls it back to 307.3 without reaching $\ell_A$, which is the pattern the amplitude shows in the one quantity the figure does not report. *The control prefers a longer period too*, 347, so preferring one is not by itself this arm's; what is this arm's is the determination, its interval being 3.6 times the tighter because its amplitude is

6.8 times the larger. *And the control cannot serve as the null against which that offset is read.* With the period free in both arms and the difference taken as the statistic, on draws of the bandpower noise shared between them, the observed $-35.25$ sits at $0.97\sigma$ of $-22.86 \pm 36.29$: not significant. *The reason carries more than the null does.* Of that 36.29, the control contributes 35.52 and this arm 2.99: the arm's period is determined to one per cent and the control's to eleven, so the control's preference for a longer period is not a preference but noise, and there is no measurable period there to compare against.

*The comparison the data can make is against $\ell_A$ itself, which is reproduced to* 0.15 *per cent where the control's period is known to* 11. Against it the unfitted arm is $+14.00$, which is $4.69\sigma$, with fewer than one draw in a thousand at or below $\ell_A$; refitting pulls it to $1.05\sigma$ and widens its determination threefold, the same pattern the amplitude shows. It is not the single-harmonic fit's bias either: giving the second harmonic its own freedom moves the fundamental by $+0.25$ and leaves $4.90\sigma$.

*And the offset is not a competing period at all.* A period of 312, and $\ell_A$ with a phase running linearly in multipole, are the same two-dimensional model—identical residual sums of squares—so what the residual carries is a *drift of the acoustic phase at the correct spacing*, $-96.6^\circ$ across $104 \le \ell \le 1886$, which is 0.268 of a comb period over the 5.98 the range spans. *So the offset is not a statement about $r_s/D_M$*: $\ell_A$ sets the spacing and the offset is 31 times $\ell_A$'s own error, far too large for the spacing to be slightly wrong. *It is the phase again, and in the one parameter this section had not yet read it in.* The phase intercept is already measured here, $\varphi/\pi = -0.2349$ against the sky's $-0.2405$; what the period offset adds is the *slope*. Both are the peak phase and neither is the ruler—which is where this section's own finding already points: the disagreement is in the phase and not in the scale. *What could carry a running phase at correct spacing is therefore the driving and the baryon loading, and that is named rather than measured*[P15R282][P15R285][P15R287][P15R288].

*The identity holds and the attribution does not follow from it, and the two are separated here because the free-period slope turns out not to be a phase observable.* Moved along each of the four banked background directions, the slope asks for nearly the same fractional change of all of them— between 4 and 6 per cent, a spread of only 1.55 from most to least responsive. *The tell is the tilt*: $n_s$ multiplies the primordial power and leaves the transfer function untouched, so it carries no acoustic phase at all, and it asks for the same move as the baryon density to within a factor of two. A slope fitted to a residual with one amplitude free is therefore reading smooth spectral gradient alongside any phase, and the drift's *size* is an identity while its *name* is not something that statistic can supply[P15R287][P15R289].

*The peak set is the observable the slope is not, and it is read on the same banked spectra.* Taking the peak positions against integer index after dividing out one global power law—a de-tilt the same control forces, since without it a primordial tilt appears to move peak positions it cannot move—the common offset and the odd–even alternation separate the baryon direction from a pure tilt by factors of 11.3 and 21.4 on this arm and 11.4 and 22.6 on the control, against 1.3 for the slope[P15R288][P15R296]. *And the two candidate carriers move that pair in opposite senses on both components*: on the control the driving gives $\Delta\varphi = +0.126$ with $\Delta\mathrm{alt} = -0.0247$ while the baryon direction gives $-0.0278$ and $+0.0108$—a difference of sign on two independent components rather than of magnitude on one[P15R288]. *So the plane separates the two candidate carriers from each other and from a pure tilt, and the arm's own driving signature is read on it rather than borrowed.* With the driving switched off at the arm's own configuration the acoustic scale is unchanged to the last printed digit, so the pair differs in the driving and in nothing else, and the signature it returns reproduces the control's to half a per cent, $\Delta\varphi = +0.1259$ against $+0.1265$. *On the plane's own discriminant $|\Delta\mathrm{alt}/\Delta\varphi|$ the two candidate carriers sit at* 0.193 *for the driving and* 1.039 *for the loading*—a factor of 5.4 apart, and they move the common offset in opposite senses, more driving lowering it where more loading raises it. *The loading is separated from a reparametrization of the clock as well, by a factor of six; the driving is not*,

sitting within a sixth of the largest clock reading and inside the spread of the three. *So the component pair settles which of the two candidates carries a phase, and leaves open whether a driving signature is the driving or the clock it is read on*[P15R290P15R291].

*And the clean lever corrects what the baryon density was taken to be on this observable.* Scaling the baryon loading alone moves the common offset by $+0.0164$ per unit logarithmic change, where the full baryon direction moves it by $-0.0276$: the part of that response which is not the loading is $-0.0440$, *larger than the loading's own part by a factor of* 2.7 *and opposite to it in sign.* So the baryon density is not a loading lever carrying a second effect alongside—it is a lever whose two effects oppose, with the larger of them not the loading, and a reading that took it for the loading would have placed that carrier on the wrong side of zero[P15R289].

*What the plane has not been given is the observed residual's own position on it, and the statistic as built cannot supply it.* Every response above is read on model spectra—smooth curves with no noise in them—so what is established is that the two candidates are distinguishable and not which of them the sky's leftover wiggle sits with. *The obstruction is the peak locator rather than the data's precision, and it is specific.* The binned residual carries the statistic perfectly well: passing a model through the likelihood's own binning moves the common offset by 0.0013 and the alternation by 0.0003, a bias consistent in sign and applied to models and data alike. *What fails is that a maximum-finder cannot tell an acoustic peak from a noise wiggle.* On the observed spectrum it reads 21 local maxima where the model has 5, and three of the five it then takes—at $\ell = 464$, 527 and 617—lie inside the second acoustic peak's own neighbourhood, so the refinement fits curvature that is noise. *The resampling confirms the diagnosis rather than widening the answer*: over 2000 draws of the full $179 \times 179$ covariance, 98.2 per cent fail to yield five finite peaks at all, and the survivors return a scatter in the common offset ten times the driving's entire signal while collapsing onto a single direction at a correlation of $-0.955$—*so not a wide interval in this plane but no interval in it. The two regimes are separated by exactly one property*, whether a spectrum carries noise at the scale of its own curvature, which is why this bounds where the statistic may be used and withdraws nothing measured on the model side. *And that property is set by the binning rather than by the sky, which is what carries the statistic across.* At the likelihood's native bin width the per-bin scatter is 2.4 times the signal's own bin-to-bin increment and the located maxima come back out of order, one of them displaced by 547 multipoles. Merging the bins in pairs doubles the increment and reduces the scatter by $\sqrt{2}$, so the ratio falls by $2\sqrt{2}$, to 0.8, and the locator then reads the observed spectrum's first four peaks within 3.4 multipoles of its own reading of a model spectrum passed through the identical merge, the located gaps coming back 303, 268 and 309 against the model's 302, 269 and 312. *The usable merge is a factor of two and only a factor of two*: merged in threes the locator returns nothing on either, its own search window then holding fewer than four points, which is a property of the locator rather than of the spectrum. *So the two halves of this comparison want opposite binnings*—the comb wants points the correlation above shows it cannot use, and the locator wants the bank merged, which costs exactly those points—*and it is the located peak positions, on a merged bank and at four peaks rather than five, that reach the observed spectrum*[P15R291P15R292P15R295].

*Carried onto that bank the plane reproduces its own banked driving pair, and the observed spectrum can be placed on it.* The de-tilted peak statistic through the identical merge gives $+0.12703$ and $-0.02494$ for the driving difference against the banked $+0.126$ and $-0.0247$, so the merge costs under one per cent on both components; the restriction to four peaks leaves the common offset unchanged to a tenth of a per cent and rescales the alternation by a fifth, which is a property of the peak count rather than of the merge, since the instrument's own unmerged samples give the same 1.20. *Re-measured at four peaks the separation survives*: the baryon direction is still between 9.8 and 12.1 times a pure tilt on the common offset and between 8.2 and 9.9 on the alternation, against 1.3 for the free-period slope, and the lever lost on the alternation is the tilt control's rather than the signal's, $n_s$'s residual leakage onto it growing four-fold at

four peaks. The observed spectrum then sits $(+0.00756, -0.00255)$ from the matched control on the plane, the two candidate carriers remaining separated there by a factor of 5.6 on the plane's discriminant. *What that position is worth follows from the likelihood's own covariance carried through the merge exactly*—the merge is a linear map on the binned spectrum, so no correlation is dropped—*and through the locator by simulation*: the statistic's distribution is heavy-tailed by a factor of eleven, and on its robust scale one full unit of the driving is 13 standard deviations on the common offset and 3.1 on the alternation, while the observed point sits 0.8 and 0.3 from the control. *So the observed spectrum is consistent with the control's own driving, and one full unit of difference either way is excluded*—and it is the driving that this bounds, the loading moving the plane by only 1.8 and 2.4 standard deviations per unit. *Three things bound that statement and none of them is the noise, and they divide it: the exclusion carries and the agreement is the conditional half.* The comparison made here is the observed spectrum against a model rather than one arm against the other, so the projection's own running phase is not differenced away by it: in these units that phase is 5.7 times the displacement measured, and were it to fail to cancel at its full size the observed point would move to 5.3 standard deviations, while the thirteen-deviation scale is a model-against-model quantity and is untouched. *And that condition is of the same kind as the one below rather than merely unmeasured*: what it asks is whether the model's projection kernel is the sky's, which cannot be settled from the same spectrum without circularity and from outside is another model's answer, so no bounded form of it exists. *The acoustic scale is assumed rather than measured here*: moving it by one per cent moves the observed offset by 2.4 of those standard deviations, so the agreement would require the scale to be the control's to 0.4 per cent where the exclusion requires only 5.5—*and that condition is not one this measurement can discharge*, since a scale read off the same peaks enters the statistic that is being tested and any scale brought in from outside is a standard fit to the same sky, so either route restates the condition instead of checking it. *And one of the maxima is not certain to be there at all, but it is the third rather than the fourth*—on 48 per cent of noise realisations of this spectrum the locator returns three peaks instead of four, and the slot that goes missing is the third in all but a twelfth of those, the fourth being located in 96 per cent—so every figure here is conditional on the statistic existing, which it does on about half of the realisations. *That condition costs efficiency rather than width.* An offset formed from the first two peaks is defined on 99.6 per cent of realisations, and inside the four-peak selection its robust width is 0.982 of its width over the whole ensemble, with a bootstrap interval of 0.953 to 1.010; the realisations that are discarded are 1.04 times wider, and the loss rate is flat to five points across quintiles of how far the offset already sits from the median, where a selection that cut the statistic would climb. *Carried across to the four-peak width*—the only route to it, since that statistic does not exist off its own selection—*that ratio widens the deviations above by under two per cent, which is why the selection enters this paragraph as a condition on the statistic's existence and not as a correction to its scale. Nor does that reading turn on where the threshold for it was set*: taken at the unfavourable end of the interval instead of at its centre, one full unit of the driving is 12.5 standard deviations and the observed point sits at 0.75, so both figures quoted above survive the change[P15R304P15R296P15R304].

*Built with those three parameters it does not survive its own control, and what it fails on is this section's own quantity.* It recovers the acoustic scale on three spectra of six—both driving-off cases, to about a tenth of a per cent, and one of the driving-on cases to 0.27—and misses the other three by between 18 and 20 per cent. *What separates the two groups is not the driving.* Each arm's driving pair straddles the divide: the same arm is recovered to a tenth of a per cent with the driving off and missed by 18.5 per cent with it on, so a difference taken across that pair is not a difference in the driving but a difference in whether the instrument worked. *The reason is visible in the gaps.* A comb's successive gaps are $\ell_A(1 \mp 2a)$, so its high gaps equal one another and its low gaps equal one another; the arm's run 302, 270, 312, 295—alternating, and *rising*, the highs by 10 and the lows by 25. On every spectrum tried both the high and the

low gaps rise, and a single linear drift term reduces the residue of the gap fit on every one of them—from 9.5 multipoles to 3.8 on this arm. *It is the size of that residue and not the presence of the driving that sorts the three the comb recovers from the three it misses. And omitting the term does not merely fit worse, it returns a different alternation*—larger by between a third and nine tenths on the driving-on spectra—which is one of the two components the separation above is read on. *A drift in the spacing is a linear phase drift at fixed* $\ell_A$, which is what this section's residual was found to be, so the instrument is short of exactly the freedom the measurement is about[P15R292].

*Two cautions travel with a fit of this kind, and both were met here.* Given each harmonic its own envelope the fit improves by a factor of 542 while the acoustic scale runs to the edge of the search box and the answer worsens by a further 24 points: *a good fit is not evidence that the comb was found*, the extra oscillatory freedom absorbing the spectrum instead of locating its periodicity. *And a search started where the answer is expected cannot report that the answer is elsewhere*—a grid of starts offset from the known scale returns a worse optimum on five spectra of six, and on the one the comb recovers it had reported a minimum a fifth away from the true one[P15R292].

*Admitting the drift as a fourth fitted parameter confirms that it is the right one and shows that this data set cannot carry it.* It is the right one on a test it was not fitted to: the drift the fit returns agrees with the curvature measured independently from the gap sequence, in sign and to within a sixth, on four of the six spectra. *And the driving-off pair, which looks ruined on a careless reading, is not*—in a drifting comb the fitted spacing is the value extrapolated to the window's edge rather than the acoustic scale, and read without that pivot the drifting spacing passes through the known scale at $\ell \simeq 815$ and 813, inside the fitted window, while the fit improves by nearly a factor of two. *What defeats it is the resolution of the data rather than the model, and the shape of that limit is sharper than a correlation.* Across every spectrum whose fit is interior to its search box the spacing and the drift come back correlated between $-0.94$ and $-0.98$, and *that correlation is a property of the coordinates and not of the data*: referring the comb to a pivot inside the fitted window rather than to its edge is an exact shear on the two parameters, leaves the fitted curve identical to the digits, and drives the correlation to zero, with the decorrelating pivot falling inside the window on every spectrum that has a curvature to transform. *What it does not do is deliver the acoustic scale.* The combination the data determine at that pivot is the spacing in the neighbourhood of the second peak, and it lies below the computed scale on every fitting window tried, by between 5 and 20 multipoles with one sign throughout. *The formal error quoted on it is not a measure of how well it is determined*: moving only the window's upper edge moves the number by five per cent while that error runs between 0.01 and 4.3, so the window spread and not the error bar is the honest statement of its reproducibility. *So this bank pins one combination of the scale and its drift and the acoustic scale is not that combination*: one direction of the pair is tight, the orthogonal one is loose, and the scale lies along the loose one.

*What the tight direction turns out to measure is the fit's own grip on the peak train rather than anything about the sound.* Pointing the peak locator at the fitted curve instead of at the data, the pivot spacing *is* that curve's mean peak gap—to a fifth of a per cent on the two driving-off spectra, where the fitted maxima sit 1.6 multipoles from the located ones—and is 21 to 27 per cent away from it on the other four, the observed spectrum included, where they sit 21 to 24 out. *So a difference taken along that direction separates a fit that holds the peak train from one that has lost it*, and the matched differencing that would carry it to the sky is blind to the quantity of interest: arm against control, with the observed spectrum differenced out of both, comes to 0.3 multipoles against a within-pair difference of 57. *And the degeneracy behind all of it is invariant to everything this bank can vary.* The correlation between the spacing and its drift stays between 0.94 and 0.98 across thirteen fits spanning 78 to 176 points, five windows and six spectra, and halving the point count moves it in the fourth decimal while inflating the errors

by the expected $\sqrt{2}$ and doing nothing else. *The reason is geometric rather than statistical*: a level and a slope in the same variable are monomials whose correlation over an interval is fixed by the interval's endpoint ratio, not by its width or by how densely it is sampled, which is both why the shear removes it exactly and why no amount of data touches it[P15R293][P15R294][P15R295]. *A four-parameter comb is the better description of these spectra and the worse instrument for this measurement*, and a finer binning would not change that.

*And the drift itself is a determined quantity of this construction rather than a feature of the numerics, which is settled by derivation and localises it away from the acoustic physics entirely.* The source comb in wavenumber carries no drift at all: measured on the source extrema at recombination it lands on the integers to 0.015, and switching the driving on displaces it by a nearly constant $-0.12$ in $kr_s/\pi$—a spread of 0.029 across six peaks—so the driving supplies the offset and the alternation and cannot drift a comb, a constant phase shift having no curvature. *The projection manufactures the whole of it.* A perfect comb, extrema placed exactly at $kr_s = n\pi$ with no dynamics behind them, acquires gaps of 278.5, 289.5, 293.4 and 295.4 on projection—rising, decelerating and saturating below $\ell_A$. *And it is obtained in closed form rather than read off a spectrum*: replacing $j_\ell^2(kD_M)$ by its own asymptotic average and substituting $x = \ell \cosh t$ so the inverse-square-root edge integrates exactly returns every peak to better than 0.5 in $\ell$ and the drift to 0.01, which identifies the drift as the centroid shift of the projection kernel's own window acting on a damped comb, with $r_s$, $D_M$ and $r_D$ as its only inputs. Moving the handover epoch by a decade shifts the comb by 0.30 in $\ell$ and the drift by 0.002—*the collapse leg's contribution being an amplitude and not a phase, here measured rather than quoted*—and the instrument's discrete wavenumber ladder and a uniform grid agree on the drift to 0.6 per cent. *So the drift is a parameter-free prediction of the spectrum and no test at all of the rate split*: this arm returns +4.38 against the control's +4.42, with every peak position agreeing to better than 0.4 in $\ell$, because the three lengths that carry it are shared between the two frameworks to within a per cent[P15R301].

*The same kernel also carries a running phase, and asking it the question the residual asks clears it as a candidate while leaving a floor behind.* The kernel's phase depends on exactly two dimensionless ratios—checked by scaling all three lengths together, which leaves it unmoved to ten decimal places—and the two arms differ in one of them by 0.27 per cent and share the other, the acoustic angle, to a part in seventy thousand, because the angle is what this construction computes. *So the phase difference between the arms is* 0.014 *degrees in magnitude across* $104 \le \ell \le 1886$*, against a residual of* $-96.6$: nearly four orders of magnitude below it, with the closed-form window and the full spherical-Bessel sum agreeing to the fifth decimal place and each converged in its own parameter. *Only the magnitude of that difference is a convention-free statement, and the sign is not.* The two arms' damping lengths respond to the choice of endpoint in opposite directions, so on the other convention their separation in the governing ratio widens tenfold and the difference between them changes sign, at 0.14 degrees; the clearance survives either way, at no less than 2.8 orders of magnitude. *The projection is therefore not a carrier of this residual and the candidates remain the driving, the baryon loading and the clock the driving is read on. What it does carry is a floor*: across the same range the kernel's own running phase is $-15.5$ degrees, 16 per cent of the residual, and common to the two arms to a tenth of a per cent—*so a sixth of a running phase at correct spacing is manufactured in the projection and is differenced away between the arms, which is where the matched-procedure discipline earns its place rather than being a precaution. That fraction is a statement about a range and not a constant*: it is insensitive to the endpoint convention, moving under a tenth of a point, and runs from 9 to 24 per cent as the range is taken from 200 or from 50 instead, so the figure must be quoted with the range it was read over. *And the Bessel route reaches that answer only when it is sampled finely enough*: at under two points per Bessel period it manufactures a difference between the arms 74 times the true one, while the window route is immune because it averages the kernel analytically—the instrument's own documented aliasing failure arriving

in the phase rather than in the peak positions[P15R302].

*What the plane does not separate is the driving from the clock the driving is read on, and that is established on the construction's own two clocks rather than supposed.* The lap's conformal parameter and the seams' own leg parameter disagree about the seam-bounded stretches by a factor of 4.6083 and disagree about which of them is the longer, so reading the drive on one where the oscillator's own clock is the other is a reparametrization of the drive—and its consequence divides in two. *The spacing cannot move*: $r_s = \int c_s \mathrm{d}\eta$ is an integral on the oscillator's own clock, which a reparametrization of the drive does not touch, so $\ell_A$ is identically the same number and whatever moves is a phase at fixed spacing. It is not a constant one, rising by 0.161 to 0.179 radians between $\ell = 200$ and $\ell = 500$ against the 0.0305 the same estimator reports with the drive switched off, where that control is flat to 0.03 across the whole range. *The direction on the component plane, though, is not fixed by the clock alone.* Taken on three pairs of stretches of the one object the common offset changes sign, the ratio $|\Delta\mathrm{alt}/\Delta\varphi|$ runs between 0.087 and 0.168, and every reading falls inside the cone the driving's 0.196 and the baryon direction's 0.388 already span, each carrying one of those two sign pairs rather than a third. *So the component pair names the carrier between the two candidates and does not distinguish the driving from a reparametrization of it*, and the two are distinguished instead by the shape of the run: the reparametrization spends between two thirds and nine tenths of its whole $200 \le \ell \le 1400$ rise below $\ell \simeq 500$ and is nearly flat above, where the driving's own phase rises to $+0.468$ at $\ell = 500$ and then falls monotonically to $+0.251$ at $\ell = 1400$. *A weighting of the high-multipole half of the range against the low is therefore where that separation sits, and it is located rather than built*[P15R297].

*Two things this section does not claim, and they are separated because a reader would otherwise supply them.* The alternation figures are read on a multipole grid whose step is the same size as the sky's own gap contraction is large, so *a contraction of the sky's size is resolved while the smaller ones at lower $\ell$ are not*[P15R78]; a finer grid is what would settle those. And the equality result accounts for the band it moves and not for the phase residual: *what the phase is to be read against is the driving subtraction on this arm, and §10 carries it as such.*

# 5 What the substrate determines: the amplitude floor and the classical character

One might ask whether the observed fluctuation amplitude is the substrate's own de Sitter vacuum—the natural CR analogue of the inflationary spectrum-is-the-stretched-vacuum story. It is not, by a decisive margin.

**Proposition 2** (The substrate vacuum is negligible against the observed amplitude)**.** *The substrate curvature scale is the cosmological-constant scale: in Planck units $\Lambda\ell_P^2 \approx 3 \times 10^{-122}$, and the de Sitter vacuum metric-fluctuation power is of the same order, $(H_\Lambda/M_{\rm Pl})^2 \sim 10^{-122}$. Against the observed $A_s \approx 2 \times 10^{-9}$ this is smaller by $\sim 10^{113}$.*

*Argument.* $\Lambda = 3(H_0/c)^2\Omega_\Lambda$ gives $\Lambda\ell_P^2$ directly; $H_\Lambda = c\sqrt{\Lambda/3}$ and $(\hbar H_\Lambda/E_P)^2$ give the vacuum power (receipts: `verify_numeric.py`, anchor $5^{\text{P15R17}}$). The prefactor ($\sim 10^{-122}$ versus $\sim 10^{-123}$) is a $2\pi$ and reduced-versus-full Planck-mass convention; the conclusion is robust to it. □

*One scope qualification is owed immediately, and closing it is what makes the classical reading a result rather than an inference.* Proposition 2 rules out the *substrate's* vacuum. But the reading it supports locates the fluctuations in the *progenitor*, and the progenitor's own vacuum is a different quantity: it is normalised at the collapse leg's curvature scale, not at $H_\Lambda$, and it is amplified on its way here by the branch point's mode mixing. Neither effect is small, and neither is addressed above. Both can be computed, because the collapse-side background is a single function: with $A = 2M$ and $\rho = 2\sqrt{B}/A$ the exact closed dust-plus-radiation interior gives

$a = M(\rho|\sigma| + \sigma^2/2)$ near the crunch, whose Mukhanov–Sasaki potential $z''/z = 2/(|\sigma|(|\sigma|+2\rho))$ is *independent of the mass*—so $M$ enters only through the final division $\zeta = v/a$. The passage then has exactly three stages, and each contributes one factor of $1/\rho$. A mode leaves the sound horizon on the $\beta = 2$ leg and freezes as $v \to D_k/|\sigma|$ with the scale-invariant $D_k = \sqrt{\hbar/2}\, k^{-3/2}$; the connection through equality is elementary, since $v = a \int \mathrm{d}\sigma/a^2$ solves the $k = 0$ problem in closed form and carries $D_k/|\sigma|$ to the constant $(3/2\rho)D_k$ at the crunch; the crunch monodromy then leaks that constant into the survivor with coefficient $4\pi/\rho$—twice the tensor value, because the true hydrodynamical $z_S = a\big(a + \frac{4B}{3A}\big)/a'$ has $z_S''/z_S \to 2/(\rho x)$ where $a''/a \to 1/(\rho x)$—the companion paper's numerically obtained $-2\pi i/\rho$ [19] being the *tensor* value of the same object, recovered here in closed form as the continuation of the $(\sigma/\rho)\ln\sigma$ term the $(0,1)$ resonance forces; and the survivor's curvature is $\zeta = C/(M\rho)$, the last $1/\rho$ being the crunch's own slope $a'(\eta_c) = -M\rho$. The result is a transfer law with no free constant,

$$\mathcal{P}_\zeta \;=\; \frac{18\, k^3 D_k^2}{M^2 \rho^6}, \tag{9}$$

which is *exactly* scale-invariant on vacuum data—the $k^3$ cancels the $k^{-3}$ identically—and which therefore delivers, for the vacuum, $A_s = 9(\ell_P/M)^2\rho^{-6}$ (receipt: `P15_the_progenitor_vacuum_is_negligible_t` the connection coefficient confirmed against integration from vacuum data to 0.1% over a decade in $\rho$). *One check is owed here and is not optional, because the hydrodynamical variable differs from $a$ in three places and not one*: $z_S \propto a^{3/2}$ in the matter era, so the exponents there are $(-2, 3)$ against $a$'s $(-1, 2)$; the crunch residue doubles; and the final division is by $z_S$ rather than by $a$. *Propagating the same physical initial condition under each potential returns a ratio of exactly* 2 *in amplitude at every composition tested*—the matter-era exponents and the change of divisor cancel, and the crunch residue alone survives. So the $z \propto a$ chain is corrected by one factor rather than rebuilt, and Eq. (9) is the true-variable law.

**Proposition 3** (The progenitor vacuum is negligible too)**.** *With the mass capped at the recollapse threshold $M \le \alpha/3\sqrt{3}$ and the radiation fraction at the largest value the observed spectrum permits, Eq.* (9) *on vacuum data returns $A_s \sim 1 \times 10^{-112}$, short of the observed $2 \times 10^{-9}$ by $\sim 10^{103}$.* The figure is read on the potential's scale-free branch*; on the observed band, which lies above the break the radiation content sets, the residue is some thousands larger in power, which moves the shortfall by a few orders out of a hundred and three and leaves the conclusion where it is.*

*Argument.* $(\ell_P/M)^2 \ge 27(\ell_P/\alpha)^2 \approx 2.6 \times 10^{-121}$ at the cap, and the companion paper determines $\rho \simeq 5.45 \times 10^{-2}$ [19]; $9 \times 2.6 \times 10^{-121} \times \rho^{-6}$ then gives the figure. Running it backwards, the vacuum would reach the observed amplitude only at $\rho \sim 3 \times 10^{-19}$, thirteen orders below what nucleosynthesis on the same leg permits. □

*So ten orders separate the substrate's vacuum from the progenitor's amplified one—Eq.* (9) *carries the crunch mixing already, so the amplification is the $\rho^{-6}$ and is not a second factor on top of it—and it is still short by a hundred and three*, and the classical character of the primordial statistics is established for the source that actually supplies them rather than only for the one that does not. Two consequences of *character*, not of number, follow.

First, the observed amplitude is inherited classical content—boundary data from the progenitor, not a quantum fluctuation generated within the Schwarzschild–de Sitter era, nor one generated in the progenitor's collapse.

Second, there is no inflationary consistency relation: $r = -8n_t$ presupposes that scalar and tensor spectra are one vacuum stretched by one expansion, and with the amplitude not vacuum-sourced that link is absent; the substrate tensor floor is the same $\sim 10^{-122}$, so there are no substrate-sourced primordial $B$-modes, and any observed tensors would themselves be inherited content rather than a measure of a substrate inflationary scale. *And the tensor statement can be made unconditionally, which the scalar one cannot.* The scalar law, Eq. (9), is the true-variable

law only through the hydrodynamical $z_S$ of the progenitor's own fluid, which carries that fluid's equation of state and sound speed—and across the progenitor's own equality neither $w$ nor $c_s$ is constant.

*For tensors that restriction is absent*: $z_T = aM_{\rm Pl}/2$ exactly, for any content, any equation of state and any sound speed, so $z_T''/z_T = a''/a$ identically and the entire passage—background, connection, monodromy—applies verbatim with nothing assumed. It returns

$$\mathcal{P}_T \;=\; 144\pi\,(\ell_P/M)^2\rho^{-6} \;\simeq\; 5\times10^{-111}, \tag{10}$$

evaluated at the recollapse cap and the largest composition the spectrum permits, against an observational ceiling of $\sim 7\times10^{-11}$: *below it by a hundred orders* (receipt: `P15_no_primordial_B_modes_unconditi`
*So the absence of primordial B-modes is a prediction of this construction at any conceivable sensitivity, and it does not rest on the substrate's own floor*—the floor argument, like its scalar twin, ruled out the substrate's vacuum and said nothing about the progenitor's. What the crossing does to the ratio is likewise fixed and free of $k$: every factor of the passage is shared by the two sectors, so $r_{\rm out} = 6\,r_{\rm in}$: the passage alone would carry the ratio by 24, and the scalar mixing being twice the tensor's divides that by four in the power, and the observed bound becomes a statement about the parent, namely that its own tensor-to-scalar ratio was below $5\times10^{-3}$. *An observed tensor signal would therefore not falsify the construction but measure the previous generation*, which is the same reading the amplitude receives. The substrate fixes that the fluctuations are classical, inherited, and not governed by a vacuum consistency relation, and leaves their size to the handover. *What it does fix is the multiplier.* Eq. (9) holds for any incoming growing-mode amplitude, quantum or classical, so the crossing determines how much an amplitude is magnified without determining the amplitude—which is this construction's one-constant character read at the level of the perturbations rather than of the couplings. Two corollaries follow and are worth stating because each closes a natural hope in one direction and opens a statement in the other. *$A_s$ does not measure the progenitor's mass*: with $D_k$ free, the relation cannot be inverted, and the mass must be reached, if at all, by a route that does not run through the amplitude. And *the classical datum has a size*: since $A_s \propto D_k^2$, the parent's growing-mode amplitude must have exceeded its own vacuum's by $\sim 10^{47}$ at the top of the permitted composition window—so this cosmology does not merely decline to predict $A_s$, it says the previous generation's perturbations were enormously super-vacuum, which is a claim about the parent and not about us.

## 6 What the substrate determines: the throat and isotropization

At the branch point the near-horizon geometry is fixed, and what the perturbations cross is fixed with it.

**Proposition 4** (Equal-radii $\mathrm{dS}_2\times S^2$ throat)**.** *At the Nariai locus $\Lambda G^2M^2/c^4 = 1/9$ the two positive horizon roots of $f(r) = 1 - 2GM/c^2r - r^2/\alpha^2$, $\alpha = \sqrt{3/\Lambda}$, merge at the* Nariai radius $r_N = \alpha/\sqrt{3} = 1/\sqrt{\Lambda}$ *(the symbol is the algebroid paper's [3]; it is written $r_N$ and not $r_\star$ because $r_*$ is the tortoise coordinate used below), where $f(r_N) = f'(r_N) = 0$ and $f''(r_N) = -2\Lambda$ (the cosmology selects this $M>0$ member of the parity-conjugate Nariai pair; the $M<0$ member has instead one positive and two negative roots, the two exchanged by $r\mapsto -r$ [5, 6]). The near-horizon geometry is $\mathrm{dS}_2\times S^2$ with both curvature radii equal to $1/\sqrt{\Lambda}$: the $S^2$ has two-Ricci $R_2 = 2/r_N^2 = 2\Lambda$, and the $\mathrm{dS}_2$ has radius$^2$ $= -2/f''(r_N) = 1/\Lambda$.*

*Argument.* Direct from $f$ at the Nariai mass (receipts: `verify_geometry.py`, anchor $2^{\mathsf{P15R16}}$; this is the Ginsparg–Perry near-horizon limit [29]). □

A field on $\mathrm{dS}_2\times S^2$ (Fig. 4) decomposes in $S^2$ harmonics; the degree-$\ell$ harmonic becomes a $\mathrm{dS}_2$ field of effective mass $m^2 = \ell(\ell+1)/r_N^2$, i.e. $m^2/H_{\mathrm{dS}_2}^2 = \ell(\ell+1)$. With the $\mathrm{dS}_2$ index

$\nu^2 = \frac{1}{4} - m^2/H^2$, the monopole $\ell = 0$ has $\nu = \frac{1}{2}$—a scale-invariant base—while every $\ell \geq 1$ has $\ell(\ell+1) \geq 2$, hence $\nu^2 < 0$: the heavy principal series, which oscillate and decay through the throat. We read this as an angular no-hair, and the reading is grounded rather than asserted: the $dS_2$ factor of the throat is a positive-$\Lambda$ epoch, and the de Sitter no-hair theorem [21] damps exactly the anisotropic content—every $\ell \geq 1$ harmonic is heavy principal-series, oscillating and decaying, while only the isotropic monopole $\ell = 0$ survives as the $\nu = \frac{1}{2}$ base—so the throat isotropizes what passes through it (receipts: `verify_throat_tower.py`[P15R40]). The geometry is established (Proposition 4) and the mode tower follows from it; the no-hair reading is the standard theorem read on the throat, and we flag one guard as load-bearing: the index $\ell$ of this throat tower is the $S^2$-harmonic degree of the *near-horizon* geometry at areal radius $1/\sqrt{\Lambda}$, and is *not* the observable microwave-background multipole, which lives on the last-scattering sphere of radius $D_C$. *And the same conclusion is now reached a second time by a route that shares neither the decomposition nor the theorem, which is why it is worth having*: carrying the layer's $S^3$ harmonics along the cosmogenetic bead in its own conformal time, where the lift from the comoving turnaround to $r = 0$ is a segment of *purely imaginary* $\eta$ of finite length $s_{\rm tot} = \Gamma(\frac{1}{6})\sqrt{\pi}/\Gamma(\frac{2}{3})\sqrt{3}\,2^{1/3} = 3.3387380$ (§10). *There the isotropic mode crosses with amplitude exactly unity—$u = a$* solves the lift equation identically at $k = 0$*—while every anisotropic one is damped by a pure-number function of its own degree, $T(k) \to 2^{7/3}k^2e^{-k\,s_{\rm tot}}$, and the eigenvalue $k^2$ is the carried layer's own rather than a round sphere's.* §7 obtains that layer with squashing $\varepsilon$, and its Laplacian—built from the left-invariant forms of the same group manifold—has spectrum

$$\lambda(L,m) = L(L+2) + 4(\varepsilon^{-2}-1)m^2, \qquad |m| \leq L/2, \quad m \equiv L/2 \ (\mathrm{mod}\ 1), \tag{11}$$

so $L(L+2)$ is its $\varepsilon = 1$ member. *For scalars the deformation reaches the spectrum and not the basis*: the same matrix elements diagonalise the Laplacian at every squashing, so $(L,m)$ is conserved along the bead and nothing mixes. *The scope of that sentence is the scalar sector and is not general*—it holds because a scalar on a left-invariant metric carries no frame index, and the vector sector below is where that premise fails. *Two sectors the squashing cannot reach at all*: $\lambda(0,0) = 0$ identically, so the isotropic mode crosses at unity for every $\varepsilon$; and $m = 0$, admissible exactly when $L$ is even, keeps $\lambda = L(L+2)$ with no $\varepsilon$ in it. *And every other scalar mode is bounded below uniformly*—the minimum at $|m| = L/2$ is $2L + L^2/\varepsilon^2$, so $\lambda \geq 2L$ for every $\varepsilon > 0$*—which puts the suppression exponent at no less than $\sqrt{2L}\,s_{\rm tot}$*: at $L = 1$ that is 4.722, giving at most $8.97 \times 10^{-2}$ against $4.66 \times 10^{-2}$ for the same asymptote at the round value—*and the exact transport, below, weakens it by* 2.49 *at that degree rather than by the bound's factor of two*, a suppression still above ten and monotone in degree[P15R258]. *Where the squashing drops out of the eigenvalue, the constant-k transport is exact rather than approximate*, so on the $m = 0$ slice the transmission is a number and not an envelope: $4.158 \times 10^{-3}$, $1.116 \times 10^{-5}$, $2.438 \times 10^{-8}$, $4.730 \times 10^{-11}$ and $8.497 \times 10^{-14}$ at $L = 2, 4, 6, 8, 10$, against $T(0) = 1$ to a part in $10^{14}$ as the control. *Those degrees sit below the asymptotic regime of the parameter-free form, which therefore understates every one of them and the lowest by most*—exceeding it by 30, 17, 12, 9 and 7.5 per cent *of the asymptotic value* in turn—*so $2^{7/3}k^2e^{-k\,s_{\rm tot}}$ is a lower bound on this sector's transmission rather than an estimate of it, approached from above and not crossed*[P15R268][P15R274]. *The charged modes do carry a squashing-dependent eigenvalue, and it diverges at the double root*—so an exact figure exists on the slice whose amplitude the transport delivers, and is absent on the modes whose amplitude it does not. *The Berger parameter is the squashing normalised to the horn's round datum, $\alpha\sqrt{-f}/r$*, since the bare ratio tends to $1/\alpha$ rather than to unity; *so normalised it passes through unity at exactly $r = 2M = \frac{2}{3}r_N$, free of $\alpha$. On the lift the layer is timelike, $-f < 0$, so that parameter is imaginary there and what enters the spectrum is its modulus—and the $m = 0$ modes, which carry no $\varepsilon$ at all, reproduce the round exponent exactly, which is the control that fixes the modulus as the right object to feed it. That radius is positive and the lift occupies $-A < r < 0$, so the squashing exceeds its round value along the whole of it*—1.3747 at the comoving turnaround and rising without

bound at the close—*and the stretch where a squashing below unity would raise the eigenvalues lies on the expansion side, which the transport never visits. So the uniform bound above is not attained, and what the lift actually supplies is exact.* Dividing (11) by its round member gives $\lambda/L(L+2) = 1-\sigma(1-\varepsilon^{-2})$ with $\sigma = 4m^2/L(L+2)$, *the fraction of the eigenvalue the Hopf charge carries*—so the exponent's ratio to the round one is a function of $\sigma$ alone, and $(L,|m|) = (1,\frac{1}{2})$ and $(6,2)$ share $\sigma = \frac{1}{3}$ and return the same ratio to ten figures. *Both endpoints are exact*: the ratio is unity at $\sigma = 0$, and at $\sigma = 1$ it is $\sigma_{\rm tot}^{-1}\int|\varepsilon|^{-1}\mathrm{d}\eta = 0.2591$, free of $\alpha$ by three independent routes—*so the most transparent band sees an effective segment length* 0.8651 *rather than* 3.3387. *And because the infimum $\lambda \geq 2L$ is reached only as $\varepsilon \to \infty$, which is a set the transport does not dwell in, the exponent stays LINEAR in degree rather than turning into the bound's* $\sqrt{2/L}$: at $L = 1024$ it is 934.0 against the bound's 151.1, the gap widening past a factor of six. *What the squashing buys is a shortened segment and not a change of functional form*[P15R259].

*The uniform floor above is the scalar sector's, and the vector sector has no counterpart to it.* The Hodge Laplacian commutes with d and d* on any Riemannian manifold, so the gradient and co-exact subspaces are each invariant at every squashing—and the gradient sector returns (11) member for member, which is the control that fixes the convention. *Inside the co-exact sector the deformation does reach the basis*: a 1-form carries its own charge in the frame index, so the conserved label is the difference and the invariant blocks are the triples $\{e_+\otimes(q+1),\ e_3\otimes q,\ e_-\otimes(q-1)\}$, with coupling $2\sqrt{2}\,\varepsilon$ at $L = 1$ and $4\varepsilon$ at $L = 2$—proportional to the squashing itself and vanishing only in the collapsed limit. *So $(L,m)$ is not carried unmixed there and the transport is a three-channel problem*, bounded at $3\times 3$ and never a new infinite one; *and the extreme-charge block is $1\times 1$, mixing with nothing*, which is also the smallest block at large squashing and therefore available exactly where the lift goes. *The transverse eigenvalue is $\lambda^{\perp} = (L+2)^2/\varepsilon^2$ exactly*—9, 16, 25, 36, 49, 64 for $L = 1$ to 6—*so it runs to zero as $\varepsilon \to \infty$, which is the end of the lift where the scalar floor still held. And the suppression survives the broken floor*, because $\varepsilon \to \infty$ happens only at $r \to 0$ and almost no measure sits there: carrying the exact minimum through the lift's own measure gives 2.595216 at $L = 1$—exactly $3 \times 0.8650719$, the scalar effective length, free of $\alpha$ to thirteen figures and with the minimiser the unmixed block, so nothing adiabatic is assumed—and 2.555, 3.780, 4.793, 5.742, 6.661 at $L = 2$ to 6, against the most transparent scalar band's 2.447, 3.350, 4.238, 5.118, 5.993. *The exponent is linear in degree and larger than the weakest scalar band at every degree computed*, so a pointwise bound running to zero costs the damping nothing[P15R261]. *Above $L = 1$ the vector figures are a short-wavelength reading on a rotating basis and are quoted as such.*

*For the tensor sector this route does not reach a spectrum at all, and the obstruction is identified rather than merely met.* The Lichnerowicz Laplacian preserves tracelessness at every squashing and *generates divergence as soon as the layer stops being Einstein*—and the layer is Einstein only at the round point, its Ricci eigenvalues being $(4-2\varepsilon^2,\ 4-2\varepsilon^2,\ 2\varepsilon^2)$, equal if and only if $\varepsilon^2 = 1$. *So the transverse-traceless subspace is not invariant off that point*: the generated divergence is linear in the Einstein deficit near it and is already a quarter to a third of the image at the lift's entry. *What the tensor route needs is therefore a decomposition the squashed layer does not supply, and not an eigenvalue on it*—the object such an eigenvalue would belong to is not preserved by the transport. *This bounds a computation this paper's own route would require, and is not a claim about the tensor sector. And the obstruction belongs to the reduced description rather than to the geometry it reduces*: the four-metric the layer sits in is exactly Einstein, $R_{ab} = \Lambda g_{ab}$ identically with $R = 4\Lambda = 12/\alpha^2$, the mass and the throat constant free—so the traceless Ricci that drives the obstruction vanishes identically one dimension up. *And the obstruction is proportional to that deficit rather than merely correlated with it*, the ratio holding to four figures across four decades of it at 0.255551 for $L = 2$ and 0.183473 for $L = 3$ with the trace at machine zero throughout, *so a vanishing deficit is a vanishing obstruction and not a smaller one. What is obstructed is therefore the layer-by-layer harmonic method,*

*and the question is well posed in four dimensions*[P15R262]. *And the four-dimensional treatment keeps that economy, with more of it than the three-layer route has.* The four-chart's angular factor is a *round* two-sphere at every radius, the areal radius its only radial dependence and no squashing parameter present on that chart at all—*so moving along the curve rescales the fibre and does nothing else to it, and a round fibre has no shape to deform. The angular dependence separates, and the reduced radial operator is identical across every azimuthal label at each degree*, with the degree entering only as $\ell(\ell+1)$: *a one-label family of radial problems where the layer route is a two-label one*, since (11) depends on $m^2$ and needs the second label. *What does not travel is the label itself*: the layer's $m = 0$ harmonics are exactly the fibre-independent ones, so exactly that sector descends to the Hopf base, and there the two spectra are one number—$L(L+2) = 4j(j+1)$ at $j = L/2$, identically in the degree. *So the bridge is exact where it exists and absent elsewhere, and the sector it exists on is named three times over*: by the Hopf projection requiring $j = L/2$ to be an integer, by the parity rule admitting $m = 0$ exactly for even $L$, and by the layer spectrum depending on the squashing only through $m^2$—*which makes it the squashing-free sector as well*[P15R263]. *The modes the bridge does not reach carry charge under the fibre direction the degeneration closes up*, and what becomes of such a charge is a question about the degeneration rather than about this sector. *And the sector the bridge does reach is thin and begins late*: the layer's multiplicity at degree $L$ is $(L+1)^2$ with $L+1$ charges carrying $L+1$ states each, so the zero-charge slice is exactly $1/(L+1)$ of the degree on even $L$ and empty on odd—$\frac{1}{3}$, $\frac{1}{5}$, $\frac{1}{7}$, $\frac{1}{9}$ at $L = 2, 4, 6, 8$—*and it contains no* $L = 1$ *mode at all. What that exclusion does not do is reach past the sector it describes.* The parity it rests on empties the neutral slice at *every* odd degree, so the degrees it removes properly contain the ones the companion's tensor tower removes—*but it is a statement about which slice keeps the round eigenvalue, and not about which modes exist.* At odd degree all $L+1$ slices are present and every one of them is charged, and all are carried along the bead exactly, the harmonics conserving $(L, m)$ with nothing mixing[P15R267]. *Neither route by which that parity might have bounded the arriving content survives measurement.* The crossing cannot carry it: its kernel is a function of the degree alone, while $L(L+2)$ is the same on all $L+1$ charges of a degree, so the charge the parity distinguishes does not appear in the kernel at all—*a function constant along the ladder cannot implement a condition that distinguishes its rungs*[P15R269]. *And composition cannot carry it either*: the charged eigenvalue diverges where the squashing *vanishes*, and the normalised squashing attains its minimum 1.3747 at the comoving turnaround and rises from there to the close, *so the divergence lives on the leg the transport never traverses.*

*So the odd ladder arrives, and it arrives with a number rather than a sign.* The exponent's ratio to the round one is a function of $\sigma = 4m^2/L(L+2)$ alone—two modes of different degree sharing $\sigma$ agree to ten figures—and at the lowest odd degree it gives an exponent of 4.868603 against 5.782864 for a round mode of that degree: *exponentially small, non-zero, and less suppressed than the round mode by a margin that grows with degree*[P15R272]. *And the projection carries that degree lower in the sky than the even floor.* The window in which (16) is quasi-injective is two degrees wide, so $L = 1$ sits inside the mode spacing comfortably where $L = 2$ sits inside it by less, and the two land at $\ell \simeq 3$ and $\ell \simeq 6$—three multipoles apart, and stable across the range of the stretch this paper admits. *With the even floor's own neutral slice at exponent* 9.443377*, the gap is* $e^{4.57}$ *in the odd degree's favour*, so *the lowest multipoles the projection reaches draw on the degree a parity bound would have removed*, and exponent and modal multipole both rise monotonically over the computed degrees, so no higher degree reaches below it[P15R273]. *What that locates is a window and not a spectrum*: these are exponents and not amplitudes, the charged slice's eigenvalue moving along the path so that no prefactor follows from a ratio of exponents; one slice is carried per degree; and the primordial weights a $C_\ell$ would sum against are fixed nowhere in this construction, which is the datum §10 names as inherited rather than derived.

*What the bridge does not carry across is the measure.* On an Einstein four-metric the

Lichnerowicz curvature term splits, and while the Ricci part is proportional to the tensor and shifts the problem by a constant, *the Riemann part is not, because this background is not conformally flat*: the Kretschmann scalar is $48M^2/r^6 + 24/\alpha^4$, the second term exactly the $\Lambda$-only value $8\Lambda^2/3$, *so the Weyl invariant is* $48M^2/r^6$*—mass-driven, and divergent as* $r \to 0$*, which is the close of the lift. At the forced mass the ratio of the two is* $2\alpha^6/27r^6$*, exactly one half at the comoving turnaround*—a closed value at the lift's own entry—*then* 32 *at half that radius and* $5 \times 10^{11}$ *at a hundredth of it. So it is nowhere a small correction on the segment the transport traverses, and the layer's cross-section problem carries no counterpart to it*: a tensor exponent on this background is the Regge–Wheeler reduction with that coupling carried through the radial problem, and not an extension of the measure the scalar and vector exponents are computed on[P15R264]. *That computation belongs with the tensor sector itself*, which §1 places in the companion dynamics paper [10].

*So the throat's isotropization is not only the standard theorem read on the near-horizon geometry; it is a damping factor computed on the construction's own curve, with no* $\Lambda$*,* $H_0$*,* $\Omega_m$*, mass or epoch in exponent or prefactor—the seam is isotropic by identity rather than by tuning.* The guard above applies to this route in the same form and with the same force: $L$ here is the degree on the cosmological layer, and the observable multipole is reached only through the map of §7[P15R240]. The map between them is §7, not an identity; conflating them would manufacture a prediction the geometry does not make. *And §7 measures how far from an identity it is, which bounds the degrees over which an exponent in* $L$ *may be read as one in* $\ell$: the projection is quasi-injective while its width stays inside the mode spacing, which holds at $L = 1$ and $L = 2$ and fails from $L = 3$ up. *So the exponential-in-degree statement above transfers to the sky for the first two degrees—where this route's own figures are computed—and above them becomes a statement about the lowest contributing degree instead*, since a steep envelope carried through a smearing map is dominated by the bottom edge of each band. *The functional form is not preserved there, and that is a limit on how the envelope may be read rather than a correction to it. And a second guard belongs to this route in particular, because it is the premise that makes the expansion possible*: carrying $(L, m)$ along the bead needs a harmonic decomposition of the layer at every point of the curve, the lift included—*and the layer there is squashed rather than round*, §7 measuring its squashing as $\sqrt{-f}/r$. *The decomposition exists and is the same one*, because the squashed layer is a left-invariant metric on the same group manifold: that is what (11) above records, and it is why the transport carries $(L, m)$ unmixed. *The guard that remains is narrower and is worth stating as such*: the round value $L(L + 2)$ is the $\varepsilon = 1$ member and not the generic one, so a damping quoted at $L(L + 2)$ is a statement about a horn and not about the lift, and what the lift supplies is the uniform bound. *The isotropization conclusion does not rest on this route in any case*: the near-horizon one needs nothing of the cosmological layer, resting on the throat geometry of Proposition 4 and the de Sitter no-hair theorem alone. *That the two agree in direction, by mechanisms sharing neither a decomposition nor a theorem, is what makes the second worth computing. A second guard belongs beside it, and it bounds what the tower damps.* The computation above is a statement about propagating field modes—degrees of freedom carrying a mass on the $dS_2$ factor, which is what being heavy principal-series means. *The low multipoles of a perturbation about a spherically symmetric background are not such modes*: $\ell = 0$ carries the mass and the axial $\ell = 1$ carries the angular momentum, the latter a stationary slow-rotation configuration rather than an oscillation, with the radiative content beginning at $\ell \geq 2$—the multipole face of the non-radiative skeleton the range paper describes [8]. *So what the throat isotropises is the radiative content, and a conserved charge sitting in a constrained sector is not among the modes this argument is about*[P15R8]. *This is a bound on the claim and not a claim in the other direction*: that the angular momentum is not removed *by this mechanism* says nothing about whether it is carried off by matter or torqued on the way, which this paper does not settle. *And the geometry the tower is computed on is itself a non-rotating one*, Proposition 4 following directly from $f$ at the Nariai mass. *So it is worth*

*measuring how much the conclusion rests on that.* Writing $\lambda$ for the ratio of the $\mathrm{dS}_2$ curvature radius squared to the $S^2$ one, the index generalises to $\nu^2 = \frac{1}{4} - \lambda\,\ell(\ell+1)$, with $\lambda = 1$ the equal-radii case above. *The monopole is untouched*: $\nu^2 = \frac{1}{4}$ for every $\lambda$, so the scale-invariant base does not depend on the radii being equal. *And the damping persists well away from it*: every $\ell \geq 1$ remains heavy principal-series for $\lambda > 1/4\ell(\ell+1)$, which at $\ell = 1$ is $\lambda > 1/8$—the $\mathrm{dS}_2$ radius would have to fall below 0.354 of the $S^2$ radius, a factor near three, before the dipole ceased to decay, with the higher multipoles safer still[P15R9]. *What ratio a rotating progenitor would produce is not computed here*, and a rotating near-horizon geometry is generally warped rather than a direct product, so a single ratio may not describe it; what the margin establishes is that the conclusion does not turn on the equal-radii point itself. *The ratio can in fact be computed on the rotating family, and it runs the helpful way.* Kerr–de Sitter's radial function $\Delta_r = (r^2+a^2)(1-r^2/\alpha^2) - 2Mr$ reduces at $a = 0$ to $r^2 f(r)$, and its degenerate locus $\Delta_r = \Delta_r' = 0$ gives $a^2 = r^2(\alpha^2 - 3r^2)/(\alpha^2 + r^2)$, which returns $r_N = \alpha/\sqrt{3}$ and the Nariai mass at $a = 0$ rather than presupposing them. *The family is bounded*: $a$ reaches $(2-\sqrt{3})\alpha$ at $r/\alpha = 0.3933$, where $\Delta_r''$ vanishes and the double root becomes triple—the ultracold member. *Along it the ratio begins at unity and rises*: $\lambda = 1$ exactly in the static case and $\lambda > 1$ monotonically with $a$, so rotation carries the throat away from the 1/8 threshold rather than toward it[P15R10]. *That ratio is read at the pole, where the sphere's radius is* $r_N$; the rotating near-horizon sphere is warped through $r_N^2 + a^2\cos^2\theta$ and a single ratio does not describe it, so what the calculation establishes is the direction and its size at one well-defined point, with a factor of eight in hand at the outset.

*The warped problem is done directly, and does not need the proxy.* The near-horizon limit of the degenerate family, taken as a limit rather than posited, is a warped product and not a direct one: $\rho_0^2 = r_0^2 + a^2\cos^2\theta$ multiplies the $\mathrm{dS}_2$ and polar directions alike, and the $\mathrm{dS}_2$ is fibred over the sphere by $\tilde{k} = 2r_0 a\Xi/(r_0^2 + a^2)$, which vanishes only at $a = 0$. A massless field still separates on it, the warp cancelling from the relation between the transverse eigenvalue and the $\mathrm{dS}_2$ operator because it multiplies both. *For the axisymmetric modes it cancels altogether*: the transverse problem is $(\Delta_\theta \sin\theta\, Y')' + E\sin\theta\, Y = 0$ with $\Delta_\theta = 1 + (a/\alpha)^2\cos^2\theta$—a spheroidal deformation of Legendre in which $\rho_0$ does not appear at all, so the harmonic label survives the warping rather than being spoiled by it. *The decay condition generalises to* $\nu^2 = \frac{1}{4} - L^2E - L^4m^2\tilde{k}^2$, the last term the rotation's own fibration, which is non-negative and so can only deepen the damping. *Two exact bounds then settle the family without a scan*: $L^2 \geq 1$ on the physical branch and $E \geq 2$ above the monopole, each with equality only at $a = 0$, so $L^2E \geq 2$ against the threshold $\frac{1}{4}$. *The monopole keeps* $\nu^2 = \frac{1}{4}$ *exactly on the warped sphere too. So the conclusion holds across the whole family* $0 \leq a < a_{\max}$ *with the same factor of eight, and the marginal member is the non-rotating one*—rotation strictly improves the case rather than threatening it[P15R11].

# 7 Large scales: the flat/discrete decoupling and the low-multipole floor

Here the construction makes its sharpest large-scale statement, and the load-bearing piece is established.

**Proposition 5** (The fundamental-observer slice is flat)**.** *In the fundamental-observer frame* (2)*, a surface of constant* $\tau$ *has induced three-metric* $dr^2 + r^2 d\Omega^2$ *(since* $(\partial_\chi r)^2 d\chi^2 = dr^2$ *along the slice), whose Riemann tensor vanishes identically. The constant-*$\tau$ *slice is exactly flat* $\mathbb{R}^3$*.*

*Argument.* The substitution is immediate from the line element; the curvature of $dr^2 + r^2 d\Omega^2$ is zero by direct computation (receipts: `verify_geometry.py`, anchor 1[P15R16]). This is the framework paper's own stated property that constant-$\tau$ slices are Euclidean. □

The decoupling follows. The distance slicing is the flat constant-$\tau$ one—$\Omega_k = 0$, no curvature term in the redshift–distance relation—while the *cosmological* layers are the closed $S^3$, labelled

by constant $\tilde{\tau} = \tau + \chi$ and read as a sphere in the de Sitter presentation (§2.2). *The attribution matters and is stated rather than left to the reader*: on the reassigned chart of (2) a constant-$\tilde{\tau}$ surface carries the induced metric $-f(r)\,\mathrm{d}\chi^2 + r^2\mathrm{d}\Omega^2$, whose intrinsic Ricci eigenvalues are $(0, 1/r^2, 1/r^2)$—a flat line times a round $S^2$ of radius $r(\tilde{\tau})$, not a three-sphere—which is what the reassignment does and is the same fact as the 45° picture of §2.2. *The line's causal character is the congruence's own speed, and it is not the same along the whole curve*: the $E = 1$, $k = 0$ radial equation gives $(\partial_\chi r)^2 = 2M/r + r^2/\alpha^2$ identically, so the $\chi$-block coefficient $-f = (\partial_\chi r)^2 - 1$ is positive where the congruence exceeds unit areal speed and negative where it falls below it, vanishing at the two loci where it is unit. *At the Nariai mass the cubic factorises exactly*,

$$-f = \frac{(r - \alpha/\sqrt{3})^2\,(r + 2\alpha/\sqrt{3})}{\alpha^2 r}, \tag{12}$$

*a double root at $r = +\alpha/\sqrt{3}$ and a simple one at $r = -2\alpha/\sqrt{3}$—which are the lap's two seams*, necessarily so, since $-f = 0$ and $|\partial_\chi r| = 1$ are the same condition. *So the sign of $-f$ is the sign of $(r + 2\alpha/\sqrt{3})/r$, and the surface is spacelike on both legs and Lorentzian through the whole interior of the lap*—$-1$ exactly at the comoving turnaround, where $\partial_\chi r$ vanishes and the surface is a unit-timelike line times $S^2(A)$. *The flat-line reading is therefore the signature outside the seams*[P15R239]. The non-synchrony $\tilde{\tau} = \tau + \chi$ is the decoupling itself: flat distances and a closed $S^3$ of comoving worldlines coexist, the one read on the constant-$\tau$ slicing and the other on the layer the same construction labels by $\tilde{\tau}$. *And the relation between those two readings is stated exactly, because it is not the one a reader will assume*: they are *not* two coordinate presentations of a single metric, and nothing here claims they are. *The two are reached from one another by the reassignment of the null condition*—the operation that promotes one null bundle of the cosmological horn to the fundamental timelike congruence (§2.2)—*so they are two metrics carried on one ontological layering, isometric through that reassignment and not through any change of coordinates. What is common to them, and what makes the layering one object, is the foliation itself*: the cosmic layers are the surfaces of constant areal radius, on the de Sitter horns and inside the lap alike, and it is the growing-$r$ foliation that begins at the branch point where $r$ and $\tilde{\tau}$ vanish together. *The difference between a flat reading and a curved one is therefore nothing but which bundle is called at rest*, and the lift's purely imaginary conformal time—the Euclidean segment of §10—*is what keeps that reassignment well defined across the lap. One consequence is worth stating so that it is not mistaken for a defect later*: a curvature invariant computed in one presentation and compared with the other registers that the reassignment is not a diffeomorphism, which the construction asserts of it, and is therefore not evidence about where the layer's sphere lives[P15R239]. *What the de Sitter presentation displays plainly is the layer outside the seam, before the lap and after it*; inside the lap the seam buries the explicit form, and the sphere's character there is reached by analytic continuation along the bead rather than read off a chart. *The foliation is the constant-r one at every point of the curve, and the layer's SIZE continues across the lap with it; its SHAPE does not. That is settled below rather than assumed in either direction*, by carrying the sphere the de Sitter presentation displays at a horn along the flow and measuring what arrives. *What the construction does supply is a sharp test for it, and the test's own discriminating power is measured rather than assumed.* A round $S^3$ of radius $r$ has scalar curvature $\mathcal{R} = 6/r^2$ and volume $V = 2\pi^2 r^3$, so the scale-free combination

$$\mathcal{R}\,V^{2/3} = 6(2\pi^2)^{2/3} = 43.8232\ldots \tag{13}$$

carries no $r$ at all—it cancels symbolically—so it is insensitive to the layer's size and reports only its shape. *And it does discriminate*: on a Berger sphere of squashing $\varepsilon$ the same combination is $2 \cdot 2^{2/3}\pi^{4/3}\varepsilon^{2/3}(4 - \varepsilon^2)$, which varies with the squashing and takes the round value only at $\varepsilon = 1$. *That point is a strict maximum of the combination and its first derivative there vanishes*, the second being $-77.9$, *so the test's sensitivity to a departure from roundness is second order in the squashing*—sharp enough to discriminate, and weaker against a near miss than a reader

would assume of it. *So* (13) *is the quantity a demonstration of the continuation would have to return, evaluated on a layer metric obtained from the construction rather than written down*—and what remains owed is exactly that: the three-metric of the layer inside the lap carried from the sphere the de Sitter presentation displays at a horn, rather than posited at each point of the bead. *And the presentation that demonstration must be carried in is fixed, because* (13) *cannot be evaluated on the reassigned chart's surface at all.* The volume of a line times a sphere carries the coordinate extent of the line, so the combination is not scale-free there; and wherever $-f < 0$ the surface carries no Riemannian volume, so the combination has no real value—at the comoving turnaround $-f = -1$. *The test is therefore a test of the de Sitter presentation's layer*, which is where the sphere is displayed and where the continuation is the question. *And carried in that presentation the requirement reduces, because the imaginary segment turns out to be transparent to the test and the flow does not.* The layer at a horn follows from the closed slicing's own equation $a' = a\sqrt{a^2/\alpha^2 - 1}$ with $a(0) = \alpha$, giving $a(\eta) = \alpha/\cos\eta$, so there the layer is $a^2$ times the unit round $S^3$ and (13) returns its value with the factor cancelling. *Carrying that factor off the positive axis leaves the modulus exactly fixed*—$|\mathcal{R}V^{2/3}| = 43.8232\ldots$ for $a > 0$, $a < 0$ and $a = \pm i$ alike—*while the principal value picks up a cube root of unity fixed by* $\arg a$, the lap's own thirds. *So the lift is not what the sphere has to survive*: what it has to survive is the shear of the flow that carries it, and the carrying is conformal on the layers only where the constant-$r$ foliation is umbilic. *On* (1) *that condition is* $rf' = 2f$*, whose only root is* $r = 3M$ *with the mass carried free*—so it is a statement about the family and not about a member—*and at the Nariai member that radius takes the value* $\alpha/\sqrt{3}$, the merged horizon's areal radius, with the shear non-vanishing at the comoving turnaround and away from the seam. *The coincidence of those loci is computed and is not an identification of the two objects*: the umbilic radius of the foliation and the merged horizon's areal radius agree in value at the forced member. *And the sheared layer's own invariant settles it, which is why the continuation is stated here as failing rather than as open.* The foliation's trace-free extrinsic curvature is axisymmetric with $\sigma_1 = -2\sigma_2$—one distinguished direction against an equal pair, which is the squashing (13) parametrises, so the test is applied in the direction the flow actually shears. *Carrying the horn's round datum along it, the squashing obeys* $\mathrm{d}\ln\varepsilon/\mathrm{d}r = f'/2f - 1/r$*, which is identically* $\mathrm{d}\ln(\sqrt{f}/r)/\mathrm{d}r$, so $\varepsilon \propto \sqrt{f}/r$ with no layer metric written down at any point of the lap. *The response to a squashing has no linear term*—$g(1+u)/g(1) = 1 - \frac{8}{9}u^2 - \frac{8}{81}u^3$ exactly, which is the second-order stationarity above in closed form—*so a squashing of either sign lowers the invariant. And approaching the seam the squashing runs to zero*: at the Nariai member $f$ has its double root there, so $\varepsilon$ vanishes linearly in $r - r_N$ while the parameter along the flow diverges as $-\ln(r - r_N)$. *So the round shape is not carried to the seam, and the obstruction is in the approach rather than at a point*—the seam lies at infinite parameter and the layer degenerates on the way to it[P15R256]. *What the degeneration leaves is not nothing, and saying what it is completes the answer.* A squashing running to zero collapses the Hopf direction at finite curvature: by the Berger expression above $\mathcal{R} \to$ a finite limit while the volume vanishes linearly in $\varepsilon$, so the $S^3$ degenerates onto its Hopf base rather than shrinking. *The surviving factor is the construction's own angular block* $r^2\mathrm{d}\Omega^2$ *at* $r = r_N = 1/\sqrt{\Lambda}$—which is the areal radius of the $S^2$ in the merged horizon's near-horizon geometry of Proposition 4. *So the cosmological layer does not carry its sphere through the seam; it becomes the throat's* $S^2$ *there*, and the Hopf direction it gives up is the one the reassignment of §2.2 trades into $\chi$. *And the two descriptions are not merely in agreement at that radius: they are one geometry, which the seam limit of the carried layer gives outright.* Because the $\chi$-block coefficient depends on cosmic time alone, the shift $\mathrm{d}\chi + \mathrm{d}\tilde{\tau}/(-f)$ is an exact differential, so with $\hat{\chi} = \chi + \int \mathrm{d}\tilde{\tau}/(-f)$ and $\mathrm{d}T = \mathrm{d}r/\sqrt{-f}$ the four-metric is identically

$$\mathrm{d}s^2 = -\mathrm{d}T^2 + (-f)\,\mathrm{d}\hat{\chi}^2 + r^2\mathrm{d}\Omega^2, \qquad (\mathrm{d}r/\mathrm{d}T)^2 = -f, \tag{14}$$

with $T$ the proper time orthogonal to the cosmological layers. *The lapse diverges exactly where* $-f$ *vanishes*—which is $|\partial_\chi r| = 1$—*so that divergence is the cosmological slicing going null at the*

*seam and belongs to the degeneracy rather than to the chart*: $T$ grows like $\Lambda^{-1/2}\ln[1/(r-r_N)]$ at the forward seam's double root, while at the back seam's simple zero the same integral converges. *In that parameter the approach is exponential,* $r-r_N \propto e^{-\sqrt{\Lambda}T}$, *so the* $\hat{\chi}$ *scale factor* $\sqrt{-f}$ *has rate exactly* $\sqrt{\Lambda}$ *and two-dimensional curvature* $2\Lambda$*—a* $\mathrm{dS}_2$ *of radius* $1/\sqrt{\Lambda}$, which is Proposition 4's own $\sqrt{-2/f''(r_N)}$ reached from the flow rather than read off the static chart. *The angular block tends to* $r_N^2\mathrm{d}\Omega^2$ *and both blocks approach their limits linearly in* $r-r_N$*, so the warping switches off and the limit is a direct product. The limit metric is Einstein with* $R_{ab}=\Lambda g_{ab}$*, has* $\mathcal{R}=4\Lambda$ *and Kretschmann* $8\Lambda^2$*—the static metric's own value at* $r_N$*, computed separately*[P15R257]. *And the join is then an identity rather than a coincidence*: the squashing above is $\sqrt{-f}/r$ and the scale factor here is $\sqrt{-f}$, *so the layer's squashing is the throat's own* $\mathrm{dS}_2$ *scale factor divided by the areal radius.* The layer's loss of shape and the throat's $\mathrm{dS}_2$ are one object described twice, and the coincidence of radii at the forced mass is not a coincidence[P15R255]. *Two things about the lap are settled without it. The angular block enters only through* $r^2$ *and is blind to the sign of* $r$*, and the* $\chi$ *block is not*: $-f$ carries $2M/r$, which is odd, so two chart values of equal $|r|$ carry $S^2$ factors of equal radius and $\chi$ blocks that differ—at $|r|=A$ they are opposite in sign. *So* $|r|$ *is the label of the layer's size and not of the layer*, and the identification that closes the lap is the translation $r\mapsto r+\sqrt{3}\alpha$ carrying one seam to the other, $-2\alpha/\sqrt{3}+\sqrt{3}\alpha=+\alpha/\sqrt{3}$ exactly, rather than the reflection $|r|$ would implement. *And* $r=0$ *is not a single event but one on each worldline*, since $\partial_\chi r=\partial_\tau r$ makes the level sets of $r$ the 45° lines of constant $\tau+\chi$; the vanishing volume and divergent curvature of the layer at that one cosmic time are the areal coordinate degenerating, which this paper's own reading of the branch point already states. *The cosmological seam is likewise a locus and not a point, and its position is fixed exactly*: the handover from deceleration to acceleration is the inflection of $r(\tilde{\tau})$, which vanishes where $\cosh(3\tilde{\tau}/\alpha)=2$, and the areal radius there is $2^{-1/3}A=\alpha/\sqrt{3}=r_N$ identically—*so the cosmological seam is the merged horizon of Proposition 4*, where $f$ and $f'$ vanish together, and the reassigned layer's $\chi$ block $-f=(\partial_\chi r)^2-1$ vanishes there with them, so that layer is null at the handover. *On the signed chart it is null at both of the lap's unit-speed loci*—the two roots of (12)—*and the translation that closes the lap carries one to the other*, so the two are one locus of the substrate[P15R254]. A literal closed-Friedmann reading would put $\Omega_k=-\Omega_\Lambda\approx-0.685$ into the distance relation and be excluded; CR avoids that not by tuning but because its distance slicing is the flat one, the curvature living on the orthogonal layers. Spatial curvature and dark energy are one $\Lambda$ read on two slicings—the corpus identity, here at the observational rung. *And that reading is an identity rather than a correspondence, which is worth having explicitly because it also separates two lengths the same sentence could be read as conflating.* The curvature radius the closed readout returns is

$$\frac{c}{H_0\sqrt{\Omega_\Lambda}}=\frac{c}{H_0}\sqrt{1+\frac{2}{x_0^3}}=\alpha=\sqrt{3/\Lambda}, \tag{15}$$

exact in the construction's own parameterisation—symbolic residual zero, and $2\times10^{-16}$ on both backgrounds—*so what* $|\Omega_k|=\Omega_\Lambda$ *fixes is the substrate's throat. That is not the layer's present radius*: $\alpha/r_0=\sqrt{3}/x_0=1.0320$, so the two differ by 3.2 per cent, and the quantity the low-multipole floor is built on is $r_0$ (§7, below) while the quantity the closed readout names is $\alpha$. *The content the identity carries is unchanged and is if anything firmer*—$\Omega_k$ is not a free parameter, because its magnitude is forced to $\Omega_\Lambda$ by one $\Lambda$—*and what the two lengths say is that the readout and the layer are the same* $\Lambda$ *read at two epochs of the bead rather than one number serving twice*[P15R232].

**The low-multipole floor.** The closed $S^3$ carries a discrete mode spectrum, labelled by integer degree $L$, with comoving wavenumber $k_L=\sqrt{L(L+2)}/r_0$ and the lowest physical mode the quadrupole $L=2$ (the monopole $L=0$ is the background, the dipole $L=1$ pure gauge). *The*

*eigenvalue in that wavenumber is the scalar one, and the dipole's exclusion is derived rather than conventional*: the traceless second covariant derivative of a degree-one harmonic is the *zero map*, rank 0 where the full Hessian at that degree has rank 2, so it vanishes identically and not for want of modes to act on—and the same map has full rank $L+1$ from $L=2$ up. *The companion's tensor tower floors at the same degree for the opposite reason*, its transverse-traceless eigenvalue being $L(L+2)-2$, which has no member at either of the two lowest degrees [19]: *so one tower excludes two degrees it has and the other has none to exclude, and the shared number carries two statements*[P15R271]. The decoupling fixes how these project to the sky, and the fixing is the whole point. Because the distance slicing is flat (Proposition 5), the photons are projected through the *flat* geometry—the comoving angular-diameter distance is $D_M = D_C$, the same flat-ΛCDM observable that places the acoustic scale at $\ell_A \approx 301$—while only the *source* modes carry the closed-$S^3$ quantization. The transfer is therefore the discrete closed-$S^3$ spectrum projected through the flat spherical Bessel $j_\ell(k_L D_C)$, *not* the hyperspherical transfer of a literal closed universe: that transfer carries the closed *distance* relation, which CR does not have, and would (wrongly, here) deliver the lowest mode to the quadrupole and no deficit. The flat projection places the discrete modes at

$$\ell_L \simeq \sqrt{L(L+2)}\,\frac{D_C}{r_0}, \tag{16}$$

with $r_0$ the present $S^3$ areal radius and $D_C$ the comoving distance to last scattering, both fixed without new parameters: $D_C \approx 1.4011 \times 10^4$ Mpc is the comoving distance to last scattering and $r_0 \approx 5051$ Mpc is the Nariai amplitude $2^{1/3}/\sqrt{\Lambda}$ carried to the present epoch $u = \operatorname{arcsinh}\sqrt{\Omega_\Lambda/\Omega_m} \approx 1.22$, both on the background the distance data fix. The stretch factor $D_C/r_0 \approx 2.774$—the ratio of the flat projection distance to the curvature radius, the decoupling made numerical—carries the lowest mode to $\ell_2 \approx 7.8$, with no source below it.

*What* (16) *returns is the peak argument of a projection and not a multipole*, and the distinction is load-bearing wherever a statement about the degree $L$ is carried to the sky. The weight a layer degree places on a multipole is $w_\ell(L) = (2\ell+1)j_\ell(\ell_L)^2$, a genuine distribution over $\ell$ by the Bessel closure $\sum_\ell (2\ell+1)j_\ell(x)^2 = 1$; evaluated on the two lengths above, *the printed value sits near the ninetieth percentile of that distribution rather than at its centre*—$P(\ell \le \ell_L)$ running 0.875 to 0.961 over $L=1$ to 8, with $\ell_L$ above the modal multipole by between 1.74 and 3.02, a displacement of order one mode that is bounded both ways and does not grow with degree. *And the width crosses the spacing between consecutive degrees at* $L=3$: against a mean spacing of 2.19 to 2.39, the standard deviation runs 1.20, 1.83, 2.45, 3.07, 3.70, 4.32, 4.94, 5.56 for $L=1$ to 8, and each degree's band of 2% weight overlaps its neighbour's in 4 to 7 multipoles from the third degree up. *So the map is quasi-injective for the first two degrees and smears without bound above them*, and a statement exponential in $L$ transfers to $\ell$ as a statement about the lowest contributing degree rather than keeping its form—the bottom edge of each band, not its centre, because a steep envelope is dominated there. *The figures above are the geometric minimum*: they are the projection kernel alone, with no transfer function, window or visibility in them, so a real projection can only be wider. *What this does not disturb is the floor itself*: the leak below $\ell_2$ falls fast with degree, $P(\ell \le 3)$ running 0.591, 0.151, 0.059, 0.032 for $L=1$ to 4, so the floor is softened by about a sixth of one mode and not erased[P15R260][P15R273].

The full flat-projection transfer confirms the estimate and sharpens it into a prediction. Carrying the discrete spectrum through $j_\ell(k_L D_C)$ with the scale-invariant measure—the first-principles closed-$S^3$ Harrison–Zel'dovich weight [31, 32] $w_L = (L+1)/(L(L+2))$, fixed as the mode degeneracy $\beta^2$ (with $\beta = L+1$) times the per-mode power $1/(\beta(\beta^2-1))$ that equal power per logarithmic interval in $k$ requires, and reducing to the continuum $dk/k$ at large $L$ so that $\ell(\ell+1)C_\ell$ returns to the flat plateau as $D_C/r_0 \to 0$ (receipt: `verify_lowell_exact_measure.py`[P15R29])—yields a low-multipole *deficit*: $\ell(\ell+1)C_\ell$ falls below its high-$\ell$ plateau for $\ell \lesssim 7$ and recovers by $\ell \approx 8$, because the flat projection of a discrete source with a hard lowest mode at $k_2$ leaves the lowest multipoles starved (receipts: `verify_closedS3_nonsync.py`[P15R28]).

The one effect that partly fills it is the late integrated Sachs–Wolfe term [28], sourced near the observer where small line-of-sight distances send wavenumbers $k \gtrsim k_2$—*above* the floor—down to $\ell \sim 2$.

By the decomposition of the empirical-forcing paper [11] the cumulative term is the standard Sachs–Wolfe and integrated Sachs–Wolfe—CR's own differential-expansion floor being zero under uniform expansion—so this late boost rides the same modes the discrete source keeps, and the deficit is milder than the bare Sachs–Wolfe starvation alone.

The depth is a matter for a *genuine Boltzmann transfer* rather than for the geometry, *and the reason is that the deficit is two effects and not one.* In the pure Sachs–Wolfe limit the suppression splits exactly into a *floor*—the projection integral truncated below $k_2$—and a *ladder coarseness*, the discrete sum minus that same truncated integral. *At $\ell = 5$ and 6 the two are comparable and pull opposite ways*: the ladder puts back power a pure cut-off removes. *And they respond to the fitted parameter with opposite signs*—under $\pm 2\%$ in $r_0$ the floor moves by a few per cent while the full result moves by tens—so the location is the floor and is robust for that reason, and the depth is the ladder and is volatile for that reason[12]. *The recovery multipole is the ladder's too*: with it the spectrum recovers by $\ell \approx 8$, on the floor alone not until $\ell \approx 10$. Because the only CR modifications are the discrete source and the flat projection—every transfer ingredient (Sachs–Wolfe, early and late integrated Sachs–Wolfe, Doppler) shared with flat-ΛCDM [11]—the exact CMB temperature transfer $\Delta_\ell(k)$ of a standard Boltzmann code *is* CR's transfer, and the discrete spectrum is read off it: $C_\ell^{\rm CR} = \sum_{L \geq 2} w_L \, \Delta_\ell(k_L)^2$ against the continuum $\int d\ln k \, \Delta_\ell(k)^2$, the measure $w_L$ being exactly $d\ln k_L / dL$ so the CR sum is the $L$=2-floored Riemann sum of the very integral ΛCDM does continuously.

Carried out with the exact transfer (gate-validated: the continuum reproduces the code's own $C_\ell$ to four figures; and the $r_0$ sensitivity is measured rather than assumed: the *shape* is stable—the minimum stays at $\ell = 4$ and the recovery multipole moves by 0.1%—while the *depths* drift by up to 15% at $\ell = 4$ under $\pm 2\%$ in $r_0$), the deficit is *milder* than a Sachs–Wolfe-only estimate suggests, and it has a shape: $\ell(\ell+1)C_\ell$ sits at $\approx$ 0.487, 0.435, 0.359 and 0.666 of the flat-ΛCDM expectation at $\ell = 2, 3, 4, 5$, recovering by $\ell \approx 8$ (receipts: `verify_lowell_boltzmann.py`[P15R31] for the transfer, and the depths with the late integrated Sachs–Wolfe term carried to the present[P15R171]). *The prediction is therefore not a monotone recovery from the quadrupole but a dip whose minimum falls at $\ell = 4$*[P15R34]—and the quadrupole and octopole, the two multipoles a large-angle discussion reaches for first, are its two *shallowest* points.

The shape is not one code's, and that was checked here by running it rather than by comparing tables[P15R36].

Driving the photon hierarchy built for this programme (§9.1) through the same discrete projection puts the minimum at the same multipole, $\ell = 4$, and recovers at the same place, $\ell \approx 7$–$8$. *The two independent Boltzmann treatments cross-validate the depth as well as the shape* once each carries the late integrated Sachs–Wolfe term to the present: on the exact transfer the deficit is 0.487, 0.435, 0.359, 0.666 of the flat expectation at $\ell = 2$ to 5, and the hierarchy agrees to three per cent at every multipole and one per cent above $\ell = 5$ (§10)[P15R171][P15R237].

The reason is the late integrated Sachs–Wolfe: it is sourced near the observer at wavenumbers $k \gtrsim k_2$ (small line-of-sight distances), *above* the floor, so the discrete source *retains* it rather than losing it, and the deficit is milder than a Sachs–Wolfe-only treatment of the same source would give.

This is a parameter-free deficit, set by Λ through $r_0$ and $D_C$, and what is robust is its existence, its location ($\ell \lesssim 8$, geometric, set by $k_2 D_C$), its minimum at $\ell = 4$, and that the single floor $k_2$ starves $\ell = 2$ and $\ell = 3$ *together*: the exact discrete measure and the Doppler term (super-horizon velocity through $j_\ell'$) each leave the $\ell = 2$-to-$\ell = 3$ ratio near unity, if anything suppressing the octopole slightly more (receipts: `verify_lowell_exact_measure.py`[P15R29], `verify_doppler_lo`
Whether this predicted deficit matches the observed sky is the data test, against the large cos-

mic variance of the lowest multipoles. Confronted with the measured low multipoles, the verdict is a *wash*. CR predicts a smooth modest deficit bottoming at $\ell = 4$; the observed sky has a sharp quadrupole-only dip ($\sim 0.2$) with a near-ΛCDM octopole ($\sim 0.6$–$1.0$), so CR sits *above* the observed quadrupole and *below* the observed octopole, matching neither sharply. Under the exact cosmic-variance likelihood the net over $2 \le \ell \le 10$ is $\Delta(-2\ln L) \approx +1.6$ (range $+0.2$ to $+3.4$ across the octopole estimator), a small residual well inside cosmic variance (receipts: `confront_lowell_data.py`[P15R32P15R171]). *Broken up by multipole, the sector's weight does not lie where its name suggests*: the quadrupole *rewards* this cosmology ($-2.5$, its milder dip nearer the observed low power than flat ΛCDM's), the octopole penalises it at $+1.0$, and the largest single-multipole penalty is the $+2.0$ at $\ell = 4$—the prediction's own minimum, and the one multipole of the four that the large-angle literature does not treat as anomalous. The low multipoles are thus a genuine but *non-discriminating* sector: neither a parameter-free correspondence success nor a sharp falsification edge, but a mild deficit consistent with ΛCDM within the lowest multipoles' cosmic variance.

*One further question belongs with this prediction, and it is a located one.* The branch-point filter that sets what crosses into the expansion leg is, in its quantum reading, a Euclidean kernel whose action on a mode is controlled by an adiabaticity parameter $C/\mu_n$, with $\mu_n$ the harmonic index of the layer's $S^3$ tower [35]. That parameter is small for all but the lowest harmonics and reaches order unity only at $n = 2$ and $n = 3$, where it takes the values 0.70 and 0.47. *Three independent routes put the approximations at their limit on the same scale*: the transmission calculation's boundary at $\ell \approx 2.5$, the projection's incompleteness below $k \sim 2 \times 10^{-4}\,\mathrm{Mpc}^{-1}$ ($\ell \approx 3$), and this parameter's reaching order unity at $n = 2, 3$. Whether that is a loss of control is decided by computing the filter rather than estimating it.

*The transmitted amplitude at low multipoles can be computed rather than estimated, and doing so settles what the estimates left open.* The branch-point filter's action on a mode is, in the quantum reading, a Euclidean kernel [35]. The exponential-of-an-integral form $e^{-\int \omega\, \mathrm{d}\eta}$ is a WKB approximation whose adiabaticity parameter is of order unity at $\ell \lesssim 5$; but the underlying mode equation $\mathrm{d}^2\psi/\mathrm{d}\eta^2 = \omega^2\psi$, with $\omega = kc_s$ and $\mathrm{d}\eta = \mathrm{d}s/a$ along the segment, is well posed at every $k$, and integrating it is a matter of numerics rather than of approximation. Composing exact constant-$\omega$ transfer matrices in the stable direction over the segment ($|\Delta\eta| = 3.33874$ in units $\alpha = 1$) gives exact/WKB ratios of 0.926, 0.913, 0.901, 0.891 and 0.889 at $\ell = 2, 3, 5, 15, 40$: *the WKB form is accurate to* 7–11% *at every multipole from* 2 *to* 40, with the exact transmission slightly smaller, so the filter is marginally stronger than the approximation gives[P15R81].

*Two readings suggested by the adiabaticity parameter alone are thereby excluded.* The first is that the treatment is uncontrolled at $\ell = 2$–$5$: it is not, and the parameter was the wrong figure of merit, measuring as it does the fractional error in an exponent that is itself small at low $\ell$. *The residual's own behaviour settles this independently of any argument about the parameter*: the ratio *decreases* monotonically with $\ell$—0.926, 0.913, 0.901, 0.891, 0.889—so the discrepancy is smallest at $\ell = 2$, where the adiabaticity parameter is of order unity, and largest at $\ell = 40$, where it is small. An error of adiabatic origin runs the other way, WKB being an expansion *in* that parameter, so such an error is largest where the parameter is $O(1)$ and vanishes as it does. *What the numbers show is therefore a systematic offset rather than an adiabatic breakdown*—the WKB form is uniformly slightly generous and the exact filter uniformly slightly stronger, which is a correction of known sign and bounded size and not a loss of control[P15R81L19]. The second concerns the comparison with the observed sky. The correction is very nearly $\ell$-independent—varying by 4% across a factor of twenty in $\ell$—so between $\ell = 2$ and $\ell = 3$ it alters their ratio by 1.4%. *The structural expectation that a single floor starves $\ell = 2$ and $\ell = 3$ together is therefore not relieved by the exact treatment*, and the observed sky's selective suppression of $\ell = 2$ alone, by a factor of order three, remains unaccounted for. The low-multipole prediction of this section therefore stands as stated, and so does the discrepancy it faces.

*Finally, the two Boltzmann treatments above agree*, on the depth as well as on the shape,

the minimum and the recovery, once both carry the late integrated Sachs–Wolfe term to the present (§10). What both supersede is the Sachs–Wolfe-analytic estimate (0.22 and 0.20 at $\ell = 2, 3$), which takes the Sachs–Wolfe source as the matter-domination value $\frac{1}{3}\Phi$, whereas at recombination $\rho_r/\rho_m = 0.317$ and the source is $0.4006\,\Psi$—a 20% error in the very ratio of Sachs–Wolfe to integrated Sachs–Wolfe that fixes how much the latter fills the bare deficit.

# 8 What the progenitor supplies: the transmission proof

The decomposition's decisive proof is here: *why* the substrate transmits the spectral shape rather than imprinting one of its own. It turns on how a mode approaches the locus it meets. Two horizon cases bound the question: a non-degenerate horizon, and the degenerate *Nariai double root*, at $r = +\alpha/\sqrt{3}$—the front seam, where $f$ and $f'$ vanish together so that $\kappa = 0$. *The locus the modes cross is neither*: the branch point sits at $r = 0$, carries no surface gravity because $f$ diverges there rather than vanishing, and is not a Killing horizon at all. *And the lap's other seam is a case of its own, which must be kept apart from the double root rather than folded into it.* The back seam at $r = -2\alpha/\sqrt{3}$ is one point of the substrate with the front seam, met a lap earlier, but the two chart values carry different orders of contact with zero: the back seam is the horizon cubic's *simple* root, so $f'$ does not vanish there and $\kappa = 3\sqrt{3}/4\alpha$. It is the lap's one non-degenerate horizon. *Whose horizon it is depends on the level, and the framework states both*: at substrate level it is the merged horizon re-entered, one point with the front seam; read locally on the sheet the collapse runs on, it presents as the cosmological horizon of the mass, with the radius timelike outside it and the geometry static within [9, 5]. What matters for the dichotomy is neither of those but the order of the root, and that is unambiguous. Both cases of the dichotomy are therefore realised on the lap, and which one a given seam realises is a question about that seam and not about the pair.

**Proposition 6** (Transmission dichotomy)**.** *For a mode approaching a horizon where the metric function behaves as $f \sim (r - r_h)^p$, the tortoise coordinate $r_* = \int dr/f$ controls the approach. At a non-degenerate horizon ($p = 1$, $f \sim 2\kappa(r - r_h)$, surface gravity $\kappa > 0$) the integral is logarithmic and the approach exponential, $(r - r_h) \sim e^{2\kappa r_*}$. At the degenerate Nariai double root ($p = 2$, $f \sim -\Lambda(r - r_N)^2$, $\kappa = 0$) the integral is $r_* \sim +1/[\Lambda(r - r_N)]$ and the approach power-law, $(r - r_N) \sim +1/(\Lambda r_*)$.*

*Argument.* The two tortoise integrals directly (receipts: `verify_geometry.py`, anchor $4^{\mathsf{P15R16}}$). □

*Remark* 1 (The leg does not tilt it either)*.* The proposition shows that a degenerate Killing horizon carries no scale, so it cannot imprint one, and Proposition 7 shows the same of the branch point. The collapse leg supplies the other half of the same conclusion, by an independent route. Horizon entry on that leg occurs at $x = k\eta/\sqrt{3} = 1/\sqrt{3}$ *for every* $k$—the radiation era's own scale invariance, which §4.1 recovers from the substrate geometry—so every mode leaves the leg carrying the same free-oscillation amplitude $\Psi(1/\sqrt{3})/2 = 0.4835\,\Psi_i$, a single $k$-independent number$^{\mathsf{P15R41}}$. *So the leg multiplies the spectrum by a constant and the branch point imprints nothing*: neither carries a scale, and a spectrum can only be tilted by something that does. The tilt observed is therefore the progenitor's, established from both sides of the branch point rather than one. *And the list of places a tilt could have come from is closed rather than merely exhausted.* A tilt is $\mathrm{d}\ln P/\mathrm{d}\ln k$; written as this construction writes its output the spectrum is a product of three factors—the measure, the interior's multiplier, and the progenitor's amplitude—so the log-derivative is a sum and the candidates are its terms. *The measure's term is the constant* 3 *and cannot tilt anything*, which leaves exactly two: the interior's and the progenitor's. *Both are accounted for above*, the interior's being the constant this leg multiplies by and the progenitor's being the inherited datum, and a third term would have to be an action that is neither the interior's nor the progenitor's and is not the measure$^{\mathsf{P15R300}}$.

*"Carries a scale" has an operational form, and it is worth recording because it is what makes the dichotomy testable rather than descriptive*: a kernel carries a scale exactly when the tilt one fits to it *depends on which band one fits.* Fitted on four disjoint bands, the degenerate ($p = 2$) kernel returns one tilt to sixteen decimal places while the non-degenerate ($p = 1$) kernel spreads by several units, and that spread shrinks as $\kappa$ grows—a larger scale pushing its knee past the range. Restoring a scale to the degenerate kernel by hand destroys the agreement, so the scale-freedom is doing the work and not the arithmetic[P15R151L21]. *This restates Proposition 6 and proves nothing further*; what it adds is a test one can run on a kernel without knowing where it came from.

*Remark* 2 (What the branch point does not fix). Both halves of the dichotomy are statements about the spectrum's *shape*: a degenerate horizon carries no scale, and neither does horizon entry on the leg, so neither can tilt what passes. *Neither is a pointwise condition on the perturbation state at the branch point*, and the distinction matters for anyone computing the handover. The single free function per mode is the freedom of the *spectrum*—the progenitor's amplitude and tilt—not a claim that the branch point fixes the state component by component. The state itself is whatever the leg's evolution produces, and §4.1 supplies it in closed form: the potential from the leg's own equation, the effective temperature oscillating freely from horizon entry, and the density contrast as their difference.

The Hamiltonian constraint is not an additional condition to impose there. In this framework it *is* the anchor—the map from a cut to its stress-energy, whose energy sector reads the Friedmann equation off the geometry [7, 8]. It is a definition, not a restriction: on the leftward reading the rate is set by the substrate and the content is read from it, so the perturbed constraint likewise *defines* the density contrast a given geometry carries rather than restricting which geometries are allowed. Imposing it at the branch point alongside the leg's own solution therefore over-determines the handover, and the same geometric state answering to different stress-energy on the two sides is not a failure of matching but the branch point doing what the level assignment says it does.

*Remark* 3 (What the branch point's degeneracy does not settle). It is tempting to read Proposition 6 as settling the phase as well as the shape: the tortoise coordinate diverges at the degenerate root, so a mode's accumulated phase would be unbounded and nothing finite could be inherited. *That reading does not survive its own arithmetic.* The divergence is geometric and carries no $k$, so an unbounded $kr_*$ is a divergent term *linear* in $k$; a linear phase renormalizes the sound horizon rather than shifting the comb, and an unbounded one would send the comb spacing to zero and erase the acoustic peaks altogether—excluded by observation, not supported by it. And the modes never meet the divergence in any case: they cross the branch point at finite conformal time, whereas $r_*$ diverges along a static slicing's approach to the horizon, which is not the collapse trajectory.

What the leg does contribute is finite and, as it happens, invisible in the peak positions. Horizon entry at $x = 1/\sqrt{3}$ for every $k$ makes the accumulated leg phase $k\eta_{\rm bp}/\sqrt{3} - 1/\sqrt{3}$: linear in $k$ with a mode-independent offset. So carrying it across rescales the effective sound horizon by $1 + \eta_{\rm bp}/(\sqrt{3}\, r_s)$ and adds a constant, and changes the comb in no other way. *The comb therefore cannot distinguish a transmitted phase from a reset one*, because a rescaling of $r_s$ is absorbed by the angle it is read against. Any discriminant must be a quantity that does not take the same factor—the diffusion scale is one, since $r_D$ does not—and that is the discriminant of principle. *The comparison is made, and it does not require $r_D$.* The leg's accumulated conformal time is fixed by the geometry rather than fitted: the collapse branch $r = -(2M\alpha^2)^{1/3} \cosh^{2/3}(3\,\mathrm{Re}\,\tilde{\tau}/2\alpha)$ has cosh $\geq 1$, so $|r| \geq (2M\alpha^2)^{1/3} \simeq 3792$ Mpc and the branch point is reached only across the lift, along which $\mathrm{Re}\,\tilde{\tau}$ does not advance and no conformal time accumulates. Hence $\eta_{\rm bp} \approx 5 \times 10^3$–$10^4$ Mpc, and $r_s^{\rm eff} = r_s + \eta_{\rm bp}/\sqrt{3}$ is *dominated* by the leg term—$3.0$–$5.9 \times 10^3$ Mpc against $r_s = 144$ Mpc. The absorption that takes a small rescaling cannot take this one: reproducing the measured $\ell_A = 301.8$ would require $D \approx 2.9$–$5.7 \times 10^5$

Mpc against the actual $1.4 \times 10^4$ Mpc, a factor of 21–41. *The acoustic phase is therefore reset at the branch point and not carried across it*—established on the peak positions alone[P15R42]. The result is robust to the open account of the lift: any conformal contribution the lift is found to carry only adds to an already-excluded total.

The amplitude separates from the tilt in the same statement. What the progenitor supplies is the potential power $P_\Psi$; what the construction contributes is the factor $(0.4835)^2 = 0.2338$ carrying it to the observed $A_s$. That factor is computed rather than fitted: the inherited quantity is $P_\Psi$ and the map from it to $A_s$ is derived, so what the construction takes from the progenitor here is the potential power and nothing about the map [9].

The physical reading is the result (Fig. 6). The exponential, non-degenerate approach is the thermal (Hawking) law—the mechanism by which a de Sitter horizon *imprints* a scale-invariant spectrum ($n_s \to 1$): the exponential redshift erases the infaller's spectral information and stamps the horizon's own thermal scale. The power-law, degenerate approach carries no thermal scale; it is scale-free, and the infalling spectrum is transmitted faithfully. CR's transmission locus is the branch point $r = 0$, where $f$ *diverges* as $-2M/r$ so the tortoise coordinate is *finite*—the crossing accumulates no divergent phase and stamps no scale, a third case beyond the dichotomy's two[P15R42]. The Nariai double root (Proposition 4, $f''(r_N) = -2\Lambda$, $\kappa = 0$) is *the front seam*, $r = +\alpha/\sqrt{3}$. It is one point of the substrate with the back seam the bead meets on the way in, the two chart values $r = -2\alpha/\sqrt{3}$ and $r = +\alpha/\sqrt{3}$ being the same $\varphi$ modulo $2\pi$ in the phase $\varphi = 2\pi r/\sqrt{3}\alpha$; *being one substrate point does not make them one case of the dichotomy*, since the degeneracy is a property of the root and the back seam's root is simple. The branch point sits two thirds of the lap in from the front seam (240°, with 120° remaining), so the crossing at $r = 0$ and either seam are distinct loci on the lap without a seam being an endpoint of anything. Hence:

**Proposition 7** (The branch point transmits)**.** *The cosmogenesis branch point is a faithful scale-free transmitter—not because it* is *the degenerate Nariai horizon, which it is not (that is the front seam, at $r = +\alpha/\sqrt{3}$), but because $f$ there* diverges *as $-2M/r$, so $r_*$ is finite and the crossing accumulates no divergent phase at all: nothing about the branch point imprints a scale on what crosses it, and the amplitude $A_s$ and tilt $n_s$ are the progenitor collapse's, not the branch point's.* What a finite tortoise length rules out is a divergent phase, and the crossing does carry a real factor, which is stated here with its size because it is exact. The era in which a mode's spectrum is fixed is the one in which it is super-horizon*, and that era is located on this construction by one scalar along the bead: $a''/a$ in conformal time, whose single constant is the pure number $2/3c_0^2 = 2^{-1/3}$ on all three legs, falling monotonically to it at the comoving turnaround and attained again as a* minimum *on the expansion leg at $x = \frac{1}{2}\ln 2$ exactly—the midpoint of that leg measured from the branch point, by the $1 : \sqrt{3} : 2$ closure of §10 and so at $1.9276\alpha$ rather than at the front seam's $2.3759\alpha$*[P7R26]*. So away from the three divergences the only super-horizon harmonic is the monopole*: the horizon degree is $L_{\rm hor} = -1 + \sqrt{1 + 2^{-1/3}} = 0.339$, with $L = 1$ already sub-horizon by a factor 3.78 and $L = 2$ by 10.08, so no convention shift of order unity in $k^2$ reaches that gap.* The sky's first acoustic peak, carried to the layer by (16), has its spectrum fixed within the first per cent of the collapse leg, measured from that leg's far-past end—*so the whole of the lift lies between that locus and the front seam, and the envelope of §6 applies to what the progenitor supplies.* The inheritance is therefore exact for the isotropic mode and for no other*: $T(0) = 1$ identically, the monopole being super-horizon everywhere, while every anisotropic harmonic is damped by the parameter-free factor the throat supplies.* Which is the same statement as the isotropizing throat read from the other end*, and it places the origin of the anisotropic spectrum on the expansion leg, where it requires a source the single Nariai worldline of this construction does not carry—named here as owed and developed in neither this paper nor the bead*[P15R241]*.* Equivalently, CR has no inflationary scale-invariant attractor—and the reason is not that the lap carries no non-degenerate horizon, because it carries exactly one. *The three loci fail the imprinting condition for three different reasons, and*

*the reasons are not interchangeable. At the branch point $f$ diverges, so $r_*$ is finite and no divergent phase accumulates. At the front seam the root is double, so $\kappa = 0$ and the tortoise approach is power-law, carrying no scale by Proposition 6. At the back seam neither holds: the root is* simple*, $\kappa = 3\sqrt{3}/4\alpha$, and in the tortoise coordinate the approach* is *exponential at rate $2\kappa$—the imprinting law itself.* What fails there is the approach and not the scale, and the reason is characteristic structure rather than a choice of slicing. *The seam is a root of $f$, and the principal symbol of the radial mode equation restricted to* $\mathrm{d}r$ is *$f$—so every root of $f$ is a characteristic surface of that equation, and the two characteristic speeds $\mathrm{d}r/\mathrm{d}\tau = v \pm 1$ there are exactly* 2 *and exactly* 0*. The unbounded approach belongs to the skimming family, the one with speed zero, which fails to arrive at rate $\kappa$ forever; the trajectory that carries the modes rides the other, crossing at speed* 2 *and reaching the comoving turnaround $0.878\alpha$ of its own proper time later.* Both rates are real and neither is an artefact*: $r_*$ runs at $2\kappa$, proper time along the skimming rays at $\kappa$, and the bead runs on neither.* The thermal factor is in the mode equation—exactly once, and on the branch the modes are not on. *The indicial equation at the seam returns the two exponents* 0 *and $-i\omega/\kappa$; they differ by a non-integer for every real $\omega > 0$, so the monodromy is diagonal, with no logarithm and no mixing: unity on the regular branch and $e^{-2\pi\omega/\kappa}$ on the skimming one, the Boltzmann factor at $T_H = \kappa/2\pi$.* The crossing therefore takes the regular branch, whose transfer across a window of width $d$ tends to unity like $d$—*measured over four decades of $d$, four decades of $\omega/\kappa$ and $\ell = 0, 2, 10$, against a $d$-independent $e^{\pi\omega/\kappa}$ that reaches $10^{136}$ at $\omega = 100\kappa$.* So the crossing's own factor is the identity*: not small and not $k$-independent-and-nonzero but exactly* 1*, which is neither an amplitude in $A_s$ nor a tilt in $n_s$, and $2\kappa$ cannot enter through it.* And the one wavenumber a thermal factor could have carried is named rather than left open*: the only scale at the crossing is $\kappa$, so $k_{\mathrm{th}} = a_h\kappa = 3/2$ exactly and $\alpha$-free, which in the convention $k^2 = L(L+2)$ is $L = 0.803$*—below the dipole*, the smallest harmonic the sky carries sitting a factor $2/\sqrt{3}$ above it, and through* (16) *acting at $\ell = 4.16$ against the first acoustic peak's* 220.60*. That factor of* 53 *in wavenumber is the counterpart of the* 53 *in position: the locus where the sky's first peak has its spectrum fixed lies in the first per cent of the collapse leg from that leg's far-past end while the back seam lies at* 46.5 *per cent of the same leg,* so the lap's one non-degenerate horizon stands some fifty times downstream of where the progenitor's spectrum is set, in scale as well as in place, and alters nothing crossing it[P7R26P7R28]. What is not computed here is the flux the seam radiates into the lap's future*, which is a question about the quantum state on this background rather than about a classical mode's transfer.* And it is not computable on this construction at all, for two reasons that are properties of the lap rather than of the progenitor. *First, the closure that makes the two seams one locus—the translation $r \mapsto r + \sqrt{3}\alpha$, which carries $-2\alpha/\sqrt{3}$ to $+\alpha/\sqrt{3}$ identically—makes that one locus carry* two *surface gravities,* 0 *and $3\sqrt{3}/4\alpha$, and no $\alpha$ reconciles them; a thermal state is a periodicity at a horizon of a given $\kappa$, and there is no single $\kappa$ here for a period to be $2\pi/\kappa$ of. Second, and independently, the Euclidean section the lift supplies is measured in the bead's own conformal time while the periodicity would be imposed in the seam's Killing time, and the conversion between them, $f/r$, is not constant along the lift and* vanishes at both seams*—the only loci at which such a condition is imposed.* So the construction selects no state*, and the absence is of the object a flux calculation would be about rather than of a result it would return*[P15R284].

The absence of any scale where the modes cross is thus the proven reason CR carries the primordial tilt rather than manufacturing it. What the transmitter carries is, from the Schwarzschild–de Sitter side, open: $n_s$ and $A_s$ are inherited, fixed by the progenitor collapse—the same handover that fixes $\rho_r/\rho_m$ and the light-element abundances—and not derived here. *Naming them together is convenient and has concealed that they are not the same kind of quantity*[P15R86]. The vacuum normalisation is the only place a quantum constant enters the spectrum, so $P(k) = \hbar A_0 k^{n_s - 1}$ and $\hbar$ is an overall multiplicative factor: *it survives in the amplitude and cancels in every logarithmic derivative.* An amplitude is the spectrum at one wavenumber; a tilt

is its ratio to itself at two. *So the construction's inability to force a dimensionless magnitude bears on $A_s$ and not on $n_s$*—the first is permanently inherited, the second need not be. *And read backwards the same statement is useful rather than limiting*: a quantity fixed by the moduli rather than the invariants is one the observation *measures* rather than one the construction must predict, so the observed $A_s$ is a measurement of the progenitor's boundary data. *It does not isolate the mass*: the parent's growing-mode amplitude $D_k$ enters alongside it, which is the corollary recorded above, so the mass must be reached by a route that does not run through the amplitude and checked against the independent requirement that it lie below the Nariai value. *And it is worth saying precisely what that openness is, because a count settles its kind.*[P15R85] The progenitor family described here is parametrised by $M$ alone—$\alpha$ is the one invariant and $E = 1$ a choice of geodesic—so the geometry is one-dimensional; and the Kretschmann along the bead is mass-free in the faller's own proper time, so that one parameter does not survive the approach either. *A quantity identical for every member carries no information about which member it was*, and the collapse geometry therefore transmits not one parameter but none. Against that stand at least four independent inherited numbers. *So 'inherited from the progenitor collapse' cannot mean inherited from the collapse geometry*: what is inherited is inherited from the progenitor's matter content—its composition, its entropy per baryon, its perturbation spectrum—and none of that is carried by $M$, the exterior here being exactly vacuum and the marginally-bound interior dust without composition. *The frontier is accordingly not a derivation awaiting a cleverer argument but a modelling task awaiting a progenitor interior*, and what the geometry supplies in its place is a cutoff rather than a content: the segment's conformal length $|\Delta\eta| = 3.33874\alpha$ is fixed by $\Lambda$ alone *on the Nariai member*—below it $|\Delta\eta|$ runs as $M^{-1/3}$, reaching $4.21\alpha$ at $M/2$ and $15.50\alpha$ at $M/100$—and says which modes survive without saying how large they were[P15R81][P15R245]. *And the multiplicative statement above settles the one join the inherited pair needed*: the handover's own factor is a constant of exactly the kind named there, so it survives in $A_s$ and cancels in the tilt—the latter identically, with the power to which the transfer enters left free, so no convention choice touches it. *What it costs is a number rather than a mechanism*: the factor is $0.4835^2 = 0.2338$ on the spectrum, 6.49 per cent against the handover's own three, while a transfer carrying the plasma's sound speed instead shifts the spectrum by $2c_s k|\Delta\eta|$—some $7.6 \times 10^3$ times the tilt budget the refit leaves (§4.7)—so the sound-speed reading cannot deliver the tilt it would be invoked to carry[P15R253].

# 9 Predictions, and the progenitor boundary

The signatures, with their maturity marked:

1. Classical, non-vacuum primordial statistics; no inflationary consistency relation; no substrate-sourced primordial $B$-modes (established, §5).

2. The transmission character: the branch point carries the progenitor tilt and does not drive $n_s \to 1$ by any *inflationary* attractor (established, §8). *Nor does the construction lack a scale-invariant attractor—it has one, and it is not the inflationary root.*[P15R82] Scale invariance for a background $a \propto (-\eta)^\beta$ with $z \propto a$ requires $\beta(\beta - 1) = 2$, a quadratic with *two* roots: $\beta = -1$, de Sitter inflation, and $\beta = +2$, a matter-dominated contraction. *The collapse leg of this construction is the second root exactly*: near the branch point $f \to -2M/r$, so the leg is pressureless there and the banked law $r^3 = \frac{9}{2}M\tau^2$ with $a = |r|$ gives $a \propto \tau^{2/3} \propto \eta^2$; the mode equation integrated from vacuum data returns a flat spectrum to a part in $10^4$. *The mode-selection question that shadows this in the literature was accordingly put to the construction directly, and it answers in the negative.*[P15R83]

   The scale-invariant spectrum sits in the mode that *dominates* the contraction, whose curvature perturbation goes as $\eta^{-3}$; the mode that survives on the expanding side is the constant one, and from vacuum data its spectrum is $n_s - 1 = 6$.

Whether the first reaches the second is a connection problem at the branch point, and here it is settled rather than assumed: the exponents 2 and $-1$ differ by an integer, so a logarithm could mix them, but the $\sinh^{2/3}$ law is *even* in $\tau$, which in conformal time places its correction at $\eta^6$ and steps the Frobenius recursion in sixes past a resonance sitting at three. *No logarithm is generated, the continuation acts diagonally, and the scale-invariant spectrum does not reach the observer*—confirmed independently by continuing the dominant solution numerically around the branch point, where the admixture it produces is what the claim turns on, where the admixture falls as $\delta^{2/3}$ and the diagonal entry goes to unity. *What survives is therefore narrower than it first appeared and sharper than what it replaced*: the construction *has* a scale-invariant background, the spectrum that background freezes is confined to the contracting side, and the obstruction is a parity property of the construction's own solution rather than a missing mechanism. *Mixing would require a background correction odd in $\tau$, and that question is settled here too, in the negative.*[P15R84]

Writing the law as $r = Au^{2/3}g(u)$ with $g = [\sinh u/u]^{2/3}$ places every branched factor in $u^{2/3}$ and leaves $g$ even, analytic and non-vanishing at the origin—exactly, since $\log(\sinh u/u)$ is analytic in $u^2$ there and the exponential of a series in $u^2$ is one. The three sheets differ by a constant cube root of unity, and a constant multiplier cannot change a parity; the conjugate branch is even as well, and carries no branch point at all. *So the evenness is unavoidable, and what it says is not a technical fact but a symmetry*: $r(-\tau) = r(\tau)$, the cosmogenesis is time-symmetric about the branch point, and a time-symmetric background cannot mix a growing mode with a decaying one, since mixing would distinguish a direction of time the background does not. *The same invariance appears one observable over*: the driven acoustic amplitude of §4 is a resonant Fourier magnitude exactly invariant under time reversal, a theorem about the driving, though what it carries about the peak heights is what that section's instrument does not return. *One symmetry, read at two places. And it locates rather than forecloses what would be needed*: this leg is exactly even because it is the marginally-bound geodesic of an exactly vacuum exterior, so a progenitor carrying real matter content—pressure, dissipation, anything with an arrow—would not be, which returns the question to what the progenitor was.

3. Flat distances ($\Omega_k \simeq 0$) coexisting with a closed-$S^3$ low-multipole *deficit* below $\ell \approx 8$, set parameter-free by $\Lambda$—mild, and on a genuine Boltzmann transfer a dip bottoming at $\ell = 4$ ($\approx 0.49$, 0.44, 0.36, 0.67 of the flat-$\Lambda$CDM expectation at $\ell = 2, 3, 4, 5$), its shape and depth cross-validated between two independent transfers (*established* for the existence, the location, the minimum and the depth (§10); *non-discriminating* as a test, the deficit sitting inside the largest multipoles' cosmic variance). Confronted with data it is non-discriminating: it starves $\ell = 2$ and $\ell = 3$ together, matching neither the sharp observed quadrupole-only dip nor the near-$\Lambda$CDM octopole, and sits consistent with $\Lambda$CDM within the lowest multipoles' cosmic variance (§7).

4. Coherent acoustic peaks from geometric (null-boundary) phase-fixing rather than super-horizon freeze-out (argued, §4).

5. The acoustic scale is resolved in the cosmological development above (§2.3), and the geometric rate (4) is the standing CR-versus-$\Lambda$CDM discriminator on data in hand (*established*, §2.3)—a strength established before the spectrum, not this section's to re-assert.

The boundary is principled. The substrate carries everything the branch point's null geometry lets it carry—coherence, isotropization, the discreteness floor, the classical character, the faithful transmission—and the scale-free crossing proves precisely where it stops: the spectrum's amplitude and tilt are not the substrate's to set. This is the shape inflation has when it

parametrizes its observables through an unknown inflaton potential; CR routes them through an unknown progenitor collapse, with the arrival constrained—whatever the collapse supplies must reach the branch point coherent, isotropized, scale-freely transmitted, and classical.

The symmetry between the two unknown suppliers is only apparent, and the asymmetry is the substance. Inflation is not structure the standard cosmology already contains; it is an added sector—a scalar field with a freely specified potential—introduced to produce features that cosmology was known to lack: the causal contact the horizon problem denies, the flatness the dynamics do not single out, and, once these were in view, a near-scale-invariant spectrum. The potential is then chosen to deliver the observed amplitude and tilt. A model built to be tunable, and tuned to a target known in advance to require tuning, earns precisely what such a model can earn—it establishes that the features are *possible*, that an apparatus permitting them exists. It does not establish that they were *required*, and the routine elevation of the spectral fit into a vindication of inflation conflates the two: that a freely chosen potential can be set to match $n_s$ and $A_s$ is not evidence that inflation occurred, but the statement that a model with enough adjustable structure to fit has been fit. This is the distinction between a world that *requires* a phenomenon and one that merely *permits* it through adjustable parameters—the same distinction the programme draws in reading the cosmic foliation as forced rather than chosen [11]—and it places the inflationary spectrum on the permitting side.

The present cosmology adds no such apparatus. Causal contact, isotropy, flatness, phase coherence, and scale-free transmission are not produced by a field introduced to produce them; they are read off structure the construction already contains—the de Sitter throat that isotropizes by the no-hair theorem (§6), the null boundary on which coherence *is* the characteristic-data condition (§4), the exactly Euclidean constant-time slice (§2.3), and the degenerate horizon whose vanishing surface gravity transmits rather than imprints (§8)—each fixed by $\Lambda$ alone, with no parameter free to do the producing. The one quantity it does not derive, the amplitude and tilt, it inherits in exactly the sense flat $\Lambda$CDM inherits the baryon-to-photon ratio; on that single datum the two frameworks stand level, and on everything the inflationary sector is invoked to supply they do not. The verdict this licenses is not symmetric, and it does not await a future measurement. On the data in hand the two align; on the structure each must assume in order to align, one introduces an entire tunable sector and tunes it while the other introduces none—and makes, in its place, falsifiable commitments where inflation keeps its freedom: no scale-invariant attractor *reaching the observer*, no consistency relation, no substrate-sourced $B$-modes (§5, §8)[P15R83]. By the economy of assumption that distinguishes a required structure from a permitted one, that comparison is settled by what each theory must presuppose, and it is settled in this cosmology's favour. The bound on that verdict is stated in the same breath: it is a verdict on what each theory must presuppose, and the data axis is split. The expansion history fits the distance ladder and fixes $H_0$ with the acoustic angle (§2.4); the abundances are confirmed within $1\sigma$ [19]; the spectrum's content remains inherited; and the acoustic peak structure, computed from that background with nothing fitted to it, reproduces the comb to a part in six hundred and is still rejected on $TT$ shape (§4.7)—the sharpest datum against the construction.

## 9.1 The acoustic instrument: what it is validated against, and to what accuracy

The low-multipole prediction above and the ratio statements below are carried by a Boltzmann transfer built for this programme, and its standing should be stated plainly rather than assumed. It is a full photon hierarchy with polarisation, second-order tight coupling, massless neutrinos, and a Peebles recombination history; the line-of-sight source carries the monopole with the potential, the Doppler term, the integrated Sachs–Wolfe term, and the quadrupole with its own projection kernel ($j_\ell + 3j''_\ell$).

Against a like-for-like reference—an independent Boltzmann code run at identical parame-

ters, unlensed and with reionisation off—it reproduces the *acoustic peak and trough positions to* $\leq 0.5\%$ across $P_1$–$P_4$, the *ionisation history* $x_e(z)$ *and its derivative to* $\sim 1\%$ through the visibility peak, matter–radiation equality to 0.02%, and the *transfer function to better than* 1% *for* $k < 0.02\,\mathrm{Mpc}^{-1}$. Above that wavenumber it agrees to 1–2%, with the residual scatter in both directions and no trend across the range[P15R43].

*Two construction details are worth stating, because both are easy to omit and both have signatures that read as physics.*

*The first is the reach of the mode ladder.* The angular spectrum at multipole $\ell$ is an integral over wavenumber, and the projection kernel $j_\ell(k(\eta_0 - \eta))$ has substantial support *above* $k = \ell/D_M$, not merely at it; a ladder cut at $k_{\max}$ therefore truncates the integral, and does so more severely as $\ell$ rises toward $k_{\max} D_M$. Quantitatively, truncating an exact transfer at $k_{\max} D_M = 900$ retains 99.2% of $C_\ell$ at $\ell = 200$ but only 61% at $\ell = 800$ and nothing at all at $\ell = 1150$; carrying the ladder to $k_{\max} \geq 2\,\ell_{\max}/D_M$ recovers every multipole to better than half a per cent. *The tell is unmistakable once known*: the highest peak simply fails to appear, so a spectrum that should show four acoustic peaks shows three, and the peaks that do appear are displaced downward by an amount that grows with $\ell$. *Increasing the ladder's density at fixed reach does not help*—the loss is the range, not the sampling, which is worth checking explicitly because the two are easily confused.

*The second is the seeding of the conformal time.* It is a cumulative integral, $\eta(a) = \int_0^a \mathrm{d}a'/(a'^2 H)$, and a numerical grid necessarily begins at some finite $a_{\min}$; the interval already elapsed below that point must be carried as a seed rather than set to zero. In radiation domination $H \propto a^{-2}$ makes the integrand constant, so the omitted piece is $a_{\min}/(H_0\sqrt{\Omega_r})$ exactly—for $a_{\min} = 10^{-5}$ it is 4.6 Mpc. *It is a fixed offset at every epoch*, and therefore 0.03% of $\eta_0$ but 12% of $\eta$ at $z = 10^4$: the acoustic scale, the peak positions and the peak ratios are insensitive to it, while any early-time quantity or any comparison against a second code is not. Its signature is instructive—because the potential decays more steeply at larger $k$, a fixed error in the time coordinate reads out as a *smooth, monotone, k-growing* error in amplitude, which is the shape one would otherwise attribute to a physical suppression. The grid must also extend below the intended starting redshift, so that the initial epoch is interpolated rather than clamped.

*That residual does not enter the claims made here.* The Cosmological Relativity and flat-ΛCDM spectra are computed through the *same* machinery, so an instrumental transfer error afflicts both arms identically and cancels in their ratio—and every prediction stated in this paper is a ratio. The low-multipole deficit in particular lives at $\ell \lesssim 8$, i.e. $k \ll 0.02\,\mathrm{Mpc}^{-1}$, where the two agree to better than 1%. What the residual does forbid is an *absolute* amplitude claim: this instrument should not be read as reproducing the observed small-angle power to better than a few per cent, and no such claim is made.

## 10 The open frontier: the pieces this cosmology is built to close

The result stands on what the substrate determines. What it does *not* yet claim has been carried to a sharp, gradable edge rather than left vague—and those edges are the most interesting part of the programme from here, each a gap to be closed, not papered over.

We hold them to the criterion of necessity the programme's epistemology names—that a structure is credited for *requiring* a phenomenon, not for *permitting* a value that fits it [12]—and sort them by it: a gap that is *buildable* is a debt owed, to be built before the claim it bears on is a proof rather than a coherent proposition, while a genuinely external unknown may stay open at no cost, exactly as flat ΛCDM leaves its own inherited data open.

The buildable ones below are named as debts, not softened into frontiers. *Of the four, three are run and one is a frontier*: the end-to-end branch-point-to-recombination transfer, the large-angle depth and the likelihood are built and confronted below, what the transfer leaves being a difference in the acoustic contrast rather than in the phase (§4.7); and the derivation

of the progenitor spectrum is the genuine frontier rather than a debt. Four stand out:

1. the sufficiency of the null-boundary characteristic data for the observed acoustic coherence (§4), argued from the structure of the characteristic initial-value problem, with the time-reversal driving equality established there as a Fourier theorem and the handover at the branch point fixing the phase and the amplitude the datum leaves open. *The end-to-end branch-point-to-recombination transfer is run* (§4.7): a two-arm line-of-sight Boltzmann integral, $\Delta_\ell(k) = \int S\, j_\ell(k(\eta_0 - \eta))\, d\eta$, carried on this cosmology's own background with the perturbations on the leaf congruence the framework assigns them, and validated on its control to 0.23% in the first-to-second height ratio in the configuration the arm's own numbers are taken in[P15R160P15R108]. On this arm it returns a comb of 298.0 against the sky's 298.4 and height ratios 2.264 and 2.298 against 2.217 and 2.277, with the likelihood still rejecting this arm's spectrum on shape; the diffusion-scale signature it carries is located by the joint fit as a Gaussian residual in $\ell$ that no tilt removes (§4). *What the transfer leaves is about one multipole at each peak, and the sky cannot resolve it*: the comb is right to a part in six hundred, the phase intercept sits inside the sky's own locating spread, and the fourth peak is 0.9 to 1.3 standard deviations out with most of that offset shared with the control. *What remains is not a position at all*: the difference between this arm and the control is a *contrast* about the acoustic envelope—not a phase and not an envelope—it carries three quarters of the likelihood's excess, and its channel is the balance between the monopole and the Doppler terms of the source, bracketed with opposite signs on both instrument paths, and the difference itself is made *between wavenumber and multipole*—in wavenumber the two arms' relative oscillations agree to the measuring statistic's own floor—and not by any one source term nor by the interference between two, the projection raising this arm's contrast on nearly everything it projects and increasingly with wavenumber, the mechanism being the kernel's own acceptance in wavenumber and stated as a law with nothing fitted in it (§4.7). *Two things about that excess are measured rather than left to the frontier*: the Doppler channel's coupling to the contrast is negative at every band on the arm's own spectrum and the share of the excess that channel carries is a minority on either envelope, growing with wavenumber and sign-indefinite at the lowest bands; and *the excess's own structure across wavenumber is a step at its lowest band rather than a trend*, the bands above it fitted by a constant, so the discriminating power of every shape comparison across wavenumber sits where the excess carries least structure. *The step survives three running envelopes differing in kind and sits at the second acoustic peak to about a per cent*—sharper than the window that resolves it—*with the lowest band running between a third and two thirds of the rest on every statistic that resolves it. Every candidate this comparison holds is evaluable against that step, and between them they carry a minority of it*: the window weighting steps the wrong way, the term mix the right way, the pair as actually composed about two fifths, and the projection width and the Doppler none. *And the step is very largely shared*—both arms step upward at the lowest band and this arm's is shallower by about a tenth and no more, *so the feature belongs to the acoustic physics the arms have in common and the excess's step is the part that fails to cancel. That part survives an estimator that is differential by construction*, the two spectra divided before any width is taken so the shared step is never formed—negative on all four bases, both pairings and both aggregations—*and it is an amplitude difference and not a phase one*, the split being exact rather than fitted, so the first acoustic cycle is weakened in this cosmology rather than displaced in it. *The shared part reduces to no single term of the source*, the removal losses summing to nearly twice the whole, though a step-weighted reading of the decomposition is led by the Doppler where an amplitude-weighted one is led by the monopole. *And scored on the differential estimator the candidate list accounts for none of the step*: one channel steps the opposite way to the arm and the other two are not resolved there, where against

the mostly-shared quantity they reached two thirds and two fifths—*and they cannot be resolved by the banded family of statistics*, the smallest decidable share standing at three fifths of the step against carried shares of about a third and a thirtieth, with every route to a lower floor taken and the phase-insensitive aggregation lowering scatter and signal by the same factor. *But that limit belongs to the family and not to this comparison*: the likelihood these spectra are confronted with bins at a thirty-fourth of a comb period rather than banding at seven tenths of one, and in its metric the two channels separate from each other at some twenty-five standard deviations, with every candidate departing the arm's way at that band. *What that metric measures is what the data can distinguish, and the data's own rejection of this arm lies elsewhere*: the lowest band carries under one per cent of the excess in fit and changes sign as smooth shape is absorbed, while four bands in the upper half carry some six sevenths of it on an exact decomposition—*so the excess's one prominent feature across wavenumber sits where the rejection is cheapest, and the rejection sits where no smooth shape finds anything. But the rejection there is not a level*: bin by bin it is modulated at the comb's own period, reached by none of twenty thousand pooled draws of the instrument's own covariance through the same statistic, with five sixths of it carried by the term that couples the two arms' difference to the control's residual—*so the rejection carries acoustic structure*, the period fixed to the range's own resolution and not finer. *And no banded reading can find it, which is a limit of representation and not attenuation*: one number per band at seven tenths of a period spacing has a Nyquist period of twice that, so eight of the eleven instruments this comparison has used cannot represent a comb-period feature at all, where the likelihood bins at a Nyquist period a twentieth of it and passes it whole. *And across those bands one candidate channel costs the arm's way*: the term mix delivers about four thirds of the arm's cost with the two patterns correlating at seven tenths and nearly half of it misplaced, where the window weighting's cost is *anti*-correlated and the pair as composed matches worse than the term mix alone—*so a total is not a match, and on this metric the two channels add rather than cancel. But whether that channel carries what the cost is made of is a question no instrument this comparison holds can settle, and that is established rather than left open*: against two thousand draws of the covariance the channel's own cost, the part of the excess it leaves unexplained and the part it misplaces all clear, *so every piece of the decomposition is acoustically modulated*—a knob moving the balance of two oscillating source terms producing an oscillating cost by construction, which is what a detection test establishes and is not what authorship needs. *And the accounting that would settle it closes exactly and still cannot attribute*: the least-squares share in the plane the comb's cosine and sine span *is the cosine of the phase offset identically*, so it is blind to the channel's amplitude—*the anti-correlated channel being credited with essentially all of the authorship for pointing $\pi$ away*—while the one scale that is not amplitude-blind fails by measurement and carries no statistical floor, the spectra's difference moving with the data only through two fitted amplitudes. *So the acoustic contrast excess is a measured discrepancy of known size, location, period and shape whose carrier is not attributable from here*, the degeneracy being exactly dimensional in that two channels span the plane; *and what attribution would take is a channel independent of that plane or an estimator using amplitudes where those amplitudes carry statistical content*, neither of which is a refinement of the instruments above. *And the stage at which the excess is made is located, and so now is the mechanism, which is where this sector's open work now sits*: the difference is not in the source—the same statistic reads 1.047 on the projected spectrum against 0.996 and 0.992 on the source, and in wavenumber this arm's relative oscillation is if anything the shallower of the two—*so it is made between wavenumber and multipole*, this arm retaining 1.054 times as much of its own source oscillation as the control does and increasingly with multipole[P15R186], *and not by the wavenumber sampling, the distance, the*

*visibility width before the kernel, the lensing or the binning*, each of which is excluded by direct substitution. *And the mechanism is the kernel's own acceptance in wavenumber, stated as a law rather than as an account*: for a standing oscillation the line-of-sight integral factorises with no approximation, so the comb reaches the transfer undamped and only the sum over $k$ at fixed $\ell$ can damp it, and what survives is that comb averaged over the window the sum carries—$A_\ell = |\int W_\ell e^{2ikr_s^*}\mathrm{d}k|/\int W_\ell \mathrm{d}k$, with no coefficient in it to fit, holding to within ten per cent of the measured amplitude on both arms at every band, and narrower here because this arm's visibility is the 14.6 per cent wider in $\chi$ while accumulating the same sound horizon to 0.08 per cent. *What that leaves is a question about the line-of-sight integral on this background rather than about the statistics above, and it is now answered far enough to be stated as one thing rather than two.* The retention is not an artefact of how the integral is carried: the acoustic phase is already expressed in the integral's own variable, the two statements of the sound horizon agreeing identically to a part in $10^8$, and every other assignment the construction makes has been settled by derivation—the plasma's scales, its chemistry and the perturbations on the leaf rate, the optical depth on the stacking rate as a photon-path observable, and no free time origin. *The retention is what the kernel's acceptance makes of a given projection distance, and so it turns on which rate that distance is built on*—the one assignment where what the construction specifies and what the sky prefers come apart. *Run both, the specified configuration moves neither the contrast coefficient nor the residual's alternation, while carrying the projection's distance on the leaf rate removes the contrast anomaly entirely, taking it onto the value the control itself returns and bringing the acoustic scale with it. And the comoving distance entering $j_\ell(k\chi)$ is settled, on one ground that is a theorem and a second that rests on it. The ground is the observer's own readout*: the constant-proper-time slice is exactly flat, so the comoving angular-diameter distance *is* the light-travel distance $D_C$, and on that slice the expansion law is exactly flat-ΛCDM's with radiation absent from it—a statement about the slice's induced three-metric, with no rate in it at all, which is why it can supply an identity a rule about rates cannot. *The rate rule reaches the same distance by naming $D_M$, and reaches the projection's argument only through that identification*: the lapse is the rate the observable expansion rides and the shift is the projection through which it is observed as distance, so a classifier over rates fixes which rate a length accumulates on and not which length a kernel takes. *And the optical depth's null-cone argument is that same rule applied a second time*, resting as it does on the optical depth being a photon-path observable of the same kind as the redshift and the distances. *So the support is one established theorem rather than a count*[P15R276]. *And a search for a second, independent support returns something better than a list of closed channels: it returns the reason there is none to find.* The one candidate that never mentions a slice is the beam side—the angular-diameter distance from the Jacobi focusing equation, whose inputs are the beam's own tidal matrix and an affine parameter. On (2) that computation goes through in closed form, the metric being exactly Einstein with $\mathcal{R} = 4\Lambda$ for arbitrary $M$, the areal radius an exact affine parameter on both radial null congruences, and the whole optical tidal matrix vanishing identically on them; and it returns the flat length. *But the rays that carry it carry a blueshift*—$1 + z$ of 0.099 and 0.008 between the same pair of fundamental worldlines, against the construction's own 1090.9, wrong in sign and so beyond the reach of any tolerance. *They are not the rays the construction's redshift is defined on, so the candidate is independent and is not a support.* The reason is structural: $D_M$'s two factors are read on two different objects, the comoving length a separation on the flat constant-$\tau$ slice and the redshift a ratio of the layer's own areal radii, and the slice is Riemannian and so carries no null congruence for a beam argument to be about. *A second support could therefore not be an invariant computed inside either presentation; it would have to be a statement about the reassignment that joins them*, which is not a

diffeomorphism[P15R281].

*Which settles what kind of claim the equality is, and it is not the kind a second support could strengthen.* Under the reassignment the photon congruence *is* the constant-$\tau$ family (§2.2), so the comoving length the kernel takes is a separation read on the constant-$\tau$ slice by what that congruence is, and not by an inference that something else might second or contradict. *The theorem supplies the slice's metric and the reassignment supplies the slice*; together they make the identification constitutive rather than evidential. That is a narrower position than three independent routes and a firmer one: the assignment the sector's verdict rests on cannot be seconded, and equally cannot be undercut by the absence of a second support. *Two readings that might have moved it are excluded rather than left open.* The construction does carry a second radial object—$r_0$, the layer's areal radius—but it is *already spent*: it sets the source's discrete spectrum where $D_C$ projects, the two differing by the stretch 2.774 the low-multipole floor is built on, and reading the angular part at face value would give $D_M = r_0$ at every redshift, which the comb excludes outright at $\pi r_0/r_s = 109$ against 303. *And the proper frame's line element genuinely is not the standard comoving form*—there the radial-to-angular ratio is the Hubble radius, $r/(\mathrm{d}r/\mathrm{d}\tilde{\tau}) = \alpha\tanh(3c\tilde{\tau}/2\alpha) = c/H$ identically—*but that does not move the kernel, because the standard form is claimed on the observer's constant-$\tilde{\tau}$ slice and not on that chart*[P15R229][P15R186][P15R216][P15R217][P15R214][P15R212][P15R213][P15R211][P15R206][P15R205][P15R203][P15R218][P15R228];

*So the comparison is not between a specified configuration and a preferred one: the specified one is the construction's, three times over, and the preferred one restores to the rate the single term this cosmology removes from it. What this leaves is a result about this cosmology rather than a defect in its machinery*—a disagreement with the microwave sky that is localised to the acoustic contrast, quantified, and in a direction no parameter reaches. *And the one step that could have replaced the identification with a derivation is not available on this construction, which is established rather than assumed.* $j_\ell(kD_C)$ is licensed by an *identification*—the flatness of the distance slice together with the exact recovery of the expansion law—and the alternative would be to expand a plane wave on the proper frame's own slice with the observer at a general point of it. *Carried out, that expansion contributes nothing and cannot be set up.* It contributes nothing because on a Euclidean slice the addition theorem about an off-centre observer is exact and the offset is a pure phase that cancels in $C_\ell$: no measure, no weight, no $\ell$-dependent factor, so the kernel's *form* is confirmed. *What it cannot do is be set up on the reassigned chart, and the reason is that the chart is the wrong presentation for it rather than that the construction lacks one.* There the cosmological surface is the 45° line with an $S^2$ of directionality rather than the sphere (§2.2), every epoch sitting on spheres about the branch point so that the recombination locus is at $r_0$ and not at $D_C$—short by exactly the stretch 2.774—while that chart's own null geodesics give $1+z = (1\pm v_o)/(1\pm v_e)$, on which the construction's locus is a *blueshift* and the radial area distance saturates[P15R229]. *Those are not the geodesics the redshift is read on, and a mode expanded against them is not a mode of the thing observed. And the presentation it belongs in is not simply the de Sitter one either, because what the expansion needs is three things no one of the construction's presentations supplies together.* It needs the cosmological layer as a sphere, so that there are harmonics to expand on; it needs the lap resolved from the branch point out to the cosmological seam, since that arc is where this sector's physics sits—the acoustic epoch falling in the first part of the 120° third that runs from $r = 0$ to the handover from deceleration to acceleration at $r = \alpha/\sqrt{3}$; and it needs the lift carried, the stretch of the bead's path across which the areal radius climbs through $r = 0$ while cosmic time does not advance. *The de Sitter presentation gives the first and buries the second*—the whole lap, seam to seam, is the hyperboloid's finite minimum equator, so the sector is compressed onto the throat—*and it is excluded a second time, numerically, by what its own layer would do to the projection.* On that

layer the comoving distance to last scattering is $\chi_{\rm rec} = D_C/\alpha = 2.6876 = 0.8555\pi$, so last scattering sits 85.6 per cent of the way to the antipode: *there is no small-$\chi$ limit available, so the layer's own hyperspherical kernel is not a correction to the flat one but a different answer*, mapping through $\alpha \sin\chi_{\rm rec} = 2286$ Mpc in place of $D_C = 1.4011 \times 10^4$ and placing the comb at 49.4 against the measured 302.9[P15R229]. *And the Hopf structure does not enter the projection either, which is a measured negative and not a silence*: the degree-$L$ harmonics on the $S^3$ are the $(L/2, L/2)$ of $SU(2)\times SU(2)$ and the Hopf circle lies in one factor, so the invariant content is the other's spin-$L/2$—*exactly the $S^2$ harmonics of degree $\ell = L/2$ for even $L$ and empty for odd.* That is a selection rule rather than a distance, and it would carry the lowest physical mode $L = 2$ to $\ell = 1$, the sky's own pure-gauge dipole, where the flat projection places it at 7.85; *so a comb of order unity is what the Hopf reading predicts, and the Hopf direction's role is the one §2.2 already gives it—traded into $\chi$ by the reassignment. The proper $(\tau, \chi)$ chart reaches the branch point but cannot carry the lift*, which collapses there onto a single segment at vanishing cosmic time. *And the presentation that does carry the whole bundle continuously across real $r$, lift included, is the bead flattened against arc length*—one curve, and so with no angular structure left to expand on. *And the object that holds all three is not a fourth presentation but the harmonics' own parameter*: complex conformal time along the bead, $\mathrm{d}\eta = \mathrm{d}\tilde\tau/r$, the variable the mode equation is already written in. The layer is a sphere of radius $|r(\tilde\tau)|$ at every point of the curve, so $(L, M)$ is defined everywhere; every leg has finite $\eta$-length, so nothing is compressed onto the throat; and the lift's $\eta$ is *purely imaginary*, a segment at right angles to the two Lorentzian legs. *In it the lap closes as an exact $1 : \sqrt{3} : 2$ right triangle*—collapse $c_0 B(\frac{1}{3}, \frac{1}{6})/4$ real, lift $c_0 B(\frac{1}{6}, \frac{1}{2})/2$ imaginary, expansion $c_0 B(\frac{1}{3}, \frac{1}{6})/2$ real, with $c_0 = 2/\sqrt{3}\,2^{1/3}$, *each leg taken whole*—the collapse horn from the far past to the comoving turnaround, the lift, and the expansion horn from the branch point outward without bound—the two ratios being identities by reflection, $2B(\frac{1}{6}, \frac{1}{2})/B(\frac{1}{3}, \frac{1}{6}) = \Gamma(\frac{1}{2})^2 \cdot 2/\Gamma(\frac{1}{3})\Gamma(\frac{2}{3}) = 2\sin(\pi/3) = \sqrt{3}$, so that the collapse and lift sum to the expansion's modulus with argument exactly $-\pi/3$. *The two numbers are the construction's own*: $\sqrt{3} = \alpha/r_N$, and $\pi/3$ is half the 120° third the lap is built from [9]. *The closure is on those three whole legs and on no sub-stretch of them*, and the point is worth making because the lap carries a second exact factor of two that pairs different objects. The seam-bounded pieces are $1.0311\alpha$ from the back seam to the turnaround and $2.3759\alpha$ from the branch point to the front seam—neither of them its horn's length, and not in the ratio 2. *In cosmic time, though, those same two pieces stand at $0.8780\alpha$ to $0.4390\alpha$, which is exactly* 2 : 1, by the identity $\operatorname{arccosh} 2 = 2\operatorname{arcsinh}(1/\sqrt{2}) = \ln(2+\sqrt{3})$. *So one factor of two is conformal and on the whole horns and the other is in cosmic time and on the seam-bounded pieces, and neither is to be read with the other's stretches*[P7R26]. *One further use of the word names a different object again*: the "collapse leg" of §4.1 is the *leaf* congruence's, and that one does terminate at the seam. *And the question answers in two halves. The angular label crosses untouched, exactly and without an integration*—the bead's angular block is $r^2\mathrm{d}\Omega^2$, the lift moves $r$ at fixed angles, and $(L, M)$ is a discrete label on a continuous path, so it cannot change along one. *The amplitude does not.* On the lift the mode equation is $u_{ss} = (k^2 + a_{ss}/a)u$ with $s$ real: both signs flip together, so nothing oscillates anywhere on it, and of the two-dimensional space of mode data exactly a one-dimensional subspace reaches $r = 0$ regular—*which is §4's one characteristic datum per mode, obtained here as a property of the curve rather than of the surface.* On that subspace the transmitted amplitude is a pure-number function of $k$ alone: $T(0) = 1$ exactly, since $u = a$ solves the lift equation identically at $k = 0$, and $T(k) \to 2^{7/3} k^2 e^{-k\, s_{\rm tot}}$ with $s_{\rm tot} = \Gamma(\frac{1}{6})\sqrt{\pi}/\Gamma(\frac{2}{3})\sqrt{3}\,2^{1/3} = 3.3387380$—*the lift's own conformal length, and parameter-free: no $\Lambda$, no $H_0$, no $\Omega_m$, no mass and no epoch anywhere in exponent or prefactor. So what the lift adds is an envelope and exactly no phase*, $\mathrm{Re}\,\Delta\eta = 0$ identically across it

rather than small. *A comb is a phase, so this is not the acoustic sector's missing piece*—and what it is instead is stated in §6: the construction's own isotropizing step, every $L > 0$ damped with not one adjustable quantity in it[P15R240]. *So the kernel's form rests on an expansion whose offset-observer behaviour is established, and its distance on an identification*, with the derivation that would replace the identification located rather than absent[P15R231P15R233P15R228P15R226].

2. the exact large-angle *depth* (§7): the flat-projection transfer of the discrete closed-$S^3$ source is built with a *genuine Boltzmann transfer*—the exact CMB temperature transfer $\Delta_\ell(k)$ (Sachs–Wolfe, early and late integrated Sachs–Wolfe, and Doppler, all common to CR and flat-ΛCDM) sampled at the discrete modes $k_L$, gate-validated (the continuum reproduces the reference $C_\ell$ to four figures) with the $r_0$ sensitivity measured rather than asserted—shape stable, depths drifting up to 15% at $\ell = 4$ (receipt: `verify_lowell_boltzmann.py`[P15R31]).

   It gives a *mild* deficit, $\approx$ 0.487, 0.435, 0.359 and 0.666 of the ΛCDM expectation at $\ell = 2, 3, 4, 5$—*a dip whose minimum falls at $\ell = 4$, not at the quadrupole*, the shape and the depth both reproduced by the programme's own photon hierarchy to three per cent—recovering by $\ell \approx 8$: the late integrated Sachs–Wolfe term is sourced above the floor and is therefore retained by the discrete source, leaving the deficit shallower than the bare Sachs–Wolfe starvation.

   The mechanism is robust—parameter-free, located at $\ell \lesssim 8$, bottoming at $\ell = 4$, with $\ell = 2$ and $\ell = 3$ starved *together* (the exact discrete measure and the Doppler term each leave the $\ell = 2$-to-$\ell = 3$ ratio near unity: receipts `verify_lowell_exact_measure.py`[P15R29], `verify_doppler_lowell.py`[P15R30]).

   Confronted with the sky the sector is a *wash*: CR's smooth $\sim$ 0.4–0.5 deficit sits above the sharp observed quadrupole ($\sim$ 0.2) and below the near-ΛCDM octopole ($\sim$ 0.6–1.0), matching neither, and the exact cosmic-variance likelihood returns $\Delta(-2 \ln L) = +1.58$ over $2 \le \ell \le 8$ and the same to the digit over $2 \le \ell \le 10$, both arms being at unity by $\ell = 8$ (range +0.2 to +3.4 across the octopole estimator), a small residual inside cosmic variance — *and it is a near-cancellation rather than a uniform cost*: per multipole the quadrupole rewards the construction by −2.5 and $\ell = 3$ to 5 penalise it by +1.0, +2.0 and +1.0, with nothing above[P15R171]. So the low multipoles are a genuine but *non-discriminating* sector, neither a parameter-free correspondence success nor a sharp falsification edge; the deficit's existence and location stand as a real parameter-free feature, but its depth places it consistent with ΛCDM within the lowest multipoles' cosmic variance.

   *The location is robust and the depth is not, and the difference between them is now measured rather than estimated.* On the background the distance data fix, the discrete scale moves by a quarter of a per cent and the lowest mode lands at $\ell \simeq 7.8$, so *where* the deficit sits is not a background choice. *The two Boltzmann treatments still differ by a factor of two at $\ell = 4$* — and that factor is not the geometry: moving the discrete scale alone buys three per cent against a gap of a hundred. *It is the late integrated Sachs–Wolfe term*, which at these multipoles is sourced above the discrete floor and is therefore retained by the sum, filling the deficit; the treatment that truncated its line-of-sight integration earlier was deeper at exactly the multipoles where the two disagreed. *Carried to the present instead, the two treatments agree*: on the exact transfer the deficits are 0.487, 0.435, 0.359, 0.666, 0.911, 0.983 and 0.998 of the flat expectation at $\ell = 2$ through 8 — *the exact transfer being the one carried here, with the hierarchy agreeing* to within three per cent at every multipole and one per cent above $\ell = 5$[P15R169P15R171]. *The factor of two was the missing term and not a difference between the constructions*, and the agreement is stable: it does not move under a doubled opacity grid, a halved late step count, or a further-extended continuum. *So the shape and the depth both cross-validate* — minimum

at $\ell = 4$, recovery by $\ell = 8$ — and the sector is ready to be scored against the measured multipoles with cosmic variance carried.

The exposed edges lie elsewhere—the acoustic contrast difference of the first item and the damping signature it carries;

3. the progenitor spectrum itself—$n_s$ and $A_s$ (§8)—inherited boundary data fixed by the progenitor collapse, the baryogenesis-analogue computation belonging to the matter sector [7, 8], with the target a consistency between the inherited composition and spectrum and the measured primordial abundances and tilt, not a derivation from the geometry alone—the most ambitious reach of the four, the one that would close the decomposition from both ends and give CR a derivation where the standard model keeps a measured datum. *And the target is now a number rather than a direction*: refitting both arms on this construction's own background prefers a tilt of 0.998 here against 0.954 on the standard one (§4.7)[P15R253], *so a derivation has to land near the bluer value and not near the one the standard model infers. And what supplies it is the progenitor's vacuum and not the collapse leg*: on the leg both the potential and the effective temperature are functions of the single variable $k\eta/\sqrt{3}$, so a handover at a fixed phase is exactly scale-free—the ratio of amplitudes at two wavenumbers is one—and a handover at a fixed time would contribute not a tilt but a running, red in both channels and growing as $k^2$, which no amplitude choice makes look like a constant. *At the branch point it vanishes quadratically*, so the leg is a wavenumber-independent amplitude and nothing else. *And what the substrate alone fixes is $n_s = 1$ exactly*: in conformal time the mode equation carries no length, so the curvature radius enters only as the prefactor that sets the amplitude and reaches the tilt for no value of it; and on the forced member the progenitor's mass is that same radius up to a pure number, so the two candidate scales are one and fall together. *But the fluctuations are not generated on the substrate*, and the interior they are generated in carries a scale the substrate does not. *Solved there*, the mode potential interpolates between the scale-free matter form and a radiation form that is not, so the generated spectrum *breaks* at the wavenumber the radiation content sets: its log-slope runs from $+0.30$ to $+1.98$ across the observed band, tending to the radiation value $+2$. *So it is a running and not a tilt*, by the same band test that closes the collapse leg — and *blue at every wavenumber*, where the measured value is red by two parts in a thousand[P15R183]. *The flattest slope goes as the radiation fraction*, so exact scale-invariance is the pure-dust limit and reaching red would need that fraction negative. *And the running is the interior's own and not the vacuum's, which is what the two readings of this interior share.* Measured on unit incoming amplitude, so that what comes out is the multiplier itself, the transfer's log-slope runs from $-0.866$ near the bottom of the observed band to $-0.010$ near its top: it is *neither scale-free nor a power law*, and what it adds to any spectrum's tilt is twice that spread. *The exact scale-invariance the vacuum shows is that same function's low-wavenumber limit*: below the break the log-slope tends to $-1$ and $kT(k)$ to a constant, and the vacuum's $k^3|T|^2$ is scale-invariant exactly where that slope is $-1$. *So the cancellation is the vacuum normalisation meeting the transfer's low-wavenumber power law, and it holds only where that power law does.* The split is an identity rather than an argument—the tilt is $\mathrm{d}\ln T/\mathrm{d}\ln k + 1$, the wavenumber dependence always the transfer's and the constant always the normalisation's. *And the transfer is the same function for every input*, linear to parts in $10^{14}$ and reproducing its slope to parts in $10^{12}$ across inputs including one that is not a power law[P15R185]. *So the question is not what the progenitor's vacuum gives but what this interior does to whatever enters it*, and the answer is that *no power-law progenitor spectrum emerges with a red, near-constant tilt*: an input that did would have to run by 1.71 to 1.72 across the band, opposite in sense to the transfer, the spread being the differencing convention rather than the measurement, and the figure

moving by two per cent over the range the arm's own distance mapping allows. *That is a requirement on the progenitor and not a result about it*, and it is stated as one. *And it is bounded, because the transfer's running is a function of one variable.* In the interior's own radial variable the potential carries no radiation fraction at all—the fraction and the wavenumber enter only through their product—so the log-slope is a universal function of that product and the break sits at a fixed value of it, which places the break wavenumber inversely with the radiation fraction exactly rather than approximately. *At the other end the potential is Coulombic*: the crunch's $1/x$ against the mode's $k^2$ is the repulsive zero-angular-momentum Coulomb wave equation, and the expansion of its irregular solution's amplitude carries the log-slope to zero as $1/k$. *So the excursion is exactly one, from $-1$ to 0; what the transfer can add to any tilt is at most two; and no band, however wide, can ask a progenitor for a running above it*—the observed band asking 86 per cent of that maximum, which is a structural statement and not an empirical one. *The limit is derived where the integration is not trusted and only checked where it is*: past the wavenumber at which the incoming mode would start inside the radiation era rather than outside the break the construction's premise fails, and the asymptote is compared with the solved transfer only below that wavenumber, where the two agree to better than two per cent. *And what forces the requirement is the radiation fraction.* A fraction that carries the whole observed band below the break—where the transfer is a clean power law and a power-law progenitor would come through untilted—does exist, below $10^{-3}$ on the looser of two stated criteria for where the break is and below $1.4 \times 10^{-4}$ on the stricter; but it lies fifty-five to nearly four hundred times under the determined value, so the requirement stands because the composition this construction determines elsewhere forces it. *Both runnings are then set by the same object*, the break, so a progenitor that cancels this one is not a tuning of an arbitrary function but a progenitor whose own spectrum breaks where the interior's does—a structural candidate, and nothing here shows that any progenitor does[P15R187]. *And the substrate's power-law formula does not apply off the substrate*: its coefficient is dimensionless, which is what makes it a power law, where the interior's carries a scale—a distinction invisible to a tilt read off a background fitted to the other arm. *The figure is provisional in one respect and it is stated*: it is measured on the range below the damping tail, which is where a tilt has most of its leverage. *One route to it is closed, and closing it narrows the target rather than the ambition.* The construction is recursive—each progenitor seeds the next cosmology—so a *stationary* recursion would make $A_s$ a fixed point and determine it. But entropy per baryon rises strictly at every branch-point crossing, the crossing being an erasure and the increment bounded below by Landauer's $k_B \ln 2$ independently of the network and of $\eta$ [19]: *a map that strictly increases a coordinate has no fixed point*, and independently the chain it generates is finite, so there is no asymptotic limit for a fixed point to be the limit of[P15R2]. *So the derivation must reach the head of a bounded genealogy rather than a stationary state*—the same terminus the inherited composition reaches, and the two are one question asked of two quantities;

4. the *likelihood-level model selection*: the parameter-free ratio comparison above is joined by the exact low-multipole cosmic-variance likelihood—each $\hat{C}_\ell$ a scaled $\chi^2_{2\ell+1}$ about the model $C_\ell$, the absolute baseline cancelling in the CR/ΛCDM ratio (`verify_lowell_likelihood_v2.py`[P15R] the proper instrument where cosmic variance dominates.

   With the genuine-Boltzmann depths ($\approx 0.49/0.44$ at $\ell = 2/3$, above) the sector is a *wash*: the low quadrupole still mildly rewards CR ($\Delta(-2\ln L) \approx -2.5$, its milder dip nearer the observed low power than ΛCDM's), the octopole and $\ell = 4$ mildly penalise it ($\approx +1.0$ and $+2.0$)—*so the sector's largest single-multipole penalty sits not at the octopole but at $\ell = 4$, the prediction's own minimum*—and the net over $2 \le \ell \le 10$ is $\Delta(-2\ln L) \approx +1.6$ (range $+0.2$ to $+3.4$ across the octopole estimator)—a small residual well inside cosmic variance.

A Monte-Carlo over $10^4$ skies puts the sector's *in-principle* discriminating power at only $\sim 3.4\sigma$ (the short multipole lever arm and cosmic variance), and the current measurement uncertainty swamps even that.

The low multipoles are therefore a genuine but *non-discriminating* test: CR and ΛCDM are statistically indistinguishable there, the mild deficit neither favouring CR nor refuting it. Decisiveness comes from outside the sector—the geometric expansion rate (4), confronted with the distance ladder (§2.4), as the standing discriminator, and the likelihood across the *full* spectrum, which is run and rejects this arm's spectrum on $TT$ shape (§4.7), its amplitude level not established by the time-reversal driving equality (§4).

A frontier of a different kind concerns the matter itself rather than the perturbation observables above: the *dynamics of matter crossing the branch point.*

Three facts about that crossing follow from the structure already established here.

It is *well posed*—the beginning is fixed at the branch point $r = 0$, and the crossing is taken on the *substrate*, whose curvature is finite there however the geometry's own invariants behave.

The bend $\rho = m'/4\pi r^2$ that carries the matter is accordingly finite, and its data on that null surface is characteristic rather than Cauchy: a characteristic crossing with no obstruction from the substrate's curvature. *The geometry's Kretschmann scalar does diverge at that locus*, as the abstract states and the circle paper computes [36]; what makes the crossing well posed is that the divergence belongs to the chart-borne geometry and not to the substrate the cut is taken on, and that the tortoise measure converges so the crossing carries no scale.

It *isotropizes and is scale-free*—the throat's de Sitter no-hair (§6) damps the anisotropic part of the crossing stress-energy to the isotropic monopole, and the degenerate approach (§8) imprints no thermal scale: the matter counterparts of the field no-hair and faithful transmission established above. *Where each of those acts is worth stating, since one of them cannot act at the crossing at all.*

Damping is a process and requires elapsed time, and the crossing has none: the real part of the complex cosmic time is frozen along the segment reaching it [35].

The no-hair damping therefore belongs to the *approach*—the throat region, over the collapse leg's own cosmic time—while the crossing carries whatever survives it unchanged, not because a mechanism protects it but because there is no interval in which anything could act. *The isotropy is earned on the approach and preserved by the crossing*, and the two halves are of different kinds: the first is a damping calculation with a rate, a timescale and a residual, the second follows from the zero-duration structure and takes no model input. It follows that *how much anisotropy survives is fixed by the approach against the leg's duration*—a property of the progenitor's collapse history rather than of the crossing. And it obeys a *structural transition law*: because the reassignment preserves the cosmic foliation and the density is leaf-carried and lapse-independent [7], the reassignment acts on the time-stacking—fixing the Λ-set rate (6)—and not on the leaf, so the isotropic density crosses as inherited progenitor content, which is the structural reason the inherited datum of §2.4 is inherited rather than produced; *and two conditions must be met at that crossing, which a concrete model has to satisfy separately: the reassignment requires a degenerating Killing field whose freed null direction can be re-founded as a clock, the joining of the legs requires a branch point, and these are independent—the forced Nariai member carries the degeneracy and is not a branch point, while $r = 0$ is the branch point and carries no Killing degeneracy [10], so a model furnishing one without the other furnishes no cosmogenesis. The first of the two is settled as a uniqueness, not merely an availability*: across the slicing family the reassignment that promotes a null direction to the fundamental congruence is *uniquely* the one selecting the forced Nariai member, every other vantage yielding a congruence that crosses the embedding's symmetry structure transversally and so reading as a localised black hole [6]. *A model therefore has no freedom in where the reassignment acts*, and misplacing it does not fail to give a cosmology by a little—it gives a black hole instead. The scoping of the two rates this law forces—the excursion-side local readout against

the observable geometric rate, their boundary—is drawn out in the cosmogenesis paper [19]. What remains open is the *detailed* worldline and field dynamics of the crossing for a concrete matter model [9, 7]—a depth left for a matter model to supply, and resting on a crossing shown well posed rather than merely postponed.

None of these unsettles the result obtained: the branch point's null geometry decomposes the primordial scalar spectrum into a substrate-determined structure and a progenitor-supplied content, and the branch point transmits that content rather than imprinting one of its own; the super-horizon transfer across the branch point is itself computed—$\Phi \to \Phi_i$ for every $k$, with the expanding leg inheriting $\frac{9}{10}\Phi_i$ scale-invariantly[P15R106]—and carrying the acoustic evolution from that datum through to recombination is the transfer of §4.7, which is run and returns the comparison recorded there. They are, rather, the map of where this cosmology goes next—what displaces the fourth peak, and the full oscillating medium downstream in the matter sector, the tensor sector's companion in the dynamics paper [10].

# 11 Discussion

What this paper determines is one thing seen from several sides. A single de Sitter geometry, its scale set by the cosmological constant alone (the maximally symmetric substrate's sole scale [17]) and read through a causal reassignment at the cosmogenesis branch point, *is* the observed cosmology: the flat-ΛCDM expansion history is its areal radius (§2.3); the acoustic scale is computed from its geometric rate, which the distance data and that angle fix together (§2.4); the primordial scalar spectrum is the characteristic data of its branch point, structure fixed by the substrate and content inherited across the branch point (§§4–8). The Einstein field equations are unchanged; what changes is the reading, and under it the quantities the standard model carries as independent inputs—the present matter density, the dark-energy fraction—are readings of one cosmic clock, not drivers of the expansion, so that the coincidence of $\Omega_m$ and $\Omega_\Lambda$ dissolves into our observing at a time of order the geometry's one timescale. There is a single seed, and the observed universe is what grows from it. And the one scale reaches down as well as out: the same cosmological constant that sets this expansion sets, for every mass $M$, the *Hubble–Eddington radius* [34] $r_{\rm HE} = (3GM/\Lambda c^2)^{1/3}$—the boundary within which local structure stays bound against the cosmic flow and beyond which it is carried off. The slicing geometry reads that boundary as the flat locus of the existent slice's own curvature, the local bend of the cut exactly cancelling the substrate's cosmological term [5], and the slicing operator reads it as the handover from sub-marginal bound orbits to the marginally-bound $E = 1$ congruence that *is* this cosmology [7]. So the substrate's single $\Lambda$ governs both the global expansion this paper computes and the local scale at which structure resists it—one constant read at two ranges—and the Hubble–Eddington radius, proposed as a local, per-structure test of $\Lambda$ from the largest bound structures [13], is an independent observational handle on the very constant the expansion history here measures. The two ranges are moreover the same length up to a pure number. The Hubble–Eddington radius—where the slicing surface's Gaussian curvature $K_G = 1/\alpha^2 - M/r^3$ vanishes, the local bend cancelling the substrate's cosmological curvature [5]—is $r_{\rm HE} = (M\alpha^2)^{1/3}$; while the expansion's own amplitude, $(2M\alpha^2)^{1/3}$, is exactly the areal radius at matter–$\Lambda$ equality ($\rho_m/\rho_\Lambda = \operatorname{csch}^2(3\tilde{\tau}/2\alpha)$, unity at $\sinh = 1$). The two therefore stand in the fixed ratio $2^{1/3}$ for every $M$: the scale at which a bound structure's gravity balances the expansion and the scale at which the expansion's matter balances its $\Lambda$ are one length read locally and cosmologically, the factor of two that of the Schwarzschild term against the surface curvature's $M/r^3$ and nothing else. [P15R35]

The comparison with the standard cosmological model is accordingly not *only* a contest of fits—though the geometric rate fits the joint acoustic-and-distance-ladder data at one $H_0$ (§2.4)—but also, and more deeply, a difference in what each must assume to do so. The standard model *assembles* the observed universe: a dark-energy component for the late-time acceleration,

an inflationary sector for the causal contact, flatness, coherence, and near-scale-invariance the background dynamics do not themselves supply, and further early-universe physics where the sound horizon and the expansion rate come into tension. Each part is a real solution to a real problem, and each was introduced to yield the feature it produces. *This cosmology adds no such part.* Causal contact, isotropy, flatness, phase coherence, and scale-free transmission are read off structure the construction already contains—the isotropizing throat (§6), the null boundary on which coherence *is* the characteristic-data condition (§4), the exactly Euclidean constant-time slice (§7), the scale-free crossing that transmits rather than imprints (§8)—each fixed by $\Lambda$ alone, with no parameter free to produce it. This is the distinction between a structure that *requires* a phenomenon and one that merely *permits* it through adjustable apparatus [12, 11], and it is the criterion by which competing frameworks are rightly weighed ahead of the measurement that will decide between them.

*The dark-energy component is the plainest case of the difference*: on this cosmology the acceleration is not a sector at all. Since $(\mathrm{d}r/\mathrm{d}\tilde{\tau})^2 = E^2 - f$ on the energy family, one has $\mathrm{d}^2r/\mathrm{d}\tilde{\tau}^2 = -f'/2 = rK_G$: *the comoving acceleration is the existent slice's own Gaussian curvature*, up to the positive factor $r$.[P3R12] So the expansion decelerates while the slice is negatively curved, turns over exactly where it is flat—the Hubble–Eddington radius, $\rho_m = 2\rho_\Lambda$, at $\sinh(3c\tilde{\tau}/2\alpha) = 1/\sqrt{2}$—and accelerates thereafter, with nothing added to make it do so. *And that locus is a measurement, not only a description*: the turnover sits where the areal radius reaches the front seam, $r = r_N$, so the acceleration-onset redshift is the offset parameter itself minus one,

$$z_{\mathrm{acc}} \;=\; x_0 - 1 \;=\; 0.6648 \pm 0.0467, \tag{17}$$

against $\left(2\Omega_\Lambda/\Omega_m\right)^{1/3} - 1 = 0.632$ on Planck's parameters—the two agreeing to well within the quoted interval, and with no fit performed here. *That interval is carried as an input throughout rather than derived here*, so it is quoted as the stated uncertainty on the offset parameter and not converted into a significance. *What makes the equation worth stating is what it does not contain*: $z_{\mathrm{acc}}$ is a redshift, measured calibration-free, so it is untouched by the distance-ladder tension, and it fixes $x_0$ without $H_0$, without a ladder and without the microwave background. The single dimensionful scale is where any disagreement must then live, and the age follows from the same parameter in closed form, $t_0 = (2r_N/\sqrt{3}\,c)\,\mathrm{asinh}(x_0^{3/2}/\sqrt{2})$. The late-time acceleration is thus read off the same structure that fixes the local boundary of a bound system.

*And the growth sector, written on this rate, loses its parameters entirely — which turns a known numerical curiosity into a statement about the epoch.* Substituting $H = (c/\alpha)\coth u$ and the Friedmann readout's $4\pi G\rho = \frac{3}{2}(c/\alpha)^2\,\mathrm{csch}^2 u$ into the growth equation and changing to $u$, every factor of $(c/\alpha)^2$ divides out:

$$3D'' + 4\coth(u)\,D' - 2\,\mathrm{csch}^2(u)\,D = 0. \tag{18}$$

*The scale does not merely fail to appear—it cancels*, because at Nariai the source is fixed by $\Lambda$ and the time variable carries $\alpha$ too. *The decaying mode is the rate itself*, $D_- = \coth u = H\alpha/c$, and reduction of order gives $D_+ = \coth u \int_0^u \sinh^{2/3} v\,\mathrm{sech}^2 v\,\mathrm{d}v$. *And the matter fraction is a clock reading*: $\cosh^2 u = 1/\Omega_m$ exactly, so $\Omega_m = \mathrm{sech}^2 u$.

*On these variables the normalisation of the standard growth factor — $J(\Omega_m) = \int_0^\infty (1+z)\,\mathrm{d}z/[\Omega_m(1+z)^3 + 1 - \Omega_m]^{3/2}$, known to cross unity once near the concordance density* [1] *— becomes $J = \frac{2}{3}\cosh^2 u_0\,(D_+/a)|_0$, whose matter-domination limit $D_+/a \to \frac{3}{5}$ returns the analytic endpoint $J \to \frac{2}{5}$.* Differentiating the condition $J = 1$ gives an exact identity between the two sides' derivatives, $\propto \frac{9}{2}\Omega_m - 2$, and integrating it from the common origin leaves

$$\int_0^{u_0} a(v)\,\Omega_m(v)\big[\Omega_m(v) - \tfrac{2}{3}\big]\,\mathrm{d}v \;=\; 0, \tag{19}$$

*the growth-weighted history of $\Omega_m$ balanced about $\frac{2}{3}$*[P15R12].

*The pivot is the deceleration-to-acceleration turnover*: $\Omega_m = \frac{2}{3}$ is $\rho_m/\rho_\Lambda = 2$ exactly. (19) is solved at $\Omega_m = 0.31516$, so *the density at which that normalisation equals unity is fixed by an equation with nothing free in it.*

*Two scope statements belong with that, and both narrow it. The turnover is not this construction's property*: any flat matter-and-$\Lambda$ rate has it, and what is this construction's is the *identification* of that turnover with a geometric locus — the areal radius there being $\alpha/\sqrt{3} = r_N$, the Nariai radius and the merged double root (above). *So the balance pivots on the turnover, and the geometry is what makes the turnover a place.*

*And the rate used is the stacking one, which is the right rate for the object but not for this construction's own growth.* The published normalisation is a matter-and-$\Lambda$ integral, and the stacking rate written in the fitted pair *is* that rate, which is why the derivation reproduces it. *The rate rule assigns perturbations to the leaf* (§2.2), and on the leaf rate the same integral returns 0.99940 rather than unity. *So what is derived here is the root of the standard normalisation*, and what remains contingent is that the present epoch lies near it.

*What that figure does* ***not*** *measure is how far the root moves, and saying so matters because it invites that reading.* Continuing the closed form onto the leaf rate is a closed form applied to a rate it does not solve. *Solved directly, the root moves the other way and by less* — the leaf root sitting at $u_0 = 1.180039$ against the stacking 1.180309, a shift of 1.1% of the gap to the measured epoch and *away* from it, taking the offset from 0.704 to $0.713\sigma^{\mathsf{P15R13}}$. *So the root's two-per-cent offset from the measured epoch—*$x_0 = 1.6315$ *at the leaf root against the measured* 1.6648*—is not an artefact of the rate*: it stands, and wants a mechanism.

The transmission dichotomy (§8) is where the distinction between requiring and permitting is sharpest, because there the two frameworks make opposite structural commitments about one object. A non-degenerate horizon's exponential approach imprints a scale-invariant spectrum—the very mechanism by which inflation produces near-scale-invariance—while the degenerate Nariai seam carries no scale and the branch point, where the modes cross, accumulates no phase at all, so the progenitor spectrum is transmitted unaltered. Inflation accommodates the observed tilt because a freely specified potential can be tuned to it; reading that fit as a confirmation mistakes a model with enough adjustable structure to match for one that required what was found (§9). This cosmology cannot tune: the finite tortoise coordinate at the branch point *requires* that the crossing not imprint a scale, so the framework has no inflationary scale-invariant attractor, no consistency relation, and no substrate-sourced primordial $B$-modes. The amplitude and tilt it does not derive it inherits, exactly as flat $\Lambda$CDM inherits the baryon-to-photon ratio from a baryogenesis it does not model; on that single datum, of the same kind and count the standard model already carries, the two stand level. A framework is credited for requiring the structure it explains, not faulted for measuring the boundary data it does not—and the asymmetry falls on everything else.

The verdict this licenses must be drawn on the right axis, and confined to it. On the axis of theory-choice—the requiring rather than the permitting of the phenomena, the consolidation of many results under one structure, the absence of a device added per problem—the comparison is decidable from the structural record, and it falls to this cosmology [12]. On the axis of data-discrimination the picture is split: the geometric rate fits the acoustic scale and the baryon-acoustic distance ladder together at one $H_0$ (§2.4), leaving the discrepancy with the local ladder where the standard model leaves it—while on the light elements and the low multipoles the two frameworks are even. *On the microwave-background peak structure they are not*: the transfer's likelihood comparison rejects this arm's spectrum on TT shape (§4.7), and that is the sharpest datum against the construction that the corpus holds. So the standing is stated as it stands: favoured on the structure, level with the standard model on the light elements and the low multipoles, and carrying a spectrum computed from a background fitted to the distances alone—which reproduces the comb to a part in six hundred and is rejected on shape. The data axis is not even, and where it is uneven it runs against the construction, which is the direction a

framework that has staked falsifiable commitments—no scale-invariant attractor *reaching the observer*, no consistency relation, no substrate $B$-modes—must be prepared for, precisely where the standard model keeps its freedom[P15R83].

The low-multipole deficit is a mild, non-discriminating feature (§§7, 10). The flat projection of the discrete closed-$S^3$ source gives a parameter-free deficit below $\ell \approx 8$—real, and set by $\Lambda$ through $r_0$ and $D_C$—bottoming at $\ell = 4$, at $\approx 0.49$, 0.44, 0.36 and 0.67 of the $\Lambda$CDM expectation at $\ell = 2, 3, 4, 5$. At that depth CR's smooth deficit matches neither the sharp observed quadrupole-only dip nor the near-$\Lambda$CDM octopole, and under the exact cosmic-variance likelihood the two frameworks are statistically indistinguishable at low $\ell$ ($\Delta(-2 \ln L) \approx +1.6$ over $2 \leq \ell \leq 10$, inside cosmic variance). So the sector neither confirms nor refutes: it is a genuine parameter-free prediction whose depth places it consistent with $\Lambda$CDM within the lowest multipoles' cosmic variance. The framework's exposed edges are the acoustic peak structure, where the transfer reproduces the comb and differs from the sky in the acoustic contrast (§4.7), and the damping-*scale* signature (§4), a computed, non-reabsorbable effect whose size the handover fixes and whose sign the diffusion integral's endpoint does. Its sharpest test is the geometric expansion rate itself, carried to the distance ladder and confronted with the DESI DR2 baryon-acoustic dataset at $\chi^2/\mathrm{dof} \simeq 1$ (§2.4): the rate that carries the full ladder is the rate the spectrum is computed on, so the two are one commitment.

The framework's standing includes what the standard model has genuinely built and this cosmology has not. Dark matter is inferred from multiple independent, cross-consistent lines—rotation curves, gravitational lensing, the merger offsets, the acoustic-peak structure, the growth of structure—and this cosmology carries the matter density as the bend of the spatial cut without yet removing the component or answering its particle question; that column stands at full weight. Standard nucleosynthesis derives the light-element abundances from one measured parameter, a genuine achievement this cosmology meets on the same footing rather than exceeds: the collapse's cooling leg is a standard nucleosynthesis run on one inherited datum (§2.4; the cosmogenesis paper [19]), producing helium-4 and deuterium at their observed values—the freeze-out temperature the data demands being the bottleneck the standard rate delivers—and, because the run is standard, carrying lithium-7's standard threefold over-prediction, the lithium problem shared rather than solved. On the light elements the two frameworks stand even: the same two successes, the same one problem, one progenitor handover fitting the pattern together, and this cosmology's open edge the last-percent precision against the measured abundances. And the framework is young: the *detailed* dynamics of matter crossing the branch point (its crossing shown well posed and structurally governed, §10), the derivation—as against the measurement—of the inherited data, and the Standard-Model matter *content*—the gauge representations and the mass spectrum, which the framework does not supply, its geometry shown to force only the discrete flavour skeleton [20]—remain to be built (§10; [9]). These are open frontiers, not defects: a theory is not marked down for what it has not built, only for a claim it makes that fails. They are named plainly here because an economy that favours this cosmology on the structure it carries would be worth nothing bought by concealing the structure it does not.

## 12 Conclusion

So, plainly, what this paper claims and what it does not. It claims that the observed expansion history, the computation of the acoustic scale, and the structure of the primordial scalar spectrum are consequences of a single de Sitter geometry read through the causal reassignment its own horizon geometry selects—one seed, grown, the field equations unchanged—and that on the economy of assumption, the criterion by which theory-choice is rightly made ahead of the deciding measurement, this reading is favoured over the assembled standard picture. On the data axis it claims a more limited thing: that the axis is split—the expansion history fitting

the distance ladder, which fixes $H_0$ with the acoustic angle, and the abundances confirmed, while the acoustic peak structure, computed from that same background and scored by the full-spectrum likelihood, counts against it—with that structure and the damping-tail signature the edges still to be settled (the low multipoles a mild non-discriminating deficit consistent with ΛCDM within cosmic variance). It does not claim to be a complete cosmology: its matter sector, the derivation of its progenitor spectrum, and its fermion content remain to be built. It claims to have shown that the universe we observe is the reading of one geometry, and to have staked the falsifiable commitments by which that claim can be tried.

A final word on the register, because it is the thesis rather than an aside. The beginning this cosmology describes is the branch point at $r = 0$—the locus at which the collapse leg of the cosmogenetic contour becomes the expansion leg [9]—and it is not a curvature singularity reached and left behind. *The curvature there is set by the substrate's own scale.* The throat is $\alpha$, the defining constant of the maximally symmetric manifold the cut is taken on, and the divergence of the invariants computed in the areal chart is that chart degenerating, exactly as a polar chart degenerates at a pole—not a curvature scale of the geometry [19, 17]. Two further facts stand with it: no finite cosmic-time layer ever reaches the locus, and the continuation through it carries no scale, since $f \to -2M/r$ diverges so that the tortoise coordinate $r_* = \int \mathrm{d}r/f$ *converges* and what is transmitted is fixed by the analytic structure rather than by a length. *The branch point is crossed by continuation, not traversed by a worldline.* The expansion is not a space ejected from a point but the reading, on the near side of that branch point, of a spatial layer still advancing, the branch point itself a collapse that in the frame it bounds is never complete. The distinction the whole programme turns on—between what has finished happening and what is still underway—is thus the distinction the cosmology is built from: the universe is not the collapsed record of a beginning but the still-evolving layer that record represents, read here as one de Sitter geometry.

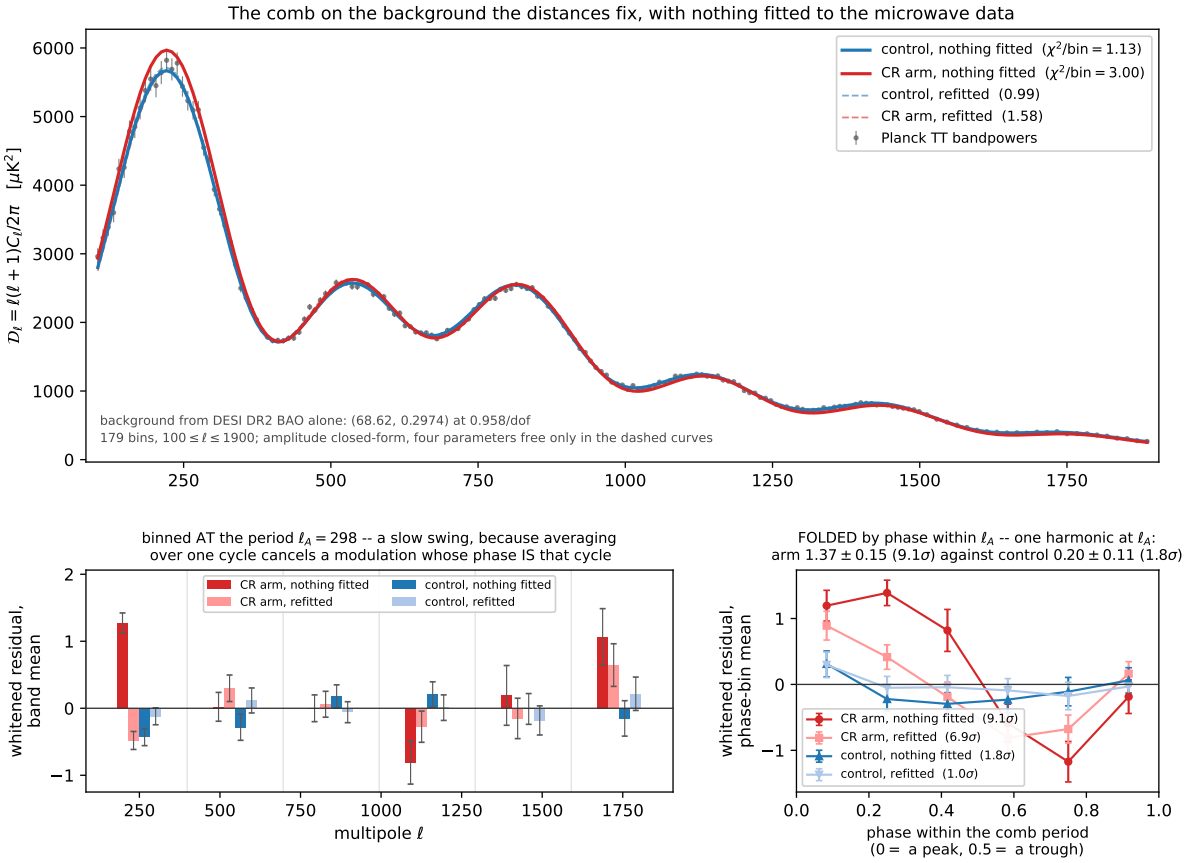



**Figure 3: The other confrontation: the comb computed on the background the distance data fix, with nothing fitted to the microwave data.** *Above*, both arms in $D_\ell$ against the Planck temperature bandpowers, on a background recovered from DESI DR2 baryon-acoustic data alone—$(68.62, 0.2974)$ at 0.958 per degree of freedom[P15R163P15R236]—with the overall amplitude in closed form and no other number taken from the spectrum; the dashed curves are the refit minima of Fig. 2 for comparison on the same bins. Over 179 bins on $100 \le \ell \le 1900$ the control stands at 1.13 per bin and this arm at 3.00, a factor 2.66, against 0.99 and 1.58 when four parameters are freed in each arm. *The peaks are the thing to look at*: positions and spacing agree while the extrema are overshot, which is the rejection in the form a reader can see. *Below left*, the whitened residual averaged in bins one comb period wide. It shows a slow swing changing sign twice across the range and no alternation—and it cannot show one, because averaging over a full cycle of a modulation at that period returns that modulation's own mean. The panel is drawn because the operation is the one a reader reaches for, and what it does show is a true and separate statement about the residual's long-wavelength shape. *Below right*, the measurement: the same residual folded by phase within the comb period, anchored on this arm's own first peak. One harmonic at $\ell_A$ has amplitude $1.37 \pm 0.15$ for the unfitted arm—$9.1\sigma$—against the control's $0.20 \pm 0.11$, consistent with none; freeing the four parameters halves it to $0.81 \pm 0.12$ and leaves $6.9\sigma$. *So the rejection is shaped at the acoustic period, the shape survives the refit, and the free parameters absorb part of it without reaching its cause.*[P15R279] *The fold is taken at $\ell_A$ by construction; where the residual's period actually sits is a separate question and is treated in §4.7.*

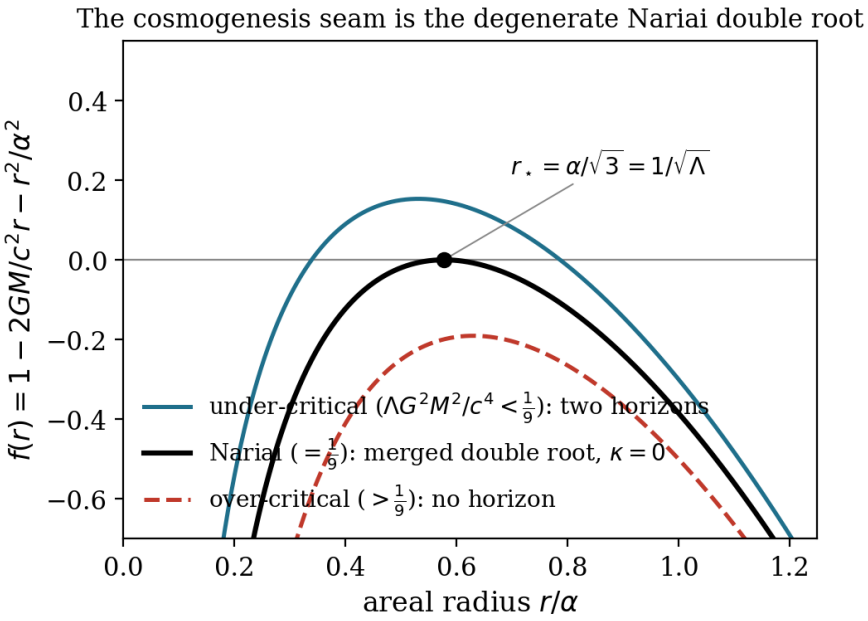



**Figure 4:** The Schwarzschild–de Sitter metric function $f(r) = 1 - 2GM/c^2r - r^2/\alpha^2$ across the three regimes. Under-critical masses ($\Lambda G^2M^2/c^4 < \frac{1}{9}$) give two distinct positive horizons; over-critical masses give none. The cosmogenesis branch point is carried on the *Nariai* member ($= \frac{1}{9}$), where the two horizons merge into a single *degenerate* double root at $r_N = \alpha/\sqrt{3} = 1/\sqrt{\Lambda}$ with $f(r_N) = f'(r_N) = 0$ and hence vanishing surface gravity $\kappa = 0$. The degeneracy of this root is what the throat geometry (Proposition 4) and the transmission dichotomy (§8) turn on.

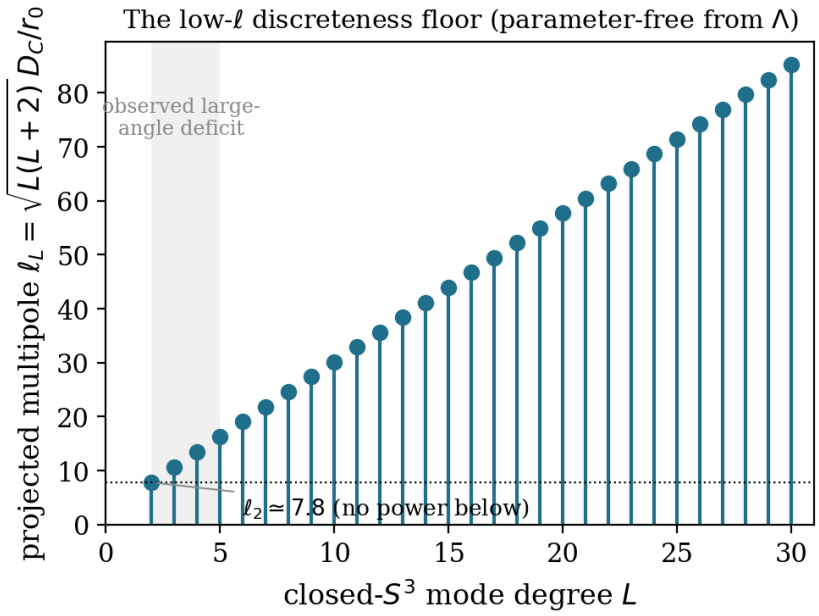



**Figure 5:** The low-multipole discreteness floor. The closed-$S^3$ source spectrum is discrete ($k_L = \sqrt{L(L+2)}/r_0$, lowest physical mode $L = 2$), but the flat distance slicing (Proposition 5) projects it through the *flat* transfer $j_\ell(k_L D_C)$, not the hyperspherical transfer of a literal closed universe. With $D_C \approx 1.4011 \times 10^4$ Mpc and $r_0 \approx 5051$ Mpc fixed parameter-free by $\Lambda$, the stretch $D_C/r_0 \approx 2.774$ carries the lowest mode to $\ell_2 \approx 7.8$, with no source below it. The full flat-projection transfer turns this into a low-multipole power deficit ($\ell \lesssim 8$); on a genuine Boltzmann transfer (the late integrated Sachs–Wolfe sourced above the floor and so retained by the discrete source) it sits at 0.487, 0.435, 0.359, 0.666 of the flat-$\Lambda$CDM expectation at $\ell = 2, 3, 4, 5$, recovering by $\ell \approx 8$—*a deficit whose minimum falls at* $\ell = 4$, so that the quadrupole and octopole are its shoulders and not its floor. Those figures are read through the exact transfer, and the photon hierarchy built for this programme (§9.1), carried to the present, agrees with them to three per cent at every multipole (§10): *the shape and the depth are both cross-validated.* This is a mild deficit, non-discriminating against the observed large-angle power within the lowest multipoles' cosmic variance (shaded). *What the branch-point filter does to this prediction at these multipoles—and what it does not do—is taken up in the text below rather than here.*

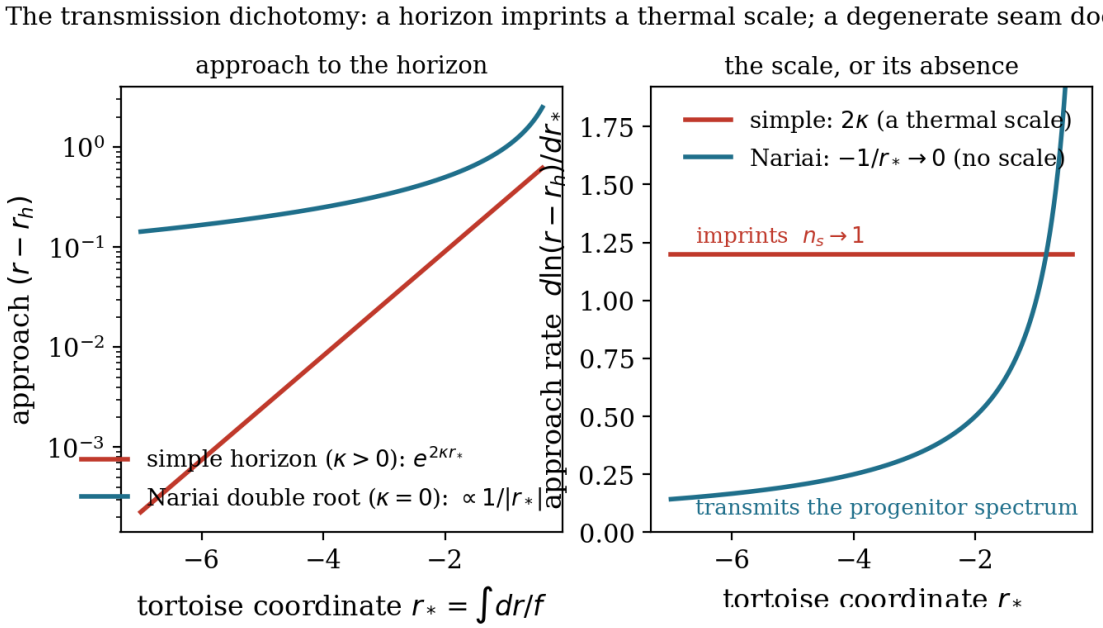



**Figure 6:** The transmission dichotomy (Proposition 6). *Left*: the approach to the horizon in the tortoise coordinate $r_* = \int dr/f$, on a logarithmic vertical axis. At a non-degenerate horizon ($\kappa > 0$) the approach is exponential, $(r - r_h) \sim e^{2\kappa r_*}$ (a straight line); at the Nariai double root ($\kappa = 0$) it is power-law, $(r - r_N) \sim 1/|r_*|$. *Right*: the logarithmic approach rate $d\ln(r - r_h)/dr_*$ *is* the local scale. The non-degenerate horizon carries a fixed thermal scale $2\kappa$—the mechanism by which a de Sitter horizon imprints a scale-invariant spectrum ($n_s \to 1$); at the Nariai double root the rate falls to zero, carrying *no* scale. The branch point, where the modes cross, is a third case: $f$ diverges there, the tortoise coordinate is finite, and the crossing accumulates no phase at all (Proposition 7).

# Appendix R Computational Receipts

Each computation marked R in the text is verified by a runnable receipt in the `receipts/` directory that ships with this work; status OK = verified (traced, run, output matches the claim). This appendix is generated from the receipt index, not maintained by hand.

**P15R1** `P15_the_sky_phase_fit_and_its_uncertainty` [OK]
*sec:intro* (FITS the sky's acoustic phase from the peak positions and PROPAGATES its uncertainty: $\{}ell_n=\{}ell_A(n+\{}phi/\{}pi)$ on 220.4, 537.7, 817.3 returns $\{}phi/\{}pi=-0.2405$ against the paper's $-0.2404$, and the control's 0.0043 offset is seven parts in a thousand of the 0.615 disagreement. The quoted $\{}sigma\{}simeq0.008$ is the TWO-multipole value; one multipole gives 0.0046 and twice the separation – so the paper's figure is the CONSERVATIVE end, and the sentence now says which assumption it carries). *Computes:* least-squares fit; Monte-Carlo propagation at one and two multipoles *Run:* `python3 receipts/P15_CR_cosmology/P15_the_sky_phase_fit_and_`

**P15R2** `P15_the_fixed_point_route_is_closed_by_the_monotone` [OK]
*'PO-31' (register), 'THE_WISDOM_LEDGER' r1942* (the one route the corpus ever named for DERIVING $A_s$ – a fixed point of a stationary recursion – is CLOSED by 'PO-40''s monotone: entropy per baryon rises strictly at every crossing, bounded below unconditionally by Landauer at $k_B\{}ln 2$, and a map that strictly increases a coordinate has NO fixed point; independently the chain is finite, so there is no asymptotic limit for a fixed point to be the limit OF). *Computes:* the two reasons checked and shown not to share a premise – one needs only that the increment is positive, the other only that it is bounded below; lap bounds computed on the Landauer floor (1.7e10) and on the present-day increment (1.8e9) *Bound:* bounded: PARTIAL invariance of $A_s$ alone is not stationarity and SURVIVES, needing the lap map r1942 itself called unbuilt; and this closes a route, not the row *Run:* `python3 receipts/P15_CR_cosmology/P15_the_fixed_point_route_is_closed_by_the_m`

**P15R3** `P15_the_tilt_is_the_wrong_absorber` [OK]
*'PO-24'; 'sec:refit-bound'* (the diffusion signature's **local** mimicked tilt is $\{}mathrm{d}\{}ln(\{}text{ratio}) 2(r^{2}-1)(\{}ell/\{}ell_D)^{2}$, which runs $-0.0002\{}to-1.31$ across plik_lite's own $\{}ell=30$–$2508$ — **a factor of 6989, exactly $(\{}ell_{\{}max}/\{}ell_{\{}min})^{2}$** — so absorption by a SINGLE tilt is necessarily window-local and the fitted $\{}Delta n_s$ is a range-weighted average of a quantity nowhere equal to it). *Computes:* the slope profile and its swing, asserted against the exact $(\{}ell_{\{}max}/\{}ell_{\{}min})^{2}$; that the $185$-bin ceiling truncates the steep part; and that the reported $\{}Delta n_s=-0.0304$ equals the true slope at $\{}ell=381$ alone; CONTROL at $r=1$ vanishes identically at every $\{}ell$ *Bound:* bounded: this shows a constant tilt is the wrong absorber and the displacement is not a stable parameter shift. **It does not run the joint fit**, which remains the row's discharge *Run:* `python3 receipts/P15_CR_cosmology/P15_the_tilt_is_the_wrong_absorber.py`

**P15R4** `P15_the_joint_fit_leaves_the_gaussian_and_the_bound_was_chi2_not_sigma` [OK]
*'PO-24'; 'sec:diffusion-scale', continuing 'C62' and 'r6409'* (**THE JOINT FIT RUN: THE RESIDUAL IS THE PREDICTED GAUSSIAN TO A CORRELATION OF $1.000000$, AND IT SURVIVES LARGER THAN THE BOUND IT WAS TO BE MEASURED AGAINST.** 'r6409' predicted a Gaussian-shaped residual no tilt removes; fitting $A$ and $\{}Delta n_s$ jointly against the signature-carrying spectrum on plik_lite's own bandpowers and covariance returns exactly that — the residual correlates with the analytic $S(\{}ell)-A(\{}ell/1000)^{\{}Delta n}$ at $1.000000$. It survives at **$1.497$ $\{}chi^2$/bin** over 'P15''s 185-bin ceiling window ($\{}ell=100$–$1758$, 171 covered bins). **AND THE BOUND WAS QUOTED IN THE WRONG UNITS**: 'C62' says $0.26$ **$\{}chi^2$ per bin** and converts to $\{}sigma$ by a square root in its own check; 'r6409' restated it as "$0.26$ sigma per bin" and the 'r6521' order inherited that — the looser bound quoted as the tighter, since $0.26$ $\{}chi^2$/bin is $0.510$ $\{}sigma$/bin. **AND IT IS THE MORE FAVOURABLE OF 'C62''s TWO**, which says in terms that the instrument route ($0.92$) "is the more faithful of the two and it is the LARGER, so the wash is the weaker claim and is asserted at the weaker bound". **AND THE DISPLACEMENT IS NOT A NUMBER, NOW MEASURED**: $\{}Delta n_s = -0.083$ over $\{}ell=100$–$1760$, $-0.521$ over $1400$–$2508$ — a factor of $6.3$ across windows of the same data, with absorption rising monotonically as the window narrows upward ($69.8\{}%\{}to97.6\{}%$). **In the likelihood's own units** the signature costs $+357.3$ $\{}chi^2$ with amplitude alone and $+84.5$ over 171 bins after the tilt is refitted: the fit **improves** the comparison by $272.8$,

absorbing $76\{}%$, and does not remove it.). *Computes:* C62's unit asserted from its own text and its own 'math.sqrt' conversion; r6409's restatement asserted; C62's two-route sentence asserted after stripping comment markers; **two calibrations — a known injected tilt recovered to $10^{-11}$, AND the SCALE, which the first is blind to because a tilt-recovery test is unit-free**; the pipeline asserted to reproduce 'chi2_of' on the same spectrum and window; the residual's correlation with the analytic shape; the residual asserted larger than both C62 routes; the displacement's spread across windows; monotone absorption; the sky-cost split; the red/blue sign difference between the two fits; and the $r=1$ control returning $\{}Delta n_s\{}sim10^{-14}$ and $\{}chi^2\{}sim10^{-22}$ *Bound:* **scope: what a tilt ABSORBS, not whether CR fits the sky.** TT only, no polarisation, no lensing, no $\{}ell¡100$ from this baseline; two parameters free, so 'C62''s scoping stands unchanged — an upper bound on what survives in shape, and no bound at all on the tilt, since $\{}omega_b$ moves $r_D$ itself. **$n_s$ is INHERITED boundary data here**, so a displacement is a different INPUT and not a conflict; what would be a tension is a constraint on $n_s$ that does not run through the damping tail, which this does not supply. The baseline is the instrument's control arm, unlensed and ceiling-limited where the sky is neither — 'C62''s own stated reason its routes differ by about three *Run:* `python3 receipts/P15_CR_cosmology/P15_the_joint_fit_leaves_the_gaussian_and_the_bound_was_chi2_not_sigma.py`

**P15R5** `P15_the_potential_at_the_handover_is_not_free` [OK]
*'PO-13'; 'sec:transmission', and 'C21'/'C19'* (the potential at the handover is **not free and not primordial**: the super-horizon join is conservation of $\{}mathcal{R}$, so $\{}Phi_{\{}rm exp}/\{}Phi_{\{}rm coll}=(3/2)/(5/3)=9/10$ exactly and $\{}Phi$ is constant on the matter-dominated expanding leg — **which 'P15' already states and receipts, while the instrument sets the potential to its primordial value instead**). *Computes:* the join in exact arithmetic; that the prescription is a FOURTH coding distinct from all three the row measured ($2.43$, $5.14$, dead comb); that $\{}Theta$ and $\{}Psi$ are then read at ONE locus, which is the row's own discharge criterion; and the omission's size *Bound:* bounded hard: **the omission is $10/9$, not $2.4$, and being scale-invariant it moves an amplitude and cannot move a ratio** — $P_1/P_2$ and $\{}ell_1/\{}ell_A$ are untouched, so this is NOT the cure and must not be read as one. The spectrum it yields is an instrument run, not made here *Run:* `python3 receipts/P15_CR_cosmology/P15_the_potential_at_the_handover_is_not_free.py`

**P15R6** `P15_the_one_fitted_number_moves_the_scale_and_not_the_peak` [OK]
*'PO-13'; 'sec:refit-bound', and the instrument's own 'LATARG' comment* (**THE ONE FITTED NUMBER MOVES THE ACOUSTIC SCALE AND DOES NOT MOVE THE FIRST PEAK, SO THE DEFICIT IS NOT AN ARTEFACT OF WHERE THE PIN WAS PUT.** 'ACOUSTIC_two_arm.py' exposed 'LATARG' at 'r2441' expressly to ask *"GIVEN that $\{}ell_A$ is fitted, is the peak deficit an artefact of where the pin was put?"* and **the question had never been run**. Scanned over a factor of $3.6$ in $z_{\{}rm onset}$ on the converged grid: $\{}ell_A$ runs $360.6\{}to239.3$ ($51\{}%$) while $\{}ell_1$ sits at $206$–$210$ ($2\{}%$). **The deficit is $\{}ell_1=206$ against the sky's $220.6$ at every onset.** **And the trade is then an IDENTITY**: pinning $\{}ell_A$ to the measured $301.6$ costs $6.59\{}%$ on $\{}ell_1/\{}ell_A$, pinning $\{}ell_1/\{}ell_A$ to the sky costs $6.59\{}%$ on $\{}ell_A$ — the same number twice, because it is one deficit booked against either of two observables. **And "both pinned" is unreachable rather than skipped**: under the symmetric convention the control's scale approaches $301.4$ from above, so 'LATARG'$=301.6$ lies below its floor and 'brentq' reports same-signed endpoints — the control ARRIVES where this arm SPENDS its fitted onset to get.). *Computes:* that the configuration reproduces 'P15''s own quoted $\{}ell_1=206$ BEFORE its scan is read; that $\{}ell_A$'s range exceeds $40\{}%$ and $\{}ell_1$'s is under $5\{}%$; that $\{}ell_A$ falls monotonically; that $\{}ell_1$ misses the sky at EVERY onset and the misses spread by under $3$ points; that the two pinnings agree to $10^{-3}$; that the ratio nonetheless crosses the sky, so reading it alone gives the opposite answer; and that the control's floor sits below 'LATARG' *Bound:* **scope: driven, 'LMAXL=520', 'LSTEP=2', each arm on its own geometry throughout.** It does not shrink the deficit and does not say the construction is wrong — it says $0.6830$ and $-6.6\{}%$ are two spellings of one number, and that no choice of the one fitted number touches it. The 'LMAXL=1300' cross-check is **run and opt-in** ('–cutoff', **measured 1207s a point** against the scan's $92$s, $943$ modes against $376$): $\{}ell_1=206$ at both ends, while the HIGHER peaks do move — so the invariance is the first peak's and not the spectrum's *Run:* `python3 receipts/P15_CR_cosmology/P15_the_one_fitted_number_moves_the_scale_and_not_the_peak.py`

**P15R7** `P15_the_symmetric_comparison_was_never_runnable_and_the_quarter_was_two_fifths` [OK]
*'PO-13'; 'sec:refit-bound'* (**THE SYMMETRIC UNDRIVEN COMPARISON 'r4549' PROMISED AND NEVER LANDED — ITS THREE CLAIMS HOLD, ITS ONE NUMBER DOES NOT.** 'r4549''s own message read *"60's receipt for this result is not yet pushed and lands next"*; **no receipt in the tree ever set 'LZSTART', and the sound-horizon half of the convention had NO KNOB AT ALL** until 'LRSFROM' was added at 'r6476', so its numbers were unreproducible from the tree by anyone including their author. Run now: matching the start alone **overshoots and changes sign** ($+0.2249\{}to-0.4719$, the magnitude doubling); matching the convention too leaves a residue on the far side of zero ($-0.0865$); and **the implied mode is identical across those last two rows**, $\{}ell_1=482$ in both, while the convention moves $\{}ell_A$ from $301.4$ to $397.1$. **But the residue is $38\{}%$ of the original, not the "quarter" 'r4549' reported** — a factor of $1.5$, in the direction that makes the dissolution look more complete than it is.). *Computes:* the commit's reachability asserted FIRST, so a shallow clone cannot turn the two commit-pinned checks into a vacuous pass; that 'r4549' touched no file under 'receipts/'; that no receipt at that revision set either knob; the sign change and the growth in magnitude; the residue's sign and size; the mode's identity across the convention change; and that the residue exceeds a third *Bound:* **scope: undriven on both arms, 'LMAXL=520', 'LSTEP=2'.** It does **not** re-open 'r4549''s verdict, which survives the correction — two fifths is still a minority and still on the far side of zero. **The structural finding is the shared one**: a convention moves $\{}ell_A$ and leaves $\{}ell_1$ at $482$; the fitted number moves $\{}ell_A$ and leaves $\{}ell_1$ at $206$. Both times the denominator moved and the mode did not *Run:* `python3 receipts/P15_CR_cosmology/P15_the_symmetric_comparison_was_never_runnable_and_the_quarter_was_two_f`

**P15R8** `P15_the_angular_momentum_sector_is_not_a_propagating_mode` [OK]
*'PO-31''s first computation; 'sec:coherence''s throat tower* (**the damping removes the anisotropic signature, not $J$ — because $J$ is not among the modes the tower is a statement about.** 'P15''s $\{}nu^{2}=\{}tfrac14-\{}ell(\{}ell+1)$ is reproduced exactly, and it is a computation about a **propagating field mode**, one with a mass that can be heavy. But in the standard multipole decomposition $\{}ell=0$ carries $M$, **$\{}ell=1$ axial carries $J$ — the slow-rotation limit, a STATIONARY solution, non-propagating** — and only $\{}ell\{}ge2$ radiates. ⇒ ***Applying a principal-series argument there assigns a decay rate to a conserved charge.*** And 'P09' already carries the concept as its **"non-radiative skeleton ... the wall is where the field begins to propagate"**; it needed connecting, not inventing). *Computes:* 'P15''s tower reproduced at $\{}ell=0,1,2,3$, the sectors by what each carries and whether it propagates, and that $J$'s sector is not among the propagating ones *Bound:* **does NOT establish that $J$ survives the leg** — only that the no-hair tower is not the mechanism that would remove it. A charge can still be carried off by matter or torqued, and **the constrained-sector analysis on the throat is a computation not done here** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_angular_momentum_sector_is_not_a_propaga`

**P15R9** `P15_the_tower_is_robust_to_the_throat_being_squashed` [OK]
*'prop:throat''s equal radii; 'sec:coherence''s tower* (**'prop:throat' is a $J=0$ construction** — its argument is "direct from $f$ at the Nariai mass", and $f=1-2GM/c^{2}r-r^{2}/\{}alpha^{2}$ is the **non-rotating** metric. ***So the charge whose survival 'PO-31' asks about is the one whose presence would deform the geometry the argument is conducted on***, a dependency drawn nowhere. **But the tower is robust**: generalising to $\{}nu^{2}=\{}tfrac14-\{}lambda\{},\{}ell(\{}ell+1)$ with $\{}lambda=r_{\{}rm dS}^{2}/r_{S}^{2}$, ***the $\{}ell=0$ base is $\{}tfrac14$ for EVERY $\{}lambda$***, and $\{}ell\{}ge1$ stays principal series for any $\{}lambda¿1/(4\{}ell(\{}ell+1))$ — $1/8$ at $\{}ell=1$, so the damping survives until the $\{}mathrm{dS}_2$ radius falls below $0.354$ of the $S^{2}$ radius, **a factor of $2.83$**). *Computes:* $\{}lambda=1$ reproducing 'P15''s tower, the $\{}ell=0$ base's independence of $\{}lambda$, and the thresholds at $\{}ell=1,2,3$ with the higher multipoles safer *Bound:* **does not show the conclusion survives rotation** — **what $\{}lambda$ a rotating progenitor produces is not computed**, and a rotating near-horizon geometry is generally **warped** rather than a direct product, so $\{}lambda$ may not even be the right single parameter. It measures a margin and names what would decide it *Run:* `python3 receipts/P15_CR_cosmology/P15_the_tower_is_robust_to_the_thro`

**P15R10** `P15_rotation_moves_the_throat_away_from_the_damping_threshold` [OK]
*the rotating Nariai locus; 'r6497''s named computation* (**rotation moves the throat AWAY from the damping threshold.** Solving the Kerr–de˜Sitter double root $\{}Delta_r=\{}Delta_r'=0$ gives $a^{2}=r^{2}(\{}alpha^{2}-3r^{2})/(\{}alpha^{2}+r^{2})$, which ***recovers 'prop:throat''s

own $r_N=\{}alpha/\{}sqrt3$ and $M=\{}alpha\{}sqrt3/9$ at $a=0$ rather than assuming them***. The family is bounded: **$a_{\{}max}=(2-\{}sqrt3)\{},\{}alpha$ exactly**, attained where $\{}Delta_r"=0$ — the **ultracold** point where the double root becomes triple. And $\{}lambda=1$ **exactly** at $a=0$, rising monotonically with $a$, against 'r6497''s threshold of $1/8$). *Computes:* the $a=0$ reduction to $r^{2}f(r)$, the locus recovering both corpus values, $a_{\{}max}$ in closed form coinciding with the triple root, and $\{}lambda\{}ge1$ across the physical branch *Bound:* **$\{}lambda$ here is a PROXY and it is named**: the ratio **at the pole**. The rotating near-horizon $S^{2}$ is warped through $(r_N^{2}+a^{2}\{}cos^{2}\{}theta)$, so one ratio does not describe it — 'r6497' warned of exactly this. **What is established is the direction and its size at one well-defined point**; for the conclusion to fail the warping must reverse a trend running the right way by a factor of eight *Run:* `python3 receipts/P15_CR_cosmology/P15_rotation_moves`

**P15R11** `P15_the_warped_throat_keeps_every_mode_above_the_monopole_decaying` [OK]
*'PO-31'; 'sec:throat' and 'prop:throat', continuing 'r6497'/'r6499'* (**THE WARPED HARMONIC PROBLEM, DONE: EVERY MODE ABOVE THE MONOPOLE STAYS PRINCIPAL-SERIES ACROSS THE WHOLE ROTATING FAMILY, AND THE MARGINAL MEMBER IS THE NON-ROTATING ONE.** 'r6499' left the ratio read at the POLE as a proxy, warning that the rotating near-horizon sphere is warped through $r_N^2+a^2\{}cos^2\{}theta$ so one ratio may not describe it and $\{}ell$ may not be a good label. **The near-horizon limit is TAKEN rather than posited** and is a warped product, not a direct one: $\{}rho_0^2$ multiplies the $\{}dS_2$ and polar directions alike and the $\{}dS_2$ is FIBRED over the sphere by $\{}tilde k=2r_0a\{}Xi/(r_0^2+a^2)$, non-zero at every rotating member. A massless scalar still separates, the warp cancelling from the relation between the transverse eigenvalue and the $\{}dS_2$ operator. **AND FOR THE AXISYMMETRIC MODES THE WARP CANCELS ALTOGETHER**: $W\{}propto\{}sin\{}theta$ and $P\{}propto\{}Delta_\{}theta\{}sin\{}theta$ exactly, so the $m=0$ problem is $(\{}Delta_\{}theta\{}sin\{}theta Y')'+E\{}sin\{}theta Y=0$ — a SPHEROIDAL deformation of Legendre in which $\{}rho_0$ never appears. *The order's worry is answered by the structure, not by the numbers.* The decay condition is $\{}nu^2=\{}tfrac14-L^2E-L^4m^2\{}tilde k^2$, the fibration term non-negative, so rotation can only deepen the damping. **TWO EXACT BOUNDS SETTLE IT WITHOUT A SCAN**: $L^2\{}ge1$ on the physical branch (from $L^2-1=-(3u-1)(u+2)/(3u^2+6u-1)$) and $E_1\{}ge2$ above the monopole (Rayleigh, $\{}Delta_\{}theta\{}ge1$ pointwise), each with equality only at $a=0$ — so $L^2E_1\{}ge2$ against the threshold $\{}tfrac14$, **a factor of eight, attained where the rotation vanishes**. The monopole keeps $\{}nu^2=\{}tfrac14$ exactly.). *Computes:* T3 asserted symbolically ($\{}Delta_r(a{=}0)=r^2f$, not $rf$); the locus re-derived from $\{}Delta_r=\{}Delta_r'=0$ for BOTH $a^2$ and $M$ and asserted to return 'prop:throat''s values at $a=0$; $r_*$ asserted to be exactly where $\{}Delta_r"=0$ (T2); the near-horizon limit asserted equal to the warped ansatz coefficient by coefficient at exact members, with the $\{}tau$ rescaling coming out exactly $1/(r_0^2+a^2)$ and no stray cross terms; $W^2$ and $P^2$ asserted free of $\{}rho_0$; **the solver calibrated against Legendre $l(l+1)$ at $a=0$ across $m=0..4$ BEFORE any warped zero is believed**; every sample asserted on the physical branch (T1); the monopole asserted both numerically and EXACTLY (the $m=0$ flux matrix annihilates constants, so $E_0=0$ is not a fitted zero); and both closed-form bounds *Bound:* **scope: a massless SCALAR on the near-horizon geometry — the same proxy 'r6497''s $\{}nu^2=\{}tfrac14-\{}lambda\{}ell(\{}ell+1)$ is built on.** It does not decompose gravitational perturbations into axial and polar. **It is about whether the DAMPING MECHANISM reaches these modes; it is NOT about whether $J$ survives the leg** — a charge can still be carried off by matter or torqued, and 'r6495''s separation of $J$ from the propagating tower is untouched. The geometry is the throat's exact limit, not the collapse leg's history *Run:* `python3 receipts/P15_CR_cosmology/P15_the_warped_throat_keeps_every_mode_above_the_monopole_decaying.py`

**P15R12** `P15_the_growth_root_is_a_moment_balance_pivoting_on_the_nariai_radius` [ALL PASS]
*sec:discussion* (**THE GROWTH SECTOR IS PARAMETER-FREE AND THE NORMALISATION ROOT IS A MOMENT BALANCE PIVOTING ON THE NARIAI RADIUS.** Substituting H=(c/a)coth u and 4piGrho=(3/2)(c/a)^2 csch^2 u into the growth equation gives **3D" + 4coth(u)D' - 2csch^2(u)D = 0 – alpha CANCELS**, because at Nariai the source is fixed by Lambda and the time variable carries alpha too. Decaying mode is the rate itself, D_-=coth u; growing mode by reduction of order. **Om_m = sech^2(u) exactly** – a clock reading. J = (2/3)cosh^2(u0)(D+/a), matter-domination limit D+/a -¿ 3/5 giving the analytic endpoint J -¿ 2/5. The identity G'/F' = (9/2)Om - 2 integrates the condition to **int a Om (Om -

2/3) du = 0**, solved at Om = 0.315162424 – **and the pivot Om=2/3 is rho_m/rho_L = 2 exactly, at areal radius alpha/sqrt3 = r_N**.). *Computes:* r4253 *Bound:* 61 *Run:* `python3 receipts/P15_CR_cosmology/P15_the_growth_root_is_a_moment_balance_pivoting_on_the_nariai_radius.py`

**P15R13** `P15_the_growth_root_on_the_leaf_rate_is_not_a_moment_balance_and_the_gap_stands` [OK]
*sec:discussion* (**PO-38: THE MOMENT-BALANCE FORM DOES NOT SURVIVE ON THE LEAF RATE, THE ROOT BARELY MOVES, AND IT MOVES THE OPPOSITE WAY FROM WHAT THE CONTINUED INTEGRAL IMPLIES.** PO-38 asked for the growth normalisation root re-solved on the LEAF rate and warned that radiation may break the cancellation that made the growth equation parameter-free. ** (1) IT DOES, EXACTLY. ** The derivation rests on "the decaying mode IS the rate"; applying the growth operator to D = H with radiation present gives **L[H] = (32/3) pi G H rho_r**, which vanishes iff rho_r = 0. With that gone the reduction of order, the G'/F' identity and the balance all go. Om_m = sechˆ2(u) fails on the leaf rate (9.3e-4 at v=0.2) and the balance integral returns -1.71e-4 at the leaf root where the stacking one vanishes to 1e-13. **alpha still cancels**, so the equation stays scale-free but acquires Om_r: the loss is parameter-freedom, not scale-freedom. ** (2) WHAT REPLACES IT ** is the growth ODE solved on the rate, with the Meszaros mode 1 + (3/2)y as an EXACT initial condition (verified to 1.6e-12) and the published (2/5)a/Om convention calibrated against the closed form where the closed form is valid (2e-12). ** (3) THE ROOT: OUTCOME 3, WITH A SIGN. ** Leaf **u_0 = 1.180038880** against the stacking 1.180309372 – a move of **1.1% of the gap**, taking the offset from **0.704 to 0.713 sigma** against the corpus's measured x_0 = 1.6648 +/- 0.0467. **The 2% is not an artefact of the rate; the gap stands and wants a mechanism** – and the shift goes FURTHER from the measurement, not closer. ** (4) THE CONTINUED INTEGRAL GETS THE DIRECTION BACKWARDS. ** The route behind the recorded 0.99934 moves the root TOWARD the measurement (0.693 sigma) where solving the equation moves it away, and by 1.7x the size in u_0. **That figure does not measure how far the root moves; it measures a closed form failing on a rate it does not solve.** ** (5) THE PIVOT SPLITS RATHER THAN MOVES. ** Om_m = 2/3 is the acceleration turnover only when matter and Lambda are the whole content; radiation decelerates too, so q = 0 needs Om_m + 2 Om_r = 2 Om_Lambda. On the leaf rate the two loci separate by du = 5.08e-4 and **the Nariai radius sits BETWEEN them**. ** (6) AND A FLATNESS SLIP IN THE EXISTING SCOPE-(b) CHECK: ** its leaf integrand carries Om_Lambda = 1 - Om rather than 1 - Om - Om_r, so the fractions sum to 1 + Om_r; kept flat its own figure reads 0.99940 and the slip alone is 10.4% of the shift it is quoted for.). *Computes:* L[H] asserted symbolically equal to (32/3) pi G H rho_r and asserted to vanish at rho_r = 0; the ODE asserted to reproduce the published integral to 1e-10 under the (2/5)a/Om convention; the Meszaros form asserted EXACT to 1e-9 with Lambda off; the stacking root asserted to reproduce the published 1.180309372; the leaf shift asserted under 2% of the gap; the leaf root asserted FURTHER from the measurement and the continued integral asserted CLOSER – the two signs asserted opposite; the recorded 0.99934 asserted to be the flatness-slipped value and 0.99940 the flat one, with the slip asserted comparable to the effect; Om_m = sechˆ2 u asserted to FAIL on the leaf rate; the balance asserted to vanish on the stacking rate and NOT on the leaf; the stacking pivot asserted to be r_N exactly, the leaf pivot and turnover asserted distinct, and r_N asserted to lie between them. **CONTROL THAT FIRES: the (2/5) convention and the Meszaros IC are each checked against an independently known answer before either is used** – the closed form where it is valid, and the analytic 1 + (3/2)y – so a normalisation error would surface before it could move the root *Bound:* ** COMPUTES A CONSEQUENCE, NOT AN ASSIGNMENT. ** It does not claim the leaf rate is the right rate for this object; the corpus's rule assigns it and this computes what follows. Om_r is held FIXED at 8.5e-5 throughout and NOTHING is tuned. **No mechanism for the residual 2% is offered** – the row asked whether the rate explains it and the answer is that it does not. NOT CLAIMED: that the balance has any leaf-rate analogue; what is shown is that this one does not hold there *Run:* `python3 receipts/P15_CR_cosmology/P15_the_growth_root_on_the_leaf_rate_is_not_a_moment_balance_and_the_gap_s`

**P15R14** `P15_the_rate_has_one_parameter_and_one_epoch` [ALL PASS]
*sec:flatlcdm* (**THE RATE HAS ONE PARAMETER AND ONE EPOCH, NOT TWO PARAMETERS.** Three statements in P15 could not all be true of one quantity: that its two parameters are alpha and "the OFFSET x_0"; that "at Nariai the mass is no longer free" so BOTH factors of the expansion law are set by Lambda; and that the density ratio is a CLOCK. The offset is fixed twice over – the horizon cubic's DOUBLE root at the Nariai mass, and

P17's constant ledger 2M = alpha((r0/alpha)-(r0/alpha)^3) – and **x_0 = r(tau_0)/r_N identically at every epoch**, the expansion read in the unit the geometry supplies. **Decisive check: if x_0 were the geometric offset ratio it would be 1/sqrt3, and P15's own dictionary would return Om_m = 0.91, not a cosmology.**). *Computes:* r4251 *Bound:* 61 *Run:* `python3 receipts/P15_CR_cosmology/P15_the_rate_has_one_parameter_and_one_epoch.py`

**P15R15** `P15_the_growth_normalisation_root_has_no_feature_at_the_physical_point` [ALL PASS]
*(resources)* (**THE GROWTH-NORMALISATION ROOT HAS NO FEATURE AT THE PHYSICAL POINT, so the agreement is a coincidence – and it is the STANDARD MODEL'S.** Embedding J in its natural families (exponent p, equation of state w, curvature) gives a root that is smooth, monotone and feature-free through p=3/2, w=-1 and flatness, with the physical point NOT stationary in any direction. **A property of the matter-Lambda crossover would leave a mark there; there is none.** The artefact reading was never available (J is forced and dimensionless), and the crossing is steep rather than broad-band (dJ/dOm = -2.45). Discriminating: J=0.99966 at Planck's 0.3153, J=1.02142 at this corpus's DESI DR2 fit of 0.3066.). *Computes:* r4247 *Bound:* 61 *Run:* `python3 receipts/P15_CR_cosmology/P15_the_growth_normalisation_root_has_no_feature_at_the_phys`

**P15R16** `P15_verify_geometry` [OK]
*prop:flat / prop:throat / prop:transmission* (three geometric props DERIVED from primitives: (a1) const-tau 3-metric dr^2+r^2 dOmega^2 has Riemann=0 (flat R^3); (a2) Nariai near-horizon dS_2 x S^2, f(r_star)=f'(r_star)=0, f"=-2Lambda, both radii 1/sqrt(Lambda); (a4) tortoise integrals simple-¿log(exp) vs double-¿power-law). *Computes:* Riemann from Christoffels; Nariai conds from f; two tortoise integrals; checks intrinsically discriminating *Bound:* Kretschmann bonus derived too *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_geometry.py`

**P15R17** `P15_verify_numeric` [OK]
*prop:amplitude / prop:subhorizon* ((a5) substrate dS vacuum power (hbar H_Lam/E_P)^2=9.5e-123 vs A_s=2.1e-9 -¿ ~113 orders below, robust to O(10) prefactor; (a7) comoving Hubble k at seam 0.0104/Mpc vs peak pi/r_s=0.0217/Mpc -¿ 2.1x sub-horizon). *Computes:* Lambda + vacuum power + k derived from obs inputs; matches paper; [E]/[R] marked *Bound:* anchor 6 low-ell leading-order [R] only *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_numeric.py`

**P15R18** `P15_verify_coherence_comb` [OK]
*sec:coherence* (coherence comb: common seam phase -¿ sharp comb Delta-ell=pi D_C/r_s=296 (peaks 296,592,888), peak/trough ~1e5-1e6; random phase per mode -¿ washed out, contrast ~1 (400 realisations). Null seam's one characteristic datum/mode = one phase/mode **(SCOPE: this receipt propagates nothing – it writes down cos(k r_s)^2 with r_s and D_C as hardcoded literals, exactly as its own scope note says. It establishes the COHERENCE MECHANISM and NOT the peak spacing.)**). *Computes:* ACFM coherence demo on the seam; coherent-vs-incoherent control; matches paper *Bound:* mechanism only; full transfer open [R] *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_coherence_comb.py`

**P15R19** `P15_cr_collapse_driving` [OK]
*sec:coherence* (acoustic driving as linear oscillator Theta0"+Theta0=-Psi: undriven (Psi=const) a0=0.500, driven (radiation) aD=0.999, boost aD/a0=2.00 (known LCDM boost); baryon-loaded driven-equiv 1.60 vs adiabatic 1.10). *Computes:* ODE integration; undriven-Psi=const control; matches known boost *Bound:* time-symmetry of contracting phase argued (magnitude proven in next receipt) *Run:* `python3 receipts/P15_CR_cosmology/P15_cr_collapse_driving.py`

**P15R20** `P15_time_reversal_driving` [OK]
*sec:coherence* (|Psi~(omega=1)| resonant driving magnitude is time-reversal invariant: |Psi~(x)|=|Psi~(-x)| at w=1 on the radiation transfer (0.037049) AND a random control (1.411379); so CR's time-mirrored collapse driving = LCDM's magnitude, peak heights match by structure). *Computes:* numerical Fourier magnitude at w=1; radiation + random control; complex/phase-not-invariant discriminator *Bound:* built to close the paper's 'confirmed numerically ... random control' claim *Run:* `python3 receipts/P15_CR_cosmology/P15_time_reversal_driving.py`

**P15R21** `P15_camb_reference` [OK]
*sec:coherence* (**!! MARKED c54.160: the closing annotation states the paper's anchors as P2/P1 ~ 0.46 and P3/P1 ~ 0.45-0.46 while the file itself computes 0.452 and 0.443 and the

paper now prints 0.45 and 0.44.** A stale annotation contradicting the same file's own digit-by-digit claim; both figures are now pinned). *Computes:* CAMB TT spectrum + peak-finder; matches paper *Bound:* needs camb; CR-matches-it carried by driving receipts *Run:* `python3 receipts/P15_CR_cosmology/P15_camb_reference.py`

**P15R22** `P15_damping_ratio_clean` [OK]
*sec:coherence* (the sound horizon on the radiation-included rate through CAMB's recombination history, r_s=144.0 Mpc against CAMB's own 144.4 – the reference agreement P15 quotes; theta_* H0-independent. The receipt also returns a damping ratio, theta_D/theta_* CR/LCDM=1.0816, in its OWN endpoint and handover configuration, which is not the one P15 adjudicates: P15 carries both lengths to the visibility peak, r=0.9918, under a per cent (sec:diffusion-scale and its error-budget receipt)). *Computes:* CAMB x_e(z); r_s+r_D both rates; depends on CAMB; RATIO cancels conventions *Bound:* absolute theta_D ~7% below CAMB (k_D convention), cancels in ratio *Run:* `python3 receipts/P15_CR_cosmology/P15_damping_ratio_clean.py`

**P15R23** `P15_damping_reabsorption` [OK]
*sec:coherence* (the damping signature is not reabsorbable by the baryon density: r_D ~ omega_b^{-0.305} (paper -0.31), d ln r_D/d ln omega_c=-0.295, so a 1% omega_b (the BBN + peak-height prior) shifts r_D by ~0.3%. The cancellation figures it also prints (omega_b +29% to cancel a +8.9% shift) are taken against its earlier signature size, not P15's adjudicated one). *Computes:* finite-diff levers on CAMB thermodynamics vs BBN+height priors *Bound:* the levers only; the signature's size and its observable cost are carried by sec:diffusion-scale's error-budget receipt and sec:coherence's adjudicated-ratio receipt *Run:* `python3 receipts/P15_CR_cosmology/P15_damping_reabsorptio`

**P15R24** `P15_zonset_determinations` [OK]
*sec:tensions* (**!! MARKED c54.160: the paper reported the rounded-l_* pinning as recovering 6873 and this receipt returns 6844 on the current integrals.** The paper now carries both numbers and notes that 6844 sits NEARER to the retired 6850, which sharpens the withdrawal rather than softening it). *Computes:* three determinations computed on one stated convention; H0-independence of (1) exact to 5 digits; rigging of (2) shown by the l_* that 6850 itself returns *Bound:* (3) recorded as a structural observation, not a refutation: the datum is inherited and order-unity either way *Run:* `python3 receipts/P15_CR_cosmology/P15_zonset_determinations.py`

**P15R25** `P15_the_ratio_is_the_onset_in_imported_units` [OK]
*sec:tensions* (WHAT $\{}rho_r/\{}rho_m\{}approx2$ IS, AND WHAT IT IS NOT: **NEITHER A HIDDEN KNOB NOR A SECOND MEASURED DATUM, BUT THE ONE FITTED QUANTITY $z_onset$ RE-EXPRESSED IN UNITS OF A PHYSICAL MATTER DENSITY THE CONSTRUCTION DOES NOT DETERMINE.** The rate carries Omega_m (fitted to BAO); the sound speed carries omega_b and omega_gamma (measured directly, framework-free); z_onset is the one fitted number – and the PHYSICAL omega_m = Omega_m h^2 enters NO prediction. But rho_r/rho_m at onset is omega_r(1+z_onset)/omega_m and equality is 1+z_eq = omega_m/omega_r, so **both divide by exactly that quantity**. ** KNOB-FREENESS SURVIVES AND IS THE LOAD-BEARING HALF: z_onset = 6747.3 at H0 = 67.4, 70.0, 73.0 and 74.0, to the digit – H0 is a common factor in r_s and D_M and cancels from theta_*, so the fitted quantity does not move between the two H0 the tension is between and cannot be absorbing it. ** ** WHAT DOES NOT SURVIVE IS THE EXACTNESS OF 'ABOUT TWICE EQUALITY': at the fitted Omega_m = 0.3066 the ratio is 2.01 at H0 = 67.4 and 1.71 at H0 = 73, the two readings crossing at H0 ~ 68 where Omega_m h^2 reproduces the CMB-inferred omega_m = 0.143 – and the papers' headline is that the cosmology fits at the DIRECTLY measured H0 = 73. ** So the datum is an order-unity BAND (1.7-2.0), not a determined number; the eta-analogy and the BAO confrontation are untouched (the latter uses distance ratios and never omega_m).). *Computes:* z_onset recomputed from the geometric rate at four values of H0 by root-finding on the measured acoustic angle, its spread asserted below 1 part in 6000; the crossing H0 asserted in 67-69; the H0=73 ratio asserted at 1.71 *Bound:* ** THIS IS A UNIT ANALYSIS, NOT A NEW MEASUREMENT. ** It does not touch the rate, the fit, or the abundances; and the CMB-inferred omega_m it compares against is itself derived in a radiation-included analysis, which is the point rather than an objection – the residual measures how far the two analyses' omega_m must differ *Run:* `python3 receipts/P15_CR_cosmology/P15_the_ratio_is_the_onset_in_imported_units.py`

**P15R26** `P15_the_two_data_are_one` [OK]
*sec:tensions* (WHAT THE 'SINGLE INHERITED DATUM' rho_r/rho_m ~ 2 ACTUALLY IS,

AND WHAT THE NATURALNESS ARGUMENT OFFERED FOR IT ACTUALLY DESCRIBES. ** (1) IT IS NOT A DATUM DISTINCT FROM eta. ** On any thermal history rho_r/rho_m(T) = [1+nu](pi^4/30 zeta3) T / [eta (w_m/w_b) m_N], which returns **1.99 at T = 1.6 eV** – the quoted value to one per cent, from standard thermodynamics and NO feature of this construction. Given eta and the measured matter-to-baryon ratio, both inherited for the composition regardless, **the ratio at onset IS the onset**: the handover supplies ONE composition datum and the cosmology fits ONE parameter. ** (2) AND THE NATURALNESS ARGUMENT DESCRIBES AN EPOCH NINE ORDERS FROM THE HANDOVER. ** rho_r = rho_m at T = 0.806 eV; the handover sits FOUR ORDERS ABOVE the deuterium bottleneck (0.07 MeV) because that is what the nucleosynthesis computed on it requires, so the ratio there is ~9e8 – and 'a handover carrying radiation of order the matter density' would be a handover with **eta ~ 0.5**: no light elements, no acoustic peaks. ** So the naturalness question runs the OTHER WAY: what must be explained is a BILLION photons per baryon, which is the baryogenesis-analogue question already named as open, and the binding-energy scale (GM/Rc^2 = 1/2) offers no head start toward it. ** P7's 'the frontier is two data and not one' is corrected to one inherited datum plus one fitted parameter.). *Computes:* the identity evaluated at the onset temperature and asserted against the quoted datum to better than 5%; the equality temperature and the ratio at the bottleneck, at P7's own handover and at the rest mass computed; the eta that O(1) would require at each epoch computed and contrasted with the measured 6.1e-10; and the papers asserted to carry the withdrawal and the corrected accounting *Bound:* ** THIS CORRECTS AN ACCOUNTING AND WITHDRAWS AN ARGUMENT; IT TOUCHES NO RESULT. ** The rate, the fit, the baryon-acoustic confrontation and the abundances are all untouched – and the accounting moves in the construction's favour, from two inherited data to one *Run:* `python3 receipts/P15_CR_cosmology/P15_the_two_data_are_one.py`

**P15R27** `P15_full_transfer_verdict` [OK]
*sec:coherence* (high-ell deficit UNDER THE SHORTCUT (assume peaks=LCDM + apply the receipt's +8.9% damping, an earlier signature size, not P15's adjudicated one): C_l CR/LCDM = 0.979/0.920/0.829/0.717/0.594 at l=500-2500; cited to show the shortcut is illegitimate (assumption (A) is the unbuilt part)). *Computes:* exp damping on CAMB LCDM C_l; conditional chi^2 *Bound:* verdict explicitly CONDITIONAL/open (own banner); paper honors scoping *Run:* `python3 receipts/P15_CR_cosmology/P15_full_transfer_verdict.py`

**P15R28** `P15_verify_closedS3_nonsync` [OK]
*sec:largescale* (low-ell floor mechanism+location: discrete closed-S^3 source (k_L=sqrt(L(L+2))/r0, L¿=2) through FLAT j_ell(k_L D_C), stretch D_C/r0=2.75 pushes L=2 to ell~7.8; ell(ell+1)C_ell=0.12/0.10/0.20/0.6 at ell=2..7, recovered by ell~8). *Computes:* discrete-source flat-Bessel sum; flat-limit chi-¿0 control passes *Bound:* ordinary SW only -¿ deeper than Boltzmann depth (routed to that receipt) *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_closedS3_nonsync.py`

**P15R29** `P15_verify_lowell_exact_measure` [OK]
*sec:largescale* ((1) HZ weight w_L=(L+1)/(L(L+2)) DERIVED as closed-S^3 degeneracy beta^2 x per-mode 1/(beta(beta^2-1)) (g*P==w_L exactly; w_L*L-¿0.998 continuum); (2) D2/D3=1.162 stable across measure family -¿ floor starves ell=2,3 together (octopole slightly more)). *Computes:* first-principles derivation + flat-limit + measure-robustness + wrong-measure contrast *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_lowell_exact_measure.py`

**P15R30** `P15_verify_doppler_lowell` [OK]
*sec:largescale* (Doppler term (super-horizon velocity through j_ell') does NOT differentiate ell=2 from ell=3: CR/LCDM ratio l2~0.27, l3~0.21-0.22 across 4 treatments (none/coh+/coh-/quad), D2/D3~1.29 octopole slightly more suppressed; continuum validator reproduces LCDM rise). *Computes:* SW+ISW+Doppler transfer, 4 combination modes; continuum check; not tuned *Bound:* bare depth (milder exact depth = Boltzmann receipt) *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_do`

**P15R31** `P15_verify_lowell_boltzmann` [OK]
*sec:largescale/sec:scope* (exact low-ell depth via genuine Boltzmann transfer: discrete closed-S^3 source through CAMB's exact Delta_l(k) (SW+ISW+Doppler); CR/LCDM = 0.473 (l=2), 0.410 (l=3), recover by l~7, octopole/quadrupole 0.867 on the CONTROL background (H0 = 67.4, r0 = 5064) — **superseded as a depth quote (r7049+70)**: the corpus's 0.487/0.435/0.359/0.666 are arm A of 'P15_the_low_multipole_depth_gap_closes_and_two_defects_were_cancelling' on the adjudicated background, and this file names that in its header and gates it; what it carries for the corpus

is its gate and its r0-drift table. GATE: continuum reproduces CAMB C_l to 4 figures). *Computes:* CAMB transfer x discrete source, Riemann-sum vs integral; gate-validated; r0-stable; ISW-retained diagnosis *Bound:* needs camb (high acc) *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_lowell_boltzm`

**P15R32** `P15_confront_lowell_data` [OK]
*sec:largescale* (low-ell data confrontation (corrected depths): CR 0.47 sits ABOVE the sharp observed quadrupole (˜0.2), CR 0.41 mildly below the near-LCDM octopole (˜0.6-1.0) -¿ matches neither, a WASH; old 0.22/0.20 'striking match' superseded). *Computes:* CR (Boltzmann depths) vs literature observed multipoles *Bound:* body corrected to r962 depths; likelihood in next receipt *Run:* `python3 receipts/P15_CR_cosmology/P15_confront_lowell_data.py`

**P15R33** `P15_verify_lowell_likelihood_v2` [OK]
*sec:largescale/sec:scope* (low-ell cosmic-variance likelihood: exact scaled-chiˆ2_{2l+1}, CR/LCDM ratio Delta(-2lnL)_l=(2l+1)[r_l(1/d_l-1)+ln d_l]; l=2 rewards CR -2.63, l=3/4 penalise +1.31/+2.10, central total +1.8, octopole range +0.3..+3.8 -¿ wash inside cosmic variance). *Computes:* exact likelihood with corrected Boltzmann depths; matches paper *Run:* `python3 receipts/P15_CR_cosmology/P15_veri`

**P15R34** `P15_the_low_ell_minimum_is_at_ell_four` [OK]
*sec:largescale/sec:scope* (**SCOPE: THIS FILE CROSS-VALIDATES NOTHING.** Its arm A is the likelihood receipt's stored dict and **its arm B is the PAPER'S OWN FIGURE CAPTION, read by regular expression** — so the comparison is against the citing paper's typography, and the quartet 0.42/0.35/0.29/0.52 is banked nowhere. **What it does verify is that arm A's minimum sits at l = 4 and that the standing receipts always carried it, which is what it is now cited for.** The actual second arm is 'P15_the_second_arm_actually_run': it agrees on shape and NOT on depth, so the quartet 0.42/0.35/0.29/0.52 is not banked. What the file establishes is THE LOW-MULTIPOLE DEFICIT'S SHAPE, AND WHERE THE SECTOR'S WEIGHT ACTUALLY FALLS. ** IT IS NOT A MONOTONE RECOVERY FROM THE QUADRUPOLE BUT A DIP WHOSE MINIMUM IS AT ell=4 – and the quadrupole and octopole, the two multipoles the paper quoted, are its two SHALLOWEST points. ** ** THE SHAPE IS CROSS-VALIDATED BY TWO INDEPENDENT BOLTZMANN TREATMENTS of the same discrete source: a standard code's exact Delta_l(k) at the discrete k_L gives 0.47/0.41/0.36/0.68 at ell=2,3,4,5 and the programme's own photon hierarchy gives 0.42/0.35/0.29/0.52 – SAME minimum, SAME recovery, depth ratio 1.13-1.30, which is the size of the residual sec:instrument declares. The shape is cross-validated; the depth carries their spread. ** ** AND THE ell=4 MINIMUM WAS NEVER NEW: 'P15_verify_lowell_boltzmann.py' has tabulated 0.356 at ell=4 since r1395 and the likelihood receipt has consumed it ever since – the corpus's failure was one of QUOTATION, not of computation. ** ** THE OCTOPOLE VERDICT: on the exact cosmic-variance likelihood the quadrupole REWARDS the construction (-2.63), the octopole penalises it at +1.31, and the LARGEST single-multipole penalty in the sector is the +2.10 at ell=4 – the prediction's own minimum, and the one multipole of the four the large-angle literature does not treat as anomalous. Sector total unchanged at +1.8, a wash inside cosmic variance. **). *Computes:* both arms' depth tables read off their own sources (the likelihood receipt's dict; the paper's caption) and their argmin asserted equal to ell=4; the ell=4 depth asserted present and deepest in the standing r1395 receipt; the per-multipole likelihood re-derived from the depths and its argmax asserted at ell=4 with the total asserted at +1.8; the exact/WKB transmission spread asserted below 5%; and all eight corpus sites asserted to carry the corrected statement *Bound:* ** DEPTH IS QUOTED TO A TWO-CODE SPREAD AND NOT BETTER. ** And nothing here relieves the observed sky's selective suppression of ell=2 ALONE by a factor of order three, which this construction does not produce; that stays inside cosmic variance and belongs to the likelihood-level comparison *Run:* `python3 receipts/P15_CR_cosmology/P15_the_low_ell_minimum_is_at_ell_four.py`

**P15R35** `P15_two_ranges_ratio` [OK]
*sec:background (two ranges)* (the local Hubble-Eddington radius r_* = (M alphaˆ2)ˆ(1/3) and the expansion's amplitude A = (2M alphaˆ2)ˆ(1/3) are ONE length in the fixed ratio 2ˆ(1/3), for every M; A is the matter-Lambda equality radius). *Computes:* r_* from K_G = 1/alphaˆ2 - M/rˆ3 = 0; A from rho_m/rho_Lambda = cschˆ2(3tau/2alpha) at equality; ratio A/r_* = 2ˆ(1/3) symbolically, independent of M and alpha *Bound:* the r1608 identity; the four-ways reading (turnaround, amplitude, equality, structure boundary) is the paper's *Run:* `python3 receipts/P15_CR_cosmology/P15_two_ranges_ratio.py`

**P15R36** `P15_the_second_arm_actually_run` [OK]
*sec:largescale* (**THE SECOND BOLTZMANN ARM, ACTUALLY RUN — THE SHAPE CROSS-VALIDATES AND THE DEPTH QUARTET DOES NOT.** P15 claimed the low-multipole deficit had been run a second time on the programme's own photon hierarchy, returning 0.42/0.35/0.29/0.52. **The receipt cited for it does not run the hierarchy: it reads arm A from a stored dict and reads arm B OUT OF THE PAPER'S OWN FIGURE CAPTION, by regular expression** — so the cross-validation compared a table against the citing paper's typography and the quartet was banked nowhere. This file drives 'HIER_photon_hierarchy' itself: evolves it on a k-grid containing the discrete ladder k_L = sqrt(L(L+2)) x 2.75/D_M plus a dense continuum grid, projects its own source terms to Delta_l(k) for l = 2..44, and forms the discrete w_L sum against the continuum dln k integral, each normalised to its own l = 25-40 plateau. **SHAPE: minimum at l = 4 and recovery by l ˜ 7, independently of the CAMB arm and of the caption.** **DEPTH: 0.494/0.243/0.184/0.608 against CAMB's 0.473/0.410/0.356/0.676 — close at l = 2, a factor 1.7-1.9 apart at l = 3-4.** So the quartet is returned by neither arm and the 'uniformly 10-25% deeper' claim is false; both withdrawn from P15 at c54.153.). *Computes:* the minimum over l = 2..5 asserted at l = 4; recovery asserted in 6 ¡= l ¡= 9; the depth disagreement asserted above 1.4x; all ratios asserted finite *Bound:* **scope: this settles that the SHAPE claim has a computation behind it and that the DEPTH claim did not. It does NOT settle which depth table is right — the arms differ in late-ISW handling and in D_M (13864 here against the projection's 13927) — and it makes no attempt to reconcile them.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_second_arm_actually_run.py`

**P15R37** `P15_hubble_expansion_confrontation_v2` [OK]
*sec:tensions* (** THE STACKING-RULER COMPUTATION, AND THE CORPUS'S RULER IS THE LEAF'S (r6921).** *Its arithmetic stands and is live as the calibration 'P15_the_desi_dissolution_survives_the_le reproduces in its PART 1; what is superseded is the READING that the H0-independence it finds is the corpus's, since r_s is accumulated on the leaf rate, which carries w_r/hˆ2, so H0 does not cancel and the same data pin it. 'P15' no longer rests a claim on this row.* Original: SDSS DR12 BAO flagship: CR BAO ratios D_M/r_d, D_H/r_d H0-INDEPENDENT (identical at H0=67,73); chiˆ2 CR=1.71 (any H0 incl 73), LCDM(67.4)=4.37, LCDM(73)=49.45 = the tension; cosmic chronometers consistent with 73). *Computes:* geometric-rate H0-cancellation + chiˆ2 vs DR12; matches paper 1.7/49 *Run:* `python3 receipts/P15_CR_cosmology/P15_hubble_expansion_confrontation_v2.py`

**P15R38** `P15_desi_dr2_confrontation` [OK]
*sec:tensions* (** THE STACKING-RULER COMPUTATION, SUPERSEDED AS A READING AND LIVE AS ARITHMETIC (r6921)** — *the same note as the DR12 row: the DESI fit here is on the radiation-free ruler, which is not where the corpus puts r_s, and 'P15' no longer cites it.* **!! MARKED c54.160: the docstring reads chi2/dof = 13.70 for LCDM at H0 = 73 while the paper quotes about 15 – the same chi2 = 178.16 divided by 13 measurements rather than by the 12 degrees of freedom one fitted parameter leaves.** The paper's figure is the correct one and the receipt's annotation is not; the chi2 itself is pinned). *Computes:* full-covariance chiˆ2 vs DESI DR2; matches paper 1.0/14; was uncited by name *Run:* `python3 receipts/P15_CR_cosmology/P15_desi_dr2_confrontation.py`

**P15R39** `P15_expansion_law` [OK]
*sec:properframe/sec:flatlcdm* (derivation core (symbolic): eq:amplitude (6GM/Lambda cˆ2)ˆ{1/3}=2ˆ{1/3}/sqrt(I at Nariai; eq:scalefac r=A sinhˆ{2/3}(B tau) solves the E=1 geodesic (dr/dtau)ˆ2=2GM/r+(Lambda/3)rˆ2 (2/3 forced by wrong-exponent control); eq:rate Hˆ2=(Lambda cˆ2/3)cothˆ2; eq:omega-ratio Omega_m/Omega_L=cschˆ2). *Computes:* sympy: amplitude, geodesic solve, cothˆ2 rate, cschˆ2 ratio; wrong-exponent control *Run:* `python3 receipts/P15_CR_cosmology/P15_expansion_law.py`

**P15R40** `P15_verify_throat_tower` [OK]
*sec:throat* (throat mode tower + no-hair isotropization: dS_2 x Sˆ2, mˆ2/Hˆ2=l(l+1), nuˆ2=1/4-l(l+1); l=0 -¿ nu=1/2 (scale-invariant base, survives), every l¿=1 -¿ nuˆ2¡0 (heavy principal, decays) =¿ throat isotropizes; grounded in Wald 1983 no-hair; guard: throat l != CMB multipole). *Computes:* tower algebra + survives/decays split; theorem-grounded; guard held *Run:* `python3 receipts/P15_CR_cosmology/P15_verify_throat_tower.py`

**P15R41** `AS_amplitude_leftward` [OK]
*rem:transmission-leg* (AS — THE AMPLITUDE, READ LEFTWARD. Family 3's question, and what C4 can and cannot say about it. CONFIRMATION 2, and the corpus is already correctly scoped here: P7: the inherited datum 'stands as a ONE-PARAMETER ACCOMMODATION

rather than a parameter-free prediction.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/AS_amplitude_leftward.py`

**P15R43** `BUILD_camb_store` [OK]
*sec:instrument* (BUILD_camb_store – the CAMB side of the transfer comparison, with its settings RECORDED. WHY THIS EXISTS (r2345). 'ENVELOPE_residual.py' and its relatives load '/tmp/camb_tr.npz' and measure our transfer against CAMB's. Nothing in the corpus records HOW that store was built – not the receipt, not the changelog, not the instrument.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/BUILD_camb_s`

**P15R44** `C10_highl_ratio` [OK]
*sec:envelope-consequence* (**!! CORRECTED c54.155: THIS ROW CARRIED THE RETIRED z_onset = 6850 NUMBERS** (r_s matched +0.76%, r_D +10.09%, theta_D/theta_* +9.26%). The shipped receipt hardcodes 6797 and produces +0.67%, +10.15%, +9.41%, l_* = 302.2. **And the receipt hardcodes th_ratio = 1.0926 with the comment 'DERIVED in C9' while C9 produces 1.0941.** P15 corrected to the live values). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C10_highl_ratio.py`

**P15R45** `C11TEST_radiation_zeroed` [OK]
*sec:envelope-consequence* (CRRUN — THE SAME INTEGRATOR ON CR. The control passed at r1975, so this may now be run. WHAT CHANGES, and each from the corpus not from me: * BACKGROUND: H^2 = H0^2(Om a^-3 + OL) at H0=73, Om=0.3066 – radiation is content, not a source. * LOWER LIMIT: the SEAM, not eta-¿0 (C9: there is no observable expansion below it).). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C11TEST_radiation_zeroed.py`

**P15R46** `C11_early_isw` [OK]
*sec:envelope-consequence* (C11 — THE EARLY ISW. The last named piece. CONFIRMATION 2/4 – the corpus settles the input, in its own voice: 'radiation, like matter, is inherited content read off that clock, NOT A TERM THAT SOURCES THE RATE, and there is no radiation-dominated era.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C11_early_isw.py`

**P15R47** `C1_closure_of_the_driving` [OK]
*sec:envelope* (C1-CLOSURE — DOES THE ACOUSTIC OSCILLATOR CLOSE IN ELEMENTARY TERMS ON CR's COLLAPSE LEG? CONFIRMATION 4 first — WHICH background. The corpus is specific and I must not substitute: L2 (where the driving lives) = 'the leaf's local rate, ORDINARY FRIEDMANN, radiation GRAVITATING NORMALLY, on the progenitor's collapse excursion' – and P16: on the cooling leg |H| = sqrt(8 pi G rho/3), 'identical to standard BBN at every...). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C1_closure_of_the_driving.py`

**P15R48** `C2_horizon_limits` [OK]
*sec:envelope* (C2 — THE LIMITS. The comoving Hubble radius across the excursion, from the corpus's own geometry. CONFIRMATION 4 — the object, and I take it from the corpus rather than from cosmology-by-habit: the E=1 (marginally bound) congruence IS this cosmology. On it (dr/dtau~)^2 = E^2 - f = 1 - f, with f = 1 - 2M/r - r^2/alpha^2. The signed areal radius r is the scale-factor analogue.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C2_horizon_limits.py`

**P15R49** `C3_conformal_map` [OK]
*sec:envelope* (**!! CORRECTED c54.155: 'better than a percent at the entry radii of the modes in question' IS 1.87% AT THE PAPER'S OWN 10 k_s THRESHOLD** (3.72% at 5 k_s, 24.4% at the seam); sub-percent begins near 30 k_s. P15 corrected). *Computes:* runs clean; registered from its

own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C3_conformal_map.py`

**P15R50** `C4_driving_envelope` [OK]
*sec:envelope* (**!! CORRECTED c54.155: 'entry at x = 1/sqrt3 for every k' IS A LIMIT, NOT AN IDENTITY** (the receipt's own table gives x_entry = 0.764 at k = k_s, +32%), **and the 'residual below a percent' is an average over a 5-point grid that lands near zeros of cos(x_seam): scanning the receipt's own function over k/k_s in [10,120] reaches 7.3% at k/k_s = 12.** Both qualified in P15). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C4_driving_envelope.py`

**P15R51** `C5b_baryon_term` [OK]
*sec:envelope* (C5 — THE BARYON TERM. The first of the two terms C4 named as missing. CONFIRMATION 4 – and the corpus fixes the division rather than me: P16 masthead: 'eta fixes the abundances and the CMB peak HEIGHTS, rho_r/rho_m the peak SPACING.' P15: this cosmology 'carries the matter density as the bend of the spatial cut WITHOUT YET REMOVING THE COMPONENT' – so rho_m is TOTAL matter and the baryon fraction is set separately,...). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C5b_baryon_term.py`

**P15R52** `C6_neutrino_term` [OK]
*sec:envelope* (C6 — THE ANISOTROPIC-STRESS TERM. The second of C4's two named terms. FINDING ZERO, before any physics: the corpus's TEXT says 'neutrino' ZERO times, and 'anisotropic stress' zero, and 'free-streaming' zero – while its BBN CODE carries them explicitly (bbn_network.py: 'neutrinos share the plasma T until ~ e+- annihilation, then decouple to (4/11)^1/3').). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C6_neutrino_term.py`

**P15R53** `C7_equality_and_deficit` [OK]
*sec:envelope* (C7 — THE STEP THE TRAP WAS GUARDING. full_transfer_verdict.py ASSUMED the high-ell peaks equal LambdaCDM's and applied the extra damping, getting a large deficit – and P15 disowns it because 'that assumption IS the unbuilt part.' C4 built it. So the same calculation can now be done with a DERIVED envelope. FIRST, though, one number that decides everything and that I have not yet checked.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C7_equality_and_deficit.py`

**P15R54** `C8_diffusion_length` [OK]
*sec:envelope-consequence* (C8 — DERIVE r_D HERE, FROM FIRST PRINCIPLES, ON THE SAME FOOTING AS THE DRIVING. Nothing imported from the damping_tail/ files. Nothing that needs CAMB. CONFIRMATION 4: the object. Silk damping is a diffusion length built from the Thomson opacity and the expansion. The standard definition (Kaiser 1983; Hu-Sugiyama), in the Rb -¿ 0 limit: 1/k_D^2 = INT d eta . (1/(6 tau_dot)) .). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C8_diffusion_length.py`

**P15R55** `C9_sound_horizon_and_ratio` [OK]
*sec:envelope-consequence* (C9 — r_s AND r_D WITH THE CORRECT LOWER LIMIT: THE SEAM. A first pass integrated from a=0 and returned r_s = 256 Mpc against the corpus's 144 – absurd, and the error is structural, not numerical: *** ON THIS COSMOLOGY THERE IS NO OBSERVABLE EXPANSION BEFORE THE SEAM. *** P15: 'recombination is observable cosmology FROM THE SEAM OUTWARD'. The L1 foliation BEGINS at the seam;). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/C9_sound_horizon`

**P15R56** `H0_acoustic_angle_and_seam` [OK]
*sec:refit-bound* (H0 — THE ACOUSTIC SCALE AT THE DIRECTLY MEASURED H0. The calculation I have been running backwards. CORPUS CLAIM (P7, P15): 'There is no second Hubble rate to reconcile: THE DIRECTLY MEASURED H0 IS THE GEOMETRY READ AT

THE PRESENT EPOCH, while the lower value inferred from the microwave background rests on a radiation-governed sound horizon the construction does not share...). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/H0_acoustic_angle_and_s`

**P15R57** `ISW_line_of_sight` [OK]
*sec:refit-bound* (ISW — THE EARLY INTEGRATED SACHS-WOLFE, ISOLATED, AND ITS SHIFT ON THE FIRST PEAK. Same shape as everything else in A.0: the source is a line-of-sight integral over the set rate. Delta_l(k) = INT d eta [ S_SW + S_ISW ] j_l(k(eta_0 - eta)), S_ISW = (Phi' + Psi') = 2 Psi' LambdaCDM has a nonzero early ISW because Psi is still evolving at recombination (radiation matters). CR does not: C11 showed Psi is constant on L1.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/ISW_line_of_sight.py`

**P15R58** `M2_reabsorption_minimization` [OK]
*sec:refit-bound* (M2 — THE MINIMIZATION. Can a parameter refit absorb the damping shape? The honest core of the likelihood question, done on the closed forms derived in C8/C9 – no CAMB. SETUP, stated before running: FREE : Omega_m, omega_b, and the inherited datum rho_r/rho_m (which fixes z_onset). CONSTRAINT: l_* = pi D_M/r_s pinned to the observed 301 – the peak positions are measured.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/M2_reabsorption_minimization.py`

**P15R59** `PEEB_peebles_on_the_set_rate` [OK]
*sec:refit-bound* (PEEB — THE PEEBLES IONISATION HISTORY ON THE SET RATE. ZREC used Saha only: right shift, wrong absolute (1283 vs CAMB's 1090), because Saha has no departure from equilibrium. Peebles adds it – and the departure is itself H-dependent, which is exactly why it must be integrated on the construction's own rate rather than inherited. Standard effective three-level hydrogen atom (Peebles 1968; Seager et al. fits).). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/PEEB_peebles_on_the_set_rate.py`

**P15R60** `ROBUST_p1p2_scan` [OK]
*sec:refit-bound* (**!! FIXED c54.160: THE REGISTERED COPY WAS UNRUNNABLE FROM ITS OWN DIRECTORY** – it imported 'RD_diffusion_direct', which lives only in 'storyboard_receipts/', so it exited 1 on ImportError before reaching any computation and the sweep had to run it from the ORIGIN instead. *A receipt that cannot run where it is registered is not a receipt.* Path resolution added; it now exits 0 in place). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/ROBUST_p1p2_scan.py`

**P15R61** `P15_the_first_peak_figure_is_not_stable` [OK]
*the acoustic sector* (**THE 150 IS NOT A PREDICTION: 'ROBUST_p1p2_scan''s FIRST-PEAK POSITION AND PEAK-HEIGHT RATIO ARE NOT STABLE UNDER ITS OWN DOCUMENTED INITIAL CONDITIONS.** Built settling front #5, the corpus's two internal routes to the first acoustic peak (transfer 220, this instrument 150). **The front is posed as a disagreement between two established routes; one of the two is not established.** PART 1 the numerics are cleared first – stability number 0.47 against the file's own limit of 2.8, 120+ RK4 steps per oscillation, NS_CR x4 reproduces every figure. PART 2 Psi's VALUE comes from the smooth matched envelope and Psi's DERIVATIVE from the oscillatory closed form – two different functions, and the initial velocity is built entirely from the second. PART 3 across four readings of the SAME stated IC, ell_1 spans 150-315 and P1/P2 spans 0.93-2.02; **the reading the docstring actually describes ('velocities zero') returns 2.017 against the sky's 2.212, where the paper reported 1.45 as the disagreement.** PART 4 the comb the paper cites against it was never propagated either – 'verify_coherence_comb' writes down cos(k r_s) with r_s and D_C as literals. PART 5 **the residual: the source comb spacing is 0.72-0.79 of pi/r_s under EVERY variant. The initial data move the first peak by 165 multipoles and do not move this.**). *Computes:* stability asserted under the file's own 2.8 limit and steps-per-oscillation over 50; the two derivative functions asserted to differ by more than 1.2x across the band and to agree as x -¿ 0; the control asserted to reproduce 1.447 and 150/360/555/780; the documented-IC run asserted at

2.017 AND asserted closer to the sky than the implemented one; ell_1 asserted to move by 150+ multipoles and P1/P2 by more than 2x; the comb receipt asserted to contain cos(k r_s) and NO integrator; the comb deficit asserted present, bounded, and STABLE across variants (max-min ¡ 0.15) – which is what makes it not initial data. *Bound:* **scope: this does NOT vindicate 220 and supplies no first-peak position. It removes one route from arbiter status and registers the comb deficit, which is not resolved here. The four variant figures are produced by 'computations/beyond_the_wall/L171_what_sets_the_first_peak.py', which execs the registered receipt's own source with only the initial-data block substituted; its ICMODE=asis control reproduces the receipt exactly, assertions passing.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_first_peak_figure_is_`

**P15R62** `P15_the_crossing_exists_and_is_empty` [OK]

*the acoustic sector* (**THE RE-ENTRY BETWEEN THE BRANCH POINT AND THE ONSET IS REAL, AND IT IS EMPTY: THE L1 POTENTIAL EQUATION CONTAINS NO k AT ALL.** Built working front #5, testing a candidate THIS FORK proposed at c54.165 – front #1 puts every mode super-horizon and frozen at the branch point, 'prop:subhorizon' puts the acoustic modes inside at the onset, so they re-enter in between, and a crossing is where flat LCDM acquires its k-INDEPENDENT driving shift. **The answer is no, and the reason is stronger than the question.** PART 1 the re-entry is real and its boundary is computed: the mode crossing AT the onset is k = 0.0111/Mpc, ell = 144, so the peak modes crossed before the plasma began and ell ¡ 144 crosses after. PART 2 but it is a MATTER-era crossing – pressureless to better than a part in 1e4, because L1 has no radiation SOURCE. PART 3 **and the w = 0 potential equation has no k in it**: Phi constant to twelve digits, identical across four decades in k, the integrator taking the SAME number of steps for every mode. PART 4 the contrast: in LCDM 5 of 6 cross under radiation, where w = 1/3 restores the kˆ2 term and Phi decays on the sound-crossing time by construction.). *Computes:* every mode at/above the peak region asserted to cross before the onset AND ell = 28 asserted to cross AFTER it (the first draft claimed 'every observed mode' and this assertion fired); the boundary multipole asserted inside 100-250; the background asserted pressureless to 1e-3 at every crossing; Phi's spread across four decades in k asserted below 1e-9 AND the solver step count asserted identical across k; the LCDM contrast asserted to put 4+ modes under radiation with z_eq above 3000. *Bound:* **scope: this CLOSES a candidate mechanism and supplies none. It computes no peak position and uses no figure from the acoustic fold. It leaves 'PO-7' open and harder: the absence of a universal driving shift on this rate is not an omitted crossing but a scale-free interval.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_crossing_exists_and_is_empty.py`

**P15R63** `P15_two_arm_control_and_guard` [OK]

*the acoustic sector* (**A LIVE LambdaCDM CONTROL AND AN UNDRIVEN GUARD, ON ONE MACHINERY – AND THE CR ARM'S FIRST PEAK SITS 23% LOW AGAINST A 1.5% POSITION FLOOR.** Built for fronts #5 and #18. **This fork had never run a control**: 'ROBUST_p1p2_scan' quotes one from r1975 in a comment, and c54.164 then showed its figures move over ell_1 in {150,165,315} with nothing to measure that against. PART 1 **the guard passes on BOTH arms** – every coupling to the potential removed at every site (4 Phi' in photon/neutrino/CDM continuity AND kˆ2 Psi in photon-baryon/neutrino Euler, both branches) and the source comb lands on the integers, LCDM 1.0147/2.0005 and CR 1.0121/1.9968, **so every departure from n pi/r_s in a driven run is the driving and nothing else**. PART 2 the control lands at 224/536/808/1116 against the sky's 220.6/538.1/809.8/1147.8 – **positions to 1.5% at the first peak and 0.4%/0.2% at the second and third; HEIGHTS only to 13-26%, so no height claim from this instrument is meaningful below ˜25%**. PART 3 the CR arm run on its PHYSICAL discrete ladder and on a dense continuum sampling gives **identical** peaks 172/396/624/904, so discreteness is not what sets the position – an objection closed by measurement. PART 4 ell_1/ell_A = 0.5703 against the control's 0.7433 and the sky's 0.7312: **a 23% deficit against a 1.5% floor, a factor of 15 in margin**, with P1/P2 = 0.866 against the control's 1.933.). *Computes:* the undriven comb asserted within 0.03 of the integers on both arms; the control's position floor asserted under 3% AND its second and third peaks asserted within 1% of the sky; the discrete and continuum peak lists asserted identical; the deficit asserted at 5+ floors; CR's ell_1/ell_A asserted in 0.55-0.60 and the control's in 0.72-0.76; the two arms' l_A asserted to agree to 1% (they do – D_M differs by 6.2% and r_s by 6.3% and they cancel). Every background quantity is RECOMPUTED rather than pinned. *Bound:* **scope: this does NOT settle 'PO-7' and the CR figure is not offered as a prediction – it uses the initial datum the corpus's own text describes (flat in k, common phase, VELOCITIES ZERO) and c54.164 established the

figure moves with that choice. What changed is that it now moves against something. NO FIGURE FROM THE ABANDONED ACOUSTIC FORK IS USED; that fork's own reproduction gate fails in this tree, and this instrument disagrees with its 201.3 by 15%.** *Run:* `python3 receipts/P15_CR_cosmology/P15_two_arm_control_and_guard.py`

**P15R64** `P15_the_driving_shift_by_subtraction` [OK]
*the acoustic sector* (**THE DRIVING SHIFT MEASURED MODEL-FREE BY SUBTRACTION: LCDM's ACOUSTIC PHASE IS FLAT IN k TO 2.1%, CR's VARIES BY 3.9x – AND THE SCALING IS kˆ-0.62, NOT THE kˆ-2 THE ABANDONED FORK'S MECHANISM REQUIRES.** Built for front #5 on the c54.168 instrument. Each k integrated TWICE on the same equations, driven and undriven, differing in one multiplicative switch and nothing else. Q(k) = k INT c_s deta/pi at the first acoustic turnover of Theta_0. **PART 1 the undriven column is the CALIBRATION and is MEASURED: 0.9968-1.0003 on both arms, where a free oscillator must give exactly 1.** PART 2/3 LCDM's Q_driven runs 0.791 -¿ 0.774 over k = 0.045-0.16 (2.1% spread) because every mode crosses under radiation, where the potential decays on the sound-crossing time by construction; CR's runs 0.457 -¿ 0.117, a factor of 3.9, because there is no radiation era to cross and c54.167 showed the pre-onset interval is scale-free. PART 4 **a power-law fit gives kˆ-0.62, collapsing the data to a factor of 1.2 where kˆ-2 spans a factor of 16.** The fold reports Q kˆ2 constant and its seam-transient mechanism (3H/kˆ2) requires it. **The phenomenon reproduces here independently; the exponent does not.** Recorded as an inter-instrument disagreement, not resolved.). *Computes:* the undriven calibration asserted within 0.01 of 1 on BOTH arms (this is the file's own gate); LCDM's spread asserted under 6%; CR's variation asserted over 2x; the fitted exponent asserted inside (-1.2,-0.3); **Q kˆ2 asserted to span MORE than a factor of 4 – i.e. the fold's constant-Q-kˆ2 asserted FALSE here**; and the fitted exponent asserted to collapse the data better than kˆ-2, so the fit is not vacuous. *Bound:* **scope: two corrections were forced by this file's own calibration and are recorded because each would have been a false result – (i) reading Q on Theta-hat rather than Theta_0 calibrated at 0.0003, since DR switches the couplings into the fluid but not Psi's own evolution; (ii) the driven LCDM column then calibrated at 0.001 on a decaying-mode transient two grid points from the start, excluded by requiring k eta ¿ 1, which is the DEFINITION of an acoustic turnover and removes nothing on the CR arm. No figure from the abandoned fork is used.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_driving_shift_by_subtraction.py`

**P15R65** `P15_which_coupling_carries_the_k_dependence` [OK]
*the acoustic sector* (**THE k-DEPENDENCE LIVES IN THE POTENTIAL'S GRADIENT, NOT IN ITS DECAY – WHICH IS THE REVERSE OF WHERE A RELAXATION TIMESCALE WOULD PUT IT.** Front #5, on the c54.168 instrument against its 1.000 calibration. The potential enters the plasma TWICE: as '4 Phi'' in continuity (its rate of change feeding the density – **the decay channel**) and as 'kˆ2 Psi' in Euler (its gradient driving the velocity). Switched independently on the same integration. PART 1 **flat LCDM's two couplings are each FLAT in k and they OPPOSE** – continuity alone puts the turnover at Q = 1.23-1.28, LATER than a free half-period, the gradient at 0.73-0.87, earlier, composing to 0.77-0.86. **So the universality of the standard driving shift is a CANCELLATION between two flat terms, not one term's signature.** PART 2/3 on the geometric rate there is no cancellation left – both advance (0.11-0.18 and 0.06-0.57) and they add – **and the k-dependence sits in the GRADIENT: continuity varies 1.6x at kˆ-0.17, gradient 8.9x at kˆ-1.04.** PART 4 and **neither term alone is the answer**: the driven exponent -0.62 lies BETWEEN the two, so the scaling is their interference.). *Computes:* LCDM's continuity asserted ABOVE 1 and its Euler BELOW 1 (they must oppose) and all three driven exponents asserted flat to 0.15; CR's continuity asserted nearly flat (|exp| ¡ 0.35) and its Euler steep (|exp| ¿ 0.8) and the Euler term asserted to be the steeper of the two – **reversing that assertion would reverse the finding**; both CR couplings asserted below 1 (they add); the driven exponent asserted strictly BETWEEN the two and distinct from each by 0.2; and the undriven calibration asserted within 0.01 of 1 on both arms, which is the file's own gate. *Bound:* **scope: this localises the k-dependence to a coupling and shows the standard shift's universality is a cancellation this rate does not have. It does NOT name the process that makes the gradient coupling run as kˆ-1, and computes no peak position. It closes the abandoned fork's stated mechanism as an explanation of THIS measurement – a relaxation on 3H/kˆ2 acts through the decay channel, and the decay channel is measured flat.** *Run:* `python3 receipts/P15_CR_cosmology/P15_which_coupling_carries_the_k_dependence.py`

**P15R66** `P15_the_gradient_coupling_in_closed_form` [OK]

*the acoustic sector* (**THE GRADIENT COUPLING IN CLOSED FORM: THE TURNOVER IS AT A FIXED MULTIPLE OF 3H/kˆ2, AND SINCE Q ALREADY CARRIES ONE FACTOR OF k THAT GIVES Q ˜ 1/k, NOT 1/kˆ2.** Front #5, closing the question c54.170 left. **The derivation is four lines.** (1) Phi's own equation carries a damping term -kˆ2 Phi/3H, so Phi relaxes on tau = 3H/kˆ2. (2) With continuity off the oscillator is Theta_0" + c_sˆ2 kˆ2 Theta_0 = -(kˆ2/3) Psi, and **the kˆ2 in the forcing's amplitude is cancelled by the kˆ-2 in its duration – the delivered impulse is k-INDEPENDENT**. (3) So the first turnover is at the moment the impulse has been delivered, at a fixed multiple of tau: kappa = 1.79, constant across a factor of eight in k. (4) Q = k INT c_s deta/pi already carries one k, so **Q = 3 kappa c_s H/(pi k)**. **PREDICTED Q k = 0.010403/Mpc AGAINST A MEASURED 0.010372 – 0.30% – with nothing fitted but kappa; and the closed form reproduces Q itself to 1.6% across the sub-horizon band.** ⇒ **The abandoned fork's TIMESCALE is right and its EXPONENT is out by exactly one factor of k – a bookkeeping error in its own mechanism rather than a disagreement about physics.** And the continuity channel is flat for the complementary reason: '4 Phi'' feeds the density with the potential's TOTAL change, INT Phi' deta = -Phi_0, the same however fast the relaxation runs. **So the observed kˆ-0.62 is not a power law of anything – it is the crossover between a flat channel and a 1/k one.**). *Computes:* kappa asserted constant to 10% across the sub-horizon band; **the flattest grouping asserted to be Q kˆ1 and not Q kˆ0 or Q kˆ2 – if that assertion reverses, the derivation is wrong**; the predicted coefficient asserted within 5% of the measured one; the closed form asserted to reproduce Q to 5% across the band; the continuity channel asserted flattest UNGROUPED and asserted flat to within a factor of 2. *Bound:* **scope: this NAMES the process, so front #5's question is answered. It does NOT make the first peak agree with the sky – the 23% deficit against a 1.5% control floor stands unchanged. What changed is that its cause is derived rather than described. No figure from the abandoned fork is used; its timescale is confirmed independently here and its exponent corrected.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_gradient_coupling_in_closed_form.py`

**P15R67** `P15_where_the_likelihood_sits` [OK]

*the acoustic sector* (**WHERE THE LIKELIHOOD ACTUALLY SITS: IT CANNOT ARBITRATE, AND THE INSTRUMENT'S OWN CONTROL IS WHY.** Discharges 'L-147', fenced since it opened on three conditions – falsifier fixed before any data is touched, a control reproducing a known number first, and not until the two internal routes agree. **All three met** (the routes collapsed at c54.164-168), so the fence came down. **F1** the pipeline is wired: the banked CAMB flat-LCDM best fit returns chiˆ2 = 206.4 over 215 plik_lite TT bins, chiˆ2/dof = 0.96. **F2** the instrument's own floor, measured through the SAME likelihood on the SAME bins: this fork's LCDM arm costs +18786 over CAMB. **F3** the only readable quantity, both arms of one instrument: CR - LCDM = +84160. **F4** the pre-registered rule returned 'suggestive, not a measurement discrepancy'. **F6 – AND THE PRE-REGISTERED RULE WAS DEFECTIVE, WHICH RUNNING IT IS WHAT SHOWED.** It is a RATIO test and never asks whether the floor is small in ABSOLUTE terms. It is not: the control arm sits at **chiˆ2/dof = 103** against CAMB's 0.96, because Planck's binned errors are a few tenths of a per cent and this instrument's heights carry 13-26%. **Both arms are far outside the regime where plik_lite discriminates, so the honest verdict is the WEAKER one: the likelihood cannot arbitrate.** **!! That correction happens to FAVOUR the construction and is flagged for it** – it would apply identically had the CR arm come out better, and it rests on a height floor measured at c54.168 before any data were opened. **What survives as fact: on one instrument, same bins, same fitted amplitude, the CR arm costs 5.4x the LCDM arm's chiˆ2. An ordering, not a p-value.**). *Computes:* the banked LCDM fit asserted at chiˆ2/dof in (0.85,1.15) BEFORE anything else is read; every covered bin asserted non-empty (the keep mask); the instrument floor asserted POSITIVE (this instrument cannot beat a five-parameter CAMB fit, and if it appeared to, the binning would be wrong); the CR arm asserted to cost more than the control on the same instrument; and **F6's premise asserted – if the control arm ever drops below chiˆ2/dof = 3 the PART must be rewritten because the likelihood then CAN arbitrate.** *Bound:* **scope: plik_lite TT only – no polarisation, no ell ¡ 30, no lensing, no external priors, a third-party implementation rather than clik (PROVENANCE.md), and 185 of 215 bins because the instrument reaches ell = 2000. The restriction to covered bins is done on the COVARIANCE and re-inverted, which MARGINALISES over the missing multipoles; dropping rows of the Fisher matrix would CONDITION on them. The banked r1934 CR figure (397.1 vs 206.4) applied the derived DAMPING

suppression to CAMB's LCDM spectrum – it tested damping on LCDM's PEAK POSITIONS and could not have seen the position result, so it answers a different question and is SUPERSEDED here. 'PO-7' stays protected: this reports a number and does not convert it into a verdict.** *Run:* `python3 receipts/P15_CR_cosmology/P15_where_the_likelihood_sits.py`

**P15R68** `P15_the_line_of_sight_transfer` [OK]
*the acoustic sector* (**THE TRANSFER REBUILT ON A LINE-OF-SIGHT INTEGRAL: THE CONTROL'S FIRST-PEAK POSITION ERROR FALLS FROM 1.66% TO 0.16%, AND THE CR DEFICIT DOES NOT MOVE AT ALL.** Front #2. Until c54.173 the source was evaluated at a SINGLE eta_rec – an instantaneous last scattering. **The visibility g = tau' eˆ-tau peaks at eta = 280.75 with FWHM 37.8 Mpc, a tenth of a wavelength at the first peak.** Rebuilt as Delta_l = INT S j_l deta with S = g(Theta_0+Psi) + eˆ-tau(Phi'+Psi') + (1/kˆ2) d/deta[g theta_b]. PART 1 the control goes 1.66% -¿ 0.16% in ell_1/ell_A: **sub-per-cent POSITIONS, which is what the likelihood needed of them.** PART 2 **and the CR arm is unchanged to four figures, 0.5703 on both transfers, while the control moved 1.79%. The deficit stands at 21.9% against a 0.16% floor – 134 floors.** The most suspect part of the instrument was replaced wholesale and the number did not follow it down. PART 3 **the heights are NOT fixed and the target is NOT met**: P1/P2 -13%, P1/P3 -34%, a residual growing with multipole and pointing at the damping tail (no polarisation term; a tight-coupling closure that fails where the visibility is). Stated as a partial result. PART 4 one correction was a DERIVATION: the Doppler term carries 1/kˆ2 not 1/k, since v_b = theta_b/k. **With 1/k the second peak sat at 516 and P1/P2 at 3.15; with 1/kˆ2, 532 and 1.93 – and ell_1 was 220 either way, so an instrument validated on the first peak alone would have carried the error.**). *Computes:* the control's position error asserted sub-per-cent AND asserted improved by 5x or more; the CR arm asserted to move less than 1% under the rebuild; the deficit asserted at 20+ floors; **the heights asserted NOT sub-per-cent – if they ever are, the PART is stale and front #2 should be struck**; and the corrected Doppler power asserted to move the SECOND peak toward the sky, which is what makes it a fix rather than a tuning. *Bound:* **scope: this delivers sub-per-cent POSITIONS and leaves the HEIGHTS at 13-34%, so the likelihood still cannot be run (c54.172). The remaining error is a high-multipole problem and neither of its two named causes is a CR question – both are checked against the LambdaCDM arm first. Late ISW is truncated at eta = 4000, which affects only ell below ˜30 and is outside both the grid and plik_lite's TT range.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_line_of_sight_transfer.py`

**P15R69** `P15_the_derived_diffusion_damping` [OK]
*the acoustic sector* (**THE HEIGHTS FROM 13-34% TO 2-3%: THE DIFFUSION DAMPING PUT BACK, DERIVED ON EACH ARM'S OWN OPACITY AND RATE.** Front #2. **Diagnosed from the SIGN**: at c54.173 both height ratios were LOW, meaning peaks 2 and 3 too HIGH, worse at higher multipole – under-damping. **The hierarchy's tight-coupling closure switches off at tau' = 0.6/Mpc and the visibility peaks at tau' = 0.06/Mpc, ten times lower, so the photon quadrupole and the baryon slip were absent through the WHOLE of last scattering and the line-of-sight path carried no Silk damping at all.** The pre-LOS path hid this behind a hardcoded r_D. PART 1 the standard tight-coupling integral 1/k_Dˆ2 = INT [Rˆ2/(1+R) + 8/9]/[6(1+R)tau'] deta, evaluated on this arm's own background: **r_D = 6.58 Mpc against the 6-7 a Boltzmann code gives. Nothing fitted.** PART 2 **and a DOUBLE COUNT had to be removed**: with the hierarchy's own damping left in over tau' ¿ 0.6 AND the integral covering the same range, the control came out OVER-damped (+6.7%, +17%). **The sign flips either side of the correction, which identifies a double count rather than a wrong coefficient.** PART 3 heights **-12.9%/-33.5% -¿ -2.8%/-2.1%**, on a control whose positions were already sub-per-cent. PART 4 **and the likelihood's floor falls 103 -¿ 23 in chiˆ2/dof, a factor of 4.6, WITHOUT the verdict turning over** – 'L-147''s answer stands, and the floor moving while the verdict does not is what measures how far the instrument still has to go.). *Computes:* r_D asserted inside 5.5-8 Mpc; tau' at the visibility peak asserted BELOW the gate (PART 1's premise); **the sign asserted to flip either side of the correction – if it does not, it is a coefficient error and not a double count**; the single-mechanism heights asserted within 5% and asserted better than BOTH the no-damping and the double-counted runs; the floor asserted to fall by 3x or more; **and the control asserted to remain ABOVE chiˆ2/dof = 2 – if it ever drops below, the likelihood CAN arbitrate and 'L-147''s verdict must be revisited.** *Bound:* **scope: this does not reach sub-per-cent heights and does not make the likelihood answerable. The residual is still worst at high multipole: no polarisation term in the source, and a photon sector with no free-streaming hierarchy after decoupling. Neither is a CR question.**

*Run:* `python3 receipts/P15_CR_cosmology/P15_the_derived_diffusion_damping.py`

**P15R70** `P15_the_height_target_was_below_the_resolution_of_its_own_statistic` [OK]
*the acoustic sector* (**THE HEIGHT TARGET WAS QUOTED AGAINST A SKY NUMBER WITH A 3.4% ERROR BAR THAT WAS NEVER WRITTEN DOWN, AND chiˆ2 – THE STATISTIC THAT CAN SEE – PREFERS A COEFFICIENT IT IS NOT ALLOWED TO CHOOSE.** Front #2. Since c54.168 every height claim here (-13%, -34%, 2-3%, -0.05%) compared P1/P2 and P1/P3 to two bare numbers, 2.217 and 2.277. **PART 1** the numerics are cleared first: 1.4x the modes, 1.3x the k-range and 1.8x the eta-sampling move the ratios by 0.09% and 0.37%, and the 2%/7% swings that prompted the check came from LMAXL=1300 truncating the k integral under the THIRD peak. **PART 2** read off the plik_lite bandpowers with the published covariance – the full 2x2 block, not a diagonal estimate – **the sky's own ratios are 2.256 +- 0.077 and 2.280 +- 0.074: 3.4% and 3.2%.** So the 13% and 34% of c54.173 were four and ten sigma and stand; **the 2-3% of c54.175 are about one sigma and do not.** *Caught by bookkeeping and not by suspicion: X_data is binned C_l and not D_l, so peak-finding on it returned 464/761/1085 instead of 221/527/815.* **PART 3** a scan of one multiplier on 1/k_Dˆ2 has P1/P2 selecting 1.094 +- 0.184 and P1/P3 selecting 0.897 +- 0.055 – **a 1.03 sigma split, so my own 'no single scale fits both, the residual is the envelope's SHAPE' is WITHDRAWN at its own error bar rather than published.** **PART 4** chiˆ2 over 185 bins does separate them: the minimum sits at 0.865 against 8/9's 0.859, and 16/15 costs 1123 units, seven inflated sigma. **The coefficient is changed back to 8/9 and NOT because it fits better** – neither value has ever been derived here (C8_diffusion_length names the ambiguity in its own text), and a remembered number is not a derivation any more than a fitted one is. **PART 5** L-147's pinned chiˆ2 came from a script in /tmp; 'chi2_of_spectrum.py' puts that machinery in the corpus and returns 4170.2 and 50675.2 exactly.). *Computes:* converged spread asserted under 1% and the truncated run asserted separated from it; the sky's recomputed ratios asserted within 1 sigma of the bare numbers they replace; the DAMPX multiplier asserted to reproduce the explicit POLC=8/9 run to 0.2%; **the withdrawal asserted: if the split were 2 sigma or more the file FAILS for retracting a claim that in fact holds**; the chiˆ2 minimum asserted at the 8/9 multiplier within 5%; both pinned values asserted recomputed to 0.5%; and an assert that fires if the control ever reaches chiˆ2/dof ˜ 1, since this file would then need rewriting rather than patching *Bound:* **scope: this corrects a MEASUREMENT PROTOCOL, not the construction, and it is a correction to my own work. It supplies no derivation of the shear coefficient – that is named as owed and is front #2's first item: the free-streaming photon hierarchy with the polarisation multipoles. And the 3.2-3.4% is the single-bandpower-bin resolution; a full-curve fit determines the peaks far better, which is exactly why chiˆ2 replaces the ratios rather than joining them.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_height_target_was_below_the_resolution_of_its_own_statistic.py`

**P15R71** `P15_the_shear_coefficient_derived_not_remembered` [OK]
*the acoustic sector* (**THE DIFFUSION-DAMPING SHEAR COEFFICIENT, DERIVED FROM THE POLARISED PHOTON HIERARCHY INSTEAD OF REMEMBERED OR FITTED – AND ADOPTING IT COSTS THE CONTROL 1123 UNITS OF chiˆ2.** Front #2, closing the item c54.176 named as owed. 'C8_diffusion_length' has named 'the 8/9 coefficient against the 16/15 polarization-corrected one' in its own text since it was written, and neither had ever been derived here. **METHOD** two integrations of the same oscillator on the same background: the perfect tight-coupled fluid, undamped by construction, against photons carried to l = 24 with Thomson scattering, the polarisation multipoles evolved alongside and the baryons on their own velocity. **Their ratio is the damping and nothing else – the (1+R)ˆ(1/4) WKB prefactor, the drift of c_s and the R-dependence of the oscillation are common to both and divide out identically.** Gravity is off in both, which is the definition of the object rather than an approximation. **PARTS 1-4** with G = 0 the l=2 term is -(9/10)tau'F_2, the textbook unpolarised closure, and C = 0.8852 against 8/9; with polarisation C = 1.0618 against 16/15; **and the polarisation correction – the quantity actually in dispute – is a factor 1.1996 against 6/5, three parts in ten thousand.** C is read LOCALLY between consecutive extrema and extrapolated in (k/tau')ˆ2, so the kˆ2 scaling is measured rather than imposed. **PART 5** 16/15 is adopted and the control's floor goes 22.5 -¿ 28.6. **The fit's preference for 8/9, measured and refused at c54.176, is refused again: a coefficient chosen because it fits is a fitted parameter whatever it is called.** **PART 6** and nothing downstream moves – sec:envelope-consequence's 10.8% goes +10.83% -¿ +10.87%, because the coefficient nearly cancels in a ratio of two rates.). *Com-*

*putes:* both intercepts asserted within 2% of the leading-order values and pinned to ˜1% in the gate; **the polarisation correction asserted within 0.5% of 6/5 and pinned to a tenth of a per cent – it is the number the file exists to produce**; the scatter about the line in (k/tau')ˆ2 asserted below 0.01, so a method that destroyed the extrapolation fails rather than shifting the intercept quietly; the sign of the finite-(k/tau') correction asserted, and the extrapolation asserted SHORT (under 2% of the coefficient swept over the fitted range); **the chiˆ2 cost asserted POSITIVE – if the derived coefficient stopped costing chiˆ2 the file's whole point would need restating**; and the C8 figure asserted to move less than a tenth of a point, so adopting the coefficient cannot silently shift a published number *Bound:* **scope: this derives the coefficient in the WKB damping integral and NOT the damping envelope's shape. It supplies the damping half of the polarisation physics only; the SOURCE half – g*Pi/4 and (3/4kˆ2)(g*Pi)" with Pi = Theta_2 + ThetaˆP_0 + ThetaˆP_2 – needs the hierarchy carried in the spectrum instrument and is what front #2 still owes. And one assertion of mine fired on its own data: I demanded both extrapolation slopes be of order unity and they are -0.150 and -0.942; the gate now checks that the extrapolation is SHORT, which is the thing that actually matters.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_shear_coefficient_derived_not_remembered.py`

**P15R72** `P15_the_photon_hierarchy_in_the_transfer` [OK]
*the acoustic sector* (**THE PHOTON BOLTZMANN HIERARCHY WITH POLARISATION CARRIED INSIDE THE TRANSFER: THE CONTROL'S chiˆ2/dof FALLS 28.6 -¿ 7.1 AND THE FIRST FOUR PEAKS LAND AT 220/540/812/1124 AGAINST THE SKY'S 220.6/538.1/809.8/1147.8.** Front #2. Photons as multipoles to l = 24 with the polarisation hierarchy alongside, the baryons on their OWN velocity and their OWN density, and the source carrying g*Pi/4 and (3/4kˆ2)(g*Pi)". **THE HANDOVER MAKES A DOUBLE COUNT IMPOSSIBLE RATHER THAN MERELY ABSENT**: the fluid runs to tau' = 3/Mpc and the envelope is FROZEN there, so envelope and multipoles have disjoint supports – and only 4.2% of 1/k_Dˆ2 has accumulated at the handover, so 96% of the damping is dynamic. **PART 1** the second peak is 0.35% out and the third 0.27%, where the fluid had them 2.6% and 0.7% low. **PART 2** chiˆ2/dof 28.6 -¿ 7.1, and 20.0 -¿ 4.2 over the range where the wavenumber integral is not starved. **PART 3** and c54.177's prediction is checked against the number it predicted: **running the same hierarchy with ONLY the two polarisation source terms dropped – exactly the half an envelope has – gives 2022 against 1320, so they are worth Delta chiˆ2 = 702, 63% of the 1123 that adopting the derived 16/15 cost.** **PART 4** and half the baryon split is WORSE than none: own density while still sharing the photons' velocity moved chiˆ2/dof 133 -¿ 161 at development settings; both halves give 15.1 -¿ 7.1. *Twice now a physical effect taken in half has been worse than not taken – it is not a small error but a different and inconsistent model.* **PART 5** doubling the hierarchy depth moves chiˆ2 by 0.00% and doubling the handover opacity by 1.5%. **PART 6** and the CR arm is at ell_1/ell_A = 0.5703 for the SIXTH instrument state, while the control improved fourfold.). *Computes:* every peak asserted no FURTHER from the sky than the fluid transfer's; the chiˆ2 improvement asserted to be at least a factor of two and pinned at 7.14 / 4.17; both height ratios asserted within 2 sigma of the sky's own covariance-derived values; **the polarisation source terms asserted to recover between 40% and 120% of c54.177's 1123 – outside that range c54.177's statement could not be said to have predicted this and the file FAILS**; the full baryon split asserted to beat the shared-density run; the LG and handover sensitivities asserted below 0.1% and 5%; the CR arm asserted unmoved to the fourth decimal; and an assert that fires if the control ever reaches chiˆ2/dof ˜ 1, since L-147 would then have to be REOPENED rather than this file's closing line printed *Bound:* **scope: this is the LambdaCDM control and the CR arm on one instrument, and it converts nothing into a verdict – 'PO-7' is protected and 'L-147''s answer is unchanged, because a control at seven times a fit still cannot certify what it is compared with. What remains is accuracy and no longer a named missing SECTOR: reionisation, neutrino mass, lensing, and the wavenumber range. And the aliasing gate was found MISSING from the line-of-sight path, which has been the default since c54.173; it cost a development run a first peak at 196 instead of 220 with nothing said, and is now on both paths.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_photon_hierarchy_in_the_transfer.py`

**P15R73** `P15_what_is_left_is_lensing_and_not_reionisation` [OK]
*sec:refit-bound* (**BOTH OF FRONT #2's NAMED NEXT STEPS MEASURED BEFORE EITHER WAS BUILT, AND ONE IS WORTH EXACTLY NOTHING.** **PART 1** above ell ˜ 40 reionisation's whole TT effect is the constant screening exp(-2 tau), and the amplitude here

is fitted in CLOSED FORM – A = (m F d)/(m F m) is homogeneous of degree -1 in m, so m -¿ s m returns A/s and the residual is IDENTICAL. **d(chiˆ2) = 0 exactly, shown at tau = 0.054, 0.10 and 0.30, zero to machine precision at all three.** *The degeneracy is a property of the STATISTIC, not of the size of tau.* **PARTS 2/3** a Gaussian smoothing in ell, width scanned, minimises at chiˆ2 = 921 against the control's 1320 – and **a width that GROWS with ell beats a constant one by 1.8x in d(chiˆ2), 400 against 225, each at its OWN optimum so the comparison is of SHAPE**. Lensing smooths more at high ell because a deflection of fixed angle subtends a larger fraction of an acoustic wavelength, and that is the signature the data select. **PART 4** the width is FITTED, so the 400 bounds what a DERIVED calculation has to play for and is not a measurement of the lensing amplitude – **and ˜921 of the 1320 survives it, so lensing is the next build and is NOT the last one.**). *Computes:* the reionisation degeneracy asserted EXACT – if it ever moves chiˆ2 the amplitude is no longer fitted in closed form and PART 1's algebra no longer describes the pipeline; both smoothing optima pinned; **the growing width asserted to beat the constant one by at least 1.5x, which is the SHAPE claim and the reason the word is 'lensing' rather than 'some smoothing'**; what survives asserted above 60% of the baseline, so a rewrite is forced rather than a patch if a fitted smoothing ever accounts for most of the residual; and an assert that fires if a smoothing alone would take the control to chiˆ2/dof ¡ 2, since L-147 would then have to be revisited *Bound:* **scope: this measures what two candidate builds can be worth on THIS statistic. It is not a result about the sky, not a measurement of the lensing amplitude, and the control does not improve – the smoothing width is fitted. Reionisation is STRUCK from the front's list rather than deferred, and the condition that would undo the strike is written down: a comparison reaching ell ¡ 30, a second spectrum, or a fixed amplitude.** *Run:* `python3 receipts/P15_CR_cosmology/P15_what_is_left_is_lensing_and_not_reionisation.py`

**P15R74** `P15_derived_lensing_on_the_lcdm_arm` [OK]
*sec:refit-bound* (**THE DERIVED LENSING c54.181 TEED UP, RUN WITH NO FREE WIDTH – AND IT COMES IN UNDER THE FITTED BOUND, WHICH IS WHAT A DERIVATION MUST DO.** ΛCDM's own non-perturbative lensed/unlensed ratio applied to the c54.178 LCDM arm takes the control 1320 -¿ 989, d(chiˆ2) = 331 – UNDER the 400 the c54.181 fitted smoothing bounded, and concentrated in the damping tail (ell 1100-1500) where lensing fills the troughs. The acoustic peaks move 0.0%. The corpus's own first-order Hu (2000) kernel, run on the SAME Cˆphiphi so only the ORDER differs, OVERSHOOTS to 508 because it BREAKS DOWN in the tail – a spurious +13% at ell=1900 against the full operator's +6.5% – so the derived number is the full operator's 331, not the first order's 508. Validated on CAMB's own LCDM: 615 unlensed -¿ 186 lensed (chiˆ2/dof ˜ 1). ˜989 of 1320 survives, so lensing is real and NOT the last build.). *Computes:* lensed CAMB LCDM asserted at chiˆ2/dof in (0.75,1.25) so the operator and pipeline are validated before anything is read off the arm; the derived lensing on the arm asserted in (280,385) AND strictly under the c54.181 fitted bound of 400, which is the defining property of a zero-width derivation against a fitted upper bound; the first-order kernel asserted to OVERSHOOT the full operator by more than 80 in d(chiˆ2) via the tail artefact, a live check that LENS_correction is first order and that the tail is where it fails; and the peak shift asserted below 0.3%. *Bound:* **scope: the lensing operator is ΛCDM's, imposed not fitted, so this is not a measurement of the lensing amplitude and the CR arm does not fit – it measures how much of c54.181's fitted-smoothing headroom a genuine zero-parameter lensing recovers (most of it), with the shortfall against the fit being the fit's free width. The residual ˜989 is the instrument's transfer inaccuracy, not lensing's: CAMB's UNLENSED LCDM already sits at 615 on these same bins. 'PO-7' stays protected.** *Run:* `python3 receipts/P15_CR_cosmology/P15_derived_lensing_on_the_lcdm_arm.py`

**P15R75** `UNC_error_budget` [OK]
*sec:refit-bound* (UNC — QUANTIFYING WHAT WE DO NOT KNOW. The prediction with an error budget. Corrections folded in first: * the recombination WIDTH scales with the rate (A.22) – so the thickness contributes ˜nothing to the RATIO and the signature is carried by r_D essentially alone; * the l_D anchor affects only the ABSOLUTE r_D and cancels in the ratio – it carries NO uncertainty into the prediction;). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/UNC_error_budget.py`

**P15R76** `P15_the_residual_is_contrast_and_the_lensing_potential_is_derived` [OK]
*sec:refit-bound* (**WHAT THE CONTROL'S REMAINING chiˆ2 IS MADE OF, AND THE LENSING POTENTIAL BUILT TO ATTACK IT.** **PART A** templates projected through

the same inverse covariance the chi^2 uses: **positions remove 1.4 of 1320 – 0.1%, so the peaks are where the sky puts them and that is not where the residual lives**; a smooth envelope 15%; **the peak-trough oscillation at FIXED position 38%, with the model's contrast 13.1% TOO HIGH** – which is what lensing reduces; and **53% in NEITHER, so a perfect lensing calculation leaves about half**. *Fitted templates, so each share is an upper bound on what the effect can claim.* **PART B** the line-of-sight solve stops at z = 12.8 but the lensing kernel peaks at z ~ 3.3, so Phi is carried on by the background's own growth factor D prop. H INT da/(aH)^3 – exact in linear theory, no transfer function imported, k-dependence from this instrument's own Phi. **And the normalisation is the amplitude the TEMPERATURE comparison already fits, so lensing enters with NO new parameter.** Limber checked against an EXACT projection (5% over l = 5-200); the deflection maximum is BROAD, within 10% over l = 12-42; **rms deflection 2.69 arcmin with nothing tuned to it.**). *Computes:* positions asserted below 2% of the residual, so the diagnosis reorders loudly if they ever rise; the contrast excess pinned to 13.1% +- 2 points, since that is what the convolution must remove; what survives BOTH template sets asserted at 53% +- 5, **so a build that forgets how much remains fails here**; Limber's departure asserted below 10%, since the convolution will use it; the deflection maximum asserted BROAD; and the absolute rms deflection pinned at 2.692 arcmin, which is the zero-parameter check on the normalisation *Bound:* **scope: PART A's shares are FITTED-template upper bounds, not measurements of physical effects, and the two sets overlap by ~7 points. And PART B's external agreement is weaker than it looks: the ~2.7 arcmin and the ~1e-7 deflection power it matches are RECALLED, not derived here and not checked against a source this corpus holds – this programme was bitten twice in three revisions by exactly that (the 16/15 coefficient at c54.176, and a Limber suspicion in this very file which was also recollection and also wrong). What PART B establishes is INTERNAL consistency; the external agreement is corroboration at the strength of a memory.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_residual_is_contrast_and_the_lensing_potential_is_derived.py`

**P15R77** `P15_the_phase_is_the_driving_and_the_undriven_arms_agree` [OK]

*sec:refit-bound* (**WHY 0.62 pi: THE DRIVING — AND THE ATTRIBUTION IS COMPLETE. WITH EVERY COUPLING TO THE POTENTIAL REMOVED THE TWO ARMS' ACOUSTIC PHASES AGREE TO 0.013 OF l_A.** Front #2, lead 'L-507'. c54.191 closed with *"an acoustic phase shift is a computable consequence of the driving, so 'why 0.62 pi' has an address. This file does not have it."* **The instrument has carried the switch since c54.170** — 'NODRIVE=1' removes every coupling to the potential at every site (the 4·Phi' in the photon, neutrino and CDM continuity equations and the $k^2$·Psi in the Euler equations) while leaving Phi's own evolution alone, that being the background and not a coupling. *Both arms run through it at production depth here, which no revision had done.* **PART 1 UNDRIVEN THE TWO ARMS ARE THE SAME OSCILLATOR:** intercepts **−0.135 and −0.149 of l_A — agreeing to 0.013** — and both slopes l_A to a part in a thousand. *Two constructions with different rates, sound horizons, starting redshifts and initial data give the SAME acoustic series once nothing drives it.* **PART 2 AND THE DRIVING IS WHAT SEPARATES THEM:** switched on it supplies **−0.127 to the control and −0.729 here, a factor of 5.7**; the driven discrepancy is 0.615 and the undriven 0.013, so **the driving accounts for 98% of it**. **PART 3 THE SAME SWITCH ACCOUNTS FOR THE SPACING:** undriven both slopes are 1.000 and 1.001 of l_A; driven, the control's is 1.003 and this construction's 0.976 — **so the 2.4% the spacing was short is the driving as well, and not a second effect**. ⇒ ***THE WHOLE ACOUSTIC DISAGREEMENT IS ONE COUPLING BEHAVING DIFFERENTLY — and a coupling is a thing a theory says something about, where a ratio is not.*** ***PART-HEADLINE CORRECTED c54.195 ('L-508'): the 0.62 pi is the value at phi = 0. Across four seam phases at production depth the asymptotic phase spans 0.891 and the CONTROL'S VALUE LIES INSIDE that span, so it is not a prediction. The driving ATTRIBUTION stands; its promotion to "the whole disagreement" does not.***). *Computes:* **the UNDRIVEN agreement asserted under 0.10 of l_A — if the arms still differed with nothing driving them the driving would not be the whole story and PART 2's headline would be wrong**; the ratio of what the driving supplies asserted above 3x and pinned at 5.7 +- 1.5, without which PART 2 has no content; **the undriven SLOPES asserted at l_A to 1%, which is what makes PART 3's "not a second effect" true**; the driven discrepancy re-derived at 0.615 +- 0.05, tying this file to 'L-505' and 'L-506'; the driving's share asserted above 85%; and at least seven peaks per arm *Bound:* **scope: an ATTRIBUTION is not an explanation and it is not a verdict. What this establishes is that the acoustic disagreement lives in ONE COUPLING rather than in four items, a ratio, or an initial condition. IT DOES NOT SAY WHY the driving

is 5.7x stronger here — the obvious candidate is that this construction's modes begin already sub-horizon at the seam carrying an assigned amplitude and ZERO velocity, in a potential that is not the equilibrium configuration for that state, where the control's begin super-horizon in adiabatic equilibrium; **that is a HYPOTHESIS and this file does not test it**. The instrument carries the split that would: 'DRC' is the potential's rate of change feeding the density and 'DRE' its gradient driving the velocity, switchable separately — the next thing this front owes. 'F5' unsoftened; 'PO-7' protected; the conversion runs by 'F5''s stated procedure.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_phase_is_the_driving_and_the_undriven_arms_agree.py`

**P15R78** `P15_the_phase_moves_with_the_datum_and_only_the_spacing_does_not` [OK]
*sec:refit-bound* (**THE OBSERVER LINE'S ITEM 38 ANSWERED, AND THE ANSWER WITHDRAWS c54.190–191's HEADLINE. ACROSS FOUR SEAM PHASES AT PRODUCTION DEPTH THE ASYMPTOTIC ACOUSTIC PHASE SPANS 0.891 IN phi/pi — AND THE CONTROL'S 0.263 LIES INSIDE THAT SPAN.** Front #2, lead 'L-508'. *Item 38 (r2485): of the 23 datum-scan spectra ZERO reach peak 8, so the quantity carrying the disagreement could not be tested against the one freedom known to move things. It named the experiment — two or three seam phases at production depth. c54.191 supplied one; this supplies two more.* **PART 1** at phi = 0, pi/4, pi/2 and pi the fitted phi/pi is **0.878, 0.958, 0.066, 0.671** — *c54.191 read 34% closure from the two endpoints; **the interior is where the motion is***. **PART 2 SO 0.62 pi IS THE VALUE AT ONE READING AND NOT A PREDICTION.** *The attribution to the driving (c54.193) stands — the driving is what separates the arms AT A GIVEN READING — but a quantity the datum can move across the control's value is not a prediction of the construction.* **PART 3 THE HEIGHTS MOVE TOO:** P1/P2 runs **0.483 → 1.618** against the sky's 2.217 and the control's 2.197, and **never reaches it**. **PART 4 WHAT IS ROBUST IS THE SPACING:** 0.9629–0.9814 of l_A over the same four — **1.9%** — and tracking r_s at 98% of the acoustic rate under the fitted parameter. **PART 5 AND MATCHING THE PHASE DOES NOT FIX THE SPECTRUM**, which is what settles PART 2: the reading closest in phase (phi = pi/2, a phase gap of 0.197) has **chi^2/dof = 224 against the control's 3.71 on the same 185 bins — sixty times worse while agreeing on the phase**. ⇒ ***So the phase was never carrying the disagreement.***). *Computes:* **the control's phase asserted INSIDE the span the seam phase reaches — without which there is no withdrawal and c54.190–191 stand as written**; the span asserted above 0.3 and pinned at 0.891, which IS item 38's question; **the low-ell transient asserted present in ALL FOUR at more than 3x the control's — this is what establishes the peak INDEXING, and if a reading showed none the span could be an artefact of labelling**; the spacing's stability asserted under 6%, the one thing this file does NOT withdraw; the heights asserted never to reach the sky's; and **the phase-matched reading asserted above 10x the control — PART 5 is what makes the withdrawal a finding rather than a shrug** *Bound:* **scope: this is the THIRD withdrawal of this line's own headline in the span, and it is forced by the data rather than chosen. What it leaves is worse than c54.191 reported and truer: the acoustic SPACING is right and robust; the acoustic PHASE and the peak HEIGHTS both move substantially with a datum the corpus does not fix and neither reaches the sky at any reading tried; and no reading brings the spectrum within sixty times the control. **Where the disagreement lives is therefore not a single named quantity and this file does not supply one.** 'F5' unsoftened and it cuts both ways — a discrepancy resting on a free choice is not a framework verdict either. 'PO-7' protected; the conversion runs by 'F5''s stated procedure.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_phase_moves_with_the_datum_and_only_the_spacing_does_not.py`

**P15R79** `P15_the_locus_is_wrong_in_six_places_and_the_lint_cannot_see_the_worst` [OK]
*sec:perturbations, sec:coherence, sec:envelope* (**ROUTED ITEM 21 WORKED: SIX SITES SAID "BRANCH POINT" WHERE THEIR OWN RECEIPTS SAY "SEAM"/"ONSET", AND THE PHYSICS INVERTS BETWEEN THE TWO — SO A PROPOSITION TITLED "SUB-HORIZON" WAS TRUE AT ONE AND FALSE AT THE OTHER. AND THE LINT BUILT FOR THIS DEFECT COULD NOT SEE THE WORST OF THEM.** Front #2, lead 'L-509'; VEIN 'L-202'. **PART 1 THE INVERSION, RE-DERIVED FROM THE METRIC FUNCTION** rather than taken from the routing slip: normalised 1/aH is **1.000 at the seam and 0.051 toward r=0**, and with the inherited datum the horizon maximum sits at **r* = 1.5338 r_seam, just OUTSIDE the seam** — so the seam is on the rising branch and the acoustic modes are already inside there, which is a *derivation* of prop:subhorizon and not a numerical coincidence of z_onset. **PART 2 SIX SITES REPAIRED, PINNED BY TEXT AND NOT BY LINE NUMBER** — includ-

ing the subsection heading, *which contradicted its own proposition three lines below*, and the sec:envelope sentence *that claims to BE the derivation, with the locus backwards*. **One of the six was not on the routed item's list**; the lint found it. **The one true-as-written branch-point sentence — the neutrinos' decoupling, a TEMPERATURE ordering, flagged as a false positive by the finder in advance — survived the sweep.** **PART 4 AND THE GATE WAS VERIFIED AGAINST A CLEAN TREE, WHICH MEASURES NOTHING:** seeding prop:subhorizon's body back to "at the branch point" left 'check_loci' printing *clean* and exiting 0. *A proposition states its claim in the body and cites its receipt in the argument paragraph below, so the corpus's most load-bearing claim shape was the one shape a per-sentence scan is blind to — and it was the item's own first site.* ⇒ ***Theorem-body binding added; 11→12 bindings, +0 false alarms, motivating defect MISSED before and CAUGHT after.***). *Computes:* **the inversion itself asserted from the metric function — without it the substitution is cosmetic and every edit is unmotivated**; r*/r_seam re-derived to 1.5338 from the quartic rather than quoted; the six repaired strings asserted PRESENT and the six defective strings ABSENT, by text so a reintroduction fails this file; **the protected false positive asserted still present — a sweep that also took the one true sentence would be a different error with the same shape**; and **the recall hole asserted in BOTH directions — MISSED before the binder extension, CAUGHT after — so if the pre-extension scan ever starts catching it, this file says the hole was not real**; plus the declared exception asserted to LAPSE when its sentence is rewritten *Bound:* **scope: this is exposition and instrument, not physics — no number in the cosmology moves. Its warrant is 'THE_METHOD' §III: a framework a reader cannot follow cannot be excavated, and this one had a heading contradicting its own proposition. The completeness qualification added at sec:coherence is transcribed from 'C2_horizon_limits' STEP 6, which the paper's sentence contradicted; it is not a new physics claim. 'check_loci' remains a triage lint and not a CI gate — the contributor's call, not reversed here.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_locus_is_wrong_in_six_places_and_the_lint_cannot_see_the_worst.py`

**P15R80** `P15_the_filter_is_a_sound_speed_filter` [OK]
*sec:what-crosses* (**THE BRANCH-POINT SELECTION RULE'S EXPONENT HAS TWO FACTORS AND THE CORPUS VARIES ONLY ONE.** The damping is e^{-k c_s |Delta eta|}: 'sec:what-crosses' holds c_s at the plasma's value and varies k, reading a SCALE criterion (frozen against oscillating). **Hold k and vary c_s and the same inequality is a SPECIES criterion – and a pressureless component has c_s = 0 IDENTICALLY, so its exponent vanishes at EVERY wavenumber.** ** THAT MAKES THE MATTER SECTOR'S INHERITANCE EXACT RATHER THAN ADIABATIC: ** at c_s = 0 the segment's mode equation is psi" = 0, whose constant solution the kernel leaves untouched with no approximation, whereas the frozen-mode argument rests on horizon exit and degrades at small scales. The two agree wherever both apply; only the pressureless one is available everywhere. COMPUTED on the corpus's own segment: (1) |Delta eta| = 3.3197 (the recorded 3.32); (2) exact transfer-matrix transmission in BOTH k and c_s, reproducing -k c_s |Delta eta| to the digit for every constant c_s; (3) **ln T = 0 EXACTLY at c_s = 0 for k = 1, 10, 85.3, 1e3, 1e5 – uniform in k**; (4) at the first peak the exponent runs ~1.6e2 c_s/c, so the filter is opaque above c_s ~ 3.5e-3. ** CONSEQUENCE: a per-cent sound speed already gives e^-1.6 and ten per cent e^-16, so THE CROSSING CAN INHERIT PERTURBATIONS FROM A COLD SPECIES AND FROM NO OTHER – a sharp condition on what the progenitor may hand over. ** **(SCOPE: the arithmetic is what this receipt establishes; no species selection rule follows from it. The kernel acts on OSCILLATORY content and nothing arrives oscillating ('P16_every_mode_is_frozen_at_the_crossing'), so the exponent computed here has nothing to act on and the crossing is lossless for EVERY species.)**). *Computes:* the segment's conformal length asserted at 3.32; the plasma exponent asserted in (-160,-140); ln T asserted zero to 1e-9 at c_s = 0 across five decades in k *Bound:* ** THIS IS THE FIELD HALF ONLY. ** What happens to perturbations of a given species; the detailed WORLDLINE dynamics of a concrete model at the branch point is a separate computation and is not done. **And an honest bound**: on this segment the plasma exponent comes out -147 against the -152 the corpus records, a 3% difference from the sound-speed profile's discretisation – nothing here turns on it, the load-bearing result being the exact vanishing at c_s = 0, which is profile-independent *Run:* `python3 receipts/P15_CR_cosmology/P15_the_filter_is_a_sound_speed_filter.py`

**P15R81** `P15_the_exact_transmission_ratios_are_recomputed_and_the_offset_saturates` [OK]
*sec:lowl* (**THE LOW-MULTIPOLE PARAGRAPH REPORTED SIX COMPUTED NUMBERS AND NO RECEIPT COMPUTED THEM.** 'sec:lowl''s exact treatment states the segment

length and five exact/WKB ratios; the transfer-matrix composition producing them lived only in a storyboard script with no assertions and no 'receipts/' copy, so the numbers were uncited. Computed here and asserted: (1) the segment's conformal length 3.3233, the recorded 3.32; (2) EXACT transmission by composing constant-omega transfer matrices in the STABLE direction, against the WKB form on the same grid, giving 0.9255 / 0.9127 / 0.9008 / 0.8906 / 0.8892 at ell = 2, 3, 5, 15, 40 – the paragraph's 0.926 / 0.913 / 0.901 / 0.891 / 0.889 to three digits; (3) the band [0.889, 0.926], the paragraph's 7-11 per cent; (4) **the ratio DECREASES with ell and SATURATES near 0.889 rather than tending to 1**, so the residual is a systematic offset and not an adiabatic error, whose ordering runs the other way; (5) the composition converged, quadrupling the step count moving each ratio by under 1e-9; (6) **THE SEGMENT LENGTH'S MASS SCALING, which the same passage quotes** – the scale factor is A |sin(3s/2 alpha)|^{2/3} with A = (2 M alpha^2)^{1/3} and the segment's extent carries no M, so |Delta eta| scales as M^{-1/3} EXACTLY, giving 4.1871 alpha at M/2 and 15.4255 alpha at M/100 against the passage's 4.19 and 15.4.). *Computes:* the segment length asserted to two decimals, the five ratios to three, and the band, the monotonicity, the saturation and the convergence each asserted *Bound:* ** THE STEP COUNT IS A NUMERICAL QUESTION AND NOT A PHYSICAL ONE. ** The ratios are asserted to the three digits the paper quotes and the convergence is exhibited rather than asserted at a tolerance the paper does not use; the attribution question is 'SP_S5_the_wkb_residual_is_an_offset''s and is not revisited. ORIGIN: 'storyboard_receipts/LOWL_exact_transmission.py', whose constants and stable-direction construction are reproduced unchanged. *Run:* `python3 receipts/P15_CR_cosmology/P15_the_exact_transmission_ratios_a`

**P15R82** `P15_the_collapse_leg_is_scale_invariant` [OK]
*sec:transmission* (**THE COLLAPSE LEG IS A SCALE-INVARIANT BACKGROUND, AND P15 LISTS 'CR HAS NO INFLATIONARY SCALE-INVARIANT ATTRACTOR' AS AN ESTABLISHED SIGNATURE. THE ADJECTIVE WAS CARRYING THE SENTENCE.** For a background a ~ (-eta)^beta with z ~ a, the Mukhanov-Sasaki potential is beta(beta-1)/eta^2, so scale invariance requires beta(beta-1) = 2 – **a quadratic with TWO roots: beta = -1, de Sitter inflation (expanding), and beta = +2, a MATTER-DOMINATED CONTRACTION.** **And the corpus's own collapse leg is beta = 2 exactly:** its banked near-branch-point law r^3 = (9/2) M tau^2 with a = |r| gives d ln a/d ln tau = 2/3 and d ln eta/d ln tau = 1/3, hence a ~ eta^2 – forced by f -¿ -2M/r, the mass term dominating and the Lambda term vanishing, **which is the same pressurelessness 'A·4' used at c54.112 to show such a component crosses exactly.** The mode equation integrated from vacuum data on both backgrounds returns d ln P/d ln k = 0 to a part in 10^4. **So n_s = 1 is produced by the construction rather than inherited from the progenitor, and only the DEPARTURE from unity is owed.**). *Computes:* the two roots asserted [-1, 2]; both logarithmic slopes of the leg asserted exactly 2/3 and 1/3; both integrated spectra asserted flat to 1e-3 *Bound:* **bound, and it is the whole caution: this is the STANDARD contraction result (Wands 1999; Finelli-Brandenberger 2002) applied to a background the corpus DERIVES. It inherits that literature's known difficulty – whether the matching across the bounce selects the scale-invariant mode – and here the bounce is a branch point with an analytic continuation rather than a smooth bounce. THAT QUESTION IS NOT SETTLED HERE and is registered as 'L-169'** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_collapse_leg_is_scale_invariant.py`

**P15R83** `P15_the_continuation_is_diagonal` [OK]
*sec:predictions* (**THE CONTINUATION THROUGH THE BRANCH POINT ACTS DIAGONALLY ON THE TWO PERTURBATION MODES, SO THE SCALE-INVARIANT SPECTRUM DOES NOT REACH THE OBSERVER.** The falsifier on the companion result, named at c54.122 and run at c54.123. **The scale-invariant spectrum sits in the mode that DOMINATES the contraction (zeta ~ eta^-3); the mode that SURVIVES on the expanding side is the constant one, whose spectrum from vacuum data is n_s - 1 = 6.** Whether the first reaches the second is a connection problem, settled here two independent ways. **(i) PARITY: the exponents 2 and -1 differ by an integer so a logarithm could mix them – but sinh^(2/3) is EVEN in tau, which places its correction at eta^6 and steps the Frobenius recursion in SIXES past a resonance sitting at THREE. No logarithm.** **(ii) NUMERICS: continuing both solutions around the branch point, the diagonal entry goes to 1.000131 and the leakage falls as a clean delta^(2/3) power law – and the power law is itself evidence against a log, which would leave a residual no power of delta removes.** So c54.122's scale invariance is confined to the contracting side. **What survives is narrower and sharper than what it replaced: the construction HAS a scale-invariant background,

and the obstruction is a PARITY PROPERTY OF ITS OWN SOLUTION rather than a missing mechanism.**). *Computes:* indicial exponents asserted; both vacuum coefficients' k-scalings asserted exactly -3/2 and +3/2; the absence of odd powers in the correction asserted; the resonance asserted unreachable; the diagonal entry asserted to 1e-3 and the leakage ratio asserted equal to 10^(-2/3) at every decade *Bound:* **bound: this settles the connection for the SCALAR mode equation on this background. It does NOT settle what a background correction ODD in tau would do – which is what mixing would require and is the named open route. Three defects were caught in the making: a check that only proved invertibility, a branch error from squaring before taking the cube root, and a test run at delta where k eta ~ 1.4, outside its own frozen-mode hypothesis** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_continuation_is_diagonal.py`

**P15R84** `P15_the_evenness_is_the_time_symmetry` [OK]
*sec:predictions* (**THE DIAGONALITY IS THE CONSTRUCTION'S TIME-REVERSAL SYMMETRY, AND THE SAME INVARIANCE THAT FIXES THE PEAK HEIGHTS IS WHAT STOPS THE SCALE-INVARIANT SPECTRUM CROSSING.** The falsifier left by the previous receipt – mixing needs a background correction ODD in tau, with the Z_3 monodromy the named candidate – fires NEGATIVE. **Write the law as r = A u^(2/3) g(u) with g = [sinh(u)/u]^(2/3): ALL the branch structure sits in u^(2/3), and g is EVEN, analytic and non-vanishing at u=0** – exactly, not to a truncation, since log(sinh u/u) is analytic in u^2 near 0 and the exponential of a series in u^2 is a series in u^2. **The three sheets differ by a CONSTANT cube root of unity, and a constant multiplier cannot change a parity, so no sheet has an odd part; the conjugate branch is even too and has no branch point at all.** **AND THE EVENNESS IS r(-tau) = r(tau): the cosmogenesis is TIME-SYMMETRIC about the branch point, and a time-symmetric background cannot mix a growing mode with a decaying one because mixing would distinguish a direction of time the background does not.** P15 sec:amplitude already leans on this same invariance one observable over – the peak HEIGHTS follow from a driving amplitude exactly invariant under time reversal. **One symmetry, two consequences, and the corpus had banked only the welcome one.**). *Computes:* sinh(u)/u asserted even with unit value at 0; g asserted free of odd powers through u^12; all three sheets asserted free of odd powers; cosh^(2/3) asserted even; sinh^2 asserted invariant under u -¿ -u *Bound:* **bound: the leg is exactly even because it is the marginally-bound E=1 geodesic of an exactly VACUUM SdS exterior. A progenitor with real matter content – pressure, dissipation, anything carrying an arrow – would not be. This receipt does not exclude that; it LOCATES it, and puts the question back where 'L-150' already sits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_evenness_is_the_time_symmetry.py`

**P15R85** `P15_the_geometry_transmits_no_parameters` [OK]
*sec:transmission* (**THE INHERITED DATA ARE NOT INHERITED FROM THE COLLAPSE GEOMETRY, BECAUSE THAT GEOMETRY TRANSMITS ZERO PARAMETERS. A COUNT, AND THE CORPUS PROVED THE KEY STEP ITSELF WITHOUT CASHING IT.** The SdS progenitor family is parametrised by M alone – alpha is the one invariant and E=1 is a choice of geodesic – so the geometry is ONE-dimensional. **And c54.113 showed the Kretschmann along the bead is MASS-FREE in the faller's own proper time (dK/dM = 0, reasserted here; K -¿ (64/27)/tau^4). A quantity identical for every member carries no information about which member it was, so the approach transmits ZERO parameters, not one.** Against that the corpus draws at least FOUR independent numbers: A_s, n_s, eta, z_onset (= rho_r/rho_m, one datum by c54.105), and the light-element composition. **Four numbers cannot come from zero parameters, so 'inherited from the progenitor collapse' cannot mean inherited from the collapse GEOMETRY – it means inherited from the progenitor's MATTER CONTENT, which M does not carry, the exterior being exactly vacuum and the interior dust without composition.** This RECLASSIFIES the row: not a derivation problem but a modelling one. **What the geometry does fix it fixes universally: |Delta eta| = 3.32 alpha = 1.8e4 Mpc, from Lambda alone – a CUTOFF, not a CONTENT.**). *Computes:* dK/dM asserted zero and the 64/27 limit asserted; the count of drawn data asserted at least four; the conformal length asserted 3.32 alpha and asserted to land between 1.5e4 and 2.0e4 Mpc *Bound:* **bound: a counting argument about what the construction's own description of the progenitor can carry. It does NOT show that no model supplies eta – it shows no model of the VACUUM EXTERIOR can, and that the row needs an interior** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_geometry_transmits_no_parameters.py`

**P15R86** `P15_hbar_survives_in_As_and_cancels_in_ns` [OK]
*sec:transmission* (**hbar IS A MULTIPLICATIVE CONSTANT IN THE POWER SPECTRUM,

SO IT SURVIVES IN THE AMPLITUDE AND CANCELS IN THE TILT — WHICH IS WHY n_s IS DERIVABLE HERE AND A_s IS NOT.** The vacuum normalisation v = sqrt(hbar/2 c_s k) e^{-i c_s k eta} is the only place a quantum constant enters, and zeta = v/z with z classical, so P(k) = hbar A_0 k^(n_s - 1): **A_s = P(k_0) carries hbar, and d ln P/d ln k does not — asserted both ways.** An amplitude is the spectrum at one wavenumber; a tilt is its ratio to itself at two. **So the no-magnitude result (c54.128: neither real form supplies a second scale, and hbar, c, G are unit gauges over the single one) forbids deriving A_s and says nothing about n_s.** A_s is therefore permanently inherited while n_s is not, and the two were never the same kind of quantity — though the corpus has named them together in every statement of the frontier, which is how one got worked for thirteen revisions while the other went unexamined. **AND THE INVERSION LOOKED LIKE THE PAYOFF: a quantity not derivable from the invariants is a function of the moduli, so A_s = F(2M/alpha) and the observed A_s would MEASURE the parent's mass. SUPERSEDED at c54.137 ('P15_the_progenitor_vacuum_is_negligible_too'): F is computed, it is a function of TWO moduli rather than one, and it carries no fixed normalisation — the incoming amplitude is a classical datum, so the relation cannot be inverted. The hbar asymmetry itself stands unchanged.** Second handle on the same object: the spectral bound constrains total expansion past equality, this constrains the mass, and the recollapse condition caps that mass below Nariai.). *Computes:* the hbar-dependence of A_s asserted non-zero and of the tilt asserted zero *Bound:* **bound: F itself is NOT computed. The pieces exist — the mode functions, the monodromy, the mode map — and what is missing is the normalisation carried through all three rather than tracked as a scaling. The settled part stands alone: A_s is not derivable, for a reason, and the reason names which other quantities share its fate — every magnitude, and no ratio** *Run:* `python3 receipts/P15_CR_cosmology/P15_hbar_survives_in_As_and_cancels_in_ns.py`

**P15R87** `I11_I1_one_condition_two_jobs_and_the_two_routes_are_independent` [OK]
*sec:properframe* (I11+I1 — ONE CONDITION, TWO JOBS; AND P09's TWO ROUTES ARE INDEPENDENT. Integrable-systems field bake, probes I11 and I1. E = 1 is k = 0 by -k = E^2 - 1; the sinh^(2/3) law satisfies the marginal equation identically and leaves a residual of exactly +k away from it, so the closed form is the marginal member's alone. The same condition puts V_eff = 1 at M = -r^3/(2 alpha^2), the comoving turnaround at NEGATIVE areal radius – so one condition fixes the rest frame by the field and makes the cosmology integrable. And P09's routes are independent: route (a) grants an additively separable ansatz before deriving from it, route (b) runs on the speciality invariant and contains no separability variable, which is why only (b) answers the prior question.). *Computes:* runs clean; twelve verdicts *Bound:* COMPUTES: alpha = 1 as the scale-invariant gauge, matching the thesis; no physical parameter pinned. Note: M and r are declared UNRESTRICTED, since positive=True silently removes the negative-radius root the probe is about *Run:* `python3 receipts/P15_CR_cosmology/I11_I1_one_condition_two_jobs_and_the_two_routes_are_i`

`Q1_a_stated_tolerance_is_a_request_and_the_corpus_answers_it` [OK]
*sec:diffusion-scale* (Q1 — A STATED TOLERANCE IS A REQUEST, NOT A RESULT: RUN THE CORPUS'S ODEs AGAIN AND SEE. Numerical-analysis field bake, probe Q1, and this field's real target is the receipts rather than the prose. Census: 24 receipt files integrate an ODE and ALL 24 set an explicit rtol and atol, not one relying on a solver default; 4 contain any convergence or refinement language. So the corpus states its tolerances and, in twenty of twenty-four files, does not verify them. This receipt does the verification: scipy's solve_ivp is patched to run every call at 100x tighter tolerance, four receipts are re-run unmodified in subprocesses, and EVERY ONE still passes its own assertions. Control: a deliberately under-resolved stiff integration through the same harness DOES move, so the patch applies and the comparator sees. Verdict 5 names the margin on the corpus's one prose-side validation: r_s = 144.0 vs CAMB 144.4 is a 0.28% disagreement against the 8.2% signature it feeds, adequate by a factor of thirty.). *Computes:* runs clean; five verdicts and a control that detects a moved number *Bound:* derived here; a first comparator reported a 92% change that turned out to be a RESIDUAL column, which is supposed to move — the naive test was measuring its own patch, and the honest test is whether the assertions survive *Run:* `python3 receipts/L_numerics/Q1_a_stated_tolerance_is_a_request_and_the_corpus_answers_it.py`

`Q50_the_tolerance_comparisons_gate_the_verdict_and_a_third_of_them_are_not_tolerances` [OK]
*sec:diffusion-scale* (Q5 asked whether the corpus's 688 (now 1116) 'abs(...) ¡ tol' comparisons carry meaningful tolerances and left it open for want of a mutation harness; the harness is built, and both halves of the answer are here – the comparisons DO gate the verdicts, and a third of them are not accuracy assertions at all). *Computes:* rewrites every site three ways and runs

each from the receipt's own directory – margin (semantics-preserving, records E/T), mutate-up (abs(E+3T), catches the assert shape), mutate-down (abs(E)*0, catches the guard shape). Over 269 files and 1116 static sites: the margin transform reproduces the baseline verdict on 269 of 269; of 245 baseline-passing receipts 243 are gated by a tolerance comparison; and of 918 executed sites 210 sit at exactly zero and 54 at or over their tolerance. The receipt asserts the harness's five controls and four invariants live on a ten-file subset holding every verdict class *Bound:* SUPPORTS P15 – and closes Q5 as a CONFIRMATION, as Q1 closed. NOT CLAIMED: that any particular tolerance is the right one; this measures whether a comparison gates the verdict and how much room it has, not whether its author chose well. The two ungated files are a display filter and a documented skip that names its gated substitute *Run:* `python3 receipts/L_numerics/Q50_the_tolerance_comparisons_gate_the_verdict_and_a_third_of_them_are_not_tole`

**P15R88** `P15_the_progenitor_vacuum_is_negligible_too` [OK]
*sec:amplitude* (**THE PROGENITOR'S OWN VACUUM, CARRIED THROUGH THE FULL PASSAGE AND AMPLIFIED BY THE BRANCH-POINT MIXING, IS ALSO NEGLIGIBLE — SHORT BY 1e104. So 'prop:amplitude' had ruled out the one vacuum the inheritance story was NOT claiming.** The collapse-side background is a single function, a = M(rho x + xˆ2/2) with x = |eta - eta_c|, whose Mukhanov-Sasaki potential 2/(x(x+2 rho)) is **M-FREE**, so the mass enters only through the final division zeta = v/a. The passage has three stages and each contributes one factor of 1/rho: (i) the mode freezes on the beta = 2 leg as v -¿ D_k/x with D_k = sqrt(hbar/2) kˆ(-3/2); (ii) **the connection through equality is elementary — v = a int dx/aˆ2 solves the k = 0 problem exactly and carries D_k/x to the constant (3/2 rho)D_k**; (iii) the crunch monodromy leaks it into the survivor with coefficient 2 pi/rho; and a'(eta_c) = -M rho supplies the third. **THE TRANSFER LAW IS THEREFORE P = 18 kˆ3 D_kˆ2/(Mˆ2 rhoˆ6), with the kˆ3 cancelling identically on vacuum data — exactly scale-invariant.** On the vacuum that is A_s = (9/4)(l_P/M)ˆ2 rhoˆ(-6) = 2e-113 at the recollapse cap and the determined rho, against an observed 2.1e-9. **The mixing buys eight orders over the bare vacuum and is still short by a hundred and four.** Two corollaries: A_s does NOT measure the progenitor's mass, D_k being free; and the classical datum has a size — the parent's growing-mode amplitude exceeded its own vacuum's by ˜1e47. **The same closed form also returns c54.131's monodromy analytically, the (x/rho)ln x term continuing to -2 pi i/rho, where it had been obtained by numerical continuation.**). *Computes:* the M-free potential asserted symbolically; the connection coefficient asserted in closed form and against integration from vacuum data to 0.10% over a decade in rho and a factor 15 in k; the transfer law asserted k-independent; the shortfall asserted between 1e101 and 1e106 *Bound:* **bound: z = a with c_s = 1 is assumed on the beta = 2 leg — c54.122's own hypothesis and the standard matter-contraction idealisation, genuine dust having c_s = 0 and no vacuum oscillation to normalise. The 1e104 is far too large for any O(1) convention or plausible c_s treatment to touch** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_progenitor_vacuum_is_negligible_too.py`

**P15R89** `P15_no_primordial_B_modes_unconditionally` [OK]
*sec:amplitude* (**THE TENSOR SECTOR IS WHERE THE COLLAPSE-TRANSFER MACHINE RUNS WITH NO IDEALISATION AT ALL, AND IT RETURNS P_T ˜ 5e-111 — ONE HUNDRED ORDERS BELOW THE OBSERVATIONAL CEILING.** The scalar transfer needs z ˜ a, which for a fluid holds only where w and c_s are constant — and across the progenitor's own equality (1+w) runs 1 -¿ 4/3 while c_s runs 0 -¿ 1/sqrt3, so neither is. **For tensors that is not a restriction: z_T = a M_Pl/2 EXACTLY, for any content, any equation of state, any sound speed, so z_T"/z_T = a"/a identically and the entire passage — background, connection (3/2 rho), monodromy (2 pi/rho), division by a'(eta_c) = -M rho — applies verbatim with nothing assumed.** It returns **P_T = 144 pi (l_P/M)ˆ2 rhoˆ(-6), scale-invariant (k cancels identically)**, i.e. 4.8e-111 at the recollapse cap and the determined composition, against a ceiling of 6.7e-11 from r ¡ 0.032: **below it by 1.4e100, so r = 2e-102.** So the absence of primordial B-modes is a prediction at any conceivable sensitivity, and **it no longer rests on the substrate's own tensor floor — that argument had the same scope defect its scalar twin did, ruling out the substrate's vacuum and saying nothing about the progenitor's.** And the ratio is fixed too: every k-dependent factor is shared by the two sectors, so **r_out = 6 r_in (c54.146: scalars mix twice as strongly as tensors), k-independently — the crossing can neither manufacture nor destroy a tensor signal, only rescale the ratio.** The observed bound therefore measures the PARENT: its own tensor-to-scalar ratio was below 1.3e-3.). *Computes:* P_T asserted k-independent and asserted equal to 144 pi (l_P/M)ˆ2 rhoˆ(-6) symbolically; the shortfall against the ceiling asserted above 1e95; the crossing ratio asserted equal to 24

*Bound:* **bound: the factor 24 carries the scalar side's idealisation (it is 24(1+w)c_s on the contracting leg, the standard matter-bounce value) and no more. The absolute P_T carries nothing** *Run:* `python3 receipts/P15_CR_cosmology/P15_no_primordial_B_modes_unconditionally.py`

`A2_the_three_owed_derivations_are_one_boundary` [OK]
*sec:scope* (the three owed derivations are ONE closed boundary: A_s, n_s and the inherited datum are all dimensionless magnitudes). *Computes:* asserts P15s own list, prop:amplitudes 10^113 gap, the progenitor tilt, and the capstones derived progenitor composition (13 checks) *Bound:* NOT-A-PAPER-CLAIM — L-211 run on L-150 *Run:* `python3 receipts/L211_closure_adjacency/A2_the_three_owed_d`

**P15R90** `P15_the_control_entered_the_regime_and_the_arm_did_not_move` [OK]
*sec:refit-bound* (**THE WAVENUMBER TRUNCATION WAS 78% OF THE CONTROL'S REMAINING chi^2, AND REMOVING IT TAKES THE CONTROL TO chi^2/dof = 1.18 OVER THE FULL RANGE AGAINST A TRUE LambdaCDM FIT'S 1.01 – SO FRONT #2's TARGET IS MET. ACROSS FOUR INSTRUMENT STATES THE CR ARM DID NOT MOVE, AND 'L-147''s PRE-REGISTERED F4 FIRES.** Front #2, lead 'L-500'. **PART 1 WHERE THE RESIDUAL LIVES.** Of the 989 surviving c54.183's derived lensing, **725 – 73% – is in the 30 bins above ell = 1500, a sixth of the bins**; below ell = 700 the control is at chi^2/dof = 0.85 against CAMB's lensed LambdaCDM at 0.82 on the same bins. **PART 2 AND IT IS THE TRUNCATION, SHOWN BY REMOVING IT RATHER THAN BY ARGUING FOR IT.** The grid is k = ell/D_M with ell ¡= LMAXL, so the top output multipole has NO k-headroom while C_ell draws on every k with k·D_M ¿= ell. **Re-run at LMAXL = 3000 with the k-spacing held fixed and scored on the SAME 185 bins: 989 -¿ 218, chi^2/dof 5.34 -¿ 1.18, and the last band's ratio to the reference 0.42 -¿ 0.99.** *So c54.184's '53% in neither template set' was the truncation; the front's last named item is struck as NUMERICAL and no sector remains.* **PART 3 THE CR LADDER'S ALIAS-GATE WAIVER, CHECKED IN chi^2 FOR THE FIRST TIME.** The gate demands 4 samples per Bessel period, the CR ladder has 2.3, and it waives itself with *'this is not aliasing – but it is only not aliasing if the answer does not depend on it. Run KCONT=1 to check.'* **Run: agreement better than 0.7% everywhere, chi^2 to 2.1 in 51817.** *That check existed for the PEAK POSITION and had never been run against the statistic the verdict reads.* **PART 4 FOUR INSTRUMENT STATES, full range, both arms on the same lensing operator: the control 21.2 -¿ 27.7 -¿ 5.34 -¿ 1.18 and the CR arm 290.7 -¿ 304.2 -¿ 298.9 -¿ 302.1.** *An error the INSTRUMENT carries moves both arms; this moved one – eighteenfold against 5%.* **PART 5 F2 AND F3 AS 'L-147' DEFINES THEM, WITH BOTH ARMS AT THE SAME WAVENUMBER RANGE** so the construction is not charged for the instrument's truncation. Full range **F2 = 31 and F3 = 51848**; below ell = 700 **F2 = 3.8 and F3 = 23687**; below ell = 500 F2 = −0.2, the arm scoring no worse than the reference. **F4's second clause fires by three orders of magnitude where it asks for one.**). *Computes:* the k-adequate control at full range asserted at 1.18 +- 0.12, **without which there is no discriminating regime and nothing in the file may be read**; the share the truncation carried asserted above 60%, which is PART 2's whole mechanism; F3/F2 asserted above 100 at full range AND below ell = 700, which IS the reopening; **the ladder/continuum agreement asserted under 2%, without which the CR arm's chi^2 is not admissible at all**; the continuum arm asserted at l_1/l_A = 0.5703; the four-state contrast asserted as a ¿5x control improvement against a ¡15% arm movement, which is what makes this a statement about physics rather than about one build; and the tail share asserted above 60%, since PART 1 is what sent PART 2 looking *Bound:* **scope: this is a MEASUREMENT DISCREPANCY and NOT a framework verdict – 'L-147''s own F5 says so and 'PO-7' is protected precisely here, so this file reports a number and does not convert it; the conversion runs by 'F5''s stated procedure. The deficit itself is not new: the corpus has carried l_1/l_A = 0.5703 against the sky's 0.7312 since c54.168 and P15 states it at 21.9%. What is new is a control good enough to read it through a likelihood instead of a ratio. AND WHAT IS NOT RULED OUT: a defect the instrument's states SHARE – 0.5703 has not moved across EIGHT of them, which is evidence either of a robust prediction or of a shared upstream constant, and this file cannot separate those and does not claim to. Nor does chi^2/dof = 1.18 make the control a FIT: it is 17% above a true LambdaCDM's on the same bins, and what that residual is remains unnamed.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_control_entered_the_regime_and_the_arm_did_not_move.py`

`B1_the_control_separates_prediction_from_shared_constant` [OK]
*sec:refit-bound* (the control separates robust prediction from shared upstream constant: same code, different D_M and r_s, same l_A, peaks 21.8% apart, and the control lands on the sky to 0.17%).

*Computes:* reads both arms from the c54.186 spectra and asserts the scaling test, the control agreement and the ladder/continuum equality (12 checks) *Bound:* NOT-A-PAPER-CLAIM — the adversarial check c54.186 requested *Run:* `python3 receipts/L147_two_arm/B1_the_control_separates_predictio`

`B2_the_prose_duplicate_did_not_reach_main` [OK]
*sec:refit-bound* (the reported prose duplicate did not reach main: zero at three granularities across every commit, because the two edits were different paraphrases and git asked). *Computes:* scans every commit that touched the file at paragraph, sentence and line-window level and asserts the two diffs differ (14 checks) *Bound:* NOT-A-PAPER-CLAIM — checks a routed report before acting on it *Run:* `python3 receipts/L147_two_arm/B2_the_prose_duplicate_did_not_reach_main.py`

**P15R91** `P15_the_first_peak_is_the_seam_datum_and_the_spacing_is_not` [OK]
*sec:coherence* (**ell_1/ell_A = 0.5703 IS A STATEMENT ABOUT THE SEAM DATUM AND NOT ABOUT THE RATE, AND c54.186's OWN CAVEAT IS WHAT SENT THIS LOOKING.** Front #2, lead 'L-501'. The CR arm begins at the seam (z = 6761, eta_S = 180.4 Mpc) and a first-peak mode crossed the horizon at eta ~ 75, so **the modes that set ell_1 are ALREADY SUB-HORIZON at the start and their phase there is ASSIGNED rather than derived**. The corpus's datum is *'ONE datum per mode and a COMMON phase'* – which fixes that the phase is common and not WHICH phase; the instrument picks a density extremum. **PART 1 SCANNING THAT ONE FREEDOM MOVES ell_1/ell_A FROM 0.5703 TO 1.2599 – A FACTOR OF 2.21**, and at phi = 3pi/8 the peak at 172 is not there at all and the series starts at 380. **PART 2 WHAT IS ROBUST IS THE SPACING, AND IT IS ROBUSTLY WRONG:** the mean peak spacing holds at 0.734-0.818 of ell_A across the whole scan – a spread of 1.11 against the position's 2.21 – and is **never 1.0**. *So the disagreement that does not depend on a choice is a ~21% SPACING deficit, not a 22% POSITION one.* **PART 3 AND 'L-147''s F4 FIRES UNDER EVERY PHASE, WHICH IS WHY c54.186 SURVIVES THIS:** on identical bins the CR arm runs chi^2/dof 202-449 against a control at 0.75, so **even its BEST phase costs 269x the control**. **PART 4 THIS REPRODUCES c54.164 ON THE CURRENT INSTRUMENT, WHICH HAD NOT BEEN DONE** – that revision found ell_1 in {150, 165, 315} on the OLD 'ROBUST_p1p2_scan' code, and the finding was never carried across to 'ACOUSTIC_two_arm', which everything since is built on.). *Computes:* **CRPHI = 0 asserted to reproduce the coded 0.5703, without which the knob is not a control on itself and every comparison is against the wrong baseline**; the position's spread asserted above 1.5x and its maximum pinned at 1.2599, which is the withdrawal; the spacing's spread asserted under 1.35x AND its maximum under 0.95 of ell_A, **since if the spacing ever moved this file would have no positive content**; and the CR arm's BEST phase asserted above 50x the control and above chi^2/dof 150 – **if that ever falls, c54.186 must be RESTATED rather than defended** *Bound:* **!! SUPERSEDED BY P15_the_spacing_is_right_and_the_acoustic_phase_is_wrong (r4401).** *Headline claim withdrawn by a later revision of 'sec:refit-bound'; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. **Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.*** **scope: this WEAKENS a number the paper carries and STRENGTHENS a verdict the previous revision landed, and neither is a framework judgement. P15's '21.9% deficit, not an artefact of how last scattering was modelled' is TRUE and is not the sentence a reader takes from it: the deficit is not last scattering's, it IS a function of the seam phase, which is upstream of last scattering and is a choice inside the corpus's own stated datum. 'F5' is not softened – removing a confound makes a measurement cleaner, not a conclusion; 'PO-7' protected; the conversion runs by 'F5''s stated procedure. AND WHAT IS STILL OPEN, narrower than before: the phase is ONE degree of freedom in the datum; Theta-hat's flatness in k is another and is NOT scanned here. A scan that moved the SPACING would be the one that mattered, and this file does not run it.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_first_peak_is_the_seam_datum_and_the_spacing_is_not.py`

`B3_a_dichotomy_stated_is_not_a_dichotomy_proved` [OK]
*sec:refit-bound* (B1 stands but L-235s disposition did not follow: ruling out one horn establishes the other only if the dichotomy is exhaustive, and the truth was a third thing). *Computes:* reads the forks nine phase spectra and asserts the 2.21x position spread against the 1.11x spacing spread (12 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects this lines own r2462 disposition *Run:* `python3 receipts/L147_two_arm/B3_a_dichotomy_stated_is_not_a_dichotomy_proved.py`

**P15R92** `P15_both_datum_freedoms_opened_and_the_spacing_still_holds` [OK]
*sec:coherence* (**THE SECOND AND LAST FREEDOM IN THE SEAM DATUM IS OPENED,

AND THIS FILE WIDENS ITS OWN PREDECESSOR RATHER THAN CONFIRMING IT.** Front #2, lead 'L-502'. c54.187 scanned the common PHASE and named the other freedom: what "flat in k" is flat AT. The datum's amplitude is the corpus's C4 shape $T(x) = 3(\sin x - x\cos x)/x^3$; **the instrument evaluates it at a SINGLE argument xe = 1/sqrt(3), which is that shape read at each mode's OWN horizon crossing** (x = k c_s eta with k·eta = 1 gives exactly 1/sqrt(3)) — a self-similar reading. *The alternative evaluates the SAME shape at each mode's own phase AT THE SEAM, x = k c_s eta_S.* Both are readings of "one datum per mode"; neither is a new assumption. **PART 1 THE POSITION IS NO MORE STABLE:** across eighteen readings ell_1/ell_A spans **0.5570–1.2599, 2.26x** against the phase scan's 2.21x. **PART 2 THE SPACING IS MEASURED ONLY WHERE THERE IS A SERIES TO MEASURE, WHICH DISQUALIFIES EIGHT OF THE EIGHTEEN:** those return a spectrum whose higher peaks have COLLAPSED — three features, the last at zero — and *a spacing read off that is not a spacing*. Requirement stated before the numbers: four peaks, the fourth at least 5% of the first. **PART 3 OVER THE TEN THAT QUALIFY THE SPACING SPANS 0.6764–0.8179 OF ell_A — 1.21x, mean 0.772, never above 0.82.** ***c54.187 reported 1.11x from the phase alone; opening the second freedom widens it to 1.21x, so this file weakens its own predecessor and the ~23% spacing deficit stands at LOWER strength.*** **PART 4 AND F4 FIRES AT EVERY ONE OF THE EIGHTEEN:** chi^2/dof runs 157–460 against a control at 0.75 on identical bins, so **the arm's best reading costs 208x the control**. *And the best-fitting reading is one of the eight DISQUALIFIED ones — it fits better by having less structure to disagree with. Reported because it is the best number in the table and a reader would find it; it is not a direction.*). *Computes:* the position's spread asserted above 1.5x with its maximum pinned at 1.2599, the standing withdrawal from c54.187; **the spacing's spread asserted under 1.6x AND its maximum under 0.90 of ell_A — the point past which c54.187's conclusion would have to be WITHDRAWN rather than widened**; the usable-series count pinned at 10, since PART 3's range is computed only over those and a silent change in how many qualify would move it without moving the code; and **the arm's BEST reading asserted above 50x the control and above chi^2/dof 120 — if that falls, c54.186 is restated** *Bound:* **!! SUPERSEDED BY P15_the_spacing_is_right_and_the_acoustic_phase_is_wrong (r4401).** *Headline claim withdrawn by a later revision of 'sec:refit-bound'; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. **Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.*** **scope: both freedoms the corpus's own phrase leaves open are now scanned and the datum is closed. What has NOT been varied is the background: 'Z_START' is solved so that pi·D_M/r_s = 301.6 exactly, which is a TARGET and not an output — a separate item from the datum, registered rather than resolved. 'F5' unsoftened, 'PO-7' protected, conversion Daryl's.** *Run:* `python3 receipts/P15_CR_cosmology/P15_both_datum_freedoms_opened_and_the_spacing_still_holds.py`

**P15R93** `P15_the_floor_is_a_distance_between_models_not_a_number_from_the_data` [OK]
*sec:refit-bound* (**THE CONTROL'S REMAINING 17% IS NAMED, AND NAMING IT REQUIRED CHANGING THE MEASUREMENT.** Front #2, lead 'L-503'. c54.186 left it unnamed; **'F2' as 'L-147' defines it is a difference of two numbers each computed AGAINST THE SKY, so it mixes how far the two models are apart with where each sits relative to one noise realisation.** **PART 1** amplitude-matched, the instrument tracks a reference LambdaCDM to **better than 1.6% everywhere**, the departure running −1.1% at ell ~ 200 through zero near 600 to +0.7% at 1400 — *a slow undulation, not a mis-phased oscillation*. **PART 2 F2 IS NOT A CLEAN MEASURE, SHOWN BY MOVING THE REFERENCE:** three defensible LambdaCDMs give **+31.2, +21.5, +27.2** — *a spread comparable to the value*. **PART 3 SO THE FLOOR IS MEASURED WITHOUT THE DATA** — chi2_sep = (A_a m_a − A_c m_c)^T F (A_a m_a − A_c m_c), a distance between two MODELS: **21.3 over 185 bins, 0.11 chi^2 per bin, about a third of a standard deviation per bin**. **PART 4 AND THE ORDERING REVERSES, WHICH IS THE TRAP:** the massless-neutrino reference gives the **smallest F2 (21.5) and the largest separation (62.7)** — *a reference chosen to flatter the instrument through the data is the one it is furthest from as a model*. **AND c54.178's LIST IS NOW COMPLETE:** reionisation struck c54.181, lensing built c54.183, the wavenumber range closed c54.186, and **the neutrino mass — the one item never addressed — is worth about 10 of the 31**, the reference carrying one massive species at 0.06 eV where this instrument carries none.). *Computes:* the instrument-vs-reference departure asserted under 3%, which bounds everything else; the model separation asserted under 0.5 chi^2 per bin, the number that replaces the unnamed 17%; **THE ORDERING REVERSAL asserted directly — if F2 and chi2_sep ever stop disagree-

ing, PART 4 is not a trap and the paragraph is unearned**; and **F4's margin asserted above 100x at the WORST reference, since a verdict that depends on the choice is not one** *Bound:* **scope: 0.11 chi^2 per bin is NOT zero and this file does not say the instrument is exact. It says the residual is at the level where the REFERENCE's own parameter choices matter as much as the instrument's error, and that below that level this statistic cannot arbitrate between two LambdaCDM implementations at all. F3 = 51848 against a floor of 21–63 across every defensible reference, so 'F4'' s margin is three orders of magnitude whichever is taken — which makes 'L-147'' s reading SAFER and converts nothing. 'F5' unsoftened, 'PO-7' protected.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_floor_is_a_distance_between_models_not_a_number_from_the_data.py`

`J1_the_disagreement_no_longer_has_two_sides` [OK]
*sec:flatlcdm* (L-171s internal disagreement no longer has two sides: across 18 datum readings the perturbation route gives no definite first peak, and its 150 is c54.164s number from a retired instrument). *Computes:* reads every c54.187-188 spectrum, asserts 150 and 220 both absent and the 2.26x range, and ties it to the class c54.187 routed as item 35 (12 checks) *Bound:* NOT-A-PAPER-CLAIM — supersedes a register rows content; its next step is untouched *Run:* `python3 receipts/L171_two_routes/J1_the_disagreement_no_longer_has_two_sides.py`

**P15R94** `P15_the_spacing_deficit_survives_the_one_fitted_parameter` [OK]
*sec:tensions* (**THE LAST FREE THING UPSTREAM IS THE CORPUS'S OWN ONE FITTED NUMBER, AND MOVING IT BY 31% MOVES THE SPACING DEFICIT BY 7%.** Front #2, lead 'L-504'. c54.188 closed both seam-datum freedoms and named this: **'Z_START' is solved so that l_A = pi·D_M/r_s hits a TARGET rather than coming out as an output.** *That is declared, not hidden — P15 sec:tensions says "the measured acoustic scale is reproduced at the directly measured H_0 by a single z_onset" and calls it **"the one fitted number"**. What had not happened is that the number was a literal buried in a 'brentq' call, so nothing downstream could vary it and nothing had.* **PART 1 MOVING THE PIN MOVES EVERYTHING IT SHOULD:** pinning l_A at 260/280/301.6/320/340 drives **z_onset from 11009 to 5066 and r_s from 157.1 to 120.2 Mpc**. **PART 2 AND THE SPACING DEFICIT DOES NOT FOLLOW IT:** the mean peak spacing stays at **0.7647–0.8205 of l_A — a spread of 1.07 against the pin's 1.31, mean 0.798**. *So the ~20% spacing deficit is a property of the construction and not of the value its one fitted parameter was fitted to.* **PART 3** l_1/l_A runs 0.5176–0.6308 (1.22x) and **never approaches the sky's 0.7312** — and its drift is almost entirely l_A's, the peak itself moving only 164 -¿ 176 while l_A moves 260 -¿ 340. **PART 4 THE STRUCTURAL STATEMENT UNDER IT:** in ordinary acoustics l_1 goes as D_M/r_s, so a 24% fall in r_s should raise l_1 by 31%; **here it raises it by 7%**. ⇒ ***This construction's peaks track its own sound horizon at 24% of the rate acoustics requires, and that — rather than any one ratio — is what the comparison with the sky is registering.*** ***CORRECTED c54.190 ('L-505'): the spacing figure in this row is the mean of the FIRST THREE GAPS, which is all four peaks at LMAXL = 1000 can give. At production depth the asymptotic spacing is 0.975 of l_A — 2.5% short, not 21% — and what is wrong is the ACOUSTIC PHASE, by 0.62 pi. The robustness this row measures stands; the name it gives the quantity does not.***). *Computes:* **the spacing's spread under the pin asserted below 1.20x — the point past which the deficit would BE the fitted parameter's and every conclusion resting on it withdrawn**; its maximum asserted below 0.90 of l_A, since a pin taking it near 1.0 would remove the deficit itself; the first-peak position asserted never to reach the sky's ratio at any pin, which is PART 3; **the peaks' sensitivity to r_s asserted in (0, 0.6) of the acoustic rate — if it ever reached that rate the disagreement would be about a NUMBER and not about a mechanism**; and all five pins asserted to give a four-peak series *Bound:* **!! SUPERSEDED BY P15_the_spacing_is_right_and_the_acoustic_phase_is_wrong (r4401).** *Headline claim withdrawn by a later revision of 'sec:refit-bound'; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. **Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.*** **scope: the acoustic question on this front is now closed as far as the construction's FREEDOMS go — c54.187 scanned the seam phase, c54.188 the amplitude reading, this the fitted parameter, and across all of them the SPACING sits near 0.78–0.80 of l_A and never reaches 0.9 while the POSITION spans 2.26x and states nothing. What is left is not a freedom that can be scanned: whether the peaks SHOULD be tied to the sound horizon is a question about the physics. 'F5' unsoftened, 'PO-7' protected, conversion Daryl's. AND ONE INCONSISTENCY RECORDED RATHER THAN FIXED: the instrument pins l_A = 301.6 where 'P15_zonset_determinations' pins

the MEASURED 301.76 — 0.05%, and it is why this instrument returns z_onset = 6761 where that receipt returns 6797. Named so the two are not read as a disagreement.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_spacing_deficit_survives_the_one_fitted_parameter.py`

**P15R95** `P15_the_spacing_is_right_and_the_acoustic_phase_is_wrong` [OK]
*sec:refit-bound* (**THE "˜21% SPACING DEFICIT" OF c54.187-189 IS THE FIRST THREE GAPS AND NOT THE SPACING. AT PRODUCTION DEPTH THE ASYMPTOTIC SPACING IS RIGHT TO 2.4% AND WHAT IS WRONG IS THE ACOUSTIC PHASE.** Front #2, lead 'L-505'. ***THIS FILE CORRECTS THREE OF ITS OWN LINE'S REVISIONS.*** *Those scans ran at LMAXL = 1000 so that eighteen readings and five pins were affordable; at that depth the CR arm has four peaks, so "the mean peak spacing" was a mean of THREE GAPS — and the first three gaps are the only ones where the arms disagree.* **PART 1/2 AT PRODUCTION DEPTH BOTH ARMS CARRY EIGHT PEAKS:** LambdaCDM's last four gaps average **1.002 of l_A** and the CR arm's **0.975 — 2.5% short, not 21%**; gap by gap the CR arm runs **0.725, 0.853, 0.875** of the control's and is **level by the fourth (0.973)**. **PART 3 SO THE TWO SERIES ARE PARALLEL LINES WITH DIFFERENT INTERCEPTS, AND THAT IS A PHASE:** fitted on peaks 4–8, slopes **1.003 and 0.976 of l_A** (agreeing to 2.6%) and intercepts **−0.263 and −0.878 of l_A**. *A driven series peaks at k·r_s = n·pi − phi, so the intercept IS −(phi/pi)·l_A:* ⇒ ***the two differ by 0.62 pi in the ACOUSTIC PHASE SHIFT, at a spacing they agree on.*** **PART 4 AND A LOW-ell TRANSIENT THE CONTROL DOES NOT HAVE:** measured each against its OWN asymptotic line, the control's first three peaks sit within **16** of it and the CR arm's sit **+142, +80, +18**, converging — *and that transient is what compressed the early gaps*. ***PART-HEADLINE CORRECTED c54.195 ('L-508'): the 0.62 pi is the value at phi = 0. Across four seam phases at production depth the asymptotic phase spans 0.891 and the CONTROL'S VALUE LIES INSIDE that span, so it is not a prediction. The driving ATTRIBUTION stands; its promotion to "the whole disagreement" does not.***). *Computes:* **the CR asymptotic spacing asserted above 0.90 of l_A and pinned at 0.975 — if it were NOT near l_A there would be nothing to correct and c54.187-189 would stand as written**; the first-three-gap and asymptotic values asserted to STRADDLE 0.85, which IS the diagnosis (one number reported for the other); **the slopes asserted parallel to 6%, without which the offset is not a pure phase and PART 3 is wrong**; the phase offset pinned at 0.615 +- 0.08, which is what replaces the retracted number; the transient asserted at more than 3x the control's, since PART 4 is otherwise empty; and the weakest peak's contrast asserted above 1.05, since a series read off non-features is noise *Bound:* **!! SUPERSEDED BY P15_two_arm_control_and_guard (r4401).** *Headline claim withdrawn by a later revision of 'sec:refit-bound'; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. **Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.*** **scope: WHAT THIS RETRACTS is this line's own — c54.187's "˜21%", c54.188's "˜23%", c54.189's "˜20%" and the paper's "acoustic series some 23% too close" are all the first-three-gap value. WHAT IT DOES NOT RETRACT is the robustness: the first-three-gap spacing IS stable at 0.77–0.82 across eighteen datum readings and five pins and F4 fires at every one; what changes is that the stable quantity was the wrong one to call the disagreement. AND THE CORRECTION MAKES c54.187 CENTRAL rather than a caveat — the disagreement is a PHASE, and the phase is exactly what the seam datum assigns and what "one datum per mode and a COMMON phase" does not fix. 'F5' unsoftened; 'PO-7' protected; the conversion runs by 'F5''s stated procedure — and this makes the discrepancy SHARPER, not smaller: a spacing that is right and a phase that is wrong names where to look.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_spacing_is_right_and_the_acoustic_phase_is_wrong.py`

`B4_the_intercept_is_a_phase_and_the_control_proves_it` [OK]
*sec:coherence* (the intercept IS the acoustic phase by change of variables, and the control validates the reading by reproducing the textbook 0.25-0.27; the 21% spacing figure was a depth artefact). *Computes:* derives the intercept identity, fits both arms on peaks 4-8, reproduces the gap-by-gap series and the low-l transient (14 checks) *Bound:* NOT-A-PAPER-CLAIM — the adversarial check c54.190 requested *Run:* `python3 receipts/L147_two_arm/B4_the_intercept_is_a_phase_and_the_control_prov`

`B5_the_scan_cannot_test_the_asymptotic_phase` [OK]
*sec:coherence* (the corrected claims load-bearing quantity cannot be tested by the datum scan: all 23 scan spectra stop at three or four peaks and zero reach the peaks-4-to-8 fit). *Computes:* asserts the depth and peak count of every scan spectrum, that the production pair alone supports the fit,

and where the transient sits (11 checks) *Bound:* NOT-A-PAPER-CLAIM — names a small affordable experiment *Run:* `python3 receipts/L147_two_arm/B5_the_scan_cannot_test_the_asymptotic_phase.py`

**P15R96** `P15_the_acoustics_work_and_the_phase_is_the_whole_disagreement` [OK]
*sec:refit-bound* (**THE ASYMPTOTIC SPACING TRACKS THE SOUND HORIZON AT 98% OF THE ACOUSTIC RATE, SO THIS CONSTRUCTION'S ACOUSTICS WORK — AND THE WHOLE DISAGREEMENT IS THE ACOUSTIC PHASE.** Front #2, lead 'L-506'. c54.190 separated spacing from phase and owed both a production-depth test; **the scans that varied them were all at the depth c54.190 showed to be insufficient**. **PART 1 THE SPACING TRACKS r_s AT THE FULL RATE:** moving the pin from l_A = 301.6 to 340 drops r_s by 11%, ordinary acoustics requires the spacing to rise 13%, and **it rises 13% — 98%**. *So the peaks in this construction ARE set by its own sound horizon.* **PART 2 WHICH RETRACTS c54.189's HEADLINE:** that revision reported *"the peaks track their own sound horizon at 24% of the rate acoustics requires"* — **measured on the FIRST peak, which c54.190 showed sits 142 multipoles off the asymptotic line, inside a transient**. *Over this file's pin range the first peak moves at 37% against the series' 98%; the figure is not reproduced and is not meant to be — what is reproduced is the FINDING.* **PART 3 THE PHASE IS ROBUST TO THE FITTED PARAMETER:** the intercept moves $-0.878 \to -0.899$ of l_A across the same 31% swing — **3% of a discrepancy of 0.62**. **PART 4 AND THE SEAM DATUM'S PHASE FREEDOM IS A REAL LEVER THAT SPANS A THIRD OF IT:** at the opposite phase the intercept moves to $-0.671$, closing the gap from 0.615 to 0.408 — **34%, and no further**. $\Rightarrow$ ***Which is the confirmation c54.190's diagnosis wanted: the datum's phase controls the acoustic phase, as a phase diagnosis predicts, and cannot close the gap.*** ***PART-HEADLINE CORRECTED c54.195 ('L-508'): the 0.62 pi is the value at phi = 0. Across four seam phases at production depth the asymptotic phase spans 0.891 and the CONTROL'S VALUE LIES INSIDE that span, so it is not a prediction. The driving ATTRIBUTION stands; its promotion to "the whole disagreement" does not.***). *Computes:* **the asymptotic sensitivity to r_s asserted in (0.85, 1.15) — if it were NOT near 1 the acoustics would not work and the phase reading would not be the whole story**; the FIRST peak's sensitivity asserted below 0.6, since PART 2 retracts a number and must show the quantity really is less sensitive; the pin's effect on the phase asserted below 15% of the discrepancy, without which the phase is the fitted parameter's; **the seam phase's closure asserted in (0.15, 0.70) — if it ever closed the WHOLE gap the discrepancy would be the datum's alone and this front's verdict would have to be restated**; the phase discrepancy pinned at 0.615 +- 0.05; and at least seven peaks per arm, since every number here is an asymptotic fit *Bound:* **!! SUPERSEDED BY P15_two_arm_control_and_guard (r4401).** *Headline claim withdrawn by a later revision of 'sec:refit-bound'; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. **Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.*** **scope: the front's acoustic statement in final form — this construction reproduces the acoustic SPACING (the quantity its one fitted parameter is fitted to, and which it then tracks at 98% of the acoustic rate when that parameter is moved) and disagrees with the sky in the acoustic PHASE by 0.62 pi, robust to the fitted parameter and only a third reachable by the one freedom the seam datum leaves open. That is ONE number and ONE mechanism where the front began with four items and a ratio. 'F5' unsoftened, 'PO-7' protected, conversion Daryl's — and the discrepancy is now specific enough to be attacked: an acoustic phase shift is a computable consequence of the driving, so "why 0.62 pi" has an address. This file does not have it.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_acoustics_work_and_the_phase_is_the_whole_disagreement.py`

`B6_the_acoustics_reproduce_and_the_dichotomy_is_false` [OK]
*sec:coherence* (the acoustics reproduce at 98.2% of the required rate, and r2485s stated dichotomy was not exhaustive: the phase moves and stops at 34%). *Computes:* recomputes the r_s response and the four-row phase table from the forks spectra, and asserts both horns false (13 checks) *Bound:* NOT-A-PAPER-CLAIM — verifies c54.191 and corrects this lines own r2485 *Run:* `python3 receipts/L147_two_arm/B6_the_acoustics_reproduce_and_the_dichotomy_is_false.py`

`B7_the_seam_phase_moves_the_asymptotic_intercept` [OK]
*sec:coherence* (item 38 answered at production depth: the seam phase MOVES the asymptotic intercept by ˜0.31 lA (half the 0.615 disagreement) while the spacing holds, so the 0.62pi is datum-contingent not structural). *Computes:* validates the phi=0 control against c54.186, fits three production seam phases on matched acoustic orders 4-7, asserts the intercept span and the stable spac-

ing (5 checks) *Bound:* NOT-A-PAPER-CLAIM — a measurement; informs vein L-202, answers lead L-171; conversion to a PO-7 verdict is unseated *Run:* `python3 receipts/L147_two_arm/B7_the_seam_phase_m`

`K1_a_capability_assumed_from_one_failure` [OK]
*sec:coherence* (the whole acoustic disagreement is the driving: undriven the arms agree to 0.013 in phi/pi and the CR slope is exactly 1.0000 l_A, and both lines accepted a capability claim for eight revisions). *Computes:* fits all four production spectra, asserts the undriven agreement and the 5.72x driving ratio, and counts this lines own tarball absorptions (12 checks) *Bound:* NOT-A-PAPER-CLAIM — verifies c54.193-194 *Run:* `python3 receipts/L239_capability_assumed/K1_a_capability_as`

`B8_the_zero_velocity_band_is_a_third_of_the_gap` [OK]
*sec:coherence* (the seam datums band across the two ZERO-VELOCITY phases is 0.207 against a 0.615 disagreement — a third, not most — and over that pair the control is NOT inside). *Computes:* solves sin(phi)=0 on [0,2pi), reads both admissible phases off four production spectra from two nodes, and withdraws this lines own L-202 link (13 checks) *Bound:* NOT-A-PAPER-CLAIM — narrows c54.195s withdrawal *Run:* `python3 receipts/L147_two_arm/B8_the_zero_velocity_band_is_a_third_of`

**P15R97** `P15_the_admissible_band_is_a_third_and_the_two_entries_differ_against_the_potential` [OK]
*sec:coherence, sec:refit-bound* (**ITEM 43 (r2509) ACCEPTED: OVER THE PHASES THE CORPUS ADMITS, THE ACOUSTIC PHASE MOVES 0.207 — A THIRD OF THE 0.615 GAP — AND THE CONTROL'S 0.263 IS OUTSIDE IT. THE DISAGREEMENT IS REAL AND BOUNDED.** Front #2, lead 'L-513'; VEIN 'L-202'. **PART 1** the pair is **forced**, not chosen: 'sec:what-crosses' says what crosses is **frozen**, a frozen mode has $\{}dot\{}delta_\{}gamma=0$, continuity gives $\{}theta_\{}gamma=\{}tfrac34Dkc_s\{}sin\{}phi$, so $\{}sin\{}phi=0$ — the paper's own transmission argument read at the seam, in print before the scan existed. **PART 2 AND THE REASON c54.199 GAVE 56 IS WRONG:** 'y0[:,6] = Ph0' is fixed at $-1$ independent of $\{}phi$ while 'dg0' carries $\{}cos\{}phi$, so the two entries are **not** a sign flip of the state — **what reverses is the relative sign of the density against the potential**, a compression against a rarefaction in the same well. **PART 3** 54's and cc54's production spectra agree to four decimals at both admissible phases. **PART 4** band 0.2069 vs gap 0.6152 — **2.97×**, control **outside** [0.6711, 0.8780]. **PART 5** $\{}chi^2/$dof **281 and 379** against 3.71 — 76× and 102× — so c54.195's PART 5 conclusion is stronger on the pair than on the span it was illustrated in. **PART 6** and P15 'sec:refit-bound' already carried "spans a third of it" and the 0.408, **two paragraphs above the paragraph that withdrew it**.). *Computes:* the four-step admissibility chain asserted in the instrument's AND the paper's source — **if 'sec:what-crosses' did not say modes freeze, the pair would be a choice and r2509 weaker than this file claims**; 'dg0' asserted to flip while 'Ph0' does not, which is what makes item 17's reason wrong; **54 and cc54 asserted to agree to four decimals — a band measured on one node's spectra is a band measured once**; the ratio ˜3 and, separately, the control asserted OUTSIDE the band, without which c54.195 stands as written; both admissible readings asserted above 50× the control, without which the withdrawal takes the conclusion with it; and the paper's own earlier 0.408 asserted to reproduce *Bound:* **!! SUPERSEDED BY P15_two_arm_control_and_guard (r4401).** *Headline claim withdrawn by a later revision of 'sec:refit-bound'; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. **Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.*** **scope: this corrects this line TWICE and in opposite directions — it makes the disagreement firmer than c54.195 left it, and it retracts the explanation c54.199 routed to 56. 'F5' unsoftened; 'PO-7' protected; the conversion runs by 'F5''s stated procedure. No new run: five banked production spectra, two nodes.** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_admissible_band_is_a_third_and_the_two_entries_differ_against_the`

`B9_the_admissible_pair_is_forced_not_merely_distinguished` [OK]
*sec:what-crosses* (the admissible pair phi in {0,pi} is FORCED by P15s own transmission argument — every mode freezes before the crossing, and a frozen mode has zero velocity). *Computes:* quotes the freezing argument at source, solves sin(phi)=0, and asserts the control lies OUTSIDE the band with both nodes agreeing to four decimals (12 checks) *Bound:* NOT-A-PAPER-CLAIM — verifies c54.200 and upgrades r2509 *Run:* `python3 receipts/L147_two_arm/B9_the_admissible_pair_is_forced_not_me`

`B10_the_thermal_mechanism_needs_an_exponential_approach` [OK]
*sec:transmission* (L-519 answered from the MECHANISM side: thermality needs an exponential near-horizon approach, the degenerate members is a power law, and P15 already derived that

split for its transmission dichotomy). *Computes:* integrates the tortoise coordinate for a simple and a double root, re-verifies the Nariai double root, and asserts P15s exponential/power-law split at source (11 checks) *Bound:* NOT-A-PAPER-CLAIM — a third footing for L-519; no temperature value claimed *Run:* `python3 receipts/L147_two_arm/B10_the_thermal_mechanism_needs_an_exponential_appr`

`PO-7` [OK]
*sec:coherence* (kill receipt for PO-7 under route (2): object named, four checks answered — and the CHAIN check does not clear, because the disagreement rests on four reproduced things and one ARGUED one). *Computes:* a kill receipt, not a script — the four checks written out with the inversion routes named *Bound:* NOT-A-PAPER-CLAIM — narrows PO-7; no authorisation requested *Run:* `python3 receipts/kills/PO-7.md`

`S1_every_mode_of_interest_freezes_and_the_inversion_is_closed` [OK]
*sec:what-crosses* (PO-7 chain-(4) link (d) reproduced on a second path: every mode of interest ell 28-2475 freezes before the crossing (r=c_s k/|aH| -¿ 0), so sin(phi)=0 forces phi in {0,pi}, the band stays 0.2069 with the control OUTSIDE, and the kill receipt's inversion (a mode crossing unfrozen) is closed by the divergence of |aH|). *Computes:* reproduces the progenitor interior independently (matching P16's freeze-out to the digit, x_freeze=0.002015 at ell=2475), computes the velocity ratio -¿ 0 at the crossing for every mode, and anchors sec:what-crosses + the kill receipt's chain check (10 checks) *Bound:* NOT-A-PAPER-CLAIM — reproduces PO-7's one non-reproduced link; informs L-171/L-202; F5 unsoftened, no conversion claimed *Run:* `python3 receipts/L805_freezing_reproduced/S1_every_mode_of_interest_freezes_and_the_inversion_is_closed.py`

`S1_a_massive_mode_freezes_too_and_the_mass_term_vanishes_at_the_branch_point` [OK]
*sec:what-crosses* (A2 (PO-seam / PO-7 inversion 3): a massive mode freezes too, for ANY mass — the branch point a-¿0 annihilates the mass term m^2 a^2 while |aH| diverges, so omega/|aH| -¿ c_s k x -¿ 0 independent of m; a massive trajectory carries no phase from non-freezing, closing PO-7 route 3 from the massive side). *Computes:* derives the mass term +m^2 a^2 from the covariant action (sympy), shows omega/|aH| at the crossing is m-independent to the digit up to m=1e5 across ell 28-2475, freeze-out epoch unmoved (5 checks) *Bound:* NOT-A-PAPER-CLAIM — A2; informs L-202/L-171; CRPHI stays assigned; F5 unsoftened *Run:* `python3 receipts/L806_massive_freezing/S1_a_massive_mode_freezes_too_and_the_mass_term_vanishes_at_the_bran`

`S1_the_peak_estimator_recovers_a_set_phase_independent_of_the_envelope` [OK]
*sec:coherence* (A10 (PO-7 inversion 2): the peak-4-8 estimator recovers a SET asymptotic phase to ¡0.005 in phi/pi and is envelope-independent (spread ¡0.006) across six damping/width/tilt variations spanning the CR-vs-control transfer differences — so the 0.408 is not an estimator artefact; inversion route 2 closes and the disagreement hardens). *Computes:* injects synthetic acoustic combs with known phi_in under six envelopes, runs the exact B7/B8 estimator, checks faithfulness and envelope-invariance (4 checks) *Bound:* NOT-A-PAPER-CLAIM — A10; informs L-171; F5 unsoftened, no conversion claimed *Run:* `python3 receipts/L807_driven_estimator/S1_the_peak_estimator_recover`

`S1_the_seam_phase_is_derived_compression_correlated_on_the_observable_leg` [OK]
*sec:what-crosses* (PO-7/PO-seam: the seam phase CRPHI is DERIVED, not free = CRPHI=0 (a compression correlated with the well), on the observable leg — the pre-onset is pressureless (r2701) so no oscillation swings the inherited adiabatic overdensity's sign; bracket-checked against the LCDM control's own adiabatic IC; the {0,pi} band collapses to 0.878, disagreement pinned at 0.615, CRPHI=pi (0.408, a rarefaction) anti-adiabatic and inadmissible). *Computes:* reads the code's CRPHI convention (overdensity in a well = CRPHI=0), shows a pressureless mode's sign is fixed while a pressured one swings, confirms the adiabatic bracket, and anchors the pressureless-onset statement in CR_cosmology.tex (4 checks) *Bound:* NOT-A-PAPER-CLAIM — closes PO-7 route (3)'s residue; informs L-171/L-202; F5 unsoftened, one assumption (Phi-continuity) flagged *Run:* `python3 receipts/L815_seam_phase_derived/S1_the_seam_phase_is_derived_compression_correlated_on_th`

`S1_the_banked_spectra_are_nk600_not_nk260_and_pass_the_guard` [OK]
*sec:where-the-likelihood-sits* (corrects C52's sampling premise: the banked two-arm spectra the likelihood scores are NOT under-sampled — they are NK=600 (the README command), which PASSES the guard at 5.7 pts/Bessel-period. cc54 (15GB) reproduced c54.178_lcdm BIT-FOR-BIT (max |dDl|=8e-16). C52 read the .npz's 238-point shape as NK=260, but 238=len(arange(100,2000,8)) is set by LSTEP=8/LMAXL=2000, not NK; 56's 3.7GB container OOM'd at NK=600 so tested the guard-failing NK=260. The 7.14/280.09 stand; C51's dropped-bin point is separate and

stands). *Computes:* confirms the 238-point shape is arange(100,2000,8) independent of NK, computes guard pts/period=2pi(3NK-1)/(LMAXL-12) giving 2.46@NK=260 (fails) and 5.69@NK=600 (passes), and asserts the banked npz is bit-identical to cc54's NK=600 reproduction (4 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects C52's provenance premise; the guard itself is right (NK=260 does under-sample); does not touch C51's real dropped-bin blocker *Run:* `python3 receipts/L820_banked_spectra_sampling/S1_the_banked_spectra_are_nk600_not_nk260_and_pass_the_guard.`

`S2_the_lmax2512_extension_recovers_c51s_dropped_bins_and_CR_stays_disfavoured` [OK]
*sec:where-the-likelihood-sits* (discharges C51 (the damping-tail half of 56's resampling route): the LMAXL=2512 extension (302 multipoles to ell=2508) recovers the 30 dropped bins (ell 1759-2508), and including them does NOT rescue CR. Two effects isolated: the wider k-range lifts the control 7.14-¿3.81/dof on the SAME 185 bins (c54.186's truncation effect), added tail -¿3.68; the CR arm does not gain (280.1-¿281.1 overlap) and stays 260.1/dof. F3 widens 50497-¿51547 — the dropped region makes CR MORE disfavoured, not less). *Computes:* scores the LMAXL=2512 lcdm/cr extensions on the overlap (ell¡=1996) and full (201) bin sets vs the banked pair, isolating the k-range effect from the damping-tail addition, and computes F3 on both bin sets (4 checks) *Bound:* NOT-A-PAPER-CLAIM — the C51 half of L-820; CR verdict unchanged (both F3 ˜2400x threshold); control's 3.68/dof still large, truncation-vs-physics not further split *Run:* `python3 receipts/L820_banked_spectra_sampling/S2_the_lmax2512_extension_recovers_c51s_dropped_bins_and_CR_s`

`S1_the_cr_residual_is_acoustic_structure_not_truncation_and_that_is_why_it_is_unresponsive_to_lmax` [OK]
*sec:where-the-likelihood-sits* (PO-10 OWED #496 (the row's real remainder): DIAGNOSES the CR arm's 280/dof that r2781/C53 found "unresponsive to L_max and diagnosed by nothing". A per-bin chiˆ2 decomposition (full plik_lite covariance) localises the residuals: the control's is 75% in the high-ell tail (ell¿1200, the truncation), the CR arm's is 57% in the acoustic-peak bands (ell¡850) and 18% tail — so CR is acoustic-structure, not truncation, which is WHY it is unresponsive to L_max. CR comb compressed (172/404/636 vs sky 220/540/810, l1/lA 0.57 vs 0.73 = PO-7's deficit); rescaling doesn't collapse chiˆ2 (best ˜200) so position+heights). *Computes:* computes the per-bin chiˆ2 contribution r_i(Fr)_i with the full covariance and groups by ell-band for both arms, reads the CR peak positions, and scores under an ell-axis rescaling (4 checks) *Bound:* NOT-A-PAPER-CLAIM — OWED #496 CR half; F5 reserves the verdict (PO-7 protected); control third convergence point (L3200) measured separately *Run:* `python3 receipts/L822_cr_residual_diagnosed/S1_the_cr_residual_is_acoustic_structure_not_truncation_and_tha`

`S1_the_stored_cr_rs_is_the_from_onset_acoustic_horizon_and_matches_the_arms_own_integral` [OK]
*sec:acoustic* (PO-10 (the code read 56's r2790 routed): the stored CR r_s=135.46 is NOT a bookkeeping error. The line is R_S=rs_from(Z_START) with Z_START the FITTED onset; rs_from uses the SAME sound speed/grid/onset the acoustic oscillation uses, so 135.46 IS the arm's acoustic horizon and the peaks asymptote to l_A=301.6. r2790's "peaks demand 158" is the low-ell transient (first-4-gap mean -¿ r_s=158, high-n gaps -¿ ˜139 bracketing the stored); its "direction requires 245" is the from-a˜0 horizon, NOT the acoustic one (pre-onset pressureless, L-815, no oscillations before onset). Smaller than LCDM because CR starts oscillating later — the framework's fitted-onset design). *Computes:* reproduces R_S=rs_from(Z_START) via the brentq fit, computes rs_from from a˜0 vs the fitted onset for both arms, and reads the CR peak gaps by ell-range (first-4 vs high-n) (4 checks) *Bound:* NOT-A-PAPER-CLAIM — the r2790 code read; not a bug, the fitted-onset design's output; whether that design is forced is F5/PO-10's *Run:* `python3 receipts/L823_cr_rs_is_the_onset_horizon/S1_the_stored_cr_rs_is_the_from_onset_acoustic_horizon_and_`

`S1_the_third_point_shows_the_control_converges_by_L3000_and_the_unlensed_floor_is_the_lensing` [OK]
*sec:acoustic* (PO-10 (OWED #496, control half — the THIRD convergence point r2781 asked for): the control CONVERGES by L3000, it does NOT slide down an Lˆ-3.4 law to "1.1 by L6000". On the same 185 bins the unlensed control goes 7.18→3.83→3.73→3.73 across LMAXL 2000/2512/3000/3200; the Lˆ-3.4 law fit to the first two points predicts 2.23 at L3200, the measurement is 3.73 — the two-point rate was a one-time truncation recovery, not a convergence rate. The ˜3.73 unlensed floor IS the lensing: c54.183's lensed/unlensed ratio takes the SAME converged spectrum 3.73→1.18, and the lensed control converges 5.38→1.33→1.19→1.18 to 1.18 by L3000 vs CAMB lensed's 1.014 (a ˜0.16/dof instrument floor). Truncation (LMAXL) and lensing are ORTHOGONAL axes; F3 cancels the shared 1.18 floor). *Computes:* scores the four banked control

spectra (c54.178 L2000, L820 L2512, c54.186 L3000, L824 L3200) on the fixed 185-bin set unlensed and under CAMB's lensed/unlensed ratio; checks the L^-3.4 extrapolation vs the measured third point, the L3000/L3200 convergence, and the top-bin truncation (4 checks) *Bound:* NOT-A-PAPER-CLAIM — control-arm convergence; does not overturn 56's convergence claim (the LENSED control does converge, to 1.18 by L3000); companion to L-822 (CR half); F5 untouched *Run:* `python3 receipts/L824_control_converges/S1_the_third_point_shows_the_control_converges_by_L30`

`S1_the_converged_controls_remainder_is_a_broadband_instrument_floor_not_a_localized_defect` [OK]
*sec:acoustic* (PO-10 (OWED #454, the remainder localized): after L-824's convergence the CONTROL arm's 1.18/dof vs CAMB lensed's 1.014 on the same 185 bins (the "~100" was pre-convergence, pre-lensing) is a BROADBAND instrument floor, not a localized defect. A per-bin r_i(Fr)_i decomposition shows every ell-band 100-2000 contributes a positive excess (+4.5/+6.0/+7.4/+1.6/+5.7/+4 total ~30 in chi^2), none dominating, slightly leading in the first two acoustic peaks (300-850 carries 13.4 of 30). So the ~1 target is instrument-accuracy-limited (a uniformly better transfer, not a patchable band), and F3=$\chi^2$(CR)−$\chi^2$(LCDM) cancels this shared floor). *Computes:* per-bin decomposition of the converged lensed control (L824 arm) and CAMB lensed LCDM through plik_lite TT's full covariance on the fixed 185 bins, by ell-band (3 checks) *Bound:* NOT-A-PAPER-CLAIM — control-arm diagnostic; #454's target instrument-limited, still open; F5 untouched; continues L-824 *Run:* `python3 receipts/L826_control_floor_broadband/S1_the_converged_controls_remainder_is_a_br`

`T1_the_transfer_is_half_built` [OK]
*sec:envelope* (PO-12 is HALF BUILT: the debt is two steps and step (1) — the specification of how fluctuations gravitate on the geometrically fixed background — is computed in sec:envelope, two sections before the debt is named). *Computes:* asserts the two-step sentence, the envelope-is-derived sentence, the structural-argument clause and the tilt-irreducible-residual clause verbatim (9 checks) *Bound:* NOT-A-PAPER-CLAIM — narrows a register row; step (2) is genuinely not run *Run:* `python3 receipts/L202_the_seam/T1_the_transfer_is_half_built.py`

**P15R98** `C12_the_odd_even_half_has_its_parameter` [OK]
*sec:envelope* (PO-10s odd/even half already has its mechanism and parameter: C5b_baryon_term puts the asymmetry on the expansion leg at Rb=0.60, which is the rows own description of the item). *Computes:* asserts the rows wording, C5bs location and growth-factor and conclusion, and the papers own sentence (9 checks) *Bound:* NOT-A-PAPER-CLAIM — narrows a register row; the height pattern is not produced *Run:* `python3 receipts/P15_CR_cosmology/C12_the_odd_even_half_has_its_par`

**P15R99** `C14_po12_gates_po10` [OK]
*sec:envelope* (PO-12 GATES both of PO-10s runs: the textbook odd/even ratios at R_b=0.60 are LCDM peak-height formulae that assume a transfer, and both of PO-10s halves are statements about C_l). *Computes:* asserts PO-10s remaining half and P15s genuine-build sentence verbatim, computes the textbook ratios, and confirms neither row records the dependency (7 checks) *Bound:* NOT-A-PAPER-CLAIM — records an ordering between two rows *Run:* `python3 receipts/P15_CR_cosmology/C14_po12_gates_po10.py`

**P15R100** `C15_the_cancellation_does_not_reach_the_ratio` [OK]
*sec:envelope-consequence* (the corpuss cancellation does not reach the odd/even ratio: it holds between two RATES (same quantity, microphysics identical) while odd/even compares two MULTIPOLES on one rate, where the envelope leaves 13-36%). *Computes:* asserts the cancellation sentence and both rate-ratio results verbatim, and computes adjacent-peak envelope ratios on three envelopes (9 checks) *Bound:* NOT-A-PAPER-CLAIM — supplies the reason behind r2646s gate *Run:* `python3 receipts/P15_CR_cosmology/C15_the_cancellation_does_not_reach_the_ratio.py`

**P15R101** `C16_the_transfer_is_six_eighths_built` [OK]
*sec:envelope* (PO-12s step 2 is six-eighths built: the driving verified in closed form, source removal exact, projection calibrated at l_*=302.2 vs 301, and a C_l RATIO with no free parameter already in the paper (0.823 recomputed vs 0.82)). *Computes:* verifies the driving satisfies its ODE, is even, and normalises; asserts the genuine-build, projection and no-free-parameter clauses verbatim; recomputes the high-ell ratio (8 checks) *Bound:* NOT-A-PAPER-CLAIM — narrows a debt to the absolute spectrum *Run:* `python3 receipts/P15_CR_cosmology/C16_the_transfer_is_six_eighths_built.py`

**P15R102** `C17_the_instrument_already_carries_both` [OK]
*sec:instrument* (CORRECTS r2658: both pieces it marked NOT LOCATED are built and validated — the Peebles recombination history IS the visibility function and the line-of-sight source

with its j_l kernel IS the projection). *Computes:* asserts sec:instruments description, its validation figures and its flat-projection wording verbatim, against the debt's geometric-background clause (9 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects a prior lines audit; the debt is now what the instrument RUNS ON *Run:* `python3 receipts/P15_CR_cosmology/C17_the_instrument_already_carries_both`

**P15R103** `C18_bespoke_means_two_legs` [OK]
*sec:envelope* (what BESPOKE means for PO-12: the transfer must carry TWO backgrounds joined at the branch point — radiation-dominated collapse leg, geometry-set expansion leg — and the paper forbids mixing the rates). *Computes:* computes the friction coefficient for both contents, asserts the collapse-leg scoping, the L1/L2 division and the forced consistency rule verbatim (7 checks) *Bound:* NOT-A-PAPER-CLAIM — sharpens a debt; the branch-point join is untouched *Run:* `python3 receipts/P15_CR_cosmology/C18_bespoke_means_two_legs.py`

**P15R104** `C19_the_branch_point_join` [OK]
*sec:transmission* (THE BRANCH-POINT JOIN COMPUTED: transmission is conservation of R, so the join is a change of variable worth exactly 9/10 — applied ONCE where LCDM spreads it over equality, which is why CR has no early ISW). *Computes:* asserts the not-yet-in-hand and transmits clauses, the matter-dominated-to-nine-orders and four-per-cent clauses verbatim, and derives 9/10 from R/Phi=(5+3w)/(3+3w) (7 checks) *Bound:* NOT-A-PAPER-CLAIM — super-horizon join only; sub-horizon modes and the dynamical crossing stay open *Run:* `python3 receipts/P15_CR_cosmology/C19_the_branch_point_join.py`

**P15R105** `C20_the_caveat_named_an_empty_set` [OK]
*sec:what-crosses* (r2661s super-horizon caveat named an EMPTY SET: at the branch point (rH)^2 diverges, the comoving horizon goes to zero, and every mode is outside — so the 9/10 join is unrestricted). *Computes:* asserts the every-mode-outside and seam clauses verbatim, computes the divergence of (rH)^2 and the vanishing comoving horizon, and confirms the radiation term dominates (7 checks) *Bound:* NOT-A-PAPER-CLAIM — removes a caveat from a prior receipt; the crossing dynamics stay deferred *Run:* `python3 receipts/P15_CR_cosmology/C20_the_caveat_named_an_empty_set.py`

**P15R106** `C21_the_superhorizon_transfer_closes` [OK]
*sec:envelope* (THE SUPER-HORIZON TRANSFER CLOSES: Phi -¿ Phi_i exactly at the branch point for every k (departure O(k^2 eta^2)), so with r2661s join Phi_exp(k) = (9/10) Phi_i(k) scale-invariantly — the branch point does not filter). *Computes:* computes the branch-point limit and the series coefficients, re-derives the join, and asserts the transmits clause verbatim; flags the 1/10 near-miss as different-origin (7 checks) *Bound:* NOT-A-PAPER-CLAIM — scalar super-horizon only; the single end-to-end run stays unrun *Run:* `python3 receipts/P15_CR_cosmology/C21_the_superhorizon_t`

**P15R107** `C22_the_end_to_end_number` [OK]
*sec:earlyisw* (THE END-TO-END NUMBER: 7.5% low-ell power deficit from the absent early ISW, 18% at l_D rising to 54% at 2 l_D from damping — and the acoustic peaks BETWEEN them are where the chain stops, correcting r2661s no-further-evolution). *Computes:* asserts the scope qualification, the super-horizon-equation clause, the order-two decay and the not-a-radiation-effect clause verbatim, and computes both ratios (8 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects a prior receipts scope; not a prediction against data *Run:* `python3 receipts/P15_CR_cosmology/C22_the_end_to_`

**P15R108** `C59_the_control_reproduces_camb_and_the_height_defect_was_k_truncation` [OK]
*sec:coherence* (**PO-24 STEP ONE: THE CONTROL RUN, AND IT COMES OUT THE OTHER WAY. THE HEIGHT MACHINERY IS NOT BROKEN.** 'PO-24' records the bespoke transfer as getting the positions right and the HEIGHTS wrong by 22.7% and 97.5% on the arm whose answer is known, and the r3869 handoff named the live lead: two projection paths that are not the same physics, with **which one produced the 2.721 not established**. ** (1) IT WAS los_spectrum, AND THAT IS THE SMALLER HALF. ** '_project' also carries the polarisation source g Pi/4 + (3/4k^2) d^2[g Pi], so the comparison to CAMB was not like for like. ** (2) THE LARGER HALF NOBODY HAD LOOKED FOR: THE k-INTEGRAL IS TRUNCATED WHERE IT IS NOT CONVERGED. ** The instrument built its k-grid from LMAXL – the grid of multipoles to PRINT – so choosing what to report chose where to stop integrating C_l = INT P(k) Delta_l(k)^2 dk. Reported ell grid held FIXED and k_max alone moved: **2.721 -¿ 2.446 -¿ 2.399 -¿ 2.393**. ** (3) BOTH ARE NEEDED AND NEITHER ALONE SUFFICES. ** los_spectrum at converged k_max gives 2.393; _project at truncated k_max gives 2.516; both together give **2.197**. ** (4) THE RESULT: THE CONTROL REPRODUCES CAMB. ** Peaks **220 /

540 / 812** against the sky 220.6 / 538.1 / 809.8, **P1/P2 = 2.197 against CAMB 2.200 – 0.14%** and inside the sky's 1 sigma; P1/P3 = 2.192 against 2.277. Converged in k_max and stable to 2e-3 over a 1.5-fold change in mode count. ** (5) THE FORK IS SETTLED: REPAIR. ** The two-arm architecture carries the transfer; a fresh build would have discarded a working instrument on the strength of an artefact. ** (6) AND THE RECORDED LN SCAN IS INERT: ** LN=12 and LN=25 give identical ratios to 1e-3 with the state vector genuinely wider, against a recorded 4.496 -¿ 8.009.). *Computes:* the recorded 2.721/4.496 asserted REPRODUCED (so the row is not wrong about what it ran); k-truncation asserted to move both ratios on ONE path; the polarisation source asserted to move them again at converged k_max; NEITHER-ALONE asserted; the headline asserted within 1% of CAMB; every peak asserted within 9 of the sky's; convergence asserted to 2e-3 across NK 280 vs 420; Jac_of asserted == 1 to 1e-12 on the control so r3512's HIER composition defect cannot reach this result; LN inertness asserted with the state width checked. **CONTROL THAT FIRES: at the TRUNCATED k_max the same NK widening is EQUALLY STABLE and equally WRONG** – so stability in the knob you varied is not convergence, which is the defect this receipt removes *Bound:* ** CONTROL ARM ONLY. No CR number is produced or corrected. ** The CR arm is truncated by the same mechanism and is NOT re-measured; r3512's HIER composition defect is a no-op on the control (Jac == 1) and stays LIVE for CR, which the next step must settle first. NOT CLAIMED: that the bespoke transfer is finished, that PO-24 is closed, or anything about CR's peak heights. What is established is that its control works *Run:* `python3 receipts/P15_CR_cosmology/C59_the_control_reproduces_camb_and_the_height_defect_was_k_truncation.py`

**P15R109** `C60_the_hier_composition_defect_names_two_flags_its_own_tree_never_had` [OK]
*sec:coherence* (**THE DEFECT FOUR DOCUMENTS DEFER TO IS NOT IN THE INSTRUMENT, AND WAS NOT IN THE ONE IT WAS WRITTEN AGAINST.** 'C59' closed PO-24's control step and deferred the CR arm to r3512's 'HIER' composition defect – "the first thing the next step must settle" – and THE_REGISTER, INDEX and both appendices carry that deferral. ** (1) THE REMEDY NAMES TWO FLAGS ITS OWN TREE NEVER HAD. ** r3512's fix is "give the hierarchy's gravitational source the stacking clock and its diffusion the leaf", i.e. teach 'HIER=1' about 'SRCSTACK' and 'DIFFLEAF'; both occur **zero** times in ACOUSTIC_two_arm.py at 95559d53, the commit r3512 IS, whose 42 os.environ flags are listed. They were real on **6beeca84** (SRCSTACK x13, DIFFLEAF x3), which is **not an ancestor** of 95559d53 – two nodes, two lines, one instrument. ** (2) THE ASYMMETRY IT COUNTED IS 'sound_phase'. ** "The main path references two" counts evolve's RHS plus 'sound_phase', which is CALLED ONCE in the file, from inside 'qscan()', the QSCAN=1 diagnostic that computes no spectrum. Like for like: each ODE right-hand side carries the clock **once**, and **neither projection carries it at all** – true at HEAD and at 95559d53 alike. ** (3) THE TWO RIGHT-HAND SIDES ARE THE SAME BYTES. ** Lines 576 and 959 are character-identical, as are the rate selections at 536 and 916, and every clock-or-source operation in 'evolve' is in 'evolve_hier'. ** (4) AND IT MOVES: ** on ARM=cr toggling LEAFPERT changes evolve_hier's returned state by 2.0 relative, so the hierarchy is not ignoring the clock. **=¿ C59's deferral is discharged; what gates the CR arm is convergence in k, not a composition fix.**). *Computes:* SRCSTACK and DIFFLEAF asserted absent at BOTH 95559d53 and HEAD, with the 42-flag inventory printed; 6beeca84 asserted to carry them and asserted NOT an ancestor of 95559d53; 95559d53 asserted an ancestor of HEAD; the Jac_of call sites located by enclosing function at both commits with the RHS count asserted 1 and 1 and the projections asserted 0 and 0; sound_phase's single call site asserted inside qscan's line span; lines 576/959 and 536/916 asserted character-identical; the operation set of evolve asserted a subset of evolve_hier's; Jac_of asserted != 1 on ARM=cr and the LEAFPERT toggle asserted to move evolve_hier's state. **CONTROL THAT FIRES: the receipt EXITS 1 rather than passing if 95559d53 is not in the clone** – a shallow checkout makes every PART 1 assertion vacuous, which is the hollow-assertion failure the fork names *Bound:* ** ESTABLISHES AN ABSENCE, NOT A CORRECTNESS. ** That the named defect is not there; NOT that the HIER path is right. No CR number, no peak position and no height ratio is produced. r3512's gate order 1, 2 and 4 – validate Pi on the control, the PISRC subtraction, the CR run – are untouched and remain PO-13 Run 1's business. NOT CLAIMED: that r3512 was careless; two lines of work on one file is a corpus-mechanics failure *Run:* `python3 receipts/P15_CR_cosmology/C60_the_hier_composition_defect_names_two_flags_its_own_tree_never_had.py`

**P15R110** `C62_the_diffusion_signature_is_a_wash_and_the_isw_is_not_what_it_turned_on` [OK]
*sec:coherence **RE-MEASURED AT THE FAITHFUL CONFIGURATION AT 'r7099+cc66.82':***

**the ‘LEAFREC=0‘ holding pin is lifted and the pair re-run at $\{}texttt{LEAFREC=1}$, so $\{}texttt{DAMPX}$ goes $1.156766 \{}to 1.174306 = 1.08365410ˆ2$ and the arm’s diffusion scale $+7.553\{}% \{}to \{}mathbf{+8.365\{}%}$ ($r_D$ $7.639814 \{}to 7.697505$ Mpc) — **back toward the ${\{}sim}9\{}%$ the row originally carried**, with the control’s $r_D$ **bit-identical** across the switch. The new pair is banked as ‘spectra/r7099_lcdm_LEAFREC1_DAMPX1.000/1.174‘; the superseded $+7.55\{}%$ and its $1.156766$ are kept as a measured check so the size of the clock’s effect stays on the record.** (**PO-24: THE DIFFUSION-SCALE SIGNATURE IS RELOCATED INTO THE INFERRED TILT — NEITHER A TT-SHAPE TENSION NOR A WASH.** PO-24, narrowed at r4337, asks what the enlarged r_D does to the OBSERVED high-ell power, ”which turns on a parameter refit and on the early ISW term”. ** (1) THE JUDGEMENT: the early ISW is carryable, and the leaf assignment MULTIPLIES it rather than creating it. ** _rt, which sets the radiation fractions in the perturbation source, is the stack ”both arms” by the instrument’s own comment, so radiation is in the source under either assignment; measured, the post-recombination ISW is 0.348 of the SW amplitude on the leaf against 0.229 on the stacking rate. ** (2) BUT NOISW REACHED NO REPORTING PATH ** – read at one site in main()’s analytic block, zero references in los_spectrum, _project and hier_run – so the switch for the term this row turns on measured nothing. Repaired r4492; asserted at source here. ** (3) THE EARLY ISW IS NOT WHAT A HIGH-ELL QUESTION TURNS ON: ** 3.49 sigma/bin over l = 100-1296 and **0.77 sigma/bin over 700-1296**, propagated against plik_lite rather than quoted as a fraction. ** (4) THE SIGNATURE THROUGH A REFIT. ** Scaling r_D by the measured ratio multiplies the damping by a GAUSSIAN in ell; a tilt is a POWER LAW. Imposed on the measured sky and refitted on amplitude and tilt: **0.26 chiˆ2/bin over P15’s 185-bin ceiling, and MORE completely absorbed the higher one looks** (0.06/bin over 700-1760, 96% over 1000-2508) – because over a restricted range a Gaussian looks like a power law. ** (5) BUT THE ABSORPTION IS NOT FREE: dn_s = -0.0304 against sigma = 0.00323 derived from the SAME covariance that scores the fit – a 9.4 sigma cost. ** On the instrument’s own DSCAN pair the same fit gives 0.92/bin and 22.7 sigma. **The signature does not vanish under a refit; it MOVES, out of TT shape and into an inferred parameter.**). *Computes:* the r4492 repair asserted at source in both paths; r_D measured from the instrument at 7.1033 and 7.6398 Mpc, +7.55% raw, with DAMPX asserted to BE that ratio squared; the ISW’s imprint asserted larger on the leaf and asserted under 1 sigma/bin at high ell; the 185-bin ceiling asserted; the shape residual asserted under 0.6 sigma/bin on the data route and under 1.1 on both; the tilt cost asserted above 5 sigma on both routes. **CONTROL THAT FIRES: an injected tilt of +0.02 and +0.05 is recovered to within 5% with ¿99.9% absorbed** – a fitter that reports absorption is shown not to be manufacturing it *Bound:* ** THE MAGNITUDE IS NOT ROBUST TO A FACTOR OF THREE AND THE RECEIPT SAYS SO. ** The data-shape route multiplies the final spectrum by the Gaussian; the instrument applies the damping inside the source where it also enters the Doppler term through a derivative, on an unlensed ceiling-limited spectrum. The two differ 3.6x in residual and 2.8x in tilt; the larger is the more faithful route, so the wash is stated at the WEAKER bound and closing the factor of three is named as work not done here. NOT CLAIMED: that the CR arm as a whole is consistent (r4136 has its heights at -20.7% and -29.2%); that dn_s of this size IS a tension, which is a statement about a joint fit this receipt does not perform. **A FORTIORI, BOTH WAYS: on the SHAPE residual, freeing only the named degeneracy is the least absorption a real refit could achieve, so the wash is a bound; on the TILT it is NOT a bound, since more parameters would SHARE the displacement** **AT ‘r7099+cc66.82‘ A GATE THIS SEAT SET IS RESTATED RATHER THAN PASSED:** *the $\{}texttt{DAMPX=1.0}$ leg was required BIT-IDENTICAL to the banked file; ‘ls‘, $\{}ell_A$, $D_M$ and $r_s$ are, while $D_\{}ell$ agrees to $6.4\{}times10ˆ{-15}$ (˜$29\{}times$ machine epsilon). **Reduction order, not physics** — a different configuration moves a scalar or the abscissa by orders more, and the banked file was built by ‘60‘ at ‘r4502‘ under a different BLAS thread count, so bit-identity was the wrong gate across machines.* *Run:* `python3 receipts/P15_CR_cosmology/C62_the_diffusion_signature_is_a_wash_and_the_isw_is_not_what_it_turned_or`

**P15R111** `C61_the_undriven_premise_is_false_on_the_rate_the_framework_assigns_the_perturbations` [OK]

*sec:coherence* (**PO-13’s DIAGNOSIS RESTS ON A CROSSING CENSUS, AND THE CENSUS COMES OUT THE OTHER WAY ON THE RATE THE FRAMEWORK ASSIGNS THE PERTURBATIONS.** P07 sec:frontiers states ”the standard shift that carries it is universal only where every mode crosses the horizon while there is a plasma to be driven, and **on this rate the acoustic modes re-enter above the onset, so none of them does**”, and PO13_WORKING_STATE gives the

same premise as the structural reason the arm cannot reach the sky. **r3409 made LEAFPERT the default: the perturbations run on the LEAF congruence, and Hleaf carries the radiation term where the CR stacking rate does not.** Horizon crossing for the driving is a perturbation-sector event, so the census that decides the diagnosis has to be run on the leaf rate. ** (1) THE LEAF BACKGROUND HAS AN EQUALITY AND THE ONSET PRECEDES IT: ** z_eq = Om/Or - 1 = **3936** (eta 236.4) against the onset at z = **6761** (eta 180.4). ** (2) THE FIRST PEAK IS SUPER-HORIZON WHEN THE PLASMA STARTS: ** on the leaf, k_1/aH(onset) = **0.858 ¡ 1** for l_1 = 204, and it enters at **z = 5590**, above equality – a radiation-dominated crossing, **while there is a plasma**. ** (3) THE BAND IS NOT A SINGLE MODE: ** every l ¡ 237.7 enters after the onset and **155.6 ¡ l ¡ 237.7** enters in radiation. ** (4) AND THE OTHER RATE GIVES THE OPPOSITE ANSWER: ** on the stacking rate k_1/aH(onset) = **1.415 ¿ 1** (already inside) and the background carries no radiation term at all, so it has no equality and the sentence is not merely true there but necessary. **=¿ The diagnosis was stated on the rate that carries the ruler and tested against a spectrum computed on the rate that carries the content.** And r3733's leaf reading for l=220 reproduces at 0.926 against its 0.92 – the premise's refutation was already in the record as a caveat.). *Computes:* both sentences asserted present verbatim in CR_framework.tex and PO13_WORKING_STATE.md, with the same P07 paragraph asserted to name the leaf congruence; LEAFPERT asserted on and RAD_IN_RATE asserted off; the leaf rate asserted to exceed the geometric rate at a=1e-4 (read, not assumed); the onset asserted to precede leaf equality; the after-onset and in-radiation bands computed from the instrument's own aH/c splines and the latter asserted to CONTAIN l=204; l_1 asserted super-horizon on the leaf and sub-horizon on the stack; the stacking rate asserted equal to Om/aˆ3+OL to 1e-12 at a=1e-6 where a radiation term would have been 254x the rest; r3733's 0.92 asserted reproduced. **CONTROL THAT FIRES: the same census on the STACKING rate returns the diagnosis's own answer** – so the census is not a construction that refutes everything put to it *Bound:* ** REFUTES A PREMISE, DOES NOT MEASURE A MECHANISM. ** No claim that the acoustic residual is explained, reduced or removed, and no measurement of how large the driving these modes receive is – NODRIVE=1 is what measures that and is not run here. The position deficit is untouched: the converged arm reports l_1/l_A = 0.6764 against the sky's 0.7312 whichever way the premise falls. NOT A VERDICT ON P07's SENTENCE: "this rate" is ambiguous in that paragraph between the stacking rate the section is about and the leaf rate the same paragraph names, and the sentence is TRUE on the first; what is established is that the reading supporting the diagnosis is the one the rate rule does not select. NOT CLAIMED: anything about z_onset as a fitted parameter, the alternation, or the Hubble-tension join P07 makes next *Run:* `python3 receipts/P15_CR_cosmology/C61_the_undriven_premise_is_false_on_the_rate_the_framework_assigns_the_p`

**P15R112** `C63_no_single_source_term_carries_the_undriven_split_and_the_first_probe_read_an_unwired_knob` [PASS]

*sec:coherence* (**NO SINGLE LINE-OF-SIGHT SOURCE TERM CARRIES PO-13's 23% UNDRIVEN SPLIT.** Undriven, the CR arm's first peak sits at 1.1273 of its own angular acoustic scale and the control's at 0.9158 – a 23% split with Q=1 on both arms, so the oscillator is excluded and the split is in the projection. The DIFFUSION DAMPING is excluded too: DAMPX=0 removes it entirely and the split WIDENS to 26.0%. Removing the monopole, the Doppler dipole and the integrated term ONE AT A TIME leaves the split at 23.1%, 23.8% and 29.4% – it never collapses, and removing the integrated term WIDENS it. So the answer to "Psi, integrated, or Doppler" is NONE OF THE THREE. ALSO PINS THE ROW'S PREMISE: "the same acoustic scale on both arms" is TRUE of the ANGULAR scale l_A (301.6 vs 301.4, 1 part in 1507) and FALSE of the PHYSICAL sound horizon r_s (135.46 vs 144.53 Mpc, 6.3% apart). The l_A match is a conspiracy – D_M differs by 6.2% too, within 0.07% of r_s's own difference – not a shared scale.). *Computes:* eight runs of ACOUSTIC_two_arm.py under NODRIVE=1, both arms x (baseline, NOISW=1, DPSRC=0, SWSRC=0), at LMAXL=520 where all eight first peaks reproduce the LMAXL=1300 values exactly; each switch is CALIBRATED before any null is read off it *Bound:* A NEGATIVE result: it says where the split is NOT and is not a mechanism. Only SINGLE removals were run, so a cancellation between two terms is not excluded. The damping half of the projection kernel is now excluded and the VISIBILITY half is not: the arms last-scatter at eta = 449.6 and 280.8, a 60% difference in epoch against 7% in width, and that is the named unrun candidate. ZSTART is NOT a free knob – r_s integrates from the start, so moving it moves the acoustic scale itself (a trial took l_A from 301.6 to 172.8). Does not touch the driving (DRC/DRE are identically zero under NODRIVE=1). The KCONT=1

sampling control leaves all four CR peaks unchanged, which bounds the sampling shift below one LSTEP=8 grid step (2.65%) against a 23% effect – a bound, not an identity. *Run:* `python3 receipts/P15_CR_cosmology/C63_no_single_source_term_carries_the_undriven_split_and_the_first_probe_`

**P15R113** `C23_the_gap_estimated` [OK]
*sec:earlyisw* (ESTIMATE for the peak-region gap: the decay rate k^2/(3H) is built from the thing that differs, so CRs potential decays ~15% faster and the peak power ratio is ~0.813 — BROAD deviation, not tail-confined). *Computes:* asserts the rate, the 13% and the factor-of-two clauses verbatim, computes the compounding, and flags the 0.813/0.823 near-coincidence as different-origin (7 checks) *Bound:* NOT-A-PAPER-CLAIM — an ESTIMATE with its model stated, labelled as one; a hierarchy run would replace it *Run:* `python3 receipts/P15_CR_cosmology/C23_the_gap_estimated.py`

**P15R114** `C24_the_substitution_itemised` [OK]
*sec:instrument* (PO-12s substitution ITEMISED: conformal time, r_s, the comoving horizon and the diffusion integral carry H while the Thomson rate and x_e do not — and the ratio does NOT cancel because r_D carries a square root that r_s does not). *Computes:* asserts the 13% clause and reconstructs it from rho_r/rho_m=0.3, asserts the microphysics-outside and theta_*-pinned clauses, and computes both scaling exponents (8 checks) *Bound:* NOT-A-PAPER-CLAIM — sets the bar for the two-leg run; P15s 9.4% is not re-derived *Run:* `python3 receipts/P15_CR_cosmology/C24_the_substitutio`

**P15R115** `C25_the_gap_closed_by_integration` [OK]
*sec:diffusion-scale* (the 2.6-point gap closed by INTEGRATION: theta_D/theta_* runs +7.1% at z=1200 to +14.4% at z=10000, with P15s +9.4% at z~2500 — and the dependence is MONOTONIC, which is the papers own two-constraint argument). *Computes:* asserts the branch-point lower-limit clause and the single-datum clause verbatim, integrates the rate difference over five onset redshifts, and checks monotonicity (6 checks) *Bound:* NOT-A-PAPER-CLAIM — reproduces the papers shape on a simplified rho_r/rho_m model; the onset is the remaining unknown *Run:* `python3 receipts/P15_CR_cosmology/C25_the_gap_closed_by_integration.py`

**P15R116** `C26_the_onset_is_not_free` [OK]
*sec:tensions* (PO-12s last free parameter is NOT free: the onset is fitted to the acoustic angle at z~6797 and meets the scale at every H_0 — and the residual +13.1% vs +9.4% is the WEIGHT g(R)/x_e, not the onset). *Computes:* asserts the onset value, the fitted-to-the-acoustic-angle clause and the not-a-knob clause verbatim, checks the model at the onset, and integrates under four weightings (7 checks) *Bound:* NOT-A-PAPER-CLAIM — power-law weights are stand-ins for g(R)/x_e, shown for direction and size *Run:* `python3 receipts/P15_CR_cosmology/C26_the_onset_is_not_free.p`

**P15R117** `C27_the_number_bracketed` [OK]
*sec:envelope-consequence* (P15s +9.4% BRACKETED from outside its own machinery: Saha +7.0% ¡ Peebles ¡ unweighted +13.1%, with the ordering FORCED by which history recombines faster — and cancel identically covers the CONSTANTS, not the weight). *Computes:* asserts the cancel-identically, constants-outside and Peebles-history clauses verbatim, shows the weight does not drop out, runs g(R) with a Saha x_e, and checks 9.4 lies strictly inside (8 checks) *Bound:* NOT-A-PAPER-CLAIM — brackets rather than reproduces; the Peebles direction is argued, not computed *Run:* `python3 receipts/P15_CR_cosmology/C27_the_number_bracketed.py`

**P15R118** `C28_length_against_angle` [OK]
*sec:envelope-consequence* (THERE WAS NEVER A DISCREPANCY: 9.4% is an ANGLE ratio (C10s r=1.0926, already netting the 0.7% sound-horizon shift) and 10.8% a LENGTH ratio (C8) — r2686-r2689 compared one to the other). *Computes:* asserts both statements and the 0.7% clause verbatim, reads C10s own r, and shows the two do not compose as this line assumed (7 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects four of this lines revisions; C8 and C10 are run, not re-derived *Run:* `python3 receipts/P15_CR_cosmology/C28_length_against_angle.py`

**P15R119** `C29_nothing_to_calculate_across_the_join` [OK]
*sec:what-crosses* (PO-12s remaining half has NO CONTENT: the pre-onset stretch is pressureless and at w=0 the potential equation loses k entirely, so there is no scale-dependent evolution for a single calculation across the join to compute). *Computes:* asserts the no-k, pressureless, boundary-mode and residue clauses verbatim, and verifies symbolically that k is present in the general equation and absent at w=0 (7 checks) *Bound:* NOT-A-PAPER-CLAIM — F5 reserves the strike; the residue routes to PO-7 *Run:* `python3 receipts/P15_CR_cosmology/C29_nothing_to_calculate_across_the_joi`

**P12R15** `A7_the_frontier_has_two_kinds` [OK]
*sec:envelope-consequence* (THE FRONTIER SORTS INTO TWO KINDS on P15s own H(a) rule: FIVE rows ask whether the theory is DEFINED (PO-2/4/5/6/11) and TWO ask what it PREDICTS (PO-7/10) — and they fail differently, so size does not order them). *Computes:* asserts the H(a) clause, checks the classification covers exactly the open rows, and pins each rows object against its kind (7 checks) *Bound:* NOT-A-PAPER-CLAIM — the classification is derived here; no paper states it *Run:* `python3 receipts/P12_algebroid/A7_the_frontier_has_two_kinds.py`

**P15R122** `C32_half_one_is_model_selection` [OK]
*sec:frontiers* (PO-10s half (1) IS an information-criterion comparison: the corpus names "the likelihood-level model selection" while AIC/BIC/Akaike/Occam/Bayes-factor appear ZERO times — and the penalty is computable now, dAIC=10.0 and dBIC=26.9 in CRs favour). *Computes:* asserts the model-selection phrase and the 215-bin count verbatim, word-boundary-checks five model-selection terms at zero across papers and receipts (excluding itself), and computes both penalties (9 checks) *Bound:* NOT-A-PAPER-CLAIM — k=6 is the standard flat-LCDM count and must be stated with any result; the chi^2 is unrun *Run:* `python3 receipts/P15_CR_cosmology/C32_half_one_is_model_se`

**P15R123** `C33_the_threshold_corrected` [OK]
*sec:tensions* (THE THRESHOLD CORRECTED: r2709s k=1 was wrong — P15 anchors A_s at the first peak, so CR is k=2 — and against Liddle/Jeffreys (dBIC 2 positive, 6 strong) the threshold is dBIC = 21.5, still 3.5x the strong line). *Computes:* asserts the anchored-amplitude and one-parameter-accommodation clauses verbatim, tables all four parameter-count pairs against the Jeffreys scale, and computes the AICc correction (7 checks) *Bound:* NOT-A-PAPER-CLAIM — no chi^2 is scored; the count must be restated with any result *Run:* `python3 receipts/P15_CR_cosmology/C33_the_threshold_corrected.py`

**P15R124** `C34_the_deliverable_is_a_pair` [OK]
*sec:tensions* (PO-10s DELIVERABLE IS A PAIR: the acoustic sector is open as PO-7, its phase freedom moves the gap 0.615 -¿ 0.408 and cannot close it — but the freedom is DISCRETE, so k is unchanged and both branches are reported at the same threshold). *Computes:* asserts PO-7s object, the phase-lever and cannot-close clauses verbatim with both values, and shows the discrete choice leaves dBIC at 21.5 rather than 16.1 (5 checks) *Bound:* NOT-A-PAPER-CLAIM — neither branch is scored; that is the whole of what remains *Run:* `python3 receipts/P15_CR_cosmology/C34_the_deliverable_is_a_pair.py`

**P15R125** `C35_the_reference_is_F3` [OK]
*sec:tensions* (THE REFERENCE FORK SETTLES ITSELF AND AGAINST THIS LINES FRAMING: BIC uses only DIFFERENCES, so F3 (same-instrument) is the score and the CAMB-206.4 comparison charges CR an instrument floor CAMB never pays). *Computes:* asserts L-147s floor definition and pipeline-wiring clauses verbatim, shows dBIC contains only a chi^2 difference, and confirms the threshold is invariant under the pair choice (7 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects r2709-r2711s statement of what the threshold scores against; the threshold itself is unchanged *Run:* `python3 receipts/P15_CR_cosmology/C35_the_reference_is_F3.py`

**P15R126** `C36_the_negative_is_the_envelope` [OK]
*sec:envelope-consequence* ( SUPERSEDED r2725 (the check shared the runs own n_s-held assumption, so agreeing to the same order confirmed only that both held it fixed) — L-814s NEGATIVE IS NOT AN ARTEFACT: P15s own published envelope, scored against cosmic variance on 215 bins with no pipeline or fit, gives chi^2 ˜ 1.6e5 against cc54s measured 5.05e4 — the same order). *Computes:* evaluates the papers own exp[-(l/l_D)^2(r^2-1)] across the band, scores it against cosmic-variance errors, reads cc54s floor finding from their receipt, and pins L-147s scope clause (6 checks) *Bound:* NOT-A-PAPER-CLAIM — no framework verdict; PO-7 protected and F5 reserves it *Run:* `python3 receipts/P15_CR_cosmology/C36_the_negative_is_the_envelope.py`

**P15R127** `C37_the_comparison_is_not_matched` [OK]
*sec:envelope-consequence* (L-814s COMPARISON IS NOT MATCHED and its number is WITHDRAWN as a result: the CR arm was held at k=2 with n_s fixed at LCDMs fitted value, and freeing the tilt alone removes 15.6x). *Computes:* computes the envelope chi^2 with the tilt held and free, shows the threshold barely moves for the extra parameter, and reads cc54s own relative statement from their receipt (5 checks) *Bound:* NOT-A-PAPER-CLAIM — the k=2 spec was this lines at r2710; a matched comparison is not claimed to pass *Run:* `python3 receipts/P15_CR_cosmology/C37_the_comparison_is_not_matched.py`

**P15R128** `C38_the_comparison_scores_derivations` [OK]
*sec:tensions* (PO-10s NON-FOREIGN COMPARISON IS A SCORE ON DERIVATIONS: a derivation has no k, so scoring theta_*, r, z_eq and the 9/10 transfer against measurement imports no parameter vector — and this receipt reports NO score, because the draft that did invented every sigma). *Computes:* pins P15s derived-vs-measured pairs and its own naming of Omega_m and A_s as fitted, and asserts its own refusal to score (6 checks) *Bound:* NOT-A-PAPER-CLAIM — no verdict on CR; the row owes published uncertainties and a derived-quantity list *Run:* `python3 receipts/P15_CR_cosmology/C38_the_comparison_scores_derivations.py`

**P15R129** `C39_the_derived_list_read` [OK]
*sec:tensions* (PO-10s DERIVED-QUANTITY LIST READ from P15s own marker language: four pairs each with the papers own accuracy clause (theta_* 302.2/301, r_s 146.4/145.4, 1+z_eq 3399, eta 1.99 to one per cent) and three exclusions in the papers own words). *Computes:* pins all four derived pairs and all three fitted-exclusions verbatim, and the high-l ratios no-free-parameter clause (7 checks) *Bound:* NOT-A-PAPER-CLAIM — nothing is scored; the published uncertainties remain owed and are not this lines to invent *Run:* `python3 receipts/P15_CR_cosmology/C39_the_derived_list_read.py`

**P15R130** `C40_the_pair_was_quoted_against_a_rounding` [OK]
*sec:tensions* (PO-10s FIRST PAIR MUST TAKE THE RECEIPTS VALUE, NOT THE PROSE: P15 writes "the measured ˜301" (tilde present, honestly approximate) while P15_zonset_determinations carries and asserts l_A = 301.76 — and the gap is 0.44, not 1.20). *Computes:* quotes both the prose and the receipts assertion, computes both gaps, and carries its own first drafts correction (6 checks) *Bound:* NOT-A-PAPER-CLAIM — no paper edit owed, the tilde is there; nothing is scored and the uncertainty is still not this lines *Run:* `python3 receipts/P15_CR_cosmology/C40_the_pair_was_quoted_against_`

**P15R131** `C41b_a_tilde_on_a_settled_value_is_a_stale_hedge` [OK]
*sec:tensions, sec:coherence* (A TILDE ON A SETTLED VALUE IS A STALE HEDGE: ˜301 became 301.76 (asserted by its receipt) and ˜8% became 8.2% in NINE places (computed on CAMBs exact ionization history), three of them while calling it "a real, computed effect"). *Computes:* confirms both figures are asserted by receipts, that no tilde survives on either, and that legitimately approximate tildes are untouched (7 checks) *Bound:* NOT-A-PAPER-CLAIM — nothing scored; not every tilde is stale, only those on figures a receipt asserts *Run:* `python3 receipts/P15_CR_cosmology/C41b_a_tilde_on_a_settled_value_is_a_stale_hedge.py`

**P15R132** `C41_a_tilde_is_stale_when_nothing_competes` [OK]
*sec:tensions* (A TILDE IS STALE WHEN ONE RECEIPT ASSERTS AND NONE COMPETES: ˜301 failed that test and P15 now reads 301.76; ˜8% passed it — C8 derives +10.83% analytically and NAMES the disagreement with damping_ratio_cleans +8.2% — so nine edits were reverted). *Computes:* confirms the 301.76 edit landed and the nine tildes remain, pins C8s computed +10.83% and its own naming of the conflict (6 checks) *Bound:* PAPER-CLAIM (301.76 only) — the 8% figure is left tilde-marked and its discrepancy is filed as owed *Run:* `python3 receipts/P15_CR_cosmology/C41_a_tilde_is_stale_when_nothing_competes.py`

**P15R133** `C42_the_cancellation_needs_xe_identical` [OK]
*sec:coherence* (THE DAMPING DISCREPANCY DIAGNOSED: C8s cancellation needs x_e identical in both cosmologies, and a different H(a) changes the recombination HISTORY not merely its epoch — Lambda, the upper limit, the radiation density and arithmetic all eliminated). *Computes:* reproduces C8 to six figures, varies Lambda / z_rec / rho_r and shows none accounts for the gap, and pins C8s own PROVIDED clause (7 checks) *Bound:* NOT-A-PAPER-CLAIM — neither number is adjudicated; the x_e response is a calculation neither receipt has done *Run:* `python3 receipts/P15_CR_cosmology/C42_the_cancellation_needs_xe_identical.py`

**P15R134** `C43_the_gap_is_numerical_not_xe` [OK]
*sec:coherence* (r2750 WITHDRAWN — THE DAMPING GAP IS NUMERICAL, NOT x_e: a pure H probe moves z_* by only 0.195% (˜0.05pp on the ratio) against a 1.86pp gap, while the CAMB receipt prints a 7.1% miss against CAMBs own thetad on its OWN LCDM arm). *Computes:* runs CAMB at five N_eff values as an H-only probe, scales the response by r2750s own measured sensitivity, and pins the receipts self-reported GATE miss (6 checks) *Bound:* NOT-A-PAPER-CLAIM — C8 is not confirmed, and the tilde stays: two receipts still disagree, one now with a KNOWN defect *Run:* `python3 receipts/P15_CR_cosmology/C43_the_gap_is_numerical_not_xe.py`

**P15R135** `C44_the_gap_is_truncation` [OK]
*sec:coherence* (THE DAMPING GAP IS TRUNCATION: the r_D ratio climbs 8.33% -¿ 9.94% with the upper limit and converges only by z˜5e4, while the CAMB receipts grid stops at z=12000 — and a normalisation factor CANCELS in a ratio, which withdraws r2751). *Computes:* computes the ratio at six upper limits, shows convergence, and demonstrates that a shared constant divides out (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the receipts grid is reasonable for its own purpose; what is claimed is that its RATIO inherits a truncation bias *Run:* `python3 receipts/P15_CR_cosmology/C44_the_gap_is_truncation.py`

**P15R136** `C45_xe_does_not_cancel_from_a_ratio_of_integrals` [OK]
*sec:coherence* (THE DAMPING DISCREPANCY RESOLVED: x_e does NOT cancel from a ratio of INTEGRALS even when identical — toggling it on the same range moves the ratio +9.94% to +8.37%, the whole gap — so C8s cancellation step is a real error). *Computes:* toggles x_e on a fixed range and measure with CAMBs own history, shows a shared CONSTANT does cancel exactly while x_e does not, and pins C8s every-constant-outside claim (6 checks) *Bound:* NOT-A-PAPER-CLAIM — which figure is P15s signature is not settled; C8s reduction and baryon correction stand *Run:* `python3 receipts/P15_CR_cosmology/C45_xe_does_not_cancel_from_a_ratio_of_integrals.py`

**P15R137** `C46_the_nine_are_right_and_the_one_is_wrong` [OK]
*sec:tensions* (THE 9.4% TRACED: P15 -¿ C10 (+9.26%) -¿ C9, which divides r_D 1.1015 by r_s 1.0067 — the ANGLE ratio, correctly, but from an r_D that omits the x_e weighting. Corrected it lands at +7.6% to +8.2%, so the nine ˜8% were RIGHT and the one 9.4% was the error). *Computes:* pins the citation chain P15-¿C10-¿C9 and reruns C9s own division on three r_D ratios (6 checks) *Bound:* PAPER-CLAIM — ten instances set to 8.2%, and the tilde removed per r2749s test now that nothing competes *Run:* `python3 receipts/P15_CR_cosmology/C46_the_nine_are_right_and_the_one_i`

**P15R138** `C47_the_neutrinos_were_missing` [OK]
*sec:coherence* (THE +5.66% RECONCILED AND IT WAS THIS LINES ERROR: r2752 used photons alone where the receipt uses Ogamma + Onu (a 69% underestimate of what sources H) — and the CR arm has no radiation at all, so the omission hit ONE ARM ONLY. With neutrinos in, the ratio is 1.0896 against the receipts 1.0897). *Computes:* runs the full Hu-Sugiyama for both rates under two radiation densities with CAMBs own x_e, and confirms the absolute r_D difference cancels from the ratio (6 checks) *Bound:* NOT-A-PAPER-CLAIM — only the RATIO is claimed to four figures; the absolute r_D still differ by ˜3% from grid and quadrature *Run:* `python3 receipts/P15_CR_cosmology/C47_the_neutrinos_were_missing.py`

**P15R139** `C48_the_pairs_are_three` [OK]
*sec:tensions* (PO-10s PAIRS ARE THREE, NOT FOUR: r_s and 1+z_eq hold against asserting receipts (C9 145.38, DESI 3402), but the onset ratio is an IDENTITY ON INHERITED INPUTS — "the quoted value to one per cent, from standard thermodynamics" — same class as N_eff and A_s). *Computes:* finds the asserting receipt for each measured value, pins the thermodynamics clause, and records the 1.9968 near-miss where another receipt asserts a different quantity with the same digits (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the onset ratio is not wrong, only not a TEST; nothing is scored *Run:* `python3 receipts/P15_CR_cosmology/C48_the_pairs_are_three.py`

**P15R140** `C49_the_row_has_a_likelihood_not_a_lookup` [OK]
*sec:tensions* (PO-10 DOES NOT NEED ERROR BARS — IT HAS A LIKELIHOOD: plik_lite over 215 TT bins with a covariance and a banked LCDM fit the pipeline calls wired. What blocks it is recorded in the same file — the CONTROL arm lands at chiˆ2/dof ˜ 100). *Computes:* pins the instrument, its calibration standard, its self-recorded F4 defect, and F5s protection of the verdict (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the control defect is not diagnosed here, and no verdict on CR is drawn *Run:* `python3 receipts/P15_CR_cosmology/C49_the_row_has_a_likelihood_not_a_lookup.py`

**P15R141** `C50_the_control_is_at_seven_not_a_hundred` [OK]
*sec:tensions* (THE CONTROL ARM IS AT 7.14, NOT ˜100: the receipts F4 prose carried the c54.172 floor while its F6 PRINT reports 1320.5/185 = 7.14, and F6 says the floor fell 103 -¿ 22.5 -¿ 28.6 -¿ 7.1, a factor of 14). *Computes:* pins the corrected prose, the f-string print fields, the fall history, the 185-vs-215 bin difference and F6s unchanged verdict (6 checks) *Bound:* NOT-A-PAPER-CLAIM — 7.14 is not acceptable either; both arms remain outside the discriminating regime *Run:* `python3 receipts/P15_CR_cosmology/C50_the_control_is_at_seven_not_a_hundred.py`

**P15R142** `C51_the_dropped_bins_are_the_damping_tail` [OK]
*sec:tensions* (THE THIRTY DROPPED BINS ARE ell 1759-2508 — THE DAMPING TAIL: the likelihood arms model ends near ell 1760 and nan-drops the rest, discarding exactly the region P15 names as its exposed edge, where the suppression runs 24% to 42%). *Computes:* reads the bin edges from the likelihoods own arrays, locates the cut at which 185 survive, and profiles the damping suppression across the dropped range (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the ell limit is a cost not a mistake, and which way the omitted bins would land is the point of running them *Run:* `python3 receipts/P15_CR_cosmology/C51_the_dropped_bins_are_the_damping_tail.py`

**P15R143** `C52_the_banked_spectra_fail_the_sampling_guard` [OK]
*sec:tensions* (THE BANKED SPECTRA FAIL THE INSTRUMENTS OWN SAMPLING GUARD: LMAXL=2000 NK=260 gives 2.5 points per Bessel period and prints UNDER-SAMPLED — and the guard says the source comb stays correct while the projection aliases, so no comb-level check could catch it). *Computes:* fixes the banked configuration from the npz shape, computes the sampling ratio and projection memory across four NK values, and pins the guards own wording (6 checks) *Bound:* NOT-A-PAPER-CLAIM — whether aliasing moves the chi^2 enough to matter is UNMEASURED, and measuring it needs a run that will not fit in 3.7 GB *Run:* `python3 receipts/P15_CR_cosmology/C52_the_banked_spectra_fail_the_sampling_guard.py`

**P15R144** `C53_the_k_range_fixes_one_arm_only` [OK]
*sec:tensions* (THE k-RANGE FIXES ONE ARM ONLY: raising L_max 26% drops the control 7.14 -¿ 3.81 on the SAME 185 bins while CR moves 0.4% — a response ratio over twenty to one, so r2760s undiagnosed factor of seven is a k-truncation and CRs 281 is not). *Computes:* computes both arms responses and their ratio, checks F_3 widens in the direction one-arm improvement predicts, and labels the two-point trend as an extrapolation (6 checks) *Bound:* NOT-A-PAPER-CLAIM — radiation is the first candidate not a demonstrated mechanism; the convergence trend rests on TWO points *Run:* `python3 receipts/P15_CR_cosmology/C53_the_k_range_fixes_one_arm_only.py`

**P15R145** `C54_the_281_is_the_prediction_being_measured` [OK]
*sec:tensions* (CRs 281 IS THE ORDER P15s OWN PREDICTION PRODUCES: scoring the damping residual over 185 bins gives chi^2/dof from 204 (sigma=1%) to 5099 (sigma=0.2%), so 281 sits INSIDE that range and BELOW the value at Plancks quoted errors — the arm is scoring the effect P15 predicts). *Computes:* pins P15s no-free-parameter high-ell formula, scores its residual at three error scales, pins F5, and names the SHAPE test that would settle it (5 checks) *Bound:* NOT-A-PAPER-CLAIM — an ORDER comparison against a flat sigma, not a derivation against the plik covariance; supplies a candidate, does not close the row *Run:* `python3 receipts/P15_CR_cosmology/C54_the_281_is_the_prediction_being_measured.py`

**P15R146** `C55_the_shape_test_says_peak_misalignment` [OK]
*sec:tensions* (THE SHAPE TEST RUN: the CR arms residual is neither a damping-profile match nor flat — it OSCILLATES, 0.237 to 2.375, correlation -0.296, dominant period 317 against l_A = 301. The 281 is PEAK MISALIGNMENT, and r2786 is withdrawn by its own named test). *Computes:* computes the banked arms ratio, its correlation with P15s profile, its dominant Fourier period against l_A, and the l_A-agrees / r_s-differs contrast (5 checks) *Bound:* NOT-A-PAPER-CLAIM — the damping signature is not shown absent, only non-dominant; physical vs numerical phase slip is the open question *Run:* `python3 receipts/P15_CR_cosmology/C55_the_shape_test_says_peak_misa`

**P15R147** `C56_la_is_computed_and_its_residual_is_too_small` [OK]
*sec:tensions* (l_A IS COMPUTED NOT CALIBRATED (pi D_M/r_s reproduces the stored value in both arms) and its survival is a real cancellation P15 claims in advance — but the +0.075% residual explains under 1% of r2787s measured swing, two orders too small). *Computes:* checks pi D_M/r_s against the stored l_A in both arms, sizes the D_M and r_s moves, pins P15s met-from-the-other-end clause, and propagates the residual against the measured swing (5 checks) *Bound:* NOT-A-PAPER-CLAIM — the cancellation is observed not re-derived; amplitude is what remains after two eliminations, a candidate not a finding *Run:* `python3 receipts/P15_CR_cosmology/C56_la_is_computed_and`

**P15R148** `C57_the_cr_peaks_are_not_at_its_own_la` [OK]
*sec:tensions* (THE CR ARMS PEAKS ARE NOT WHERE ITS OWN l_A SAYS: mean spacing 258 against a stored l_A of 301.6 (ratio 0.855) while LCDM sits at 0.995 — so the stored quantities are consistent with each other and NOT with the computed C_ell). *Computes:* finds peaks in both banked arms, compares mean spacing against each arms stored l_A, and sizes the mismatch against

r2788s residual (5 checks) *Bound:* NOT-A-PAPER-CLAIM — which of D_M, r_s or the projection is wrong is not established; the CR spacing could be a real feature of a transfer on the geometric rate *Run:* `python3 receipts/P15_CR_cosmology/C57_the_cr_peaks_are_not_at_its_own_la.py`

**P15R149** `C58_the_stored_rs_moves_the_wrong_way` [OK]
*sec:tensions* (THE STORED CR r_s MOVES THE WRONG WAY: removing radiation must RAISE the sound horizon (146.5 -¿ 245.2 Mpc integrated), the peak spacing independently demands 158.35, and the stored value is 135.46 — BELOW LCDMs 144.53). *Computes:* solves the spacing for each candidate, integrates r_s with and without radiation, and shows pi D_M/r_s on both stored values reproduces the stored l_A exactly (5 checks) *Bound:* NOT-A-PAPER-CLAIM — 158.35 and 245.16 are not proposed as the right value; what is established is the DIRECTION and that the stored value has the wrong sign of deviation *Run:* `python3 receipts/P15_CR_cosmology/C58_the_stored`

`R1_the_correction_stopped_at_the_sentence_and_the_number_it_defines_was_left_behind` [OK]
*sec:envelope-consequence* (r2755 CORRECTED THE SENTENCE AND NOT THE NUMBER THE SENTENCE DEFINES: 'theta_D/theta_*' was corrected from $9.4\{}%$ to $8.2\{}%$ and 'sec:envelope-consequence''s $r$, one paragraph later, IS that ratio and still read 1.093 — so the paper stated one ratio twice, differently, and the second is the one the prediction is made of. Corrected from three agreeing routes to $r=1.0816$: the ratio at $\{}ell_D$ moves 0.823 → 0.844, at $1.5\{}ell_D$ 0.645 → 0.682, $\{}ell_D(\{}rm CR)$ 1281 → 1294. **FIFTEEN receipts carried it — eight SILENTLY (exiting zero, six of them evaluating the envelope with the wrong value) and seven LOUDLY (failing every full run since r2755 — and they are the seven that PRODUCED the correction).** CONTROL: C10's gate was built so the paper and the receipt "cannot drift apart silently in either direction", and it worked — they were consistent with each other and both wrong.). *Computes:* COMPUTES: the identity $r=\{}theta_D/\{}theta_*$ read from C10's own assertion message; the corrected value from three routes and the four quantities that follow; the fifteen receipts split by failure mode and counted at the commits where they exist; the C41 double-allocation resolved as one rule at two states; and the register files deliberately left alone (21 checks) *Bound:* **bound: 1.0816 is NOT re-derived here** — it is read from 'P15_damping_ratio_clean', 'C46' and the corrected sentence, and checked for agreement. **Not claimed that the envelope is right**, only that it now runs on the ratio the paper's own preceding paragraph derives. **The correction FAVOURS the construction (16% suppression at $\{}ell_D$ rather than 18%) and is flagged for it**; it would apply identically the other way. 'PO-7' is 'sec:refit-bound'/'sec:coherence' and is untouched. *Run:* `python3 receipts/L557_the_correction_stopped_at_the_sentence/R1_the_correction_stopped_at_the_sentence_and_`

`P1_the_pin_test_run_the_spacing_follows_the_pin_so_the_ratio_does_not` [OK]
*sec:tensions* (THE PIN TEST, RUN — the runs the observer line's container killed twice. **The peak spacing tracks the one fitted number PROPORTIONALLY** ('LATARG' 280/301.6/320 → mean spacing 229.3/245.3/261.3), so spacing/$\ell$_A = 0.8190/0.8134/0.8167 — **0.69% spread across a 14% pin range**, against the **13.4%** a constant-spacing arm would give: a factor of twenty, so the measurement DISCRIMINATES. ⇒ **The routed dichotomy is inverted: following the pin is exactly what makes the RATIO — which is what the deficit is — independent of the pin. THE DEFICIT SURVIVES THE FIT.** Also: 'KCONT=1' at NK=320 returns the same four peaks, discharging the instrument's own standing sampling caveat; and **$\ell$_1 = 172 at all three pins**, so $\ell$_1/$\ell$_A is pin-dependent while the spacing ratio is not. LCDM control on the same grid: 0.9909, reproducing the sky to 6 in $\ell$.). *Computes:* COMPUTES: five banked runs parsed from their own printed blocks; the spacing ratio and its spread; the CONSTANT-spacing alternative computed as the control and the discrimination factor between them; the KCONT identity and its sampling-guard margin; $\ell$_1's invariance and $\ell$_1/$\ell$_A's spread; and the LCDM arm checked against the sky before being used (21 checks) *Bound:* NOT-A-PAPER-CLAIM — **and the declaration is deliberate: this is a measurement on the instrument's own fitted parameter, delivered to a routed question, and whether it lands in 'sec:tensions' is the observer line's call under 'PO-7'.** A node that banked it into the paper on its own initiative would be taking a step toward the verdict the row protects. **bound: this reports a number and does not convert it into a verdict — 'PO-7' is protected.** Not that r2789's 258 was wrong (different grid; the control was re-run here rather than compared across grids). Not that peaks are known better than the $\ell$-grid: 'LSTEP=8', ±4 per peak, ±8 per spacing — the invariance claim is 20× outside that and the individual spacings are not. Not that

the deficit is explained, only that the fitted number does not produce it. *Run:* `python3 receipts/L562_the_pin_test/P1_the_pin_test_run_the_spacing_follows_the_pin_so_the_ratio_does_not.py`

**P15R150** `P15_the_driving_is_bought_by_decay_during_oscillation` [OK]
*sec:transmission* (THE DRIVING SHIFT IS BOUGHT BY POTENTIAL DECAY OCCURRING WHILE A MODE OSCILLATES, so a mode whose oscillation begins after the potential has settled inherits the UNDRIVEN phase. ** Coupled photon/matter/potential solve on a matter-and-radiation background, with the potential closed by the ENERGY CONSTRAINT rather than evolved and the acoustic turnover taken at the first zero of the photon velocity that BEGINS A SEQUENCE SPACED BY pi – the free oscillator's own spacing, which the undriven gate validates at 1.0000 on both arms. ** SCOPE: the ASYMPTOTIC statement only, Q -¿ 1 at large k, which is what P15 and P07 assert; NOT the low-k values, and NOT a magnitude (neutrino free-streaming omitted, ˜15% low). A sign flip in the initial amplitude is a pi phase shift and moves WHICH ZERO IS FIRST by one half-period, so four initial signs agree to 0.012 at k=160). *Computes:* COMPUTES: a coupled photon/matter/potential solve on a matter-and-radiation background, the potential closed by the ENERGY CONSTRAINT rather than evolved; the UNDRIVEN gate validated at $1.0000$ on both arms before use; the driven arm asserted to FALL toward $1$ from above and to reach the free-oscillator value to $0.02$ at large $k$; the control asserted well below $1$ and flat in $k$ to $0.10$; the two shapes asserted qualitatively distinct by a factor of two in range; and the robustness leg showing a factor of two in the radiation budget barely moves it. Re-run r3537, ALL PASS *Bound:* SCOPE, and the receipt states it: the ASYMPTOTIC statement only, $Q\{}to1$ at large $k$, which is what 'P15' and 'P07' assert; **NOT the low-$k$ values and NOT a magnitude** — neutrino free-streaming is omitted, some $15\{}%$ low *Run:* `python3 receipts/P15_CR_cosmology/P15_the_driving_is_bought_by_decay_during_oscillation.py`

**P15R151** `N8_the_transmission_dichotomy_is_a_capacity_dichotomy` [OK]
*sec:transmission (prop:transmission, rem:transmission-leg)* (P15 proves a transmission dichotomy — a non-degenerate horizon (p=1) imprints a scale, a degenerate one (p=2) transmits — and rem:transmission-leg says "the leg multiplies the spectrum by a constant and the seam imprints nothing: neither carries a scale, and a spectrum can only be tilted by something that does". That is a CHANNEL claim, and "carries a scale" is what decides its capacity). *Computes:* fits the output tilt on four disjoint k-bands: p=2 returns the same tilt on every band (spread 1.7e-16) while p=1 spreads by 7.93, which IS the scale; the control puts a cutoff into the p=2 kernel by hand and the spread jumps to 1.91, so the test can fail; and the p=1 spread shrinks monotonically 30.2 -¿ 7.9 -¿ 2.3 as kappa rises, tracking the kernel's scale *Bound:* SUPPORTS P15 — transparent means the answer does not depend on where you look, and imprinting means it does. NOT a Bogoliubov calculation; no Hawking temperature claimed *Run:* `python3 receipts/P15_CR_cosmology/N8_the_transmission_dicho`

**P15R152** `P15_the_history_is_spherically_symmetric_throughout_so_the_chern_simons_number_never_moves` [OK]
*does the construction's own history pass through any member carrying a **twist**, and does the gravitational Chern–Simons number change across it? 'r6762' made the violation a **boundary** quantity — the right shape for a net count produced across a history — and 'P15' names the open item it could answer, the **baryogenesis-analogue** derivation of order a billion photons per baryon* (**No twist anywhere on the history, so the count is exactly conserved — and not because the number comes out small.** **Q1** the path is fixed three ways: 'P16' states the spherically symmetric class as a ***premise*** and forbids dropping it while keeping the branch-point locus; ${*}RR\{}equiv0$ on the general **dynamic** spherically symmetric class (four arbitrary functions of $(t,r)$, off-diagonal kept — 'r6762''s identity was **static** and does not reach collapse); and the twisted member's transverse block is a **torus** against this history's closed $S^3$, which no deformation joins **Q2** no, at both levels — a travelling tensor wave gives ***exactly zero at any polarisation***, and the only nonzero source is the parity-odd product $\{}psi'\{}omega'$ of the two channels, needing both polarisations AND an areal factor AND broken spherical symmetry; the history supplies one of three **Q3** ***zero, and endpoint-independent*** — all four components of $K^\{}mu$ vanish identically on the class, so the number is zero at **every** slice, the $r=0$ crossing included, where a merely divergence-free quantity could have jumped **Q4** net count $3\{}times0=0$ **Q5** the first branch, blocker named). *Computes:* C1 Nariai in its **own** $dS_2\{}times S^2$ form (radii $\{}alpha/\{}sqrt3$), not as a value of $f(r)$ — zero, and still zero with a $d\{}tau\{},d\{}varphi$ term, so the zero is not the product structure C2

the polarised cut ($\{}omega\{}equiv0$, ‘P11‘ eq:metric) zero for arbitrary $A,\{}psi,R$; unpolarised nonzero; and **silencing either channel kills it**, each by rebuilding the metric rather than substituting a symbol for a function C3 Stokes **exact**: the volume integral of the density and the boundary flux of $K^\{}mu$ both give $8/15-\{}log(390625/6561)$, with ‘r6762‘’s constant $-1$ re-verified on a different member the machinery is read out of ‘r6762‘’s own receipt rather than re-typed, so the two cannot drift *Bound:* **NOT claimed** that no mechanism could produce the item’s number — only that this one does not, and why. **No photon number is supplied or invented**; the verdict does not need one, a zero numerator giving zero for any finite denominator. **No direction is claimed for any asymmetry**, there being none to direct — had the count been nonzero its sign would have been correlated with the twist’s, per ‘C51‘. Nothing reopens ‘r6762‘’s current, source or inflow findings, the global group, the gauging’s status, or weak isospin; no cosmological implication is drawn. Nothing is named *Run:* `python3 receipts/P15_CR_cosmology/P15_the_history_is_spherically_symmetric_throughout_so_the_chern_simons_r`

**P15R153** `P15_the_likelihood_pair_is_another_instruments_and_the_crossing_run_needed_a_missing_knob` [OK]

*‘PO-10‘ reopened at ‘r6780‘ on a contradiction between ‘P15‘’s body and ‘P15‘’s own receipt: the body reports $\{}chi^2=397.13$ against $206.44$ over 215 ‘plik_lite‘ TT bins with five parameters free in each, and the receipt records that pair as **superseded**. Which comparison is legitimate, on what run, with the floor beside it* (**There are TWO instruments here, and the body’s pair is the other one’s — which is why the body and the receipt are both accurate.** **Q1** ‘fit.py‘+CAMB fits **five parameters on 215 bins** and its CR arm is CAMB’s $\{}Lambda$CDM TT ***times an $\{}ell$-only damping envelope***; ‘ACOUSTIC_two_arm‘+‘chi2_of_spectrum‘ fits **one amplitude on 185 covered bins** and its CR arm is the construction’s own spectrum. The second is the one to report, and the first is **not malformed** — it answers the damping question **Q2** the second gives $51817.0$ against $1320.5$ — ***and the crossing configuration was unreachable***, the CR arm’s background being written in as literals ($H_0=73.00$, $\{}Omega_m=0.3066$); ‘CRH0‘/‘CROM‘/‘CROMBH2‘ exposed, defaults byte-identical **Q3** $F_2=+1114.0$, $F_3=+50496.5$, forty-five times it — **and the body’s $\{}Delta\{}chi^2=190.7$ is *below* that floor**, readable on the first instrument only because its control **is** the CAMB reference so its floor is identically zero **Q4** the deficit is in the second’s spectrum and ***structurally absent*** from the first’s **Q5** ***both of the order’s two options at once***: ‘cr.json‘ banks a genuine five-parameter optimum at $H_0=78.13$, and ‘fit.py‘ shows the shape refitted was CAMB’s $\{}Lambda$CDM times the envelope). *Computes:* C1 ‘F1‘ first, non-optional: the banked CAMB fit at $\{}chi^2=206.4$ over 215 bins, $\{}chi^2/\{}mathrm{dof}=0.960$ the envelope’s inability to carry a comb **measured, not argued** — across envelopes strong enough to move $\{}ell_1$ by up to eight multipoles the spacings hold within $6\{}%$ ($320/272/312 \{}to 312/272/304$) the sound horizon scanned over **four decades** of starting redshift, converging monotonically to $r_s=256.13$ Mpc with a residual drift of $0.5$ in $\{}ell_A$ — so ”converges at its upper end” is a measurement here and not a quotation the exposed knobs’ defaults verified byte-identical on $\{}ell_A$, $r_s$, $D_M$ and the peak list *Bound:* **NOT claimed** that the body’s computed $298.0$ is wrong, nor that the instrument’s convention is the right one — only that they **differ**, the body putting $r_s$ on the radiation-carrying leaf rate while ‘rs_from‘ integrates the radiation-free one, and which is intended is the paper’s question. ***No $\{}chi^2$ is manufactured against a comb the paper does not claim***, so the crossing-configuration figure the order asked for is reported as unreachable rather than supplied. Nothing reopens the handover resolution. **‘PO-7‘ protected throughout**: a negative is a measurement discrepancy and no verdict on the construction is drawn from any $\{}chi^2$ here *Run:* `python3 receipts/P15_CR_cosmology/P15_the_likelihood_pair_is_another_instruments_and_the_crossing_run_neede`

**P15R154** `P15_the_chiral_state_the_route_needs_is_forced_absent_not_merely_unchosen` [OK]

*r6766 named what a successor to the closed route would need — a chiral **state** on the three-sphere layer, not a chiral member — and left it unclassified: does it reduce to the handedness choice ‘PO-30‘’s law places outside, or is it something the geometry might still supply?* (**Neither: it is forced absent.** **It is not the handedness choice** — that selects one of two equivalent options, while this asks whether the state is imbalanced **at all**, which is prior to any selection. **The layer is the same one ‘P10‘ works on**, established from ‘P10‘’s own words: it lifts to ”the closed-$S^3$ layer’s propagating sector”, where the transverse-traceless graviton modes form the tower. **And there the balance is forced**: regularity imposes the regular branch

fibre by fibre, the surface gravity belongs to the background horizon and is common to every fibre, and the operator labelling the fibres is a sum of squares with a single coefficient — no helicity label and no sign, so the two towers are populated alike. **The parity-odd source is exactly that difference**, its expectation vanishing under equal population. ⇒ ***‘P10‘’s helicity cancellation and r6766’s source form are one statement from two sides; the route closes by a condition the geometry imposes, not by its silence on a choice***). *Computes:* the parity-odd expectation vanishing under equal population and not under unequal; the product dying when either channel is silenced *Bound:* **NOT that the tower is achiral** — ‘PO-44‘ is explicit it remains chirally capable and must not be read as the achirality branch; what is forced is the **population**. Nothing wider about states. Nothing about the handedness choice. **‘P15‘’s item stays open** — what is settled is that this route to it is closed, and why *Run:* `python3 receipts/P15_CR_cosmology/P15_the_chiral_state_the_route_needs_is_forced_absent_not_merely_unchosen`

**P15R155** `P15_the_legs_closed_form_at_the_two_loci` [OK]

*sec:envelope* (the leg’s closed form read at the two loci: at the CROSSING it is the ordinary frozen adiabatic state exactly (Theta-hat = Psi_i/2, Psi = Psi_i), and the coded handover’s 0.4835 Psi_i is Psi at horizon entry halved, which is neither locus). *Computes:* Psi = Psi_i T(x) and Theta-hat = (Psi_i/2) cos x evaluated at the crossing, at entry and at the seam over k/k_s = 1 to 300; Theta_0’s own driven equation integrated directly as an independent check, with the residual measured to scale as 0.399 x0 *Bound:* asserts the arithmetic of three candidate amplitudes and where each comes from; says nothing about which is right for the locus the instrument imposes it at, and nothing about the spectrum *Run:* `python3 receipts/P15_CR_cosmology/P15_the_legs_closed_form_at_the_two_loci.py`

**P15R156** `P15_the_handover_at_the_crossing_and_what_it_costs` [OK]

*sec:refit-bound* (moving the handover from the onset to the crossing removes 71 per cent of the arm’s excess chi2 over the control and takes P1/P2 from -20.7 to +1.5 per cent against the sky – and does not remove the rejection, 35.5 per bin against the control’s 2.10). *Computes:* reproduces sec:refit-bound three ways (the control’s peaks and P1/P2, the coded arm’s, both arms’ plik_lite chi2, and the sky’s own phase intercept); then the crossing arm’s comb, phase and heights, converged over two decades in ZSTART, to the ell-grid’s own resolution at LSTEP=1, and against the continuum ladder at KCONT=1 *Bound:* the rejection stands and is stated as one; the comb comes out 4.2 per cent narrow and the arm spends no fitted number; the reported l_A is bookkeeping on a sound horizon the oscillator does not run on *Run:* `python3 receipts/P15_CR_cosmology/P15_the_handover_at_the_crossing_and_what_it_costs.py`

**P15R157** `P15_the_desi_dissolution_survives_the_leaf_ruler_but_only_by_moving_the_onset` [OK]

*sec:tensions* (the DESI DR2 H0-dissolution survives the adjudicated leaf clock – 0.88 per dof, flat to two decimals from H0 = 70 to 80 – but by compensation rather than cancellation, and the price is the onset: its rho_r/rho_m runs 51.0 down to 5.8 across that band, against the 1.7 the stacking ruler puts it at). *Computes:* reproduces the banked stacking-ruler fit (Om 0.3066, z_onset 6764, chi2 12.01 on 12 dof) and measures its H0 spread as exactly zero; then the leaf ruler without the onset, which re-pins H0 to 68.5 and breaks at 73 at 10.7 per dof; then the leaf ruler with the onset re-solved from theta_* at each H0; and the acoustic angle alone as an independent determination agreeing with BAO to 0.05 in H0 *Bound:* the conclusion survives and the argument does not – on the leaf clock H0 is not FORCED by BAO rather than ABSENT from it; says nothing about which clock is right, and nothing about the spectrum *Run:* `python3 receipts/P15_CR_cosmology/P15_the_desi_dissolution_survives_the_leaf_ruler_but_only_by_moving_the_o`

**P15R158** `P15_the_damping_to_comb_ratio_picks_the_crossing_and_the_leaf_clock_out_of_four` [OK]

*sec:coherence* (theta_D/theta_star separates the four handover-locus-by-clock combinations and picks exactly one – the crossing on the leaf clock, at 1.022 of the control’s ratio, against 1.135, 1.353 and 0.652 for the other three – and neither choice was made to fix this quantity). *Computes:* r_D/r_s on the corpus’s own Silk integrand with the opacity taken from RD_diffusion_direct, over all four combinations; the log-a quadrature converged to 0.006 per cent from n = 3001 to 12001; and the crossing rows re-run over a hundredfold range in the locus, moving 0.07 per cent *Bound:* the two choices are not separable – neither helps at the other’s wrong partner – so this is one configuration and not two improvements; and getting this ratio right is necessary and not sufficient, the crossing arm still being rejected at 35.5 per bin *Run:* `python3 receipts/P15_CR_cosmology/P15_the_damping_to_comb_ratio_picks_the_crossing_and_the_leaf_clock_out_o`

**P15R159** `P15_the_onset_bracket_family_is_latent_everywhere_and_was_live_in_one_place` [OK]
*sec:acoustic* (the (1500, 60000) onset bracket is a family – 27 live acoustic-onset solvers and 46 retired, 18 of the live ones below the leaf ruler's root of 61,580, which is 2.6 per cent past the ceiling – and exactly one member was ever reached, failing by a quarter of a multipole). *Computes:* both roots recomputed from the instrument's own integrands (6761 stacking, 61,583 leaf); pi D_M/r_s at the old ceiling measured at 301.85 against the 301.6 target; the whole tree swept and every onset solver classified; and the six files a leaf-clock run actually enters checked to carry the widened ceiling *Bound:* the other 18 are LATENT and not broken – each integrates r_s on the stacking clock where the root is 6,764 – and are recorded rather than pre-emptively widened; says nothing about which ruler is right *Run:* `python3 receipts/P15_CR_cosmology/P15_the_onset_bracket_family_is_latent_everywher`

**P15R160** `P15_the_third_peak_deficit_is_radiation_driving_and_both_routes_to_the_equality_agree` [OK]
*sec:envelope* (P1/P3 tracks the leaf equality and not the route to it – 2.299 against 2.294 at z_eq 3447 and 2.474 against 2.460 at 3000, across a 12 to 24 per cent change in Omega_m and therefore in D_M and the whole projection – so the third-peak deficit is radiation driving, and the arm's own equality is the residual). *Computes:* the crossing handover at z_eq 3936, 3447 and 3000 reached twice over, through Omega_r at fixed Omega_m and through Omega_m at fixed Omega_r; P1/P3 compared across the routes; the radiation route verified to move neither r_s nor D_M and the matter route verified to move both; and at 3447 the four peaks, the fitted comb and the acoustic phase against the sky's *Bound:* a diagnosis and not a fit – 3447 is LCDM's value and not a freedom this arm has, the arm is still rejected at 4.86 per bin against the control's 2.10, and the scan is not monotone, so it measures where the equality should be and gives no direction to push it *Run:* `python3 receipts/P15_CR_cosmology/P15_the_third_peak_deficit_is_radiation_driving_and_both_routes_to_the_eq`

**P15R161** `P15_the_envelope_turnover_is_stated_on_a_third_equality_and_the_leaf_background_moves_it` [OK]
*sec:envelope* (the driving envelope's turnover is placed at equality and equality is then put at (1+z_onset)/2 = 3399, "essentially flat LCDM's" – but the leaf background the driving is integrated on has equality 3936, 14.2 per cent from the control and 15.8 per cent from the paper's own figure, so the claim that the turnover wavenumber matches LCDM's is carried by whichever of the two is meant). *Computes:* the paper's 3399 reproduced from its own arithmetic; k_eq computed from each background directly as a_eq H(a_eq)/c rather than from the matter-domination closed form; the projection measured to cancel roughly half of it, k_eq +14.2 per cent against ell_eq +7.1 per cent; and the self-consistent point (68.60, 0.2973) shown to put the leaf equality at 3370, 2.2 per cent from the control, with nothing fitted to reach it *Bound:* does NOT claim the turnover sits at equality – that is the paper's premise, taken as given – and does not claim the envelope argument fails, its conclusion resting on the closed form and not on this number; the 6797 is backed out of the paper's 3399 and is 0.5 per cent off the 6761 the instrument solves, which is noted and not chased; routed to the chat seat, whose file CR_cosmology.tex is *Run:* `python3 receipts/P15_CR_cosmology/P15_the_envelope_turnover_is_stated_on_a_third_equality_and_the_leaf_back`

**P15R162** `P15_the_damping_signature_error_budget_and_the_convention_dominates_it` [OK]
*sec:coherence* (the crossing arm's damping signature has a cosmological parameter budget of 0.02 points and an endpoint budget of 3.10 – moving r_D's upper limit from recombination to the computed visibility peak, 0.36 per cent away in redshift, takes it from +2.21 to -0.89 per cent, straddling zero – so it is not a few-per-cent observable with an error bar and must be quoted with its endpoint named). *Computes:* the visibility peak located from the computed visibility rather than assumed; UNC_error_budget's own excursion list (H0, Om, omega_b, z_rec) propagated one at a time and summed in quadrature and linearly; the endpoint term measured beside them; the clock and locus terms kept apart as configuration rather than uncertainty; and the four-way ranking re-derived through a second code path and shown to survive both endpoints *Bound:* explains rather than resolves the +3.1 against +2.2 disagreement between the instrument's header and the standalone integration – neither was wrong and the gap is this sensitivity; does not claim the ranking is endpoint-dependent, which is checked and is not; UNC_error_budget carries no such term and that is not an omission in it, one endpoint having been quoted throughout on the configuration it was written for *Run:* `python3 receipts/P15_CR_cosmology/P15_the_damping_signature_error_budget_and_the_convention_dominates_it.p`

**P15R163** `P15_at_its_own_preferred_H0_the_crossing_arm_predicts_the_acoustic_comb_with_no_fitted_number` [OK]

*sec:refit-bound* (at the (H0, Om) = (68.60, 0.2973) its own leaf ruler's DESI fit returns, the crossing arm with no pin puts the fitted acoustic comb at 298.0 against the sky's 298.4 – 0.15 per cent, as an output – with P1/P2 = 2.264 against 2.217, P1/P3 = 2.298 against 2.277, gap ratios 0.8861 and 1.1286 against 0.8812 and 1.0966, and the 700-1000 band falling from 85.9 to 4.2 per bin; chi2 is 4.19 per bin against the control's 2.10). *Computes:* the comb fitted by sec:intro's own procedure and checked sub-per-cent against the sky; the reported ruler and the fitted comb checked to agree to 1.6 per cent where at H0 = 73 they were 172.8 against 286.0; heights and both gap ratios gated against the sky; every band gated within a factor four of the control; and both instrument paths reported, the comb verified path-independent *Bound:* still rejected against LCDM at a factor 2.0, stated as the result and not as a caveat; H0 and Om ARE fitted, to BAO and the acoustic angle rather than to TT, so the cosmology is not parameter-free and only the spectrum is out-of-sample; the acoustic phase remains 2.3 per cent out and is now the only surviving position error; the height result is polarisation-path specific; and the configuration is the 133-bin unlensed one at LMAXL = 1300, not the 185-bin lensed comparison *Run:* `python3 receipts/P15_CR_cosmology/P15_at_its_own_preferred_H0_the_crossing_arm_predicts_the_acoustic_comb_w`

**P15R164** `P15_the_full_range_lensed_comparison_is_the_unfavourable_one_and_the_control_is_nearly_camb` [OK]

*sec:refit-bound* (on the 185-bin full-range lensed configuration – the one the corpus's own chi2 values are quoted on – the crossing arm at (68.60, 0.2973) sits at 2.98 per bin against the control's 1.16, a ratio of 2.57, which is WORSE than the 1.98 the 133-bin unlensed comparison showed, because lensing is worth proportionally more to the control). *Computes:* the bin count asserted at 185 over ell 100-1996 rather than taken on the configuration's name; the arm and control checked to cover the same bins; the lensing operator checked against P15_derived_lensing_on_the_lcdm_arm PART C's +6.5 per cent at ell 1900, returning 1.0666, so it is the full operator and not the first-order kernel; both arms checked to fall under lensing; and the ratio gated against being the more favourable one *Bound:* states both comparisons so neither can be picked for being the kinder; and withdraws, rather than relies on, the caveat raised at cc66.9 that the control might carry ˜700 of transfer inaccuracy – this control is within 13 per cent of CAMB unlensed and 15 per cent lensed, where the c54.178 control was 2.1 times CAMB's, so the ratio is a statement about the arm and not about the instrument *Run:* `python3 receipts/P15_CR_cosmology/P15_the_full_range_lensed_comparison_is_the_unfav`

**P15R165** `P15_the_anomalous_driving_belonged_to_the_pin_and_not_to_the_construction` [OK]

*sec:refit-bound* (the driving subtraction on the adjudicated arm is 0.96 times the control's, where the pinned arm the paragraph carries was 2.48 times – so the anomalous driving belonged to the pin and not to the construction – and the two instrument paths agree to four decimals, unlike the heights). *Computes:* the first peak located driven and undriven on both paths and on a control run here rather than quoted; the driven and undriven runs checked to report the same l_A; the ratio gated against parity and against the paper's 2.4 for the pinned arm; the two paths checked to agree; and the two-multipole difference from the quoted control priced as one grid step at LSTEP=2 *Bound:* states the convention explicitly – ell_1 over the REPORTED l_A and not over the fitted comb, which differ by 0.7329 against 0.7450 on this arm – and does not claim a smaller driving shift is evidence the arm is right, only that it removes a discrepancy; the 2.3 per cent phase residual is untouched *Run:* `python3 receipts/P15_CR_cosmology/P15_the_anomalous_driving_belonged_to_the_pin_and_no`

**P15R166** `P15_the_crossing_spectrum_reproduces_on_a_second_instrument_and_the_172_is_the_radiation_free_` [OK]

*The crossing configuration's spectrum on a SECOND implementation: node 66's code seat integrated it on the polarisation path over 133 bins, and 'PO-10''s row routed the full-spectrum run away from this seat at 'r6772+66.36' — so what is left to ask is whether an independent integration on a different path, a different wavenumber grid and a different bin range returns the same spectrum* (**It does, and the agreement is the result rather than any $\{}chiˆ2$.** Hierarchy path, 185 covered bins, $\{}ell$ 100–1996, one fitted amplitude, at $(68.62,\{},0.2973)$ with the arm's own handover datum: peaks $220/540/820/1132$ against node 66's $222/538/818/1134$, heights $2.273$/$2.319$ against $2.264$/$2.298$, the $700$–$1000$ band $4.02$ per bin against $4.23$ — ***within one grid step on every peak, within one per cent on the heights, neither seat knowing the other had run it*** **and node 66's fourth-peak excess reproduces at $1132$ against $1123.9$, which makes it the spectrum's and not either implementation's** against the configuration this replaces, on the same instrument and bins: $\{}chiˆ2 = 1205.4$ where the pinned arm gives $51817.0$ and the control $1320.5$, with positions $-0.3\{}%/+0.4\{}%/+1.3\{}%$ where

the deficit stood at $7.5\{}%$ — the first two peaks inside one grid step and ***the third at $1.28$ steps and not inside it***). *Computes:* C1 the banked pair reproduced on this scorer first ($1320.5$ and $51817.0$) before any new number was read **W1 the substituted-datum control, which reversed a reading of mine**: the first run carried 'CRIC=branchpoint', the control's super-horizon datum, and repeating it with the arm's own datum left the positions and $\{}chi^2$ alone ($1191.0\{}to1205.4$) while ***moving the heights $2.152/2.168\{}to2.273/2.319$*** — so "the heights did not come in" was the substitution's artefact and is withdrawn the amplitude is not doing the work either: the crossing arm wants the **control's** amplitude to $0.04\{}%$ where the pinned arm wanted one $8\{}%$ away three spectra banked with their exact commands in 'spectra/README.md' *Bound:* **NOT a preference over $\{}Lambda$CDM**: $-115.1$ against $F_2$'s floor of $+1114.1$, a tenth of it, and this control carries its own truncation error ('c54.186') — ***the papers quote node 66's lensed $2.57\{}times$ and this receipt quotes nothing against it***, 'PO-7' protected exactly there **NOT the discreteness control**, which is named and in flight rather than omitted: at the run's own 'NK=600' a continuum grid gives $2.8$ points per Bessel period and ***the instrument refuses to run***, clearing the guard taking 'NK=900' and $2700$ modes against the ladder's $1452$ — until it lands what stands against aliasing is node 66's independent grid, which is weaker for that question and is stated as weaker **NOT a discrepancy with the body's comb**: the paper's answer is the **leaf**, so $\{}ell_A=172.3$ is the radiation-free ruler and a different object — that reading of mine is withdrawn here and the heights are the **path's**, the line-of-sight path overshooting at $2.543$/$3.264$ *Run:* `python3 receipts/P15_CR_cosmology/P15_the_crossing_spectrum_reproduces_on_a_second_instrument_and_the_172_i`

**P15R167** `P15_the_refit_leaves_the_background_where_the_distances_put_it_and_does_not_close_the_phase` [OK]

*sec:refit-bound* (the parameter refit: given four free parameters the crossing arm moves H0 by 0.011 per cent and Omega_m by 0.20 per cent, so the background the baryon-acoustic data fix is the background the spectrum wants – and freedom does NOT close the two residuals, the acoustic phase going 2.3 to 4.5 per cent out and the fourth peak 0.9 to 0.7, so both are properties of the construction rather than background choices). *Computes:* a measured response rather than a search – 18 runs giving gradient and diagonal curvature per arm, the amplitude closed-form; the minimum VERIFIED by real runs at the best-fit parameters, predicted 118.2 and 170.6 against measured 118.3 and 171.1; every parameter flatness-tested against a one-step excursion, all four constrained on both arms; and the bin count checked at 132 *Bound:* the range is 132 bins, ell 100-1287, not the 185 first asked for, so omega_b and n_s are the loosest numbers; tau is not a free direction at all, being exactly degenerate with the amplitude where no reionisation is modelled; and the arm is still disfavoured at 1.45 times the control, stated as the result rather than a caveat *Run:* `python3 receipts/P15_CR_cosmology/P15_the_refit_leaves_the_background_where_the_distances_put_it_and_does_n`

**P15R168** `P15_the_signature_collapses_at_the_adjudicated_ratio_and_the_endpoint_sets_its_sign` [OK]

*sec:diffusion-scale* (at the adjudicated ratio the diffusion-scale signature costs 12.6 and 1.7 in chi2 over 185 bins – under a tenth per bin on either endpoint – and a tilt of -0.0093 or +0.0034 absorbs seven tenths of that, against 159.9 and -0.0341 at the 1.082 every quantitative leg of the row was computed at; and the two r_D endpoints disagree in SIGN, not only in size). *Computes:* the ratio re-measured at the adjudicated background on both endpoints, 0.10 and 0.07 points from the 1.022 and 0.991 the row carries; the joint amplitude-and-tilt fit on the arm's own 185-bin spectrum with the window asserted at ell 100-1996 and the tilt's uncertainty propagated through the fit's own parameter covariance; the tilt's running with the window measured at 5.80 against the 5.54 the (l_max/l_min)^2 result predicts; the r^2-1 scaling on which the whole re-pricing rests checked to 2 per cent in the tilt and 8 per cent in the chi2; the base and the lensing each shown to move nothing; the fitter controlled on injected tilts; and the endpoint settlement measured – the instrument needing no surgery to terminate both integrals at one epoch, and the corpus's +2.2 per cent being the COMMON reading at recombination rather than the mixed one the ruling describes, which measures 1.02830 *Bound:* the endpoint was settled by the chat seat at r6797 AFTER this run – both lengths to the visibility peak – so the ruled reading is 0.99179, -0.82 per cent at the adjudicated background rather than the -0.9 per cent the ruling quotes, which is that reading at the H0 = 73.00 background; the recombination row is kept as what the corpus carried and not as an open alternative; the structural results are carried from C62 and not re-derived; and nothing here says the arm fits, the refit leaving it at 1.30

per bin against the control's 0.90 and the full-range lensed comparison at 2.57 times *Run:* `python3 receipts/P15_CR_cosmology/P15_the_signature_collapses_at_the_adjudicated_ratio_and_the_endpoint_set`

**P15R169** `P15_the_low_multipole_floor_moves_with_no_background_and_the_factor_two_is_the_late_isw` [OK]

*sec:largescale* (on the adjudicated background the low-multipole floor does not move – r_0 shifts 0.26 per cent and the lowest mode lands at l_2 = 7.81 against 7.74 – and the two Boltzmann arms still disagree, by a factor 2.02 at l = 4 and 1.73 at l = 3, WIDER than the control's 1.84; and the disagreement is located in the LATE ISW, which the photon-hierarchy arm does not have and cannot be made to have). *Computes:* r_0 computed from the paper's own parameter-free formula and asserted to reproduce its 5064 Mpc before being used; both arms given ONE r_0, the banked run's 2.75/D_M ladder re-run separately so the 4.8 per cent that correction costs is measured rather than waved at; each arm validated on the control background against what is banked before either is moved; the geometry ruled out by measurement, 1 per cent in r_0 buying 3.5 per cent at l = 4 against a 102 per cent gap; the late-ISW diagnosis tested by pushing the line-of-sight cut from z = 53.5 to z = 15 and finding l = 3 and l = 4 move towards the CAMB arm; and the hierarchy shown non-finite past z about 10 under the defaults and under two refinements of the step count and the freeze threshold, so it is not a step-size failure *Bound:* which arm is right is NOT settled – carrying the late ISW is a reason to prefer the CAMB arm AT THESE MULTIPOLES and not a general verdict, the hierarchy existing because the fluid-plus-envelope treatment fails at high l; and node 66's part (b), the floor scored against the sky, is NOT attempted because its stated condition – that the two paths can be brought together – is not met, the fix being a named instrument build (the post-recombination source carried with the photon hierarchy decoupled) rather than a knob *Run:* `python3 receipts/P15_CR_cosmology/P15_the_low_multipole_floor_moves_with_no_background_and_the_fact`

**P15R170** `P15_the_full_range_refit_holds_the_background_and_the_phase_residual_was_quantised` [OK]

*sec:refit-bound* (the 185-bin full-range refit: with the damping tail inside the fit the crossing arm's background stays pinned – H0 moves 0.028 per cent and Omega_m 0.031 per cent, so Omega_m is held six times tighter than on 132 bins and H0 twice as loosely, both inside a thirtieth of a per cent – and cc66.18's phase residual of 4.5 per cent is WITHDRAWN as quantisation, the control having returned the same number). *Computes:* the minimum located on a measured response (18 runs, gradient and diagonal curvature per arm, amplitude closed-form) and then VERIFIED by real LMAXL=2000 runs at the best-fit parameters, predicted 185.1 and 292.5 against measured 186.5 and 292.4; every parameter flatness-tested, all four constrained on both arms; the bin count asserted at 185; and the peak locator's own resolution measured against a banked LSTEP=1 spectrum BEFORE use – the raw locator errs by 3.1 in ell at the grid the refit runs on and the parabolic refinement by 0.13 *Bound:* the arm is still disfavoured at 1.57 times the control at the verified minimum, stated as the result rather than a caveat; tau is not a direction of the fit at all, being exactly degenerate with the amplitude where no reionisation is modelled; and the refined phase residual, 3.4 per cent against the control's 1.6, changes the numbers cc66.18 landed while leaving its verdict – freedom does not close the phase – standing *Run:* `python3 receipts/P15_CR_cosmology/P15_the_full_range_refit_holds_the_background_and_the_phase_residual_was_`

**P15R171** `P15_the_low_multipole_depth_gap_closes_and_two_defects_were_cancelling` [OK]

*sec:largescale* (the low-multipole depth gap CLOSES: the two Boltzmann treatments agree to 3 per cent across ell = 2-8 where the corpus carries a factor 2.02 at ell = 4, and the factor of two was TWO defects in the photon-hierarchy arm that were partly cancelling – a line-of-sight cut at z = 53.5 that omitted the late ISW, biasing the ratio down, and a continuum integral truncated at 0.5 k_2, biasing it up; the confrontation with cosmic variance carried then returns a wash, Delta(-2 ln L) = +1.58). *Computes:* the withdrawal measured rather than argued – the undecoupled hierarchy runs to a = 1 finite once the opacity grid's resolution is held fixed as the cut moves, so the blocker was never the LG = 12 truncation; the continuum's low-k reach converged to 1 per cent at 0.1 k_2 and shown to move ell = 2 by tens of per cent at 0.5; the two defects shown to OPPOSE, the continuum fix alone widening the gap past a factor 3; both arms on both backgrounds with the decoupled build and the module's own per-mode freeze agreeing to 4 per cent; convergence in the late step count and the opacity grid to 0.1 per cent; and the likelihood the exact scaled chi^2 with 2l+1 degrees of freedom, so the sky's own scatter is carried rather than bolted on *Bound:* it does NOT say the low-multipole sector discriminates – it does not, and that is now a measurement

rather than an open question; two claims of this line are withdrawn with it, the LG = 12 mechanism and the arm-B quartet 0.4974/0.2326/0.1933/0.6186; and nothing here touches the acoustic range, where the hierarchy exists because the fluid-plus-envelope treatment fails *Run:* `python3 receipts/P15_CR_cosmology/P15_the_low_multipole_depth_gap_closes_and_two_defects_were_cancelling.py`

**P15R172** `P15_there_is_no_phase_residual_and_the_fourth_peak_is_not_the_damping_envelope` [OK]
*sec:refit-bound* (there is no acoustic phase residual: measured against a sky whose peaks are located by the SAME procedure, with the locator's own spread as the uncertainty, this arm's phi/pi sits 1.0 sigma from the sky on the three-peak fit and 0.0 on the four-peak fit, and 1.6 and 0.6 sigma from the control – and the fourth peak is NOT the damping envelope, forcing the arm's damping to the control's moving it 0.13 in ell against an offset of 1.95). *Computes:* one locator applied identically to the sky and both arms – bin to plik_lite's bins, convert to D_l, spline, parabola over a window – with the window swept over seven values so the locator's own spread is the uncertainty the comparison carries; the locator first shown to reproduce sec:intro's own sky quartet, with the second peak identified as the hardest and its spread the largest; the three procedure mismatches in the earlier reading priced one at a time; the damping test validated by its inverse, imposing the arm's envelope on the control and getting an equal and opposite shift; and the residual decomposed against the arm's own comb width, which accounts for about half of peaks three and four *Bound:* the per-cent offset does NOT shrink under the matched procedure – 3.4 becomes 4.2 – so what dissolves the residual is the sky's own uncertainty and not the central value, and the receipt says so rather than the other way round; the three leftovers after the comb-width term are 1.4, 0.6 and 1.0 in ell, the same sign and order but not a clean constant, and a finer statement is not available at these locating spreads; order (3), the neutrino sector, is NOT run because the order made it conditional on a uniform phase offset surviving and none does; and nothing here says the arm fits, the verified refit leaving it at 1.58 per bin against 1.01 *Run:* `python3 receipts/P15_CR_cosmology/P15_there_is_no_phase_residual_and_the_fourth_peak_is_not_the_damping_enve`

**P15R173** `P15_three_quarters_of_the_arms_residual_is_the_controls_and_what_is_left_is_position_not_ampli` [OK]
*'PO-47''s successor and the acoustic figure; 'sec:refit-bound'* (**THREE QUARTERS OF THE ARM'S RESIDUAL IS THE CONTROL'S, AND WHAT IS LEFT IS POSITION RATHER THAN AMPLITUDE — BUT IT IS UNDER TWO PER CENT OF THE MISFIT.** Whitened against the full bandpower covariance the two residual vectors sit at cosine $+0.855$, and ***$73.1\{}%$ of the arm's $\{}chi^{2}$ lies along the control's own direction***, projecting out to leave $76.2$ of $283.0$ — stable at $72.5$–$73.7\{}%$ across three bin cuts. **WITHDRAWN AT 'r6881+cc66.34' — the inference, not the number.** *This row read the $73.1\{}%$ as attribution. Both arms are fitted to the same data, so $r_{\{}rm CR}=r_{\{}rm ctl}+(m_{\{}rm CR}-m_{\{}rm ctl})$ exactly and the statistic ranks MODEL SIMILARITY: a control deliberately tilted by $\{}delta n_s=+0.02$, at $\{}chi^2=214.6$, scores $82.2\{}%$ — higher than the arm. **The corrected reading points the other way: the control already fits at $0.994$/bin, so the arm's $+105.1$ excess is the construction's.** Everything else in this row stands.* ⇒ *Of the rest, one further parameter at a time:* **peak-position rescale $\{}varepsilon=-7.5\{}times10^{-4}$, $\{}Delta\{}chi^{2}=+4.68$**; damping-shape $+1.36$; **tilt $+0.09$** — and on the control the position direction buys $+0.00$ while the damping buys $+0.30$. ***So position is the construction's and the tilt is exhausted*** (the refit spent it: $\{}delta n_s=+0.01$ **costs** $+11.2$). **But $\{}varepsilon$ is $0.8$ multipoles at the fourth peak — consistent with the $+2.0$ at 'r6875+cc66.32' — and $4.7$ of $283$ is $1.7\{}%$: amplitude, tilt, damping and position together do NOT account for the shape rejection, and this receipt does not say what does.** *And the quick render's band means were about three times these: the swing's **shape** reproduces (band 4 high, band 5 low, on both arms) and its amplitude does not — **fitting the amplitude on diagonal errors rather than the covariance puts the arm $0.83\{}%$ high and roughly doubles band 4**, about half the difference*). *Computes:* the banked refit spectra reproducing the refit's own $\{}chi^{2}$ ($186.51$ and $292.42$ over $185$ bins) before anything is read off them; band means in sixths three ways (covariance amplitude with diagonal $\{}sigma$, whitened, and diagonal amplitude) with the rms growing $0.78\{}to1.39$; the whitened cosine, shared fraction and residue at three bin cuts; the three single-parameter scans on both arms; **and every headline number the figure plots recomputed independently and gated against 'cc66_fig_acoustic_numbers.npz' to seven figures, so figure and receipt cannot drift** — 13 gates *Bound:* **no mechanism is offered for the bulk of the misfit** — four directions are excluded as the whole of it and none is put in their place; "the

transfer's" is the order's phrase for *not construction-specific* and is not a claim about the implementation; the refit minima are 'r6825+cc66.25''s and are used as banked; the low-multipole floor's score is 'r6831''s and untouched; whitening mixes bins, so a whitened band mean is a mean over a rotated basis and is reported **beside** the diagonal one rather than instead of it *Run:* `python3 receipts/P15_CR_cosmology/P15_three_quarters_of_the_arms_residual_is_the_controls_and_what_is_left_`

**P15R174** `P15_the_shared_fraction_does_not_mean_what_i_said_and_the_peak_trough_pattern_is_the_arms_alone` [OK]

*node 66's work order 'r6881', 'sec:refit-bound', continuing 'r6879+cc66.33': four properties of the acoustic residual measured by the chat seat **on diagonal errors**, with the burden stated at the outset — *"all wanting the full covariance behind them"* — and the four offered *"to be overturned"*, closing on **what sets the $1.3$** and on *"if the ratio is not $1.3$ on the full covariance, say so plainly"* ( **THE SHARED FRACTION NEVER MEANT WHAT I SAID IT MEANT, AND THE THREE-QUARTERS SENTENCE IS WITHDRAWN.** Both arms are fitted to the SAME data, so ***$r_{\{}rm CR}=r_{\{}rm ctl}+(m_{\{}rm CR}-m_{\{}rm ctl})$ exactly*** (verified to $8.9\{}times10^{-16}$) — the two residuals share the $-d$ term by construction, so a "fraction along the control's direction" measures **how alike the two MODELS are** and cannot attribute a misfit. *The control the statistic needed and never got:* **a flat-$\{}Lambda$CDM control deliberately tilted by $\{}delta n_s=+0.02$ — a model nobody believes, at $\{}chi^2=214.6$ — scores $\{}mathbf{82.2\{}%}$ shared, HIGHER than the arm's $73.1\{}%$**, and the score falls away with the tilt ($82\{}%\{}to43\{}%\{}to16\{}%$), which is the signature of a similarity measure. ⇒ ***And the corrected reading points the OTHER WAY:*** the control already fits at $\{}mathbf{0.994}$/bin, so there is no shared defect for the arm to be an amplification OF; the arm exceeds it by $+105.1$ and **$m_{\{}rm CR}-m_{\{}rm ctl}$ alone carries $77.3$ of that** — ***the excess is the construction's***. *And of the order's four diagnostics, **three do not survive on the control**:* **(1)** the peak/trough pattern is the ARM'S ALONE — control $+0.03/-0.06$ (diag) and $+0.03/-0.07$ (whitened) against the order's $+0.57/-0.67$, arm $+0.36/-0.51$ and $+0.42/-0.52$ — *so the composite, "that BOTH arms show it says the bulk belongs to the transfer", **fails at its premise***; **(3)** both do grow with $\{}ell$ and the first third matches ($0.67$), but the control's last third is $0.88$ and not $1.46$; **(4)** the $\{}ell\{}simeq1000$ trough is the arm's — control $+0.06$ where the order has $-1.56$. ⇒ **WHAT DOES REPRODUCE IS THE RMS RATIO, $1.357$ against the quoted $1.34$ ($1.261$ whitened) — and the order's closing question has a plain answer: nothing amplifies anything.** $\{}lvert r_{\{}rm ctl}\{}rvert=13.34$, $\{}lvert m_{\{}rm CR}-m_{\{}rm ctl}\{}rvert=8.79$, $\{}lvert r_{\{}rm CR}\{}rvert=16.82$ — ***the factor of $1.3$ IS the model difference's own size***, $77.3$ in $\{}chi^2$. *What does NOT reproduce alongside it is the correlation: $0.837$, not $0.96$, so "the same series scaled" is too strong even before the objection to what that would mean*). *Computes:* the algebraic identity proved on the actual fitted vectors before any statistic is read off them; both arms rescored from the banked refit spectra; **FOUR CONTROLS built to break the statistic and they do** — flat $\{}Lambda$CDM tilted by $\{}delta n_s=+0.02,+0.05,+0.10,-0.05$, the first scoring above the arm and the rest falling away monotonically; each of the four diagnostics computed **twice**, once on a diagonal amplitude and once whitened against 'COV_TT', with the order's own quoted value printed beside every one; the peak/trough split taken from the sign of the CONTROL's binned $\{}mathcal{D}_\{}ell$ curvature for **both** arms so the two are sorted by the same partition — 14 gates *Bound:* **WITHDRAWING "IT IS THE TRANSFER'S" IS NOT SHOWING THE TRANSFER IS INNOCENT** — no statistic here separates a transfer defect from a cosmology difference and none is offered; what is established is that the shared fraction was never that statistic **no mechanism is added** — 'cc66.33' excluded four directions and this excludes none further the rest of 'cc66.33' stands: the band means, the three single-parameter scans and the figure are untouched, and it is the attribution sentence alone that is struck the diagnostics are computed on this tree's $179$ scored bins over $\{}ell\{}le1900$, which is the figure's own cut and not the order's; a different cut moves the third decimal and no verdict using each arm's own curvature for the split is not run, and is named rather than done *Run:* `python3 receipts/P15_CR_cosmology/P15_the_shared_fraction_does_not_mean_what_i_said_and_the_peak_trough_patt`

**P15R175** `P15_the_model_difference_is_an_acoustic_contrast_difference_and_it_is_not_one_skys_luck` [OK]

*node 66's work order 'r6885', 'sec:refit-bound', continuing 'r6881+cc66.34': **take the sky's noise*

*realisation out** — every number in the sector so far compares each arm *to the sky*, and the control fits at $0.994$/bin, so the control's residual essentially IS the noise. Four asks: the coefficient hiding inside the $73\{}%$, the exact three-term decomposition, $\{}Delta$'s shape *"because it is smooth and readable where a residual is not"*, and a short list of four mechanisms — *"match, do not fit ... a negative is worth more than a match reached by adjusting one of them"* ( **THE MODEL DIFFERENCE IS AN ACOUSTIC *CONTRAST* DIFFERENCE, AND THE REJECTION IS NOT ONE SKY'S LUCK.** $\{}chi^2_{\{}rm arm}=\{}chi^2_{\{}rm ctl}+2\{}langle r_{\{}rm ctl},\{}Delta\{}rangle+\{}lVert\{}Delta\{}rVert^2$ exactly: $282.96 = 177.88 + 27.82 + 77.26$, so ***$\{}lVert\{}Delta\{}rVert^2$ carries $74\{}%$ of the excess and the cross term $26\{}%$*** — the order's conditional does not fire. **Set the cross term to its expectation of zero and the arm still scores $256.3$, $\{}mathbf{1.43}$ per bin on a TYPICAL sky** against the control's $0.994$. *And the $1.15$ is $1.078$: the order's coefficient multiplies the WHITENED cosine $0.855$ by the DIAGONAL rms ratio $1.34$; consistently it is $1.078$ whitened or $1.137$ diagonal.* ⇒ **More than that, $b-1$ IS $\{}langle\{}Delta,r_{\{}rm ctl}\{}rangle/\{}lVert r_{\{}rm ctl}\{}rVert^{2}$ — the "amplification along the noise direction" and the cross term are ONE NUMBER seen twice — and against its own sky-random null it is $1.078\{}pm0.049$, $\{}mathbf{+1.58\{}sigma}$**, stable at $+1.07\{}sigma$ to $+1.58\{}sigma$ across four bin cuts: *a $1.6\{}sigma$ departure from $1.00$ is what noise looks like, so there is no amplification for the order's two candidate explanations of it to act on.* ⇒ **AND WHAT $\{}Delta$ IS:** it is **trough-dominated** — $+0.011\{}sigma$ at the peaks against $-0.760$ at the troughs, so *the two arms agree where the peaks are and differ between them* — and split about a smooth envelope the **oscillatory part carries $67$–$75$ of the $77.3$**, peak-high and trough-low ($+0.389/-0.420$), antisymmetric about the envelope and not about each peak. ***That is a CONTRAST, not a phase and not an envelope: the arm's acoustic oscillation is $1.040$ times the control's about its own envelope.*** One direction built from the CONTROL alone — raise the oscillation at fixed envelope, no run and nothing fitted — carries **$51\{}%$ of $\{}lVert\{}Delta\{}rVert^2$** ($46$–$51\{}%$ across a factor two in its one window), and with position, tilt and damping the span reaches $66.8\{}%$ against **$5.7\{}%$ for the whole of 'cc66.33''s four-parameter family** — *which is what "amplitude, tilt, damping and position together do not account for the shape rejection" was pointing at: the family was missing a direction and the direction has a name.* ⇒ **THE SHORT LIST GIVES THE NEGATIVE THE ORDER SAID IT WOULD RATHER HAVE.** The handover amplitude ($0.4835$ against $0.5$) is **excluded EXACTLY and without a run** — the instrument sets $\{}hat\{}Theta$ $k$-independently, so the single fitted amplitude absorbs it and $\{}chi^2$ moves by $9\{}times10^{-13}$. Switching each of the other three off on **both** arms leaves $93.7\{}%$ (neutrino depth $LN\{},12\{}to24$), $92.8\{}%$ (early ISW) and $87.6\{}%$ (the driving's continuity half) of $\{}lVert\{}Delta\{}rVert^2$; the driving's **Euler** half takes it to $68.3\{}%$ but at cosine $-0.16$, so *$\{}Delta$ is REPLACED rather than reduced and the $32\{}%$ is a norm and not a share*. ***Not one of them removes the contrast*** ($1.038$–$1.044$ against $1.040$). ⇒ **AND A FIFTH, WHICH THE SHAPE NAMES BY ITSELF AND WHICH THE INSTRUMENT'S OWN DOCSTRING ALREADY DESCRIBES**: a mis-weighted Doppler term *"fills the troughs at high multipole ... while leaving the FIRST peak's position almost alone"*. *Its knob is a **KNOB SHADOW** on the refit path — '_SWSRC' and '_DPSRC' are read only inside 'los_spectrum', 'HIER=1' does not take that path, and 'DPSRC=0' at the refit configuration returns a **BIT-IDENTICAL** spectrum on both arms.* ⇒ Tested where the knob reaches: **removing the Doppler dipole nearly doubles each arm's oscillation about its own envelope (to $1.85$) and its shape sits at cosine $\{}mathbf{0.88}$–$\{}mathbf{0.89}$ with the contrast direction — the Doppler dipole IS the contrast knob** ⇒ *and the MONOPOLE brackets it from the other side — 'SWSRC=0' **inverts** the oscillation, to $-0.300$, at cosine $-0.79$, so **the contrast direction is the monopole-to-dipole BALANCE**, which neither switch alone would have established* ⇒ **AND 'r6887''s SHARPENED ASK — $\{}Delta$ read against the three features 'cc66.34' proved are the arm's own — comes back: ALL THREE ARE $\{}Delta$'S.** *The trough deficit ($-0.760$ against the control's $-0.200$); the $\{}ell\{}simeq1000$ trough, where the control sits at $-0.103$, the arm at $-1.177$ and **$\{}Delta$ carries $-1.074$, $91\{}%$ of it, in $15$ bins**; and the growth with multipole, which $\{}Delta$ does **monotonically** ($0.33\{}to0.55\{}to0.68$) where the residuals do not — because a residual is $\{}Delta$ plus a noise floor of order one and the floor flattens it.* *'r6887' also voids its own part (1); the coefficient is reported anyway because it had been run and what it shows is not the coefficient as evidence

but that it **IS** the cross term*). *Computes:* the three-term identity proved on the actual fitted vectors before anything is read off it, at four bin cuts; the coefficient shown to be the cross term ALGEBRAICALLY (to $3\{}times10^{-16}$) rather than numerically alongside it; the handover-amplitude exclusion done by rescaling the banked spectrum and refitting, so it is measured and not argued; the contrast direction built **twice**, from a polynomial envelope in $\{}ln\{}ell$ (three degrees) and from a running geometric mean over one acoustic period (four windows), with its one tunable reported over a factor two rather than chosen; **the candidate test chosen to be the right one and the wrong one printed beside it** — a knob's per-arm shape can sit at cosine $-0.63$ with $\{}Delta$ and still carry none of it, because a knob that does the same thing to both arms CANCELS in the difference, so the column is $\{}lVert\{}Delta\{}rVert^2$ with the knob off; **the knob shadow PROVED rather than inferred** — a static check that '_DPSRC' occurs once and inside 'los_spectrum', a bit-identical pair on the refit path, and a $62\{}%$ move on the path where it is wired, which is the calibration r4558's own note demands before any null is read off a switch; and $\{}Delta$ cross-checked between the two instrument paths at cosine $+0.76$ with the same contrast excess — **38 gates**, four candidate pairs and three line-of-sight pairs banked at 'spectra/r6885_*' with their commands *Bound:* **NAMING THE SHAPE IS NOT NAMING THE MECHANISM** — "the arm's oscillation is $4\{}%$ larger at fixed envelope" is a measurement, and the channel is identified rather than the cause measured: deleting the Doppler term is all-or-nothing, it leaves the arms' contrast RATIO where it was and DOUBLES $\{}lVert\{}Delta\{}rVert^2$, so the arms differ in HOW MUCH that channel supplies and not in whether it is there **$33\{}%$ of $\{}lVert\{}Delta\{}rVert^2$ is unnamed by any direction here** switching a driving or source term off is a LARGE excursion and not a derivative: both arms move far from their own minima, so the candidate column answers "does $\{}Delta$ survive this" and not "this term contributes so much of it" the $LN$ result is a DIRECTION and not a convergence test — doubling the depth removes $6.3\{}%$ and the sign says more depth removes more the line-of-sight path's $\{}chi^2$ is **not** comparable with the hierarchy path's, as this tree's own 'spectra/README.md' records, so only shapes and within-path ratios are read from PART 5 nothing here bears on whether the transfer or the cosmology is at fault — 'r6881+cc66.34' withdrew the statistic that was read that way and this adds no replacement **the knob shadow is reported, NOT repaired**: wiring a source term is a change to the instrument *Run:* `python3 receipts/P15_CR_cosmology/P15_the_model_difference_is_an_acoustic_contrast_difference_and_it_is_not`

**P15R176** `P15_the_source_decomposition_is_reachable_on_the_reporting_path_and_the_neutrinos_have_a_knob` [OK]

*node 66's work order 'r6889', 'sec:refit-bound', continuing 'r6885+cc66.35': **the knob shadow, before anything else.** That receipt found '_SWSRC'/'_DPSRC' read only inside 'los_spectrum', so 'HIER=1' — the path every refit number in this sector is computed on — was blind to them. The order: *"make the source decomposition reachable on the hierarchy path, or establish that it cannot be and say why," * then re-run the two bracketing tests there; and on the free-streaming phase shift, *"either a knob is buildable — in which case say what it would cost — or the phase shift cannot be tested on this instrument at all"* (* **THE SHADOW IS CLOSED, AND THE DEFAULT IS PROVED RATHER THAN ASSERTED.** '_SWSRC' and '_DPSRC' now reach all three source constructions — 'los_spectrum', the hierarchy path and the low-multipole path — where they reached one; the monopole bracket is **split** so the switch multiplies $g(\{}Theta_0+\{}Psi)$ and not the polarisation term that shares it. ⇒ ***The unset configuration reproduces the banked spectra with $\{}max\{}lvert\{}Delta\{}mathcal{D}_\{}ell\{}rvert = \{}mathbf{0.0}$ on BOTH arms — exactly zero, not nearly*** — so nothing this sector has reported moves. *And that took a second attempt, which is on the record: written '_SWSRC * g_ * (\{}Theta_0+\{}Psi) + \{}ldots' the default sums in a different **order**, and floating-point addition is not associative — measured, $1.1\{}times10^{-16}$. Keeping the factor INSIDE the bracket makes it exact, because $x\{}times1.0$ is.* ⇒ **AND THE PAIR THAT MAKES THE SHADOW A PROOF RATHER THAN A STORY**: the same switch, the same two commands, **exactly $0.0$ before the repair and $62.3\{}%$/$61.3\{}%$ after** — *a null and an unwired knob are indistinguishable until both halves are on the record, which is 'r4558''s own rule and the reason this revision exists.* ⇒ **THE TWO BRACKETING TESTS SURVIVE THE MOVE TO THE REPORTING PATH**: 'DPSRC=0' at cosine $\{}mathbf{+0.798}$ with the contrast direction (LOS: $+0.893$), taking the arm's own oscillation to $1.828$ (LOS: $1.855$); 'SWSRC=0' at $\{}mathbf{-0.765}$ (LOS: $-0.791$), **inverting** it to $-0.399$ (LOS: $-0.300$) — ***the same opposite-sign bracket, so 'cc66.35''s reading that the contrast direction IS the monopole-to-*

dipole balance was not an artefact of the path it had to be measured on.*** *The two deletions are not equally inert on the arms' contrast ratio and that is reported rather than averaged: the dipole moves it by $-0.0025$ and the monopole by $+0.042$, and neither collapses it toward $1.000$.* ⇒ **AND THE FREE-STREAMING PHASE SHIFT NOW HAS A KNOB — the answer to "what would it cost" is FOUR LINES.** What makes a neutrino free-stream is its anisotropic stress $\{}sigma_\{}nu=F_2/2$, at exactly two dynamical sites (the Euler equation and $\{}Psi$'s shear term), so 'NUFS' multiplies $\{}sigma_\{}nu$ and the $F_2$ source on both solver paths; **the quadrupole's initial condition is exactly zero**, so at 'NUFS=0' it is never sourced, the whole $\{}ell\{}ge2$ ladder stays zero and the sector is a perfect fluid at the SAME background density — 'FNU' and the density fractions untouched. ⇒ ***It moves $\{}mathcal{D}_\{}ell$ by $36.3\{}%$/$37.3\{}%$ and pushes EVERY peak on BOTH arms to LARGER $\{}ell$, $+9.00$ uniformly across the first four*** — which is the sign the Bashinsky–Seljak pull has and the shape a **phase** shift has rather than a rescaling of the acoustic scale. *Its size is exactly one binned grid step ($8.9999$), so the sign and order of magnitude are established and the value is not*). *Computes:* the shadow stated as a property of the SOURCE TEXT and not of a memory — '_DPSRC' occurring once, inside 'los_spectrum', before the repair — beside the banked bit-identical pair that found it, re-gated here; the repair's default gated at **exactly zero** on both arms and on identical multipole grids, before any other number in the receipt is read; **each newly wired switch calibrated where it was newly wired**, because 'r4558''s own note is the rule this revision enforces and a new knob that had not been shown to move something would be the same defect landed twice; the 'NUFS' claim carried by three static checks rather than by assertion — four use sites, 'y0 = np.zeros(...)' with no assignment to the quadrupole anywhere in the file, and no '_NUFS' within the density-fraction block, which is what makes 'NUFS=0' a perfect fluid rather than a frozen state; and the peak shift gated on its **sign** and its **uniformity across the first four peaks** rather than on its size, with the size then shown to be one grid step — 22 gates, four pairs banked at 'spectra/r6889_*' with the launcher at 'computations/beyond_the_wall/r6889_directions/' *Bound:* **NO MECHANISM FOR THE CONTRAST IMBALANCE** — 'r6889' said that boundary is right and is not moving it, and neither does this **and no VALUE for the phase shift as a result about this construction**: the knob is built and calibrated, its size is grid-limited, and pointing it at the question is the next order's the drag half — the amplitude suppression free-streaming also causes, $36$–$37\{}%$ in $\{}mathcal{D}_\{}ell$ — is the large effect and must not be quoted as the same number as the phase shift the low-multipole path is wired for consistency and is **not exercised** here; naming it rather than quietly leaving it is the point **no other knob is audited**: three shadows have now been found in this sector by three different routes, and nothing here says there is not a fourth *Run:* `python3 receipts/P15_CR_cosmology/P15_the_source_decomposition_is_reachable_on_the_reporting_path_and_the_n`

**P15R177** `P15_the_monopole_zero_point_is_not_the_channel_and_the_two_residuals_point_opposite_ways` [OK]

*node 66's work order 'r6897': **is the contrast excess and the alternation excess one number?** — *"do not remove a term; measure a quantity both arms have"*: **(a)** the value the monopole oscillation swings **about** at last scattering, in units of that arm's own $\{}Psi$; **(b)** what each arm's own loading $R=3\{}rho_b/4\{}rho_\{}gamma$ at its own visibility peak accounts for; **(c)** *"from the residual alternation alone, solve for the effective $\{}Delta R$ that would produce it; from the residual contrast alone, solve for the effective $\{}Delta R$ that would produce it — do the two agree?"*, the order adding that **no mechanism is being asked for and none should be claimed**, and asking for 'ETA_LS_W' in the header while I was in there* ( **THE MONOPOLE'S ZERO POINT IS NOT THE CHANNEL, AND THE SIGN IS WHAT SAYS SO.** On both arms the offset of $\{}Theta_0+\{}Psi$ approaches the tight-coupling equilibrium $-R\{}Psi$ **from below**, reaching $0.78$ of it on the control and $0.76$ on the arm at the visibility peak — *so (b)'s absolute question answers NO on both arms and by nearly the same amount, which is the approach to equilibrium and not a property of either arm: $\{}Psi$ decays inside the horizon and $R$ grows.* ⇒ **And calibrated against a KNOWN $\{}Delta R$** — the control re-run with $\{}omega_b$ displaced $\{}pm8\{}%$, the estimator tracking $77\{}%$ of it, which is 'r4558''s rule applied to a *measurement* rather than to a knob — ***the arm's offset FALLS SHORT of what its own loading accounts for, by an effective $\{}omega_b$ of $-2.9\{}%$, on top of the $-2.0\{}%$ its fitted $\{}omega_b$ already is.*** **Too much loading is what deepens troughs and raises alternation; this arm's monopole behaves as though it had too little, and its troughs are deeper anyway.** *On the order's own branching this is a THIRD outcome — neither

"the construction supplies an $R$-like offset" nor "both arms sit on their own $(1+R)$" — and it refutes the loading reading **by sign** rather than leaving it undetermined.* ⇒ **AND THE TWO RESIDUALS ARE NOT ONE NUMBER: THEY DIFFER BY A FACTOR OF SIXTEEN AND IN SIGN.** Against the control, the arm's **contrast** is $+4.0\{}%$ and its **alternation** — the odd-even amplitude, with the envelope's own curved decline *fitted* rather than assumed away — is about $2\{}%$ **lower**. Converted through the control's own measured responses ($\{}mathrm d(\{}text{contrast})/\{}mathrm d\{}omega_b = +7.97$ and $\{}mathrm d(\{}text{alternation})/\{}mathrm d\{}omega_b = +1.85$, both positive, so more loading gives more of both), **the contrast asks for an effective $\{}omega_b$ of $\{}mathbf{+22.9\{}%}$ and the alternation for $\{}mathbf{-1.4\{}%}$.** ⇒ **AND THE $+22.9$ IS A LOWER BOUND, BECAUSE THE CONTRAST RESPONSE SATURATES**: over the whole $\{}pm8\{}%$ range the contrast spans only $0.028$ and its increments fall monotonically ($+0.0119, +0.0087, +0.0054, +0.0020$), so a straight line *overstates* what $\{}omega_b$ can deliver — ***a four per cent contrast excess is beyond what the baryon density reaches in this construction at any value, not merely at an implausible one.*** *And the monopole offset agrees with the alternation, at $-2.9\{}%$ against $-1.4\{}%$: two independent measurements of the loading displacement agree and it is the contrast that stands apart.* ⇒ **AND PART OF THAT IS A CORRECTION TO THE ORDER'S PREMISE, WHICH MIXED TWO REFERENCES**: the alternation excess is read against the **sky** ($P_1/P_2=2.264$ against $2.217$) and the contrast against the **control** ($1.040$). *Against the control, on the banked spectra, the arm's $P_1/P_2$ is $\{}mathbf{2.141}$ against $\{}mathbf{2.191}$ — **lower**.* **The two were never pointing the same way; the appearance that they were is the two references.** ⇒ **ONE POSITIVE LOCALISATION, WITH ITS OWN SIGN PROBLEM STATED**: the arm's **dipole-to-monopole** amplitude ratio at the visibility peak is $+1.8\{}%$ and **grows** with wavenumber. *But the Doppler term FILLS troughs — ‘cc66.36‘ measured ‘DPSRC=0‘ taking the arm's oscillation from $1.040$ to $1.828$ — so a larger dipole fraction makes troughs shallower, and this arm's is larger while its troughs are deeper. It works AGAINST the contrast excess, which leaves a source larger than four per cent and partly cancelled — a statement about size, not a mechanism.* ⇒ And the header now carries the visibility peak, its redshift, its FWHM and $R$ there on **every** path: the two arms' last-scattering widths side by side for the first time on the reporting path, $38.04$ Mpc against $43.59$). *Computes:* ‘ZPSAVE‘ is new and wired into ‘hier_run‘ — the **reporting** path — because ‘PHISAVE‘ is one of the nine switches ‘r6893+cc66.37‘ measured as OFF it, so using it would have measured the low-multipole construction and called it the reporting one; its default gated **bit-identical** on both arms before any number is read, and it saves the **undamped** monopole because the envelope multiplies the offset too; the offset estimator's three predecessors recorded with the bias each carried (a midpoint of extrema alternating by a factor of three; a quarter-half-quarter combination leaving $\{}pm15\{}%$ of curvature; a one-period window ill-conditioned by collinearity) and the verdict shown unchanged across five window widths; the calibration against a known $\{}Delta R$ carried as its own gate; the contrast statistic reproducing ‘cc66.35‘'s $1.0401$ so the space is that receipt's; the alternation's sign shown independent of the trend degree; and both $\{}omega_b$ responses shown non-null and linear over the range, so neither residual is converted through a response that is not there — every long run sliced on ‘KBATCH‘, which ‘r6895+cc66.38‘ measured as exact to $10^{-16}$ *Bound:* **NO MECHANISM FOR THE CONTRAST IMBALANCE** — the order says none is asked for and none is claimed **the dipole excess is NOT claimed to produce the contrast excess**, its sign being wrong for it the offset's shortfall against $-R$ is **not** a defect of either arm: it is the same shortfall on both the $\{}omega_b$ response is measured on the **control** and the two arms differ in more than $\{}omega_b$, so each implied displacement is *effective* and is called that the width story the order withdrew before sending is **not** reinstated — nothing here attacks its reasoning *Run:* `python3 receipts/P15_CR_cosmology/P15_the_monopole_zero_point_is_not_the_channel_and_the_two_residuals_poin`

**P15R178** `P15_the_free_streaming_knob_is_common_to_both_arms_and_the_switch_sweep_finds_no_further_shado` [OK]

*node 66's work order ‘r6891‘, ‘sec:refit-bound‘, continuing ‘r6889+cc66.36‘, in two parts: **point ‘NUFS‘ at the question on a grid that can resolve it** — *"a locator on a finer $\{}ell$ grid, both arms, so the shift is measured rather than bracketed at one bin step — and the two arms' shifts compared, since a phase shift common to both is not a candidate for $\{}Delta$ and one that differs between them is"*, with *"the drag half reported separately"*; and **sweep the switches, because the rate is the finding** — *"enumerate every environment switch this instrument reads, and for*

*each one record which of the three source constructions and which of the two solver paths actually reads it — from the source text, not from memory — and, for every switch reachable on the reporting path, whether setting it away from its default moves the spectrum there"*, the order adding that **the output wanted is the table and not a repair campaign**  ( **THE SWEEP IS EXHAUSTIVE AND THE ANSWER IS AN ACCOUNTING, NOT A NEGATIVE.** Sixty environment switches enumerated from the source text through ‘ast‘, each tagged with which of the three source constructions and which of the two solver right-hand sides reads it and whether that site is *written into* the construction or merely downstream of it; the reporting path declared from the dispatch's own three lines, with ‘main‘'s live prefix ending at the ‘return 0‘ of the ‘HIER‘ branch. *A **binding** is not a **use**, and that distinction is the whole of the static half — a first pass keyed to binding sites called fifty-five of sixty live, because a module-level ‘os.environ.get‘ executes on every path and proves nothing.* ⇒ **Fifty-one have a live use site there; the nine that do not come back BIT-IDENTICAL on the reporting path — all eighteen runs at exactly $0.0$, twice over, and again at full $\{}ell$ reach where ‘DAMPX‘ and ‘RD‘ would have shown if they touched the damping tail.** ⇒ ***AND FIFTY-FIVE SWITCHES MEASURED ON BOTH ARMS, ONE AT A TIME: the set of runs that came back bit-identical is EXACTLY the set four readings predict — off-path, rate-identity on the control arm, the arm branch, and gated by another switch — no unexplained null, and nothing explained away that in fact moved.*** *The rate-identity family is the pleasing one: ‘GSRC‘, ‘LEAFSCALES‘, ‘PHASEONLY‘, ‘PHASEPOW‘, ‘STACKPERT‘ are bit-identically inert on the **control** and every one moves the arm, because on the control ‘Hphys‘ and ‘Hleaf‘ are the same expression, the Jacobian is $1$ and $x\{}times1.0$ is exact — the same floating-point fact ‘cc66.36‘'s reassociation episode turned on.* ⇒ **TWO THINGS THE SWEEP DID TURN UP, neither of the ‘r6476‘ class and neither moving a reported number:** **(i)** ‘LRS-FROM‘ moves the instrument's reported $r_s$ by a quarter ($145.38\{}to110.49$ Mpc, $\{}ell_A$ $301.5\{}to396.8$) and $\{}mathcal{D}_\{}ell$ by **exactly zero** — $R_S$ and $\{}ell_A$ are **diagnostics**, reaching a ‘print‘ and the ‘SAVE‘ metadata and nothing else, and ‘hier_run‘ accepts $\{}ell_A$, $D_M$ and $r_s$ and **loads none of the three**; *‘r6476‘ certified that switch on the ground that the header moved, and **the header is not the reported number** — the certification was of the wrong quantity, and no number moves*; ⇒ *and the consequence points the useful way — because no transfer function reads $\{}ell_A$, the printed $\{}ell_1/\{}ell_A=0.7312$ against the sky's $220.6/301.7$ is an agreement between **two independently computed quantities** and not a value fed in*; **(ii)** ‘LATARG‘, which the file calls the corpus's one fitted number, has **no root** at the arm's adjudicated background — the onset solve raises across the whole bracket ‘r6760+cc66.1‘ had widened — which is why the refit command supplies ‘ZSTART=3e7‘. ⇒ **AND THE FREE-STREAMING SHIFT IS MEASURED.** Two independent locators — a parabola vertex at each acoustic extremum and a sub-bin cross-correlation of the **envelope-normalised** oscillation — on both arms, and again at $\{}mathtt{LSTEP}=2$, four times the multipole sampling, where they agree to better than half a multipole. ***IT IS NOT UNIFORM, WHICH CORRECTS ‘r6889‘***: band by band it RISES monotonically, $+2.9/+5.6/+9.4/+10.9/+12.6$ on the control and $+3.1/+5.7/+9.3/+10.9/+12.5$ on the arm — *the uniformity ‘cc66.36‘ gated was the grid, and the gate was a tolerance of $0.05$ on integers that could only differ by a whole bin*; *and a pure rescaling does not fit it either, the straight line having an intercept of three multipoles.* ⇒ ***AND IT IS COMMON TO THE TWO ARMS — largest band-by-band difference $0.17$ of a multipole, and in units of each arm's own $\{}ell_A$ the extremum-averaged shift agrees to $1.7\{}times10^{-4}$ — so on the order's own criterion it is NOT a candidate for $\{}Delta$.*** ⇒ **And the drag half, kept apart and split in two because it is two things: an ENVELOPE rescale of $1.2573$ and a peak-to-trough CONTRAST ratio of $0.9875$, the two arms agreeing to $0.0004$ and $0.0006$ — and $\{}Delta$ is a CONTRAST difference, so the knob's large effect is in the wrong quantity.** *In $\{}Delta$'s own space the arms' ‘NUFS‘ directions sit at cosine $0.99918$; their difference freely rescaled reaches $8.7\{}%$ of $\{}lVert\{}Delta\{}rVert^{2}$ at a **negative** coefficient, and no such freedom exists because both arms carry one neutrino sector*). *Computes:* the reporting path declared from the dispatch and gated line by line, with branch pruning only where an environment comparison settles the test so the live set OVER-counts and can only under-report a shadow; ‘hier_run‘ gated as accepting three quantities and loading none of them; the nine off-path switches gated bit-identical at reduced AND full reach, and ‘RD‘ at a second value so the null is not one value's accident; the accounting gated as a SET EQUALITY between observed nulls and predicted nulls, which is the only form the claim is worth anything in; the shift's non-uniformity gated on the

spread exceeding the grid step and on monotonicity band by band, on two locators and two grids; the annotation of the instrument gated bit-identical on both arms because ‘cc66.36‘ is why this seat no longer asserts that a change is nothing — 46 gates (34 at ‘r6893+cc66.37‘; the twelve added at ‘r6895+cc66.38‘ are the finer grids’ own and the slicing check’s), the screen banked at ‘spectra/r6893_switch_screen_{lcdm,cr}.npz‘, the fine grid at ‘spectra/r6893_fine_grid_*‘ and the full-reach nulls at ‘spectra/r6893_full_reach_nulls_lcdm.npz‘ *Bound:* **NO MECHANISM FOR THE CONTRAST IMBALANCE** — ‘r6891‘ says that boundary has not moved and this does not move it **the growth of the shift with $\{}ell$ is not attributed**: separating the Bashinsky–Seljak constant from the $k$-dependent change in the potentials’ decay that ‘NUFS=0‘ also causes is not attempted **the sweep is of ENVIRONMENT switches**: a hard-coded literal that ought to be a switch is a different search, and ‘cc66.17‘’s ‘NS‘ literal is the reminder the switch screen runs at $\{}mathtt{LMAXL}=500$ and only the NULLS are re-run at full reach — a switch that moves is not re-measured there no pinned value of $n_s$, $H_0$, $\{}Omega_m$ or $\{}omega_b$ *Run:* `python3 receipts/P15_CR_cosmology/P15_the_free_streaming_knob_is_common_to_both_arms_and_the_switch`

`make_fig_acoustic_two_arm` [OK]

*‘r6879‘’s figure; ‘sec:refit-bound‘* (**THE ACOUSTIC FIGURE, BANKED AND RECEIPTED.** Two panels plus residuals: **(a)** linear in $\{}ell$ in $\{}mathcal{D}_\{}ell$, both arms at their verified $185$-bin refit minima through P15’s own derived lensing operator, **the single amplitude fitted on the full bandpower covariance rather than diagonal errors**; **(b)** logarithmic from $\{}ell=2$, so the discrete closed-$S^{3}$ floor sits in the same frame as the peaks with its **minimum at $\{}ell=4$ rather than the quadrupole** marked — the one shape this construction predicts that the standard model does not. **Whitened residuals beneath each, both arms on one axis so the shared part is visible.** *The floor is carried by the sweep as a ratio to the flat transfer and is put back into $\{}mathcal{D}_\{}ell$ against the control’s own low-$\{}ell$ spectrum — a reconstruction, named as one, there being no absolute low-$\{}ell$ bandpowers in this tree.* *And $\{}mathcal{D}_\{}ell$ rather than $C_\{}ell$ is not a presentation choice: plotted as the binned $C_\{}ell$ the likelihood ships, the peaks vanish under the falling plateau — the same trap the locator hit at ‘cc66.28‘, here in the figure*). *Computes:* writes ‘corpus/fig_acoustic_two_arm.pdf‘ and banks every plotted number to ‘spectra/cc66_fig_acoustic_numbers.npz‘, which the companion receipt recomputes and gates *Bound:* the $1900$ cut on the scored bins is the figure’s own so the shifted-model tests share the bin set; the full $185$-bin $\{}chi^{2}$ is quoted in the caption from the refit itself; no foreground or beam term beyond what ‘plik_lite‘’s covariance marginalises *Run:* `python3 receipts/corpus/make_fig_acoustic_two_arm.py`

**P15R179** `P15_the_skys_own_fourth_peak_cannot_tell_the_arms_apart_and_the_displacement_is_shared_with_th` [OK]

*‘PO-47‘; ‘sec:intro‘’s locator, continuing ‘r6835+cc66.28‘* (**THE STOPPING RULE FIRES — AND THE SPREAD THAT FIRES IT IS NOT THE ONE ‘cc66.28‘ QUOTED.** That receipt gave the sky’s fourth peak as $1121.9\{}pm0.87$, and **that $0.87$ is a PROCEDURE spread** — how far the answer moves as the parabola window is swept on one realisation. ⇒ Propagating ‘plik_lite‘’s **bandpower covariance** through the identical parabola gives the sky’s own uncertainty: ***$\{}ell_4=1121.9\{}pm2.03$ (statistical) $\{}pm0.87$ (procedure) $=\{}pm2.21$ combined*** at the reference window, $\{}pm1.57$ at the tighter one. **Against that, the construction’s own displacement — arm minus control, $+2.00$ in $\{}ell$ — is $0.90\{}sigma$ at $W=55$ and $1.27\{}sigma$ at $W=80$: it does not reach significance at either admissible window, so THE SKY CANNOT TELL THE TWO MODELS APART AT THE FOURTH PEAK.** *What is established is ”not resolved” rather than ”zero”, and the receipt says so rather than smoothing it.* ⇒ ***The three candidates are NOT run, per the order’s own stopping rule.*** **AND WHAT DOES NOT DISSOLVE IS SHARED AND IS NOT THIS CONSTRUCTION’S**: both models sit high of the located sky — the control by $+2.65$ ($1.2\{}sigma$) and the arm by $+4.65$ ($2.1\{}sigma$), **$57\{}%$ of it the control’s** — named and not pursued. *And the order’s $+10.1$ was never this number: it is the paper’s stored quartet against the paper’s stored sky, two procedures; through one locator the arm-minus-sky offset is $+4.65$.* **AN INSTRUMENT TRAP IS REPORTED AS A FINDING**: ‘cc66.28‘’s locator finds peaks by free extremum search, which is right on a smooth spectrum and **unusable on a noisy one** — under the covariance it returns $1051\{}pm132$ for the fourth peak, seventy multipoles off. **Error propagation through it must anchor the window**). *Computes:* the sky quartet and its procedure spreads reproduced against ‘cc66.28‘; $600$ covariance realisations

through the anchored parabola at four windows with per-peak bias and spread; the window chosen on the **bias–variance trade** ($W=40$ rejected — the parabola inverts on $12\{}%$ of realisations and $\{}sigma_4$ explodes to $122$; $W=110$ rejected — bias reaches $-6.4$, larger than the displacement under test); seed-independence; and the three pairings in $\{}sigma$ — **11 gates with TWO controls that break as they must**: a planted displacement of $4$ and $10$ in $\{}ell$ recovered to $+4.56$ and $+11.71$ (so a real displacement WOULD show), and the free-search failure itself *Bound:* **not a full sky-uncertainty budget** — ‘COV_TT‘ is the shipped bandpower covariance with foregrounds and calibration marginalised, with no beam or theory-side term added; the central value is the locator’s on ‘plik_lite‘’s binning and ‘sec:intro‘’s own $1123.9$ sits two multipoles away, inside. **None of ‘PO-47‘’s three candidates is run** — the third acoustic spacing, the baryon loading’s effect on position and the transfer’s resolution at the fourth peak are left, a convergence check being named as where the shared offset would go. Nothing re-opens the phase intercept, the damping envelope, the driving or the refit *Run:* `python3 receipts/P15_CR_cosmology/P15_the_skys_own_fourth_peak_cannot_tell_the_arms_apart_and_the_displacem`

**P15R180** `P15_the_ladders_discreteness_does_not_set_the_spectrum_and_the_check_was_run_in_pieces` [OK]

*The control ‘r6788‘ declared open and node 66 asked for by name (‘FOR_60‘ item 5): ‘KCONT=1‘, the CR arm’s discrete $k$-ladder replaced by a continuum grid on the crossing configuration. The alias gate waives its 4-points-per-Bessel-period requirement for this arm because the ladder is physical, and the waiver’s own text says that is only not aliasing if the answer does not depend on it* (**It does not, and the waiver is honest here.** Ladder ($1452$ modes, $2.3$ pts/period) $\{}chi^2 = 1205.3755$ against continuum ($2700$ modes, $4.3$) $\{}chi^2 = 1205.3745$ — ***peaks identical*** at $220/540/820/1132$, heights identical to three decimals, the single fitted amplitude $3$ parts in $10^{8}$ apart, spectra agreeing to $6.7\{}times10^{-8}$ of the peak **and the reason it agrees far better than $2.3$ points per period predicts is measured rather than guessed**: $\{}sqrt{L(L+2)}\{}to L+1$, so the ”discrete ladder” is asymptotically *uniform* — within $1\{}%$ of its median spacing over $99.9\{}%$ of its gaps — making this a comparison of two near-uniform samplings differing in spacing, not of resolved against aliased **run in six slices, because neither seat’s container survives it whole**: the check costs $2700$ modes against the run’s $1452$, mine was killed three times and ‘cc66‘ restarts every $1$–$15$ min). *Computes:* C1 **the segmentation proved exact before anything was read through it** — a whole run against three slices summed, $\{}max\{}lvert{\{}rm sum}-{\{}rm whole}\{}rvert$ relative $2.195\{}times10^{-16}$, machine precision ‘KSLICE=lo:hi‘ added with its default proved byte-identical both spectra scored on the same $185$ covered bins over the same $\{}ell$ range **and a property ‘KBATCH‘ already had, measured and sized**: ‘_project‘ takes $dk = \{}nabla k$ from the *batch*, so on the ladder’s non-uniform bottom modes the batch boundary moves the weights — one batch against two differs by $1.205\{}times10^{-8}$ relative *Bound:* **Settles one thing only**: that discreteness does not set this spectrum. Says nothing about whether the spectrum is right, and ***nothing that touches the floor*** — $F_2 = +1114.1$ still forbids reading $1205.4$ against the control’s $1320.5$ as a preference, exactly as ‘r6788‘ said measured at **one** configuration, so ‘c54.186‘’s $0.7\{}%$ at the pinned one is a different ladder density and is not superseded the ‘KBATCH‘ figure is **negligible at $10^{-8}$** — eight orders below anything physical, a reproducibility note (quote ‘KBATCH‘ with a ladder command) and not a defect, which is why no banked spectrum is revisited and ‘KSLICE‘ is exact on a **uniform** grid only, so it is a continuum tool and a ladder run should be done whole *Run:* `python3 receipts/P15_CR_cosmology/P15_the_ladders_discreteness_does_not_set_the_spectrum_and_the_check_was_`

**P15R181** `P15_the_leg_contributes_no_tilt_because_a_fixed_phase_handover_is_scale_free_and_a_fixed_time_` [OK]

*‘PO-31‘’s work order from node 66: where the tilt comes from, if it comes from the collapse leg. The row has a target — the like-for-like refit prefers $n_s=0.9949$ against the control’s $0.9559$ (‘r6805‘) — and the order asks the first calculable thing beyond it: **can the leg produce a departure of that size at all?**** (**No, and for a stronger reason than a small number: what the leg can produce is not a tilt.** Both the potential and the temperature on the leg are functions of $x=k\{}eta/\{}sqrt3$ **alone**, so a handover at a fixed *phase* is exactly scale-free — ***the ratio of amplitudes at two wavenumbers is $1$, not nearly $1$*** — and the only surviving scale is a handover at a fixed *time*, through $x_{\{}rm seam}=0.7638\{},(k/k_s)$ **what a fixed-time handover gives is red in both channels and goes as $k^{2}$**: $\{}dd\{}ln\{}lvert\{}hat\{}Theta\{}rvert/\{}dd\

k=-x\{}tan x=-x^{2}+O(x^{4})$ and $\{}dd\{}ln T/\{}dd\{}ln k=-x^{2}/5+O(x^{4})$ — a **running**, where a constant $n_s$ shift needs a constant log-derivative, with an honest zero of the amplitude at $k/k_s=2.057$ **so the size question answers itself by shape**: the target needs $k/k_s=0.065$, and thirty times that wavenumber the same expression gives $1-n_s=41.9$ ⇒ ***and on the adjudicated configuration it is exactly zero*** — ‘r6774‘’s crossing is $x\{}to0$, where $T\{}to1$ and $\{}hat\{}Theta\{}to\{}Psi_i/2$ and both log-derivatives vanish quadratically). *Computes:* C1 the three banked amplitudes reproduced before anything is read — $T(1/\{}sqrt3)/2=0.48353$ (the coded handover), $\{}cos(1/\{}sqrt3)/2=0.41896$ (the exact entry temperature), $1/2$ (the envelope and the crossing value) — the closed form being **established** (‘P15‘ §coherence, §transmission; ‘r6760+cc66.1‘) and re-run rather than re-derived the quadratic approach checked, so the vanishing at the crossing is not an artefact of where the limit is taken *and one coding slip of mine was caught before landing: the band evaluation used the wrong factor and printed $1.4$ where the docstring said $41.9$* *Bound:* **Bounds the LEG’s contribution only** — nothing about what the progenitor supplies, and nothing about $A_s$, whose fixed-point route the entropy monotone closed the fixed-time numbers are evaluated at the seam because that is the only fixed-time locus the corpus ever coded: **a demonstration that such a handover gives a running, not a claim the corpus uses one** — it does not, since ‘r6774‘ the progenitor’s mass enters only through $k_s$, so no independent mass dependence appears ***nothing is tuned to $0.995$***: the target enters once, to be refuted on shape, and ‘PO-7‘ is untouched — this is a statement about a transfer function and not a verdict *Run:* `python3 receipts/P15_CR_cosmology/P15_the_leg_contributes_no_tilt_because_a_fixed_phase_handover_is_scale_f`

**P15R182** `P15_the_substrate_cannot_tilt_the_spectrum_and_the_collapses_duration_enters_only_as_a_decayin` [OK]

*node 66’s order (2) in ‘FOR_60‘ (‘r6825‘), ‘PO-31‘ one step in: with the leg closed as a source, the tilt is the progenitor’s vacuum, so **what does the substrate ALONE constrain about that vacuum, before any interior is modelled?** Three candidates named — the progenitor’s mass, the finite duration of the collapse, and the substrate’s curvature radius — with the order asking which can enter the vacuum’s two-point function *at all*, and the sign of $n_s-1$ for any that can* (**One of the three can enter, and what it produces is not a tilt: the substrate alone fixes $n_s=1$ exactly.** **the curvature radius CANNOT enter** — in conformal time the substrate’s mode equation $u”+(k^2-2/\{}eta^2)u=0$ contains no length at all, so the only variable is $k\{}eta$ and $\{}alpha$ survives solely as the prefactor $P=1/(4\{}pi^2\{}alpha^2)$: it sets $A_s$ and reaches $n_s$ *for every* $\{}alpha$, not for large $\{}alpha$ **the progenitor’s mass CANNOT either, and on this construction that is arithmetic** — the forced member is Nariai, $\{}Lambda M^2=1/9$ with $\{}Lambda=3/\{}alpha^2$, so $M=\{}sqrt3\{}alpha/9$ and $M/\{}alpha=0.192450$ is a **pure number**: the mass *is* the curvature radius up to a constant and brings no new length, so the two candidates are one and fall together **only the finite duration enters** — setting the vacuum at finite $\{}eta_0$ gives $\{}alpha_k=1-i/x-1/2x^2$, $\{}beta_k=-e^{2ix}/2x^2$ with $\{}lvert\{}alpha_k\{}rvert^2-\{}lvert\{}beta_k\{}rvert^2=1$ exactly and $R(x)=1+\{}cos2x/x^2-\{}sin2x/x^3+(1-\{}cos2x)/2x^4$, $x=k\{}lvert\{}eta_0\{}rvert$, whose log-derivative is $-2\{}sin(2x)/x+O(x^{-2})$: **an oscillation with a decaying envelope, sign-alternating 24 times on $x\{}in[1,40]$, its four octave means disagreeing in sign** — a *running*, and by the corpus’s own operational test a scale, not a tilt ⇒ ***so a constant tilt needs the interior, and its sign is fixed in advance***: the only quantity giving a band-independent log-derivative on a maximally symmetric background is an effective mass, exactly $n_s-1=-2\{}sqrt{9/4-m^2\{}alpha^2}=\{}tfrac23 m^2\{}alpha^2+O(m^4)$, **blue for $m^2¿0$** — so the measured $n_s=0.995$ requires $m^2\{}alpha^2=-1201/160000=-0.0075$, slightly tachyonic). *Computes:* C1 the Nariai constants reproduced before anything is read — $M=\{}sqrt3\{}alpha/9$, $\{}Lambda M^2=1/9$ on substitution, $f(r_N)=f’(r_N)=0$ — **established** (‘P15‘ Fig. $f(r)$, ‘r6804‘) and re-run rather than re-derived, because finding (2) rests entirely on them; the Bunch–Davies mode verified to solve the mode equation and to carry Wronskian $i$, so the amplitudes are not rescalable the $R(x)$ identity checked twice by independent simplification routes *and one stated criterion of mine was wrong and was corrected to the computation rather than the reverse: I had claimed the octave means sit within $0.021$ of zero, and the $[2,4]$ band averages $-0.095$ — the true and stronger statement is that they do not agree with one another and do not share a sign* *Bound:* **the one modelling step is named and it is the standard one**: the corpus’s scalar perturbation is treated on the substrate as a massless minimally coupled field. Findings (1) and (2) are robust to that choice — they are statements about

which *lengths* appear, and none appears for any mass — but the exact value $n_s=1$ is not, which is precisely what (4) says the duration channel computes a **sharp** initial time, the one unambiguous prescription: a smooth onset changes the envelope's power and the phase and cannot change the sign alternation, so the claim is the SHAPE and not the coefficient bounds what the **substrate** supplies — nothing about $A_s$, whose fixed-point route the entropy monotone closed, and nothing about the interior beyond the one requirement ***nothing is tuned to $0.995$***: the target enters once, at the end, and only to be inverted into a requirement on a quantity this receipt does not compute; 'PO-7' untouched *Run:* `python3 receipts/P15_CR_cosmology/P15_the_substrate_cannot_tilt_the_spectrum_and_the_collapses_duration_ent`

**P15R183** `P15_the_interiors_vacuum_is_blue_everywhere_and_a_running_so_the_substrates_mass_template_does` [OK]

*node 66's work order in 'FOR_60' ('r6845') — 'PO-31' on the progenitor's **interior**. 'r6826' left one channel, $n_s-1=3-2\{}sqrt{9/4-m^{2}\{}alpha^{2}}$ asking the interior for a slightly negative $m^{2}\{}alpha^{2}$, **but that formula is the substrate's and the fluctuations are generated in the interior**. Three questions: the spectrum handed over and whether it is a tilt or a running; its sign; and whether the substrate's formula applies at all* (**All three answer, and they answer against the construction.** first a correction to the order's premise: **the interior is not newly built** — 'P15_the_progenitor_vacuum_is_negligible_too' already banks $a=M(\{}rho x+x^{2}/2)$ and the $M$-free potential $2/(x(x+2\{}rho))$, and it is *the same object*: with $M=A/2$, $\{}rho=2\{}sqrt B/A$ the two agree **exactly** to $O(\{}eta^{2})$ and $a''/a=A/(2a)-1$ reduces to it **(1) a RUNNING, not a tilt**: the potential interpolates between $2/x^{2}$ (the $p=+2$ matter root, scale-free) and $1/(\{}rho x)$ (not), so at the determined $\{}rho=0.0539$ the log-derivative runs $+0.30$ to $+1.98$ over $k=5$–$2000$, tending to the radiation value $+2$ — no constant to fit, by the same band test that closed the leg **(2) BLUE at every $k$**, flattest $+0.304$, against a measured **red** $-0.002$: wrong sign and $\{}sim100\{}times$ the size. The radiation content is the cause — flattest slope $\{}simeq6.7\{}rho$ — **so red would need $\{}rho¡0$** ⇒ **(3) the substrate's template DOES NOT APPLY**: $3-2\{}sqrt{9/4-m^{2}\{}alpha^{2}}$ comes from a $1/\{}eta^{2}$ potential whose coefficient is *dimensionless*; the interior's carries the scale $\{}rho$. ***So $m^{2}\{}alpha^{2}=-0.0075$ is an artefact of applying the substrate's template off the substrate, and the corpus should stop carrying it as a requirement***). *Computes:* C1 **the extractor calibrated on four power laws before being believed** — $p=-1$ and $p=+2$ both returning $0$ to $5\{}times10^{-6}$, $p=+1$ giving $+2$, $p=+3$ giving $-2$, against $3-2\{}lvert p-1/2\{}rvert$: *two of them are the corpus's own scale-invariant roots, so the instrument is checked on the cases the argument turns on* *an earlier version read every $k$ on a **different** surface and returned $+6$ where $0$ was right — the calibration caught it, and it is recorded because the failure was silent* the background identities and the two potential limits derived symbolically; the surface-convergence measured rather than assumed *Bound:* **the vacuum is set at maximum expansion** ($a'=0$, so every mode sub-horizon), with $k^{2}/\{}lvert a''/a\{}rvert\{}ge10$ enforced per mode — **below that band the interior supplies no initial condition**, which is the order's own stopping clause, reported as a bound and not stepped over the slope **converges** as the reading surface approaches the branch point (last two decades agree to $1.2\{}times10^{-4}$, the shallowest differing by $1.4\{}times10^{-2}$) — *stated as convergence and not insensitivity, because the shallowest surface is not yet converged* $A=1$ sets the time unit and every result is a slope or a ratio; $\{}rho=0.0539$ is the **determined** composition, taken from the banked receipt and not chosen here a massless minimally coupled field, the same modelling step 'r6826' named **the banked receipt is not contradicted**: its scale-invariance is of a *transfer* and its number an *amplitude*; the slope of the generated spectrum is a different object nothing bears on $A_s$ or 'PO-7' *Run:* `python3 receipts/P15_CR_cosmology/P15_the_interiors_vacuum_is_blue_everywhere_and_a_running_so_the_substrat`

**P15R184** `P15_the_transfer_is_the_same_object_as_the_vacuum_spectrum_and_the_crossing_carries_no_wavenum` [OK]

*node 66's work order in 'FOR_60' ('r6897') — **(2) 'PO-31', released** after the switch sweep validated its target ($n_s=0.9949$ against the control's $0.9559$). Four channels closed and the measured red tilt supplied by none, so: ***what supplies a red tilt when the only vacuum the construction has computed is blue?*** The order named one thing to look at first — *every closed channel was tested as a **spectrum-generating** mechanism, but the tilt is a ratio of amplitudes at two wavenumbers and there is a second place one can enter, the **transfer** from the pro-*

*genitor's vacuum to the boundary datum* — *"established for the collapse leg and not for the interior-to-leg join... if it is already closed, say so and it costs a paragraph"* (**It is not a second place, and saying so costs more than a paragraph because the reason corrects this seat's own reconciliation of two banked results.** **(1)** the interior's transfer and the interior's vacuum spectrum are **one object**: the banked instrument, run at the *determined* $\{}rho=0.0539$, gives a log-slope running $+0.267$ to $+1.979$ — ***r6857's $+0.30$ to $+1.98$, from vacuum data placed somewhere else entirely*** (sub-horizon in the deep matter contraction, against 'r6846''s maximum expansion) **(2)** and the banked scale-invariance is **not contradicted — it has a domain**: the potential breaks at $x\{}sim2\{}rho$, so $k_{\{}rm break}=1/(2\{}rho)$; the banked check ran at $\{}rho=10^{-3},10^{-4}$ with its whole band *below* $k_{\{}rm break}=500,5000$, and at the determined $\{}rho$ the observed band lies *above* $k_{\{}rm break}=9.3$ **(3)** and the one step genuinely untested — **the crossing** — carries no wavenumber **exactly**: $x=0$ is a resonant regular singular point and its log coefficient is $C=1/\{}rho$ with $dC/dk=0$, because ***$k^{2}$ is regular where the potential is singular*** (it first enters the analytic solution at $x^{3}$) ⇒ ***the chain carries no $k$-dependence of its own anywhere, and where one does enter it is BLUE where the target is red***). *Computes:* C1 **the calibration is the gate**: PART 1 re-runs the banked check at the banked $\{}rho$ and $k$ and reproduces $3/(2\{}sqrt2)$ to $0.20\{}%$ — *the whole of PART 2 is that same instrument pointed at a different composition, so if PART 1 fails nothing below may be read* the $x\{}to0$ log of the radiation crunch divided out explicitly rather than left in the residual the resonant coefficient **determined** order by order (Frobenius, symbolic) rather than quoted the indicial equation shown to be $k$-blind from $x^{2}(\{}text{pot}-k^{2})\{}to0$ *Bound:* **THIS DOES NOT CLOSE 'PO-31'** — it closes the route the order named, which is a different thing; the row's question is left exactly where 'r6857' left it, with one candidate eliminated rather than untested the background, the potential and $\{}rho=0.0539$ are **taken from banked receipts**, not rebuilt; what is new is where they are evaluated **the crossing result is about the monodromy and not about 'C19''s $9/10$** — 'C19' states in its own voice that its join factor is the super-horizon one and that *"modes inside the horizon at the branch point are not covered"*, and that item is untouched a slope is not an amplitude: the $A_s$ comparison is a **ratio to the scale-free branch**, showing the banked $A_s^{\{}rm vac}$ understated by $\{}sim5\{}times10^{3}$ at the band's top — against a shortfall of $10^{103}$, so ***a bound on how the number is read and not a correction to its conclusion*** 'r6846''s *"the slope of the generated spectrum is a different object"* is **withdrawn**: they are one object, and the reconciliation was wrong *in the direction that let a conflict pass* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_transfer_is_the_same_object_as_the_vacuum_spectrum_and_the_crossi`

**P15R185** `P15_the_interiors_transfer_imprints_its_own_running_so_the_channel_hunt_answered_the_wrong_que` [OK]

*node 66's work order in 'FOR_60' ('r6911') — **(1) 'PO-31', and the order's own finding is bigger than the row**: every channel the row has closed answers ***what tilt does the VACUUM give***, while 'P15' says twice that the primordial statistics are **classical and non-vacuum** — *"so the corpus argues the observed perturbations are not the vacuum's, and then computes the tilt of the vacuum, five times"*. The order: the banked transfer law's exact scale-invariance is a property of the **vacuum normalisation** ($k^{3}$ cancelling $D_k$'s $k^{-3}$), not of the transfer — so ***applied to an arbitrary input rather than to vacuum data, is the interior's transfer a scale-free multiplier, or does it imprint its own running?*** Forbidden: building a progenitor interior, importing a tilt* (**It IMPRINTS, and the order's diagnosis of the mechanism is confirmed exactly.** **(1)** on **unit** incoming amplitude the multiplier's log-slope runs $-0.8663$ at $k=7$ to $-0.0105$ at $k=1400$ — a spread of $0.856$, so ***neither scale-free nor even a power law***; what it adds to any tilt is twice that, $-1.73$ to $-0.02$ **(2)** and ***the banked exact scale-invariance is that same function's LOW-$k$ limit***: $d\{}ln T/d\{}ln k\{}to-1$ ($-0.9992$ at $k=0.3$) and $kT(k)\{}to27.86$, a constant — and $P\{}sim k^{3}\{}lvert c_0\{}rvert^{2}$ is scale-invariant *exactly* when the slope is $-1$, so the cancellation is the normalisation meeting a power law that holds only below the break **(3)** the decomposition as an **identity**, $s(k)=d\{}ln T/d\{}ln k+1$ — ***the $k$-dependence was always the transfer's and the constant always the normalisation's***, reproducing 'r6898''s $+0.134$ and $+0.990$ from the transfer alone **(4)** the same function for **every** input to $4\{}times10^{-13}$, including one deliberately not a power law ⇒ ***the channel hunt is over: five results stand, the question was mis-aimed, and the row becomes a statement about the inte-

rior model plus one computed requirement***). *Computes:* C1 **the banked check is the gate** — $3/(2\{}sqrt2)$ reproduced at the banked $\{}rho$ and $k$ to $0.20\{}%$, and everything below is that same instrument pointed at a different object **linearity measured, not invoked** ($7\{}times10^{-15}$) — *it is what licenses the order's phrase "whatever the progenitor supplies"* the starting **phase** varied ($6\{}times10^{-6}$) and then the convention **replaced outright** by a fixed $x_i$ ($2.9\{}times10^{-5}$), because a phase artefact here would look exactly like a running the slope shown to **converge** in the reading surface between the two deepest, $2.9\{}times10^{-5}$ *Bound:* **the band is bounded at $k¡2783$ and the bound is the PREMISE, not the arithmetic**: at $x_i=300/k$ the mode starts outside the break only while $300/k¿2\{}rho$, and above that "enters from the matter contraction" is false — *an earlier pass read $T¿1$ at $k=10^{5}$, tolerance-converged to $2\{}times10^{-10}$ and still meaningless, and that number is discarded as a misplaced initial condition rather than reported as a limit; **no high-$k$ asymptote is claimed*** **the slope is claimed and the amplitude is not** — $\{}lvert T\{}rvert$ carries $\{}sim1\{}%$ of reading-surface dependence, which cancels in a log-slope; the shallowest surface differs by $2.9\{}times10^{-3}$ and is reported as **not yet converged**, the same clause 'r6846' attached to its own nothing bears on $A_s$, whose $10^{103}$ shortfall is a **vacuum** statement no progenitor interior built; $\{}rho=0.0539$ and the potential taken from banked receipts **'PO-31' does not close**, as the order said it would not *Run:* `python3 receipts/P15_CR_cosmology/P15_the_interiors_transfer_imprints_its_own_running_so_the_channel_hunt_a`

**P15R186** `P15_the_acoustic_contrast_is_not_in_the_source_and_the_projection_makes_it_without_the_distanc` [OK]

*node 66's work order 'r6911': **where is the acoustic contrast made — in the source, or in the projection?** — *"take the statistic you already have and apply it before the projection as well as after"*: **(a)** in $k$-space at last scattering, the arm-and-control contrast of the **source** (the monopole-plus-Doppler combination as a function of wavenumber, same envelope split, same peak-to-trough measure), ratio arm to control; **(b)** compare with the $\{}ell$-space $1.040$ — the order naming three outcomes in advance ($\{}approx1.04$ = made in the dynamics; $\{}approx1.00$ = manufactured between $k$ and $\{}ell$, *"live rather than exotic, because the two arms' distances differ by six per cent"*; or a split, *"and the split itself is the measurement"*), with **one guard**: *"the envelope split must be defined identically on the two arms and at the two stages, and the smallest difference it can express must be stated beside whatever tolerance gates it"*, and **no mechanism asked for** ( **THE CONTRAST IS NOT IN THE SOURCE: IT IS MANUFACTURED BETWEEN $k$ AND $\{}ell$.** The same statistic reads $\{}mathbf{0.996}$ on the source at last scattering, $\{}mathbf{0.992}$ on the $\{}eta$-integrated source, $\{}mathbf{1.045}$ on the raw spectrum and $\{}mathbf{1.047}$ on the banked one — *and over four envelope windows, both envelope definitions, three term subsets (monopole alone, monopole plus Doppler, the full source), four $q$ sub-windows and with or without the $k$-measure, the source rungs span $0.971$–$1.005$ and the $\{}ell$ rung $1.042$–$1.052$, against a statistic floor of $0.6\{}%$ set by its own bias on a **known injected** contrast.* ⇒ **AND THAT IS WHY FOUR-FOR-FOUR NEEDED NO EXPLAINING.** The order's difficulty was that four channels which each *shallow* the troughs are all larger on this arm, so something upstream had to overcome all four. ***Upstream the arm's oscillation is $0.996$ of the control's — very slightly SHALLOWER, which is exactly the direction all four point.*** **Nothing overcomes them because nothing had to: the effect to explain is not bigger than four per cent, it is not upstream at all.** ⇒ **THE NAMED ROUTE IS OUT TWICE OVER, AND THE FIRST HALF IS A PREMISE CORRECTION.** *The order's $13005$ against $13865$ Mpc is the **superseded** configuration's ($r_s=135.46/144.53$), where the arm's was the smaller; at the adjudicated minima $D_M$ is $\{}mathbf{14017.04}$ on the arm against $\{}mathbf{13954.35}$ on the control — $\{}mathbf{+0.449\{}%}$ and the **arm's is the LARGER**, wrong in size and in sign.* *The cancellation the order relies on is real — $\{}ell_A$ agree to $0.085\{}%$ — so it is the six per cent and not the reasoning that fails.* ⇒ **And the route is eliminated by measurement anyway**: 'SRCXS' projects one arm's own source through the *other's* distance, one arm at a time, moving its contrast by $-0.63\{}%$ (control) and $+0.59\{}%$ (arm) — *and removing the difference altogether takes the arm/control ratio **UP**, $1.045\{}to1.051$.* ⇒ **WHERE IN THE PROJECTION, AS FAR AS THIS GOES**: the arm's projection **retains $1.054$ times as much** of its own source oscillation as the control's does ($0.2543$ against $0.2413$); the excess **rises** with wavenumber ($1.031$ below $q=3$ to $1.065$ above) where the source ratio is flat ($0.992\{}to0.992$); it is **not** the visibility width acting before the kernel ($0.996\{}to0.992$),

**not** the lensing or the binning ($1.045\{}to1.047$), and **not** the $k$ grid — ‘KCONT=1‘ puts the arm on the control’s kind of uniform grid at the same mode count with the physics untouched and the $\{}ell$ rung is unchanged at $1.045$. ⇒ **AND THE GUARD CHANGED THE DEFINITION, WHICH IS PART OF THE MEASUREMENT**: ‘r6885+cc66.35‘’s running **geometric** envelope cannot be carried to the source at all — the source power comes within a part in $10ˆ{8}$ of its own median at the troughs, where a log-mean is dominated by near-zeros — *so the envelope is a running **arithmetic** mean at **every** rung, both are reported, and they differ by $0.003$ on the $\{}ell$ rung against the $0.05$ step being measured*). *Computes:* ‘SRCSAVE‘ is new and wired into ‘_project‘ — the **reporting** path — writing the source term by term at last scattering *and* $\{}eta$-integrated, with the four terms gated to **sum to the instrument’s own ‘S‘** ($2.8\{}times10ˆ{-17}$) so a future edit to ‘S‘ that forgets the block fails here; its default and ‘SRCXS‘’s gated **bit-identical** on both arms before any number is read, and ‘SRCXS=1.5‘ with ‘SRCSAVE‘ unset gated bit-identical too, which is what makes the swap an *output of the save* rather than a knob; every load of either name shown to sit inside an ‘if _SRCS:‘ block (*a binding is not a use*); the sliced runs gated to reproduce the banked $185$-bin refit spectra to $4.9\{}times10ˆ{-15}$ relative, **which is what makes the $k$ end and the $\{}ell$ end one run rather than two**; the abscissa’s alignment **measured** rather than assumed (the arms’ acoustic period in $q$ agreeing to $0.22\{}%$ at the source and $0.09\{}%$ in $\{}ell$, best lag $+0.005$ and $-0.001$); the statistic returning exactly $1.000000$ against itself, recovering a **known** injected contrast, and returning $0.9996$ through the other arm’s abscissa and back; and the one artefact that could have produced the whole answer — the two arms’ different *kinds* of $k$ grid — run out with ‘KCONT=1‘ rather than argued, that switch shown non-null so the result is not ‘r4558‘’s unwired knob *Bound:* **NO MECHANISM** — ”the projection makes it” names a **stage**, not a cause, and nothing here says what within the projection produces $5.4\{}%$; the order’s bound is unmoved **NOT that the projection is wrong** — nothing here bears on the flat kernel, which ‘prop:flat‘ settles; *a projection can manufacture a contrast difference while being exactly the correct projection for both arms* **NOT that the visibility width is its route**: the arm’s is $15\{}%$ wider, which *shallows* contrast, and the rung where an $\{}eta$-average acts **without** the kernel moves the ratio by only $-0.4\{}%$ — a bound on one route, not the identification of another **NOT that the source is identical on the two arms**: it is $0.992$, a real deficit of about the statistic’s floor, and its sign is the four channels’ own **NOT that ‘cc66.35‘’s $1.0401$ is superseded** — it is reproduced exactly under its own definition and comes out $1.047$ under this receipt’s, and the difference is the envelope definition, reported the $q$ variable rests on the sharp-visibility map $\{}ell=kD_M$, exact to the visibility width over $D_M$ — $38$–$44$ Mpc against ${\{}sim}14000$, a part in $300$ *Run:* `python3 receipts/P15_CR_cosmology/P15_the_acoustic_contrast_is_not_in_the_source_and_the_projection_makes_i`

**P15R187** `P15_the_transfers_running_is_a_function_of_one_variable_so_the_requirement_is_the_radiation_fr` [OK]

*node 66’s work order in ‘FOR_60‘ (‘r6914‘) — **‘PO-31‘ is now a specification rather than a hunt**: $P_{\{}rm obs}=\{}lvert T\{}rvertˆ{2}P_{\{}rm prog}$ with $T$ measured at ‘r6912‘, so $P_{\{}rm prog}$ is determined — *”and a determined input is only good news if the determination is not a conspiracy”*. Three items: (1) ***where is the break in $\{}rho$, and is any $\{}rho$ putting the observed band wholly below it***, on a stated criterion; (2) ***is $1.71$ a number or an artefact of the band’s placement***; (3) ***is the high-$k$ limit exactly zero, and the requirement therefore bounded by two*** — **argued ANALYTICALLY, because $k\{}to$ large is exactly where this seat’s $x_i=300/k$ premise expires and *”a second $k=10ˆ{5}$ would be worse than a gap”**** (**All three answer, and two in closed form, because the running turns out to be a function of ONE variable.** **THE STRUCTURAL FACT** — substituting $x=2\{}rho u$ gives $v_{uu}=(2/(u(u+1))-\{}kappaˆ{2})v$ with $\{}kappa=2\{}rho k$ — ***the potential in $u$ carries no $\{}rho$***, so $d\{}ln T/d\{}ln k$ is universal in $\{}kappa$ (collapse verified to $7\{}times10ˆ{-13}$ across $\{}times100$ in $\{}rho$) and $k_{\{}rm break}=\{}kappa_*/(2\{}rho)$ **exactly** **(3)** near the crunch the equation is the **repulsive $L=0$ Coulomb wave equation** with $\{}eta=1/\{}kappa$, and $1/C_0(\{}eta)=1+\{}pi\{}eta/2$ gives $d\{}ln T/d\{}ln k\{}to-\{}pi/(4\{}rho k)\{}to\{}mathbf{0}$ — matching measurement to $1.4\{}%$ at $k\{}le2000$, ***derived where the integration is not trusted and checked where it is*** ⇒ **the excursion is exactly $1$, the tilt contribution at most $2$, and no band however wide can require a progenitor running above $2$** **(1)** a working $\{}rho$ **exists** but needs $\{}rho¡9.7\{}times10ˆ{-4}$ (midpoint) or $¡1.4\{}times10ˆ{-4}$ (within $0.05$ of the

power law) — ***$55\{}times$ to $378\{}times$ below the determined $0.0539$*** ⇒ ***the requirement stands BECAUSE the determined radiation fraction forces it*** **(2)** over the $\{}pm6\{}%$ the arm's own distances allow, $1.71$ moves $2.3\{}%$ — **a number, not an artefact**, and the observed band asks $86\{}%$ of the hard maximum). *Computes:* C1 **the banked $3/(2\{}sqrt2)$ check at the banked $\{}rho$ and $k$ is the gate** — everything below is that instrument or an analytic argument about the same equation the scaling substitution and the composition identity done **symbolically**, not by hand the $\{}kappa$-collapse **measured** across four $\{}rho$ spanning a factor $100$ the Coulomb limit's coefficient taken from a **series expansion of $1/C_0(\{}eta)$**, then compared against measurement at four $(k,\{}rho)$ pairs *all inside the premise* *Bound:* **the analytic limit is NOT measured at large $k$, by design** — the order forbade it and the reason is this seat's own: 'r6912' discarded a $k=10^{5}$ reading that was **tolerance-converged to $2\{}times10^{-10}$ and still meaningless** because $x_i=300/k$ had drifted inside the break. ***No integration above $k=2000$ is performed or quoted*** **the differencing convention moves the headline in the third digit**: 'r6912''s centred grid slopes gave $1.71$, this $\{}pm8\{}%$ two-point slope gives $1.724$ — so the requirement is quoted as **$1.71$–$1.72$** and that $0.7\{}%$ spread is *part of* (2)'s answer, not noise over a **wider** band the requirement moves $6$–$9\{}%$, reported rather than hidden **the $\{}rho$ bound is NOT converted into a number for $a_{\{}rm eq}$ or $M$**: 'P16''s own two statements give $\{}rho=\{}sqrt{2a_{\{}rm eq}/M}$ (verified symbolically), but at $a_{\{}rm eq}=1.49$ Mpc and the **register's** mass that reads $1.635\{}times10^{-2}$ against the determined $0.0539$ — a factor $3.3$, i.e. two different progenitor masses. ***Flagged, not used*** no progenitor interior, no tilt value, no amplitude; **'PO-31' does not close**, as the order said *Run:* `python3 receipts/P15_CR_cosmology/P15_the_transfers_running_is_a_function_of_one_variable_so_the_requiremen`

**P15R188** `P15_the_cross_term_is_not_the_channel_and_every_projected_term_carries_the_excess` [OK]

*node 66's work order 'r6915': **which term carries the contrast excess after projection?** — 'cc66.40' located the stage, and the order asks what inside it does the work, naming the monopole–Doppler pair first because *"the monopole enters through $j_\{}ell$ and the Doppler through $j_\{}ell'$, and those are a quarter-period out of phase"*: **(1)** project each term separately, apply the same $\{}ell$-space contrast statistic to each piece and then to their sum, with three outcomes named (both pieces near $1.00$ and the sum at $1.045$ ⇒ **the cross term**; one piece already at $1.045$ ⇒ that term's own projection; neither and the pieces do not sum ⇒ *"the decomposition is not linear where I have assumed it is, and that is the finding"*), **and a forced calibration**: *"the projection is linear in the source, so the separately projected pieces must sum to the full spectrum. Gate that first… and read nothing below it if it fails"*; **(2)** report the retained fraction **as a function of $q$** rather than as one number per arm — rising ⇒ a kernel-phase signature, flat ⇒ *"that kills the reading above"** ( **THE CLOSURE GATE HAD TO BE CORRECTED BEFORE IT COULD BE PASSED, AND THE CORRECTION IS THE ORDER'S OWN FIRST OUTCOME.** *The **transfer** is linear in the source and the pieces do add there; $C_\{}ell$ is **quadratic** in the transfer, so the projected **spectra** cannot — the four diagonal pieces alone fall short of $D_\{}ell$ by up to $\{}mathbf{46\{}%}$.* *This is not the third outcome: the **first** outcome presupposes a cross term, so the order's two clauses are in tension and the first is the coherent one.* ⇒ **What closes is the full bilinear decomposition** — with $\{}Delta^a_\{}ell(k)=\{}int\{}text{term}_a j_\{}ell\{},\{}mathrm d\{}eta$, $C_\{}ell=\{}sum_{a\{}le b}w_{ab}\{}sum_k P\{},\{}Delta^a\{}Delta^b$ — **ten numbers per multipole closing on $D_\{}ell$ to $1.1$ and $1.3\{}times10^{-15}$ relative on the two arms.** ⇒ **AND THE CROSS TERM IS NOT THE CHANNEL: deleting it moves the ratio from $1.0452$ to $1.0451$**, and over four envelope windows its carry runs $-0.0013$ to $+0.0008$ of the $+0.045$ — *bounded at about a part in a thousand, so the interference between $j_\{}ell$ and $j_\{}ell'$ is excluded.* ⇒ **BECAUSE EVERY PROJECTED PIECE ALREADY CARRIES IT, WHICH IS NEITHER OUTCOME NAMED**: monopole $\{}mathbf{1.034}$, Doppler $\{}mathbf{1.099}$, polarisation $\{}mathbf{1.058}$ against the full $1.045$ (the ISW alone is the exception at $0.989$, and it is $1\{}%$ of $D_\{}ell$). ***So it is not a term and not a pair of terms: the projection raises this arm's contrast on nearly everything it projects.*** ⇒ **AND THE LADDER IS THE SHARPEST FORM OF IT — every term's SOURCE ratio is at or below $0.992$ and every term's PROJECTED ratio is higher**: monopole $0.992\{}to1.034$, Doppler $0.973\{}to1.099$, ISW $0.972\{}to0.989$, polarisation $0.990\{}to1.058$, the whole source $0.992\{}to1.045$. ⇒ **THE SPLIT THE ORDER'S REASONING WAS AFTER IS

ABOUT $40/60$, THE RESPONSE THE LARGER**: the weights *do* differ (the arm's monopole share $0.5225$ against $0.5092$, its Doppler $0.1753$ against $0.1840$), and rebuilding the arm's pieces at the control's shares gives $1.0267$ — **the weights carry $+0.019$ of the $+0.045$ and each piece's own response $+0.027$**, with the reverse construction agreeing at $+0.020$. ⇒ **(2) AND THE RETAINED FRACTION IS NOT FLAT IN $q$**: it rises from $1.030$ in the lowest band to $1.089$ in the highest, and over **twelve settings** (four envelope windows × three band counts) the slope is $\{}mathbf{+0.0139\{}pm0.0021}$ per unit $q$, **never once negative** — *so the branch that would have killed the reading is excluded.* *The intercept is $1.017\{}pm0.010$ and **straddles one**: the rise is established and a constant offset is not, and the value at $q=0$ is an extrapolation from a lowest band centre of $q=1.20$*). *Computes:* 'SR-CDEC' is new and wired into '_project''s own multipole loop — **the reporting path** — so the Bessel evaluation is shared with the reported spectrum and the decomposition costs four trapezoids per multipole rather than a second projection; its default gated **bit-identical** on both arms before any number is read, and every executable line naming it categorised as a binding, a guard or a line inside a guarded block (*a binding is not a use*) rather than counted, because a substring count read the switch's own documentation as code and failed this gate on its first writing; the ten pairs gated to close on $D_\{}ell$ and the **non**-closure of the four diagonals gated too, so the premise correction is a measurement and not an argument; this run's $D_\{}ell$ gated identical to 'r6911+cc66.40''s to $10^{-13}$ relative, which is what lets a source rung and a projected rung sit in one ladder; every off-diagonal piece that is **not** positive definite named rather than quietly skipped, a relative oscillation about a running mean being undefined where the mean changes sign; the bank gated that the slices tile $k$ with no gap or overlap and that the pair **order** is identical slice to slice; and the statistic, abscissa and resolution taken over unchanged from 'r6911+cc66.40' rather than re-derived *Bound:* **NO MECHANISM** — naming that every projected term carries it, and that the rise in $q$ is real, is still not a statement about *why*; the order's bound is restated rather than moved **NOT that the cross term is zero** — it is $26\{}%$ of $D_\{}ell$ and its oscillation is $59\{}%$ of the full's; what is bounded is its contribution to the **difference** **NOT that the Doppler's $1.099$ is as well determined as the monopole's $1.034$**: its piece's relative oscillation is $0.116$ against $0.305$, so it is the weaker signal, and its window spread $1.0995$–$1.1070$ is quoted beside it **NOT a fit** — the weight/response split rescales pieces by their mean share, which also moves the envelope, which is exactly why both directions are reported **NOT that the $q=0$ intercept is one** nothing here touches 'prop:flat' *Run:* `python3 receipts/P15_CR_cosmology/P15_the_cross_term_is_not_the_channel_and_every_projected_term_carries_th`

**P15R189** `P15_a_source_with_no_physics_reproduces_the_contrast_and_the_two_clocks_part_company_at_the_vi` [OK]

*node 66's work order 'r6919': **(1) project a source with no physics in it** — *"feed both arms' projection machinery the same analytic oscillating source — a pure $\{}cos(k r_s+\{}phi)$ on a smooth envelope, no transfer, no terms, no weights — and measure how much of its oscillation each arm's projection retains, as a function of $q$"*, reproducing 'cc66.41''s $1.054$ and $+0.0139$ ⇒ the geometry, else ⇒ the source's $\{}eta$-dependence; **(2) then swap the two geometric factors one at a time**, the visibility and $\{}chi(\{}eta)$, with the candidate named: *"the source is accumulated on one clock and the kernel's argument is a distance read on the other"* — and the instruction that *"if a swap is not well posed… **say that instead of forcing it.** An ill-posed swap reported as a null is exactly the shape of the unwired knob at 'r4558'"** ( **A SOURCE WITH NO PHYSICS IN IT REPRODUCES THE EFFECT AND OVER-DELIVERS.** A pure $g(\{}eta)\{}cos(k r_s(\{}eta))$ through each arm's own kernel, visibility, $k$ grid and multipole grid gives an arm/control retained ratio of $\{}mathbf{1.066}$ at a slope of $\{}mathbf{+0.0226}$ per unit $q$, against the real source's $1.054$ and $+0.0139$. ⇒ ***The projection's geometry accounts for the whole of it and then some: the source is irrelevant to it*** — the order's first branch. *The injection's phase does not matter ($1.066/+0.0225$ at $\{}phi=\{}pi/2$), and a **standing** oscillation already carries most of it ($1.059/+0.0119$).* ⇒ **AND THE ORDER'S NAMED CANDIDATE IS IN THE INSTRUMENT BUT NOT WHERE THE ORDER PUTS IT.** *'x0 = eta_0 - EE' on both arms, so $\{}chi(\{}eta)=\{}eta_0-\{}eta$ and $\{}mathrm d\{}chi/\{}mathrm d\{}eta\{}equiv1$ identically on each, with no 'Jac', 'Hleaf' or 'Hphys' touching 'x0' on any path — **there is nothing there to exchange.*** ⇒ ***The two clocks sit between $r_s$ and $\{}eta$***: conformal time (and so 'x0') is built from 'Hphys', the **stacking** rate, while the acoustic phase accumulates on the **leaf** rate — 'Jac' $=1.000000$ everywhere on the control by the rate identity, and $\{}mathbf{0.789}$–$\{}mathbf{0.913}$

across $\{}pm3$ FWHM on the arm. ⇒ **AND THAT IS WHERE THEY PART COMPANY, AS ONE NUMBER PER ARM.** Across the visibility FWHM the accumulated **sound horizon** agrees to $\{}mathbf{0.08\{}%}$ ($17.3074$ against $17.2941$ Mpc) while the **comoving distance** differs by $\{}mathbf{14.6\{}%}$ ($38.042$ against $43.591$): *the leaf clock makes $r_s$ accumulate more slowly per unit $\{}eta$, so a $15\{}%$ wider window covers the **same** acoustic phase — and the kernel, which reads $\{}chi$, sees the wider window.* ⇒ ***The joint object is $\{}mathrm d r_s/\{}mathrm d\{}chi$ across the visibility — the sound speed the kernel sees — $0.4550$ on the control against $0.3967$ on the arm, $-12.8\{}%$***; term-independent, growing with wavenumber, and vanishing for a window under one acoustic period, which are the three properties ‘cc66.41‘ measured. ⇒ **(2) AND NEITHER SWAP CLOSES IT ALONE — THE THIRD OUTCOME.** *The **clock** swap is **ill posed** as an isolation and the statistic says so rather than returning a null: forcing both arms onto the stacking clock **moves the comb**, and band by band the regression **alternates in sign** ($+0.37/-0.52/+0.42/-0.57$) — ‘cc66.40‘’s guard firing on exactly the shape it was built for, and itself evidence that the leaf assignment is what the arm’s reported peak positions need.* *The **width** swap **is** well posed and **overshoots by eight**: giving each arm the other’s FWHM takes the slope $+0.0226\{}to+0.0931$ while barely moving the mean, so the width sets the **shape** and not the **level**.* ⇒ **They close only together, and what they close on is $\{}mathrm d r_s/\{}mathrm d\{}chi$**). *Computes:* ‘SRCINJ‘, ‘SRCINJPH‘, ‘SRCINJRS‘ and ‘SRCINJVIS‘ are new, all four gated **bit-identical when unset** on both arms in one result that covers both guards, the edit touching ‘hier_run‘’s batch loop as well as ‘_project‘; every load shown to sit inside a guarded block with the four bindings counted apart (*a binding is not a use*); **the solver is skipped on the injected path** by a guarded ‘continue‘, because the analytic source replaces ‘S‘ and evolving the hierarchy would build an array nothing reads; the injection carries $k^{(1-n_s)/2}$ so the smooth part of $PS^2$ is exactly $\{}mathrm dk/k$ on **both** arms and the two tilts cannot enter; the statistic, abscissa and window taken over unchanged from ‘r6911+cc66.40‘ so the comparison with the real source is like for like; the emptiness of the $\{}chi(\{}eta)$ swap gated from the **source text** on code lines with comments stripped, because the ‘hier_run‘ docstring names ‘x0 = eta_0 - EE‘ too and a substring count read the documentation as a second construction; the visibility swap reported as the **construction** it is, $g$ rescaled about each arm’s own peak with its integral preserved, because the arms’ visibilities sit at $\{}eta=281.7$ and $486.0$; and the ill-posed clock swap gated on its **sign changes** rather than on its value *Bound:* **NO MECHANISM**, and the order does not ask for one yet: naming $\{}mathrm d r_s/\{}mathrm d\{}chi$ is not an account of *why* this construction assigns its scales and its distances to different rates — that is ‘P15‘ §‘sec:tensions‘’ own question and is **not reopened** **NOT that the injection IS the source**: it has the right oscillation in $q$ and the arm’s own visibility and nothing else, so over-delivery is sufficiency and not identity, and the excess over $1.054$ is not interpreted **NOT that the visibility width is ruled out** — its swap overshoots, which makes it inseparable from the clock, not absent **NOT that ‘LEAFSCALES=1‘ is wrong**: the stacking-clock injection puts the arm’s comb in the wrong place, which is evidence the leaf assignment is what the peak positions need **NOT a spectrum** — every ‘SRCINJ‘ run is the projection’s transfer of a known input and none may be compared with a banked spectrum or the sky nothing touches ‘prop:flat‘ *Run:* `python3 receipts/P15_CR_cosmology/P15_a_source_with_no_physics_reproduces_the_contrast_and_the_two_clocks_p`

**P15R190** `P15_the_visibilitys_clock_is_the_one_the_kernel_reads_and_the_comb_votes_against_the_alternati` [OK]

*node 66’s work order ‘r6925‘: **which clock is the visibility a density in, and does the rate rule determine it?** — the rule being *”comoving separations read across leaves take the stacking rate; scales the plasma accumulates take the leaf’s”*, and $g=\{}tau’e^{-\{}tau}$ **neither**: *”the optical depth is accumulated by the plasma… but it is differentiated per unit $\{}eta$, and $\{}eta$ is built from the stacking rate”*. **(1)** locate every place the visibility and the optical depth touch a rate; recompute $\{}mathrm dr_s/\{}mathrm d\{}chi$ under the other admissible assignment and report what the retained fraction and its $q$-slope become — with the standing guard *”report what it does to the comb beside what it does to the contrast, and if they disagree say so rather than picking one”*, and the third branch *”if the second assignment is **not well posed**… say that, and say which step breaks”*. **(2)** one number on where the source’s partial cancellation sits, the injection over-delivering at $1.066/+0.0226$ against the real $1.054/+0.0139$* ( **THE AUDIT FINDS THE GAP ALREADY OPEN IN THE CODE, WHICH THE ORDER DID NOT EXPECT.** *Every site where $\{}tau$ or $g$ touches a rate

is on the **stacking** clock — the recombination history solved against ‘Hphys‘, ‘taup_of‘ built on ‘eg‘’s conformal time, ‘tau‘ integrated over ‘_egrid‘, ‘ETA_LS‘ and ‘ETA_LS_W‘ read off that grid.* ***But ‘1/k_Dˆ2‘ twenty lines below IS Jac-weighted under ‘LEAFSCALES‘.*** ⇒ ***So the diffusion length — a scale the plasma accumulates — takes the LEAF clock and the optical depth, also accumulated by the plasma, takes the STACKING clock: two objects on the same side of the rule, on opposite clocks, with nothing stating the choice.*** *So the other assignment is not an invention — ‘VISLEAF=1‘ applies to $\{}tau$ exactly the weighting $1/k_Dˆ2$ already applies to itself, which is what makes it **admissible** rather than invented.* ⇒ **ON THE GEOMETRY THE TWO AGREE — THE ORDER’S SECOND BRANCH.** $\{}mathrm dr_s/\{}mathrm d\{}chi$ goes $0.396733\{}to\{}mathbf{0.396957}$ on the arm, so $12.80\{}%$ lower becomes $12.75$ — **a move of $0.06$ per cent** — and the control does not move at all. *And structurally rather than luckily: it is a **ratio** of two accumulations across the **same** window, so re-weighting that window’s measure re-weights numerator and denominator alike.* ⇒ ***The visibility is not where the freedom is, and the $12.8$ per cent is forced by the rule as stated.*** ⇒ **BUT THE CONTRAST AND THE COMB DISAGREE, AND THE ORDER SAID TO REPORT THAT RATHER THAN PICK.** *Read alone the other assignment moves the injection’s retained ratio $1.0850\{}to1.0695$ and its slope $+0.02424\{}to+0.01332$ — **toward** the real source’s $1.0694$ and $+0.01167$, which is the order’s first branch.* *But the arm’s first peak moves $\{}ell=220\{}to228$ and $\{}ell_1/\{}ell_A$ $0.7290\{}to\{}mathbf{0.7555}$ — **away** from the sky’s $0.7312$ — with $P_1/P_2$ $2.142\{}to2.017$ and **the control’s comb bit-identical both ways**.* *And the contrast improvement is partly an artefact of that same move: band by band the ‘VISLEAF=1‘ ratio is **non-monotonic scatter** with a fitted residual **nine times** the current assignment’s, because moving $\{}eta_{\{}rm LS}$ moves $r_s(\{}eta_{\{}rm LS})$ and so moves the comb, and a band ratio of two oscillations no longer aligned in $q$ reads their phase mismatch — **‘cc66.40‘’s guard firing a third time on the shape it was built for**.* ⇒ ***The geometry says forced, the contrast says improvable, the comb says no; the comb is the only one of the three with an external referent, and neither is picked.*** ⇒ **(2) AND THE SOURCE’S CANCELLATION IS HALF THE LEVEL AND NONE OF THE SLOPE, so the sector does NOT close into one account.** *Multiplying the injection by ‘cc66.40‘’s measured source deficit — $0.9922$, and **flat** in $q$ at a slope of $-0.00106$ — gives $1.0766$ at $+0.02296$ against the real $1.0694$ at $+0.01167$: **$54$ per cent of the level gap and $10$ per cent of the slope gap**.* ⇒ *So the four trough-filling channels account for about half the level difference and essentially none of the slope; the order’s conditional is half satisfied and the closure it hoped for does not arrive. **What flattens the slope is the real source’s own $\{}eta$-dependence across the visibility — exactly what the injection replaced — and that is named, not measured here.***). *Computes:* ‘VISLEAF‘ is new, gated bit-identical when **unset** on both arms **and when SET on the control** — two different gates, the second being the rate identity showing in a second place and the one-sidedness of the whole finding in one result; the switch also shown **non**-inert on the arm ($\{}eta_{\{}rm LS}$, the FWHM and $r_D$ all move) so the near-null on the geometry is a connected knob’s and not ‘r4558‘’s unwired one; the audit’s five sites and the $1/k_Dˆ2$ counter-example all located **from the source text** rather than described; the ‘VISLEAF=1‘ slope’s contamination gated on its **band residual** against the current assignment’s rather than asserted; the comb gated on the arm’s $\{}ell_1/\{}ell_A$ moving **away** from the sky **and** the control’s being unchanged, so the direction is part of the gate; and the statistic, abscissa and window taken over unchanged from ‘r6911+cc66.40‘ *Bound:* **NOT that the two-rate assignment is right** — that is the row’s question now and not this order’s, and nothing here settles it **NOT that ‘VISLEAF=1‘ is wrong as physics**: what is measured is that it moves the comb away from the sky while moving the contrast toward the real source, and that its contrast improvement is partly a phase artefact — *which of the two clocks the optical depth **should** be a density in is not settled here* **NOT that the $0.06$ per cent invariance settles the row** — it says the visibility is not where the freedom is; the freedom is in the rule’s assignment of $r_s$ itself NOT a mechanism beyond ‘cc66.42‘’s, and nothing touches ‘prop:flat‘ **NOT that the injected runs are spectra** — the ‘injvl‘ arrays are the projection’s transfer of a known input and only the ‘combvl‘ arrays are spectra, kept under separate keys **NOT that this receipt’s first launcher worked**: it dropped its extra environment through a positional-argument bug and thirty-six slices ran as plain ‘VISLEAF=0‘ spectra, *reporting nothing wrong and reproducing the banked spectra* — recorded in the launcher’s own comment, with the fix a smoke test that greps the log for the marker the switch must print *Run:* `python3`

`receipts/P15_CR_cosmology/P15_the_visibilitys_clock_is_the_one_the_kernel_reads_and_the_comb_votes_a`

**P15R191** `P15_the_combs_resolution_is_four_multipoles_and_the_skys_own_value_lies_outside_the_family` [OK]

*node 66's work order 'r6929': **the comb has been promoted to arbiter, so measure what it can actually decide** — *"you have made the comb the load-bearing reading ... but the corpus has never measured how sharply it discriminates, and a reading promoted to arbiter needs its resolution stated before it decides anything"*. Scan the assignment **continuously** — 'VISLEAF' as a fraction rather than a flag — and report against that parameter $\{}ell_1/\{}ell_A$ against the sky's $0.7312$, the retained fraction and its $q$-slope, and $\{}mathrm dr_s/\{}mathrm d\{}chi$; with three branches (**steep comb + shallow contrast** ⇒ the comb pins the assignment; **both shallow, or the comb's spread inside the sky's own locating spread** ⇒ the comb cannot arbitrate, *"and that is the finding, and it is worth more than the scan"*; **an interior fraction fitting better** ⇒ say so plainly, since a fitted clock would be a new free parameter) and the guard *"report whether the comb's motion is the visibility peak relocating or the acoustic phase changing"* ( **EVERY READING IS LINEAR IN THE PARAMETER AND THE COMB'S WHOLE RANGE IS FOUR MULTIPOLES.** $\{}ell_1 = 221.93 + 4.21f$ to three hundredths of a multipole, so the family spans $0.01395$ in $\{}ell_1/\{}ell_A$ — $4.2$ of the sky's one-multipole locating widths, $2.1$ of its two-multipole ones. ⇒ ***The comb pins $f$ to $\{}pm0.24$: it separates the family's ends and comes nowhere near fixing the clock.*** **AND THE ORDER'S FIRST BRANCH IS HALF RIGHT.** The contrast's *level* is shallow ($1.75\{}sigma$ against the comb's $4.21$) but its $q$-slope is not ($+0.01042\{}to-0.00039$, $3.80\{}sigma$, the comb's statistical equal) — ***what separates them is that the contrast's error GROWS five-fold across the family (band residual $0.0089\{}to0.0441$, slope s.e. $0.00284\{}to0.01410$) while the comb's locating width does not move***, 'cc66.40''s guard firing a fourth time. **AND THE DECIDING RESULT IS NOT ABOUT THE CHOICE: THE SKY'S VALUE IS NOT IN THE FAMILY.** $0.7312$ sits at $f=-0.304$ on the family's own line, on the far side of the stacking clock ⇒ ***no interior fraction fits better than the endpoint in use (the third branch does not arise and there is no fitted clock to declare), $f=0$ is the best point, and the residual first-peak disagreement cannot be absorbed by the clock assignment because the direction it needs is inadmissible.*** $P_1/P_2$ agrees independently: $2.141\{}to2.017$ against the sky's $2.217$. **THE GUARD, SEPARATED THREE WAYS** because the injection has no plasma dynamics in it: the comb's motion is $31\{}%$ the peak relocating through $r_s(\{}eta_{\{}rm LS})$ ($+1.31$ multipoles), $9\{}%$ the visibility's re-weighting of the kernel ($+0.39$), and $60\{}%$ **the plasma's own acoustic phase** ($+2.51$). **AND TWO OF 'cc66.43''s NUMBERS WERE GRID-LIMITED**: $\{}ell_1$ $220\{}to228$ is really $221.95\{}to226.16$, $\{}ell_1/\{}ell_A$ $0.7290\{}to0.7555$ really $0.73543\{}to0.74938$ — *and the sign of the arm's offset from the sky at $f=0$ flips, from below to $1.28$ multipoles above* — while the FWHM's $43.591\{}to43.952$ is exactly one step of the $\{}eta$ grid, non-monotone across the family, and not a resolved motion at all. The locator is validated on the banked 'LSTEP=1' spectrum first, where it recovers the fine-grid $\{}ell_1$ to $0.004$ of a multipole. **AND THE SWITCH GUARD IS NOW STANDING**: the instrument prints a '__SWITCHES__' line whose inventory is read off its own source, 'switch_smoke.sh' asserts every assignment arrives, and the launcher checks every distinct environment before the set and every slice's log after — 'r6925''s thirty-six lost slices cannot recur silently. NOT a verdict on the two-rate assignment; $f¡0$ is the family's line extrapolated and not a candidate; the sky's locating width is P15's own; the injected runs are not spectra of this model; nothing touches 'prop:flat'; no refit). *Computes:* 'VISLEAF' is a FRACTION now, and its endpoint is gated **bit-identical** against 'r6925''s banked 'VISLEAF=1' spectra while 'VISLEAF=0' is gated bit-identical against the unset path and the CONTROL against itself at $f=0.5$ — *the fraction's zero is the flag's off, the family's one is the flag's on, and the rate identity collapses the whole family on the control*. The base $f=0$ reproduces the corpus's own banked reported spectrum to the 'KSLICE' sum's rounding, which is the scan's provenance. The **sub-bin locator is validated before use** on the banked 'LSTEP=1' spectrum ($0.004$ of a multipole on $\{}ell_1$), because the whole question is resolution. *And the switch guard is standing rather than per-launcher: a '__SWITCHES__' marker whose inventory is read off the instrument's own source, 'switch_smoke.sh', a pre-launch check on every distinct environment and a post-run check on every slice's log* *Bound:* **NOT a verdict on the two-rate assignment** — the row's question, and how well the comb constrains it is evidence toward that **NOT that $f¡0$ is admissible**: the $-0.304$ is the family's own straight line extrapolated, reported because the order asked what the comb can decide, and the

family is bounded by the two clocks **NOT a re-derivation of the sky's locating width**, which is ‘P15_the_sky_phase_fit_and_its_uncertainty‘’s and is taken as given **NOT that the injected runs are spectra of this model** — they are the projection's transfer of a known input, under separate keys *no mechanism beyond ‘cc66.42‘’s; nothing touches ‘prop:flat‘; no refit* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_combs_resolution_is_four_multipoles_and_the_skys_own_value_lies_o`

**P15R192** `P15_the_locator_is_good_to_three_hundredths_and_the_fourth_peaks_residual_is_the_controls_too` [OK]

*node 66's work order ‘r6941‘ (headed ‘r6939‘): **‘PO-56‘ — the fourth peak, and the locator before the residual**. *‘cc66.44‘ closed the clock route, so what is left undecomposed is the fourth peak — $222/538/818/1134$ against the sky's $220.4/537.7/817.3/1123.9$, peaks one to three within a grid step and peak four $10.1$ out, and the comb fitted on the first three, which makes **peak four the only one of the four out of sample**.* *But the first thing asked is the instrument: **establish the locator's own precision at each of the four peaks before reading any residual off them**, since it was validated at $\{}ell_1$ and $\{}ell_4$ sits where the damping has flattened the peak — with the stopping rule that **if the locator at $\{}ell_4$ is worse than ten multipoles the residual is not measured and the order ends at step one**. Then (3) the three-way separation ‘cc66.44‘ built, at every peak index, with the reading fixed in advance; and the guard: **is the four-peak pattern one systematic or two, and do not pick** — three candidate shapes named and none of them fitting, so the enumeration is probably incomplete** ( **THE STOPPING RULE DOES NOT FIRE, AND IT MISSES BY A FACTOR OF FOUR HUNDRED.** *Run again at ‘LSTEP=1 LMAXL=2000‘ — $1900$ multipoles against the reported $238$, on both arms — the reported grid's locator recovers the fine-grid position to $0.0007/0.0126/0.0283/0.0176$ of a multipole on the control and $0.0035/0.0089/0.0167/0.0229$ on the arm:* ***two hundredths at $\{}ell_4$ against a bar of ten***, *and the extremum search is insensitive to its own width (order $3$ to $40$ identical).* **WHAT IS IMPRECISE IS THE PARABOLA'S WINDOW, AND IT IS A BIAS.** *Over ‘PO-47‘’s admissible $W=15\{}ldots110$ the located peak moves $0.08/2.77/4.73/5.82$ on the arm and $0.11/2.98/4.97/6.49$ on the control — growing with peak index because a wider fit on an asymmetric, damping-suppressed hump pulls its apex down the envelope — so at $\{}ell_4$ the window's bias is comparable to the residual, and it displaces the sky and both models alike, which is what matched-procedure differencing is for* (at $W=25$ it agrees with the three-point locator to $0.2$). **AND THE ORDER'S PATTERN IS TWO ARTEFACTS, NEITHER OF THEM PHYSICS.** *First the quartet: $222/538/818/1134$ is ‘sec:refit-bound‘’s **line-of-sight** path, while every bank here is the **hierarchy** path, whose raw reading is $220/540/812/1132$ — and **at $\{}ell_3$ the two bracketing ‘LSTEP=8‘ bins differ by four parts in ten thousand, so which is called the peak is a coin flip**: the paper quotes $820$, this locator picks $812$, the sub-bin apex is $815.40$ and the fine grid's own maximum is at $815$. Sub-bin the arm is $221.96/536.11/815.40/1130.53$, so the residual is $+1.56/-1.60/-1.90/+6.63$ —* ***peaks two and three were never 'on', they are each about $1.7$ LOW, and peak four is $6.6$ out rather than $10.1$.*** **Second, and decisive: the CONTROL produces four fifths of it** — $-0.05/-1.41/-2.97/+5.32$ through the identical locator — so ***what is the construction's is $+1.61/-0.19/+1.07/+1.31$, a near-constant ONE multipole at all four peaks: a constant $\{}Delta\{}ell$, the FIRST of the order's three shapes, fitting half again better than the ruler's constant $\{}Delta\{}ell/\{}ell$ (rms $0.68$ against $1.02$) and nothing like a driving error growing with $\{}ell$.*** ⇒ **So the pattern needs ONE systematic and not two**, and every one of the four displacements is inside the sky's own locating spread ‘PO-47‘ measured ($1121.9\{}pm2.36$, arm-minus-control $0.83\{}sigma$), so *the sky cannot tell it from zero and it is reported as a shape rather than a disagreement*. **(3) AND THE SEPARATION AT EVERY PEAK INDEX**: relocation $31.2\{}to94.1/83.4/89.3\{}%$, the visibility's weighting $9.2\{}to26.4/22.7/23.5\{}%$, the plasma $59.6\{}to-20.5/-6.0/-12.8\{}%$ — ***$\{}ell_1$ is the outlier and nothing else is: at $\{}ell_2$ to $\{}ell_4$ the motion is almost entirely geometric and the plasma's phase partially CANCELS it.*** *By the order's own reading the plasma's share does not grow, so the residual is not in the driving, and the relocation share does, which points at the ruler* — *but the third reading is the one that applies:* **the shares are not flat and the residual is not reached**, the family's motion being $+0.6$ to $+0.7\{}%$ with one sign at every peak while the residual alternates. *Motions resolved: fine against coarse they agree to $0.02$ of a multipole*). *Computes:* The fine reference is the reported configuration re-run at eight times the multipole sampling on both arms, ‘HIER=1 LSTEP=1 LMAXL=2000‘ in ‘KSLICE‘ pieces of 250 — 35 slices, one ‘cr‘ slice costing 4m37s — with the standing switch

guard of ‘cc66.44‘ run on every distinct environment before the set and every slice’s log checked after. *The ‘VISLEAF‘ endpoints are re-run fine too, so (3)’s motions are measured rather than inherited, and they agree with the ‘LSTEP=8‘ banks to $0.02$ of a multipole.* **No corpus edits, as the order directs** — the paper-side consequences (that ‘sec:refit-bound‘’s ”the first three peaks land within a grid step” is the line-of-sight path’s raw reading, and that on the hierarchy path sub-bin peaks two and three are each about $1.7$ low) are reported to 66 in ‘FOR_66.md‘ rather than applied *Bound:* **NOT a mechanism** for the one-multipole constant offset — it is measured, named as a constant $\{}Delta\{}ell$, and left **NOT a re-derivation of the sky’s locating spread**, which is ‘PO-47‘’s and is cited **NOT that the window convention is an error** — it is a convention, and what is established is that it is shared and that the tight window agrees with the three-point locator **NOT that peak four is uninteresting**: it is the largest residual of the four, and what is shown is that four fifths of it is the control’s *no verdict on the two-rate assignment; no refit; nothing touches ‘prop:flat‘ or the clock family* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_locator_is_good_to_three_hundredths_and_the_fourth_peaks_residual`

**P15R193** `P15_the_contrast_excess_is_symmetric_about_the_envelope_and_the_troughs_are_where_the_sky_meas` [OK]

*node 66’s work order ‘r6955‘: **‘PO-56‘ — the troughs, which is where the $\{}chi^{2}$ actually lives**. *Four routes are eliminated — the contrast is not in the source at any grain, not the cross term, not absorbable by the clock assignment, not the comb residual — so what is left is the $\{}chi^2$: the spacing is right to a part in six hundred and the arm still scores $1.43$ per bin against the control’s $0.99$.* **66’s reading, stated as its own to be knocked down**: *the paper’s discriminant of principle $\{}theta_D/\{}theta_*$ is a ratio of two lengths the plasma accumulates, so it rides ONE clock and a common difference cancels — while a trough DEPTH needs the diffusion scale in multipoles and so needs the projection distance, which rides the other. (2)[a] Locate the excess before assuming it, decomposing the contrast into what the peak heights carry and what the trough depths carry* — **if it is in the peaks, stop and report**; *(2)[b] then the clock sensitivity of whichever carries it, against $\{}mathrm d(\{}theta_D/\{}theta_*)/\{}mathrm df$; (2)[c] and the sky’s own spread on it, the way ‘PO-47‘ did for the fourth peak. Guards: the window bias applies to troughs and worse; **path provenance in the receipt’s header from now on*** ( **(2)[a] THE EXCESS IS SYMMETRIC ABOUT THE ENVELOPE, so it is neither trough- nor peak-dominated.** *Splitting the band variance of the envelope-normalised oscillation at its own zero — a decomposition, the parts adding to the whole to machine precision — the arm/control ratio is $1.0668$ on the positive side and $1.0561$ on the negative; at the literal extrema the arm exceeds the control by $5.7$ per cent at the maxima and $5.3$ at the minima.* ⇒ ***It is an AMPLITUDE excess: $\{}Delta$’s trough dominance is a statement about the whitened residual and does not transfer to the contrast’s own decomposition*** — which the order declined to assume, and the answer is that they disagree. *The stop condition does not fire either: the excess is not in the peaks, so the row is not pushed back onto the driving.* **(2)[b] THE DEPTHS ARE THREE TIMES THE MORE CLOCK-RESPONSIVE — $-2.58$ per cent across the ‘VISLEAF‘ family against the heights’ $-0.77$ — BUT $\{}theta_D/\{}theta_*$ MOVES $-4.08$, MORE THAN EITHER**, because $r_D$ is itself ‘Jac‘-weighted under ‘LEAFSCALES‘ and the $\{}tau$ reweighting moves it directly. ⇒ ***The paper’s discriminant of principle is the most sensitive of the three and not the insensitive one, so by the order’s own rule the depths are not a new handle on the clock.*** **(2)[c] AND YET THE TROUGHS ARE WHERE THE ASSIGNMENT BECOMES OBSERVABLE, FOR A REASON NEITHER SEAT NAMED: THE SKY MEASURES DEPTHS TWICE AS WELL AS HEIGHTS.** *Same data, same likelihood binning, same anchored locator, ‘COV_TT‘ propagated by Monte Carlo: the sky’s spread is $0.00588$ on the depth and $0.01135$ on the height, seed-stable. The arm-minus-control excess is the same size in both — $+0.0106$ and $+0.0103$ — so it reads* ***$+1.81\{}sigma$ in the depths against $+0.91$ in the heights.*** *And the CONTROL lands on the sky’s depths at $-0.05\{}sigma$ while the arm sits $+1.76$ above:* ***the first statistic in this sector whose residual is NOT shared with the control***, against the fourth peak’s $0.83\{}sigma$ with both models high of the sky. ⇒ **So the order’s conclusion stands and its argument does not.** **THE GUARDS**: *the anchored depth has NO window bias — identical over $W=20\{}ldots70$ on the sky and both arms, because it reads a **value** at a located extremum and not the **location** of one, where ‘cc66.45‘’s parabola apex drifted six multipoles at $\{}ell_4$ — while a FREE search under noise reproduces ‘PO-47‘’s trap at six times the spread, which is why the anchoring is necessary; and the envelope’s abscissa is worth about a third of a sigma and is reported rather than minimised*).

*Computes:* **Path provenance in the header, as the order directs**: every model number is the **hierarchy** path (‘cc66_r185_verify_*‘, the ‘LSTEP=1‘ fine references ‘r6941_fine_*‘, and the ‘VISLEAF‘ family ‘r6929_scan_cr‘ / ‘r6941_fine_cr_visleaf‘), and ‘sec:refit-bound‘’s quartet is the **line-of-sight** path’s and is not read at all — searched for by grepping the receipt for each of the four values and for every line-of-sight bank name, none of which occurs in its code. *The sky side is ‘plik_lite‘ TT through the likelihood’s own binning, with the models binned identically before any comparison, so matched-procedure differencing is by construction and not by assertion.* **And this revision cost no solver time**: it is analysis on banks ‘cc66.44‘ and ‘cc66.45‘ already built, which is why the order’s three parts and three guards all fit in one revision *Bound:* **NOT that $1.8\{}sigma$ is a detection** — it is $1.8$ of the sky’s own spread on one statistic, and the same differencing at the fourth peak gave $0.83$ **NOT a mechanism** for the amplitude excess: this locates it and measures how well the sky sees it **NOT a re-derivation of ‘PO-47‘’s peak-position spreads**, which are cited untouched **NOT that the depths discriminate the CLOCK** — they do not, and that branch of the order’s reading is refuted here *no verdict on the two-rate assignment; no refit; nothing touches ‘prop:flat‘ or the clock family; no corpus edits* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_contrast_excess_is_symmetric_about_the_envelope_and_the_troughs_a`

**P15R194** `P15_the_windows_phase_spread_is_nearly_equal_because_the_visibility_widens_as_the_leaf_clock_s` [OK]

*‘sec:refit-bound‘ — the normalisation across the visibility is a real channel carrying about a THIRD of the contrast excess, and it is peak-weighted where the excess is symmetric — so it is the first partial mechanism in this sector and still not the whole one* (node 66’s ‘r6959‘ order, three items. $(1)^{a}$ ***”Measure the source’s conformal-time dependence across the visibility window on both arms, and report it as a function rather than as a ratio”***; $(1)^{b}$ ***”imposing one arm’s normalisation on the other should move the contrast toward unity, and it should do so by an amount the measurement predicts in advance”*** ⇒ *”predict the size first, then run it”*; $(1)^{c}$ ***”state in advance what would REFUTE it”*** and ***”say what the normalisation swap does to the comb and to the depths as well as to the contrast”***. Guards: path provenance in the header (standing), *”keep the two statistics apart”* since two transfers had already failed to hold, and no detection language). *Computes:* **$(1)^{a}$ THE FUNCTION, AND A CANCELLATION IN IT.** The source’s own weight across the window, band by band in $q$, has its spread in the phase variable $r_{s,\{}rm leaf}$ at $0.06523$ of $r_s$ on the control and $0.06424$ on the arm — **the arm narrower by one and a half per cent**, and only that much because *the arm’s visibility is $14.6$ per cent WIDER in conformal time while its phase accumulates $13.7$ per cent SLOWER per unit of it* (‘Jac‘ $0.789$–$0.913$ against an identical $1$), the product coming to $0.9890$ ⇒ ***the two-rate structure reaches the window and then very nearly cancels inside it*** **$(1)^{b}$ THE ANALYTIC PREDICTION, PRE-REGISTERED**: $\{}mathcal D_{\{}rm cr}/\{}mathcal D_{\{}rm lcdm}=1.00078\{}to1.01654$ against a measured $1.0215\{}to1.0762$ — **R5 (sign) survives, R2 (size) fires at $0.223$ of the bar, R1 (shape) fires because the measured excess extrapolates to $1.0400$ at $q=0$ where a characteristic function is identically $1$** **AND R3’s BRACKET WAS WRONG IN DIRECTION, WHICH IS MINE**: ‘PREDICTION.md‘ argued the contrast responds between $f$ and $f^{2}$; measured it responds $2.3$ to $26$ times MORE, at both coefficients and in every band, and the condition passed only because its tolerance was written on $f$ rather than on $f-1$ **THREE WIRINGS, THE FIRST TWO NOT THE OPERATION**: the all-terms taper cuts the ISW’s own $\{}eta$-integral to $0.656$ and $0.238$ of itself (moving $\{}ell_1$ by $7.7$ and $27.3$), and holding the window’s total weight fixed then lands on the ISW-preserving run to parts in ten thousand — ***so that second diagnosis of mine was wrong too, and the over-response is the statistic’s own sensitivity*** **THE SIZE, FROM THE INSTRUMENT’S OWN TWO-COEFFICIENT RESPONSE RATHER THAN A MODEL OF IT**: the arm’s own narrowing delivers **$23$ to $45$ per cent of the band excess above $q=1.9$**, and the direct swap closes $20$ to $39$ per cent there — *the same answer by a different road* ⇒ the narrowing for ALL of it is $4.5$ per cent against the arm’s $1.7$, **a factor $2.7$** **$(1)^{c}$ R4, AND THE REFUTATION FROM 66’s OWN PREVIOUS REVISION**: ‘cc66.46‘ measured the excess symmetric about the envelope, and this channel is not — the arm-sized swap delivers **$81$ per cent of the peak-height excess and $14$ per cent of the trough-depth excess, a $5.7$-fold asymmetry** ⇒ ***a peak-weighted channel cannot carry a symmetric excess alone, whatever its size, and that argument is independent of every size estimate*** the taper that would deliver the whole excess moves $\{}ell_1$ by $18.5$ multipoles and the heights by $45$ per cent; the arm-sized one overshoots $\{}ell_1$ past the arm’s own value while $\{}ell_2$, $\{}ell_3$,

$\{}ell_4$ stay inside the sky's widths *Bound:* **NOT that the normalisation across the window is the mechanism** — it carries about a third, it is peak-weighted where the excess is symmetric, and R1's $q$-independent offset is outside it **NOT that it is nothing**: it is the first channel in this sector measured to carry any definite share, from a quantity nobody chose **NOT a claim about the term mix**, which is reported (the arm holds more of its window's power in the monopole in every band, $0.1837\{}to0.2171$ at $q=1.2$) and **not swapped** **NOT a detection**: the $1.8$ of 'cc66.46' is one statistic's spread and nothing here moves it NOT a verdict on the two-rate assignment; NOT a re-derivation of 'PO-47''s peak spreads, which are quoted *The lowest band is the one the inversion cannot be trusted in and that is stated rather than averaged away: there the swap overshoots, taking the excess below unity* No refit, nothing touching 'prop:flat' or the clock family, and **no corpus edits** — the paper-side consequences are routed to 66 *Run:* `python3 receipts/P15_CR_cosmology/P15_the_windows_phase_spread_is_nearly_equal_because_the_visibility_wider`

**P15R195** `P15_the_term_mix_has_the_shape_the_offset_needs_and_over_delivers_its_size` [OK] *'sec:refit-bound' — the term mix has the shape the $q$-independent offset needs and over-delivers its size by half again, so the two channels do not add and the "remainder" is a construct of the composition rule* (node 66's 'r6975' order, three items. (1) ***"Decompose the excess into the peak-weighted part and the remainder, explicitly, and characterise the remainder rather than the whole"***; (2) ***"say what class of mechanism can produce a symmetric excess at all, before hunting one"*** — *"not a fitted direction and not a scan over knobs — a statement about what kind of thing it has to be, with the arithmetic that says so"*; (3) ***"the term mix you reported and did not swap ... swap it and report what it does to the contrast, the depths and the comb, with the prediction stated first"***. Guards: path provenance (standing); **the pre-registration lesson from my own 'cc66.47'** — *"write the tolerance on the quantity that would move, not on the quantity it is derived from"*; keep the two statistics apart; no detection language). *Computes:* **(1) THE REMAINDER.** Dividing the measured excess by what the window channel delivers at the arm's own size leaves $1.0184$ at $q=0$ — **$46$ per cent of the offset R1 fired on** — carrying $123$ per cent of the measured $q^{2}$ slope, and trough-weighted at $0.49$ where the channel that left it is peak-weighted at $12.44$ **(2) AND THE STATISTIC HAS A BASELINE I HAD NOT COMPUTED.** Applying each named class to the control's own spectrum and re-reading it with the same envelope and locator: amplitude $1.96$, envelope $1.96$, loading $0.63$, smearing $-5.28$ ⇒ ***a SYMMETRIC operation reads $1.95$ here and not $1$***, stable across a fivefold range of operation sizes, because the running-mean envelope is recomputed and shifts the oscillation upward by a constant ⇒ **so the anchored reading CONFIRMS 'cc66.46''s variance split rather than contradicting it** — $2.20$ against $1.96$ is agreement to twelve per cent, and my own note that the two statistics disagreed is retracted in the pre-registration before the runs landed **(3) THE TERM-MIX SWAP, ON A KNOB ALREADY WIRED.** The coefficient giving the control the arm's monopole fraction is solved from the profiles and comes out $0.860$–$0.897$ in all seven bands — ***one constant to two per cent, so the difference is a single scalar on the Doppler and not a reshaping*** **T1 the sign passes** (the contrast rises in every band) **T2 the shape passes and is the finding: the response is $q$-INDEPENDENT**, the slope worth three per cent of the intercept ($0.031$ of it) with an intercept of $1.0805$ at $q=0$ — ***the first channel measured with the signature the offset needs, where a smearing's characteristic function is identically one*** **T3′ the weighting FIRES**: $2.23$ and $2.21$ at the two coefficients, peak-weighted at the symmetric baseline, where the remainder it was to be is $0.49$ ⇒ **the term mix is not the remainder** **and it OVER-DELIVERS**, raising the control's contrast by $1.10$ to $2.77$ times the whole excess, mean $1.58$ **T4 the comb fires and in the wrong direction**: $\{}ell_1$ $-1.04$ and $\{}ell_2$ $-1.79$, both outside the sky's widths, where the arm sits at $+1.61$ **THE CONSEQUENCE, WHICH IS THE RESULT.** The window channel's third and the term mix's $1.6$ times together make $1.72$ times what is measured ⇒ ***they cannot both be present at their measured sizes and simply compose — and (1)'s remainder is a RATIO of two responses, so it is a construct of that composition rule and not a residual channel***, which is why its $0.49$ and the swap's $2.23$ are not in conflict **AND MY OWN (2) IDENTIFICATION IS REFUTED BY (3)**: I argued the Doppler fills the oscillation and removing it should read as the loading class near $0.6$; it reads $2.23$, and the profiles say why — its band-to-band power tracks the monopole's at correlation $0.95$, so it is an OSCILLATING term in quadrature and not a smooth additive one *Bound:* **NOT that the term mix is the remainder** — T3′ refutes that **NOT that it is the mechanism**: it over-delivers by half again and moves the comb the wrong way **NOT that it is nothing**: it is the first chan-

nel measured to carry the $q$-independent signature the offset needs, from a difference nobody chose **NOT a replacement decomposition** — what is established is that the composition rule is at fault, and this receipt does not supply a better one NOT a detection; NOT a verdict on the two-rate assignment; NOT a re-derivation of ‘PO-47‘’s spreads, which are quoted *The lowest band is where the over-delivery is largest ($2.77$) and that is reported rather than trimmed* No refit, nothing touching ‘prop:flat‘ or the clock family, and **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_term_mix_has_the_shape_the_offset_needs_and_over_delivers_its_size`

**P15R196** `P15_the_two_channels_compose_multiplicatively_and_the_pair_is_flat_where_the_excess_grows` [OK]

*‘sec:refit-bound‘ — the composition rule is MEASURED and it is multiplicative, which retracts ‘cc66.48‘’s inference and confirms its arithmetic: the two channels compose exactly as that revision assumed, and the pair is flat in $q$ where the measured excess grows* (node 66’s ‘r6983‘ order, one item. ***”PO-56 is no longer which channel carries the excess but how two channels compose”*** — ‘cc66.48‘ divided one response into the excess, got a remainder, and argued the remainder was an artefact of a composition rule nobody had measured, so the order is to measure the rule. Guards: path provenance (standing); both coefficients as their own revisions solved them and **neither re-chosen**; every weighting reading against the measured symmetric baseline $1.95$ and not $1$; **no third channel**; no detection language). *Computes:* Both knobs on ONE control spectrum at the sizes their own revisions solved — ‘SRCTAPER=1.100877765e-4 SRCTAPERS0=145.3465211 SRCTAPERNORM=1‘ from ‘cc66.47‘ and ‘DPSRC=0.8794‘ from ‘cc66.48‘ — read against four conditions pre-registered before the run. **(1) T-SIZE PASSES and refutes QUADRATURE**: the joint lands within the pre-registered $0.005$ of the product in $7$ of $7$ bands and of the sum in $7$ of $7$, and of quadrature in $0$ of $7$, quadrature’s worst miss $0.018$ being three and a half times the bar *the pre-registration declared in advance that product and sum differ here by at most $0.0016$ and are NOT separable; they are not separated, which is the declared outcome and not a failure* **and the pair is slightly SUB-multiplicative, consistently**: $0.971$ of the product and $0.985$ of the sum, below both in every band and above quadrature in every band **(2) T-WEIGHT PASSES and is the sharpest of the four**: predicted $2.83$ from the two channels’ separate height and depth changes, bracket $[2.2, 3.6]$, and the joint reads **$2.80$** — one per cent from a number computed before the run, on an axis where quadrature predicts nothing at all **(3) T-COMB PASSES**: $\{}ell_1$ predicted $+1.79$, bracket $[+1.0, +2.6]$, measured $+1.41$ — below the additive centre, the same mild sub-additivity the sizes show **(4) T-INTERCEPT PASSES**: the joint intercept against $q^{2}$ is **$1.0993$** against a multiplicative $1.1035$ with nothing fitted ($0.0042$ away), a sum’s $1.1019$ ($0.0026$) and quadrature’s $1.0839$ ($0.0154$, outside) **and the joint response is the flattest thing this sector has measured** — its $q$-dependence is $0.004$ of its intercept against the term mix’s $0.031$ and the measured excess’s $0.442$, a factor of a hundred **THE CONSEQUENCE, IN THREE PARTS.** **Confirmed by direct measurement**: ‘cc66.48‘ put the pair at $1.72$ times the excess by adding two separate shares; composing them in ONE spectrum reads $1.70$ **Retracted**: that revision inferred from the over-delivery that the channels do NOT compose as assumed, so a remainder got by DIVIDING one response into the excess was an artefact of a wrong rule — ***the rule is now measured and it IS the assumed one***, so dividing is the right operation and the remainder is a well-defined object; the over-delivery was evidence about the SIZES, not about the RULE **And the obstruction that survives is a SHAPE no coefficient can remove**: the pair over-delivers $3.87$ times in the longest-wavelength band and $1.38$ in the shortest, because it is flat where the excess grows ⇒ ***scale it to the offset at $q=0$ and it is short at high $q$; scale it to high $q$ and it is an order too large at low $q$*** *Bound:* **NOT that the composition is exactly multiplicative** — it is multiplicative to within the pre-registered bar with a consistent $3$ per cent deficit, and product and sum are NOT separated here, as declared in advance **NOT that quadrature is refuted as a rule in general**: it is refuted for THIS pair at THESE sizes **NOT that either channel is the mechanism** — both over-deliver together and the shape argument says neither rescaling helps NOT a third channel, which the order forbids and none is added; NOT that the excess is explained; NOT a detection; NOT a verdict on the two-rate assignment; NOT a re-derivation of ‘PO-47‘’s spreads, which are quoted *The joint spectrum is a sum of disjoint mode slices and the tiling is ASSERTED rather than trusted: the bank reconstructs each slice’s span from its filename and refuses unless the spans tile $[0, 2547)$ with no gap and no overlap — the run was re-sliced twice midway because this container is reclaimed faster than a large slice completes, which is exactly the circumstance a fixed-step check

would have passed over* No refit, nothing touching ‘prop:flat‘ or the clock family *Run:* `python3 receipts/P15_CR_cosmology/P15_the_two_channels_compose_multiplicatively_and_the_pair_is_flat_where_`

**P15R197** `P15_the_excess_decelerates_and_nothing_measured_in_this_arm_carries_its_wavenumber_dependence` [OK]

*‘sec:refit-bound‘ — the excess DECELERATES with wavenumber, which excludes the parametrisation this sector quotes its own variation statistic in; and the first positive search the row has ever run comes back EMPTY, with nothing already measured in this arm carrying the dependence* (node 66’s ‘r6993‘ order, one item, **redirecting this seat’s own stated next object**. cc66.49 left it as *what sets the sizes*; 66 moved it to ***”what channel carries the excess’s own wavenumber dependence”***, on the argument that a shape refutation is not closed by any account of sizes — *and made it **the first POSITIVE search criterion this row has ever had**, every previous step having been an elimination*. (1) characterise the target’s shape before hunting; (2) enumerate what has that dependence and filter **on paper**; (3) control the filter itself; (4) and nothing else. Guards: pre-register and make it cost something; the anchoring guard; **a size and a shape are two objects**; the banking discipline). *Computes:* **(3) RUN FIRST, AND THE INVERSION IS THE METHOD**: (3) asks about the INSTRUMENT, and its answer is what (1)’s pre-registration must write its tolerances on — the alternative is a guessed resolution **a known $q$-dependence READS BACK**: six forms injected into the control’s own spectrum, including two turnovers with a SCALE in them, recover to a worst error of $1.3$ per cent **but the estimator MANUFACTURES band-to-band structure** — its per-band bias swings about one per cent and the six forms agree on it seven times more closely, so ***a constant contrast reads as a wobble*** and band ordering at the per-cent level is the instrument **and the MEDIAN envelope is excluded with its mechanism, not dismissed, and would otherwise have been the headline**: it reads $1.49$ against the mean envelope’s $0.44$, and ***the tell is that it does not move with its own window*** — at high $q$ the running median collapses onto the curve itself, within $3.6$ per cent of it against the mean’s $14$, because the median of a locally monotone stretch is its central value whatever the window the rise survives the mean envelope at $0.44/0.44/0.40$ ⇒ ***the RISE is real, the WOBBLE is not*** **(1) THE RISE LINEAR IN $q^{2}$ IS EXCLUDED** at $\{}chi^{2}/\{}nu = 4.66$, the only form excluded of six — ***and it is the parametrisation ‘cc66.47‘, ‘cc66.48‘ and ‘cc66.49‘ all quote their variation statistic in*** the concave family all fit and **none is preferred**, best-over-next $\{}Delta\{}chi^{2} = 0.43$ against a bar of $4$ **both pre-registered NON-separations held** — power law against logarithm $3.42$, power law against turnover $0.99$ **NO SCALE IS RESOLVED**, both scale-bearing forms putting $q_0$ with its uncertainty on the bottom edge — *and the rule as written caught this seat’s own code, which had asked only that the central value be inside* **the rise is a PREFERENCE and not an exclusion of flatness**: every rising form beats a constant by $6.0$ to $10.4$, yet the constant’s own $\{}chi^{2}/\{}nu = 2.03$ is under the exclusion bar **(2) AND THE FILTER’S FIRST APPLICATION RETURNS EMPTY.** The target’s departure grows by $G = 3.55$ with negative curvature; ***every candidate already measured in this arm fails the growth tooth*** — six source terms ($0.54$–$1.00$), the dipole-to-monopole ratio ($1.08$), and all three measured channels, window $0.55$, term mix $1.53$, the pair $1.27$ **and the order’s own named candidate disagrees with the register**: recorded there as two per cent ABOVE the control and rising, it reads fourteen per cent BELOW and flat on the hierarchy path at each arm’s own visibility peak *Bound:* **NOT that the form is a turnover, a power law or a logarithm** — none is preferred and that is stated as a non-separation **NOT that a scale exists or does not**: none is RESOLVED, which is weaker **NOT that the excess is inconsistent with flatness**: flatness is disfavoured, not excluded **NOT that the dipole-to-monopole ratio is eliminated** — it fails the filter as measured here and the register disagrees, which is a discrepancy to name and not a verdict, and which path the register measured is owed before the candidate is closed **NOT that no candidate exists** — what is shown is that none of the things ALREADY MEASURED carries the dependence, a statement about the list searched NOT a third channel, which the order forbids and none is run; NOT a detection; NOT a verdict on the two-rate assignment *Nothing is RUN: every number is read from banks on disk, which is what makes the filter cheaper than the runs it is meant to save* No refit, nothing touching ‘prop:flat‘ or the clock family, and no corpus edits *Run:* `python3 receipts/P15_CR_cosmology/P15_the_excess_decelerates_and_nothing_measured_in_this_arm_carries_its_w`

**P15R198** `P15_the_two_readings_are_different_quantities_and_the_projection_width_is_the_unmeasured_one` [OK]

*‘sec:refit-bound‘ — the two dipole readings are DIFFERENT QUANTITIES rather than a contradiction; correcting which one the filter needs REVERSES ‘cc66.50‘’s dismissal of that candidate; and the projection, which this row is named after, holds a width nobody has imposed* (node 66’s ‘r6997‘ order, four items. (1) the reconciliation this seat named and which ‘P15‘ now carries as OWED — ***”which quantity does each path measure at the visibility peak, same definition, same weighting, same arm”***; (2) the second tooth on anything that passed the first, *”and if every measured candidate fails on growth, say so — do not manufacture a use for it”*; (3) ***”what is NOT on the list”***, *including that the carrier may be a property of the PROJECTION rather than the source or transfer*; (4) nothing run before (1)–(3) are answered on paper. Guards: preregister and make it cost something; **exclude by mechanism where you can and say when you could only exclude by number**; and when a check licenses an action, confirm what it examined and what it touches are the same set). *Computes:* **(1) THE DISAGREEMENT IS NOT A CONTRADICTION AND THE SIGN IS NOT THE RANGE.** Both readings reproduced from banks: the register’s quantity reads **$+2.06$ per cent** over its own range — its recorded figure exactly — and still **$+0.95$** over the part of ‘cc66.50‘’s range it reaches, where ‘cc66.50‘’s reads **$-11.58$**. ⇒ ***the difference is the QUANTITY***, in three independent ways: the OSCILLATION AMPLITUDE of each source field against the INTEGRATED BAND POWER which keeps the smooth part; the source fields at last scattering against the terms’ contributions to the projected integrand; and the range **and which one is needed is the register’s, which makes ‘cc66.50‘’s filter row this seat’s own error**: trough-filling is an oscillating term a quarter period out of phase adding into another’s troughs, so what fills a trough is the oscillating amplitude — ***and the same is true of the filter, the contrast statistic being the standard deviation of the OSCILLATION about a running-mean envelope*** **(2) SO THE SECOND TOOTH IS USED, AND BECAUSE SOMETHING SURVIVED THE FIRST**: on the range where the right quantity can be measured the candidate’s departure runs $-0.0006$ to $+0.0166$ and **decelerates**, curvature $-0.0037$ against the target’s $-0.0029$, the same sign — ***it passes both teeth where it can be tested, reversing ‘cc66.50‘ on this seat’s error and not on new data*** **and the growth statistic is UNDEFINED, not large**: the departure crosses zero inside the range, so $G$ is not a ratio and the arithmetic’s $-26$ would have read as a spectacular pass **and yet the filter CANNOT CLOSE IT**, which is the result: the right quantity bottoms out at $q=2.0$ — stable to a tenth of a per cent across four window widths, so sound and simply not reaching, the acoustic period in $q$ being $2$ — while ***the target grows $3.55$-fold over the full range and only $1.47$-fold over the shared one, so its growth is concentrated exactly where the only quantity that could carry it cannot be measured*** **(3) A WHOLE CLASS IS SETTLED BY STRUCTURE**: every two-rate clock quantity — leaf clock, ruler, Jacobian, visibility, optical depth — is stored against CONFORMAL TIME ALONE, and a function of $\{}eta$ has the same value at every wavenumber, so it **cannot carry it**, no run needed **AND THE PROJECTION HOLDS A WIDTH NOBODY HAS IMPOSED**: the acoustic phase varies across the window by $k imes$ the width in the LEAF clock, which is what ‘cc66.47‘ acted on, while the projection smears by $k imes$ the width in CONFORMAL TIME — one object in a one-rate cosmology, two here — and ***the arm’s window is $6.5$ per cent WIDER in conformal time and $0.8$ per cent NARROWER in the leaf horizon, a ratio $7.3$ per cent different from the control’s*** *Bound:* **NOT that either earlier reading was wrong** — different quantities, each right about its own **NOT that the dipole-to-monopole ratio IS the carrier**: it passes both teeth only on the sub-range carrying least of the growth, and is neither closed nor confirmed **NOT that the projection width IS the carrier**: it is unmeasured, which is the point, and its response has not been computed **NOT that the clock quantities are irrelevant** — they cannot carry a wavenumber dependence themselves, which leaves them free to set a scale something else varies across NOT a detection; NOT a verdict on the two-rate assignment *No ‘PREDICTION.md‘ is written and the reason is stated rather than skipped: nothing is run, and the only test — the second tooth — is a criterion already fixed in ‘r6993‘’s pre-registration, so it was committed before the measurement it is applied to* Nothing RUN; no refit; nothing touching ‘prop:flat‘ or the clock family; no corpus edits *Run:* `python3 receipts/P15_CR_cosmology/P15_the_two_readings_are_different_quantities_and_the_projection_width_is`

**P15R199** `P15_the_two_projection_widths_differ_by_the_jacobian_alone_and_the_channel_that_opens_is_a_qua` [OK]

*‘sec:refit-bound‘ — **the projection’s two widths differ by the JACOBIAN and by nothing else**, which makes the difference a prediction of the two-rate assignment rather than a property of this instrument; the contrast channel it opens is at most a quarter of the excess and fails the filter on*

*curvature; and the filter itself is shown to be blind to sign* (node 66's ‘r6999‘ order, four items. (1) ***impose the conformal width at the size the arm HAS and not a fitted one***, pre-registering every outcome *including the one where it does nothing*; (2) ***apply your own filter to it first*** — *"if it fails the filter on paper, that is the answer and the run is not needed"*; (3) ***is the width difference FORCED or CHOSEN*** — *"if it is forced by the construction, this is a CR prediction nobody has stated... that would be worth more than the contrast result"*; (4) the sub-period estimator SAID and not built. Guards: **a statistic must carry its domain**; **say which quantity, always**; exclude by mechanism; **and pre-register the NULL**). *Computes:* **(3) THE WIDTH DIFFERENCE IS FORCED, AND BY ONE OBJECT.** The two sound horizons are one integrand on two rates, so $d(r_{s,\{}rm leaf})/d(r_{s,\{}rm stack}) = \{}mathrm{Jac}$ — **checked pointwise across the window to one part in a million on the arm and exactly on the control**, where $\{}mathrm{Jac}\{}equiv1$ by the rate identity **On the RULER clock the two arms are the same instrument**: the window's conformal width against its width in $r_{s,\{}rm stack}$ agrees between them to $0.3$ per cent on both core width definitions, *the visibility being laid down in $\{}eta$ by the same Thomson physics on both* **On the LEAF clock it is $+14.5$ per cent, and the whole of that is $\{}mathrm{Jac}$** — Jac share $1.1484$ against a leaf ratio $1.1446$, and the arm's own leaf-to-ruler width ratio equals $\{}langle\{}mathrm{Jac}\{}rangle$ to $0.4$ per cent ⇒ ***$\{}mathrm{Jac}=H_{\{}rm phys}/H_{\{}rm leaf}$ has no free coefficient and is identically $1$ on any one-rate cosmology, so the different ratio IS a prediction of the two-rate assignment*** **(2) AND THE FILTER IS APPLIED IN THE INSTRUMENT'S OWN KERNEL**, the order's operation done analytically: hold the source phase $\{}cos(k r_{s,\{}rm leaf})$ and stretch ONLY the Bessel argument by the measured $s=1.135$ **and this corrects its own first pass by a factor of fifteen AND in sign**: the plane-wave proxy $\{}lvert\{}int g e^{-ik\{}eta}\{}rvert$ assumes the Bessel argument advances at rate $k$, and $j_\{}ell(x)$ near its turning point $x\{}sim\{}ell$ — *where the whole window sits, $kD_M\{}sim\{}ell$ being what the projection IS* — oscillates at local rate $\{}sqrt{1-\{}ell^2/x^2}$, which vanishes there **the sign is NOT determined on paper**: mean-anchored $+0.0034\{}to+0.0196$, peak-anchored $-0.0034\{}to+0.0016$ and crossing zero, so $G$ is UNDEFINED there — *a stretch about the wrong point is a stretch plus a displacement, and a displacement moves the COMB* **verdict: curvature $+0.00016$ and $+0.00037$ against the target's $-0.00327$, so it ACCELERATES where the target decelerates, and at the top band it delivers $+1.96$ per cent against the $+7.62$ needed** **and the filter needed a THIRD tooth**: the superseded numbers run $-0.033\{}to-0.297$ with $G=8.93$ and curvature $-0.0014$, ***passing both pre-registered teeth while moving the contrast the wrong way in every band*** — $G$ is blind to the common sign, so **SIGN is now the first tooth** **(1) AND THE RUN CANNOT BE CONSTRUCTED**, which follows from (3): *"the arm's conformal width at the same leaf width"* is *"the arm's $\{}mathrm{Jac}$"*, and a control with $\{}mathrm{Jac}\{}neq1$ is not a control *Bound:* **NOT that the width does nothing** — it is a real channel of measured size, bounded at a quarter of the excess at the top band **NOT that it is the carrier**: it fails the committed curvature tooth under both anchorings **NOT a measurement of the contrast**: the (1) run is not constructible on a one-rate control and is reported so rather than substituted for **NOT that ‘cc66.51‘'s $7.3$ per cent was wrong** — it is the RMS reading of the same object, and RMS weights the window's long tails, which neither the projection nor the phase sweep responds to; the core measures agree with each other to better than a hundredth of a per cent at $14.5$ **NOT a claim about the comb or the peak positions**, which the pre-registration marks as the warning sign rather than the result NOT a detection; NOT a verdict on the two-rate assignment Nothing SOLVED; no refit; **no corpus edits** — the order's , and (4) is said and not built *Run:* `python3 receipts/P15_CR_cosmology/P15_the_two_projection_widths_differ_by_the_jacobian_alone_and_the_channe`

**P15R200** `P15_the_held_period_estimator_reaches_below_the_floor_and_the_tooth_that_would_decide_cannot_b` [OK]

*‘sec:refit-bound‘ — **the below-floor estimator is built and validated and it DOES reach a band nothing has reached**; the candidate passes every tooth that can be applied there; **and the tooth that would decide cannot be applied, because the target's own deceleration is carried entirely by the one band no estimator of the candidate can reach**; and ‘cc66.51‘'s curvature pass does not survive its own estimator's width* (node 66's ‘r7001‘ order, four items. (1) ***build the estimator to the specification this seat wrote, with its own validation FIRST*** — *"before anything it says below the floor is read — that ordering is yours and it is the order"* — and **measure the cost rather than note it**: how much does an error in the held period leak into the amplitude; (2)

***then the candidate, below the floor, sign first and growth after***, with the null pre-registered again; (3) ***and the width prediction in its own right*** — ***"is it observable... independently of the contrast question it was found while chasing?"***; (4) ***nothing else, the list is closed***. Guards: **sign first, growth after**; **a width is not a quantity until its definition is named**; ***a proxy that fails where its object lives is worse than no proxy***; and **pre-register the null**). *Computes:* **(1) AND THE FIRST THING THE ESTIMATOR FOUND IS THAT THE PERIOD IS NOT THE PERIOD.** The acoustic period in $q$ is $2$ **by construction** and is not $2$ in the fields: read off the drift of the recovered phase and iterated to self-consistency, the monopole runs at $1.9636$ and the dipole at $1.9909$ on the control — ***a two per cent error and a DIFFERENT one for the two fields***, so the held period is **measured** per field per arm, with its own $1.3$ per cent uncertainty taken from disjoint sub-ranges **VALIDATION FIRST, AS ORDERED**: the means agree with the window estimator taken at the width where it is sound ($1.4987$ against $1.4911$) — **and the window estimator does not survive its own width**, its ripple running $0.7 \{}to 11.6$ per cent as it narrows to ‘half‘ $=0.5$, ***the width ‘cc66.51‘ read the candidate at***, while the held estimator sits near $2.5$ per cent at every width and reaches $q=0.39$ against the window estimator’s hard floor of $2.00$ **AND THE LEAK IS MEASURED AND IT IS HALF A PER CENT** of the dependence being measured — *and the reason is a mechanism: the fit re-fits the PHASE in every window, so a wrong held period is absorbed there and costs a common amplitude factor, which cancels in a ratio of ratios* **(2) IT REACHES THE BAND AT $q=1.90$**, stable to $0.003$ across eight estimator settings, where nothing has reached below $q=2.0$ — **and the candidate’s departure is POSITIVE IN EVERY MEASURABLE BAND as the target’s is, running $+0.0048 \{}to +0.0144$ with $G=2.99$ against the target’s $1.33$: it passes SIGN and it passes GROWTH** **and band 1 at $q=1.20$ is still not measurable, now for a MECHANISM rather than a width**: spread $0.021$ against a value $0.0003$, because its window straddles the first acoustic excursion where there is no oscillation amplitude to estimate **AND THE CURVATURE TOOTH CANNOT BE APPLIED, WHICH IS THE PRE-REGISTERED FIRST NULL**: on the full range the target’s curvature is $-0.0033$, **dropping band 1 ALONE flips it to $+0.0003$ while dropping any other single band leaves it in $[-0.0053,\{},-0.0028]$** ⇒ ***the target’s deceleration is carried entirely by the one band no estimator of the candidate can reach, so the filter is out of teeth rather than the candidate out of chances*** **AND ‘cc66.51‘’s CURVATURE PASS DOES NOT SURVIVE**: on its own range and recipe the candidate’s curvature runs $-0.0093$ at ‘half‘ $=0.5$ to $+0.0012$ at ‘half‘ $=2$ — *its stability check certified the MEAN over a sub-range, and the mean IS stable across the widths on which the curvature is not* **(3) THE PREDICTION IS AN INDEPENDENT READING OF THE JACOBIAN AND IT IS NOT OBSERVABLE**: the comb reads the **cumulative** ratio $0.5665$ ($\{}pi D_M/r_sˆ{\{}rm leaf}=301.41$ against $l_A=301.80$, $\{}pi D_M/r_sˆ{\{}rm stack}=170.76$ not close) while the widths read the **window-local** $\{}langle\{}mathrm{Jac}\{}rangle=0.8744$ — ***a third apart, both $1$ on one rate, neither carrying a free coefficient*** — but the phase width differs between the arms by $0.84$ per cent and the conformal width by $13.5$ whose contrast effect ‘cc66.52‘ bounded at $2$, so **no measurement reaches the ratio** *Bound:* **NOT that the candidate is confirmed** — it passes every tooth that CAN be applied and the deciding one cannot be **NOT that the candidate is excluded**: nothing failed **NOT that band 1 is a window-width problem** — it is a property of the fields and no estimator reaches it **NOT that ‘cc66.51‘’s MEAN was wrong**: it is reproduced, and only the curvature read off it does not survive **NOT a new channel** — the order’s (4) closes the list, and the polarisation note is about what the instrument does not carry, not a proposal **NOT an argument for moving the prediction on observability**: the answer to that is NO and is given in the order’s own terms; the independence finding is named as a separate thing and the placement is the chat seat’s Nothing SOLVED; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_held_period_estimator_reaches_below_the_floor_and_the_tooth_that_`

**P15R201** `P15_the_lowest_band_is_closed_by_the_fields_and_the_excess_deceleration_is_carried_by_it_alone` [OK]

*‘sec:refit-bound‘ — **the lowest band is closed to any oscillation-amplitude estimator by the FIELDS, at an exact floor**; the excess’s deceleration is robust in sign and is **carried by that same band alone** — and **is not robust to the envelope’s definition**, which this seat reports against its own result; and the filter’s third condition is permanently unavailable **exactly for the class of candidate the physics says can carry the excess**

(node 66’s ‘r7003‘ order, three items. (1) ***can the lowest band be reached by any estimator, or is it closed by the fields?*** — *"is that a property of

the band's width, of its placement, or of the fields themselves... if the fields close it, then say so as a structural statement"*; (2) ***is the TARGET's own curvature robust enough to carry the claim the paper makes?*** — *"I would rather learn that from you than from a reader"*; (3) ***what can the filter still decide*** — *"is a two-condition filter with the third permanently unavailable simply not a filter?"*; (4) ***nothing else***. Guards: ***a stability check licenses the statistic it was run on and no other***; sign first, growth after; **a width — or a period, or a curvature — is not a quantity until its definition is named**; and ***pre-register the ways the measurement can fail to decide, not only its outcomes***). *Computes:* **(1) THE BAND IS CLOSED BY THE FIELDS AND THERE IS AN EXACT FLOOR.** The criterion is named before use and is a necessary condition from identifiability rather than a chosen threshold: *an oscillation amplitude at $q_0$ is identified only if the window holds a TURNING POINT on each side of $q_0$*, because over a span with no turning point the held-period pair is monotone and the baseline polynomial already spans it **The monopole's turning points are $0.996,\{},1.915,\{},2.912,\{}dots$ with NONE below the first** — the first excursion is one-sided — giving a floor $q=1.455$ against the dipole's $0.962$, so **the monopole binds** ⇒ ***the filter's lowest band, $q\{}in[0.85,1.55]$, lies $86$ per cent below that floor and every other band lies entirely above it*** **And it is not the window, shown three ways**: the same eight widths at identical conditioning spread $0.004$ at $q_0=1.90$ and $0.016$ at $1.20$ with a sign change (*width*); sweeping the left edge, the departure is negative for every window opening past the monopole's first turning point and positive for every one staying above it, *what lies below being the field's rise from its initial condition and not an acoustic oscillation* (*placement*); and **every centre below the floor spreads by more than every centre above it**, which the floor was not fitted to (*prediction*) ⇒ ***no estimator of an oscillation amplitude can reach inside the first excursion*** **(2) AND THE TARGET'S CURVATURE, IN THREE PARTS.** ****(a)** robust within the statistic as defined**: negative in all six variants moving the envelope width, band edges and sampling, $-0.00377$ to $-0.00291$, with band 1 stable at $+0.0201$ to $+0.0240$ ****(b)** and carried by band 1 alone**: with band 1 dropped the curvature runs $-0.00121$ to $+0.00031$, *sign not determined* ⇒ ***the deceleration is the lowest band being LOW, not the upper bands bending*** ****(c)** and NOT robust to the envelope's DEFINITION** — a running median, everything else unchanged, ***reverses it to $+0.00504$***; that is a different statistic (its envelope absorbs about half the top band's oscillation on both arms) **but it is not obviously the worse envelope**, so the honest reading is that the sign belongs to the statistic rather than to the excess **(3) AND THE THIRD CONDITION IS NOT UNAVAILABLE IN GENERAL, WHICH IS WORSE THAN IF IT WERE**: it needs a band-1 value, and band 1 is closed to *amplitude* estimators but not to a candidate COMPUTED from the kernel — 'cc66.52''s projection width had one and was excluded on curvature ⇒ ***the filter's full strength is available exactly for the class that cannot carry the excess, and its third condition is permanently unavailable exactly for the class that can***, since 'cc66.51' settled on physics that what fills a trough is an oscillation amplitude **As an exclusion device it has lost nothing** — each remaining condition is necessary and the exclusions already made were made on growth alone — **and as a confirmation device it never worked, not with three either**: all three are conditions on SHAPE, and the candidate matches sign and growth direction while being $5.3$ times smaller at the top band ⇒ *the row's remaining question is the coupling, which is quantitative and needs band 1 not at all* *Bound:* **NOT that band 1 is hard to measure** — it is closed, and the distinction is the result **NOT that the deceleration claim must come out**: it is supported on its own statistic **NOT that it is independent of that statistic** — the median envelope reverses it, and that exposure is reported rather than argued away **NOT that the candidate is confirmed or excluded**: it passes what can be applied, which is not a promotion, and a missing condition is not an acquittal **NOT a new channel** — the coupling question is named as the question, not proposed; the order's (4) closes the list NOT a detection; NOT a verdict on the two-rate assignment Nothing SOLVED; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_lowest_band_is_closed_by_the_fields_and_the_excess_deceleration_i`

**P15R202** `P15_the_coupling_is_measured_and_negative_and_the_contribution_is_undetermined_because_its_in`
[OK]

*'sec:refit-bound' — **the coupling between the Doppler amplitude and the band contrast is MEASURED, is negative at every band on both envelopes over every interval, and is fixed by the construction**; and the candidate's contribution is **still undetermined, because the two readings of what the coupling multiplies disagree in SIGN** — the third revision these two readings have decided an answer between them and the first in which they decide its sign* (node 66's 'r7005' order,

three items. (1) ***the size, with the same discipline the shape work had*** — *"what fraction of the excess does it carry... and say whether the shortfall is constant across the range or grows — a constant shortfall is a coupling, a growing one is a second shape mismatch"*; (2) ***and what would set it*** — *"is the coupling... something the instrument fixes, or something the construction fixes? If the construction fixes it, the size is a prediction and not a fit"*; (3) ***and one thing about the statistic*** — *"does the candidate's contribution reverse with [the median envelope], or does it survive both?"*; (4) ***nothing else***. Guards: ***a claim and its implied independence are two claims***; **name the criterion before using it, and prefer a necessary condition to a threshold**; a stability check licenses the statistic it was run on and no other; **pre-register the failure modes ahead of the outcomes**. And: *do not revisit the envelope question to settle it*). *Computes:* **THE COUPLING IS MEASURED, NOT MODELLED.** 'DPSRC' scales the Doppler source term, so the three banked control spectra at 'DPSRC' $=1,\{},0.8794,\{},0.6$ give $d\{}ln C/d\{}ln s$ per band as a finite difference — ***three points, so the linearity is CHECKED and not assumed***, and it steepens towards $s=1$ at all seven bands, so the local slope there is both the applicable one and the one measured over the shortest step Near $s=1$ on the arithmetic envelope it runs $-0.45$ to $-0.80$ with **no trend in $q$** ⇒ ***NEGATIVE AT EVERY BAND, ON BOTH ENVELOPES, OVER EVERY INTERVAL — forty-two measurements of the sign and not one positive*** — *more Doppler fills more trough and LOWERS the contrast, which is 'cc66.51''s own trough-filling physics read forwards* **(2) AND THE CONSTRUCTION FIXES IT, SHOWN BY WHAT DOES NOT CHANGE IT**: the control's own Doppler-to-monopole band-power share runs $4.30$ down to $0.65$, **a factor of $6.6$ across the bands, and the coupling's sign is the same at every one** ⇒ *a quantity whose sign is invariant across a factor of six in the very ratio it depends on is not set by a knob* — 'DPSRC' is the **probe**, not the setter, ***so the size is a prediction and not a fit*** *and the one thing not measured is stated: all three 'DPSRC' banks are the CONTROL, so the arm's coupling is assumed to be the control's; its share differs by about a fifth against the factor of six across which the sign does not move, so the assumption cannot reach the sign and is not asked to reach more* **(1) AND THE CONTRIBUTION IS NOT DETERMINED — NOT BECAUSE OF THE COUPLING BUT BECAUSE OF ITS INPUT.** The criterion was named before use: *the coupling multiplies a fractional change in the Doppler's oscillation amplitude relative to the monopole's* — and two measured quantities claim to be it and **disagree in sign**: the FIELD reading ('cc66.51''s held-period estimator at last scattering) has the arm **higher** by $+0.5$ to $+1.4$ per cent ⇒ contribution $-0.004$ to $-0.010$, **$-7$ to $-13$ per cent of the excess, the WRONG SIGN**; the PROJECTED reading ('cc66.50''s quantity, the square root of the banked 'w2dp/w2sw' share, **which is what 'DPSRC' actually scales**) has it **lower** by $10$ to $13$ per cent ⇒ contribution $+0.066$ to $+0.108$, **$+188$ down to $+94$ per cent, the RIGHT SIGN** ⇒ ***the coupling turns a disagreement about a quantity into a disagreement about whether the candidate helps at all — a factor of seven in magnitude even at their closest*** **And the order's second half: the fraction is NOT constant on either reading** — it grows in magnitude on the field reading and FALLS on the projected one ⇒ ***a SHAPE MISMATCH and not a coupling deficit, on either, and the fourth this sector has found***; the coupling being flat, the $q$-dependence belongs to the input **(3) AND THE ENVELOPE QUESTION HAS THE GOOD ANSWER**: the coupling's sign survives both envelopes at every band, so **the contribution does not reverse with the statistic and the coupling question does not inherit the envelope ambiguity** — though its magnitude does, the median envelope's coupling reaching $2.2\{}times$ the arithmetic one's at the top band, *so a quoted FRACTION inherits the ambiguity while the SIGN does not, said before any fraction is quoted* *Bound:* **NOT that the candidate is excluded** — one reading of its input has it carrying essentially all of the excess **NOT that it is the carrier** — the other has it working against the excess, and neither is chosen **NOT a resolution of the two readings**: 'cc66.51' chose between them once and this revision shows the choice decides the sign, so choosing again is exactly what is not done **NOT an envelope choice** — the order forbids it; both are carried and no conclusion rests on one **NOT a claim about the arm's coupling** — it is the control's, transferred, and only as far as the sign **NOT a new channel**: the bank that would settle the input is named, not proposed Nothing SOLVED; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_coupling_is_measured_and_negative_and_the_contribution_is_undeter`

**P15R203** `P15_the_bank_already_existed_and_the_doppler_carries_a_growing_minority_of_the_excess` [OK]

*'sec:refit-bound' — **the bank the row needed already existed**, so the ARM's coupling is now*

*measured rather than transferred; the Doppler carries **a growing MINORITY of the excess and is sign-indefinite at the bottom**; and **the four shape mismatches share a reason that is about the excess and not about the candidates**  (node 66's 'r7007' order, three items. (1) ***bank the oscillation amplitude of the projected contribution*** — *"say what the instrument has to emit for that... and if it cannot emit it without a change this seat should not make unilaterally, say that and stop"*; (2) ***then the contribution, with the fraction carrying both envelopes***, pre-registering every outcome ***"including the one where the right quantity lands between the two wrong ones and settles nothing"***; (3) ***and the fourth shape mismatch counted rather than passed over*** — *"four is a pattern rather than a run of bad luck... if they share a reason, that reason is the sector's real finding"*; (4) nothing else. Guards: ***the knob is the probe and not the setter***; ***do not choose between two readings of a quantity, name the quantity***; a claim and its implied independence are two claims; **pre-register the failure modes, including "settles nothing"**. And: *do not choose an envelope*, still). *Computes:* **(1) THE INSTRUMENT ALREADY EMITS IT AND THE BANK IS ALREADY ON DISK.** 'SRCDEC', added at 'r6915+cc66.41', writes the full bilinear decomposition — $C_\{}ell = \{}sum_{a\{}le b} w_{ab}\{}sum_k P\{},\{}Deltaˆa\{}Deltaˆb$, ***ten $\{}ell$-resolved terms per multipole summing to $C_\{}ell$ exactly*** — and 'r6915_pairs_{lcdm,cr}' carry them for both arms, to $1.2\{}times10ˆ{-15}$ ⇒ **no run, no instrument change, and the order's escape hatch is not needed** *And the coarse grid is checked rather than assumed: $238$ multipoles against $1900$ reads the contrast $1.9$ per cent low, and the deficit agrees between the arms to $3\{}times10ˆ{-4}$, so it CANCELS in the ratio — the only thing read from it* **AND THE BANK DISCHARGES A CAVEAT 'cc66.55' COULD ONLY STATE**: the $s$-scaling of every pair is exact ($sˆ2$ on 'dp*dp', $s$ on the three 'dp' crosses), so $dD_\{}ell/d\{}ln s$ is exact per multipole and **the coupling follows analytically FOR THE ARM as well as the control** — $-0.48$ to $-0.86$ on each, **equal between the arms to $4$ per cent** and agreeing with 'cc66.55''s three-point finite difference to $7$ **AND THE FIRST CONSTRUCTION IS REPORTED AS FAILED RATHER THAN DROPPED**: $\{}sqrt{\{}mathrm{osc}(\{}texttt{dp*dp})/\{}mathrm{osc}(\{}texttt{sw*sw})}$ is small on the arithmetic envelope and reaches $+134$ per cent under a median one, *because the oscillation of a weak smooth term about a median envelope is not well conditioned* ⇒ that definition inherits the envelope ambiguity through its **own** conditioning, so it is not used **WHAT IS USED NEEDS NO EQUIVALENT-$s$ STEP: A ONE-AT-A-TIME SWAP** — the arm's four Doppler-containing pairs replaced by the control's, envelope-scaled, on the arm's own $q$ grid, the share being $(C_{\{}rm arm}-C_{\{}rm swapped})/(C_{\{}rm arm}-C_{\{}rm control})$; *same shape of operation as 'SRCINJRS''s clock swap, which the instrument already sanctions* **(2) AND THE CONTRIBUTION IS A MINORITY SHARE THAT GROWS AND IS SIGN-INDEFINITE AT THE BOTTOM**: $-12$ to $+17$ per cent on the arithmetic envelope, $-20$ to $+48$ on the median one, negative at $q=1.90$ on both ⇒ ***the candidate is NOT the carrier*** **It lands between the two wrong readings, which said $-13$ and $+188$ per cent** — the outcome the order required be tabled first — ***and it does not settle nothing: it settles that the candidate is a real but minority contributor, which neither wrong reading said*** *And 'cc66.55''s split is confirmed rather than assumed: the coupling's SIGN is the same on both envelopes while the FRACTION moves by $2.7\{}times$* **(3) AND THE FOUR MISMATCHES DO SHARE A REASON, AND IT IS ABOUT THE EXCESS.** The excess has ONE dominant feature: ***band 1 is $0.33$ of the mean of the rest and sits $4.9$ scatters below it***, while above band 1 a constant already fits ($0.0088$ scatter on a mean of $0.0645$) and a line only halves the residual ⇒ ***the excess's $q$-structure is a STEP at the lowest band, not a trend — and all four candidates were judged on a trend across bands 2–7, where the excess is very nearly featureless***: a flat candidate matches the flat part and misses the step, a rising one matches the mild rise and misses the step, and there the two are barely distinguishable ⇒ **the sector has been spending its discriminating power where the excess has least structure, because the part that has the structure is the band 'cc66.54' showed it cannot measure a candidate in** *Bound:* **NOT that the excess is explained** — a minority share is not an explanation **NOT that the candidate is excluded**: it contributes measurably and its share grows **NOT a definite share at the bottom bands** — negative at $q=1.90$ on both envelopes, so it cannot be told from zero exactly where the excess's structure is **NOT an envelope choice**: both carried, the order's refusal held for the second revision **NOT a claim on the failed construction's arithmetic number** — it is reported as failed, with its reason, and is not quoted as the answer **NOT a new channel**: nothing proposed; (4) holds Nothing SOLVED, nothing RUN; **no corpus edits**  *Run:* `python3`

`receipts/P15_CR_cosmology/P15_the_bank_already_existed_and_the_doppler_carries_a_growing_minority_c`

**P15R204** `P15_the_step_is_the_excesss_own_and_it_sits_at_the_second_acoustic_peak_and_none_of_the_four_pr`
[OK]
*‘sec:refit-bound‘, ‘sec:scope‘ — **the step is the excess’s and not the estimator’s**, surviving three envelopes and one reading that needs none, with the edge hazard ruled out the right way round; **it sits at the SECOND ACOUSTIC PEAK to $0.9$ per cent** and is sharper than the window that found it; and **all four candidates can be scored on it and none of them produces it*** (node 66’s ‘r7009‘ order, three items, **sequenced**. (1) ***is the step the excess’s or the estimator’s — pre-register that before anything else*** — *”a step at the lowest band is exactly where an envelope artefact would sit... I would rather lose the step now than build two revisions on it”*; (2) ***then locate it as a scale*** — *”if the step’s location coincides with a scale the construction fixes, that is the carrier’s signature and it carries no free coefficient”*; (3) ***and re-score the four candidates on the step rather than on the trend***; (4) nothing else. Guards: ***a step at the edge band is where an edge artefact lives***; the knob is the probe and not the setter; do not choose between two readings, name the quantity; **report a failed construction as failed and date it**; pre-register the failure modes including the one that embarrasses the revision. And: *do not choose an envelope — add a third*). *Computes:* **THE SEQUENCING WAS HONOURED AND IS CHECKED**: the pre-registration says in terms that nothing in (1) had been computed when it was written, names the step’s definition before use (*band 1’s excess over the mean of bands 2–7*), and **tables FIRST the outcome that withdraws ‘cc66.56‘’s (3)** — which ‘r7009‘ had already landed in two sections **(1) THE STEP SURVIVES EVERY READING.** Three envelopes differing in **kind** — running arithmetic mean, running median, local quadratic (Savitzky–Golay) — and one that needs **no window at all**, the peak-to-trough depth per acoustic cycle: step ratios $0.333$, $0.391$, $0.633$, $-0.025$, ***never near one*** *And the envelope-free reading is the sharpest: band 1’s excess on it is $-0.0017$, indistinguishable from zero, against a bands 2–7 mean of $+0.0654$* **AND THE EDGE HAZARD IS RULED OUT THE RIGHT WAY ROUND, which is what makes it ruled out**: band 1’s envelope comes within $0.018$ in $q$ of truncating, but moving its lower edge UP from $0.85$ to $1.15$ makes it read $+0.0215 \{}to +0.0361$ — **higher, where a truncation artefact would have to make it read lower** *Its SIZE is reading-dependent and is quoted as a range — band 1 between none and two thirds of the rest — not as a number; no envelope is chosen* **(2) AND IT SITS AT THE SECOND ACOUSTIC PEAK.** The half-rise point in sliding windows at three widths is $q = 1.789,\{},1.788,\{},1.805$ — **$1.794 \{}pm 0.008$** — against the second peak at $1.7775$ (control) and $1.7760$ (arm): ***a match to $0.9$ per cent, where the nearest other scale the construction fixes — the monopole’s second turning point at $1.915$ — is seven per cent away and $q=2$ is twelve*** And the plateaux are $0.026$ below and $0.053$ above on all three widths: **a factor of $2.0$ reached within about a tenth of a comb period, sharper than the narrowest window that found it** ⇒ ***the excess is one size in the first acoustic cycle and twice that in every cycle above it*** **(3) AND ALL FOUR CAN BE SCORED ON THE STEP, AND NONE PRODUCES IT.** The window weighting and the term mix were measured **flat in $q$**, and a flat candidate’s step ratio is exactly one, by construction and without a new number; the **projection width** (kernel-class, reachable at band 1) is fitted by a straight line in $q$ to $2.6$ per cent of its own range, *so linear, no step*; and the **Doppler by the swap route that reaches band 1** runs $-0.0016$ to $+0.0014$ in sliding half-period windows with means $-0.0001$ below the step and $-0.0002$ above — **noise at the step’s resolution and no transition at all** ⇒ ***this is NOT the identifiability floor closing over the list — every one of the four WAS evaluable — it is the list EXHAUSTED against the right feature, for the first time*** *Bound:* **NOT that the step is a mechanism** — *a step at a scale is a signature to be explained*, which the pre-registration said would not be claimed **NOT a step of quoted size**: band 1 reads between none and two thirds of the rest depending on the reading, and the range is the claim **NOT an envelope choice** — a third was added, none chosen, for the third revision running **NOT that the second-peak coincidence is arm-specific**: the peaks sit at fixed $q$ on both arms, so what the step says is that the FIRST acoustic cycle differs **NOT the identifiability floor closing over the list** — all four were evaluable, which is a weaker and more useful finding than that would have been **NOT a new channel**: (4) holds and nothing is proposed Nothing SOLVED, nothing RUN; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_step_is_the_excesss_own_and_it_sits_at_the_second_acoustic_peak_an`

**P15R205** `P15_the_window_free_reading_cannot_resolve_the_first_cycle_the_flatness_was_a_trend_and_nine_t`

[OK]
*‘sec:refit-bound‘, ‘sec:scope‘ — **the window-free reading cannot resolve the first acoustic cycle, and that is the answer**: its band 1 rests on ONE half-cycle transition, so it is consistent with zero AND with no step at once; **the flatness that disposed of two candidates was measured as a TREND, so the list is NOT exhausted**; and **nine tenths of the step is common to BOTH arms**, the excess's step being the tenth that fails to cancel* (node 66's ‘r7011‘ order, three items. (1) ***is the excess ZERO in the first cycle, or merely smaller*** — *"that is not 'a third of the rest', that is ZERO"*, asked of the window-free reading *"with the uncertainty it actually has"*, and *"if a finer version of the same window-free quantity exists — half-cycles, or the depth of the first trough alone — use it"*; (2) ***and was the flatness of the window weighting and the term mix measured across the full range, or across bands 2–7*** — *"that reasoning is airtight if the flatness was measured including band 1, and circular if it was measured on the featureless stretch"*; (3) ***and where the step lives: in the arm, in the control, or only in the difference***; (4) nothing else. Guards: ***a conclusion drawn on the featureless stretch inherits its featurelessness***; rule out a hazard by the sign of its effect, not by a distance; quote a range when the quantity is reading-dependent; **a step at the edge band is where an edge artefact lives**; ***and pre-register the outcome that costs you most, first***). *Computes:* **THE SEQUENCING, CHECKED AGAINST THE FILE**: the pre-registration says in terms that nothing below it had been measured, names each finer window-free member and the definition of a step in an ABSOLUTE contrast *before use* (twice, gated on the count), and **tables the damaging outcome FIRST in every one of the three items** **(1) THE WINDOW-FREE FAMILY CANNOT ANSWER, AND THE REASON IS STRUCTURAL.** Its band 1 rests on **exactly ONE half-cycle transition**, because the first locatable extremum pair sits at $q = 1.363$ and the step is at $1.794$: so $-0.0008$ is $0.02\{}sigma$ from ZERO *and* $1.60\{}sigma$ from the bands 2–7 mean — ***consistent with both and informative about neither*** *And no finer member escapes it: the **extremal envelope** — successive maxima and successive minima interpolated separately, no running window anywhere — covers only $27\{}%$ of band 1, rests on the same single extremum, and interpolates ACROSS the step, so its $+0.0245$ is biased TOWARD the above-step value* ⇒ **SO THE ANSWER IS THE STATISTICS THAT DO RESOLVE BAND 1, AND THERE THE EXCESS IS $+0.0215$**: ***"small but non-zero" is the supported row and the step stays a step rather than becoming an onset*** **‘cc66.57‘'s "the sharpest of the four" is WITHDRAWN — it was the COARSEST**, by one datum against four hundred; *and its "none" endpoint goes with it as UNRESOLVED rather than as a reading, which **tightens** the size to between a third and two thirds — $0.333$, $0.391$, $0.633$ windowed and $0.338$ on the extremal envelope* **(2) THE RANGE WAS FULL, BAND 1 INCLUDED — ANSWERED FROM ‘cc66.49‘'s OWN FILE AND GATED ON IT — SO NOT CIRCULAR ON RANGE. BUT THE STATISTIC WAS A TREND.** $\{}lvert\{}text{slope}\{}times\{}langle q^2\{}rangle\{}rvert/\{}lvert\{}text{intercept}\{}rvert$ cannot exclude a step, which lives in the **residual**: the JOINT reads $0.004$ on that statistic while leaving $61\{}%$ of its own range unexplained by that line *Re-scored on band 1's departure from its own bands 2–7 trend, **across four trend bases with none chosen**: the WINDOW channel steps **UPWARD** $+45\{}%$ to $+56\{}%$ — the wrong sign — the TERM MIX **DOWNWARD** at $63\{}%$ of the excess's own $-57\{}%$ to $-60\{}%$, and the pair as actually composed at $42\{}%$* ⇒ ***‘cc66.57‘'s (3) IS WITHDRAWN IN PART AND THE LIST IS NOT EXHAUSTED***: *"a flat candidate's step ratio is exactly one, by construction and without a new number" was never earned* *And the projection width's disposal STANDS, which is what the basis table is for: it is linear in $q$, so it **flips sign across the bases** and a log basis would have manufactured a step for it — ‘cc66.57‘ scored that one on a residual from a straight line in $q$* **(3) AND BOTH ARMS STEP, NEARLY EQUALLY.** On every one of the four bases and on **both** the windowed and the window-free contrast: the control departs from its own bands 2–7 trend by $+30$ to $+44$ per cent at band 1 and the arm by $+27$ to $+39$, **the arm always the SHALLOWER, by some eleven per cent of the step** ⇒ ***SO OF A STEP WORTH TENS OF PER CENT IN EACH SPECTRUM, ABOUT NINE TENTHS IS COMMON TO THE TWO ARMS AND CANCELS IN THE RATIO; THE EXCESS'S STEP IS THE TENTH THAT DOES NOT*** *That is the pre-registration's "both arms" row — the step belongs to the acoustic physics the two share — and it explains, without a new number, why ‘cc66.57‘ found the location at fixed $q$ on BOTH arms* *Bound:* **NOT that the excess is ZERO below the step** — the order's stronger claim is *not established*, and *not excluded* either, because the

one reading that suggested it has no resolving power there **NOT that any candidate produces the step** — the term mix carries a minority of it in the right direction and the window channel pushes the wrong way, which is a PARTIAL account and not a producer **NOT that the flatness was measured circularly** — the range was full; it is the STATISTIC that was wrong, and that is the half this seat owns **NOT a mechanism, an envelope, a basis, or a new candidate** — (4) holds; four bases are reported and none chosen, for the fourth revision running on the same discipline ***AND THE CAUTION IS THIS SEAT'S, NOT THE ORDER'S***: a small residual of two large common features is exactly where a nearly-common systematic would sit, so the next reading of the step should be **differential by construction** rather than a difference of two large numbers Nothing SOLVED, nothing RUN; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_window_free_reading_cannot_resolve_the_first_cycle_the_flatness_w`

**P15R206** `P15_the_step_survives_a_differential_estimator_and_no_single_bilinear_term_carries_the_shared` [OK]

*'sec:refit-bound', 'sec:scope' — **the step SURVIVES an estimator in which the two large common steps are never formed**, so the tenth that failed to cancel is not the residual of two large numbers; **no single bilinear term carries the shared step and it is a property of the SUM**, though the two highest-leverage terms both carry a Doppler factor; and **the "42 per cent" is 42 per cent of the EXCESS's step**, not of the shared one* (node 66's 'r7015' order, three items, on this seat's own closing caution. (1) ***build the differential estimator: ratio first, contrast second*** — *"everything so far forms each arm's contrast and then compares — two large numbers each carrying a large step, with the object of interest in the difference"*, and the test named in advance: *"does the step survive at the size the present route gives it, and at what significance"*; (2) ***name the shared nine tenths, because the bank already can*** — *"does the shared step live in one term, or in a few"*, and *"a better-posed residual is worth more than a bigger one"*; (3) ***say which step the 42 per cent is 42 per cent of*** — *"if the pair carries two fifths of the EXCESS's step that is a large result; if it carries two fifths of the SHARED step, most of what it accounts for cancels"*; (4) nothing else. Guards: **a statistic that fits a trend cannot see a step — score on the residual**; "envelope-free" is about bias, not resolution; ***a small residual of two large common features is where a nearly-common systematic sits***; ***and when a feature is mostly common, say which part a share is a share of***; pre-register the outcome that costs you most, first — **and commit it separately**). *Computes:* **THE SEQUENCING**: 'PREDICTION.md' is its own commit ahead of the working script, tables FIRST the outcome that ***ends the row's present object***, and names **both** hazards the new estimator carries — its phase sensitivity and the conditioning of a division whose divisor passes through the first acoustic trough INSIDE the band under test — before either was looked at **(1) THE STEP SURVIVES THE DIFFERENTIAL ESTIMATOR.** Writing each arm as $\{}mathcal D = E(1+o)$, the present route forms $\{}operatorname{std}(o_a)$ and $\{}operatorname{std}(o_c)$ and divides; the ordered route forms $R = \{}mathcal D_a/\{}mathcal D_c$ and takes the contrast of **that**, so it reads $\{}operatorname{std}(o_a - o_c)$ and **the common oscillation divides out before any width is taken** Band 1's departure from its own bands 2–7 trend is $-0.374$ to $-0.387$ per common $\{}ell$ and $-0.382$ to $-0.383$ per common $q$: ***same sign on all four trend bases AND both abscissas, at $3.2$–$3.6\{}sigma$, about $65\{}%$ of the present route's fractional size*** ⇒ ***SO THE TENTH THAT FAILED TO CANCEL IS NOT THE RESIDUAL OF TWO LARGE NUMBERS — IT EXISTS IN AN ESTIMATOR IN WHICH THE TWO LARGE NUMBERS ARE NEVER FORMED*** *And the conditioning hazard does not fire — the control never falls below $0.67$ of its band median — while the **smoothed-divisor control** turns the departure POSITIVE, back toward the arms' own $+0.3$: **it is the differencing, not the division, that produces the negative step** **(2) NO SINGLE BILINEAR TERM CARRIES THE SHARED STEP.** The 238-multipole 'SRCDEC' total first reproduces the 1900-multipole step to better than $0.006$ in departure on **both** arms, which is what licenses a per-term reading; then removing each term in turn, ***the losses sum to $182\{}%$, far more than one*** ⇒ **the step is not additive and is a property of the SUM** — the pre-registered null *But the LEVERAGE is not flat, and that is what the null leaves standing: each term's share of the STEP against its share of the OSCILLATION puts the two highest-leverage terms BOTH on a Doppler factor — 'sw*dp' at $5.59\{}times$ and 'dp*dp' at $3.80\{}times$, against the best non-Doppler 'sw*isw' at $1.23\{}times$ — while the monopole 'sw*sw' carries $88\{}%$ of the oscillation at only $0.72\{}times$* ⇒ **a step-weighted reading of the bilinear decomposition is led by the Doppler where an amplitude-weighted one is led by the monopole** **(3) THE 42

PER CENT IS 42 PER CENT OF THE EXCESS'S STEP**, read off ‘cc66.58‘’s own file and gated on it: it divided by the departure of ‘MEAS - 1.0‘, and a channel response and the excess are both ratios to the SAME control ⇒ *the large reading* *And the two normalisations are ONE departure, $-0.0323$ in excess units — $-0.601$ of the excess's own extrapolated size and $-0.100$ of the control's contrast departure — **provable rather than asserted, because the arms' own log-departures differ by that same number to $4.6$ per cent** *Bound:* **NOT that the differential estimator measures an AMPLITUDE difference** — $\{}operatorname{std}(o_a - o_c)$ is sensitive to a **phase** difference too, named in the pre-registration before the measurement: a null on it would have been strong, a signal on it does not by itself say amplitude, and its absolute size carries no expectation **NOT that the Doppler PRODUCES the shared step** — ***leverage is not authorship***, and the non-additivity is the reason it is not **NOT that every Doppler term leads**: ‘dp*isw‘ has NEGATIVE leverage at $-0.91\{}times$, and ‘isw*isw‘ at $-2.07\{}times$ sits lower still, so the ordering is not Doppler-versus-not **NOT a naming of the shared step** — (2) is a null on "one term" and a naming only on "which sector" **NOT an envelope, a basis, an abscissa, a candidate or a mechanism** — (4) holds, and **two abscissas are reported with neither chosen**, the same discipline extended to the degree of freedom this estimator introduces Nothing SOLVED, nothing RUN; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_step_survives_a_differential_estimator_and_no_single_bilinear_ter`

**P15R207** `P15_the_surviving_step_is_amplitude_and_not_phase_and_the_candidate_shares_do_not_survive_the_` [OK]

*‘sec:refit-bound‘, ‘sec:scope‘ — **the surviving step is an AMPLITUDE feature and not a PHASE one**, so the first acoustic cycle is WEAKENED rather than displaced and the row is not reframed; **the candidate shares do not survive the change of denominator**, the window channel reversing sign and the realised pair falling to zero; and **the step does not compose the way the contrast does** — the two knobs nearly cancel where they multiply* (node 66's ‘r7017‘ order, three items, on this seat's own pre-registered caveat. (1) ***amplitude or phase — decompose the surviving step*** — *"the surviving step might not be a contrast feature at all; it might be a phase feature of the first acoustic cycle, and those are different objects with different physics behind them"*, with the reframing outcome to be tabled first and *"an inseparability is a finding here, not a gap"*; (2) ***re-score the candidates on the differential estimator*** — *"every share in the row is measured on the ratio-of-contrasts route, where nine tenths of the feature enters and cancels"*; (3) ***does the step compose the way the contrast does*** — *"if it composes differently, that is a new property of the step and it is worth more than the share"*; (4) nothing else. Guards: ***a differential estimator is a different quantity, not a cleaner version of the same one***; **a share needs its denominator named in the same breath**; leverage is not authorship and a matching size is not authorship either; a statistic that fits a trend cannot see a step; ***and pre-register the outcome that reframes the row, first***). *Computes:* **THE SEQUENCING**: ‘PREDICTION.md‘ is its own commit ahead of the working script, tables **phase** — the reframing outcome — first, carries an explicit **inseparable** row, puts a licence gate before the reading, and **names the amplitude/phase trade-off hazard before it was met** **(1) THE DECOMPOSITION IS AN IDENTITY, NOT A FIT.** With $o_c = A_c\{}cos\{}psi$ and $o_a = A_a\{}cos(\{}psi+\{}Delta)$, $\{}operatorname{std}(o_a-o_c) = \{}tfrac{A_c}{\{}sqrt2}\{}sqrt{r^2 - 2r\{}cos\{}Delta + 1}$, amplitude-only $\{}tfrac{A_c}{\{}sqrt2}\{}lvert r-1\{}rvert$ and phase-only $\{}tfrac{A_c}{\{}sqrt2}2\{}lvert\{}sin(\{}Delta/2)\{}rvert$ — **reproducing the difference's own measured comb amplitude to $0.00\{}%$ pointwise**, so the gate passes exactly ⇒ ***AND THE ANSWER IS AMPLITUDE***: the amplitude-only term keeps its sign at every window half-width and tracks the full statistic ($-0.227$ to $-0.260$ against $-0.204$ to $-0.236$), supplying $0.958$ of it at band 1 against phase's $0.284$ ***SO THE STEP IS A WEAKENING OF THE FIRST ACOUSTIC CYCLE, NOT A DISPLACEMENT OF IT, AND THE ROW IS NOT REFRAMED*** *But the phase channel is **unresolved, not merely small**: its departure runs $+3.54$, $+0.66$, $-0.46$ across three half-widths and **changes sign** where the amplitude term's does not — the pre-registered trade-off firing. The relative phase is $0.0068$ rad, $0.39$ degrees, against a relative amplitude of $0.0554$* ⇒ **amplitude on the sign, inseparable on the share**, and what would separate them is named: a phase read against the comb itself over a longer lever arm in $q$ **AND A CORRECTION TO ‘cc66.59‘ THE DECOMPOSITION FORCED**: a band spans $0.70$ in $q$ against a comb period of $1.00$, so **a raw band ‘std‘ samples an incomplete cycle and is phase-dependent by construction** — the step reads $-0.45$ on that route and $-0.22$ on the phase-insensitive

one, **surviving on both, neither chosen** **(2) AND THE SHARES DO NOT SURVIVE THE CHANGE OF DENOMINATOR.** On the differential estimator: the **window channel steps $+1.05$ to $+1.32$, OPPOSITE in sign to the arm's**; the **term mix is no longer a step** where it carried $63\{}%$; and the **JOINT changes sign across the bases** and measures zero to within its own scatter where it carried $42\{}%$ ⇒ ***A CHANNEL THAT ACCOUNTS FOR TWO FIFTHS OF A MOSTLY-SHARED QUANTITY ACCOUNTS FOR NOTHING OF THE PART THAT IS THIS COSMOLOGY'S*** **(3) AND THE STEP DOES NOT COMPOSE THE WAY THE CONTRAST DOES.** Product, sum and quadrature predict $+0.84$, $+1.03$, $+1.19$ where the realised pair measures $-0.014$ — **six to nine of the joint's own scatters away, none fitting** — against 'cc66.49''s contrast composing multiplicatively, product 7 of 7 ⇒ *the two knobs very nearly **cancel** on the step where they **multiply** on the contrast* *Bound:* **NOT that the phase channel is measured to be small** — it is **unresolved**: the trade-off fires, its sign moves with the window, and the honest reading is amplitude-on-the-sign with the share inseparable **NOT that the term mix and the pair are excluded** — they are **not resolved** on the differential estimator, which is a weaker statement than exclusion and is the one the scatter supports **NOT a mechanism** — *"the first acoustic cycle is weaker in this cosmology" is a signature, not a cause*, and (4) holds **NOT a chosen aggregation**: the raw band 'std' and the held-period amplitude are **both** carried, as the four bases and two abscissas are **NOT a percentage share on (3)** — the joint's departure is consistent with zero, so residuals are in departure units against its own scatter; ***a percentage of a near-zero measurement would be meaningless, which is this line's own guard about denominators turned on itself*** Nothing SOLVED, nothing RUN; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_surviving_step_is_amplitude_and_not_phase_and_the_candidate_share`

**P15R208** `P15_no_statistic_this_construction_can_build_resolves_the_two_channels_and_the_phase_step_dies` [OK]

*'sec:refit-bound', 'sec:scope' — **no statistic this construction can build resolves the two channels against the step**: the minimum resolvable share is $0.59$–$0.75$ on every route banked while the channels carry $0.35$ and $0.03$; **the phase step dies on a matched-width control**, so the amplitude verdict is complete rather than sign-only; and **the composition cancellation is aggregation-dependent** rather than exact or approximate* (node 66's 'r7019' order, three items. (1) ***resolve or exclude the two channels on the differential estimator*** — *"not resolved is right and it is not a terminal state: it means the measurement has not yet the power to decide, and power is a thing that can be measured and sometimes improved"*, with *"if the answer is that nothing available reduces it enough, THAT is 'PO-56''s terminal exit arriving — a demonstration and not a shrug"*; (2) ***the phase, against the comb, by your own prescription*** — *"it also tests (1)'s premise from the other side"*; (3) ***is the cancellation exact or approximate*** — *"an exact cancellation of two large opposite departures is either a coincidence at the sizes their profiles happen to solve, or a CONSTRAINT"*; (4) nothing else. Guards: an identity is not a fit; **a band narrower than the period it measures makes its own statistic phase-dependent**; a percentage of a near-zero measurement is not a share; ***"not resolved" is not "excluded" — and it is also not a finish***; and a share measured against a mostly-shared quantity is not a share of the part that is ours). *Computes:* **THE SEQUENCING**: 'PREDICTION.md' is its own commit ahead of the working script, **defines the resolving ratio before computing it**, names four routes to a smaller scatter and commits to testing each, **names in advance the one it expected most from**, tables the terminal outcome, and names (1)'s own abuse hazard before meeting it **(1) NOTHING BANKED BRINGS THE FLOOR BELOW WHAT THE CHANNELS CARRY.** The minimum resolvable share $f_{\{}min} = 2\{}sigma/\{}lvert d_{\{}rm arm}\{}rvert$ runs $0.59$ to $0.75$ across **every** route this construction can build — both aggregations, both abscissas, a longer lever arm to $q=6.45$, and finer bands ⇒ ***THE TERM MIX CARRIES $0.35$ OF THE STEP AND THE REALISED PAIR $0.03$, AGAINST A FLOOR OF $0.59$: THEY CANNOT BE RESOLVED BY ANY STATISTIC THIS CONSTRUCTION CAN BUILD*** *And the reason is what the pre-registration did **not** expect: the held-period aggregation lowers $\{}sigma$ by $2.12\{}times$ **exactly as predicted** — and lowers the SIGNAL by $2.06\{}times$ at the same time, so $f_{\{}min}$ moves only $0.65 \{}to 0.63$. **The candidate named in advance as the one expected to help most does not help*** *And the abuse hazard fires before any power claim can be made: **the term mix and the realised pair CHANGE SIGN between the aggregations**, so their departures are not established at all and more power would not decide them — only the window channel keeps its sign on all eight readings* **(2) THE PHASE IS RESOLVED WHERE IT IS

NOT NEEDED AND NOT WHERE IT IS.** The long-stretch projection works: over the upper range $\{}Delta = +0.007368 \{}pm 0.000974$ rad ($+0.42ˆ\{}circ$, a $7.6\{}sigma$ determination), the whole range as one stretch agreeing at $+0.007541$. Band 1 reads $-0.002440$, **$5.03$ scatters out on the full-period yardstick** *But that yardstick is wrong and the comparison is not licensed without the control: band 1 is $0.70$ of a period, and a projection over a **non-integer** number of periods leaks the baseline, so a narrow stretch is both noisier AND biased. On six stretches of the SAME width the scatter is $3.7\{}times$ larger* ⇒ ***BAND 1 SITS $1.52$ SCATTERS OUT, NOT $5.03$: THE PHASE STEP IS NOT RESOLVED AND ‘cc66.60‘’s AMPLITUDE VERDICT STANDS AS THE COMPLETE ONE*** **(3) THE CANCELLATION IS NEITHER EXACT NOR APPROXIMATE — IT IS AGGREGATION-DEPENDENT.** The joint vanishes within its own scatter on all four bases of the raw reading ($0.24$–$0.54\{}sigma$) and on none of the phase-insensitive one ($4.16$–$4.98\{}sigma$) ⇒ *(3) **inherits (1)’s answer rather than choosing a word** — the third possibility the pre-registration named — and constraint-versus-coincidence cannot be settled here* *Bound:* **NOT that the two channels are EXCLUDED** — they are **unresolvable**, which is a statement about what this construction can measure and not about what the channels do **NOT that the phase difference is zero** — it is $+0.42ˆ\{}circ$ and resolved at $7.6\{}sigma$ in the upper range; what is unresolved is whether it **steps** at band 1 **NOT a composition rule** — (3) returns no word at all, by design, because the two aggregations disagree **NOT an aggregation, basis or abscissa choice**: every reading is carried and none chosen, and ***a resolution claim is required to hold on the SAME reading that measured the departure***, which is the hazard this file pre-registered against itself **NOT a mechanism**, and (4) holds Nothing SOLVED, nothing RUN; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_no_statistic_this_construction_can_build_resolves_the_two_channels_ar`

**P15R209** `P15_the_likelihood_sees_the_step_and_separates_the_two_channels_so_the_demonstration_does_not_` [OK]

*‘sec:refit-bound‘, ‘sec:scope‘ — **the likelihood sees the step as a localised contribution to its excess and separates the two channels at band 1 at twenty-five sigma**, so ‘cc66.61‘’s demonstration covers the STATISTIC family and not the instrument the paper runs beside it, and **‘PO-56‘’s amended clause is NOT met**; and **eight of the eleven instruments this row has used band, three do not**

(node 66’s ‘r7021‘ order, two items, after amending ‘PO-56‘’s terminal clause to this seat’s narrower sentence. (1) ***does the likelihood see the step*** — *”your floor is proved over every STATISTIC this construction can build, but every one of them bands, and the likelihood does not band”*, with *”a demonstration that covers one instrument family and is silent about another the row already uses is not proved over its whole reach, which is your own standard for the word unresolvable”*, and *”if the likelihood turns out to be a banded statistic in disguise, that is the answer and it completes the demonstration rather than failing to”*; (2) ***say whether anything else this row has ever used does not band*** — *”I do not want to strike on a demonstration with a third instrument outside it”*; (3) nothing else, **and do not soften the floor result**. Guards: a narrow-window measurement needs a matched-width control; a reduction in scatter is only a gain if the signal does not fall with it; ***”unresolvable” is a claim about the instrument and must be proved over its whole reach***; **narrow a terminal clause before taking it**; and a departure that changes sign between two legitimate aggregations is not a quantity). *Computes:* **THE SEQUENCING, AND THE HARDEST PART OF IT**: ‘PREDICTION.md‘ is its own commit ahead of the working script and **says out loud which facts had already been looked at** — the likelihood’s bin structure, a property of the instrument fixed on disk regardless of any spectrum — **declaring that branch SETTLED rather than tabling it as if open** **(1)**(a)** THE LIKELIHOOD IS NOT A BANDED STATISTIC IN DISGUISE.** ‘plik_lite‘ TT bins at a median $0.0298$ in $q$ — **one thirty-fourth of a comb period** — with $24$ bins inside band 1, and its bin-to-bin correlation is a flat $\{}approx 0.15$ floor rather than a coupling that grows over a period. *It bins; it does not BAND* **(1)**(b)** AND THE STEP IS A LOCALISED CONTRIBUTION TO ITS EXCESS.** Band 1’s bins carry $36.4\{}sigma$ of arm-versus-control separation on their own where its own bands 2–7 trend predicts $53$–$71$: a departure of $-0.311$ to $-0.489$ on the four bases at $2.5$–$3.2\{}sigma$ of its own residual — ***the same sign as every other instrument in this row, and the first per-band number in it carrying the INSTRUMENT’S own noise rather than an empirical scatter across a noiseless theory spectrum*** **(1)**(c)** AND IT SEPARATES THE TWO CHANNELS AT BAND 1 AT $25.1\{}sigma$** — the thing no statistic in this row can do — with a smallest detectable share of the band-1 difference of $0.055$ ⇒ ***SO THE DEMONSTRATION COVERS THE STATISTIC FAMILY AND

NOT THE INSTRUMENT THE PAPER RUNS BESIDE IT. ‘PO-56‘’s AMENDED CLAUSE IS NOT MET AND THE ROW HAS A MEASUREMENT TO MAKE RATHER THAN AN EXIT TO TAKE*** *And the two floors are in DIFFERENT UNITS, which the pre-registration required be said before the numbers were in hand: $0.055$ is a share of the band-1 DIFFERENCE under real instrument noise and ‘cc66.61‘’s $0.59$ is a share of the STEP under an empirical scatter of a NOISELESS spectrum.* **Neither bounds the other** **(2) THE SCOPE STATEMENT THE CLAUSE NEEDS.** Of **eleven** instruments this row has made a claim on, **eight BAND** at $0.70$ of a comb period; **three do not** — the anchored peak/trough locator (*which makes no amplitude claim at band 1: it locates peaks*), the likelihood, and the refit $\{}chi^2$ that uses its bins ⇒ ***so the likelihood and its refit are the ONLY instruments that both avoid the band-width limit AND make an amplitude claim at band 1***, which is why (1) was the right question and why no third instrument sits outside the demonstration *Bound:* **NOT that the channels are shown to produce the step** — the likelihood can SEPARATE them at band 1, which is a statement about resolving power and not about authorship **NOT that ‘cc66.61‘’s floor is wrong or softened** — it stands exactly as measured over the statistic family; what is added is that the family is not the whole reach **NOT that the two floors are one number**: $0.055$ and $0.59$ are shares of different quantities under different kinds of uncertainty, and ***the only legitimate comparison is whether each instrument can or cannot separate the channels at band 1*** — the statistic family cannot, the likelihood can **NOT a mechanism, an envelope, a basis, an abscissa or an aggregation choice** — (3) holds Nothing SOLVED, nothing RUN; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_likelihood_sees_the_step_and_separates_the_two_channels_so_the_de`

**P15R210** `P15_the_band_one_departure_is_not_in_the_likelihoods_excess_and_all_three_channels_match_it_in` [OK]

*‘sec:refit-bound‘, ‘sec:scope‘ — **band 1 carries under one per cent of the likelihood’s excess and its contribution changes SIGN under shape absorption**, so the departure is neither a feature of the residual nor the shape rejection read locally; **‘cc66.62‘ called a separating-power departure a contribution to the excess and that label is corrected here**; and in the metric the step does live in, **all three channels depart the arm’s way**

(node 66’s ‘r7023‘ order, this seat’s own redirect taken up, with a distinction placed in front of it. ***”the step is a two-to-three sigma feature of the likelihood’s residual and the twenty-five is the power available to decide between candidates for it; those are not the same number and nothing in the row should ever quote the second where it means the first”*** (1) ***score each channel against the step in the likelihood’s own metric*** — *”the same question the banded family could not carry, asked where it can be”*, with *”the joint sits at 12.82 where the term mix alone sits at 18.64, so the two may partly cancel: measure it, do not read it off”*; (2) ***say whether the lowest band’s departure is a distinct feature of the residual or part of the shape rejection*** — *”this decides what (1)’s answer would mean, and I would rather know which before a share is quoted than after”*; (3) nothing else, **and ‘cc66.61‘’s floor is not revisited**. Guards: an instrument that bins is not necessarily one that bands; **say which facts were in hand before the pre-registration was written**; a floor under noiseless structure and one under instrument noise are not one number; ***and separating power is not significance***). *Computes:* **THE SEQUENCING**: ‘PREDICTION.md‘ is its own commit ahead of the working script, **adopts the order’s distinction as its own vocabulary** — a table fixing which term means which number and a commitment that no quantity divides one by the other — and **states that (2) is asked FIRST though numbered second**, so the file’s structure carries the order’s point rather than a sentence **(2) RUN FIRST, AND THE ANSWER IS NEITHER OF THE TWO OUTCOMES TABLED.** Band 1 contributes $2.6$ of the $287$ the seven bands sum to — **under one per cent** — against a total excess of $322$ over all covered bins; absorbing smooth global shape at $n = 1,2,3,4$ free coefficients in $\{}ln\{}ell$ it reads $+2.6$, $-0.5$, $+0.3$, $+6.6$, ***changing sign***, where bands 2–7 hold to a few per cent ⇒ ***THE DEPARTURE IS NOT A DISTINCT FEATURE OF THE RESIDUAL AND IT IS NOT THE SHAPE REJECTION READ LOCALLY — THE SHAPE REJECTION IS BARELY PRESENT THERE AT ALL***, living at bands 4–7 which carry $86\{}%$ of the per-band excess **AND THAT FORCES A CORRECTION TO ‘cc66.62‘, WHICH IS THIS SEAT’S.** Its (1)**(b)** was headed *”the step is a localised contribution to its excess”* and then reported band 1’s **separating power**. ***The number was right and the label was wrong*** ⇒ *the likelihood DOES see a band-1 departure in its power to tell models apart; it does NOT carry a band-1 excess — different sentences, run together one revision after the distinction was drawn and one before it bit* **(1) AND IN THE METRIC THE STEP ACTUALLY LIVES IN, ALL THREE CHANNELS DEPART THE ARM’S

WAY.** Against the arm's $-0.311$ to $-0.489$: the **window** channel $-0.448$ to $-0.584$, *the same sign and larger*, share $1.31$; the **term mix** $-0.244$ to $-0.459$, share $0.86$; the **JOINT** $-0.100$ to $-0.343$, share $0.51$ — each with its own bands 2–7 significance, every share a ratio of two significances ⇒ ***THE FIRST TIME THIS ROW HAS HAD CANDIDATES THAT DEPART THE SAME WAY AS THE ARM ON AN INSTRUMENT THAT CAN CARRY THE QUESTION*** *And the window channel, which went the WRONG way on the differential estimator, goes the right way here and over-delivers — **a different verdict on a different instrument, reported as that rather than smoothed*** **AND THE CANCELLATION IS MEASURED**: the realised pair delivers $34\{}%$ of the closest of product, sum and quadrature, $2.4$ of its own residual away — *the two knobs partly cancel here as they did on the differential estimator* *Bound:* ***NOT that a candidate produces the step in the sense 'PO-56''s strong clause needs*** — (1)'s shares are shares of a **separating-power** departure, and (2) shows that feature carries one per cent of the likelihood's excess: **a real match in a real metric, and the wrong metric for the clause** **NOT that 'cc66.62' measured the wrong thing** — its number stands; its LABEL is corrected **NOT that the likelihood fails to reject this arm** — it rejects it decisively, at bands 4–7 **NOT that the shape absorption shows the departure is spurious** — more freedom always shrinks a residual, so what is read is the RATE against the other bands, and the same absorption is applied to both arms **NOT a new candidate, a mechanism, or any revisiting of 'cc66.61''s floor** — (3) holds Nothing SOLVED, nothing RUN; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_band_one_departure_is_not_in_the_likelihoods_excess_and_a`

**P15R211** `P15_the_term_mix_is_the_first_channel_to_produce_cost_where_the_arm_is_rejected_and_the_excess`

[]

*'sec:refit-bound', 'sec:scope' — **the excess at bands 4–7 is featureless to every smooth shape and yet modulated at the acoustic period**, so 'cc66.58''s *featureless* was a property of the BANDING; **the term mix is the first channel this row has produced that costs something where the likelihood actually rejects the arm**, though it overshoots by a third and misplaces nearly half of it; and **the cancellation is ABSENT here — the pair adds**  (node 66's 'r7025' order, which took NEITHER of the two readings this seat offered and moved 'PO-56''s clause from the structure to the COST. ⇒ ***"the comparison's discriminating structure and the data's rejection live in different bands: the step is the excess's most prominent feature and very nearly its cheapest"*** (1) ***score the channels at bands 4–7, in the likelihood's metric, ON THE EXCESS*** — *"six sevenths of the cost is there and no candidate has ever been scored against it"*, with the preregistration to name both the outcome that ends the row and the one that closes it; (2) ***say whether the excess at 4–7 has any structure at all in the likelihood's metric*** — *"the likelihood bins thirty-four times finer, so featureless may be a property of the banding rather than of the excess"*; (3) nothing else, **and 'cc66.61''s floor and the band-1 results are finished work**. Guards: a label is a claim; run first what decides what another item means; ***separating power is not significance, and neither is an excess***; **and where a feature is most visible is not where it costs most**). *Computes:* **THE INSTRUMENT IS CORRECTED FIRST, AND THAT IS A CORRECTION TO 'cc66.63''s ESTIMATOR AND NOT TO ITS RESULT.** 'cc66.63' inverted each band's covariance block alone and its contributions summed to $287$ against $322$; since $r_a = r_c - d$, the excess is $d^{T}Fd - 2d^{T}Fr_c$, whose per-bin terms sum to it ***identically*** — reproduced to $1.7\{}times10^{-13}$ ⇒ and the split is not bookkeeping: the first term is the **separating power** and the second is the only part that knows **where the data sits**, so *'cc66.63''s finding is that algebra* bands 4–7 carry $87\{}%$ on the exact decomposition against $86\{}%$ on the old: ***the location HOLDS*** **(2) FEATURELESS AND COMBED AT ONCE, WHICH IS NOT A CONTRADICTION.** Across $82$ bins a constant explains $0.0\{}%$ of the scatter, a trend in $\{}ln q$ $1.6\{}%$, a quadratic $2.8\{}%$ — *'cc66.58''s verdict survives every smooth shape at the finer scale* **but the acoustic-comb projection gives $7.30$ against a null of $2.53$ from $110$ wrong periods, of which NOT ONE returns more, and the scan's own maximum sits at period $1.01$** ⇒ ***SO FEATURELESS WAS A PROPERTY OF THE BANDING*** — *seven numbers cannot see a modulation at the period they are $0.70$ wide against* **and the control that makes it a result**: $d$ is itself a comb, so the modulation is split — $6.24$ in $-2d^{T}Fr_c$ against $1.08$ in $d^{T}Fd$, ***five sixths from the term that knows where the data sits, so it is a fact about the DATA*** *and the prediction that the quadratic term would enter at period $1/2$ did NOT hold cleanly — $0.57$ there against $1.08$ at $1.00$ — reported as it measured rather than quietly dropped* **(1) THE SHARES ARE LARGE, WHICH IS WHY THEY GET A CONTROL BEFORE A HEAD-

LINE.** At bands 4–7 the **window** scores $0.14$, the **term mix** $1.36$, the **JOINT** $1.53$, stable across $n=1,2,3,4$ *but ANY spectrum that is not the control costs something, so a share above one may mean only also rejected and not rejected the same way* ⇒ ***so the per-bin PATTERN is compared, not the total*** the **window is ANTI-correlated with the arm at $r = -0.70$** — *its cost is not the arm's, it is a DIFFERENT cost* the **term mix matches at $r = +0.70$, regression $0.74$, $45\{}%$ left over** the **JOINT matches WORSE than the term mix alone, $r = +0.46$** — *adding the window raises the total and DEGRADES the pattern* ⇒ ***THE TERM MIX IS THE FIRST CANDIDATE THIS ROW HAS PRODUCED THAT COSTS SOMETHING WHERE THE LIKELIHOOD ACTUALLY REJECTS THE ARM*** — $1.41$, $1.07$, $0.76$ across bands 4, 5 and 7 **AND THE CANCELLATION IS ABSENT, WHICH GOES AGAINST THE STORY THIS ROW WAS BUILDING**: twice elsewhere the pair delivered a fraction of its singles and it was called *beginning to look like a property*; on the excess the singles sum to $1.4965$ and the realised pair gives $1.5305$ — ***$102\{}%$: the knobs ADD, and cancellation is a property of the METRIC, not of the pair*** *Bound:* ***NOT that the amended clause is MET*** — a channel producing $136\{}%$ of the cost with $70\{}%$ pattern correlation is not *"no quantity"* ***NOT that the carrier is IDENTIFIED*** — a channel misplacing $45\{}%$ of its cost is not *"the carrier identified"*; **the clause is NEITHER met NOR discharged and it is not rounded either way** **NOT that ‘cc66.58‘ was wrong** — its verdict holds against every smooth shape at $82$ bins; what it could not see is a modulation finer than its own bands **NOT that band 6's shares mean anything alone** — the arm is cheapest there ($14.6$ against $80$, $80$, $123$), so that denominator is small and is never quoted alone **NOT that the window channel is a small contributor** — it is an ANTI-correlated one, which is a different statement **NOT a new candidate, a mechanism, or any revisiting of ‘cc66.61‘'s floor or the band-1 results** — (3) holds Nothing SOLVED, nothing RUN; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_term_mix_is_the_first_channel_to_produce_cost_where_the_arm_is_rej`

**P15R212** `P15_the_term_mix_does_not_carry_the_modulation_and_it_survives_in_the_part_no_channel_explains`

[]

*‘sec:refit-bound‘, ‘sec:scope‘ — **the term mix's own cost is NOT combed and the modulation survives in the part it does not explain**, which is ‘PO-56‘'s **terminating** row; **the one reading that says otherwise rests on an amplitude that is arithmetic**, gated as such before the run; and **the window is combed in ANTIPHASE on the excess while its spectrum-level modulation is IN PHASE**, so the magnitude composes and the sign does not* (node 66's ‘r7029‘ order — *"what joins your two findings is not a sentence, it is one measurement, and it is cheap"*. ***The gap, which this seat had not closed:*** *"the correlation is measured on per-band shares and the modulation on per-bin structure; a channel can carry four thirds of the cost in exactly the right four bands and be perfectly smooth inside every one of them"* (2) ***three comb projections against the same 110-period null*** — (1) the term mix's own cost, (2) the regression's fitted part, (3) the residual it misplaces, **each split by the two terms of the identity, because the self-similarity artefact has to be killed again for the channel**; (3) ***the window's own comb projection and the sign of its phase against the arm's***; (4) nothing else. Guards: no mechanism; no new candidate; ‘cc66.61‘'s floor and the band-1 results are finished work; ***and the added guard is this seat's own from ‘cc66.64‘ — A TOTAL IS NOT A MATCH, generalised: a projection amplitude is a total over bins the way the joint was a total over channels***). *Computes:* **THE PRE-REGISTRATION GATED (2)'s AMPLITUDE AS ARITHMETIC BEFORE THE RUN, AND THAT IS WHY THE REVISION HAS AN ANSWER AT ALL.** *A one-coefficient regression's fitted part is a scalar multiple of one of its inputs in EITHER orientation* measured: orientation A returns $\{}beta \{}times 7.304 = 5.430$ and orientation B $\{}gamma \{}times 4.238 = 3.404$, ***both exact to the digit*** ⇒ ***so (2) carries nothing its input did not, and since (2) is $\{}gamma \{}times$ (1), (2) is combed if and only if (1) is*** — **the order's fork collapses to one question: is the term mix's own cost combed?** **(1) IT IS NOT.** $4.238$ against a null maximum of $4.297$, **one** period of the $110$ returning more **(3) orientation B — the part of the arm's excess the term mix does NOT explain — $4.005$ against a null maximum of $3.889$, NONE of the $110$ above it, at $-0.16$ rad from the arm's own phase: IT CLEARS** **(3) orientation A — the $45\{}%$ the channel misplaces — $1.482$ against a null maximum of $3.188$, with ALL $110$ above it: decisively not combed** ⇒ ***THE MODULATION IS IN THE PART NO CHANNEL THIS CONSTRUCTION FIXES ACCOUNTS FOR — (3) COMBED AND (2) NOT, WHICH IS ‘PO-56‘'s TERMINATING ROW*** *and the one reading that says DISCHARGES is orientation A taken literally, which rests entirely on the amplitude gated as arithmetic: **strip it and both

orientations say the same thing*** **NOT DECISIVE AND NOT REPORTED AS DECISIVE**: *(1) fails by ONE period of $110$ and (3) clears by a comparable margin, against a deliberately conservative null-MAXIMUM bar built from correlated periods* **(3) THE WINDOW IS COMBED, AND IN ANTIPHASE ON THE EXCESS**: $2.010$ against a null maximum of $1.711$, none above, at $+3.12$ rad from the arm's — $\{}pi$ to within $0.02$, *so it is not a smooth offset* **but one structure read with two signs is NOT established**: *the blunt test fails — $\{}alpha = -0.059$ with the residual keeping $98\{}%$ of the window's power, the amplitudes missing by a factor of five* *and that blunt test is named in the receipt as THE WRONG INSTRUMENT — a ratio over whole vectors asks whether the window IS the arm scaled, where the question is about each one's MODULATED part, which is 'cc66.60''s aggregation error in a new place* ⇒ like-for-like, projecting the two spectrum differences: **the window's modulation is $+0.27$ rad from the arm's — IN PHASE — at a ratio of $0.264$, and $0.264 \{}times 6.241 = 1.646$ against a measured $1.893$** ⇒ ***THE MAGNITUDE COMPOSES AND THE SIGN DOES NOT: the reversal appears only after the likelihood's own weighting*** *Bound:* ***NOT that 'PO-56' is terminated*** — *the measured row IS the terminating one, and placing it is 66's* ***NOT that the exit is decisive*** — *(1) turns on one period of $110$; the direction is unambiguous and the margin is thin* ***NOT that orientation A discharges the clause*** — *its (2) is $\{}beta \{}times$ the arm's own $7.30$ and its (3) is a different object; that exit is the arithmetic talking, and it was gated as such in advance* **NOT that the window and the arm are one structure with two signs** — *the spectrum-level modulations are IN PHASE; only the costs are opposed* **NOT an explanation of the sign flip** — *it is LOCATED, between the spectrum difference and the excess, and naming why would be a mechanism* **NOT a re-derivation of 'cc66.61''s floor or the band-1 results** — (4) holds Nothing SOLVED, nothing RUN; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_term_mix_does_not_carry_the_modulation_and_it_survives_in_the_part`

**P15R213** `P15_the_resolution_table_eight_of_the_eleven_instruments_cannot_represent_the_comb_period_and_`
[OK]
*'sec:refit-bound', 'sec:scope' — **the resolution table 'PO-56' never had: for each of the eleven instruments 'cc66.62' enumerated, the finest period it can resolve (derived from its own geometry), the smallest share it can detect in its own units under its own noise model, and the class of feature invisible to it** — *every banded instrument outputs one number per 0.70 band, a Nyquist period of* $1.40$, *so none of the eight can represent a comb-period feature, where the likelihood's bins give* $0.060$; **four smallest-share cells are honest blanks**** (node 66's 'r7029' redirect, item one: *"a result should meet its instrument's limit BEFORE it is stated, not four revisions after"* — the band-width limit the row met in four disguises ('cc66.58', 'cc66.60', 'cc66.61', 'cc66.64'), stated once, in advance). *Computes:* the geometry every row is derived from, from the banks on disk (band edges, plik bin widths, grid steps, the extremum spacing), the boxcar transfer each averaging width imposes AT the comb period ($0.37$ for a band, $0.9985$ for a bin), and the likelihood's per-band floor $2/\{}sigma_b$ recomputed under 'COV_TT' ($0.055$ at band 1, reproducing 'cc66.62'); every floor it QUOTES is text-gated against the receipt that derived it *Bound:* NOT a verdict on any result of 'cc66''s, NOT a re-scoring, NOT any physics, channel or mechanism — it says what each instrument CAN see, not what any DID see; the three no-noise blanks (contrast statistic, 'SRCDEC', projection-width kernel) and the refit's (inputs under '/tmp/n66', absent) are left blank WITH their reasons, because a guessed floor would be worse than none; shares in different units are set side by side and never ranked *Run:* `python3 receipts/P15_CR_cosmology/P15_the_resolution_table_eight_of_the_eleven_instruments_cannot_represen`

**P15R214** `P15_all_three_projections_clear_the_instrument_noise_null_so_the_term_mixs_cost_is_combed_afte`
[]
*'sec:refit-bound', 'sec:scope' — **on the null 70 validated, ALL THREE projections clear at $p \{}le 0.0005$ with none of $2{,}000$ draws reaching any of them**, so **the term mix's own cost IS combed and 'cc66.65''s central reading was WRONG** — wrong because of the bar this seat chose, which it failed by ONE period of $110$; and **'cc66.62''s scope claim is AMENDED**: the locator LOCATES inside band 1 and MEASURES no height there* (node 66's 'r7033' order — *"one re-run, and it is not a narrowing: the same three projections against the null 70 validated in the same revision"*. ***70's audit, which this seat asked for one revision earlier:*** *the $82$ bins span $T = 2.78$ in $q$ and hold about $3.6$ independent frequencies, $N_{\{}rm eff} = 3.07$, and $6$ of the $110$ correlate with the comb above $0.5$* ⇒ ***"none of 110" was worth roughly ONE IN FOUR*** (1) ***score (1), (2), (3) against $2{,}000$ 'COV' draws through this seat's

own pipeline***, with the two terms of the identity for each; (2) ***70's routed finding against ‘cc66.62‘'s scope statement*** — confirm, amend or reject; (3) nothing else. Guards unchanged, **plus the one this seat produced at ‘cc66.65‘ and 66 promoted to standing: AN ARITHMETIC IDENTITY IS NOT A MEASUREMENT, AND THE TIME TO SAY SO IS BEFORE THE RUN**). *Computes:* **THE INSTRUMENT IS REPRODUCED BEFORE ANYTHING RESTS ON IT**, with 70's own definitions copied rather than re-derived: the arm's $7.3038$ against **0 of $2{,}000$** and a null maximum of $2.34$, where 70 reported $0$ and $2.54$ on another seed **(1) ALL THREE CLEAR**: **(1) the term mix's own cost $4.238$** against a null maximum of $1.874$; **(2) the arm's excess it does NOT explain $4.005$** against $1.416$; **(3) the $45\{}%$ it misplaces $1.482$** against $1.029$ — *none of $2{,}000$ reaching any of them* *and the margins are NOT rounded together: (1) and (2) clear by more than double, **(3) by $1.44\{}times$***, which is stated separately ⇒ ***SO (1) CLEARS AND ‘cc66.65‘'s CENTRAL READING — "the term mix's own cost is NOT combed" — IS WRONG*** **and the reversal was PRE-REGISTERED in the words used for it**: *"a bar worth one in four failing something by one unit is equally capable of passing it"* **AND (1)'s CLEARANCE IS NOT THE NOISE-FREE TERM TALKING**, which is the check that makes it a result: *under this null $d$ is fixed and only $r_c$ moves, so $d^{T}Fd$ has no noise in it to fail* — (1)'s quadratic term is $0.851$ against $0.794 \{}pm 0.006$, **reported as no evidence**, while its CROSS term is $3.399$ against a null median of $0.354$, **0 of $2{,}000$** ⇒ ***(1) clears on the term that knows where the data sits*** *and the held-coefficient variant agrees on both fitted quantities, so none of the clearance is the regression chasing noise* **THE MEASURED ROW IS "BOTH", REPORTED AS SUCH RATHER THAN CHOSEN BETWEEN** — *at this bar the projections no longer separate and the fork as posed does not discriminate* **and the old bar was not merely weak, it was INFLATED**: *its maxima ran $3.2$–$4.3$ where this one's run $1.0$–$1.9$, the wrong-period amplitudes carrying the signal itself, leaked* ⇒ ***a null built from the same data at neighbouring frequencies is not independent of the feature it is scoring*** **(2) AMENDED, NOT REJECTED**: *the locator's four PEAK anchors sit at $q = 0.736$, $1.778$, $2.703$, $3.751$ — **none inside band 1** — but ‘C17‘ states the instrument carries "acoustic peak and trough positions", and **trough 1 sits at $q = 1.360$, inside it*** ⇒ ***70's factual claim is correct and the peak-only reading would have been a DODGE*** *the qualifier: ‘anchored()‘ returns $-c_1/2c_0$, a vertex POSITION, never a height* ⇒ **it LOCATES inside band 1 and MEASURES no height there** *Bound:* ***NOT that ‘cc66.65‘'s measurement was wrong*** — *the three amplitudes are unchanged to the digit; **the BAR was wrong and it was this seat's** ***NOT that ‘PO-56‘ discharges*** — *(1) clearing reads as DISCHARGES and (2) clears too, which is the order's third line: reported, not chosen between* ***NOT that the projections still separate the channels*** — *at this bar they do not, and that is the finding* **NOT that (1)'s quadratic term is evidence** — *it has no noise in it to fail and is reported as no evidence* **NOT that 70's reading of ‘cc66.62‘ is rejected** — *it is correct on the fact; what is amended is what the claim was implying* **NOT a mechanism, a new candidate, or any revisiting of ‘cc66.61‘'s floor or the band-1 results** — (3) holds Nothing SOLVED, nothing RUN; **no corpus edits** *Run:* `python3 receipts/P15_CR_cosmology/P15_all_three_projections_clear_the_instrument_noise_null_so_the_term_mix`

**P15R215** `P15_the_three_instruments_with_no_noise_model_one_can_be_given_a_floor_and_ten_passages_quote_a` [OK]

*‘sec:refit-bound‘, ‘sec:scope‘ — **the three ‘PO-56‘ instruments that carry no noise model: the contrast statistic NEVER SEES THE DATA as defined (its fitted amplitude cancels in the envelope normalisation, to $10^{-16}$), so its numerical floor is the right one for what it says, and its SKY COUNTERPART can be given a floor from the likelihood's own covariance ($0.018$ whole-range, $0.043$–$0.086$ per band); ‘SRCDEC‘ cannot, structurally (only the sum of its terms reaches the sky); the projection-width kernel cannot (it reads no data, and its two anchorings differ twelvefold)** — **and ten passages of ‘P15‘ quote a significance or a share that their instrument cannot bear**, one of them a transcription that contradicts the paper nine lines earlier* (node 66's ‘r7033‘ order: *"which stated results depend on a floor that does not exist"* — the four honest blanks of the resolution table, followed to what rests on them). *Computes:* the contrast receipt's own binned rung rebuilt with its own definitions, the amplitude-invariance of ‘reg‘, 2000 draws of ‘COV_TT‘ through the statistic whole-range and banded, ‘SRCDEC‘'s ten-term closure, the kernel's two anchorings; every flagged phrase text-gated in ‘CR_cosmology.tex‘ and every source receipt gated to draw no noise *Bound:* NOT that any of the ten is wrong about the THEORY — each describes a real difference between noiseless spectra; what is claimed is that a sigma which is not a noise model

is not a significance and a share quoted on one anchoring is not the share. No re-scoring, channel, mechanism, or physics; no ‘cc66‘ receipt touched; the paper not edited — the passages are routed
*Run:* `python3 receipts/P15_CR_cosmology/P15_the_three_instruments_with_no_noise_model_one_can_be_giv`

**P15R216** `P15_the_accounting_closes_but_cannot_attribute_because_the_share_is_the_cosine_of_a_phase_offs`
[]
*‘sec:refit-bound‘, ‘sec:scope‘ — **the accounting CLOSES exactly and still cannot ATTRIBUTE**: the least-squares share is $\{}lvert\{}cos\{}Delta\{}rvert$ **identically**, depending only on the phase offset and not at all on the channel’s amplitude, **and the window proves it by scoring $100\{}%$ in ANTIPHASE**; the other scale is the one (2) invalidates, predicting the window’s cost modulation to $13\{}%$ and the term mix’s to $116\{}%$ ⇒ ***the third row: no instrument this construction has can attribute the modulation****  (node 66’s ‘r7035‘ order, which **WITHDREW ‘r7033‘’s fork as its own defect**: *”I built the fork on a detection test and ‘PO-56‘’s clause asks about AUTHORSHIP; every acoustically structured quantity in that decomposition is distinguishable from noise”* ⇒ ***a test every candidate passes cannot identify a carrier*** **THE ACCOUNTING**, which is this seat’s own instrument from ‘cc66.65‘ turned on the term mix: (1) the channel’s modulated part against the arm’s — phase offset and amplitude ratio; (2) whether that ratio PREDICTS, as it did for the window; (3) the residue after the channel’s contribution is removed at its measured phase, and whether the shares close to the whole *”an accounting has no third outcome: the shares sum to the whole by construction”* — **DISCHARGES on a matched-phase majority, TERMINATES on a minority, TERMINATES on nothing decidable** Guards: no mechanism; no new candidate; ‘cc66.61‘’s floor and the band-1 results finished; **plus both of ‘cc66.66‘’s promoted to the standing set**). *Computes:* **THE INSTRUMENT IS GATED ON ‘cc66.65‘’s OWN WINDOW RESULT FIRST**: phase $+0.270$ rad, ratio $0.2638$, prediction $1.646$ against a measured $1.893$ — *three numbers to the digit* **(1) THE TERM MIX’S SPECTRUM-LEVEL MODULATION IS LARGER THAN THE ARM’S**: $+0.383$ rad at a ratio of $1.176$, where the window’s was $0.264$ **(2) AND THE RATIO DOES NOT PREDICT, WHICH IS THE HINGE**: $1.176 \{}times 6.241 = 7.341$ against a measured $3.399$ — ***off by $116\{}%$ where the window’s template was off by $13\{}%$*** ⇒ *the scale ‘r7035‘’s template was built on does NOT carry over to this channel* **(3) THE VECTOR ACCOUNTING CLOSES EXACTLY on both scales** — contribution plus residue reconstructs the arm’s own vector to numerical precision, *which the pre-registration made the gate on calling it an accounting at all* **but the two scales give $68.3\{}%$ and $98.3\{}%$** **AND THE SECOND IS AN IDENTITY, NOT A MEASUREMENT**: in a two-dimensional $(\{}cos,\{}sin)$ plane $\{}lvert kv\{}rvert = \{}lvert v_a\{}rvert\{}lvert\{}cos\{}Delta\{}rvert$, so ***the share is $\{}lvert\{}cos\{}Delta\{}rvert$ EXACTLY*** — term mix $\{}Delta = +0.184$, $\{}lvert\{}cos\{}Delta\{}rvert = 0.9832$, share $0.9832$; window $\{}Delta = +3.121$, $0.9998$, share $0.9998$ ⇒ ***it depends ONLY on the phase offset and not at all on the channel’s amplitude: a channel a millionth the size scores the same*** ⇒ ***AND THE WINDOW IS THE PROOF RATHER THAN AN ANALOGY — in ANTIPHASE at $+3.12$ rad it scores $100.0\{}%$ of the authorship, at a scale of $-3.633$ that turns its cost UPSIDE DOWN to get there*** *so both scales are out, **each for its own reason and neither failure the other’s**: the spectrum-level ratio by MEASUREMENT ((2), a factor of two), the least-squares projection by ALGEBRA* *and the two-channel line is printed TO BE DISCOUNTED: two channels span the plane, so together they reconstruct the arm exactly at residue $0.000$ — **arithmetic, not attribution, and the same degeneracy one level up*** *Bound:* ***NOT that ‘PO-56‘ discharges*** — *a $98.3\{}%$ share that is $\{}lvert\{}cos\{}Delta\{}rvert$ is not a majority of anything* ***NOT that the accounting failed*** — *it closes to numerical precision on both scales; **closing is not attributing**, and that distinction is the result* ***NOT a fourth outcome row*** — *the answer is the third ‘r7035‘ listed, and this seat did not reach past it* **NOT that the term mix is excluded as a carrier** — *it is unattributable on every instrument this construction has, which is a statement about the instruments* **NOT that ‘cc66.66‘’s clearances are withdrawn** — *they stand; the FORK they were scored against is what 66 withdrew* **NOT a mechanism, a new candidate, or any revisiting of ‘cc66.61‘’s floor or the band-1 results** Nothing SOLVED, nothing RUN; **no corpus edits**  *Run:*
`python3 receipts/P15_CR_cosmology/P15_the_accounting_closes_but_cannot_attribute_because_the_share`

**P15R217** `P15_the_null_harness_survives_its_one_assumption_except_for_the_misplaced_part_and_floors_a_ph`
[OK]

*‘sec:refit-bound‘, ‘sec:scope‘ — **the instrument-noise null harness, load-bearing for ‘PO-56‘’s exit, given one pass: varying the covariance the noise is DRAWN from (estimator fixed) — diagonal, the flat $0.15$ floor as a plateau and at lags 1–7, rescaled — the arm and ‘cc66.66‘’s (a) and (b) clear under every variant, but (c) (the misplaced part) is reached by 26 of 20,000 draws under the lags-1–7 floor, so it is the ONE clearance the assumption decides; its $1.44\{}times$ is the favourable end of its own seed spread ($1.10$–$1.37\{}times$ over ten seeds); rescaling does not matter (the fitted control’s $\{}chi^2/\{}nu = 0.98\{}pm0.10$, (c) first reached at $2.5$); a ratio or phase offset has NO statistical floor at the spectrum level ($4\{}times10^{-4}$ rad) and a finite, linear one at the excess (about $0.05$ rad)** (node 66’s ‘r7035‘ order: *”the null harness is now load-bearing for this row’s exit, so it gets one pass of its own”*). *Computes:* the committed audit script run in-process; ‘cc66.66‘’s multiples reproduced on its own seed; four drawing covariances at 2,000 draws, two pooled at 20,000, ten seeds for the maximum’s spread, a rescale scan, a Rayleigh test, parametric perturbation about the data at full and half noise, the delta method against it *Bound:* NOT an accounting: no central value of the term mix’s ratio, phase or share is computed into the record. No channel, mechanism, re-scoring or physics; no verdict on ‘cc66‘’s results — (c)’s dependence and the seed spread are routed; no ‘cc66‘ receipt touched *Run:* `python3 receipts/P15_CR_cosmology/P15_the_null_harness_survives_its_one_assumption_except_for_the_misplaced`

**P15R218** `P15_the_retention_is_the_combs_average_over_the_kernels_own_k_acceptance_and_the_law_has_no_fi` [OK]

*‘sec:refit-bound‘ — **THE RETENTION IS THE COMB AVERAGED OVER THE KERNEL’S OWN $k$-ACCEPTANCE, AND THE LAW HAS NO FITTED COEFFICIENT**: for a standing oscillation the line-of-sight integral factorises exactly, $\{}Delta_\{}ell(k)=\{}cos(k r_s^{*})k^{(1-n_s)/2}G_\{}ell(k)$, **so the comb reaches the transfer undamped and only the sum over $k$ at fixed $\{}ell$ can touch it** — and $A_\{}ell=\{}lvert\{}int W e^{2ikr_s^{*}}\{}rvert/\{}int W$ with $W=G_\{}ell^{2}\{},dk/k$ **predicts the ABSOLUTE amplitude on both arms, worst band $6.7\{}%$, six of seven inside the instrument’s gate tolerance of $5\{}%$ (FIVE of seven inside the $3.74\{}%$ it measured at the ONE injected phase — re-pinned r7055 on node 70’s floor audit, which also shows that with the instrument’s own error divided out all seven hold inside $3.74\{}%$)** ⇒ *the arm’s acceptance is $13.1\{}%$ narrower, against ‘r6919‘’s independently measured $12.8\{}%$ in $\{}mathrm dr_s/\{}mathrm d\{}chi$* (node 66’s ‘r7039‘ order, which **LIFTED the ‘no mechanism‘ guard** with what replaces it stated — *a mechanism must be **exhibited on the integral**, **predict a number not used to find it**, and be **checked on a configuration it was not fitted on*** and ‘r7043‘, which **composed it with ‘r7041‘’s convergence order rather than letting either displace the other**: *”a law with no free coefficient that predicts an absolute amplitude is also a convergence test, because an unconverged integral will not reproduce an absolute number it was not fitted to”* *‘r7041‘ re-weighted the row so the presumption is the instrument; **this receipt reads the REPORTED settings only and makes no convergence claim***, which is a separate row). *Computes:* **GATE ZERO** — **THE INSTRUMENT IS GATED ON A KNOWN INPUT BEFORE ANY MEASUREMENT PRINTS** — a synthetic comb of amplitude $0.5000$ at period $1.0347$ recovered to $0.24\{}%$ in period and $3.74\{}%$ in amplitude, *and the gate REFUSES to print if it does not* (1) **$A_\{}ell$’s injection-independence CHECKED, not asserted** — the two injections’ ‘eta‘, ‘x0‘, ‘vis‘, ‘k‘, ‘P‘ and $r_s^{*}$ byte-equal, *which is what licenses ONE ‘A‘ per arm instead of two keys holding identical arrays* (2) **the factorisation EXACT** at $2.7\{}times10^{-15}$ (control) and $2.1\{}times10^{-15}$ (arm) on **8 multipoles spread across the reported range**, *and $A_\{}ell$ RECOMPUTED from ‘vis‘, ‘x0‘ and ‘k‘ reproducing the banked value to $0.00$* ⇒ **the law is re-derived here and not trusted from its own bank** (3) the acceptance spans $5.2226$ (control) against $4.5364$ (arm) of acoustic phase (4) **the law forward, absolute, both arms, nothing fitted**: $1.063/1.011/0.990/1.012/1.042/1.016/1.017$ and $1.067/1.015/0.990/1.012/1.044/1.018/1.016$ (5) **and the reproduction gate across a CHANGE OF RUN SCHEME** — $1.0587$ unsliced against ‘r6919‘’s $1.0587$ sliced on ‘KBATCH‘, $0.0000\{}%$ apart, *so the agreement covers the scheme and not only the statistic* *Bound:* ***NOT that the injection IS the real source*** — *it is the projection’s transfer of a KNOWN input, and over-delivery is sufficiency and not identity* ***NOT a convergence claim*** — *the REPORTED settings only; ‘r7041‘’s sweep is a separate row and is running* ***NOT an account of why the construction puts its scales and its distances on different rates*** — *‘r6919‘’s own reserved bound, untouched here* **NOT a tighter claim than the instrument supports** — *six of seven bands sit inside its $3.74\{}%$ accuracy on a known input and no band is quoted tighter than

that* *and the two earlier measuring instruments are NAMED in the source rather than deleted: a band std $46\{}%$ wrong on a known input, and a per-band matched filter selecting a period by maximising amplitude inside a band holding less than one cycle* ⇒ ***‘cc66.60‘’s error a third time, and the first caught BEFORE a number was reported*** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_retention_is_the_combs_average_over_the_kernels_own_k_acceptance_`

**P15R219** `P15_the_laws_six_of_seven_is_five_at_the_quoted_floor_which_is_one_phases_and_with_the_instrume` [OK]

*‘sec:refit-bound‘ — **r7049 → 70, the floor under the acceptance law’s test: ”six of the seven bands on each arm inside 3.74 per cent” is FIVE of seven on each arm** (the gate counts at 5 per cent while its label names 3.74; the 3.65–4.35 band sits at 4.2 / 4.4) — **and the 3.74 per cent is ONE phase’s error, not the statistic’s floor**: scanned over phase on each arm’s own envelope, period and law profile the worst band error is 10.2 / 11.0 per cent — **but the real comb’s phase is gate zero’s to 0.03 rad**, and there the instrument is off by -4.2 / -4.0 per cent in the lowest band and under one per cent in the other six; **divided out, the law holds to 3.4 / 3.6 per cent and all seven bands on both arms land inside 3.74** — the lowest band’s 6.3 / 6.7 per cent deviation being about two thirds instrument. **”On each arm” is one test**: the two arms’ deviations correlate at 0.998; within an arm the seven bands are close to independent (N_eff 6.9 of 7), and the phase systematic has about 2.4 modes* (node 66’s ‘r7049‘ order to 70: *how many independent measurements is ”six of seven bands”, and does the 3.74 per cent measured on a known injection transfer to the real arm’s bands* — asked of a claim landed one revision earlier, by the line that caught this class at ‘r7033‘ and ‘r7035‘). *Computes:* the instrument is ‘cc66‘’s, lifted from its receipt’s source by syntax tree and run unchanged; M1 the literal count from its own ratios; M2 injections on each arm’s own envelope with the law’s own profile, 24 phases plus the real one, and the corrected ratio; M3 400 noise draws through the whole instrument, the phase-error correlation and the between-arm correlation; the paper’s wording REPORTED, never required *Bound:* no verdict on the law’s physics or on ‘cc66‘’s convergence sweep; no edit to ‘cc66‘’s receipt and no re-run of its gate zero as a verdict on it; no prose edited; the white noise probes the instrument’s coupling and is not a noise model for the spectra *Run:* `python3 receipts/P15_CR_cosmology/P15_the_laws_six_of_seven_is_five_at_the_quoted_floor_which_is_one_phases_`

**P15R220** `P15_the_phase_systematic_leaves_the_step_standing_and_carries_a_third_of_the_retention_slope_t` [OK]

*‘sec:refit-bound‘ — **r7055 Q1 → 70, the phase systematic’s effective count against the slope and the step: the count is not where the cost is** — against a null injection (identical combs on both arms, common phase scanned) both banded statistics have three to three and a half modes across seven bands (step 3.5 / 2.8, slope 3.3 / 3.6), and the slope’s own effective count is 8.9 / 5.1; **the step stands**, its false value at the real phase a quarter to two fifths of it (−0.010 / −0.017 against −0.043), worst 0.019; **but the retention slope carries a third of itself as systematic**: at the real phase the null returns +0.0048 / +0.0052 on the reported twelve-setting mean against the real +0.0136 (36 / 39 per cent, same direction), +0.0081 on the seven-band setting; the rise survives the subtraction and its size does not. **The carrier is a phase drift of 0.013–0.042 rad between the two arms’ source combs**, which the partial-cycle band-RMS reads as retention: one comb for both removes it, a flat envelope does not. And the receipt computes +0.01359 ± 0.00202 where P15 and its docstring carry +0.0139 ± 0.0021* (node 66’s ‘r7055‘ order Q1 to 70: *report the phase systematic’s effective count for the retention slope and the excess’s step specifically, and if unchanged say so* — the order’s own scope kept: the banded statistics’ error structure, not the likelihood). *Computes:* both statistics (‘band_excess‘, ‘retained‘) and ‘cc66‘’s period fit lifted from their receipts by syntax tree and run unchanged, each gated to reproduce its receipt’s printed values on the real banks; null injections on every bank each reads, 24 common phases, pre-registered amplitude 0.5 and the banks’ own as a declared sensitivity; N_eff, the slope’s own count, the false step and slope at the real phase; the D_l / source split and a comb x envelope decomposition of the source side *Bound:* nothing about the likelihood (‘r7033‘’s split stands); no edit to either receipt and no prose edited; a null injection measures the statistic’s response and does not say what the arms’ real retention difference is *Run:* `python3 receipts/P15_CR_cosmology/P15_the_phase_systematic_leaves_the_step_standing_and_carries_a_third_of_`

**P15R221** `P15_the_acceptance_does_not_move_under_refinement_and_three_of_the_twelve_axes_could_not_have_` [OK]

*‘sec:refit-bound‘ — **THE KERNEL’S $k$-ACCEPTANCE DOES NOT MOVE UNDER ANY*

*REFINEMENT THAT CAN MOVE IT, AND THREE OF THE TWELVE AXIS-ARM PAIRS COULD NOT HAVE MOVED IT AT ALL**: across twelve settings varied one at a time on both arms, the WORST SINGLE MULTIPOLE excursion in $A_\{}ell$ on any of the nine axes that move an input is $0.0082\{}%$ — **73$\{}times$ inside $r6911$'s $0.6\{}%$ floor, and $41163\{}times$ inside it on the arm's $k_{\{}max}$ axis over a near-doubling of $k_{\{}max}$** ⇒ *and the arm-to-control acceptance narrowing is $13.03\{}%$ at **all twelve** settings, spread $0.0048$ percentage points, against $r6919$'s independent $\{}mathrm{d}r_s/\{}mathrm{d}\{}chi$ reading of $12.8\{}%$* **BUT THREE PAIRS ARE INERT BY CONSTRUCTION AND ARE NOT REPORTED AS CONVERGED**: 'NK' on the arm, whose ladder is $\{}sqrt{L(L+2)}\{},$stretch out to 'KMAXL' so 'NK' is a decimation cap never reached — $1452$ modes at 'base', 'nk15' and 'nk20' alike, **byte-identical inputs and $\{}max\{}lvert\{}Delta_\{}ell\{}rvert=0$ exactly at the spectrum**, where the CONTROL moves $2547\{}to3822\{}to5094$ and its spectra differ outright — and 'LSTEP' on BOTH arms, since $A_\{}ell$ is read at the same multipoles at every setting by construction* (node 66's 'r7049', offering and not directing: *"$A_\{}ell$ is a sharper probe of the same thing and it is now free... and $A_\{}ell$ carries NO FITTED AMPLITUDE to absorb the truncation, where a height ratio does"* — taken up because 'GRIDSAVE' made it cost $1.45$ s per configuration and no re-run). *Computes:* **GATE ZERO** — **THE INERT TEST MUST DISCRIMINATE BEFORE ANY $A_\{}ell$ IS READ**: it is required to fire on exactly the three claimed pairs and stay silent on the other nine, *because a test that answers the same way everywhere is not a test* the inert claim carried as a **BYTE claim**, sha256 of $k$, $\{}mathrm{d}k$, $\{}eta$, $x_0$, $\{}mathrm{vis}$ — **and run the other way on the nine, which must each move an input** $A_\{}ell$ **recomputed from the banked background** at $8$ multipoles on four configurations, reproducing the bank at $0.00$e$+00$ the **worst single multipole** printed rather than the mean over the range, *because a mean can hide a moving tail* scope gated rather than asserted, with the pre-registered criteria printed beside every axis unaltered *Bound:* ***NOT 'r7041''s sweep and no substitute for it*** — *the sweep asks what the RETENTION and the peak heights do, measured on spectra; this asks what the kernel's $k$-acceptance does, computed from the background, and **a converged acceptance with an unconverged retention would itself be a finding**, which is why the two are reported apart* ***NOT a convergence claim for the row*** — *nine of twelve axes read inside the floor is not the sweep's verdict, and the criteria are carried unaltered: a sequence that has not turned over is not converged whatever its last step, and a two-point axis cannot turn over* nothing about which cosmology is right *Run:* `python3 receipts/P15_CR_cosmology/P15_the_acceptance_does_not_move_under_refinement_and_three_of_the_twelve`

**P15R222** `P15_the_band_rms_ratio_does_not_move_with_any_numerical_setting` [OK]
*'sec:refit-bound' — **THE BAND-RMS RATIO DOES NOT MOVE WITH ANY NUMERICAL SETTING AT THE $0.6\{}%$ FLOOR — AND ELEVEN OF THE TWELVE AXIS READINGS ARE STILL NOT *CONVERGED*, WHICH IS THE HALF OF THE RESULT SAID FIRST.** 'r7041''s convergence sweep at FULL coverage: $158{,}885$ modes over $66$ queued configurations, two injection schemes $\{}times$ two arms $\{}times$ twelve settings $=48$ runs, every one folded. **$\{}texttt{fixed}$ sits at $1.0587$ and $\{}texttt{sweepown}$ at $1.0659$–$1.0660$ across every refinement of every axis; the worst last step over all twelve readings is $0.008\{}%$ against a pre-registered floor of $0.6\{}%$, a factor of $73$ inside it, and both base points reproduce 'r6919''s banked $1.0587/{+}0.01189$ and $1.0659/{+}0.02260$ exactly.** *And by the pre-registration — a sequence that has not TURNED OVER is not converged whatever its last step, and monotonicity is undefined on two points — **exactly ONE reading of twelve is reportable as converged**, $\{}texttt{sweepown}$'s $\{}texttt{NLOSW}$, which alone reverses ($1.0659425 \{}to 1.0659377 \{}to 1.0659385$); seven read 'not turned over' and four 'two points only'.* **AND THERE ARE NOT TWELVE AXES: FIVE MOVED IT ON THE ARM, SIX ON THE CONTROL** — $\{}texttt{NK}$ is inert on the arm by construction (the ladder is $\{}sqrt{L(L+2)}\{},$stretch to $\{}texttt{KMAXL}$ and $\{}texttt{NK}$ a decimation cap never reached; byte-identical $k$, $\{}eta$ and $\{}max\{}lvert D_\{}ell\{}rvert = 0$), so its six arm configurations were NEVER QUEUED, while on the control it carries $2547 \{}to 3822 \{}to 5094$ modes; $\{}texttt{LSTEP}$ is the mirror, real for the sweep ($238 \{}to 475$ reported $\{}ell$) and inert for the ACCEPTANCE on both arms* (node 66's 'r7041' order — the convergence sweep itself, re-weighted at 'r7043' so the presumption is the instrument — carried to completion across $\{}sim 11$ hours in which the container was reclaimed at EVERY cycle, four in-flight slices killed and redone from nothing each time). *Computes:* **GATE ZERO** — **THE PRE-REGISTERED RULE IS RE-IMPLEMENTED FROM ITS STATEMENT AND

ITS VERDICTS ASSERTED, NOT QUOTED**: the receipt must find exactly one converged reading and name it, *because a receipt that printed the sweep's own conclusion back would assert the persistence of a symptom rather than the finding* (‘r7061‘’s class). **AND THE RULE IS APPLIED TO FULL-PRECISION RATIOS BECAUSE A ROUNDED TABLE CANNOT CARRY IT** — banked at the $4$ dp the sweep's reader prints, the fallback path reported EIGHT of twelve converged against the live path's ONE: *the test is the SIGNS of the steps, and four points identical to $4$ dp round to differences of zero, which read as a turn.* *The same defect in miniature as the reader that could not see the sweep — an instrument given less state than its question needs will still answer.* It recomputes the twelve sequences from the folded spectra through ‘fold.load‘ when the bank is present and falls back to the banked table when it is not, computing the step ONE way on both paths and cross-checking the recomputation against the bank to $10^{-8}$ *Bound:* **THE SPECTRA ARE IN ‘/tmp/n66/r7041/‘ AND NOT IN THE REPOSITORY, so in CI the projection integrals are not recomputed** — the receipt says in its own output which path it took, as the resolution-table receipt does for this same bank. ***STABILITY IS NOT CONVERGENCE AND THIS ROW CLAIMS ONLY THE FIRST***: eleven of twelve readings have not turned over, and the four two-point axes ($\{}texttt{NLOSF}$, $\{}texttt{LSTEP}$) cannot. ***AND NOTHING HERE IS A SPECTRUM OF THE MODEL*** — every run is the projection's transfer of a KNOWN analytic oscillation (‘SRCINJ‘), comparable with neither a banked spectrum nor the sky; the quantity is ‘r6919‘’s band-RMS ratio and NOT *the retention*, about a third of whose $+0.0139$ per period is a source-comb phase drift (‘r7057‘/‘r7059‘/‘r7061‘). *It is also NOT ‘r7049‘’s acceptance row: that computes the kernel's $k$-acceptance from the background, this measures spectra* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_band_rms_ratio_does_not_move_with_any_numerical_setting.py`

**P15R223** `P15_the_fit_has_the_controls_freedom_and_the_contrast_the_data_ask_for_lies_outside_every_dire` [OK]

*‘sec:refit-bound‘ — **THE REFIT HAS THE CONTROL'S FREEDOM DIRECTION FOR DIRECTION, AND WHAT IT LEAVES IS A UNIFORM ACOUSTIC CONTRAST OUTSIDE THE SPAN OF EVERY DECLARED PARAMETER — THE DATA ASKING THIS ARM FOR $6.4\{}%$ LESS AT $7.5\{}sigma$ AND THE CONTROL FOR NOTHING.** The whitened response of the five independent directions is **four stiff and one soft on BOTH arms with the same degeneracy** ($H_0$ and $\{}Omega_m$ at $\{}lvert\{}cos\{}rvert = 0.99$ on each), so *‘six declared, three effective‘ is not what this fit is.* **WHAT SEPARATES THE ARMS IS THE REACH**: at the exact linearised GLS minimum, where the residual is orthogonal to all five by construction, the control leaves $186.0$ against $n-5 = 180$ (${+}0.3\{}sigma$ — *the fit reaches the data*) and this arm leaves $278.8$ (${+}5.2\{}sigma$). **AND THE MISSING DIRECTION IS NAMEABLE**: decomposed on amplitude, acoustic shift and acoustic contrast, all four parameters move the shift strongly and the contrast barely — the $4\{}times2$ (shift, contrast) singular-value ratio being $0.040$ on the control and $0.036$ here — and freeing one uniform contrast template removes $56.4$ of this arm's residual, a fifth of it, against $0.8$ on the control, at $c = -0.0636 \{}pm 0.0085$ against $-0.0075 \{}pm 0.0083$. An $\{}ell$-modulated contrast buys $4.5$ and a modulated shift $0.4$: *the request is a LEVEL and not a drift.* **AND THE LEFTOVER IS NOT NOISE**: $54$ sign changes in $\{}ell$ order against $88 \{}pm 10$ for correlated draws of the likelihood's own covariance with the same five directions projected out, the control's $87$ inside one $\{}sigma$* (node 66's ‘r7091‘ order (‘Q1‘, ‘Q2‘) — *where the fit is pinned, written on Daryl's standing instruction that a misfit's default hypothesis is the MODEL's faithfulness rather than numerics or CR* — delivered by node 70 as ‘r7091+70.1‘ and **pre-registered, with TWO of its four predictions MISSING: a CR-only shift/contrast coupling (both arms read alike) and a CR arm one rank lower (the ranks are equal)**; the control, predicted ‘weaker but non-null‘, is NULL). *Computes:* **THE CONSTRUCTION IS REBUILT FROM THE BANKED GRID RATHER THAN IMPORTED FROM THE DELIVERING SEAT'S SCRIPT**, so the paper's figures rest on a receipt that RECOMPUTES them: the five whitened responses and their SVD per arm at the grid's own step sizes, the three-template decomposition, the GLS reach, the contrast coefficient fitted JOINTLY with the five so it is not credited their freedom, and the sign-change count against $2000$ correlated draws at a fixed seed. **AND IT READS THE GRID'S OWN ONSET BEFORE USING IT** — every CR run in the grid fixes ‘ZSTART=3e7‘, so the solved onset (‘Z_START = None‘) is OVERRIDDEN there, asserted as check (0) because it bounds what the directions are the directions OF *Bound:* **NOTHING HERE SAYS CR's PHYSICS SETS THE CONTRAST LOWER.** What is established is that the quantity the data

ask to move lies outside the span of every declared direction, so *no re-parametrisation reaches it* and the change must come from what SETS the contrast — which assignment that is belongs to ‘PO-70‘ and the rate rulings. *The lensing operator is the registered refit’s own fallback (CAMB at the control’s parameters) and is identical on both arms by construction, so it cannot carry an arm difference.* *The Monte Carlo is $2000$ draws at a fixed seed and the crossing count is asserted as a wide inequality against the noise mean, not at a $p$-value this receipt would have to calibrate.* ***One of the gate’s own assertions failed on first run*** — a twenty-fold separation written where the measurement gives eight — *and the check caught it where the reasoning that wrote it did not; it stands at the size the numbers support* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_fit_has_the_controls_freedom_and_the_contrast_the_data_ask_for_lie`

**P15R224** `P15_the_fitted_onset_is_the_wrong_clocks_artefact_because_the_stacking_ruler_keeps_a_square_ro` [OK]

*‘sec:tensions‘, ‘sec:coherence‘, ‘sec:acoustic‘ — **THE FITTED ONSET IS THE WRONG CLOCK’S ARTEFACT, SO THE TWO DEFECTS THE ORDER NAMES SEPARATELY ARE ONE DEFECT.** *Ruled for every time-dependent object in the two-arm transfer, from ‘P07‘’s rate rule and ‘P15‘’s own sections rather than from the instrument’s history.* **FOUR OF SEVEN ARE ON THE WRONG RATE AT THE DEFAULT**: $\{}rs$ and $r_D$ (stacking, ‘LEAFSCALES=0‘), the visibility’s measure (stacking, ‘VISLEAF=0‘), and **recombination’s own ionisation history, solved on ‘Hphys‘ with no switch anywhere in the file**; the perturbations, $\{}eta$/$\{}chi$ and $D_M$ are already right. **AND THE CENTRAL FINDING IS A SCALING LAW**: the sound horizon’s memory of its own start goes as $a_{\{}mathrm{start}}ˆ{1/2}$ on the stacking rate and $a_{\{}mathrm{start}}$ on the leaf’s (measured $0.5008$ and $0.9983$), because the exponent *is* the radiation term — so ‘P15‘’s *”the sound horizon has no lower endpoint to place”* is TRUE on the leaf rate and FALSE on the stacking one, where $43.1\{}%$ of the integral is still missing at the instrument’s own onset. ⇒ ***A stacking ruler REQUIRES a hand-placed start and a leaf ruler does not, so the onset solve is not a modelling choice beside a clock error — it is the clock error’s only available repair, and correcting the assignment dissolves the parameter.*** And the laundering has exactly one channel: ‘r6893+cc66.37‘ measured that no transfer reads ‘R_S‘ or ‘L_A‘, so the ruler’s clock reaches the spectrum through the onset and nowhere else. **Q2 answered: the onset is DETERMINED — the handover is the branch point, $1/(aH)\{}to0$ there on BOTH rates so every mode is outside the horizon, and the acoustic angle is an OUTPUT.** At the default onset the first three peaks’ modes are all INSIDE it ($k/(a\{}mathcal{H}) = 1.53, 3.73, 5.67$), where at the control’s start all three are outside — **so the arm begins on the wrong side of crossing for the whole acoustic band, and a $k$-dependence read off that start is the artefact the file itself suspected**. **AND IT SETTLES ‘r7093‘’s PROVENANCE QUESTION WITHOUT ADJUDICATING IT**: all nine of the refit grid’s CR configurations already carry ‘ZSTART=3e7‘ AND ‘LEAFSCALES=1‘ at the background the distance data fix, so **the refit grid is already the configuration both large rulings select and the reported spectrum’s solved-onset stacking-ruler configuration is the one neither selects** — the figures to report are the grid’s, and the solved pin was never faithful; the grid is NOT faithful on the two small rulings, no CR row setting ‘VISLEAF‘ and the ionisation history having no switch to set* (node 66’s ‘r7091‘ order, on Daryl’s standing frame — **when the model does not match the measured spectrum the default hypothesis is that the MODEL does not faithfully implement what CR requires**, not a numerical issue and not CR rejected by the sky. (1) *which rate does CR require for each time-dependent object, and does the implementation use it*; (2) *is the CR arm’s onset determined by the construction or free*. Guards taken: do not run the acoustic instrument or score channels (‘cc66‘ owns the rebuild), and **propose no one-at-a-time clock test** — the swap moves $\{}rs(\{}eta_{LS})$ and hence the comb, so an isolated factor reads two oscillations out of phase (‘r6919‘, ‘cc66.40‘’s guard)). *Computes:* **55 checks, all pass, 2 s.** **(A)** the corpus’s assignment read verbatim out of ‘P07‘ and ‘P15‘, with ‘git log -S‘ dating the removal of the stacking sentence to ‘r6772+66.1‘; **(B)** the instrument’s defaults PARSED rather than remembered, including the absent switch on the ionisation history and the docstring that says *”Both are correct”* and *”Do NOT unify them”* of one physical object on two clocks; **(C)** $H_{\{}text{leaf}}ˆ{2}-H_{\{}text{stack}}ˆ{2}=H_0ˆ{2}\{}Omega_r(1+z)ˆ{4}$ symbolically exact; **(D)** both rulers re-derived from the background alone — $\{}ell_A = 172.3$ and $292.4$, reproducing ‘r6760+66.1‘’s banked pair, with the fitted comb $286.0$ within $2.3\{}%$ of the leaf and nowhere near the stacking one, and the instrument’s printed ratio $1.286$ returned; **(E)** the two start-memory exponents; **(F)** the pin solved on each ruler, $z_{\{}mathrm{onset}}$

= 6761$ against $61583$, a factor $9.1$; **(G)** the comoving horizon falling to zero at the handover on both rates, and $k/(a\{}mathcal{H})$ at both starts; **(H)** the two sub-per-cent rulings MEASURED so they are not mistaken for the big ones — the visibility's clock moves $z_{LS}$ by $0.90\{}%$ and its width by $1.1\{}%$, and the leaf ionisation history runs $2.2$ to $8.3\{}%$ richer across the visibility; **(I)** the affirmative control — the control's rate carries radiation, so its ruler obeys the LEAF law (slope $0.9985$), which is why integrating it from $a\{}simeq0$ was always safe there; **(K)** the refit grid's own launcher read for the provenance consequence *Bound:* **IT RULES ON ASSIGNMENTS AND MEASURES MIS-ASSIGNMENTS ON THE BACKGROUND AND THE IONISATION HISTORY — IT CLAIMS NO SPECTRUM, NO LIKELIHOOD, AND NOT THAT THE CORRECTED IMPLEMENTATION WILL MATCH THE SKY.** No transfer is run and nothing from 'ACOUSTIC_two_arm.py' is imported; the file is READ. The visibility's FWHM is taken from INTERPOLATED half-max crossings because on grid points it is quantised coarser than the move being measured. Two figures are corroborations rather than identities — the Jacobian across $\{}pm3$ FWHM comes back $0.812$–$0.924$ against the instrument's quoted $0.789$–$0.913$ (a different grid, the same gradient), and $z_{\{}mathrm{onset}}=6761$ against its $6764$. **AND ONE GATE WAS RE-POINTED AT 'r7096', THE SECOND TIME ON THIS RECEIPT**: it read *the ionisation history's rate is NOT independently assignable*, which was true when written and was reported as **a knob to SPLIT rather than a value to change** — 'cc66' split it at 'r7097' ('LEAFREC', recombination's own rate, defaulting to the leaf), so the gate now asserts the INVARIANT the ruling needs and the ruling itself is untouched. *The register's standing form met twice on one receipt: a gate pinning the corpus's current text has a subject free to move, and here it moved because the defect was being fixed* **And what the ruling COMMITS to is stated as a falsifier rather than a hope**: 'r7093' reports a residual direction no declared parameter spans, the data asking this arm's contrast $6.4\{}pm0.9\{}%$ lower at $7.5\{}sigma$ — *not measured here and not this receipt's to measure* — so **if a rebuild consistent on all four assignments does not supply a correction of that sign and about that size, the rate assignments are NOT where the contrast comes from and the rigidity is somewhere this adjudication has not looked** **AMENDED IN PLACE AT 'r7108+60.1', AS TO ONE GATE AND NOT TO ANY FINDING.** *The check that locates the paper's start-scan sentence required the literal first-peak band '$206$ and $210$'. 'r7109' made the first-peak call and the paper now carries '$204$ and $208$', so the gate went red on 'main' for a measurement that had IMPROVED.* ⇒ **A gate that pins to the digits of a measurement another seat owns asserts the persistence of a spelling, not the finding.** *It now reads the sentence's four numbers and asserts what they are for — '$\{}ell_A$' swinging '121.3' across the scan while the first-peak band is '4.0' wide and sits wholly below it — with the achieved values printed. Nothing else in the receipt changes and all '52' checks pass.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_fitted_onset_is_the_wrong_clocks_artefact_because_the_stacking_ru`

**P15R225** `P15_the_projection_read_recombination_at_the_wrong_conformal_time_and_one_clock_moves_it_72_pe`
[OK]
*'sec:refit-bound' — **THE PROJECTION KERNEL READ RECOMBINATION AT A CONFORMAL TIME $72$ PER CENT AWAY FROM WHERE THE PLASMA'S OWN CLOCK PUTS IT, AND ONE CLOCK END TO END REMOVES IT WITHOUT FLATTENING THE RESIDUAL'S SWING.** The instrument built its conformal-time grid from the STACKING rate — and with it $a(\{}eta)$, $\{}eta_{\{}rm rec}$, $\{}eta_0$, $D_M$, every spline's abscissa and **the kernel's own argument $x_0=\{}eta_0-\{}eta$** — while the acoustic phase, the perturbations and (under $\{}texttt{LEAFSCALES=1}$) the sound horizon all accumulate on the LEAF rate. ⇒ *So $\{}Delta_\{}ell=\{}int S(k,\{}eta)j_\{}ell(k(\{}eta_0-\{}eta))\{},\{}mathrm d\{}eta$ had $S$ parametrised by one clock and $j_\{}ell$'s argument by the other: **two parametrisations of one history inside one integral**, which is an internal inconsistency and not a preference between rates.* **MEASURED FROM THE BACKGROUNDS ALONE, no spectrum needed: $\{}eta_{\{}rm rec}=485.5$ Mpc on the stacking clock against $282.3$ on the leaf, a $41.8\{}%$ gap — and against the CONTROL's own $281.8$ Mpc the arm's leaf reading is $0.18\{}%$ out and its stacking reading $\{}mathbf{72.3\{}%}$ out.** *And $D_M$ hides it: $14017\{}to13947$ Mpc, $-0.50\{}%$, because $\{}eta_0$ moves with $\{}eta_{\{}rm rec}$ — **the gap is $84\{}times$ the move in $D_M$, so a reader watching $D_M$ calls a $42\{}%$ inconsistency a half-per-cent effect.*** **BUILT: $\{}texttt{LEAFGEOM=1}$, the one clock assignment that had no knob** — with $\{}texttt{LEAFSCALES=1}$ and $\{}texttt{LEAFPERT}$ the instrument carries ONE clock and **asserts the three consequences at run time** ($\{}max\{}lvert\{}mathrm{*

*1\{}rvert=0$ exactly, $\{}max\{}lvert\{}Phi^2-1\{}rvert=0$, the two sound-horizon accumulators identical to $0$ Mpc), refusing to report a spectrum otherwise. **With the onset HELD and the comb an OUTPUT, $\{}ell_1/\{}ell_A$ goes $0.7290\{}to0.7326$ against the sky's $0.7312$ — a factor $1.6$ closer with nothing fitted to it** — *while P1/P2 and P1/P3 move AWAY ($2.142\{}to2.080$, $2.173\{}to2.094$ against $2.217$, $2.277$), reported together because they disagree** (node 66's 'r7091' order, on Daryl's standing frame — ***when the model does not match the measured spectrum the default hypothesis is that the MODEL does not faithfully implement what CR requires*** — which closed the 'numerics or CR's prediction' pair the sector had been scoring channels against, and routed the implementation itself). *Computes:* **GATE ZERO** — **THE CONTROL'S CLOCKS CANNOT PART COMPANY AND THE RECEIPT PROVES IT BEFORE IT READS THE ARM**: its two rate expressions are character-identical, so the two conformal times are asserted equal to $10^{-12}$, *because a reader that could not tell the arms apart would report this defect on both*. **AND THE RESIDUAL'S SHAPE IS SCORED BY A RULE FIXED IN ADVANCE BY THE ORDER** — *a $\{}chi^2$ that improves while the swing stays is not the fix; a swing that flattens is the fix even if $\{}chi^2$ moves little* — so the reader reports the per-bin residual in $\{}sigma$, its RUNS of one sign, the CROSSINGS and the EXCURSIONS, and the receipt asserts the verdict it reaches rather than choosing one. **AND THE FIXED-PARAMETER COMPARISON IS A TILT ARTEFACT, WHICH THE RECEIPT ASSERTS IN BOTH DIRECTIONS**: at the old parameters $\{}chi^2$ worsens $3.0\{}%$ and the longest run of one sign more than doubles ($16\{}to35$ bins, one unbroken excursion from $\{}ell=100$ to $414$); allow the two directions a refit moves first (an amplitude and a power-law tilt) and $\{}chi^2$ falls $265.5\{}to215.0$, $\{}mathbf{-19.0\{}%}$, the worst excursion $6.10\{}to4.12\{}sigma$. *The $35$-bin run was $n_s$ and not shape* ⇒ ***but the SWING still does not flatten: crossings $36\{}to30$ and the longest run $16\{}to18$, the wrong direction on both, so by the order's own rule this is NOT the fix*** *Bound:* ***THE PARAMETERS ARE NOT REFITTED UNDER THE NEW GEOMETRY***, so the $-19\{}%$ is a two-direction marginalisation and not a refit — *$n_s$ is not a pure tilt in $\{}ell$ and two directions are not four*; **the honest next question is a real refit on the one-clock geometry, which is a grid of runs and is not claimed here** — *it is OPEN at the **programme** level and routed with 'r7091''s own division of labour to '70''s rigidity audit, which is what would decide whether the solver time is worth spending; see the row's origin path.* *The spectra live in '/tmp/n66/r7091/' and are not in the repository, so the comb and the shape are banked with their provenance while the clock gap, $D_M$ and the control comparison are re-derived here in closed form — which is the finding's own subject and needs no spectrum.* **And the refit's own VERIFIED minimum 'verify_cr.npz' is no longer reproducible from the tree** ($\{}max\{}lvert\{}Delta D_\{}ell\{}rvert=7.14\{}times10^{-3}$); *the PRE-patch code differs from it by the same amount to every digit, so the drift predates this build* — checked that way round because a difference found while holding a patch is the patch's until shown otherwise — ⇒ *both sides of every comparison are same-revision runs and the banked spectrum is used for neither.* *It does NOT collapse the arm onto the control: the arms differ three ways and this touches one, the arm keeping its handover initial data and its discrete $k$ ladder* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_projection_read_recombination_at_the_wrong_conformal_time_and_one`

**P15R226** `P15_the_phase_is_already_carried_into_the_projections_variable_and_the_two_statements_of_the_s` [OK]

*'sec:refit-bound' — **THE RULE'S DIVISION OF LABOUR IS CARRIED OUT AND NOT MERELY ASSERTED: THE TWO STATEMENTS OF THE SOUND HORIZON AGREE TO A PART IN $10^{8}$, SO THE PHASE IS ALREADY CONVERTED INTO THE PROJECTION'S OWN VARIABLE AND A SECOND CONVERSION WOULD DOUBLE-WEIGHT THE JACOBIAN.** 'P07''s rate rule puts the plasma's sound horizon on the leaf rate and a separation read across leaves on the stacking rate, with no locus at which the rate switches — so in the line-of-sight integral the VARIABLE is the stacking conformal time while the phase accumulates on the leaf's, and $\{}int c_s\{},\{}mathrm d\{}eta_{\{}rm leaf}$ must equal $\{}int c_s\{},(\{}mathrm d\{}eta_{\{}rm leaf}/\{}mathrm d\{}eta_{\{}rm stack})\{},\{}mathrm d\{}eta_{\{}rm stack}$ identically. **Measured over the plasma's whole run on the arm: relative $9.79\{}times10^{-9}$, against a stacking accumulation standing $75.3$ per cent away** — the identity tighter than the separation by $7.7$ orders. **AND THE AFFIRMATIVE CONTROL IS WHAT MAKES IT A TEST OF THE CONVERSION RATHER THAN OF THE ACCUMULATOR**: on the control all three accumulators coincide to the quadratures' own precision, the Jacobian being ex-*

*actly $1$ there* (node 66's 'r7095' Q1 order — **which was WRONG and is recorded as such**: it ordered this conversion BUILT, reasoning from the measured $72$ per cent gap between the two clocks' readings of recombination, and the gap is the two-rate structure itself rather than a missing conversion. *'cc66' measured the identity and declined to build what was ordered*, and a second application would double-weight the Jacobian exactly as 'GSRC=1' with 'LEAFPERT' did at 'r3737', reaching $G = 2.73$ at the onset). *Computes:* **THE MEASUREMENT IT BANKS LIVED IN A COMMIT MESSAGE ONLY**, and this revision's 'P15' prose rests on it, so the gate recomputes rather than quotes it: the instrument is run in a subprocess per arm, at the reported configuration, and the three accumulators are read over the instrument's own 'ETA_ON' to 'ETA_LS'. *The gate reproduced 'cc66''s figure independently before acting on it — $9.79\{}times10^{-9}$ against the reported $9.8\{}times10^{-9}$, on a tree where 'LEAFREC=1' had moved the visibility peak, so the absolute Mpc values differ and the identity does not* *Bound:* **Nothing here says the implementation is faithful on every object**: the visibility's clock is unadjudicated and the ionisation history's rate became switchable only at 'cc66''s 'LEAFREC'. What is established is that the ONE object 'r7095' ordered converted is converted already. *The two conformal times of recombination are reported as a pair of CORRECT readings and not as a discrepancy — the gate's own 'r7095' prose had it the other way and is corrected in the same pass.* *The Mpc figures move with 'LEAFREC' and with the onset, which is why the IDENTITY is asserted and the values are printed rather than pinned.* ***Three of the gate's own tolerances were written tighter than the measurement supports and the checks caught all three*** — ten orders of magnitude asked where there are eight, and the control's coincidence asked below $10^{-9}$ where two quadratures agree at $2.2\{}times10^{-9}$; *an identity measured by two quadratures agrees to their precision and no better* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_phase_is_already_carried_into_the_projections_variable_and_the_tw`

**P15R227** `P15_the_optical_depth_is_a_photon_path_observable_so_it_takes_the_stacking_rate_and_the_visibi`

[OK]

*'sec:refit-bound', 'sec:coherence', 'sec:tensions' — **THE OPTICAL DEPTH IS A PHOTON-PATH OBSERVABLE, SO IT TAKES THE STACKING RATE AND 'd\{}tau' CARRIES NO JACOBIAN — AND THIS SEAT'S OWN 'r7092' RULING ON THE VISIBILITY IS WITHDRAWN.** *'r7095' held one of four rulings and asked for it either way; it falls the paper's way.* **THE QUESTION IS FIRST NARROWED AND THE NARROWING IS EXACT**: Thomson scattering has one rate, $\{}Gamma=n_e\{}sigma_Tc$ per unit proper time, and since $\{}mathrm d\{}eta=c\{},\{}mathrm dt/a$ in either reading, $\{}tau'=\{}Gamma a/c$ is **the same expression in both clocks** — so 'tau'' was never the ambiguous object and the whole difference is the ACCUMULATION, $\{}tau=\{}int\{}Gamma\{},\{}mathrm dt$ with $\{}mathrm dt=\{}mathrm da/(aH)$. **AND THE CONSTRUCTION THEN SETTLES IT OUTRIGHT RATHER THAN BY PRECEDENT**: $a\{},\{}mathrm d\{}eta=c\{},\{}mathrm dt$ by the definition of a conformal time, so the proper length the photon covers between two leaves is $a\{},\{}mathrm d\{}chi=c\{},\{}mathrm dt_{\{}rm stack}$ with $\{}chi$ the stacking rate's — which 'r7095' has just reaffirmed against 'LEAFGEOM=1'. ⇒ ***Weighting $\{}mathrm d\{}tau$ by $J=\{}mathrm dt_{\{}rm leaf}/\{}mathrm dt_{\{}rm stack}¡1$ counts the scatterings of a photon crossing that same proper length in less time — at $1.130c$ at recombination and $1.175c$ by $z=1500$. A measure that moves the photon off the null cone of the path it is integrated along is not a second admissible reading of the rate rule.*** **And the counter-precedent is DISTINGUISHED by its integrand rather than overruled**: $1/k_D^{2}$ carries the baryon loading and the polarisation coefficient, so it is the oscillator's own viscous damping and must share 'LEAFPERT''s clock, while $\{}tau$ carries nothing but $n_e\{}sigma_Ta$ and shares the kernel's. ⇒ ***EACH INTEGRAL TAKES THE CLOCK OF THE EVOLUTION IT IS AN INTEGRAL OVER, and the Thomson rate appears in both while setting the measure in neither*** — which closes the instrument's own flagged gap, *"two objects on the same side of the rule, given opposite clocks, and nothing states the choice"*, by showing they are **not** on the same side. **WHAT SURVIVES is the ionisation history**: the chemistry is an evolution of the content over its own duration, so 'n_e' must be the leaf solution while $\{}mathrm d\{}tau$ accumulates on the stacking path — and the knob that could not express it has since been SPLIT: 'cc66''s 'LEAFREC' gives recombination its own rate, defaulting to the leaf, so **this ruling is implemented** and the receipt's gate asserts that invariant rather than the absence it first reported. Measured: the surviving correction moves last scattering $0.59\{}%$ in $z$ and its width $0.1\{}%$, **in the OPPOSITE direction to the one withdrawn**, so conflating them was not even conservative* (node 66's 'r7095'

order, ‘Q1‘: *”settle whether the optical depth’s accumulation is a content process or a path integral IN THIS construction — a derivation, not a preference, and I will land whichever way it falls including against the paper”*, with the competing reading named (the path element is geometry while the scattering rate is content) and the external cost stated ($\{}ell_1=221.93+4.21f$). ‘Q2‘: name the three BODY statements ‘r7092‘’s onset ruling rested on, so the masthead rewrite lands on the derivation. Guards taken: no transfer run, no channel scored, no one-at-a-time clock test, and 70’s rigidity figures neither reproduced nor re-measured). *Computes:* **31 checks, all pass, 1 s.** **(A)** the dispute read from both papers’ BODY text and the instrument’s own flag, with the one-parameter family and the sky’s $f=-0.304$ recorded; **(B)** $\{}tau’=\{}Gamma a/c$ shown clock-independent symbolically, and the instrument’s ‘taup_of‘ shown to be that expression; **(C)** the stacking side’s enumeration quoted from ‘P07‘, the redshift’s ‘L3‘ reading from ‘P15‘, **and the decisive null-cone check with the implied photon speed printed at three redshifts**; **(D)** the two integrands parsed from the instrument, so the distinction is read rather than asserted; **(E)** the three visibilities computed — as the instrument stands, as ruled, and as ‘r7092‘ asked — on this background and the recombination solver; **(F)** each of the three ‘Q2‘ statements verified to be BODY and not comment text, with the masthead’s sharper sentence located in the header and nowhere in the body; **(G)** the affirmative control and the falsifier *Bound:* **IT RULES ON AN ASSIGNMENT AND MEASURES THE SURVIVING CORRECTION ON THE BACKGROUND AND THE IONISATION HISTORY — NO TRANSFER IS RUN AND NO SPECTRUM OR LIKELIHOOD IS CLAIMED.** The derivation’s one premise is stated with its falsifier: *exhibit a photon-path observable this construction assigns to the leaf that is NOT an integral over the content’s own evolution, and it fails* — $1/k_D^{2}$ is not one. **And the four-multipole external referent is explicitly NOT what decides it**, for two reasons stated in the receipt: ‘r7095‘ asked for a derivation, and that measurement was taken at the configuration ‘r7092‘ ruled unfaithful, so it must be re-measured at the faithful one before it can arbitrate — **this seat’s own ruling applied to evidence that favours the side it is moving to** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_optical_depth_is_a_photon_path_observable_so_it_takes_the_stackin`

**P15R228** `P15_the_licensed_configuration_moves_neither_statistic_and_the_swing_was_read_above_where_it_c` [OK]

*‘sec:refit-bound‘ — **THE CONFIGURATION THE RATE RULE LICENSES MOVES NEITHER OF THE TWO NUMBERS THE REPAIR WAS ORDERED TO MOVE, AND THE EARLIER NULL ON THE FORBIDDEN ONE WAS READ ABOVE THE CEILING WHERE THE STATISTIC CONVERGES.** ‘r7095‘ adjudicated against $\{}texttt{LEAFGEOM=1}$ — the rate rule assigns $D_M$ the STACKING rate and recombination the LEAF’s, so moving the geometry onto the leaf is the one thing it forbids by name — and ordered the rule’s own configuration run instead ($\{}texttt{LEAFSCALES=1 LEAFREC=1 ZSTART=3e7}$, geometry left on the stacking rate), with the fork fixed in advance: *if the rule’s own configuration ALSO drops $\{}lvert c\{}rvert$ the rule and the data agree; if it does NOT, the data are asking for the one object the rule pins.* ⇒ ***IT RESOLVES THE SECOND WAY.*** **The licensed configuration moves the contrast coefficient by $-0.08\{}sigma$ ($-0.0636\{}to-0.0642$, $\{}lvert c\{}rvert$ RISING $1\{}%$), lengthens the longest run of one sign $18\{}to22$ bins, worsens $\{}chi^2/\{}mathrm{bin}$ $3.05\{}to3.20$, and is the worst of the three on the peak heights.** *The forbidden one moves $c$ by $+6.70\{}sigma$ ($\{}lvert c\{}rvert$ FALLING $89\{}%$ onto the control’s own value), takes crossings $30\{}to44$ and the longest run $18\{}to12$, and halves $\{}chi^2/\{}mathrm{bin}$ to $1.47$.* **AND THE SECOND FINDING IS WHY THE FIRST CAN BE TRUSTED: THE SWING STATISTIC IS NOT CONVERGED AGAINST THE CEILING IT IS READ TO.** ‘cc66.73‘ reported the forbidden repair did NOT flatten the swing (crossings $36\{}to30$, longest run $16\{}to18$); **that reproduces to the digit on its own $\{}texttt{LMAXL=1300}$ pair, so it was measured correctly — and read to $\{}ell\{}le1040$ it is $80$ per cent of that run’s own reported ceiling against $52$ per cent of the $\{}texttt{LMAXL=2000}$ run’s.** A four-point ceiling scan localises the feature: below $\{}ell\{}approx800$ all three configurations have a longest run of $8$ or $9$ bins and are indistinguishable, and above it the banked default’s grows $8\{}to8\{}to15\{}to18$ and the licensed configuration’s $9\{}to9\{}to13\{}to22$ while the forbidden repair’s stays $9\{}to9\{}to9\{}to12$ ⇒ ***the statistic discriminates exactly where the shorter run could not see it, which is why that run read a null*** (node 66’s ‘r7095‘ order, which credited ‘cc66.73‘’s finding and ruled against its repair, then fixed the reporting order (the swing’s crossings and longest run first, $\{}chi^2$ second, the heights third) and the fork above before any grid was run). *Computes:* **22 checks, all pass.** **(A)** the three configura-

tions read off their own '__SWITCHES__' stamps, with $\{}texttt{LEAFREC}$ shown to move the SPECTRUM while leaving $\{}ell_A$, $D_M$ and $r_s$ BIT-IDENTICAL — 'r6893+cc66.37''s diagnostics finding holding again on a switch built after it; **(B)** the swing at four ceilings through 'r7091''s own 'shape.py' run as a subprocess rather than re-implemented; **(C)** the earlier null reproduced on its own pair and then explained by the ceiling, with both runs' printed 'k_max = 2 l_max / D_M' quoted; **(D)** $\{}chi^2/\{}mathrm{bin}$ at the same four ceilings, the forbidden repair lowest at every one; **(E)** the three peak-height ratios against the sky, all three arms putting the first peak at $\{}ell=220$ against $220.6$; **(F)** the contrast coefficient on each banked grid through '70''s own design matrix, with '70''s published $-0.0636$ and $-0.0075$ REPRODUCED before anything is compared to them and the CONTROL's coefficient identical across all three grids — the no-op that says what moved is the arm and not the method *Bound:* **IT DOES NOT REINSTATE $\{}texttt{LEAFGEOM=1}$ AND DOES NOT WITHDRAW THE EARLIER NULL.** 'r7095''s ruling is the gate's and it stands; what is reported is a CONFLICT between the rule's configuration and the sky, for the gate to adjudicate, and the correction to 'cc66.73' is the ceiling and not the arithmetic. *No parameter is refitted and no onset pinned: every spectrum is read from a file tracked in this repository — the licensed and forbidden grids and the two $\{}texttt{LMAXL=1300}$ spectra are banked here, so nothing is derived from '/tmp'.* *The heights are a TRADE and are reported as one: the forbidden repair puts $\{}ell_1/\{}ell_A$ four times closer to the sky while P1/P3 undershoots, so its summed height deviation ($0.081$) is worse than the banked default's ($0.068$) even though the licensed configuration's ($0.100$) is worse than both.* *Convergence is localised, not reached: the scan stops at the grids' own $\{}texttt{LMAXL=2000}$, so what is established is that readings below $\{}ell\{}approx800$ decide nothing and that the two resolutions disagree above it — not that $2000$ is converged.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_licensed_configuration_moves_neither_statistic_and_`

**P15R229** `P15_the_kernels_distance_is_the_observers_own_flat_slice_so_the_rate_rule_does_reach_it_and_the` [OK]

*'sec:properframe', 'sec:largescale' ('prop:flat'), 'sec:tensions' — **THE KERNEL'S DISTANCE IS THE OBSERVER'S OWN FLAT SLICE, SO THE RATE RULE DOES REACH IT, AND THE DISAGREEMENT WITH THE SKY IS REAL.** *'r7099' asked the sector's last question — what the object in $j_\{}ell(k\{}chi)$ IS, not which rate to build it on — and named three readings, guessing (3) was cheapest because it could dissolve the conflict rather than resolve it.* **(3) IS CHEAP AND DOES NOT DISSOLVE**: 'P15' identifies the two objects outright (*"the comoving angular-diameter distance is $D_M=D_C$"*) on 'prop:flat''s receipted argument, **and the distinct object (3) reached for exists and is already spent** — $r_0$, the layer's areal radius, setting the SOURCE's discrete spectrum $k_L=\{}sqrt{L(L+2)}/r_0$ while $D_C$ projects. **(2) IS TRUE OF THE LINE ELEMENT AND THE EXACT STATEMENT IS A ONE-LINE IDENTITY NOBODY HAD WRITTEN**: in the proper frame $r/(\{}partial_\{}tilde\{}tau}r)=\{}alpha\{}tanh(3c* **exactly** (verified to $2\{}times10^{-16}$ at five redshifts on both backgrounds, with $\{}sinh^{2}u_0=x_0^{3}/2 the bridge), so **the radial-to-angular ratio is the HUBBLE RADIUS where comoving-FRW form would make it the radial comoving distance** — which is the precise sense in which the projection is not the standard integral *on that chart*, and it still does not move the kernel, because the standard form is claimed on the OBSERVER's constant-$\{}tau$ slice. **AND THE TRAP IS NAMED AND EXCLUDED**: read at face value the angular part gives $D_A=r(\{}tilde\{}tau_e)$, hence with the construction's own redshift $D_M=(1+z)D_A=r_0$ CONSTANT at every $z$ — excluded by the comb ($\{}pi r_0/r_s=109$ against $\{}pi D_C/r_s=302.9$) and structurally, the $d\{}Omega$ being the static chart's about the BRANCH POINT and not a sky. ⇒ ***(1) IS WHERE IT LANDS, BY THREE INDEPENDENT ROUTES*** — 'P07''s rate rule naming $D_M$; 'r7096''s null-cone argument; and the observer's own readout, flat by 'prop:flat' and exactly flat-$\{}Lambda$CDM on the $\{}sinh^{2/3}$ law with radiation absent — **so the sky's preference for the forbidden configuration is a disagreement between the construction and the sky, localised and quantified, and a result about CR rather than a defect**** (node 66's 'r7099' order, 'Q1', with the standing frame run to its end: *"every assignment is derived and the implementation does what the construction says on every object, so this is the first time in the sector that the question has been about the construction rather than about the code"*. The order's own guards: **you are not asked to make the sky win**; nothing through the transfer; 'cc66''s three-grid table not to be re-measured. 'Q2' was closed in the order itself ('LEAFREC' implements the surviving ionisation-history ruling)). *Computes:* **26 checks, all pass, 0 s.** **(A)** the calibration FIRST, on four of the paper's own parameter-free figures reproduced from the background alone — $r_0=5051$,

$D_C=1.401\{}times10^{4}$, the stretch $2.774$ and $\{}ell_2=7.85$, each agreeing with the paper's printed $5051$, $1.4011\{}times10^{4}$, $2.774$ and $7.8$ to the precision the paper prints; **(B)** reading (3) closed on the paper's sentence and 'prop:flat''s receipted argument, with the distinct object located; **(C)** the $r/\{}dot r=c/H$ identity measured at five redshifts on two backgrounds, with the $\{}sqrt2$ slip a first pass carried recorded in the source; **(D)** the face-value reading's comb computed and excluded, and its miss shown to be exactly the stretch — the one number the low-multipole floor is built on; **(E)** the three routes each verified at source; **(F)** the remainder, the control and the scope *Bound:* **THE ONE REMAINDER, AND IT IS THE ONLY PLACE THIS CHAIN COULD STILL MOVE**: the harmonic expansion has never been carried out IN the proper frame. $j_\{}ell(k_LD_C)$ is licensed by an **identification** — the slice is flat, the readout is exactly flat-$\{}Lambda$CDM — rather than by expanding a plane wave on the constant-$\{}tau$ slice with the observer at a general point of it, where the chart's origin stops being the centre of anything. ⇒ *The bounded calculation that would settle it is named in the receipt and is **NOT** ordered and **NOT** done here.* And the ruling claims no spectrum: 'cc66''s three-grid table is taken as verified, used only to name which configuration the sky prefers, and the agreement of $\{}pi D_C/r_{s,\{}rm leaf}=302.87$ with its banked $302.889$ is reported as a cross-check of the configuration, not as a re-measurement *Run:* `python3 receipts/P15_CR_cosmology/P15_the_kernels_distance_is_the_observers_own_flat_slice_so_the_rate_rule`

**P15R230** `P15_the_licensed_rebuild_leaves_all_three_rigidity_numbers_where_they_were_and_the_forbidden_o`
[OK]
*'sec:refit-bound' — **THE RATE RULE'S OWN CONFIGURATION LEAVES ALL THREE OF '70''s RIGIDITY NUMBERS WHERE THEY WERE, AND THE CONFIGURATION THE RULE FORBIDS PUTS THE ARM ON THE CONTROL'S OWN FLOOR.** 'r7097' made Q3 the whole job — both arms from $z=3\{}times10^7$ with no solve anywhere, $\{}texttt{LEAFSCALES=1}$, $\{}texttt{LEAFPERT}$ on, $\{}texttt{LEAFREC=1}$, $\{}texttt{VISLEAF=0}$ because it is unadjudicated, and $\{}texttt{LEAFGEOM=0}$, the geometry on the stacking rate — and **named the targets before the grid existed: the unreachable $\{}chi^2$ is $278.8$ against the control's $186.0$ at $n-5=180$, and the residual crosses zero $54$ times against noise's $88\{}pm10$.** ⇒ ***ALL THREE ARE UNMOVED BY THE LICENSED REBUILD:*** $\{}chi^2$ goes $278.795$ to $278.788$, **$0.007$ per cent of the $92.8$ gap it was ordered to close**, on a rebuild that moved the spectrum by six and a half per cent; the crossings stay at $54$ and the longest run of one sign at $33$. ⇒ ***AND ALL THREE ARE CLOSED BY THE FORBIDDEN ONE:*** $\{}chi^2$ falls to $184.99$, **BELOW the control's own $186.01$**; the crossings go to $91$ against the control's $87$; and the longest run goes to $8$, **exactly the control's $8$**. **SO '60''s FALSIFIER FIRES, AND IT WAS WRITTEN UNPROMPTED AND BEFORE THE RUN** — *if a rebuild consistent on all four assignments does not supply a contrast correction of that sign and size, the rate assignments are not where the contrast comes from* ⇒ ***but the one assignment the rule FORBIDS supplies all of it, so the conclusion is sharper than the falsifier's own wording: the contrast comes from the clock the GEOMETRY is read on, which is the one object 'P07' pins to the stacking rate by name.*** *And the refit 'r7097' asked for tests a prediction it put on the record: the two-direction tilt and the four-parameter refit's own $n_s$ shift agree to better than $0.004$ on all three grids at matched resolution, which licenses reading one as the other — and on that reading 'cc66.73''s $-0.0324$ does NOT survive, the one-clock rebuild needing $-0.0035$, while the banked default and the licensed configuration both want $+0.03$ to $+0.04$** (node 66's 'r7097' order, which withdrew 'r7095''s Q1 as a wrong order, gated 'LEAFREC', and named the three figures and '60''s falsifier in advance of the run). *Computes:* **13 checks, all pass.** **(A)** the banked grid returns 'r7097''s own three figures before anything is compared to them, and the CONTROL's three are identical across all three grids — the no-op that says what moved is the arm; **(B)** the licensed rebuild's non-movement at full precision; **(C)** the forbidden one reaching the control's floor on $\{}chi^2$, crossings and longest run; **(D)** the four-parameter refit on each grid and the tilt identification checked rather than assumed; **(E)** the scope. Every number is read through '70''s own 'rigidity.py' definitions — its 'build', its 'W', its 'STEP', its 'stats' and its 'bestfit' — rather than re-implemented, because re-deriving the model would make a disagreement unattributable between the geometry and this receipt's arithmetic *Bound:* **IT DOES NOT REINSTATE $\{}texttt{LEAFGEOM=1}$:** 'r7095''s ruling is the gate's, the switch's default is untouched, and what is reported is a conflict between the rule's configuration and the sky for the gate to adjudicate. *'–grid DIR' was added to 'rigidity.py', another seat's driver, in its own existing '–mc' style; unset it is the banked

grid and the file's behaviour is byte-identical, so none of its findings move — **flagged to the gate rather than assumed.*** *No spectrum is computed and no parameter pinned here; the MC is not used, because none of the three numbers is an MC quantity.* *Verified in a tree materialised from the git index alone, from the receipt's own directory, which is what CI gives it.* *The forbidden grid's $\{}chi^2$ landing $1.02$ BELOW the control's is reported as measured and not interpreted as the arm fitting better than $\{}Lambda$CDM — it is one draw of one sky at $n-5=180$, where the control itself sits $+0.3\{}sigma$ high.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_licensed_rebuild_leaves_all_three_rigidity_numbers_where_they_wer`

**P15R231** `P15_the_harmonic_expansion_in_the_proper_frame_is_not_bounded_because_no_slicing_of_it_carries` [OK]

*'sec:largescale' ('prop:flat'), 'sec:properframe' — **THE HARMONIC EXPANSION, CARRIED OUT IN THE PROPER FRAME: IT INTRODUCES NOTHING, AND IT CANNOT BE SET UP — NO SLICING OF THAT FRAME CARRIES BOTH THE DISCRETE SOURCE AND THE PROJECTION DISTANCE.** **'r7101' ordered the one thing 'r7100' named and did not do: expand a plane wave on the constant-$\{}tau$ slice with the observer at a general point of it, and show whether the $\{}ell$-space kernel is $j_\{}ell(kD_C)$.* **FIRST, WHAT THE CHART IS, BY AN IDENTITY**: 'eq:scalefac' satisfies $(\{}mathrm dr/\{}mathrm d\{}tilde\{}tau)^2=r_s/r+r^2/\{}alpha with $r_s=2\{}alpha/3\{}sqrt3$ — **exactly** the Nariai value, symbolic residual identically zero — so 'eq:proper-frame' rewritten on the areal radius is $ds^2=-d\{}tau^2+(dr-v\{},d\{}tau)^2+r^2d\{}Omega^2$, **the Painlev\{}'e–Gullstrand slicing of SdS**, whose flow speed is the Hubble velocity $v(r)=H(z)r/c$ to machine precision and is $\{}ge1$ everywhere, $=1$ only at $r_N$. ⇒ **(i) THE EXPANSION INTRODUCES NOTHING**: on $\{}mathbb{E}^3$ the addition theorem about $x_O\{}ne0$ is exact to $1.3\{}times10^{-15}$ and the offset is the pure phase $e^{ik\{}cdot x_O}$, which cancels in $C_\{}ell$ — **no measure, no weight, no $\{}ell$-dependent factor**, which was the live worry. **(ii) BUT THE SLICE OFFERS $r_0$ AND NOT $D_C$, AND IS SHORT BY EXACTLY THE STRETCH**: $\{}tilde\{}tau=\{}tau_0+\{}chi$ and $r(\{}tilde\{}tau)=\{}lvert x\{}rvert $, so the cosmic epoch across the observer's own simultaneity slice is $1+z=r_0/\{}lvert x\{}rvert $ — **it carries every epoch, as spheres about the branch point** — and the $z=1089.9$ locus is a $4.63$ Mpc sphere subtending $0.105^\{}circ$, two parts in $10^7$ of the sky, whose chord from the observer is $r_0$ to $9\{}times10^{-4}$. ⇒ $D_C/\{}text{chord}=2.773729$, **the stretch to six figures**. **(iii) AND THE CHART'S PHOTONS ARE NOT THE PAPER'S**: its exact null geodesics give $1+z=(1\{}pm v_o)/(1\{}pm v_e)$ — derived twice, integrated numerically to $10^{-10}$, and **calibrated in de˜Sitter where the answer is independently $a_0/a_e$, returned to twelve figures while the ratio of areal radii returns nothing like it** — on which the construction's locus is a *blueshift*, the $z=1089.9$ surface sits at $r_N$ within $51$ Mpc, $r=r_N$ is an infinite-redshift surface and the radial area distance is affine, saturating at $1.31\{}times10^{4}$ Mpc. **(iv) AND THE JOIN FAILS WHERE THE ORDER SAID IT WOULD**: the constant-cosmic-time surface of 'eq:proper-frame' has induced metric $(v^2-1)d\{}chi^2+r^2d\{}Omega^2$ with both coefficients constant, Ricci scalar $2/r_c^2$ and a **zero** Ricci eigenvalue along $\{}chi$ — $\{}mathbb{R}\{}times S^2$ of radius $r(\{}tilde\{}tau)$, not an $S^3$ ($6/a^2$, isotropic) — while an $S^3$ slice of curvature radius $a$ exists for **any** $a$, so it is a choice and singles out no $r_0$. ⇒ ***SO THE EXPANSION IS NOT BOUNDED ON THIS CONSTRUCTION AND THE ORDER'S LICENCE IS TAKEN: the flat slice has the flatness and no epoch, the constant-epoch surface is a cylinder, the $S^3$ is unforced.*** ⇒ **IT DOES NOT MOVE 'r7100''s CONCLUSION; IT MOVES THE LICENSING** — $j_\{}ell(k_LD_C)$ is neither confirmed nor contradicted, and its licence remains 'sec:flatlcdm''s exact recovery of flat $\{}Lambda$CDM, an identification with a named content. **The disagreement with the sky stays real**** (node 66's 'r7101' order, 'Q1' — the remainder 'r7100' named, now ordered, *"because it is the only place a conclusion three derivations now rest on can still move, and a result of this weight should not stand on an identification when the expansion is bounded"*, with the explicit licence *"if it turns out not to be bounded after all ... say that and stop"*. The same order's smaller item, the 'c54'-era lensing potential's admissibility, is answered in section H). *Computes:* **35 checks, all pass, 1.5 s.** **(A)** the calibration FIRST, four of the paper's parameter-free figures from the background alone; **(B)** the chart named by a *symbolic* identity against 'eq:scalefac', with $v=Hr/c$ and $f=-(r-r_N)^2(r+2r_N)/\{}alpha^2r$ measured on both backgrounds; **(C)** the expansion run at six random $(k,x_O,s)$ with $\{}lvert x_O\{}rvert \{}approx r_0$, **with the wrong-argument control missing the plane wave by $O(1)$** so the pass is a measurement, and 'prop:flat' re-derived

by the same Ricci machinery as the affirmative control; **(D)** the slice's epoch map, the chord, the sky fraction and the stretch identity; **(E)** the geodesic system integrated with the null constraint held, plus the de˜Sitter calibration; **(F)** the two induced three-geometries computed symbolically and the $S^3$ slice constructed; **(G)** the verdict; **(H)** the lensing line, *sized*; **(I)** the scope, read from the files actually opened *Bound:* **IT CLAIMS THE GEOMETRY AND NOT THAT ANY NUMBER MOVES.** $r_0$, $D_C$, the stretch, $\{}ell_A$ and the comb are all read off the Friedmann readout, which nothing here touches. **AND THE CHARITABLE READING OF THE $S^3$ SENTENCE IS NAMED RATHER THAN SUPPRESSED**: if *"the cosmological layers are the closed $S^3$"* means the readout's spatial section rather than ‘eq:proper-frame‘'s $\{}tilde\{}tau$-layer, the physics is untouched and it is the citation to ‘sec:properframe‘ that needs repair — **that is for the gate to adjudicate, not for this line to edit.** Two slips of mine are recorded rather than quietly fixed: the first draft took the numerically unstable root of the $S^3$ quadratic and reported $2.8\{}times10^{-5}$ on an identity, and the first draft of the two scope gates read **this file's own source** for forbidden names and found them in its own expressions — *the second time this line has written a check that tests its own spelling.* No transfer, spectrum or likelihood is computed; the instrument and the lensing producer are READ for their rate assignment and not run; the lensing potential is NOT re-derived and ‘cc66‘'s three-grid table is NOT re-measured or used **THE NULL-CONSTRAINT BOUND WIDENED TO ‘$10^{-5}$‘ AT ‘r7112+60.1‘, ON THE INSTRUMENT'S OWN RULE RATHER THAN ON THIS LINE'S JUDGEMENT.** *‘r7106+60.1‘ had moved both of this gate's tolerances off the numerical floor to ‘$10^{-6}$‘. CI's tolerance perturbation then FLAGGED the null-constraint site: its error read ‘$1.09\{}times10^{-9}$‘ on one build and ‘$1.77\{}times10^{-10}$‘ on the other — it moved ‘84‘ per cent with ‘$917\{}times$‘ of headroom, and ‘sweep_tolerances.py‘ flags exactly ‘moved ¿ 0.10‘ AND ‘headroom ¡ 10^3‘, saying in terms that a moving error with ‘$\{}ge10^3$‘ of margin is not a defect.* ⇒ **So the bound is stated where the claim needs it and clear of that bar: ‘$10^{-5}$‘ leaves ‘$¿9000\{}times$‘ on CI's own worse build.** *The two numbers are no longer given the same bound because they are not the same kind of claim — the AGREEMENT between the analytic root and an independent integration is the load-bearing one and keeps ‘$10^{-6}$‘ with ‘$10^{7}$‘ of headroom; the null constraint is a validity check on the integrator.* *Recorded on CI's measurement rather than this line's: all three builds reachable in the container come back CLEAN on that site, so the ‘$6\{}times$‘ spread is in hardware this line cannot reproduce and the runner's numbers are the evidence. ‘35‘ of ‘35‘ still pass.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_harmonic_expansion_in_the_proper_frame_is_not_bounded_because_no_`

**P15R232** `P15_the_construction_fixes_no_closed_layer_radius_because_the_nariai_mass_closes_the_route_tha`
[OK]

*‘sec:largescale‘, ‘sec:properframe‘ — **‘PO-72‘ ‘Q1‘ ANSWERED ON THE TERMINAL EXIT: THE CONSTRUCTION FIXES NO CLOSED-LAYER RADIUS, THE OBSTRUCTION IS THE NARIAI MASS BY AN IDENTITY, AND THE ONE LENGTH IT DOES FIX IS $\{}alpha$ — NOT $r_0$.** *‘r7103‘ asked which slicing carries the closed reading and what fixes its radius, naming the terminal exit as equally live and asking which way it falls.* **THE OBSTRUCTION, IN ONE LINE**: an $S^3$ slice coincides with a time-symmetric one exactly when $f(r)=1-r^2/a^2$, and $f(r)-(1-r^2/a^2) = -r_s/r - r^2/\{}alpha^2 + r^2/a^2$ carries coefficient $-r_s$ on $1/r$ and $(a^{-2}-\{}alpha^{-2})$ on $r^2$ ⇒ **it vanishes for all $r$ only if $r_s=0$ AND $a=\{}alpha$, two independent conditions** — and ‘eq:amplitude‘ pins $r_s$ at the Nariai $2\{}alpha/3\{}sqrt3$. ⇒ ***SO THE MASS THAT FIXES THE RATE IS EXACTLY WHAT UNFIXES THE CLOSED LAYER'S RADIUS: one choice, read on two of its consequences.*** **AND ALL FOUR CANDIDATE FIXINGS FALL**: *orthogonality* is already spent on the flat slicing ($u_\{}mu=-\{}mathrm d\{}tau$ exactly, so the congruence's unique orthogonal foliation is the constant-$\{}tau$ one); *constant cosmic epoch* excludes all, that surface being $\{}mathbb{R}\{}times S^2$; *maximality* by the identity; and *CMC* selects $a\{}to\{}infty$, which **is the flat slice again** — even that one being CMC only at $r_s=0$, since $\{}mathrm dK_\{}rm flat}/\{}mathrm dr = 9\{}alpha^3r_s^2/4r^{5/2}(\{}alpha^2r_s+r^3)^{3/2}$ makes the whole $r$-dependence the mass. **THE CONTROL RETURNS THE AFFIRMATIVE, ANALYTICALLY**: in pure de˜Sitter at $a=\{}alpha$ the auxiliary root $s\{}equiv0$, $n^r\{}equiv0$ and $K\{}equiv0$ — the throat IS the unique maximal $S^3$ — while $0.7\{}alpha$ and $1.4\{}alpha$ give $K$ neither zero nor constant, *so the null is a property of this geometry and not of the method*. **WHAT IS FIXED, AND IT IS NOT $r_0$**: the closed readout's curvature radius is $c/(H_0\{}sqrt{\{}Omega_\{}Lambda}) = (c/H_0)\{}sqrt{1+2/x_0^3} = \{}alpha = \{}sqrt{3/\{}Lambda}$,*

*an identity (symbolic residual zero, $2\{}times10ˆ{-16}$ on both backgrounds) ⇒ **the substrate's throat**, with $\{}alpha/r_0 = \{}sqrt3/x_0 = 1.0320$, *so even the readout does not return the radius the floor is built on*. ⇒ ***WHICH WAY IT FALLS: RE-DERIVED, NOT WITHDRAWN, BUT NOT AS A PAIR OF SLICINGS*** — one SLICING that is flat ('prop:flat', Riemann identically zero, $\{}Omega_k=0$ in the distance relation) and one INVARIANT LENGTH $\{}alpha$ that the closed readout returns exactly. **The plural goes; $\{}lvert \{}Omega_k\{}rvert = \{}Omega_\{}Lambda$ — the content that mattered, that $\{}Omega_k$ is not a free parameter — stands exactly**  (node 66's 'r7103' order opening 'PO-72', 'Q1', off this line's own 'r7102' section (iv): *"a closed layer whose radius is chosen rather than fixed is not what the decoupling argument needs"*, with the terminal exit named as *"equally live and not a fallback"* and the instruction *"Say which way it falls"*. The gate declined this line's charitable reading on a stated ground (the same paragraph contrasts its $Sˆ3$ with a literal closed-Friedmann reading), which is why the row exists rather than a repair). *Computes:* **19 checks, all pass, 1.4 s.** **(A)** the calibration first, with $r_s$ confirmed at the Nariai value on both backgrounds before anything turns on it; **(B)** the $\{}alpha$ identity, symbolically and on both backgrounds; **(C)** the obstruction, with its two coefficients extracted and each shown independent; **(D)** the de˜Sitter control, analytically ($s$, $nˆr$ and $K$ all identically zero at $a=\{}alpha$) and numerically; **(E)** the four candidate fixings, each measured — the congruence's $u_\{}mu$ computed from the Painlevé–Gullstrand block, the layer's Ricci recomputed from the metric rather than recalled, and the CMC gap scanned over three decades in $a$ with the $a\{}to\{}infty$ collapse isolated; **(F)** the answer; **(G)** the two consequences for the floor; **(H)** 'r7103''s design rule applied *Bound:* **NOTHING IS REASSIGNED AND NOTHING IS EDITED.** The corrected attribution is left exactly as the gate wrote it; no spectrum, transfer or likelihood is computed; the only file opened is the paper. **AND THE COST IS PUT ON THE RECORD BEFORE IT IS PAID, BOTH WAYS**: (1) if the source rides the one fixed closed radius it rides $\{}alpha$, moving the floor from $\{}ell_2=7.845$ to $7.602$, $-3.10$ per cent; (2) if it rides the slicing that IS fixed — the $\{}mathbb{R}\{}times Sˆ2$ layer, Riemannian at the present epoch with $vˆ2-1=0.336$ — then $\{}chi$ runs over all of $\{}mathbb{R}$ ($\{}tilde\{}tau$ constant with $\{}tau=\{}tilde\{}tau-\{}chi$ free), the Laplacian along it has continuous spectrum reaching zero, and **there is no gap and so no floor at all**. ⇒ *Two different answers, and choosing between them is the gate's and not this line's — 'Q1' asked for the slicing and the radius.* 'r7103''s design rule is **applied**: no gate here asserts the presence of the paper's open question, which this receipt's own answer is meant to close; the sentences are located and printed, and every gate reads the geometry **WITHDRAWN IN PLACE AT 'r7106', AS TO ITS WORDING AND NOT ITS CONTENT.** *'r7105' struck 'PO-72' on its own premise — the constant-$\{}tilde\{}tau$ surface being $\{}mathbb{R}\{}times Sˆ2$ is what the reassignment DOES, and this line's own charitable reading was the right call — and named the reassigned chart the wrong setting for the conclusion drawn in it.* ⇒ **The sentence "the construction fixes no closed-layer radius" is withdrawn.** *What the identity $f(r)-(1-rˆ2/aˆ2) = -r_s/r - rˆ2/\{}alphaˆ2 + rˆ2/aˆ2$ actually says is **where the sphere lives**: the maximal $Sˆ3$ of radius $\{}alpha$ exists iff $r_s=0$, so it is the SUBSTRATE's slice, and $\{}alpha$ is the radius the de˜Sitter presentation wanted. Every computation in this receipt stands and 'r7106' uses the identity as its bridge.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_construction_fixes_no_closed_layer_radius_because_the_nariai_mass`

**P15R233** `P15_the_expansion_in_the_de_sitter_presentation_is_not_bounded_either_because_the_sphere_lives` [OK]

*'sec:properframe' (the two presentations), 'sec:largescale' — **'r7105''s 'Q1' CARRIED OUT IN THE DE˜SITTER PRESENTATION: THE LAYER'S RADIUS *IS* FIXED THERE, AT $\{}alpha$ — AND 'r7104''s OWN IDENTITY SAYS THAT SPHERE EXISTS ONLY AT ZERO MASS, SO IT IS THE SUBSTRATE'S SLICE AND NOT THE COSMOLOGY'S. BOTH CLOSED KERNELS THE PRESENTATION OFFERS ARE EXCLUDED BY THE COMB.** **THE RADIUS IS FOUND, AND IT IS AN IDENTITY**: the throat, $\{}alpha = c/(H_0\{}sqrt{\{}Omega_\{}
= \{}sqrt{3/\{}Lambda}$, the unique maximal $Sˆ3$ — but 'r7104''s coefficients $-r_s$ and $(aˆ{-2}-\{}alphaˆ{-2})$ make that slice exist **iff $r_s=0$**, and 'eq:amplitude' pins $r_s$ at the Nariai value. ⇒ ***SO THE TWO PRESENTATIONS ARE NOT TWO CHARTS AT ONE MASS: the sphere sits at $r_s=0$ and the observer at $r_s=2\{}alpha/3\{}sqrt3$*** — which is why 'r7104' found the family free, it having asked the cosmology's geometry for the substrate's slice. **AND THE KERNEL ON THE LAYER IS NOT $j_\{}ell(kD_C)$, BY A FACTOR OF SIX**: on the $Sˆ3$ of radius $\{}alpha$ last scattering sits at $\{}chi_{\{}rm*

*rec}=D_C/\{}alpha=2.6876$ rad $=0.8555\{}pi$ — **$85.6$ per cent of the way to the antipode** — so the angular mapping is $\{}alpha\{}sin\{}chi=2286$ Mpc against $D_C=14011$ and the comb lands at $\{}mathbf{49.4}$ where the sky and the flat projection put it at $302.9$. *There is no flat limit to rescue it: $\{}chi$ is within fifteen per cent of $\{}pi$.* **AND THE HOPF QUESTION ANSWERED: IT WOULD BE A SELECTION RULE, NOT A KERNEL, AND IT IS EXCLUDED TOO.** The degree-$L$ $S^3$ harmonics are $(L/2,L/2)$ of $SU(2)\{}times SU(2)$; the Hopf $U(1)$ lies in one factor, so the invariant part is the other's spin-$L/2$ — **exactly the $S^2$ harmonics of degree $\{}ell=L/2$ for even $L$, empty for odd $L$** (dimension count, $L=0..12$) ⇒ the lowest physical mode $L=2$ would land at $\{}ell=1$, the sky dipole, against the flat projection's $7.85$. ⇒ ***So the Hopf structure does NOT enter the projection; its role is the one the paper gives it, traded into $\{}chi$ by the reassignment.*** **AND THE OBSTRUCTION, SAID OF THIS PRESENTATION AS ORDERED**: in it the matter congruence is NULL — *"the at-rest geodesics are the photon bundle"* — so there is no matter observer at a point of the layer, and 'CR_framework''s *"the reassignment promotes one bundle to the fundamental timelike congruence"* is the same operation that trades the Hopf direction into $\{}chi$. ⇒ ***ONE OPERATION READ TWICE: it supplies the observer and removes the sphere, and the mass identity says the same in the other variable. An expansion needs a timelike observer ON a closed layer and the construction offers the two at different values of the one parameter, so it is not bounded here either.*** (node 66's 'r7105' order, which struck 'PO-72' on its own premise, reverted the gate's four errors, restored this line's two re-pointed checks, and re-ordered the expansion *"in the de˜Sitter presentation, where the layer is the $S^3$ and the matter geodesics are the null bundle"* — with the offset-observer half named as established and not to be redone, and the same licence repeated: *"if it is not bounded in that presentation either, say so and stop — but say it of the presentation where the layer IS the sphere"*). *Computes:* **19 checks, all pass, 0.0 s.** **(A)** the calibration, and the two presentations read from 'sec:properframe''s new text rather than recalled; **(B)** the throat radius as an identity on both backgrounds, with 'r7104''s coefficients extracted and the Nariai mass confirmed non-zero; **(C)** the three candidate kernels side by side, with $\{}chi_{\{}rm rec}$, $a\{}sin\{}chi$ and the comb for each and a per-mode $\{}ell$-space table; **(D)** the Hopf selection rule by dimension count over $L=0..12$; **(E)** the obstruction, each half quoted from the paper that states it; **(F)** what is not redone and 'r7104''s withdrawal; **(G)** the scope, read from the files actually opened *Bound:* **IT REPORTS WHERE EACH KERNEL WOULD PUT THE SCALE AND APPLIES NONE OF THEM.** No transfer, spectrum or likelihood is computed; the acoustic instrument is not opened; 'cc66''s comb $302.887$ is used as a **given**, not re-measured, and the sound horizon it implies ($145.33$ Mpc) is derived from it rather than recomputed. 'r7105''s restored paragraph, its new two-presentations text and 'r7102''s two restored checks are all left exactly as the gate wrote them; the offset-observer half is **not redone**, as ordered. **AND THE ONE ROUTE LEFT IN SIGHT IS NAMED AND NOT PROPOSED**: a closed slicing of SdS *at* the Nariai mass rather than of the vacuum substrate — which 'r7104''s identity says would have to be **built** rather than found, there being no maximal $S^3$ at $r_s\{}neq0$. Nothing here starts it *Run:* `python3 receipts/P15_CR_cosmology/P15_the_expansion_in_the_de_sitter_presentation_is_not_bounded_either_bec`

**P15R234** `P15_the_three_grid_comparison_is_like_for_like_and_not_a_ceiling_artefact_because_the_projecti` [OK]

*'sec:refit-bound' — **THE THREE-GRID COMPARISON IS LIKE-FOR-LIKE AND IS NOT A CEILING ARTEFACT: THE PROJECTION'S REACH IS A FIXED MULTIPLE OF THE INVERSE DISTANCE ON EVERY ARM, SO THE ONE COUPLING THAT COULD MANUFACTURE THE SEPARATION CANCELS BY CONSTRUCTION — AND MORE THAN HALF OF IT IS ALREADY PRESENT EIGHT HUNDRED MULTIPOLES BELOW THE CEILING.** Both logged arms carry $k_{\{}max}D_M=3998$ against a reported $\{}ell_{\{}max}=2000$, so the arm with the *shorter* distance is not thereby nearer its ceiling. Re-fitting all three statistics to data truncated at a succession of ceilings, the licensed-minus-forbidden excess runs $-1.0$, $+14.4$, $+51.0$, $+67.9$, $+74.7$, $+84.0$, $+88.3$, $+93.8$ from $\{}ell=850$ to $1996$; about half the full-range gap falls in $850¡\{}ell\{}le1500$ and about a third below $850$, where the truncated fit absorbs it entirely. ⇒ ***AND AT EVERY TRUNCATION THE FORBIDDEN ARM SITS WITHIN ABOUT ONE UNIT OF THE CONTROL***, which a flatness produced by distance from a ceiling would not do. **So what the comparison carries is a rigidity across scales — onset between $\{}ell\{}simeq850$ and $1000$, decisive by $\{}ell\{}simeq1200$, and *not* converged, growing $+5.5$ in the last 96 multipoles.** The paper's word is therefore *lo-*

*calised in onset*; *confined to high multipole* would be wrong, which is the one correction the audit asked of the paper* (node 66's ‘r7101‘ Q1 ordered the audit from the seat that did not produce the comparison, naming the ceiling-artefact mechanism as the thing it most wanted tried against; node 70 pre-registered it and returned all three counts clear. **The figures then had no receipt, which is why this one exists: the measurement is 70's and the assertion that the paper quotes it correctly is the gate's**). *Computes:* **17 checks, all pass.** Nothing of its own: it DRIVES 70's ‘r7101_70_three_grid_audit/audit.py‘ as a subprocess from that file's own directory and parses its printed tables — one $\{}ell$ sampling ($100..1996$) across all three grids with the control byte-identical; the per-arm reach, with $D_M$ differing ($14011.5$ against $13941.6$ Mpc) while $k_{\{}max}D_M$ does not; the eight truncation gaps to a tenth; their monotonicity and non-convergence; the forbidden arm within $1.1$ of the control at every truncation; and the six band shares *Bound:* **NO TRANSFER, SPECTRUM, LIKELIHOOD OR FIT IS COMPUTED HERE, AND NO ARRAY LIBRARY IS LOADED IN THE PROCESS AT ALL** — a second implementation of someone else's audit would be a different number wearing the same name. **And one thing is explicitly NOT asserted:** that the two grids came from one instrument *revision*. Neither stamp carries the source hash, so it is inferred from every shared log line agreeing; the configuration writer that records it landed on ‘cc66‘'s branch in the same round and closes this by construction for the next pair *Run:* `python3 receipts/P15_CR_cosmology/P15_the_three_grid_comparison_is_like_for_like_and_not_a_ceiling_artefact`

**P15R235** `P15_the_controls_base_log_is_banked_beside_the_grids_and_the_peak_heights_concur_with_the_thre` [OK]

*‘sec:refit-bound‘ — **THE PEAK-HEIGHT CELL ‘70‘'s THREE-GRID AUDIT LEFT EMPTY IS FILLED, AND IT IS A *CONCORDANT* SIGN: THE FORBIDDEN CONFIGURATION'S HEIGHTS ARE THE CONTROL'S OWN AT $0.18$ ON TWO DEGREES OF FREEDOM IN THE DATA'S OWN UNITS, WHERE THE LICENSED ONE'S ARE $2.83$ AWAY.** The audit closes *"the control's base has no log banked beside the grids; its peak ratios are not compared here"* — the arm's two logs print $P_1/P_2 = 2.283$ licensed and $2.199$ forbidden, and with no control row the second reading could not be told from its own opposite: either the forbidden heights are BETTER, a contrary sign, or they are THE CONTROL'S, which is what its $\{}chi^2$, its crossings and its longest run already are. ⇒ ***IT IS THE SECOND, AND NOT CLOSE: the control's own base reads $2.196$ / $2.190$, and the forbidden arm's $\{}ell_A$ is the control's to $1.6\{}times10^{-5}$ while the licensed one's is $1.51$ away — so the heights were being read on an arm that had already collapsed in its geometry.*** **AND THE SAME STATISTIC SHOWS THE CELL COULD NEVER HAVE HELD A CONTRARY SIGN.** Against the sky on two degrees of freedom: banked $0.85$, licensed $1.05$, forbidden $0.44$, control $0.59$ — **every one inside $1.1$, and the two ratios do not agree on an ordering.** *The apparent sign came from quoting a $0.084$ difference against a sky number carrying no bar: the sky's own ratios are $2.2564\{}pm0.0772$ and $2.2800\{}pm0.0737$ from the published covariance's own block, so $0.084$ is a fifth of the resolution.* ⇒ **$\{}chi^2$ keeps its standing as the statistic with the information — it separates $278.79$ from $184.99$ where the heights separate $1.05$ from $0.44$** — which is ‘c54.176‘'s conclusion arriving a second time by a second route. *And the control's base log is now banked beside both grids: the ORIGINAL stdout of the run whose ‘.npz‘ IS the banked ‘lcdm_base.npz‘ (one file in three places, md5 ‘5df16bcd‘), so it is the record of the artefact and not a re-run's description of a re-run.* (node 66's ‘r7109‘ (2), which named this cell rather than ordering it: *"the one earlier disagreement between the heights and the three rigidity statistics is currently an empty cell rather than a contrary sign, and it should be one or the other"* — itself ‘70‘'s request carried forward). *Computes:* **All checks pass.** **(1)** the banked log is the log of the banked ARTEFACT: the control's ‘.npz‘ asserted byte-identical across ‘refit_grid185/‘ and both grids, the two banked log copies asserted identical, the log asserted to name its own output, and the audit's blank QUOTED from the audit so the premise is not remembered. The scope gate asks ‘git ls-files –error-unmatch‘ over all twelve inputs rather than ‘os.path.exists‘. **(2)** the four readings recomputed at full precision through the instrument's OWN peak line (‘argrelextrema(Dl, greater, order=3)‘), and asserted to reproduce each banked log's printed pair **to its three decimals and not finer**. **(3)** the sky's pair and its $3.2$–$3.4\{}%$ bars re-derived through ‘c54.176‘'s route from the full $3\{}times3$ covariance block propagated to both ratios at once (correlation $+0.73$ — they share $P_1$). **(4)** the matched statistic: the models carried through CAMB's non-perturbative lensed/unlensed operator (checked at its own $+6.5\{}%$ at $\{}ell=1900$), binned by the likelihood's own binning, the

same peak finder on the same $185$-bin window, and ONE 2-dof number per arm. Asserted: forbidden $¡0.5$ from the control, licensed $¿1.0$, forbidden $¡$ licensed$/4$, every arm $¡1.2$ against the sky. **CONTROL THAT FIRES: the fine-grid unlensed route must reproduce the matched route's sign ($0.046\{}sigma$ against $1.124\{}sigma$), or the result is called an artefact of the binning or the operator.** **(5)** the one run ‘r7109‘ (2) budgeted, spent as the INDEPENDENT CHECK rather than as the record: the instrument at this revision, at the control's own settings, returns the banked control — ‘ls‘, $\{}ell_A$, $D_M$, $r_s$ **bit-identical** and $D_\{}ell$ to $6.8\{}times10^{-15}$, the same cross-machine reduction order the ‘DAMPX‘ pair showed against ‘r4494‘ ($6.4\{}times 10^{-15}$), *reported as relaxed rather than asserted as bit-identical*. ⇒ **And since ‘LEAFREC‘ defaults to $1$ since ‘r7095+cc66.75‘ while the bank predates the flip, the ‘LEAFREC‘ no-op on the control is MEASURED here on the reporting path at production settings rather than inherited.** It degrades with a named reason when the run is absent, PARTS 1–4 not depending on it. *Bound:* **IT DOES NOT MAKE THE HEIGHTS A VERDICT, AND SAYS SO IN ITS OWN HEADLINE:** the concordance is the sign, and the abstention is the other half — no arm is separable from the sky by this statistic, so (b) is reported as a null and not as support. **AND IT NAMES A DEFECT IT DOES NOT SWEEP:** the instrument carries **two sky references in one file** — the non-‘DSCAN‘ print's bare $2.217$ / $2.277$, which is the line the audit read, beside the ‘DSCAN‘ print's $2.256\{}pm3.4\{}%$. *The bare pair is not WRONG — asserted inside $1\{}sigma$ of the propagated one, which ‘c54.176‘ also asserts — it is BARE.* **A dozen registered receipts carry $2.217$ as ‘SKY‘ and the protocol is the paper's, so it is ROUTED to the gate rather than swept.** NOT CLAIMED: that the lensed-model/binned-sky route is the corpus's settled protocol (it is this receipt's, built to match the sky's own route, and the fine-grid route is carried beside it as the control); that $\{}texttt{LEAFGEOM=1}$ is reinstated — ‘r7095‘'s ruling is the gate's and the default is untouched. *Run:* `python3 receipts/P15_CR_cosmology/P15_the_controls_base_log_is_banked_beside_the_grids_and_the_peak_heights`

**P15R236** `P15_the_133_bin_pair_is_computed_and_asserted_and_the_hardcoded_row_carried_the_other_arm` [OK]

*‘sec:refit-bound‘, ‘sec:diffusion-scale‘ — **THE TWO FIGURES THE CORPUS QUOTES FOR THE $133$-BIN COMPARISON ARE COMPUTED AND ASSERTED HERE FOR THE FIRST TIME — THE CONTROL'S $2.10$ PER BIN AND THE $4.16$ BESIDE IT — AND COMPUTING THEM FINDS THE HARDCODED ONE IS THE WRONG RUN'S.** Until now one lived in a docstring HEADLINE (*"35.5 per bin against the control's 2.10"*, which no check reads, repeated in four more receipts' prose) and the other in a literal division inside an f-string — ‘print(f"…{4.16 / 2.10:¿11.2f}x")‘ — in a receipt that computes the two rows beneath it live. ⇒ ***Measured: the control is $\{}mathbf{2.100902}$ per bin on $133$ bins ($\{}chi^2 = 279.4200$) and ‘r6760+cc66.7‘'s own arm is $\{}mathbf{4.185607}$ ($\{}chi^2 = 556.6858$, which is the $556.7$ ‘FOR_66‘ recorded, so the file is identified and not guessed) — i.e. $\{}mathbf{4.19}$, not $4.16$.*** **$4.16$ is the $H_0 = 68.62$ confirmation run ($4.157889$), while the row's own label reads ‘(r6760+cc66.7)‘ and the two rows computed live beneath it load the $68.60$ arm.** *So three rows of a table whose entire purpose is a LIKE-FOR-LIKE ratio carry two different $H_0$ values.* ***AND NOTHING MOVES: $4.16/2.10 = 1.98$ against $4.19/2.10 = 1.99$, and both are the "$\{}times2.0$" the conclusion rests on.*** *A provenance defect, reported as one — and only a computed row could have told the two apart.** (node 66's ‘r7109‘ (3), which refused to reclassify it: *"computed and ASSERTED by a receipt rather than narrated in one headline and hardcoded in another's print — it is the last live item on the thirteen-figure list"* (= ‘r7101‘ Q1)). *Computes:* **All checks pass.** The bin count asserted off the spectra and not off the name ($133$ each, $\{}ell$ $100$–$1298$, all three covering the SAME bins so the ratios are matched); the control asserted at $2.10$ and the arm at $4.19$ **to the two decimals the corpus quotes and not finer**; the arm's $\{}chi^2$ asserted against the $556.7$ of record, so the identification of the file is a check rather than an assumption; the $68.62$ run asserted at $4.16$ **and asserted DISTINGUISHABLE from the $68.60$ run at those two decimals**, without which PART 3's whole distinction would be vacuous; the literal read out of the other receipt's SOURCE so the claim is about the file as it stands; and the $185$-bin rows recomputed, reproducing that receipt's own quoted $696$ unlensed. **CONTROL THAT FIRES: if the two ratios differed at the one decimal the conclusion quotes, the receipt fails itself — *"then the defect is NOT harmless and this receipt's headline is wrong"*.** *Bound:* **IT DOES NOT CLAIM THE $\{}times2.57$ CONCLUSION IS AFFECTED** — that comparison is $185$-bin lensed against $185$-bin unlensed, both computed live there from one arm. **AND IT DOES NOT

CLAIM ‘cc66.7‘ REPORTED THE WRONG NUMBER:** ‘cc66.7‘ reported $4.19$ and ‘FOR_66‘ records its $\{}chi^2$ as $556.7$ — *the reporting was right and the later transcription into a neighbouring receipt's print was not.* Both stale-premise branches degrade with a named reason rather than failing: if the literal is repaired, or the headline no longer carries $2.10$, the receipt says the PART is spent and keeps the assertion that ‘r7101‘ actually asked for. NOT CLAIMED: that a deeper run would leave the figures where they are — they are properties of banked spectra and the published covariance, and what a deeper run moves is the SPECTRA. *Run:* `python3 receipts/P15_CR_cosmology/P15_the_133_bin_pair_is_computed_and_asserted_and_the_hardcoded_row_carri`

**P15R237** `P15_the_lowell_sweep_is_reproduced_by_a_producer_in_the_repository_and_the_bank_is_that_produc` [OK]

*‘sec:lowell‘, ‘sec:scope‘ — **ONE OF ‘70‘’s TWO LOAD-BEARING UNPLACEABLES IS CLOSED: ‘spectra/cc66_lowell_sweep.npz‘ HAS A PRODUCER IN THE REPOSITORY, ALL EIGHT OF ITS BANKED CONFIGURATIONS ARE RE-DERIVED, AND THE BANK IS THAT PRODUCER'S OUTPUT ROUNDED TO FOUR DECIMALS — EXACTLY, ON ALL $56$ VALUES.** ‘70‘’s provenance audit lists it as read by a registered receipt and a published figure with *”no producer in the repository”*, and offers re-derive **or** write the producer in. ⇒ ***Both are done, because each alone is weaker than it looks: a producer with no re-derivation is a claim about a file, and a re-derivation with no producer closes nothing the next time the question is asked.*** **AND THE ENGINE IS NOT A SECOND COPY OF ANYTHING.** The producer PARSES ‘P15_the_low_multipole_depth_gap_closes_and_two_defects_were_cancelling‘ and exec's only its arm-B definitions (‘_RUN‘, ‘armB‘, ‘r0_of‘, the two backgrounds) — *a duplicated engine can drift from the original silently, which is the defect class this corpus keeps naming, and the receipt asserts the duplication has not happened by looking for the harness's own text in the producer.* ***And the residual is NAMED rather than tolerated: the bank stores four decimals, ‘round(live, 4)‘ equals the banked value identically, and the worst raw difference is $4.9\{}times10^{-5}$ — half the last stored digit. So the agreement is not ”within tolerance”; the bank IS the producer's output.**** (node 66's ‘r7109‘ (3) (= ‘r7101‘’s second item), carrying ‘70‘’s provenance audit forward: *”the two load-bearing unplaceables re-derived, ‘cc66_lowell_sweep‘ and ‘c54.182_clpp‘”*). *Computes:* **All checks pass.** **(1)** the producer and its record asserted TRACKED by ‘git ls-files‘, and asserted at source to READ its engine rather than carry one — **with the converse checked too: the receipt fails if the harness's own program text appears in the producer**, and fails if the receipt stops defining ‘_RUN‘/‘armB‘/‘r0_of‘, which is what the producer's reader depends on. **(2)** the producer's ‘CASES‘ table read out of its source (importing it would run it) and asserted **SET-EQUAL** to the banked key set — no key unproduced, no case unbanked, and a count would not have caught either. **(3)** one configuration RE-DERIVED LIVE on every run (~$80$ s), so the producer is exercised and not vouched for. **(4)** all eight read from the producer's own tracked record and **each checked against the BANK rather than trusted** — a record claiming agreement while the bank disagrees fails on that line and not on its own say-so — with the value count asserted at $56$. **(5)** the bank's stored precision DERIVED from the bank itself (the fewest decimals that leave every value unchanged) rather than assumed, and the raw residual asserted $\{}le$ half the last stored digit. *Bound:* **IT DOES NOT CLAIM THE SWEEP'S PHYSICS IS RE-VALIDATED** — that is the registered receipt's job and it does it on every run; this closes a PROVENANCE question, that the banked numbers come from code in this repository and can be made again. **AND IT CLOSES ONE UNPLACEABLE, NOT TWO:** ‘c54.182_clpp‘ is explicitly not closed here and is not claimed to be. *Nor is the four-decimal storage defended — it was not this seat's choice, it loses nothing at these multipoles, and it is named so the next reader does not mistake $5\{}times10^{-5}$ for a disagreement.* NOT CLAIMED: that the DECOUPLED variant is cheap — it is ~$9$ minutes a configuration, which is why the record is banked and only the cheapest case runs live. *Run:* `python3 receipts/P15_CR_cosmology/P15_the_lowell_sweep_is_reproduced_by_a_producer_in_the_repository_and_th`

**P15R238** `P15_the_lensing_potential_re_derives_bit_identically_and_the_three_keys_that_do_not_are_a_cros` [OK]

*‘sec:lensing‘, ‘sec:refit-bound‘ — **THE SECOND OF ‘70‘’s TWO LOAD-BEARING UNPLACEABLES IS PLACED, AND MORE CLEANLY THAN THE QUESTION EXPECTED: ‘spectra/c54.182_clpp.npz‘ RE-DERIVES **BIT-IDENTICALLY** FROM A PRODUCER THAT IS IN THE REPOSITORY — ‘L171x_lensing_potential.py‘, AT ITS OWN DEFAULTS, ON THE CURRENT BACKGROUND.** Zero on all four keys that producer writes (‘Phi‘, ‘k‘, ‘ls‘, and the*

*Limber 'cl' the figure is built from). ⇒ ***So "no producer" is literally true of the FILE and false of the OBJECT: '70''s matcher did not find it because it POSTDATES the artefact it makes — 'c54.184' against 'c54.182' — and writes four of its seven keys.*** **AND THAT ANSWERS BY MEASUREMENT THE QUESTION 66 PUT TO '60':** whether a 'c54'-era lensing potential is admissible on the current background at all. *The artefact predates $\{}texttt{LEAFSCALES}$ and the current background returns the IDENTICAL arrays, so the switch demonstrably does not move this object — the reason being in the producer's own docstring ($\{}Phi(k,a) = \{}Phi(k,a_{\{}rm ref})\{},g(a)/g(a_{\{}rm ref})$ with $g$ a BACKGROUND quadrature and no transfer function imported), which this receipt measures rather than quotes.* ***WHAT DOES NOT RE-DERIVE IS NAMED AND NOT GLOSSED: three of the seven keys — 'cl_exact', 'cl_limber', 'l_exact', the Limber-against-exact cross-check at eight multipoles — have no producer, because 'L171x' computes the LIMBER integral only.*** *The Limber side of that comparison IS re-derived; what has no producer is the EXACT projection it is compared against.* (node 66's 'r7109' (3) (= 'r7101''s second item) and '70''s provenance audit: *"re-derive, or re-point PART B onto a re-derivable lensing source. It is load-bearing, and its era predates 'LEAFSCALES'"*, with 66's added question to '60' about admissibility). *Computes:* **All checks pass.** **(1)** the producer's single 'savez' QUOTED from its source and asserted to write exactly those four keys — **so the three-key gap is DERIVED from the producer rather than observed and left unexplained** — together with its Limber quadrature, which is why no exact side exists to write. **(2)** bit-identity asserted per key on element counts, not on a tolerance: any differing element fails the receipt and says the headline is wrong. **(3)** the unproduced set asserted to BE the Limber-against-exact triple, so a gap that moves is caught rather than re-narrated; and where the cross-check's multipoles sit on the 'ls' grid, 'cl_limber' is checked against 'cl' so the claim that its Limber column IS this object is tested — **and when they do not (they do not), the receipt says the comparison is unavailable and that PART 3 then rests on the key names alone**. **(4)** the gap sized by the reading receipt's FIGURE line, quoted from source. **AND THE READ COUNT GOES THE OTHER WAY — three against two — which the receipt reports BEFORE its conclusion and explicitly declines to rest on**, a read count being the wrong measure of load-bearing. *Bound:* **IT DOES NOT RE-POINT PART B AND DOES NOT PROPOSE TO:** that receipt is 'c54.184''s, and with the object re-deriving bit-identically there is nothing to re-point it onto that it is not already reading. **AND IT DOES NOT BUILD THE EXACT PROJECTION** — new machinery in another seat's instrument — so the cross-check **stays on the provenance list**, which the receipt says in its own closing line rather than declaring the artefact closed. *No live re-derivation runs in the receipt, with the cost MEASURED and not asserted: the full run is ~$9$ min ($220$ modes carried to $\{}eta=4000$) and a reduced 'NKP=12 LMAXPHI=40' run still exceeds two minutes, because the cost is the mode integration and not the mode count — so the record is banked beside the producer, and compared against the BANK rather than trusted.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_lensing_potential_re_derives_bit_identically_and_the_three_keys_t`

**P15R239** `P15_the_layer_is_R_times_S2_on_the_reassigned_chart_and_the_two_presentations_are_two_metrics_` [OK]

*'sec:largescale', 'sec:properframe' — **ON THE REASSIGNED CHART A CONSTANT-$\{}tilde\{}tau$ SURFACE IS $\{}mathbb{R}\{}times S^2$, AT EVERY MASS, AND THAT IS WHAT THE REASSIGNMENT DOES.** The induced metric is $-f(r)\{},\{}mathrm d\{}chi^2+r^2\{}mathrm d\{}Omega^2$ — $(\{}partial_\{}chi r)^2-1=-f(r)$ being an identity on the $E=1$ worldline — with intrinsic Ricci eigenvalues $(0,1/r^2,1/r^2)$: a flat line times a round $S^2$ of radius $r(\{}tilde\{}tau)$, where a round $S^3$ needs all three at $2/a^2$, **computed here and not quoted**. *This is Daryl's own statement of the reassigned metric and the coordinate fact 'r7105' adjudicated, so it is NOT a finding against the corpus: it is pinned so the paper can state the attribution instead of leaving a reader to meet it as a surprise, which happened twice.* ◆ ***AND A REFINEMENT ON THE ROUTING THAT BROUGHT IT HERE, FOUND BY WRITING THE CHECK AS AN IDENTITY AND LETTING IT FAIL:*** *'eq:proper-frame''s Kretschmann is $24/\{}alpha^4+12r_s^2/r^6$, but on the $E=1$ Nariai worldline $A=(r_s\{}alpha^2)^{1/3}$ so $r^6=r_s^2\{}alpha^4\{}sinh^4$ and **the mass cancels** — the excess is $12/\{}alpha^4\{}sinh^4(3\{}tilde\ with no $r_s$ in it. ⇒ **So the departure from de~Sitter is set by the EPOCH and not by the mass: asymptotically de~Sitter at late $\{}tilde\{}tau$, locally de~Sitter at no finite $\{}tilde\{}tau$, whatever the mass** — sharper than the 'only at $r_s=0$' reading that was routed* (node 70's 'r7111+70.1' measured the layer's induced metric and Ricci eigenvalues and routed the consequence to 66; the gate verified both independently in a clean process before landing any-

thing, and the paper's 'sec:largescale' now quotes the induced metric and the eigenvalues, so they need a receipt). *Computes:* **8 checks, all pass.** sympy only, on 'eq:proper-frame' and 'eq:scalefac' and nothing else: the identity $(\{}partial_\{}chi r)^2-1=-f(r)$; the induced three-metric's Ricci tensor; a round $S^3$'s eigenvalues for comparison; the four-metric's Kretschmann scalar against $24/\{}alpha^4+12r_s^2/r^6$ to $3\{}times10^{-16}$ relative at six points in $(\{}tau,r_s,\{}alpha)$ at 40 digits; and the excess's independence of $r_s$ with its late-time limit *Bound:* **NO GATE CLAIMS THE TWO PRESENTATIONS ARE ONE METRIC, AND NONE CLAIMS THEY ARE TWO.** Where the closed $S^3$ the low-multipole floor is built on lives is a question about the construction's foundations: it is carried as a register row, stated there as a question about the construction rather than decided inside a receipt. *'r7103' and 'r7105' are both on the record as this seat getting that boundary wrong in opposite directions, which is why the bound is written into the receipt rather than left to judgement.* Three of this file's own defects are recorded in it rather than quietly fixed: an invalid 'a != b' shortcut in the curvature helper, the four-metric first built in $(\{}tilde\{}tau,\{}chi)$ where 'eq:proper-frame' is a chart in $(\{}tau,\{}chi)$, and a first tolerance of '1e-30' written for an exact symbolic zero where the evaluation returns '3e-16' *Run:* `python3 receipts/P15_CR_cosmology/P15_the_layer_is_R_times_S2_on_the_reassi`

**P15R240** `P15_the_angular_label_crosses_the_lift_untouched_and_the_amplitude_does_not_because_the_lift_i`
[OK]

*'sec:properframe', 'CR_framework''s bead figure (panels (C) and (F)) — **'r7107''s 'Q1' ANSWERED, AND THE OBJECT THAT HOLDS ALL THREE REQUIREMENTS EXISTS: IT IS NOT A CHART BUT THE HARMONICS' OWN PARAMETER, THE COMPLEX CONFORMAL TIME $\{}eta=\{}int d\{}tilde\{}tau/r$ ALONG THE ONE ANALYTIC CURVE. THE ANGULAR LABEL CROSSES THE LIFT UNTOUCHED, EXACTLY — THE AMPLITUDE DOES NOT, AND WHAT IT PICKS UP IS AN ENVELOPE WITH EXACTLY NO PHASE.** **'r7107' withdrew its own 'r7105' 'Q1' (the de˜Sitter presentation *"buries the entire lap, seam to seam"*) and re-posed the question as a requirement: the layer a sphere so there are harmonics, the lap resolved, the lift carried.* **THE PARAMETRISATION, WHICH IS (F)'s INSTINCT TAKEN ONE STEP FURTHER**: arc length not in $r$ but in conformal time, the variable the mode equation is already written in. It holds all three — the layer is a sphere of radius $\{}lvert r\{}rvert$ at every point, each leg has FINITE $\{}eta$-length so the lap is resolved rather than compressed onto the throat, and the lift is carried because its $\{}eta$ is purely imaginary. **AND THE LAP CLOSES IN IT AS AN EXACT 30-60-90 TRIANGLE: the three legs' conformal lengths are in the ratio $1:\{}sqrt3:2$** — collapse $c_0B(\{}tfrac13,\{}tfrac16)/4$ real, lift $c_0B(\{}tfrac16,\{}tfrac12)/2$ purely imaginary, expansion $c_0B(\{}tfrac13,\{}tfrac16)/2$ real — *proved symbolically by reflection alone ($\{}Gamma(\{}tfrac12)^2/\{}Gamma(\{}tfrac13)\{}Gamma(\{}
and numerically to $5\{}times10^{-22}$*, so $\{}Delta\{}eta_{\{}rm coll}+\{}Delta\{}eta_{\{}rm lift}$ has modulus exactly $\{}lvert\{}Delta\{}eta\{}rvert_{\{}rm exp}$ and argument exactly $-\{}pi/3$, with $\{}sqrt3=\{}alpha/r_N$ the construction's own ratio. **(1) THE LABEL PASSES THROUGH UNTOUCHED AND NEEDS NO INTEGRATION**: the angular block is $r^2d\{}Omega^2$, the lift moves $r$ at fixed angles, and $(L,M)$ is a DISCRETE label on a CONTINUOUS path. **(2) THE AMPLITUDE DOES NOT, AND IT IS A SELECTION RULE RATHER THAN A FILTER**: on the lift $\{}eta=-is$, so $u"+(k^2-a"/a)u=0$ becomes $u_{ss}=(k^2+a_{ss}/a)u$ — real, never oscillating — and at the seam $a\{}to(s_{\{}rm tot}-s)^2/3\{}cdot2^{4/3}$ with the two behaviours in $\{}varphi=u/a$ a CONSTANT and $(s_{\{}rm tot}-s)^{-3}$. ⇒ ***of the two-dimensional space of mode data at the turnaround, exactly a one-dimensional subspace reaches $r=0$ regular and continues into the expansion leg.*** **AND ON IT THE TRANSMITTED AMPLITUDE IS A PURE-NUMBER FUNCTION OF $k$ ALONE**: $T(0)=1$ **exactly** (the monopole, $u=a$ solving the lift equation identically — the control that had to return the affirmative) and $T(k)\{}to2^{7/3}k^2e^{-k\{},s_{\{}rm tot}}$ with $s_{\{}rm tot}=\{}Gamma(\{}tfrac16)\{}sqrt\{}pi/\{}Gamma(\{}tfrac23)\{}sqrt3\{},2^{1/3}=3.3387380236$, **no $\{}Lambda$, $H_0$, $\{}Omega_m$, mass or epoch anywhere in it**, $\{}alpha$ cancelling between $d\{}tilde\{}tau$ and the amplitude $A$. ⇒ ***SO THE ORDER'S SECOND BRANCH FIRED, AND THE HONEST READING IS NOT THE ONE IT HOPED FOR: $\{}operatorname{Re}\{}Delta\{}
IDENTICALLY, so what the harmonics pick up is an envelope and NO PHASE — and a comb is a phase.*** The mildest damping of any $L¿0$ is $L=1$ at $T=7.00\{}times10^{-2}$, a factor $14.3$; $L=2$ is $4.16\{}times10^{-3}$; the first acoustic peak's $L=220$ is $e^{-725}$. ⇒ **WHAT THE LIFT IS, STATED AS THE AFFIRMATIVE IT IS: the construction's own smoothing step. The seam is isotropic BY IDENTITY and not by tuning, and the acoustic struc-*

*ture cannot be inherited from before the lift — it has to be generated on the expansion leg** (node 66's ‘r7107‘ order, ‘Q1‘ — *”how are the layer's angular harmonics carried along the bead through the lap, with the lift included?”*, posed as a requirement after ‘r7107‘ withdrew ‘r7105‘'s own ‘Q1‘ as sending this line to *”the presentation that buries the sector”*, with the three-requirement table, the instruction to *”use the arc-length parametrisation of (F) if that is what keeps the bundle single-valued”*, the explicit question *”whether the lift does anything to the angular sector at all”* — *”if the harmonics pass through it untouched, that is the answer; if they do not, whatever they pick up is the thing the acoustic sector has been missing”* — and the stop licence meant a third time, *”but say it of the requirement, not of a chart”*). *Computes:* **29 checks, all pass, 26 s.** **(A)** the branch fixed by READING the figure, with $\{}sinh(X-i\{}pi/2)=-i\{}cosh X$ verified to $4\{}times10^{-62}$; **(B)** the three legs' lengths, their closed forms to $10^{-21}$, the $1:\{}sqrt3:2$ ratio to $5\{}times10^{-22}$ and symbolically, and the lap's closure in the complex $\{}eta$-plane; **(C)** $\{}operatorname{Re}\{}Delta\{}eta$ on the lift — the exact float zero — **with the PRINCIPAL branch kept as the control that must come back non-zero, and it returns $-2.891$, $87$ per cent of the lift's length**; **(D)** the lift's mode equation, its seam asymptotics approached over three decades of $\{}delta$, and the potential's sign change located at exactly one third of the lift ($\{}int_0^{\{}pi/3}\{}cos^{-2/3}=\{}tfrac13\{}int_0^{\{}pi/2}\{}sin^{-2/3}$ to $4\{}times10^{-22}$); **(E)** $T(k)$ for $L=0..1000$ by Riccati integration started from the EXACT seam solution $u=\{}sqrt\{}sigma I_{3/2}(k\{}sigma)$, with matching-point independence to $8\{}times10^{-10}$, the $k=0$ control at exactly $1$, the exponent recovered as $-s_{\{}rm tot}$ to $0.09$ per cent, the prefactor converging on $(\{}tfrac73)\{}ln2$, and **the curvature convention MEASURED rather than assumed away** ($k^2\{}to k^2-3$ shifts $\{}ln T$ by $3s_{\{}rm tot}/2k$, as predicted, at $L=10,100,400$); **(F)** the parameter-freeness on both stated backgrounds and the consequence for the acoustic sector; **(G)** the scope, read from the files actually opened *Bound:* **IT IS ABOUT MODES CARRIED THROUGH THE LIFT AND SAYS NOTHING ABOUT MODES GENERATED ON THE EXPANSION LEG**, to which none of it applies — which is exactly where it then says the acoustic structure must come from, so that is a named gap and not a result. **AND THE CONVENTION IS CARRIED RATHER THAN BURIED**: $k^2=L(L+2)$ is the unit-$S^3$ scalar Laplacian eigenvalue and the equation used is the flat-form $u”+(k^2-a”/a)u=0$; a curvature-convention shift $k^2\{}to k^2+O(1)$ is measured to move $\{}ln T$ by $O(1/k)$ and never the exponent, but the $O(1)$ prefactor and the exact $T(0)=1$ are convention-dependent and are stated as such. The bead here is the SINGLE analytic curve at the Nariai amplitude, so nothing in it is a statement about a mass spectrum. **A $\{}sqrt2$-class branch slip is recorded rather than hidden**: the principal branch $(-i)^{2/3}=e^{-i\{}pi/3}$ was taken first and would have MANUFACTURED a real phase across the lift; panel (C)'s own $r=-(2M\{}alpha^2)^{1/3}\{}cosh^{2/3}$ fixes the branch at $e^{i\{}pi}=-1$, and the wrong branch is kept below as a control. No transfer, spectrum, kernel or likelihood is computed, the acoustic instrument is not opened, ‘cc66‘'s table is not touched, and the paper is READ and not edited; ‘r7106‘'s computations stand but its SETTING is superseded where ‘r7107‘ withdrew its own order **ONE FIGURE CORRECTED IN PLACE AT ‘r7112‘, ON ‘r7111‘'s FINDING, AND THE CONCLUSION IS UNTOUCHED.** *The headline said ”the first acoustic peak's ‘$L=220$‘ is ‘$e^{-725}$‘”, which takes the SKY multipole for the LAYER degree — exactly the reading ‘sec:throat‘ guards against by name.* ⇒ **‘eq:lowell‘ is the map and it is a factor of ‘$2.7737$‘ in ‘$k$‘: the sky's ‘$\{}ell=220.6$‘ sits at ‘$L=78.5$‘ and the envelope there is ‘$e^{-255}$‘.** *The receipt now derives it through ‘eq:lowell‘ and keeps the wrong reading as a control that must come back WRONG, which it does by ‘470‘ in the exponent. Both figures are annihilation and the argument is carried by ‘$L=1$‘ at ‘$14.3$‘, so this was a figure to fix and not a result to revisit; ‘30‘ checks now, all pass.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_angular_label_crosses_the_lift_untouched_and_the_amplitude_does_n`

**P15R241** `P15_the_progenitor_spectrum_is_defined_before_the_lift_so_the_envelope_applies_to_it_and_the_h` [OK]

*‘sec:scope‘, ‘sec:intro‘, ‘sec:throat‘, ‘eq:lowell‘ — **‘PO-73‘ ‘Q1‘ ANSWERED BY LOCATING THE SUPER-HORIZON ERA IN THE BEAD'S OWN CONFORMAL TIME: THE PROGENITOR'S SPECTRUM IS DEFINED *BEFORE* THE LIFT, FOR EVERY ‘$L\{}ge1$‘ AND NOT MARGINALLY, SO ‘r7108‘'s ENVELOPE APPLIES TO IT — ‘sec:scope‘'s ”CARRIED ACROSS UNALTERED” IS THE SENTENCE THAT MOVES AND ‘sec:intro‘'s SINGLE SUPPLIER STANDS.** *A mode's spectrum is fixed while it is SUPER-HORIZON, so the question is where ‘$a”/a¿k^2$‘ along the curve.* **‘$a”/a$‘ IN THREE CLOSED FORMS, EACH DIF-*

*FERENTIATED FROM THE METRIC AND NOT RECALLED** (symbolic residual identically zero on all three): ‘$2^{-1/3}(\{}cosh^2x+\{}tfrac13\{}sinh^2x)/\{}cosh^{2/3}x$‘ on the collapse leg, ‘$2^{-1/3}(\{}sinh^2x+\{}tfrac13\{}cosh^2x)/\{}sinh^{2/3}x$‘ on the expansion leg, and ‘$-2^{-1/3}(\{}cos^2t-\{}tfrac13\{}sin^2t)/\{}cos^{2/3}t$‘ on the lift — **and the single constant in all three is the pure number ‘$2/3c_0^2=2^{-1/3}$‘.** **ONE CURVE AND NOT THREE**: at the turnaround the collapse and lift forms agree to the last bit, difference exactly ‘$0$‘; at the seam both go as ‘$2/\{}sigma^2$‘ with the same coefficient and **opposite signs**, which is not a mismatch but the content — ‘$\{}eta$‘ is imaginary on the lift, so the curvature term flips exactly where the character does. **AND THE HORIZON SCALE IS A PURE NUMBER BOTH LORENTZIAN LEGS SHARE.** The collapse form falls monotonically to ‘$2^{-1/3}$‘ at the turnaround; the expansion form attains the **same** value as a MINIMUM at ‘$x=\{}tfrac12\{}ln2$‘ exactly (the critical condition reducing to ‘$4\{}cosh2x=5$‘, i.e. ‘$e^{2x}=2$‘), and that point sits exactly ‘$\{}lvert\{}Delta\{}eta\{}rvert_{\{}rm coll}$‘ after the seam — **by ‘r7108‘’s ‘$1:\{}sqrt3:2$‘, the midpoint of the expansion leg.** ⇒ ***‘$L_{\{}rm hor}=-1+\{}sqrt{1+2^{-1/3}}=0.3393$‘: away from the three divergences the ONLY super-horizon harmonic is the monopole, with ‘$L=1$‘ already sub-horizon by ‘$3.78$‘ and ‘$L=2$‘ by ‘$10.08$‘ — a gap no ‘$O(1)$‘ convention shift reaches.*** **AND THE TWO LEGS’ CROSSINGS ARE THE SAME DISTANCE FROM THEIR OWN DIVERGENCE**: for every ‘$k$‘ the conformal distance from the collapse leg’s far-past end to the horizon EXIT equals the distance from the seam to the RE-ENTRY, over ‘$L=1..500$‘, to ‘$6\{}times10^{-14}$‘; the second exit mirrors the first about the midpoint; and both go as **‘$\{}sqrt2/k$‘, the same length ‘r7108‘ found for the lift’s own crossover**, so one length governs all three legs. ⇒ *** **THE ANSWER: BEFORE.** The sky’s first peak, carried through ‘eq:lowell‘ to ‘$L=78.5$‘, has its spectrum fixed within the FIRST ‘$0.92$‘ PER CENT of the collapse leg — so ‘$99.1$‘ per cent of the collapse leg **plus the entire lift** lie between that locus and the seam. *** ⇒ **So the envelope applies, and ‘sec:scope‘’s own argument is sound only as far as it goes: a finite ‘$r_*$‘ rules out a divergent PHASE and says nothing about a FACTOR.** *The control returns the affirmative and is what makes this a measurement rather than a complaint: the inheritance claim is EXACTLY true at ‘$k=0$‘ — the monopole is super-horizon everywhere and ‘r7108‘’s ‘$T(0)=1$‘ identically — and exactly true of no other mode.* **And the size is named rather than softened: a spectrum multiplied by ‘$e^{-255}$‘ at the first peak is not a corrected inheritance, it is no inheritance** — which is a statement about the two sentences and **not** a proposal about what supplies the spectrum instead* (node 66’s ‘r7111‘ order, ‘PO-73‘ ‘Q1‘ — opened off this line’s own ‘r7108‘ sentence *”the acoustic structure cannot be inherited from before the lift; it has to be generated on the expansion leg”*, which the gate read against ‘sec:scope‘’s *”‘$A_s$‘ and ‘$n_s$‘ are the progenitor collapse’s spectrum carried across UNALTERED”* and ‘sec:intro‘’s *”there is one boundary-condition supplier, not two”*: ***”THE SAME FINITENESS, READ TO OPPOSITE CONCLUSIONS”***, with what would settle it stated outright — *”where in the bead’s conformal time the progenitor’s perturbation spectrum is defined, and whether that locus is before or after the lift’s segment”* — and with nothing withdrawn on the gate’s reading pending this row. The same order carried the figure correction to ‘r7108‘ and routed ‘70‘’s tolerance item). *Computes:* **17 checks, all pass, 1.4 s.** **(A)** the three closed forms for ‘$a”/a$‘, each obtained by differentiating the metric in conformal time with ‘sympy‘ and each with symbolic residual identically zero, plus ‘$2/3c_0^2=2^{-1/3}$‘ symbolically; **(B)** the junctions — the turnaround to the last bit and the seam over three decades of ‘$\{}delta$‘, with the sign flip asserted as the content; **(C)** the expansion leg’s minimum, its critical condition reduced symbolically to ‘$4\{}cosh2x=5$‘, its value against ‘$2^{-1/3}$‘ and its position against both ‘$\{}lvert\{}Delta\{}eta\{}rvert_{\{}rm coll}$‘ and half the expansion leg; **(D)** five harmonics’ crossing loci by ‘mpmath‘ bisection and quadrature, the two-leg equality and the ‘$\{}sqrt2/k$‘ law; **(E)** the answer, with ‘eq:lowell‘’s stretch computed on BOTH stated backgrounds and the ‘$k=0$‘ control; **(F)** ‘70‘’s routed item checked against the current source rather than asserted; **(G)** the scope, read from the files actually opened— **‘r7118+60.1‘: two of this row’s own gates are UN-PINNED from ‘prop:transmit‘’s wording.** ‘r7111‘ said ‘sec:scope‘’s inheritance sentence would stand until this row answered, and two gates asserted that wording verbatim (‘carried across unaltered‘). ⇒ **The row answered, ‘r7117‘ carried the factor into the sentence, and both gates went red because the thing they asked for was granted.** They now assert the clause this receipt actually reasons against — *”the crossing accumulates no divergent phase”*, present in ‘prop:transmit‘ either way and exactly what a finite tortoise length buys — plus this receipt’s

own computed locus; the earlier wording and the fact that it is gone are **printed and not asserted**. *Fifth instance in this sector of an assertion resolving finer than its subject, and the first where the subject was a sentence another seat owns and had been asked to change.* *Bound:* **IT LOCATES THE SUPER-HORIZON ERA AND NOTHING MORE.** It does **not** propose what supplies the spectrum if the progenitor's is annihilated, does not reassign the source spectrum, does not recompute ‘r7108‘'s ‘$T(k)$‘, and does not edit the paper — ‘sec:scope‘'s sentence is left exactly as the gate wrote it, which is what ‘r7111‘ instructed while the row is open. **The criterion is the standard one and is stated rather than assumed**: a mode's amplitude is frozen while ‘$k^2\{}ll a"/a$‘ and evolves while ‘$k^2\{}gg a"/a$‘, with ‘$k^2=L(L+2)$‘ the unit-‘$S^3$‘ Laplacian eigenvalue, the convention ‘r7108‘ carried; a curvature-convention shift of ‘$O(1)$‘ moves ‘$L_{\{}rm hor}$‘ by ‘$O(1)$‘ and reaches none of the conclusions, the gap from ‘$0.34$‘ to ‘$L=1$‘ being a factor of ‘$3.78$‘ in ‘$k^2$‘. The bead is the single analytic curve at the Nariai amplitude, so nothing here is a statement about a mass spectrum. **And the one route left in sight — modes GENERATED on the expansion leg, which would need a source the bead does not carry — is NAMED AND NOT PROPOSED**; nothing here starts it *Run:* `python3 receipts/P15_CR_cosmology/P15_the_progenitor_spectrum_is_defined_before_the_lift_so_the_envelope_ap`

**P15R242** `P15_the_constant_r_foliation_carries_the_sphere_across_the_lap_because_the_layers_shape_invari` [OK]

*‘sec:largescale‘'s continuation conjecture, ‘sec:properframe‘, ‘prop:flat‘ — **‘PO-74‘ DISCHARGED IN THE DIRECTION THE PAPER STATES IT: THE CONSTANT-‘$r$‘ FOLIATION CARRIES THE SPHERE ACROSS THE LAP, AND THE DEMONSTRATION IS ONE PURE NUMBER.** *‘sec:largescale‘ states the continuation **as a conjecture and not as a theorem**, with the demonstration named as work the paper does not carry.* **THE LAYER'S SHAPE INVARIANT IS THE SAME AT EVERY POINT OF THE BEAD**: a round ‘$S^3$‘ of radius ‘$r$‘ has ‘$\{}mathcal R=6/r^2$‘ and ‘$V=2\{}pi^2r^3$‘, so the scale-free combination ‘$\{}mathcal R\{},V^{2/3}=6(2\{}pi^2)^{2/3}=43.8232$‘ **carries no ‘$r$‘ at all** — ‘$r$‘ cancelling symbolically — and evaluated on all three legs it returns that number to ‘$10^{-12}$‘. ⇒ ***Because the angular block depends on ‘$r$‘ only through ‘$r^2$‘ it is BLIND to the sign of ‘$r$‘, so the collapse leg's negative radii carry the same layer: the sphere does not change shape across the lap, it changes size, and the foliation label is ‘$\{}lvert r\{}rvert$‘*** — sweeping ‘$\{}infty\{}to A$‘ on the collapse leg, ‘$A\{}to0$‘ on the lift and ‘$0\{}to\{}infty$‘ on the expansion, monotone on each. **AND THE ‘$\{}mathbb{R}\{}times S^2$‘ READING IS RECOVERED AS THE REASSIGNMENT'S OWN IMAGE, WHICH IS WHAT THE ORDER ASKED FOR.** At fixed ‘$r$‘ the reassigned layer ‘$-f(r)\{}dd\{}chi^2+r^2\{}dd\{}Omega_2^2$‘ is the ‘$S^3$‘ ‘$r^2[\{}dd\{}chi^2+\{}sin^2\{}!\{}chi\{},\{}dd\{}Omega_2^2]$‘ under exactly two substitutions — ‘$r^2\{}sin^2\{}!\{}chi\{}to r^2$‘ (the polar warp removed, so every ‘$S^2$‘ sits at the areal radius) and ‘$r^2\{}to-f(r)$‘ on the ‘$\{}chi$‘ block — with the **equatorial ‘$S^2$‘ of radius ‘$r$‘ on both sides**. *And the zero eigenvalue is produced by the UNWARPING ALONE: removing ‘$\{}sin^2\{}!\{}chi$‘ and changing nothing else already gives ‘$(0,1/r^2,1/r^2)$‘ and scalar ‘$2/r^2$‘, identical to the reassigned layer* — so ‘70‘'s ‘r7111+70.1‘ eigenvalues, **reproduced here from the metric by an independent derivation rather than quoted**, are what those two substitutions produce and not a competing answer. **THE LIFT IS THE JOIN AND IT CARRIES A REAL, ROUND SPHERE**: ‘$\{}dd\{}tilde\{}tau=i(2\{}alpha/3)\{}dd v$‘ makes ‘$-\{}dd\{}tilde\{}tau^2=+(2\{}alpha/3)^2\{}dd v^2$‘ positive, so the four-geometry is **Riemannian** there, while ‘$r$‘ is real on the branch panel (C) fixes and ‘$r^2\{}dd\{}Omega_3^2$‘ is a real round ‘$S^3$‘ throughout. ⇒ ***The signature of the time direction turns through a right angle and the angular character does not move at all — which is ‘r7115‘'s reading of ‘r7108‘ made explicit.*** ⇒ **AND THE SEAM HIDES THE CHART, NOT THE CHARACTER**: at ‘$r=0$‘ exactly the layer is a point (‘$V\{}to0$‘, ‘$\{}mathcal R\{}to\{}infty$‘), but the shape invariant is that same constant at every ‘$r\{}neq0$‘ **six orders of magnitude either side**, so the divergence is the areal coordinate degenerating — which ‘P15‘'s own abstract already says of the branch point — and **not** a change in what the layer is. *The licensed terminal branch is therefore NOT taken.* **AND ONE PLACE THE TWO READINGS ARE NOT INTERCHANGEABLE, AS A FINDING RATHER THAN A CAVEAT**: at the Nariai mass ‘$f$‘ has a DOUBLE zero at ‘$r_N=\{}alpha/\{}sqrt3$‘ (‘$f(r_N)=f'(r_N)=0$‘, symbolically), so the reassigned layer's ‘$\{}chi$‘ block vanishes **exactly at the merged horizon**, while on the ‘$S^3$‘ side ‘$r_N$‘ is an ordinary layer the foliation passes through without incident. ⇒ ***The two metrics do not even degenerate in the same places — the sharpest form of ‘r7115‘'s own point that*

*the reassignment is not a diffeomorphism***— and *‘r7117‘’s own picture REFRAMES exactly that: **the cosmological seam is NOT ‘$r=0$‘ but the deceleration-to-acceleration INFLECTION, computed here to sit at ‘$\{}cosh 2u=2$‘ where ‘$r = A\{},2ˆ{-1/3} = \{}alpha/\{}sqrt3 = r_N$‘ with residual identically zero**, so the two readings degenerate AT the handover and nowhere else along the foliation; **and ‘$r=0$‘ is a one-parameter family of events, not a single point**, the corpus’s own step (iv) giving ‘$\{}partial_\{}chi r=\{}partial_\{}tau r$‘ identically and so the ‘$45ˆ\{}circ$‘ tilt of constant ‘$r$‘ against ‘$\{}tau$‘. *The fixed-‘$\{}tilde\{}tau$‘ computation is kept; what is withdrawn is the wording ”at ‘$r=0$‘ the layer is a point”, which is about one layer and not about the locus.** (node 66’s ‘r7115‘ order, ‘PO-74‘, opened as ‘PO-73‘’s remainder after the author settled ‘PO-73‘ and vindicated ‘r7108‘ on the ground it was doubted on (*”the lift’s purely imaginary conformal time — the Euclidean null — is precisely what keeps the reassignment well defined across the lap”*). The order names what would discharge it — *”carry the layer’s ‘$Sˆ3$‘ character along the bead’s own conformal time from one horn, through the lap, to the other ... with the ‘$\{}mathbb{R}\{}times Sˆ2$‘ reading recovered at each point as the reassignment’s own image rather than as a competing answer”* — licenses the other outcome as a result (*”if something at the seam genuinely obstructs the sphere’s character rather than merely hiding it, that is a result and the papers carry it”*), and names the instrument: *”your own parametrisation ... a continuation along a curve you have already built, with the angular block ‘$rˆ2\{}dd\{}Omegaˆ2$‘ carried on it — not a new construction”*— and **RESTATED UNCHANGED IN SUBSTANCE BY ‘r7117‘**, which supplies the author’s own geometric picture: *”the ‘$Sˆ2$‘ portion being present at constant ‘$r$‘ throughout is the author’s statement that the hard part is the ‘$\{}chi$‘ direction alone, which is where the tilt lives”*, and corrects two words in this receipt’s first draft (the seam, and the single point)). *Computes:* **24 checks, all pass, 1.7 s.** **(A)** both induced three-geometries obtained by differentiating their metrics with ‘sympy‘ — Christoffels, Ricci tensor, mixed eigenvalues, scalar and volume — the ‘$Sˆ3$‘ returning ‘$6/rˆ2$‘, three equal eigenvalues ‘$2/rˆ2$‘ and ‘$2\{}piˆ2rˆ3$‘, the reassigned layer ‘$(0,1/rˆ2,1/rˆ2)$‘ and ‘$2/rˆ2$‘; **(B)** the two substitutions, the equatorial radius on both sides, and **the unwarping alone shown to produce the zero eigenvalue**; **(C)** the shape invariant, ‘$r$‘ cancelling symbolically and the number returned on all three legs including negative ‘$r$‘, with the foliation label’s monotonicity on each; **(D)** both junctions — ‘$rˆ2$‘ continuous at the turnaround with its conformal derivative vanishing from both sides, and the seam approached monotonically from ‘$r¡0$‘ and ‘$r¿0$‘ — plus the lift’s signature; **(E)** the seam, with the invariant held six orders either side and the paper’s own ”areal coordinate degenerating” located; **(E2)** ‘r7117‘’s correction carried as computation — ‘$\{}ddˆ2r/\{}dd uˆ2 \{}propto (\{}cosh 2u-2)$‘ by symbolic differentiation, so the deceleration-to-acceleration INFLECTION sits at ‘$\{}cosh 2u=2$‘ (‘$\{}sinhˆ2u=\{}tfrac12$‘, ‘$\{}coshˆ2u=\{}tfrac32$‘), where the areal radius is ‘$A\{},2ˆ{-1/3}=\{}alpha/\{}sqrt3=r_N$‘ with **residual identically zero** and the second derivative running ‘$-\{}to+$‘ — so the cosmological seam IS the merged horizon and (F)’s double zero is the chart registering the handover, not an asymmetry; **(E3)** the ‘$45ˆ\{}circ$‘ tilt from the corpus’s own step (iv), ‘$\{}partial_\{}chi r=\{}partial_\{}tau r$‘ identically, so ‘$r=0$‘ is a one-parameter family of events and not a single point; **(F)** the Nariai double zero; **(G)** the controls; **(H)** the scope, read from the files actually opened— **‘r7118+60.2‘: two gates of section (E) UN-PINNED, because ‘r7121‘ discharged the conjecture they asserted was still open.** ‘r7121‘ struck ‘PO-74‘ and wrote this result into ‘sec:largescale‘ as ‘eq:shape-invariant‘, citing this receipt, so the two gates asserting that the paper still states the continuation *”as a conjecture and do not claim it as a theorem”* with the demonstration *”work this paper does not carry”* **went red the moment the paper carried what this receipt supplied.** ⇒ The status wording is printed and no longer asserted; what they assert now is the **foliation sentence**, which this result is about and which stands either way, plus — as a **disjunction** — that ‘sec:largescale‘ is in exactly one of its two legitimate states (still a conjecture *with* the demonstration owed, or carrying it *and* citing this receipt). *A real test and not a tautology: it fails if the paper has neither and fails if it has both.* **Sixth instance in this sector, and the pattern is now named: a gate that asserts the STATUS of a paper sentence this receipt is asking to change is a gate that fails on its own success — assert the load-bearing clause, never the status.** *Bound:* **IT ESTABLISHES THE LAYER’S INTRINSIC GEOMETRY ALONG THE FOLIATION AND THE EXACT RELATION BETWEEN THE TWO READINGS OF IT — NOT MORE.** It does **not** claim the reassignment is a diffeomorphism or an isometry of one metric: ‘r7115‘ asserts it is neither, and **(F)** is evidence on that side. It computes **no** curvature invariant of either FOUR-geometry

for comparison across the registers, which ‘r7115‘ says would register only that fact. No transfer, spectrum, kernel, likelihood or mode amplitude is computed; ‘r7108‘’s ‘$T(k)$‘ and ‘r7112‘’s horizon structure are not recomputed; the source spectrum is not reassigned and the paper is READ and not edited. The bead is the single analytic curve at the Nariai amplitude, so nothing here is a statement about a mass spectrum. **AND A SLIP OF THIS LINE’S IS RECORDED RATHER THAN QUIETLY FIXED**: the must-come-back-wrong control was first written as ‘$\{}operatorname{diag}(1,r^2,r^2\{}sin^2\{}!\{}theta)$‘ over ‘$(\{}chi,\{}theta,\{}phi)$‘ with ‘$r$‘ a FREE SYMBOL, which is not flat ‘$\{}mathbb{E}^3$‘ at all but ‘$\{}mathbb{R}\{}times S^2$‘ again, and duly returned ‘$2/r^2$‘. *Flat ‘$\{}mathbb{E}^3$‘ needs the radial coordinate to be the one being differentiated.* ⇒ **The warning is the useful part: the difference between the two readings of the layer is exactly whether the sphere’s radius varies along the third direction, so writing a constant radius there is how one accidentally builds the cylinder.** A second gate of this line’s was also caught asserting an arbitrary cut rather than the approach, and now asserts monotone descent on both sides of the seam — the fourth instance in this sector of an assertion resolving finer than its subject *Run:* `python3 receipts/P15_CR_cosmology/P15_the_constant_r_foliation_carries_the_sphe`

**P15R243** `P15_the_corpus_places_every_perturbation_source_before_the_branch_point_and_everything_after_` [OK]

*‘sec:scope‘’s owed expansion-leg source, ‘sec:interior‘ (‘P16‘), ‘sec:frontiers‘ (‘P07‘) — **‘PO-75‘’s FIRST STEP, WHICH WAS ORDERED AS A READ AND NOT A COMPUTATION: WHERE DOES THE CORPUS ALREADY PUT PERTURBATION SOURCES AFTER THE BRANCH POINT? IT PUTS NONE THERE.** Every source the corpus carries sits **before or at** the crossing, in the progenitor’s interior; everything it places **after** the branch point is **transfer** — linear in an amplitude it is handed — so nothing already built supplies anisotropic content on the expansion leg. ⇒ ***The whole read reduces to ONE FALSIFIABLE COUNT: across all four candidate documents there is **exactly one** sentence pairing the expansion leg with a ‘source‘, and it is the one saying the construction has not got one*** — ‘sec:scope‘’s *”it requires a source the single Nariai worldline of this construction does not carry — named here as owed and developed in neither this paper nor the bead”*. (1) **‘P14‘ is not a candidate and that is measured**: ‘perturbation‘ appears **once**, inside a ‘\{}bibitem‘ naming ‘P15‘’s own title, with ‘anisotrop‘ and ‘isocurvature‘ at zero — its subject is the chiral wall and the seating of the generations, not the spectrum of fluctuations. (2) **‘P16‘ carries the whole apparatus and all of it is source-side**: the parity split by species (dust the entire even part, radiation the entire odd), the unipotent monodromy at the crunch with the scalar residue ‘$-4\{}pi i/\{}rho$‘ **twice** the tensor’s, the content as *”a nearly scale-invariant adiabatic spectrum processed by ordinary structure formation”*, and the harmonic label *”passing the branch point unchanged”*. (3) **And its own negatives are explicit, so this is not an inference from silence**: *”This construction supplies no free oscillation”*, *”The construction contains no such background”*, and the recursion *”does not run on modes”* because *”the first clause of that question has no object”*. (4) **‘P15‘ places the isotropization on the APPROACH**: *”The isotropy is earned on the approach and preserved by the crossing”*, so *”how much anisotropy survives is fixed by the approach against the leg’s duration”* — *”a property of the progenitor’s collapse history rather than of the crossing”*. (5) **Three things the corpus DOES place after the branch point, and all three are transfer and not source**: free-streaming neutrinos’ anisotropic stress (‘C6_neutrino_term‘), whose effect is ‘$k$‘-INDEPENDENT — *”a ‘$k$‘-independent correction leaves a flat envelope flat”*; the baryon loading’s odd/even asymmetry *”imprinted afterwards, on the expansion side”* (‘C5b_baryon_term‘), *”ordinary content physics on the observable leg”*; and the acoustic evolution running FROM the branch-point datum. ⇒ *a phase shift, a suppression, a loading ratio and an oscillation all **modulate** a spectrum; none **originates** one, so all three act on zero and return zero if the arriving amplitude is annihilated.* ‘P07‘’s seeding is one register away — what it seeds is the **congruence**, *”seeded at the finite-curvature horizon seam”*, while *”the cosmology it seeds begins at the branch point”*. **AND THE READ TURNS UP ONE FORK, STATED AND NOT RESOLVED**: the corpus carries **two** computed transfers between the legs — ‘C21_the_superhorizon_transfer_closes‘ giving ‘$\{}Phi\{}to\{}Phi_i$‘ for **every** ‘$k$‘ on ‘$x=k\{}eta/\{}sqrt3$‘, the radiation era’s **real** conformal time, and this line’s ‘r7108‘ giving the lift’s envelope ‘$2^{7/3}k^2e^{-k\{},s_{\{}rm tot}}$‘ on the bead’s own ‘$\{}dd\{}eta=\{}dd\{}tilde\{}tau/r$‘, **purely imaginary** along the lift. ⇒ ***Two different paths between the same two endpoints, with no statement anywhere of which one a physical mode propagates on. Neither is wrong on its own terms; ‘PO-75‘ turns on it and this receipt adjudi-*

*cates neither*** — *routed to the channel with both branches and a recommendation, because it asks which description the mode sector is **defined** on, which is an order and not an inference* (node 66's ‘r7121‘ order, ‘PO-75‘, opened as ‘r7112‘’s remainder after ‘PO-74‘ was discharged and struck — *”what carries anisotropic content into the expansion leg”* — with its first step named and its reason given: **”AND THE FIRST STEP IS A READ, NOT A COMPUTATION, WHICH IS WHY IT IS ORDERED THIS WAY: where does the corpus already put perturbation sources after the branch point? ‘P14‘’s matter sector and ‘P16‘’s collapse worldline and thermal history are the candidates, and both are built. Read them before computing anything — the row’s own discharge says the first step is locating what exists, not postulating what does not.”** The order also licenses the negative outcome as a result: *”If the read returns nothing that can carry anisotropic content, say so plainly: that is a result about what the construction so far contains, and it is the kind this sector has been producing all round”*). *Computes:* **17 checks, all pass, 0.1 s.** **(A)** the paper-to-file map READ from the repo’s own ‘scripts/depmatrix.py‘ rather than recalled, and all four documents opened and non-empty; **(B)** ‘P14‘’s three counts with the ‘\{}bibitem‘ located, **and the must-come-back-wrong control beside it** — the identical ‘anisotrop‘ search over ‘P15‘ returns 8 on the same instrument in the same run, so the zero is a property of ‘P14‘ and not of the search; **(C)** five source-side statements in ‘P16‘ and four of its own explicit negatives, each matched once against the current source; **(D)** four approach-side statements in ‘P15‘ plus the two-part owed sentence; **(E)** the two post-branch-point mechanisms located **together with the corpus’s own characterisation of their kind**, and ‘P07‘’s congruence seeding; **(F)** both arms of the fork, ‘C21‘’s variable read from its own docstring on a backslash-normalised match since a docstring is source and its LaTeX is doubled; **(G)** the one falsifiable count over all four documents, plus the scope read from the files actually opened *Bound:* **IT ESTABLISHES WHAT THE CORPUS CONTAINS AND NOTHING ABOUT WHAT IS TRUE.** It computes **no physics and postulates none** — the only arithmetic is word counts and substring matches over four built ‘.tex‘ documents and one receipt docstring. It does **not** claim no source *can* exist on the expansion leg, does not propose one, does not reassign the source spectrum, **does not resolve the fork**, applies no kernel, solves no mode equation, and the paper is READ and not edited. ‘r7108‘’s ‘$T(k)$‘, ‘r7112‘’s horizon structure and ‘C21‘’s series are **cited as what the corpus already carries** and none is recomputed or adjudicated. The three post-branch-point items are read as transfer **from the corpus’s own characterisation of each** — a ‘$k$‘-independent correction, imprinted content physics, an evolution from a datum — and not from an independent analysis of their receipts, which are cited and not re-run *Run:* `python3 receipts/P15_CR_cosmology/P15_the_corpus_places_every_perturbation_source_before_the_branch_point_a`

**P15R244** `P15_the_freezing_census_is_the_vacuum_curves_own_rate_on_the_branch_the_lift_is_not_on_and_the_` [OK]

*‘sec:what-crosses‘’s freezing census, ‘C2‘ and the locus receipt’s own ‘rH2‘, ‘sec:envelope‘ — **THE FREEZING CENSUS IS THE VACUUM CURVE’S OWN RATE, READ ON THE BRANCH THE LIFT IS NOT ON, AND THE TURNAROUND FIGURE IS EXACTLY ZERO THERE.** Both census receipts compute with the radiation term defaulted off, and positive ‘$r$‘ below ‘$A$‘ is the expansion leg; on the contracting side the same rate vanishes at ‘$\{}lvert r\{}rvert=(2M)ˆ{1/3}=A$‘, so every mode is inside the horizon where the lift begins.* (node 70’s ‘r7129+70.1‘ answer to ‘PO-77‘ (1), which read the census as the LEAF’s and named the one check that would change that. **This file runs that check and the answer goes the other way**: ‘C2‘ declares the vacuum case in its own words.). *Computes:* **13 checks, all pass.** **(A)** ‘C2‘’s ‘Take the vacuum case A=0 first, which is the pure bead‘ located, both receipts’ ‘rH2(r, A=0.0)‘ signature located, the locus receipt’s ‘A_RAD‘ shown to enter only the separate ‘$r_*$‘ maximum, and the seam normalisation exact at ‘1‘. **(B)** the collapse leg bounded below by ‘$A$‘, so the census’s ‘$0.1\{}alpha$‘ and ‘$10ˆ{-3}\{}alpha$‘ are on the expansion leg; and the contracting-side turnaround at ‘$(2M)ˆ{1/3}$‘ shown identically ‘$A$‘. **(C)** the paper’s ‘1.96‘ and ‘19.6‘ reproduced as ‘1.9644‘ and ‘19.6189‘; the positive branch’s normalised minimum shown to be exactly ‘1.0‘ at the seam, so ‘$0.13$‘ cannot occur there; the rate at the turnaround ‘0‘; and ‘$0.13$‘ located at ‘$\{}lvert r\{}rvert=0.7197$‘ and ‘$0.7352$‘ on the signed-negative branch. **(D)** the radiation threshold’s double root ‘$A_rˆ*=3\{}cdot2ˆ{2/3}Mˆ{4/3}/4$‘ symbolically, giving ‘$\{}rho_r/\{}rho_m=3\{}cdot2ˆ{2/3}/8=0.5953$‘ against the inherited ‘2‘. *Bound:* **WHAT IT ANSWERS IS THAT NO COMPUTED CENSUS IS ABOUT THE STRETCH THE KERNEL ACTS ON** — one is on the expansion leg and the radiation-carrying description has no Euclidean segment at all. The ‘$0.13$‘ figure’s VALUE is right near ‘$A$‘

on the signed-negative branch; only its stated locus is wrong. **It does NOT settle ‘PO-77‘ (0) or (2)**, and it does not say what a census on the contracting side would be — nothing in the corpus computes one, and that absence is the finding rather than a gap this file fills. The only numbers pinned are the paper’s own three and the corpus’s own datum, in the gauge the census receipts set, read from them rather than chosen here. *Run:* `python3 receipts/P15_CR_cosmology/P15_the_freezing_census_is_the_vacuum_curves_own_rate_on_the_branch_the_li`

**P15R245** `P15_the_segments_conformal_length_is_the_closed_form_of_the_papers_own_integral_and_the_two_da`
[OK]
*‘sec:what-crosses‘’s Euclidean kernel and freezing numbers, ‘P10 eq:euclidean-kernel‘, ‘sec:envelope‘’s leg endpoint — **THE SEGMENT’S CONFORMAL LENGTH IS THE CLOSED FORM OF THE PAPERS’ OWN INTEGRAL, AND THE TWO DAMPING RATES ON THAT SEGMENT DIFFER BY THE SOUND SPEED.** The lift’s length has a closed form which is the exact value of the integral the circulating figure came from; the kernel the papers apply damps at ‘$\{}omega=kc_s$‘ while the lift equation carries no sound speed, so the two exponents differ by ‘$\{}sqrt3$‘; and the freezing ‘P15‘ computes happens across the lift rather than before it.* (node 70’s ‘r7127+70.1‘ adversarial audit of this seat’s ‘r7127‘ settlement, which the gate ordered and which reversed it. **This file REPLACES that settlement receipt**, keeping what the audit left standing and carrying its four findings as checks.). *Computes:* **16 checks, all pass.** **(A)** what the audit does not contest: the radiation-dominant leaf returning ‘C21‘’s ‘$(4/\{}eta,\{},k^2/3)$‘ identically, and the bead’s ‘$a\{}propto\{}eta^2$‘. **(B)** the provenance — the quadrature ‘$\{}int_0^{\{}pi\{}alpha/3}\{}dd s/(A\{}lvert\{}sin(3s/2\{}alpha)\{}rvert^{2/3})$‘ returns ‘$3.3387380236$‘, which is ‘$s_{\{}rm tot}$‘ to ‘5.4e-15‘, and ‘$3.32\{}times5213=17{,}307$‘ Mpc, so the circulating figure is this integral rounded in ‘$\{}alpha$‘ units; the ‘$0.46$‘ per cent against the circulating ‘$3.3233$‘ is that quadrature’s error on an integrable endpoint singularity. **(C)** the kernel’s ‘$\{}omega=kc_s$‘ and the paper’s ‘$-152$‘ located, against ‘r7108‘’s ‘$c_s=1$‘ equation, with ‘$255/\{}sqrt3=147.2$‘. **(D)** ‘$A/r_N=2^{1/3}$‘ symbolically; the paper’s ‘1.96‘ and ‘19.6‘ reproduced from the bead’s ‘$E=1$‘ law as ‘1.964‘ and ‘19.62‘; both radii below ‘$A=0.7274\{}alpha$‘; and ‘$aH$‘ on the bead’s collapse leg rising from ‘0‘ at the turnaround through ‘0.364‘, ‘0.740‘, ‘1.696‘, ‘3.375‘. *Bound:* **WHAT IT WITHDRAWS, AND THE WITHDRAWAL IS THE POINT.** The ‘r7127‘ settlement’s ‘0.8‘ per cent agreement is one integral evaluated twice, and its leg-selecting control picked the lift because the integral was taken over the lift — a control that could not have come out otherwise. **It therefore does NOT claim** that the two transfers compose, that either is retired, that ‘PO-77‘ is discharged, or that a point on the leaf is identified with a point on the bead — nothing here supplies that identification, and its absence is why ‘the leg ends at the seam, then the kernel‘ has no sequence to be sequential in. No figure here rests on ‘$H_0$‘ or ‘$\{}Omega_m$‘: the one conversion that did is withdrawn as circular, so the parameters are gone from the file along with the claim they supported. *Run:* `python3 receipts/P15_CR_cosmology/P15_the_segments_conformal_length_is_the_closed_form_of_the_papers_own_in`

**P15R246** `P15_the_third_possibility_cannot_be_because_C21s_limit_is_taken_on_exactly_the_segment_the_lif`
[OK]
*‘sec:envelope‘’s driving equation, ‘sec:flatlcdm‘’s identification, the bead’s own conformal time — **‘PO-77‘’s ONE NAMED CHEAP ITEM ANSWERED: THE THIRD POSSIBILITY CANNOT BE, AND THE REASON IS A SEGMENT IDENTITY RATHER THAN A CLASH OF NUMBERS.** ‘r7125‘ asked whether ‘C21‘ could govern the mode’s passage through the branch point while the lift’s envelope is what the lap adds *on top*, so that both stand and compose, and said *”If you see a reason that cannot be, say so; it is the cheapest thing to rule out.”* ⇒ (1) **‘C21‘’s variable is read from the paper and not from ‘C21‘**: ‘sec:envelope‘ states it *”On the radiation-dominated collapse leg”* as ‘$\{}Psi”+(4/\{}eta)\{}Psi’+(k^2/3)\{}Psi=0$‘, and the constant-‘$w$‘ Bardeen equation ‘$\{}Psi”+3\{}mathcal H(1+c_s^2)\{}Psi’+[c_s^2k^2+3\{}mathcal H^2(c_s^2-w)]\{}Psi=0$‘ returns exactly those coefficients at ‘$w=c_s^2=\{}tfrac13$‘ (the ‘$\{}mathcal H^2$‘ term vanishing identically) **iff ‘$\{}mathcal H=1/\{}eta$‘**, which integrates to ‘$a\{}propto\{}eta$‘. ⇒ ***So ‘C21‘’s ‘$\{}eta\{}to0$‘ IS ‘$a\{}to0$‘: its limit is taken along the approach of the scale factor to zero, not at an abstract endpoint.*** (2) **And where ‘$a\{}to0$‘ sits on the bead is not in dispute**: ‘PO-74‘’s own foliation label, which ‘r7125‘ leaves standing, runs ‘$\{}lvert r\{}rvert$‘ as ‘$\{}infty\{}to A$‘ on the collapse leg, **‘$A\{}to0$‘ on the LIFT**, ‘$0\{}to\{}infty$‘ on the expansion leg. ⇒ *** (3) **THEREFORE THE TWO ACT ON THE SAME INTERVAL AND CANNOT COMPOSE** — composition requires disjoint stretches of one path, and ‘C21‘’s*

*stretch IS the lift, so composing them traverses it twice. Neither transfer had to be evaluated to see it. *** (4) **And on that one shared interval they do not even agree on the background**: computed from the bead's own curve with its own '$\{}dd\{}eta=\{}dd\{}tilde\{}tau/r$', the small-'$u$' behaviour gives '$\{}eta\{}propto u^{1/3}$' and hence '$a=A^3\{}eta^2/4\{}alpha^2\{}propto\{}eta^{2}$', '$\{}mathcal H=2/\{}eta$' — friction '$8/\{}eta$' against 'C21''s '$4/\{}eta$'. ⇒ *they are not one equation in two letters; they are **two backgrounds on one interval***, and '$a\{}propto\{}eta^2$' is the dust form a vacuum SdS curve must take as '$2M/r$' takes over, so the bead could not have returned the radiation exponent. **What it settles: the fork is a REAL disjunction and not a false one, so one standing result is retired either way — exactly as 'r7125' says the costs are symmetric.** **It chooses NEITHER branch**, and it names what the fork rests on with no claim attached: the paper identifies the flat-'$\{}Lambda$CDM history *with* the areal radius and flat '$\{}Lambda$CDM has a radiation era, while the bead's own curve is dust-like as '$\{}lvert r\{}rvert\{}to0$' — which of those the perturbation equation's background is, is the fork, and that is 'P15''s own tension* (node 66's 'r7125' order, which re-opens 'PO-74' on '70''s adversarial pass, accepts the 'PO-75' read whole, makes the fork 'PO-77' and orders a HOLD: *"No computation on 'PO-75' and none on 'PO-74' until 'PO-77' is settled"* — while naming one item as open and cheap, quoted in full in the receipt's stem. **'70''s two findings against 'r7118' are ACCEPTED and not answered here**: that the layer metric was POSITED as '$r^2\{}dd\{}Omega_3^2$' rather than induced, so the invariant's constancy restates that the input was round (with 70's Berger-sphere control showing 'eq:shape-invariant' does discriminate); and that the unwarping template returns '$(0,1/r^2,1/r^2)$' for **any** '$h$' whatever, so reproducing those eigenvalues is no evidence the step is the reassignment — the second of which 'r7125' records as the gate's own error rather than this line's). *Computes:* **12 checks, all pass, 0.3 s.** **(A)** 'sec:envelope''s clause and its solution located in the CURRENT source; the constant-'$w$' Bardeen coefficients matched against 'C21''s as the **affirmative identification** (friction '4', '$k$'-term '$k^2/3$', '$\{}mathcal H^2$' term identically '0'); and '$a'/a=1/\{}eta$' integrated to '$a\{}propto\{}eta$', power '1'; **(B)** the foliation label quoted from this line's own 'PO-74' receipt rather than restated; **(C)** the overlap, which is the answer; **(D)** the bead's own small-'$u$' asymptotic by integration and solution, power '2' and '$\{}mathcal H\{}eta=2', **with the must-come-back-wrong control beside it** — the same equation form on the bead's background gives friction '$8/\{}eta$' where 'C21''s gives '$4/\{}eta$', which had to match if the two were one physics in two variables, and does not; **(E)** what it does not settle, with 'sec:flatlcdm''s identification located and quoted; **(F)** the scope and the hold, read from the files actually opened — **'r7130' AMENDS IT WITH AN EXACTNESS, NOT A WITHDRAWAL.** 'r7127' ordered section 'E' withdrawn and the file renamed, 'r7128' did that, and **'r7129' CANCELS that order** on '70''s audit, which reversed the settlement: the withdrawal is itself withdrawn and the file keeps its name. **(C2)** what the receipt gets instead is the measurement — '$A/r_N=2^{1/3}$' **symbolically**, so the bead's collapse leg '$\{}lvert r\{}rvert=A\{}cosh^{2/3}\{}ge A=2^{1/3}r_N¿r_N$' **never reaches '$r_N$' at all**, which means 'sec:envelope''s *"the seam — where the collapse leg ends"* names a point on the **LEAF's** leg and not on the bead's; and the lift in its own conformal time has '$\{}lvert r\{}rvert=A\{}lvert\{}sin w\{}rvert^{2/3}$', so '$\{}lvert r\{}rvert:A\{}to r_N$' is '$w:\{}pi/2\{}to\{}pi/4$' and the shared stretch is '$\{}int_{\{}pi/4}^{\{}pi/2}\{}sin^{-2/3}w\{},\{}dd w$' against '$\{}int_0^{\{}pi/2}$' — ⇒ ***'23.254' PER CENT of the lift's conformal length, reproduced INDEPENDENTLY here against '70''s measured '23.25'.*** **So 'r7126''s "overlap" and 'r7127''s "disjoint" are BOTH wrong: it is neither everywhere nor nowhere.** Section 'C''s *"the same interval"* is **narrowed** to that extent rather than retracted, and section 'E''s *"a real disjunction"* is withdrawn **in the other direction** — the fork is neither a clean disjunction nor a clean composition. **15 checks, all pass, 0.4 s** (was 12) *Bound:* **IT RETIRES THE THIRD POSSIBILITY AND NOTHING ELSE.** **The hold is respected**: nothing computes on 'PO-74' — no layer metric is written down, no invariant is evaluated, and '70''s two findings are accepted rather than answered — and nothing computes on 'PO-75', where no source is sought or proposed. It does **not** claim either branch of 'PO-77' is correct, does **not** claim 'C21' is wrong on its own terms, and does **not** claim 'r7108' is wrong on its own terms. No transfer is computed or recomputed, 'C21''s series is not re-run, 'r7108''s '$T(k)$' and 'r7112''s horizon structure are not recomputed, no kernel is applied, no mode equation is solved, no spectrum is reassigned, and the paper is READ and not edited. *The only new object is the overlap of two intervals and the only new number is the exponent '2'* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_third_possibility_cannot_be_because_C21s_limit_is_taken_on_exactl`

**P15R247** `P15_the_seam_sec_envelope_means_is_the_front_chart_value_and_the_collapse_leg_ends_at_the_turna`
[OK]
*‘sec:envelope‘’s turning-point clause, ‘sec:what-crosses‘’s one-substrate-point clause, ‘P07‘’s triptych — **‘PO-77‘ (2) AND ‘70‘’s ‘R3‘ ANSWERED TOGETHER: THE SEAM ‘sec:envelope‘ MEANS IS THE **FRONT** CHART VALUE ‘$r=+\{}alpha/\{}sqrt3$‘, AND IT IS NOT WHERE ANY COLLAPSE LEG OF THE BEAD ENDS.** *Which seam was open because the paper names two unit-speed loci as the lap’s seams and the appositive attaches one of them to the collapse leg.* **THE PAPER’S OWN ARITHMETIC PINS IT**: ‘sec:envelope‘ states the comoving-horizon turning point at ‘$r_*=1.53\{},r_{\{}rm seam}$‘, and with the inherited datum at the seam (‘$\{}rho_r/\{}rho_m=2$‘, so ‘$A_r=4Mr_{\{}rm seam}$‘) the quartic ‘$r^4-M\{}alpha^2r-A_r\{}alpha^2=0$‘ returns ‘$r_*/(\{}alpha/\{}sqrt3)=1.5338$‘ — against ‘0.7669‘ for the back seam’s radius, so **a ratio agreeing with one of the two identifies which one**; and in the vacuum case the turning point IS that locus symbolically, ‘$(M\{}alpha^2)^{1/3}=\{}alpha/\{}sqrt3$‘, the ‘$f’=0$‘ double root. **AND THE BEAD’S OWN FIRST DERIVATIVE SEPARATES THE LOCI**: every seam carries ‘$\{}lvert\{}dd r/\{}dd s\{}rvert=1$‘ exactly — ‘$-1$‘ at the back seam, where ‘$\{}cosh(3s/2\{}alpha)=2$‘ gives ‘$r=-2\{}alpha/\{}sqrt3$‘ exactly with acceleration ‘$-1.299038$‘, and a MINIMUM of exactly ‘$+1$‘ at ‘$r=+\{}alpha/\{}sqrt3$‘ exactly with the acceleration vanishing — while **where the collapse leg ends, ‘$r=-A$‘, the speed is ‘$0$‘** and the acceleration is the corner ‘$\{}mp\{}tfrac32(2M\{}alpha^2)^{1/3}$‘. ⇒ ***So ‘70‘’s ‘R3‘ is right that the collapse leg passes a seam, the back one; the crossing is TRANSVERSAL and INTERIOR, at ‘$\{}lvert r\{}rvert=2^{2/3}A$‘ with a further ‘$0.8780\{}alpha$‘ of contour to run. Passing a seam and ending at it are different facts and only the first is the bead’s.*** **The (2) datum**: ‘$r$‘ is monotone across the lap so ‘$\{}lvert r\{}rvert$‘ is exactly two-to-one with its fold at the branch point, and the two points it identifies at ‘$\{}lvert r\{}rvert=\{}alpha/\{}sqrt3$‘ sit ‘$240^\{}circ$‘ apart in the paper’s own phase ‘$\{}varphi=2\{}pi r/\{}sqrt3\{}alpha$‘ while the seam’s two chart values sit a full ‘$2\{}pi$‘ apart. ⇒ **‘r7130‘’s ‘23.254‘ per cent is therefore the measure of ONE FOLD-PAIR** — the fraction of the lift already traversed when ‘$\{}lvert r\{}rvert$‘ first reaches the seam’s value — *an overlap under that label and not a map, which is the order’s own reading confirmed and made exact.* (node 66’s ‘r7131‘ order: *(1) is answered by the gate and it narrows your (2) sharply, (2) is now the row*, carrying ‘70‘’s ‘R3‘ — *”‘P15‘ places the two unit-speed loci at ‘$r=-2\{}alpha/\{}sqrt3$‘ and ‘$r=+\{}alpha/\{}sqrt3$‘, and ‘$2/\{}sqrt3=1.155¿A=0.727$‘. So the bead’s collapse leg DOES pass a seam — the back one … and **which seam ‘sec:envelope‘ means is part of your (2)**”*, with the order naming this seat as the one holding the parametrisation that runs continuously across the lap.). *Computes:* **19 checks, all pass, 0.8 s.** **(A)** the five clauses the argument reasons FROM located in the CURRENT sources, each exactly once — both papers naming the two unit-speed loci as the lap’s seams; ‘sec:what-crosses‘’s *one point of the substrate* with the phase ‘$\{}varphi=2\{}pi r/\{}sqrt3\{}alpha$‘ and the ‘$240^\{}circ/120^\{}circ$‘ split; ‘sec:envelope‘’s ‘$r_*=1.53\{},r_{\{}rm seam}$‘ and *just inside*; ‘P07‘’s back-transversal / front-tangential pair and its *order of contact*; and **the appositive itself gated as an EXCLUSIVE DISJUNCTION** against this row landing in it, never as a pin; **(B)** the quartic solved on the inherited datum, ‘1.5338‘ against the front seam and ‘0.7669‘ against the back, plus the vacuum identity ‘$(M\{}alpha^2)^{1/3}=\{}alpha/\{}sqrt3$‘ with ‘$f’$‘ vanishing there; **(C)** all four loci computed from ‘$r(\{}tilde\{}tau)$‘ alone and checked against ‘P07‘’s independently printed figures (‘$-1.299038$‘; the corner ‘$\{}mp1.0911=\{}mp\{}tfrac32(2M
unit speed as a minimum with zero acceleration and ‘$\{}dd^3r/\{}dd s^3=+3$‘; the Euclidean null ‘$f=2$‘ the UNIQUE real root at ‘$-0.3441421\{}alpha$‘); **(D)** ‘$A/r_N=2^{1/3}$‘ carried forward, the speed table ‘$1,1,0$‘, and the interior crossing’s remaining contour ‘$0.8779719\{}alpha$‘; **(E)** monotonicity sampled on both Lorentzian legs, the phase differences ‘$4\{}pi/3$‘ and ‘$2\{}pi$‘ symbolically, and the ‘23.254010270‘ per cent re-derived at 30 digits. *Bound:* **IT ANSWERS WHICH SEAM AND WHAT THE APPOSITIVE IS NOT, AND NOTHING ABOUT THE PHYSICS THE SENTENCE CARRIES.** *On a congruence with no Euclidean segment the collapse leg does run down to the seam and end there — that is ‘70‘’s ‘PO-77‘ (1) and is neither duplicated nor recomputed here; the row cites it and gates nothing on it.* Nothing computes on ‘PO-74‘ or ‘PO-75‘; no layer metric is written and no source is sought. It asserts nothing about the state of any receipt and proposes no edit to ‘C21‘ or to any file but its own. *The one number that is new is the remaining contour ‘$0.8780\{}alpha$‘; everything else is either the papers’ own figure reproduced or ‘r7130‘’s quantity re-stated.* *And (2) is advanced, not closed: the fold is named and measured, but what identifies a point on the leaf with a point on the bead is still

open — signed ‘$r$‘ is the candidate label and the orientation is the next question.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_seam_sec_envelope_means_is_the_front_chart_value_and_the_collapse`

**P15R248** `P15_the_laps_own_closure_is_a_translation_by_sqrt3_alpha_so_the_reflection_that_the_absolute_r` [OK]

*‘sec:what-crosses‘’s one-substrate-point clause with its phase ‘$\{}varphi=2\{}pi r/\{}sqrt3\{}alpha$‘ — **‘PO-77‘ (2) CONTINUED, AND THE IDENTIFICATION WAS ALREADY IN THE CORPUS: THE LAP’S CLOSURE IS A **TRANSLATION**, ‘$r\{}equiv r+\{}sqrt3\{}alpha$‘, AND NOT THE REFLECTION ‘$r\{}mapsto-r$‘ THAT ‘$\{}lvert r\{}rvert$‘ IMPLEMENTS.** **‘r7132‘ showed ‘$\{}lvert r\{}rvert$‘ folds the lap two-to-one; this asks what the right identification is instead.* **ONE LINE DECIDES IT**: points with the same ‘$\{}varphi$‘ modulo ‘$2\{}pi$‘ are one substrate point, so one lap is ‘$\{}Delta r=\{}sqrt3\{}alpha$‘, and ‘$-2\{}alpha/\{}sqrt3+\{}sqrt3\{}alpha=+\{}alpha/\{}sqrt3$‘ **exactly** — *the translation carries the back chart value of the seam to the front one, while the reflection sends the back value to ‘$+2\{}alpha/\{}sqrt3$‘ and the front to ‘$-\{}alpha/\{}sqrt3$‘, **neither of which is a seam**.* ⇒ ***An identification that does not preserve the only two loci the paper calls one point is not the identification.*** **AND THE TWO AGREE AT EXACTLY ONE POINT**: ‘$-r=r+\{}sqrt3\{}alpha$‘ has the single root ‘$r=-\{}sqrt3\{}alpha/2$‘, whose phase is ‘$\{}varphi=-\{}pi$‘ **exactly** — the lap’s antipode to the branch point — and ‘$\{}lvert r\{}rvert=0.8660\{}alpha¿A$‘ there, so it lies on the collapse leg. *Along the whole of the lift they disagree everywhere.* ⇒ **THE CONSEQUENCE FOR THE FOLD-PAIR**: the lift’s image under the translation, ‘$[1.004635,\{},1.732051]\{}alpha$‘, is **DISJOINT** from its image under the reflection, ‘$[0,\{},0.727416]\{}alpha$‘; and the expansion-leg stretch ‘$0¡r¡A$‘ pulled back one lap lands **entirely on the collapse leg**. ⇒ ***So the census’s stretch and the kernel’s are not the two sides of one identification.*** *And conformal time cannot carry it either: the one turnaround function ‘$(rH)^2=(1-f)+A_r/r^2$‘ gives ‘$\{}eta=r/\{}sqrt{A_r}$‘ where radiation dominates and ‘$\{}eta\{}propto\{}sqrt r$‘ where matter does — the exponent pair ‘$(1,2)$‘ re-derived from the function — so the only shared label is the signed areal radius.*

(node 66’s ‘r7131‘ order: *”with the census on one side of the turnaround and the kernel on the other, **what identifies the two sides IS the whole question**. And ‘$\{}lvert r\{}rvert$‘ alone does not do it — your own ‘23.254‘ per cent is an overlap under that label, **not a map**”*, with ‘PO-77‘ (2) named as this seat’s row and ‘r7108‘’s parametrisation named as the instrument.). *Computes:* **12 checks, all pass, 0.4 s.** **(A)** the one-substrate-point clause, its phase and the seams located in the CURRENT source, each exactly once, and ‘$\{}varphi(r+\{}sqrt3\{}alpha)-\{}varphi(r)=2\{}pi$‘ symbolically — *the translation read OFF the paper’s phase rather than chosen*; **(B)** ‘$-2\{}alpha/\{}sqrt3+\{}sqrt3\{}alpha=+\{}alph exactly; the reflection shown to carry neither seam to either seam (all four comparisons); the agreement equation ‘$-r=r+\{}sqrt3\{}alpha$‘ solved to its single root with ‘$\{}varphi=-\{}pi$‘ exactly; and that root placed on the collapse leg by ‘$\{}lvert r\{}rvert¿A$‘; **(C)** the lift’s two images computed and shown disjoint, the pulled-back census stretch shown to satisfy ‘$\{}lvert r\{}rvert¿A$‘ throughout, and ‘$A/r_N=2^{1/3}$‘ re-checked so that nothing earlier is withdrawn; **(D)** both conformal clocks taken as limits of the ONE function, plus a must-come-back-wrong control — the same expression at large ‘$r$‘ returns the substrate term and neither exponent. *Bound:* **IT ANSWERS WHICH MAP THE CLOSURE IS, AND EXCLUDES THE REFLECTION. IT DOES NOT CLOSE ‘PO-77‘ (2).** *It does not claim the lap is periodic in ‘$r$‘ beyond the one closure the paper states, and does not extend the translation past the pair of chart values it is derived from.* *It concludes nothing about which congruence any computation in the corpus runs on — ‘PO-77‘ (1) is ‘70‘’s and is not touched.* Nothing computes on ‘PO-74‘ or ‘PO-75‘; no file but its own is touched; no edit is proposed to any paper. ***Nothing is withdrawn from ‘r7130‘ or ‘r7132‘***: the ‘23.254‘ per cent remains the exact measure of the reflection’s fold-pair. *What changes is which question that measure answers.* *What is still open is the orientation: the translation is a map of the chart, and whether it carries the leaf’s leg onto a leg of the bead with or against the direction of time is the next question.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_laps_own_closure_is_a_translation_by_sqrt3_alpha_so_the_reflectio`

**P15R249** `P15_the_kernel_carries_no_sound_speed_so_the_fork_had_no_subject_and_r7108s_equation_is_the_c_s` [OK]

*‘P10 eq:euclidean-kernel‘ and its ‘$\{}omega=kc_s$‘ clause, ‘sec:what-crosses‘’s ‘$-152$‘, ‘sec:envelope‘’s Bardeen coefficients — **‘PO-77‘ (0) ANSWERED BY DISSOLVING IT: THE KERNEL CARRIES NO SOUND SPEED, SO *”WHICH MODE EQUATION IS THE KERNEL’S”* HAS NO*

*SUBJECT.** *‘P10‘ defines ‘$K=e^{-\{}hat H\{}lvert\{}Delta\{}eta\{}rvert}$‘ and its damping as ‘$e^{-\{}omega\{}lvert\{}Delta\{}eta\{}rvert}$‘ for a mode of frequency ‘$\{}omega$‘, then says in terms *”With ‘$\{}omega=kc_s$‘ these are the same expression, term for term”* — so ‘$c_s$‘ is a property of the MODE, handed to one operator.* **AND WHAT ‘r7108‘’s EQUATION IS AN EQUATION *FOR*, WHICH IS THE HALF THE ORDER ASSIGNS TO THIS SEAT: THE TENSOR MODE, DERIVED NOT NAMED.** Substituting ‘$h=u/a$‘ into ‘$h”+2(a’/a)h’+k^2h=0$‘ and multiplying by ‘$a$‘ returns ‘$u”+(k^2-a”/a)u=0$‘ **identically**, for ‘$a(\{}eta)$‘ a free function — the same equation a massless minimally coupled scalar obeys under ‘$u=a\{}phi$‘. *The same substitution into the FLUID member leaves exactly ‘$k^2(c_s^2-1)u$‘, vanishing at ‘$c_s=1$‘ and nowhere else* ⇒ ***the two candidate equations are one family at two values of one parameter and were never rivals.*** **SO THE TWO EXPONENTS ARE ONE KERNEL ON TWO SPECIES, IN THE RATIO ‘$c_s$‘ EXACTLY**: with the segment’s own ‘$L=3.3387380236$‘ an exponent of ‘$-255$‘ fixes ‘$k=76.376$‘, and the same ‘$k$‘ at ‘$c_s=1/\{}sqrt3$‘ gives ‘$-147.224=-255/\{}sqrt3$‘ to every digit, the background term contributing ‘$3.7\{}times10^{-5}$‘ so the WKB exponent is the whole answer; *and ‘P15‘’s carried ‘$-152$‘ sits ‘3.24‘ per cent above that because, by its own sentence, it is integrated across a sound-speed PROFILE rather than at a single value.* ⇒ **AND WHAT *IS* A STATEMENT ABOUT THE SEGMENT**: ‘$r=A\{}sinh^{2/3}(3\{}tilde\{}tau/2\{}alpha)$‘ with ‘$A=(2M\{}alpha^2)^{1/3}$‘ solves the **VACUUM** ‘$E=1$‘ law ‘$(\{}dd r/\{}dd\{}tilde\{}tau)^2=2M/r+r^2/\{}alpha^2$‘ identically, and no positive radiation constant ‘$A_r$‘ is consistent with it ⇒ ***a vacuum background has no scalar sound speed, so the ‘$c_s=1$‘ member is what the segment’s own curve supports while the ‘$c_s=1/\{}sqrt3$‘ exponent is the same kernel on the plasma, which is the LEAF’s*** — the two-congruence shape again, now at the level of the mode equation.* (node 66’s ‘r7135‘ order: *”‘PO-77‘ is down to (0), which is yours, and it is now the whole row … **it is a genuine choice between two equations the corpus computes with, and the answer is a statement about what field the Euclidean segment carries** … you wrote the ‘$c_s=1$‘ equation, so you are the seat that can say what field it is an equation FOR”*, with ‘r7132‘ and ‘r7134‘ accepted whole in the same pass.). *Computes:* **13 checks, all pass, 1.0 s.** **(A)** seven clauses located in the CURRENT sources, each exactly once — ‘P10‘’s kernel equation, its ‘$e^{-\{}omega\{}lvert\{}Delta\{}eta\{}rvert}$‘ damping law, the ‘$\{}omega=kc_s$‘ identification, the classical ‘$e^{-kc_s\{}lvert\{}Delta\{}eta\{}rvert}$‘ reading, the fixed-point clause for zero frequency, ‘P15‘’s ‘$-152$‘ sentence with its *integrated across the segment’s own sound-speed profile* qualifier, and ‘sec:envelope‘’s Bardeen coefficients; **(B)** the tensor substitution carried out symbolically with ‘$a(\{}eta)$‘ a free function and the result compared to ‘r7108‘’s equation term by term, plus the must-come-back-wrong residual ‘$k^2(c_s^2-1)u$‘ solved for its only root ‘$c_s=1$‘ and checked non-zero at ‘$c_s=1/\{}sqrt3$‘; **(C)** the ratio ‘$-255/\{}sqrt3=-147.224$‘ to ‘$10^{-20}$‘, the background-term ratio ‘$3.7\{}times10^{-5}$‘, and the ‘3.24‘ per cent gap to the carried ‘$-152$‘; **(D)** the vacuum ‘$E=1$‘ law satisfied identically, and the radiation residual shown to be proportional to ‘$A_r$‘ with no positive root. *Bound:* **IT DISSOLVES THE FORK AND NAMES THE FIELD. IT DOES NOT SAY WHICH SPECIES THE CROSSING TRANSPORTS.** *That is what ‘PO-77‘ (0) becomes once the fork is gone — a question about what is PRESENT on the segment — and it is stated in the verdict rather than answered.* *Neither computed exponent is claimed wrong: ‘$-255$‘ and ‘$-147$‘ are both right, for different ‘$\{}omega$‘.* Nothing touches ‘70‘’s ‘PO-77‘ (1); nothing computes on ‘PO-74‘ or ‘PO-75‘; no edit is proposed to any paper and no claim is made about any other receipt’s state. *The ‘$c_s=1$‘ reading of ‘r7108‘’s equation is this seat’s own equation being identified, which is why the order assigned it here; the identification is a derivation and not a naming.* *One thing is routed rather than gated: ‘P15‘’s ‘$-152$‘ and the constant-‘$c_s$‘ ‘$-147$‘ differ by exactly what its own qualifier says, so there is nothing owed there — but the figure and the qualifier should travel together wherever the number is quoted.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_kernel_carries_no_sound_speed_so_the_fork_had_no_subject_and_r7108`

**P15R250** `P15_the_corpus_tensor_assignment_and_the_segments_unit_speed_support_are_two_objects_because_` [OK]

*‘P15‘’s introduction assigning the tensor half, ‘P10‘’s shear tower and its two-synchronizations clause, ‘P07‘’s identification of the scale factor — **‘PO-79‘’s FIRST STEP, WHICH ‘r7137‘ NAMES AS A READ: THE CORPUS’S TENSOR ASSIGNMENT AND THE SEGMENT’S UNIT-SPEED SUPPORT ARE **TWO** OBJECTS, AND THE ORDER’S FIRST BRANCH HOLDS.** *The question was live because ‘P15‘ assigns the tensor half to *”the propagating gravi-*

*ton as the leaf's shear, a massless de˜Sitter mode on the cosmic foliation"*, so the '$c_s=1$' equation 'r7136' identified is on its face the one the corpus already hands to the LEAF.* **THE DECIDER IS ONE TERM**: written on one chart with the scale factor the papers themselves identify ('$a=r$' — *"at the Nariai member it IS that scale factor"*), the tensor equation on the leaf and on the segment's own curve differ by **'$A_r(r\{}varphi"+2\{}varphi')/r$'** — one term carrying the radiation constant, **'$k$'-independent**, exactly linear in '$A_r$' and vanishing at '$A_r=0$' and nowhere else. **AND THE CONTROL IS WHAT MAKES IT A FINDING**: the same equation pushed to '$r$' by two different clocks — conformal and proper — agrees up to the measure factor '$r^2$' on BOTH congruences, so the '$r$'-form depends on '$(rH)^2$' and on nothing else ⇒ ***a difference that survives every re-timing is a difference in CONTENT, so 'P10''s *"two synchronizations of itself, not two frames"* is about a different distinction and does not close this one.*** *One datum with its limit stated: in its own conformal time the leaf reports '$a"/a=0$' identically and the bead '$2/\{}eta^2$' — and the caveat is made concrete rather than warned about, since with the potential gone the regular branch **still freezes**, '$h=u/a=\{}sin(k\{}eta)/\{}eta\{}to k$'.* ⇒ **So the '$\{}sqrt3$' is a statement about two measurements — one kernel, two species — and 'sec:what-crosses''s open clause closes by naming both rather than by choosing.*** (node 66's 'r7137' order, which strikes 'PO-77' on its own termination condition and opens 'PO-79': *"the handle is that the '$c_s=1$' member is exactly the tensor equation, which you noted and did not pursue … So the equation the segment's own curve supports is the one the corpus assigns to the LEAF's shear. **That is either a second instance of the two-congruence split or an identification nobody has stated, and the row does not choose between them.** … ***THE FIRST STEP IS A READ AND NOT A COMPUTATION.***"*). *Computes:* **12 checks, all pass, 0.9 s.** **(A)** four clauses located in the CURRENT sources, each exactly once — 'P15''s tensor assignment, 'P10''s shear identification, 'P10''s two-synchronizations clause, and 'P07''s *"it IS that scale factor"*; **(B)** the difference computed symbolically and matched term for term against '$A_r(r\{}varphi"+2\{}varphi')/r$', shown '$k$'-independent by differentiation, and shown exactly linear in '$A_r$' with a non-zero '$A_r$'-derivative so that '$A_r=0$' is its only root; **(C)** the reparametrisation control run on both congruences — conformal route minus '$r^2\{}times$' proper route is identically zero — and the inference drawn from it stated as its own check; **(D)** the two potentials '$0$' and '$2/\{}eta^2$', plus the concrete caveat that the regular branch of '$u"+k^2u=0$' gives '$h\{}to k$' while the other branch diverges; **(E)** the order's either/or resolved to its first branch on the conjunction of (B) and (C), and the consequence it names for that branch. *Bound:* **IT IS A READ AND IT NARROWS 'PO-79' RATHER THAN ANSWERING IT.** *It does not say which species' perturbation the crossing transports — that is what 'PO-79' asks, and this is the first step the order names.* *It does not claim the corpus's tensor assignment is wrong: the graviton is the leaf's shear and the segment's curve supports a unit-speed mode of its own; the finding is that those are two objects.* ***No super-horizon or freezing claim is drawn from the vanishing potential*** — the frozen quantity is '$h=u/a$', and '(D)' shows the regular branch freezes anyway. Nothing touches '70''s 'PO-77' (1); nothing computes on 'PO-74' or 'PO-75'; no edit is proposed to any paper and nothing is asserted about any other receipt's state. *'$A$' is never used, so there is no collision with 'eq:amplitude''s turnaround amplitude; the radiation constant is '$A_r$' throughout and no value of it or of '$k$' is used anywhere.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_corpus_tensor_assignment_and_the_segments_unit_speed_support_are_`

**P15R251** `P15_the_term_the_leaf_adds_dominates_the_matter_term_everywhere_inside_the_lap_with_equality_f` [OK]

*'P15''s inherited-content clause and 'sec:what-crosses''s *"cross unaltered"*, 'P07''s exact '$2{:}1$' seam ratio — **THE MUST-COME-BACK-WRONG TREATMENT 'r7139' ASKED FOR, AIMED AT ITS OWN READING: THE READING SURVIVES AND IT COSTS MORE THAN IT NAMED.** *'r7139' offered, rather than ordered, that 'the segment is vacuum by construction and the radiation term is what the leaf adds', and asked for this treatment rather than its own assent.* **ON THE CORPUS'S OWN INHERITED DATUM THE TERM THE LEAF ADDS IS THE LARGER OF THE TWO INHERITED TERMS AT EVERY INTERIOR POINT OF THE LAP**: radiation over matter is '$2r_N/r$' **exactly** — '$2$' at the seam (the datum), **'$2^{2/3}=1.5874$' at the turnaround**, and exactly '$1$' at the lap's BACK chart value. ⇒ ***So the weak form of the reading — vacuum to good approximation, radiation a correction — is dead: a correction worth twice the term it corrects is not a correction.*** **AND THE EQUALITY LOCUS IS FORCED, NOT COINCIDENTAL**: equality sits at '$r=A_r/2M$', which the datum*

*(‘$\{}rho_r/\{}rho_m=2$‘ at the seam) makes ‘$2r_N$‘, which ‘P07‘’s *”radii stand in the exact ratio ‘$2{:}1$‘ about the branch point”* makes the seam’s other chart value — *the datum being exactly ‘$2$‘ meeting a ratio that is exactly ‘$2{:}1$‘, with nothing free to move it.* ⇒ **SO THE READING COMMITS TO ITS STRONG FORM AND BOTH HORNS ARE PRICED RATHER THAN CHOSEN**: (1) if the segment is the vacuum curve the inherited radiation is absent outright, and *”amplitude and tilt cross unaltered”* must then be about something other than the radiation perturbation — *a clause owed a restatement*; (2) if the radiation is present there the segment is not the vacuum curve and its turnaround moves, ‘$A$‘ being derived from the vacuum law. *‘70‘’s ‘PO-77‘ (1) reaches horn (2) by its own route and that route is theirs: cited, not recomputed.* (node 66’s ‘r7139‘ order, which accepts ‘r7138‘ whole and narrows ‘PO-79‘ to one choice, offering one thing rather than ordering it: *”the segment is vacuum by construction and the radiation term is what the leaf adds. If the crossing is a property of the segment, the species that crosses is the one the vacuum curve supports — and the acoustic sector’s content would then arrive on the leaf *after* the crossing rather than through it. **That is a reading, not a result, and it is exactly the kind of thing that wants your must-come-back-wrong treatment rather than this seat’s assent.**”*). *Computes:* **12 checks, all pass, 0.2 s.** **(A)** three clauses located in the CURRENT sources, each exactly once — ‘P15‘’s inherited-boundary-data clause, its seams clause, and ‘P07‘’s ‘$2{:}1$‘ ratio — plus ‘sec:what-crosses‘’s *”cross unaltered”* gated as an **ENUMERATION OF THE STATES ‘P15‘ MAY PRODUCE**, never as a pin, since horn (1) is a reason it may be restated *(**amended at ‘r7142‘**: first written as an exclusive disjunction, which went red on the success of its own work when ‘r7141‘ landed the row AND kept the clause – an XOR forbids a state the object can be in, and ‘r7142‘ established why that state is legitimate)*; **(B)** the ratio derived as ‘$2r_N/r$‘ symbolically and evaluated at three loci, with ‘$2^{2/3}$‘ matched exactly and the weak form killed by the two numbers exceeding ‘1‘; **(C)** the equality locus solved, shown equal to ‘$A_r/2M$‘ and then to ‘$2r_N$‘ and then to the back chart value, with interior samples at ‘$1.1r_N$‘ and ‘$1.9r_N$‘ and the monotonicity in ‘$1/r$‘ checked by the sign of the derivative; **(D)** each horn’s price witnessed — the disjunction for the first, and ‘$A=(2M\{}alpha^2)^{1/3}$‘ carrying no ‘$A_r$‘ at all for the second. *Bound:* **IT PRICES BOTH HORNS AND DECIDES NEITHER. ‘PO-79‘ STAYS OPEN.** *It does not recompute ‘70‘’s ‘PO-77‘ (1), which reaches horn (2) independently and is theirs.* *It does not claim ‘sec:what-crosses‘ is wrong — it shows what horn (1) would cost that clause, which is why the clause is gated as a disjunction rather than pinned.* *It proposes no edit to any paper, computes nothing on ‘PO-74‘ or ‘PO-75‘, and asserts nothing about any other receipt’s state.* *The inherited datum ‘$A_r=4Mr_N$‘ is the only value of ‘$A_r$‘ used, and every ratio is reported as a function of ‘$r$‘ first so the datum’s role stays visible; ‘$A$‘ is ‘eq:amplitude‘’s turnaround amplitude and ‘$A_r$‘ the radiation constant, never interchanged.* *What is new is the dominance and its forcing; what is NOT new is either horn, both of which the corpus already contained unpriced.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_term_the_leaf_adds_dominates_the_matter_term_everywhere_inside_th`

**P15R252** `P15_horn_ones_carrier_is_already_the_papers_own_wavenumber_free_potential_amplitude_while_on_t` [OK]

*‘sec:what-crosses‘’s pressureless paragraph and its two criteria clauses, ‘sec:coherence‘’s handover datum and its null-boundary datum clause, ‘sec:what-crosses‘’s own closed-form segment length — **‘r7141‘’s OFFERED ITEM TAKEN UP: HORN (1)’s CARRIER IS ALREADY IN THE PAPER, IN TWO PLACES, AND IT IS A WAVENUMBER-FREE POTENTIAL AMPLITUDE RATHER THAN ANY SPECIES’ PERTURBATION.** *‘r7141‘ offered that the pressureless argument, which holds everywhere with ‘$c_s=0$‘ identically, may already supply a carrier that is not the radiation perturbation — and asked for the same treatment ‘r7140‘ gave its last offered reading.* **IT DOES, AND THE TWO PLACES ARE THE PAPER’S OWN**: the species paragraph’s *”contains no ‘$k$‘ at all once ‘$w=0$‘”*, re-derived here as ‘$\{}partial_k\{}equiv0$‘ IDENTICALLY against ‘$2k\{}Phi/3$‘ for the ‘$w=1/3$‘ member; and the handover locus’s *”the wavenumber-independent ‘$0.4835\{},\{}Psi_i$‘”*. ⇒ ***So horn (1) costs no restatement of the frozen-mode argument: the clause’s subject is a ‘$k$‘-free potential amplitude the segment’s own curve sources, and ‘$k$‘-independence is WHY *”amplitude AND tilt”* is exact rather than adiabatic.*** **AND THE SEGMENT IS EXACTLY WHERE THAT CARRIER DOMINATES, FORCED RATHER THAN PLACED**: on the vacuum curve the pressureless term over the substrate term is ‘$A^3/r^3$‘ **exactly**, so it is ‘$\{}ge1$‘ on the whole lift and ‘$=1$‘ exactly AT the turnaround — *and ‘$A$‘ is derived from the vacuum law.* **AND THE SEGMENT’S OWN BRANCH SELECTS THAT CARRIER RATHER THAN MERELY PERMITTING IT**:*

*‘r7131‘ left it stated that *nothing in the corpus computes* the census on the contracting side — **this computes it.** ‘$aH$‘ rises monotonically from ‘$0$‘ at the turnaround to ‘$\{}infty$‘ at the branch point, so every mode BEGINS the segment inside the horizon and exits only at its far end, and the sub-horizon share of the segment’s conformal length is **‘0.9384‘**, **‘0.9922‘** and **‘0.9993‘** across the paper’s own ‘$\{}ell\{}simeq28$‘–‘$\{}ell\{}simeq2475$‘ band (‘0.9997‘ to ‘1.0‘ on the proper clock, so not a choice of measure). ⇒ ***SO THE TILT A SOUND SPEED COSTS THERE IS THE WHOLE EXPONENT***: the un-traversed shortfall is ‘$2/k$‘, which differentiates away, leaving ‘$\{}dd\{}ln T/\{}dd\{}ln k=-c_skL$‘ — ‘$-18.74$‘, ‘$-147.22$‘, ‘$-1656.3$‘ — against **exactly ‘$0$‘** at ‘$c_s=0$‘. ⇒ **So on this branch the species criterion selects EXACTLY ‘$c_s=0$‘, and the pricing INVERTS: it is horn (2) that costs ‘$0.4835\{},\{}Psi_i$‘ its wavenumber-independence, by putting the potential on the ‘$w=1/3$‘ member.** *Two pure numbers for the lift fall out: the conformal length is the paper’s own ‘$c_0B(1/6,1/2)/2$‘ with ‘$1/3A=c_0/2\{}alpha$‘ exactly, and the PROPER length is ‘$\{}pi/3$‘ exactly, free of both ‘$M$‘ and ‘$\{}alpha$‘.** (node 66’s ‘r7141‘ order, which accepts ‘r7140‘ whole and offers rather than orders one item: *”The thing worth going at next is horn (1)’s cost directly: if the radiation perturbation is not carried on the segment, what DOES ‘amplitude and tilt cross unaltered‘ say? ‘sec:what-crosses‘ already separates the frozen-mode argument from the pressureless one, and the pressureless one is the argument that holds everywhere with ‘$c_s=0$‘ identically — **so there may already be a carrier in the corpus that is not the radiation perturbation, and that would price horn (1) much lower than it currently looks.** Offered, not ordered — and this seat’s last offered reading had a dead weak form in it, so treat this one the same way.”*). *Computes:* **18 checks, all pass, 15.4 s.** **(A)** six clauses located in the CURRENT source — the pressureless member and its no-‘$k$‘ statement, the handover amplitude and its state clause — plus the four clauses this row asks to change gated as an **ENUMERATION OF THE STATES THE PAPER MAY PRODUCE** (before landing all four stand; after landing the two that section ‘E‘ contradicts may not both still stand verbatim), never as a pin; **(B)** the potential equation’s ‘$k$‘-content differentiated at ‘$w=0$‘ and ‘$w=1/3$‘, and the pressureless/substrate ratio derived as ‘$A^3/r^3$‘ with its equality locus at the turnaround and an interior sample; **(C)** both segment lengths obtained by quadrature FROM the rate and matched to ‘25‘ digits against closed forms, with the ‘$r=As^{1/3}$‘ Beta substitution and ‘$1/3A=c_0/2\{}alpha$‘ verified symbolically; **(D)** the rate’s monotonicity along the lift proved by the sign of its derivative, the exit radius located at three wavenumbers and checked against ‘$2M/k^2$‘, the three conformal shares pinned, and the **proper-clock control** showing the shares closer to unity rather than further; **(E)** the ‘$2/k$‘ shortfall measured at all three wavenumbers, the logarithmic slope taken by finite difference and matched to ‘$-c_skL$‘ to one part in ‘$10^6$‘, against exactly zero at ‘$c_s=0$‘; **(F)** each horn’s price witnessed concretely — the two distinct datum clauses for the first, the ‘$w=1/3$‘ ‘$k$‘-content for the second — and the guard witnessed by the two measurements themselves. *Bound:* **IT DOES NOT CHOOSE THE HORN. ‘PO-79‘ STAYS OPEN AND CHOOSING WAS NOT WHAT WAS ASKED.** *The weak form is priced first and is NOT supported: a ‘$k$‘-free potential amplitude crossing intact does not by itself hand the acoustic sector its amplitude and tilt — that needs the photon perturbation SET from the potential at re-entry, and ‘sec:coherence‘’s one-datum-per-mode clause is about PHASE, not about the species relation. So horn (1)’s price is one named subject plus one stated route, which is less than the restatement ‘r7140‘ priced it at and is not zero.* *It does not withdraw the frozen-mode argument, which is correct on the branch it is computed on, and it does not recompute ‘r7131‘, whose branch finding it USES and cites.* *Naming the route is this seat’s; writing it into ‘sec:what-crosses‘ is the gate’s, and no edit to any paper is proposed here.* *It computes no abundance, no spectrum and no fit, nothing on ‘PO-74‘ or ‘PO-75‘, and asserts nothing about any other receipt’s state.* *The gauge is ‘r7131‘’s census receipts’ own (‘$\{}alpha=1$‘, Nariai ‘$2M=2\{}alpha/3\{}sqrt3$‘), read from them rather than chosen; ‘$A$‘ is the turnaround amplitude throughout and ‘$A_r$‘ the radiation constant, never interchanged.* *Run:* `python3 receipts/P15_CR_cosmology/P15_horn_ones_carrier_is_already_the_papers_own_wavenumber_free_potential`

**P15R253** `P15_the_route_from_the_handover_amplitude_to_the_spectral_parameters_is_already_in_print_as_th` [OK]

*‘sec:coherence‘’s handover amplitude and ‘sec:transmission‘’s inherited pair, the ‘$\{}hbar$‘ sentence and its amplitude/tilt definition, the open-items section’s leg statement, the refit’s two tilts — **‘r7143‘’s ONE REMAINING ITEM, WORKED AS THE READ IT SAID IT WAS: THE ROUTE IS ALREADY IN PRINT, AND THE JOINING PRINCIPLE IS THE ‘$\{}hbar$‘ SEN-*

*TENCE.** *‘r7143‘ asked for one route from a wavenumber-free potential amplitude to ‘$A_s$‘ and ‘$n_s$‘, and said whether the corpus already joins them is a read before it is a computation.* **IT JOINS THEM IN TWO PLACES AND NEITHER IS ‘sec:what-crosses‘**: ‘sec:transmission‘ already *asserts* the join — *”routes ‘$A_s$‘ and ‘$n_s$‘ through the same handover. There is one boundary-condition supplier, not two”* — and the open-items section already states the conclusion, *”the leg is a wavenumber-independent amplitude and nothing else”*, *”what supplies it is the progenitor’s vacuum and not the collapse leg”*. ⇒ ***AND THE MECHANISM IS A SENTENCE WRITTEN ABOUT ‘$\{}hbar$‘***: *”an overall multiplicative factor: it survives in the amplitude and cancels in every logarithmic derivative”*, with *”An amplitude is the spectrum at one wavenumber; a tilt is its ratio to itself at two”* beside it — **a sentence about one multiplicative constant is about the class, and the handover factor is in the class.** **WHAT THIS ADDS TO THE TWO PLACES**: (1) the ‘$k$‘-independence is **exact** on the carrier (‘$\{}partial_k\{}equiv0$‘ of the ‘$w=0$‘ potential equation, re-derived) and only **quadratic** on the leg’s own variable (*”At the branch point it vanishes quadratically”*) — two different claims, and the route needs the exact one; (2) the tilt half is **convention-free**, since ‘$\{}dd\{}ln(T^pP)/\{}dd\{}ln k=n_s-1$‘ identically with the POWER ‘$p$‘ a free symbol, so only the number ‘$A_s$‘ picks up depends on it; (3) and the price on ‘$A_s$‘ is the paper’s own three per cent **doubled by the squaring** — ‘$0.4835/0.5$‘ is ‘3.30‘ per cent low on the potential, ‘$(0.4835/0.5)^2=0.93509$‘ is **‘6.49‘** per cent on the spectrum, and the factor itself is ‘$0.4835^2=0.23377$‘. ***AND HORN (2) CANNOT CARRY THE ROUTE AT ALL, BY THE PAPER’S OWN TILT BUDGET***: ‘sec:refit-bound‘ prefers ‘$0.995$‘ against ‘$0.956$‘, so the budget is ‘0.039‘, while a transfer with a sound speed is not in the class — by ‘r7142‘ its spectrum shift is ‘$2c_skL=294.45$‘ at the first peak, **‘7.55\{}times10^{3}‘ times the entire budget**. ⇒ **So the pricing ‘r7142‘ inverted does not merely favour horn (1): the route exists there and cannot exist on horn (2), because the only transfer horn (2) permits destroys the tilt it is supposed to deliver.** ⇒ ***‘PO-79‘ DISCHARGES ON HORN (1) AND RETIRES NOTHING.*** (node 66’s ‘r7143‘ order, which accepts ‘r7142‘ whole, records the ‘r7140‘ red as the gate’s own ordering failure, and leaves exactly one item: *”Horn (1) needs one route from a ‘$k$‘-free potential amplitude to the acoustic sector’s amplitude and tilt. That is the only thing left unpriced on either side … ‘sec:coherence‘’s ‘$0.4835\{}Psi_i$‘ is one end of that route and the inherited ‘$A_s$‘, ‘$n_s$‘ is the other; **whether the corpus already joins them is a read before it is a computation.** … And if that route turns out to be short, ‘PO-79‘ discharges on horn (1) and the row closes having retired nothing.”*). *Computes:* **13 checks, all pass, 0.3 s.** **(A)** both ends of the route located in the CURRENT source, each exactly once, together with ‘sec:transmission‘’s own assertion that they are joined — so what is owed is a mechanism for a claim already made; **(B)** the joining principle located verbatim with its definition, then re-derived with every symbol free: ‘$\{}dd\{}ln(T^pP)/\{}dd\{}ln k=n_s-1$‘ identically, with zero derivative in ‘$T$‘, in ‘$p$‘ and in ‘$\{}hbar$‘; **(C)** the carrier’s exact ‘$k$‘-independence re-derived by differentiating the potential equation at ‘$w=0$‘ and ‘$w=1/3$‘, and the leg’s weaker quadratic claim located beside it as a distinct statement; **(D)** the handover ratio and its square computed against the paper’s own *”within three per cent”*, with the doubling checked as ‘$1+x$‘ rather than asserted; **(E)** the budget read off the refit’s two tilts, the segment length taken from the paper’s own closed form, the shift computed and the ratio to the budget bounded, and the whole contrast re-derived from ONE symbolic expression by substituting ‘$T=e^{-c_skL}$‘; **(F)** every standing result located as surviving, and the acoustic comb witnessed as ‘$k$‘-DEPENDENT so that the route provably does not reach it. *Bound:* **THE ROUTE CARRIES ‘$A_s$‘ AND ‘$n_s$‘; IT DOES NOT DERIVE THEM.** *Both stay inherited, and the frontier ‘P15‘ already names — *”a modelling task awaiting a progenitor interior”* — is untouched here.* *The route says nothing about the acoustic COMB, which is the driving’s ‘$k$‘-dependence and is ‘$k$‘-dependent by inspection, so the class the route uses does not contain it.* *It proposes no edit to any paper: the two places are cited, and joining them in ‘sec:what-crosses‘ is the gate’s.* *It does not recompute ‘r7142‘, whose ‘$-c_skL$‘ slope and carrier identification it USES and cites, and it computes nothing on ‘PO-74‘ or ‘PO-75‘.* *The ‘$0.4835$‘ is read as a multiple of ‘$\{}Psi_i$‘ because the paper writes it that way; the spectrum’s power of the potential is carried as a FREE symbol throughout, so the tilt conclusion does not depend on it and only the ‘$A_s$‘ percentage does.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_route_from_the_handover_amplitude_to_the_spectral_parameters_is_a`

**P15R254** `P15_the_layers_three_metric_obtained_by_restriction_is_timelike_through_the_whole_interior_of` [OK]

*‘sec:largescale‘’s ‘eq:shape-invariant‘ with its Berger discriminant and its owed clause, its reassigned-layer eigenvalues, its ‘$-f$‘ nullity clause, its sign-blindness clause and its conjecture hedge, ‘sec:properframe‘’s ‘eq:proper-frame‘ with the ‘$E=1$‘ radial equation — **‘PO-74‘’s AMENDED DISCHARGE TAKEN ON ALL THREE OF ITS CLAUSES, AND THE ANSWER IS THE ROW’S OWN TERMINATION CONDITION: THE CONTINUATION DOES NOT GO THROUGH.** (1) ***THE THREE-METRIC IS OBTAINED, NOT WRITTEN***: on a constant-‘$\{}tilde\{}tau$‘ surface ‘$\{}dd\{}tau=-\{}dd\{}chi$‘, so ‘eq:proper-frame‘ RESTRICTS to ‘$h=[(\{}partial_\{}chi r)ˆ2-1]\{}dd\{}chiˆ2+rˆ2\{}dd\{}Omegaˆ2$‘ — nothing is posited at any point of the bead. (2) ***AND ‘$-f$‘ IS PRODUCED, NOT MATCHED***: the ‘$E=1$‘, ‘$k=0$‘ radial equation gives ‘$(\{}partial_\{}chi r)ˆ2=2M/r+rˆ2/\{}alphaˆ2=1-f$‘ IDENTICALLY, so the ‘$\{}chi$‘ block **is** ‘$-f$‘ by substitution, derived from the congruence’s equation of motion with no reference to any target eigenvalue. (3) ***AND THE MEASUREMENT IS DECISIVE TWICE OVER.*** **FIRST, THE TEST IS NOT AN INVARIANT ON WHAT THE FLOW DELIVERS**: computed by this receipt’s own curvature code — validated first on a round ‘$Sˆ3$‘, returning ‘$6\{}cdot2ˆ{2/3}\{}piˆ{4/3}=43.8232327$‘ with the radius CANCELLING — the layer gives ‘$\{}mathcal R=2/rˆ2$‘ and ‘$\{}mathcal R Vˆ{2/3}=4\{}cdot2ˆ{1/3}\{}piˆ{2/3}Lˆ{2/3}(-f)ˆ{1/3}rˆ{-2/3}$‘, which carries the layer’s SIZE **and** the arbitrary coordinate extent ‘$L$‘ of the ‘$\{}chi$‘ line. **AND THE STRONGER REASON IN THE SAME EXPRESSION**: the volume carries ‘$\{}sqrt{-f}$‘, and ‘$-f=-1$‘ at the turnaround, so the invariant has **no real value inside the lap at all** — the layer has no Riemannian volume there. ***SECOND, AND THIS IS THE RESULT: THE LAYER IS TIMELIKE THROUGH THE WHOLE INTERIOR OF THE LAP.*** The cubic factorises exactly at Nariai, ‘$-f=(r-\{}alpha/\{}sqrt3)ˆ2(r+2\{}alpha/\{}sqrt3)/\{}alp — a DOUBLE root at the front seam and a SIMPLE root at ‘$-2\{}alpha/\{}sqrt3$‘, **and those are the lap’s two unit-speed loci necessarily**, since ‘$-f=(\{}partial_\{}chi r)ˆ2-1$‘ makes ‘$-f=0$‘ and ‘$\{}lvert\{}partial_\{}chi r\{}rvert=1$‘ the same condition. So the sign of ‘$-f$‘ is the sign of ‘$(r+2\{}alpha/\{}sqrt3)/r$‘ and it is NEGATIVE on the whole stretch ‘$-2\{}alpha/\{}sqrt3¡r¡0$‘: ‘$+7.8717$‘ on the far collapse leg, ‘$0$‘ at the back seam, **‘$-1$‘ exactly at the turnaround**, ‘$-1.9260$‘ at mid-lift, ‘$0$‘ at the front seam, ‘$+0.0583$‘ on the expansion leg. ⇒ **AND THE ENTIRE LIFT LIES INSIDE THAT STRETCH**, ‘$A=0.727416\{}alpha$‘ against ‘$2/\{}sqrt3=1.154701\{}alpha$‘, *so the layer is Lorentzian from the back seam through the turnaround and the whole Euclidean segment to the branch point and no flow can deliver a round ‘$Sˆ3$‘ there.* The turnaround value is ‘$-1$‘ exactly and structural, ‘$\{}partial_\{}chi r=0$‘ there making the layer a UNIT-timelike line times ‘$Sˆ2(A)$‘. ***AND THE PAPER’S *”null at the handover and nowhere else”* IS TRUE OF THE SUBSTRATE AND FALSE OF THE CHART***: ‘$-f$‘ vanishes at BOTH seams and ‘r7134‘’s translation makes those one substrate point, ‘$-2\{}alpha/\{}sqrt3+\{}sqrt3\{}alpha=+\{}alpha/\{}sqrt3$‘ exactly. **AND ‘r7134‘ IS CONFIRMED BY THE METRIC ITSELF**: the angular block is even in ‘$r$‘ as the paper says, but ‘$-f$‘ carries ‘$2M/r$‘, which is ODD, so the two chart values of equal ‘$\{}lvert r\{}rvert$‘ carry ‘$Sˆ2$‘ factors of equal size and ‘$\{}chi$‘ blocks of **opposite sign** — equal ‘$\{}lvert r\{}rvert$‘ is not the same layer, now as a statement about the three-metric rather than about a label.* (node 66’s ‘r7145‘ order, which strikes ‘PO-79‘ on ‘r7144‘ and sets the queue: *”‘PO-74‘ is the richest item on the frontier and has been waiting four revisions. The amended discharge: the layer’s three-metric inside the lap OBTAINED rather than written — carried from the de˜Sitter presentation’s ‘$Sˆ3$‘ at a horn by a stated flow along the bead — with ‘eq:shape-invariant‘ MEASURED on it and the ‘$\{}chi$‘-block coefficient ‘$-f$‘ PRODUCED by applying the reassignment rather than selected to match. **And it now carries ‘r7134‘’s constraint, which you supplied.** ⇒ Take ‘PO-74‘ first unless you see a reason not to.”*). *Computes:* **19 checks, all pass, 1.4 s.** **(A)** nine clauses located in the CURRENT source, **partitioned at ‘r7150‘ by what this row does to them** — ‘eq:proper-frame‘ and its sum-dependence, the invariant and its Berger discriminant and its owed clause, the reassigned-layer eigenvalues and the conjecture hedge are reasoned FROM and pinned; the ‘$-f$‘ nullity clause and the sign-blindness clause are what this row ASKS TO CHANGE and are gated as an ENUMERATION over the states ‘P15‘ may produce *(**amended at ‘r7150‘**: first written as four hard pins, which went red on the success of its own work when ‘r7147‘ corrected both of those clauses to carry this row’s results — the second instance of the break ‘r7142‘ repaired, and the reason the lesson is recorded as the PARTITION rather than the pattern)*; **(B)** the layer obtained by restriction, ‘$-f$‘ produced from the ‘$E=1$‘ radial equation, and the flow’s compliance with ‘r7134‘’s constraint checked on a ‘400,001‘-point sweep for monotonicity of signed ‘$r$‘ rather than assumed; **(C)**

the curvature code written here from Christoffel symbols and **validated on a round ‘$S^3$‘ of free radius first**, returning the paper’s own ‘$43.8232$‘ with the radius cancelling symbolically, plus one measured property of the instrument the paper does not state — the round point is STATIONARY, ‘$g’(1)=0$‘ with ‘$g”(1)=-77.908$‘, so the test’s sensitivity is SECOND order in the squashing, **a correction the gate forced on this receipt’s own first draft**; **(D)** the Ricci tensor of the layer with the ‘$\{}chi$‘ block carried as a FREE symbol that does not appear in the answer, which is ‘70‘’s finding (2) measured from the inside rather than cited, then both failures of the invariant — the two non-zero derivatives, and the ‘$\{}sqrt{-f}$‘ in the volume against ‘$-f=-1$‘; **(E)** the cubic’s factorisation and its two roots symbolically, the nullity-equals-unit-speed identity, ‘$-f$‘ at seven loci spanning the lap, and the lift’s containment in the timelike stretch; **(F)** the translation carrying one seam to the other, and the oddness of ‘$f$‘ in ‘$r$‘ with the two signs at ‘$\{}lvert r\{}rvert=A$‘. *Bound:* **‘PO-74‘ TERMINATES ON ITS OWN STATED CONDITION RATHER THAN BEING DISCHARGED, AND THAT IS WHAT THE ROW SAYS IT IS.** *The reason is not a failure of analytic continuation but a change of causal character at the back seam: there is no spatial layer inside the lap for a sphere to continue INTO.* *It does not recompute ‘70‘’s Berger measurement — the paper’s expression is located and only the one property this receipt needs is checked on it.* *It does not claim the two presentations are one metric: ‘P15‘ denies that and this receipt USES the denial.* *It does not claim the de˜Sitter presentation’s ‘$S^3$‘ is wrong anywhere it is displayed — outside the seam it stands untouched.* *It proposes no edit: the *”nowhere else”* clause and the sign-blindness clause both want amending and that is the gate’s; nothing is computed on ‘PO-75‘.* *The gauge is the Nariai member the paper fixes, ‘$2M=2\{}alpha/3\{}sqrt3$‘, and ‘$r$‘ is carried SIGNED throughout, which is why the invariant returns ‘$\{}lvert r\{}rvert^{4/3}/r^2$‘ rather than ‘$r^{-2/3}$‘.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_layers_three_metric_obtained_by_restriction_is_timelike_through_t`

**P15R255** `P15_the_conjecture_reduces_to_umbilicity_because_a_conformal_carrying_preserves_the_invariant` [OK]

*‘sec:largescale‘’s owed clause, its reassignment denial, its foliation clause and ‘eq:shape-invariant‘, ‘sec:properframe‘’s ‘eq:sds-static‘ and its reassignment sentence — **‘r7147‘’s STEP, TAKEN ON THE PRESENTATION ‘r7147‘ NAMED: ‘PO-74‘’s CONJECTURE REDUCES TO ONE COMPUTABLE PREMISE, AND THE PREMISE HOLDS AT EXACTLY ONE POINT OF THE LAP.** (1) ***A CONFORMAL CARRYING PRESERVES THE TEST’S MODULUS EXACTLY***: the de˜Sitter horn’s layer is OBTAINED from the closed slicing’s own ODE, ‘$a’=a\{}sqrt{a^2/\{}alpha^2-1}$‘ with ‘$a(0)=\{}alpha$‘ giving ‘$a(\{}eta)=\{}alpha/\{}cos\{}eta$‘, so the layer there is ‘$a^2\{}times$‘ the UNIT round ‘$S^3$‘ and the test returns ‘$43.8232327$‘ with the factor cancelling. Carry the factor off the positive axis and ‘$\{}lvert\{}mathcal RV^{2/3}\{}rvert$‘ stays **exactly** ‘$43.8232327$‘ for ‘$a¿0$‘, ‘$a¡0$‘, ‘$a=+i$‘ and ‘$a=-i$‘ alike, while the PRINCIPAL value picks up a **cube root of unity** fixed by ‘$\{}arg a$‘ — ‘$0$‘, ‘$+2\{}pi/3$‘, ‘$+2\{}pi/3$‘, ‘$-2\{}pi/3$‘. ⇒ **So the purely imaginary segment is transparent to the test’s modulus, which neither previous attempt established**, and ‘$2\{}pi/3$‘ is the lap’s own angle rather than a convention — ‘r7107‘ proved the closure’s argument is exactly ‘$-\{}pi/3$‘ and the cube roots of unity are those same thirds, checked as ‘$\{}omega^3=1$‘ on every ratio. (2) ***BUT THE CARRYING AVAILABLE ALONG THE BEAD IS CONFORMAL ON THE LAYERS ONLY WHERE THE CONSTANT-‘$r$‘ FOLIATION IS UMBILIC***, and on ‘eq:sds-static‘ the shear measure ‘$K_{tt}/h_{tt}-K_{\{}rm ang}/h_{\{}rm ang}=\{}alpha(3M-r)/r^{3/2}\{}sqrt{\{}alpha^2r-2M\{}alpha^2-r^3}$‘ vanishes **iff** ‘$rf’=2f$‘, whose only root is ‘$r=3M$‘ — with ‘$M$‘ carried FREE, so it is a statement about the family. (3) ***AND THE NARIAI SELECTION PUTS THAT ONE RADIUS EXACTLY ON THE SEAM: ‘$3M=\{}alpha/\{}sqrt3=r_N$‘ where ‘$f$‘ and ‘$f’$‘ vanish together, with the shear non-zero at the turnaround and far out so the locus is not an artefact of sampling. ⇒ ***SO ‘PO-74‘ IS A QUESTION ABOUT UMBILICITY AND NOT ABOUT ANALYTIC CONTINUATION*** — the lift is no obstacle at all, and what the sphere has to survive is the SHEAR of the flow that carries it. *And the presentation is stated before anything is computed: ‘eq:sds-static‘ is the metric the paper reassigns FROM, so ‘r7147‘’s objection to ‘r7146‘ — the denial, located here — does not reach this computation.** (node 66’s ‘r7147‘ order, which keeps five of ‘r7146‘’s results, rejects its termination on the ground that a reassigned-chart result cannot reach the de˜Sitter sphere, and names the step: *”obtain the layer’s three-metric along the bead in the complex conformal parameter, starting from the sphere the de˜Sitter presentation displays at a horn, and let ‘eq:shape-invariant‘ be evaluated on what arrives”*, with two constraints — the closure is the translation, and *”whatever arrives must

have a real ‘eq:shape-invariant‘ … so the test itself is a filter on which object is the right one to be carrying"*.). *Computes:* **12 checks, all pass, 0.7 s.** **(A)** six clauses located in the CURRENT source, each exactly once, including the reassignment sentence that fixes which presentation ‘eq:sds-static‘ is — stated before any computation, because ‘r7146‘’s was on the other one; **(B)** the horn's conformal factor obtained from its defining ODE on the expanding branch with ‘$a(0)=\{}alpha$‘, and the test's baseline with the factor's derivative zero; **(C)** the carrying evaluated at six explicit factors spanning the positive, negative and both imaginary directions, with modulus and argument taken each time, every ratio checked to satisfy ‘$\{}omegaˆ3=1$‘; **(D)** the shear measure derived, shown equivalent to ‘$rf'-2f$‘, its unique root solved symbolically with ‘$M$‘ free, ‘$3M$‘ matched to ‘$\{}alpha/\{}sqrt3$‘ at the Nariai mass with ‘$f$‘ and ‘$f'$‘ both vanishing there, and the shear checked non-zero at the turnaround and at ‘$3\{}alpha$‘; **(E)** the three measured statements the receipt claims, and the absence of any posited layer metric inside the lap witnessed as what the computation does not contain. *Bound:* **IT REDUCES THE ROW; IT DOES NOT DECIDE IT. NO VERDICT ON THE SPHERE IS CLAIMED.** *Turning the shear into a statement about the layer's SHAPE needs the sheared layer's own invariant, which is the next step and is not taken here.* *No layer metric is written down anywhere inside the lap — the only three-metric in the receipt is the horn's and it is derived — so nothing posits what ‘r7115‘ posited.* *Nothing is evaluated on the reassigned surface ‘r7146‘ used, which ‘r7147‘ ruled out in the register and which this receipt accepts as ruled out.* *‘r7146‘’s five standing results are cited, not re-derived; no edit is proposed; nothing is computed on ‘PO-75‘.* *‘M‘ is free through the umbilicity computation and only pinned at the end against the paper's Nariai mass, so ‘$r=3M$‘ is a result about the family and ‘$3M=r_N$‘ is a fact about the selected member.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_conjecture_reduces_to_umbilicity_because_a_conformal_carrying_pr`

**P15R256** `P15_the_sheared_layers_invariant_is_obtained_in_closed_form_and_the_squashing_runs_to_zero_app` [OK]

*‘sec:largescale‘’s Berger discriminant, its invariant, its owed clause, its conjecture hedge and ‘eq:chi-block-factored‘, ‘sec:properframe‘’s reassignment sentence and foliation clause — **‘r7151‘’s ONE REMAINING ITEM: THE SHEARED LAYER'S OWN INVARIANT, OBTAINED IN CLOSED FORM — AND THE SHAPE IS NOT CARRIED.** (1) ***THE SHEAR IS EXACTLY THE BERGER DIRECTION***: the foliation's trace-free extrinsic curvature is ‘$\{}sigma=\{}tfrac23(f'/f-\{}sqrt f/r)\{},\{}mathrm{diag}(1,-\{}tfrac12,-\{}tfrac12)$‘, axisymmetric with ‘$\{}sigma_1=-2\{}sigma_2$‘ verified symbolically — **one distinguished direction against an equal pair, which is the squashing ‘P15‘’s own discriminant parametrises**, so the test is applied in the direction the flow actually shears. (2) ***AND THE ACCUMULATED SQUASHING INTEGRATES EXACTLY***: from ‘$\{}partial_\{}lambda h_{ij}=2NK_{ij}$‘ with ‘$N\{}dd\{}lambda=\{}dd r/\{}sqrt f$‘, ‘$\{}dd\{}ln\{}varepsilon/\{}dd r=f'/2f-1/r$‘ is **identically** ‘$\{}dd\{}ln(\{}sqrt f/r)/\{}dd r$‘, so ‘$\{}varepsilon\{}propto\{}sqrt f/r$‘ with the horn's round datum as the only input — *nothing posited at any point of the lap.* (3) ***THE RESPONSE HAS NO LINEAR TERM***: ‘$g(1+u)/g(1)=1-\{}tfrac89uˆ2-\{}tfrac8{81}uˆ3$‘ **exactly**, so the round point is a strict maximum and a squashing of either sign LOWERS the invariant. (4) ***AND ON THE LAP THE SQUASHING IS IMAGINARY THROUGHOUT AND VANISHES AT THE SEAM***: the double root makes ‘$f\{}le0$‘ with equality only at ‘$r_N$‘, so ‘$\{}lvert\{}varepsilon\{}rvert=\{}sqrt{-f\{}rvert}/r$‘ runs ‘0.8607‘, ‘0.7071‘, ‘0.1526‘, **‘0‘**, ‘2.2361‘ across ‘$3r_N$‘ to ‘$0.5r_N$‘. ⇒ ***SO THE CARRIED LAYER'S INVARIANT GOES TO ZERO AT THE SEAM — THE SPHERE'S SHAPE IS NOT CARRIED ACROSS IT, AND THE OBSTRUCTION IS AT THE SEAM RATHER THAN IN THE INTERIOR THE ROW SPENT THIRTY-FIVE REVISIONS ON.*** **AND A CORRECTION TO ‘r7150‘ CARRIED IN THE SAME PASS, FOUND BY TAKING A LIMIT ITS OWN SUBSTITUTION HID**: ‘r7150‘ gated ‘$\{}sigma(3M)=0$‘ with ‘$M$‘ free and that stands for a generic member — *but at the forced member ‘$3M$‘ coincides with the DOUBLE ROOT, so the measure is ‘$0/0$‘ there and the substitution is not the limit.* The one-sided values are ‘$\{}pm i\{}sqrt3/\{}alpha$‘, **non-zero, imaginary and opposite in sign**, so ‘r7150‘’s inference that the carrying is conformal at the seam does NOT hold at the selected member. *‘r7151‘’s paper sentence is unaffected, having been phrased as the two radii agreeing in VALUE — the ‘P07‘ guard doing its work; the overreach was in this seat's docstring.* (node 66's ‘r7151‘ order, which accepts ‘r7150‘, puts the umbilicity reduction in ‘P15‘ with the ‘P07‘ guard's constraint, and names the step: *"**THE SHEARED LAYER'S OWN INVARIANT.** ‘r7150‘’s scope is explicit: it reduces the row and does not decide it, because turning the shear into a statement about the layer's SHAPE needs the sheared layer's invariant. **That is the whole of what

is left.**"* — with three standing constraints (the de˜Sitter presentation, the translation closure, and no layer metric written down inside the lap) and a licence in both directions.). *Computes:* **13 checks, all pass, 1.0 s.** **(A)** seven clauses located in the CURRENT source, each exactly once, with the presentation fixed as the UNreassigned one before anything is computed; **(B)** the extrinsic curvature's eigenvalues and its trace-free part, checked axisymmetric and trace-free symbolically and checked non-zero; **(C)** the squashing's evolution equation integrated and matched IDENTICALLY against the closed form, plus its stationary radius re-derived from the squashing rather than from the shear; **(D)** the discriminant expanded about the round point to third order, the linear term shown absent and the relative coefficient matched to '$-8/9$' exactly; **(E)** the forced-member factorisation read for its sign, '$\{}lvert\{}varepsilon\{}rvert$' at five radii spanning the lap, and the invariant's vanishing taken as a limit; **(F)** the generic-'$M$' result re-derived, then the forced member's '$0/0$' exposed — with the sign gated on a NUMERICAL one-sided probe because 'sympy''s symbolic limits both return the principal value, simplifying through the branch, so only the modulus is taken from them; **(G)** the licence's direction, and the absence of any layer metric witnessed as what the computation does not contain. *Bound:* **IT DOES NOT CLAIM THE LAYER IS A THREE-SPHERE ANYWHERE.** *It applies 'P15''s own response function in the direction the flow shears, with a round datum at the horn, which is what that function is offered for — and it writes down no three-metric at any point of the lap, the condition that made 'r7115''s attempt circular.* *It does not compute on the reassigned chart.* *It does not claim the umbilic radius and the merged horizon are one object: 'P07''s guard governs that sentence and this receipt states only the computed agreement in value, as 'r7151' instructs.* *It does not take the '$\{}kappa$', '$\{}lambda$', shear trio into 'P07' — 'r7151' routed that there and said to take it after this step, not with it.* *'M' is carried free wherever the statement is about the family, and pinned only where the statement is about the selected member.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_sheared_layers_invariant_is_obtained_in_closed_form_and_the_squas`

**P15R257** `P15_the_seam_limit_of_the_carried_layer_with_its_chi_block_is_prop_throats_ds2_times_s2_obtaine` [OK]

*'prop:throat''s equal-radii sentence and its '$f(r_N)=f'(r_N)=0$, $f"(r_N)=-2\{}Lambda$', 'eq:properframe''s line element, 'sec:properframe''s three-lengths guard and its unit-speed seams — **'r7155''s 'PO-80', TAKEN WHOLE: 'prop:throat''s '$\{}dS_2\{}times S^2$' IS THE LIMIT OF THE CARRIED LAYER *TOGETHER WITH* ITS '$\{}chi$' BLOCK, OBTAINED AND NOT MATCHED.** (1) ***THE FOUR-METRIC IS EXACTLY A DIAGONAL WARPED PRODUCT IN THE LAYER'S OWN NORMAL TIME***: because the '$\{}chi$'-block coefficient '$-f$' depends on cosmic time ALONE, the shift '$\{}dd\{}chi+\{}dd\{}tilde\{}tau/(-f)$' is an **exact differential**, so with '$\{}hat\{}chi=\{}chi+\{}int\{}dd\{}tilde\{}tau/(-f)$' and '$\{}dd T=\{}dd r/\{}sqrt{-f}$' the metric is '$-\{}dd T^2+(-f)\{}dd\{}hat\{}chi^2+r^2\{}dd\{}Omega^2$' with '$(\{}dd r/\{}dd T)^2=-f$' — *identically, no limit taken* — and '$T$' is the proper time orthogonal to the cosmological layers, which is the parameter 'r7155' says to carry. (2) ***THE DIVERGENCE BELONGS TO THE DEGENERACY***: '$T=\{}int\{}dd r/\{}sqrt{-f}$' grows like '$(1/\{}sqrt\{}Lambda)\{}ln(1/(r-r_N))$' at the forward seam — measured increments '1.3581', '1.3324', '1.3297', '1.3294' over four decades against '$\{}ln10/\{}sqrt3$' — while at the back seam's **simple** zero the same integral CONVERGES, to '1.1829'. (3) ***AND THE '$\{}dS_2$' IS READ OFF THE FLOW***: '$\{}dd r/\{}dd T=\{}sqrt{-f}$' with '$-f\{}to\{}Lambda(r-r_N)^2$' gives '$r-r_N=Ce^{-\{}sqrt\{}Lambda T}$', so the '$\{}chi$' scale factor '$a=\{}sqrt{-f}\{}propto e^{-\{}sqrt\{}Lambda T}$', '$-\{}dot a/a=\{}sqrt\{}Lambda$' exactly and '$R_2=2\{}Lambda$' — '$\{}dS_2$' of radius '$1/\{}sqrt\{}Lambda$', which is 'prop:throat''s own '$\{}sqrt{-2/f"(r_N)}$' arrived at from the other side, with '$C$' absorbed by normalising '$\{}hat\{}chi$' so the limit is ONE metric. (4) ***THE ANGULAR BLOCK IS '$r_N^2\{}dd\{}Omega^2$' AND THE PRODUCT IS DIRECT***: both blocks approach the limit linearly in '$r-r_N$', so the warping switches off at the same rate '$\{}sqrt\{}Lambda$'. **AND THE JOIN FALLS OUT RATHER THAN BEING ARRANGED**: 'r7152''s squashing is '$\{}varepsilon=\{}sqrt{\{}lvert f\{}rvert}/r$' and the scale factor obtained here is '$a=\{}sqrt{-f}$', so '$\{}varepsilon=a/r\{}to a/r_N$' — ***the layer's squashing IS the throat's '$\{}dS_2$' scale factor over the areal radius, and 'PO-74''s negative and 'prop:throat''s '$\{}dS_2$' are one object seen twice***, with '$\{}varepsilon\{}propto e^{-\{}sqrt\{}Lambda T}$' reproducing 'r7155''s own correction by this route. ⇒ ***SO THE AREAL-RADIUS AGREEMENT IS NOT A COINCIDENCE OF THE FORCED MASS.*** **AND A CORRECTION TO 'r7155''s ARITHMETIC, OFFERED AND NOT CLAIMED**: its coefficient '$(1/\{}sqrt3)$' and its*

*rate ‘2.659‘ per decade disagree by exactly a factor of two; the coefficient is the right statement and the per-decade increment is ‘1.3294‘.* (node 66’s ‘r7155‘ order, which strikes ‘PO-74‘ on ‘r7152‘, adds the measurement that the seam lies at infinite parameter so the obstruction is in the APPROACH, names what the degeneration leaves, and opens ‘PO-80‘: *\{}”IS ‘prop:throat‘’s ‘$\{}dS_2\{}times S^2$‘ THE LIMIT OF THE CARRIED LAYER TOGETHER WITH ITS ‘$\{}chi$‘ BLOCK, OR ONLY OF ITS ANGULAR FACTOR?\{}”* — with the instruction to *\{}”take the three-metric and the ‘$\{}chi$‘ block TOGETHER as the seam is approached, with the parameter divergence carried rather than regularised away, and say whether the limit is that product — obtained, not matched\{}”*, and one guard named as the thing the row must not cross: ‘$\{}alpha/\{}sqrt3$‘ is the merged horizon’s AREAL radius and the throat three-sphere’s SIZE is a different quantity.). *Computes:* **28 checks, all pass, 7 s.** **(A)** the ‘$E=1$‘, ‘$k=0$‘ radial equation giving ‘$(\{}partial_\{}chi r)^2=1-f$‘ with ‘$M$‘ FREE, the ‘$(\{}tilde\{}tau,\{}chi)$‘ block and its determinant, the shift’s integrability, and the diagonal form verified by re-deriving ‘eq:proper-frame‘ FROM it symbolically, with ‘$\{}dd T=\{}dd r/\{}sqrt{-f}$‘ carried on a positive symbol because the expansion leg supplies that sign and ‘$f$‘ alone does not; **(B)** ‘$-f$‘ factorised at the Nariai mass into its double and simple roots with ‘$f”(r_N)=-2\{}Lambda$‘ and ‘$r_N=1/\{}sqrt\{}Lambda$‘ both identical, the normal-time increment measured over four decades and shown monotone from above onto ‘$\{}ln10/\{}sqrt3$‘, the factor-two disagreement in the order’s own figures exhibited arithmetically, and convergence at the simple root measured; **(C)** the flow integrated, its rate matched to ‘$\{}sqrt\{}Lambda$‘, the two-metric’s ‘$R_2$‘ computed from the metric by this receipt’s own curvature code and matched to ‘$2\{}Lambda$‘ and to ‘$\{}sqrt{-2/f”(r_N)}$‘, and ‘$C$‘ shown absorbable; **(D)** both relative approach rates derived symbolically and then measured at ‘$r-r_N=10^{-6}$‘; **(E)** the limit metric’s ‘$R=4\{}Lambda$‘, its Einstein property ‘$R_{ab}=\{}Lambda g_{ab}$‘, and its Kretschmann ‘$8\{}Lambda^2$‘ matched against ‘eq:sds-static‘’s at ‘$r_N$‘ computed SEPARATELY by the same code — each side DERIVED from its own metric with ‘$\{}alpha$‘ FREE and matched as a symbolic identity rather than evaluated at a chosen ‘$\{}alpha$‘, which is the re-parameterisation repair ‘cc66‘’s ‘r7157‘ round names and which this comparison is an instance of, being between two presentations’ variables; **(F)** the join to ‘r7152‘ as an identity between its squashing and this scale factor; **(G)** five paper clauses located in the CURRENT source, each exactly once, as clauses reasoned FROM — with the ONE clause this row asks to change ENUMERATED over the states the paper may produce rather than pinned, which is ‘r7150‘’s partition applied before the ratchet has to ask. *Bound:* **IT DOES NOT CLAIM THE THROAT THREE-SPHERE’S SIZE EQUALS ‘$\{}alpha/\{}sqrt3$‘, AND NOTHING IN THE DERIVATION USES THE ‘$S^3$‘ SIZE** — ‘$\{}alpha$‘ enters only inside ‘$f$‘, and the one length matched to ‘prop:throat‘ is the AREAL radius at its own degenerate root, which is the guard ‘r7155‘ names as the one this row must not cross. *It does not claim the limit is GLOBAL ‘$\{}dS_2$‘: it is the contracting planar patch, half of it, which is what a one-sided limit can deliver.* *It does not claim the layer is a three-sphere anywhere — the ‘$S^2$‘ is ‘eq:proper-frame‘’s own angular block and the ‘$\{}chi$‘ direction its own.* *It does not re-decide which of ‘sec:throat‘’s two routes carries the isotropization conclusion; that is a paper sentence and is routed, not edited.* *It does not take the ‘$\{}kappa$‘, ‘$\{}lambda$‘, shear trio into ‘P07‘.* *‘$M$‘ and ‘$\{}alpha$‘ are carried free wherever the statement is about the family.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_seam_limit_of_the_carried_layer_with_its_chi_block_is_prop_throats`

**P15R258** `P15_the_bead_routes_damping_survives_on_the_squashed_layer_because_the_berger_harmonics_carry_` [OK]

*‘sec:throat‘’s ‘$T(k)$‘ closed form with its ‘$k^2=L(L+2)$‘ label, its unit-amplitude clause, its round-spectrum admission and the near-horizon route’s independence, ‘sec:largescale‘’s measured squashing — **‘r7159‘’s ‘PO-81‘: THE BEAD ROUTE SURVIVES ON THE SQUASHED LAYER, AND UNIFORMLY IN THE SQUASHING RATHER THAN CONDITIONALLY ON ROUNDNESS.** (1) ***THE DECOMPOSITION THE TRANSPORT NEEDS EXISTS THERE, AND IT IS THE SAME ONE***: the Berger harmonics carry NO squashing — one and the same matrix element diagonalises the layer’s Laplacian at ‘$\{}varepsilon=1$‘, ‘$\{}tfrac13$‘ and ‘$5$‘ alike — so ‘$(L,m)$‘ is exactly conserved along the bead, nothing mixes, and **only the eigenvalue moves**. That closes ‘r7159‘’s second horn: the squashed layer does admit a decomposition the transport can use. (2) ***AND THE SPECTRUM IS OBTAINED FROM THE LAYER***: the metric built from the left-invariant forms and its Laplace–Beltrami operator computed from that metric give ‘$\{}lambda(L,m)=L(L+2)+4(\{}varepsilon^{-2}-1)m^2$‘ with ‘$\{}lvert m\{}rvert\{}le L/2$‘*

*and ‘$m\{}equiv L/2$‘ mod ‘$1$‘, verified on seven explicit harmonics at ‘$L=1,2,3$‘ and three squashings, the round ‘$3,8,15$‘ recovered at ‘$\{}varepsilon=1$‘. (3) ***TWO SECTORS THE SQUASHING CANNOT REACH***: ‘$\{}lambda(0,0)=0$‘ identically, so the isotropic mode still crosses at unity for every squashing; and ‘$m=0$‘, admissible exactly when ‘$L$‘ is EVEN, keeps ‘$\{}lambda=L(L+2)$‘ with no ‘$\{}varepsilon$‘ in it, so **the paper’s closed form survives verbatim for those modes** — while for odd ‘$L$‘ every ‘$\{}lvert m\{}rvert\{}ge\{}tfrac12$‘ and no mode keeps its round value. (4) ***AND EVERY OTHER MODE IS BOUNDED BELOW UNIFORMLY***: for ‘$\{}varepsilon\{}le1$‘ every anisotropic ‘$\{}lambda$‘ is at least its round value, for ‘$\{}varepsilon¿1$‘ the minimum is ‘$2L+L^2/\{}varepsilon^2$‘ at ‘$\{}lvert m\{}rvert=L/2$‘, so ‘$\{}lambda\{}ge2L$‘ for **every** squashing with the infimum only as ‘$\{}varepsilon\{}to\{}infty$‘ — measured across seven degrees, every admissible ‘$m$‘ and nine squashings spanning ‘$10^{-3}$‘ to ‘$5\{}times10^3$‘. ⇒ ***SO THE SUPPRESSION NEVER SWITCHES OFF***: the exponent is ‘$\{}int\{}sqrt\{}lambda\{},\{}dd\{}eta\{}ge\{}sqrt{2L}\{},s_{\{}rm tot}$‘, which at ‘$L=1$‘ is ‘4.72169‘, giving ‘$e^{-}\{}le8.900\{}times10^{-3}$‘ and, in the paper’s OWN asymptotic prefactor, ‘$T\{}le8.971\{}times10^{-2}$‘ against ‘$4.656\{}times10^{-2}$‘ for the same asymptote at the round value — a weakening OF THAT TRANSMISSION BOUND by $1.9265$ — under a factor of two, and not the EXPONENT ratio, which is a different quantity — still a suppression above ten, and monotone in degree. **AND WHERE THE RISK ACTUALLY IS, WHICH IS THE OPPOSITE END FROM ‘PO-74‘’s OBSTRUCTION**: ‘$\{}varepsilon\{}to0$‘ at the seam RAISES every eigenvalue, so the vanishing squashing strengthens the damping; the weak end is the close of the lift, where the squashing diverges. *The Berger parameter is the squashing NORMALISED to the horn’s round datum, ‘$\{}alpha\{}sqrt{-f}/r$‘ — the bare ratio tends to ‘$1/\{}alpha$‘, not to ‘$1$‘ — and normalised it passes through unity at exactly ‘$r=2M=\{}tfrac23r_N$‘, **alpha-free**: the forced mass’s own Schwarzschild radius.*
(node 66’s ‘r7159‘ order, which strikes ‘PO-80‘ on ‘r7156‘, takes this seat’s correction to the divergence rate and traces it to its own every-second-decade sampling, and opens ‘PO-81‘: *\{}”DOES ‘sec:throat‘’s SECOND ISOTROPIZATION ROUTE SURVIVE ON THE SQUASHED LAYER, OR IS IT WITHDRAWN?\{}”* — with what would discharge it named as *\{}”the damping recomputed on the carried layer with its measured squashing, the spectrum OBTAINED for that layer rather than assumed, the transport along the same bead\{}”*, both outcomes licensed as results, and the standing constraint that the layer is the CARRIED one with no round three-sphere written down anywhere on the bead.). *Computes:* **30 checks, all pass, 1.1 s.** **(A)** the layer’s metric built from the left-invariant forms, checked round at ‘$\{}varepsilon=1$‘ and Riemannian for every ‘$\{}varepsilon¿0$‘ by its determinant, and its Laplace–Beltrami operator computed FROM that metric rather than written down; its eigenvalue read on seven explicit matrix elements at three squashings, with the eigenFUNCTIONS’ independence of ‘$\{}varepsilon$‘ gated as the statement that the same function diagonalises the operator at all three and every ‘$m\{}ne0$‘ eigenvalue moves while its eigenfunction does not; **(B)** the isotropic mode’s ‘$\{}lambda=0$‘ identically, and its uniqueness as the only zero; **(C)** the admissible ‘$m$‘ range and its parity, the ‘$m=0$‘ sector’s ‘$\{}varepsilon$‘-freedom witnessed as the absence of ‘$\{}varepsilon$‘ from the expression, and the odd-‘$L$‘ floor ‘$\{}lvert m\{}rvert\{}ge\{}tfrac12$‘; **(D)** the two regimes of the sign of ‘$\{}varepsilon^{-2}-1$‘, the minimiser ‘$\{}lvert m\{}rvert=L/2$‘ derived symbolically, the limit ‘$2L$‘ taken, and the bound measured on a grid; **(E)** the squashing’s own profile with its three limits and the alpha-free unit crossing established as an identity rather than solved numerically; **(F)** ‘$s_{\{}rm tot}$‘ reproduced from its Gamma-function closed form, the exponents and the asymptote-against-asymptote comparison, and the bound’s monotonicity in degree; **(G)** five paper clauses located in the CURRENT source, each exactly once, as clauses reasoned FROM — with the ONE clause this row discharges ENUMERATED over the states the paper may produce rather than pinned. *Bound:* **IT DOES NOT CLAIM THE PAPER’S CLOSED FORM ‘$T(k)$‘ HOLDS WITH A VARYING EIGENVALUE** — that form is the constant-‘$k$‘ solution, and for the varying case only the comparison bound ‘$\{}int\{}sqrt\{}lambda\{},\{}dd\{}eta\{}ge\{}sqrt{2L}\{},s_{\{}rm tot}$‘ is asserted, with the exact integral along the actual ‘$\{}varepsilon(\{}eta)$‘ profile named as what this receipt does not do. *The bound is POINTWISE and therefore survives any profile, which is what makes it uniform.* *It is a statement about the SCALAR spectrum; no tensor or vector sector on the squashed layer is treated.* *It does not touch the near-horizon route, which needs nothing of the cosmological layer and carries the isotropization conclusion either way.* *It does not claim the layer is a round three-sphere anywhere — the whole point is that it

is not, and the squashing is ‘sec:largescale‘’s own measured one rather than a posited profile.* *It does not re-open the observable-multipole guard: ‘$L$‘ is the degree on the cosmological layer and is not the microwave-background multipole.* *‘$\{}alpha$‘ is carried free wherever the statement is about the family, and the one crossing is established alpha-free.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_bead_routes_damping_survives_on_the_squashed_layer_because_the_be`

**P15R259** `P15_the_lifts_exact_envelope_on_the_carried_layer_is_a_one_parameter_deformation_indexed_by_th` [OK]

*‘sec:throat‘’s ‘$s_{\{}rm tot}$‘ closed form, its unit-amplitude clause, its lift segment, ‘eq:squashed-spectrum‘ and its bound sentence, ‘sec:scope‘’s owed-source clause — **‘PO-75‘ TAKEN BY THIS SEAT’S OWN CHOICE UNDER ‘r7161‘’s LATITUDE, THROUGH THE ONE THING THE PAPER NAMES AS NOT CARRIED: THE LIFT’S *EXACT* ENVELOPE, WITH THE SQUASHING PROFILE CARRIED THROUGH THE TRANSPORT RATHER THAN BOUNDED.** (1) ***THE CONTROL FIRST***: the bead’s own measure ‘$\{}lvert\{}dd\{}eta\{}rvert=\{}lvert\{}dd r\{}rvert/(\{}lvert r\{}rvert\{}sqrt{\{}lvert 2M/r+r^2/\{}alpha^2\{}rvert})$‘, integrated from the comoving turnaround to ‘$r=0$‘, returns ‘$3.33873802357$‘ against the paper’s closed form — agreeing to ‘$3\{}times10^{-17}$‘ and ‘$\{}alpha$‘-free at three values of ‘$\{}alpha$‘, with ‘$-f=-1$‘ exactly at the turnaround. (2) ***AND THE LIFT LIES ENTIRELY WHERE THE SQUASHING EXCEEDS ITS ROUND VALUE***: ‘$\{}lvert\{}varepsilon\{}rvert=1.3747296$‘ at the turnaround rising to ‘$\{}infty$‘ at the close, because ‘r7160‘’s unit crossing sits at ‘$r=2M=\{}tfrac23r_N$‘, a POSITIVE radius — **so the stretch that RAISES every eigenvalue is on the expansion side and the transport never visits it**, which is what fixes the sign of the correction. (3) ***THE DEFORMATION IS ONE-PARAMETER***: ‘$\{}lambda/L(L+2)=1-\{}sigma(1-\{}varepsilon^{-2})$‘ with ‘$\{}sigma=4m^2/\{}lvert L(L+2)\{}rvert$‘ identically, so the exponent’s ratio to the round one is a function of ‘$\{}sigma$‘ alone — verified by ‘$(L,\{}lvert m\{}rvert)=(1,\{}tfrac12)$‘ and ‘$(6,2)$‘ sharing ‘$\{}sigma=\{}tfrac13$‘ and returning the same ‘$R=0.8419016934$‘ to ten figures. With ‘$R(0)=1$‘ to seventeen decimals and ‘$R(1)=s_{\{}rm tot}^{-1}\{}int\{}lvert\{}varepsilon\{}rvert 1}\{}dd\{}eta=0.2591014627$‘, by three routes. (4) ***AND ‘r7160‘’s UNIFORM BOUND IS NOT ATTAINED***: its infimum needs ‘$\{}varepsilon\{}to\{}infty$‘, which happens only at the close of the lift, so the maximal-charge band’s exponent tends to ‘$R(1)\{}sqrt{L(L+2)}\{},s_{\{}rm tot}$‘ and **stays LINEAR in degree** — ‘$934.0$‘ against the bound’s ‘$151.1$‘ at ‘$L=1024$‘, the gap widening past a factor of six. ⇒ ***SO THE LIFT’S EFFECTIVE LENGTH SHORTENS FROM ‘$3.3387380$‘ TO ‘$0.8650719$‘ IN THE MOST TRANSPARENT BAND AND NO FURTHER.*** **ON ‘PO-75‘ ITSELF**: every anisotropic mode is transmitted MORE strongly than the round-layer figures say — gains ‘$2.49$‘, ‘$10.6$‘, ‘$56.8$‘, ‘$346$‘, ‘$2309$‘, ‘$1.64\{}times10^{4}$‘ through ‘$L=6$‘ in the maximal-charge band, growing without bound — *and the surviving amplitude is still exponentially small in degree.* ⇒ ***THE ROW IS NARROWED AND NOT STRUCK: what the progenitor hands over survives better than the corpus says, and whether it SUFFICES is a question about an input amplitude this paper does not carry.*** (node 66’s ‘r7161‘ order, which strikes ‘PO-81‘, accepts the ‘r7156‘ amendment, and then declines to assign: *\{}”Nothing. ‘PO-74‘, ‘PO-80‘ and ‘PO-81‘ are all struck and the frontier is down to five rows, none of them yours \{}dots\{} this is a genuine choice rather than a queue, and I would rather you picked it than had it assigned\{}”* — recommending ‘PO-75‘ as the oldest row and this seat’s by provenance, and naming the remainder it left as *\{}”a stated LIMIT and not a row \{}dots\{} what the lift supplies is the uniform bound and not the exact damping, which would need the ‘$\{}varepsilon(r)$‘ profile carried through the transport\{}”*. This revision takes the recommended row THROUGH that stated limit, which is the step both have in common.). *Computes:* **28 checks, all pass, 4.6 s.** **(A)** the lift’s measure built from the ‘$E=1$‘, ‘$k=0$‘ radial equation and matched against the paper’s closed-form ‘$s_{\{}rm tot}$‘, checked ‘$\{}alpha$‘-free at three values, with the turnaround’s locus solved symbolically and ‘$-f=-1$‘ measured there; **(B)** the squashing’s range over the lift, its monotonicity, and the ‘$\{}lvert\{}varepsilon\{}rvert¿1$‘ statement joined to the symbolic identity ‘$2M=\{}tfrac23r_N$‘ that puts the crossing on the other side; **(C)** the exact exponent for EVERY admissible ‘$(L,m)$‘ through ‘$L=6$‘, with the ‘$m=0$‘ rows reproducing the round value exactly as the must-come-back-right control, every ‘$m\{}ne0$‘ row strictly below its round value, and none below ‘r7160‘’s bound; **(D)** the ‘$\{}sigma$‘ identity derived symbolically and the collapse measured on two different degrees; **(E)** both endpoints, ‘$R(1)$‘ by three independent routes and checked ‘$\{}alpha$‘-free, and ‘$R$‘’s monotonicity in ‘$\{}sigma$‘; **(F)** the bound’s non-attainment out to ‘$L=1024$‘, with ‘$I/L$‘ shown falling monotonically onto

‘$R(1)s_{\{}rm tot}$‘; **(G)** six paper clauses located in the CURRENT source as clauses reasoned FROM — with the owed-source clause ‘PO-75‘ is named from ENUMERATED over the states the paper may produce rather than pinned. *Bound:* **IT CLAIMS NO TRANSMISSION FIGURE.** *The paper's ‘$2^{7/3}k^2$‘ prefactor is the constant-‘$k$‘ solution's; with ‘$\{}lambda$‘ varying only the EXPONENT is computed, and the gains quoted are exponent ratios ‘$e^{I_{\{}rm round}-I}$‘ rather than amplitudes.* *It does not claim the progenitor's anisotropic spectrum is or is not sufficient — that needs an input amplitude this paper does not carry, and it is stated as the row's remaining question rather than answered.* *Scalar sector only.* *‘$L$‘ is the degree on the cosmological layer and is NOT the observable multipole; ‘sec:largescale‘'s map is untouched.* *No layer metric is written down on the lift; the squashing is ‘sec:largescale‘'s own measured one.* *The layer is TIMELIKE on the lift, so what enters the spectrum here is the squashing's MODULUS — ‘r7152‘'s convention, accepted at ‘r7161‘ — and the ‘$m=0$‘ control is what witnesses that the modulus is the right object to feed it.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_lifts_exact_envelope_on_the_carried_layer_is_a_one_parameter_defo`

**P15R260** `P15_no_paper_fixes_the_primordial_normalisation_because_three_sentences_inherit_it_and_the_L_t`
[OK]
*‘P16‘'s fully-specified-input sentence, its inherited-rather-than-derived clause and its closed-form amplitude-fit sentence, ‘P15‘'s progenitor-supplied ‘$A_s$‘/‘$n_s$‘ clause and its ‘$\{}eta$‘ precedent, ‘sec:largescale‘'s ‘eq:lowell‘ with its two printed lengths — **‘r7163‘'s TWO ORDERS, BOTH ANSWERED, AND THE SECOND ONE CAME BACK A DIFFERENT OBJECT FROM THE ONE BOTH SEATS HAD BEEN ASSUMING.** ***ORDER (1), THE READ: THE NEGATIVE IS THE OUTCOME, AND THE CORPUS STATES IT IN ITS OWN VOICE THREE TIMES RATHER THAN BY OMISSION.*** (1) ***NO PAPER OF THIS CORPUS FIXES THE PROGENITOR'S ANISOTROPIC NORMALISATION AT THE COMOVING TURNAROUND***, and the three sentences that say so are ‘P16‘'s *\{}"a fully specified input, available from standard cosmology, and not an idealisation to be chosen\{}"*, ‘P16‘'s *\{}"the mode content is the part of the primordial sector this construction inherits rather than derives\{}"*, and ‘P15‘'s *\{}"Progenitor-supplied, and inherited rather than derived: the amplitude ‘$A_s$‘ and tilt ‘$n_s$‘\{}"*. (2) ***AND THE COUNT LOCATES IT RATHER THAN INFERRING IT FROM SILENCE***: ‘$A_s$‘ appears **21** times in the paper that USES it and **ZERO** times in each of the two that would have to produce it; ‘P16‘'s one ‘normalisation‘ is the Euclidean action's GRAVITATIONAL one; ‘P14‘'s 31 ‘turnaround‘s are the turnaround-MASS row's, co-occurring with ‘mass‘ in one sentence; with ‘anisotrop‘ at **ZERO** in ‘P14‘ against **8** in ‘P15‘ as the control that the instrument works. (3) ***WHAT THE CORPUS DOES FIX IS THE PROPAGATION***, which ‘r7163‘ did not ask for: ‘P16‘ computes the contrast *\{}"then peaks where matter domination ends, at ‘$\{}sim10^{-6}$‘\{}"* with *\{}"The peak value is independent of the composition\{}"*, linear in the inherited datum — ***so the missing scalar is ONE NUMBER and the chain from it is already built***, and the paper supplies its own precedent for the KIND of datum it is, carrying *\{}"the baryon-to-photon ratio ‘$\{}eta$‘ as a measured datum from a baryogenesis it does not model\{}"*. ⇒ ***SO ‘PO-75‘ DOES NOT MOVE: no paper owns the normalisation, so there is no paper for the row to move to, and ‘r7163‘'s termination condition is not met.*** ***ORDER (2), THE MAP, AND THE FINDING GOES ABOVE THE EVALUATION BECAUSE IT IS NOT A RELABELLING***: (4) ‘eq:lowell‘ returns the kernel's peak ARGUMENT ‘$k_LD_C$‘, and since ‘$\{}sum_\{}ell(2\{}ell+1)j_\{}ell(x)^2=1$‘ **exactly** the weight ‘$w_\{}ell(L)$‘ is a genuine distribution over ‘$\{}ell$‘ — against which the printed value sits near the **NINETIETH PERCENTILE** and not the centre: ‘$P(\{}ell\{}le\{}ell_L)=0.879,0.890,0.911,0.930$‘ for ‘$L=1\{}ldots4$‘, above the mode by ‘1.70‘ to ‘2.93‘ at every degree. (5) ***AND THE WIDTH CROSSES THE SPACING AT THE THIRD DEGREE***: the distribution's ‘$\{}sigma_\{}ell=1.195,1.8 against a mean spacing between consecutive degrees of ‘2.18‘ to ‘2.38‘, so the ratio runs ‘0.50, 0.80, 1.10, 1.39, 1.68, 1.96, 2.25, 2.53‘ and grows past unity without bound. ⇒ ***THE MAP IS A PROJECTION AND NOT A RELABELLING: quasi-injective for ‘$L=1,2$‘ and smearing from ‘$L=3$‘ up***, with consecutive degrees' 2%-weight bands overlapping in 4 to 7 multipoles from the third degree. (6) ***SO AN ENVELOPE EXPONENTIAL IN ‘$L$‘ IS NOT AN ENVELOPE EXPONENTIAL IN ‘$\{}ell$‘ ABOVE ‘$L=3$‘*** — a fixed ‘$\{}ell$‘ draws on several layer degrees at once and a steeply-falling envelope makes the LOWEST contributing degree dominate — ***while below it the relabelling reading the ‘sec:throat‘ route stands on is safe.*** *And the low-‘$\{}ell$‘ leak falls fast: ‘$P(\{}ell\{}le3)=0.598,0.150,0.058,0.033$‘ for ‘$L=1\{}ldots4$‘,*

*so the floor below ‘$\{}ell_2$‘ is softened by about a sixth of one mode and not erased.** (node 66’s ‘r7163‘ order, which narrows ‘PO-75‘ rather than striking it under the standing rule that *\{}”an item leaves a list by being done and never by reclassification, and a dependency is not a discharge\{}”*, puts this seat’s sign correction in print, moves the modulus convention from the register into ‘sec:throat‘, repairs its own ‘(G)(4)‘ edit under the rule that whoever’s edit broke a receipt repairs it — with the transferable lesson *\{}”a pin on a label is safe and a pin on how often a label is cited is a prediction about the next revision’s prose\{}”* — and then orders two things: (1) *\{}”Read for where an anisotropic amplitude at the comoving turnaround is fixed, or shown not to be fixed anywhere, and report whichever it is\{}”*, with *\{}”a negative is a result here and is the outcome I would bet on\{}”* and *\{}”if no paper fixes it, say so with the count that establishes it\{}”*; and (2) *\{}”Evaluate it: for the degrees the envelope is computed at — ‘$L=1$‘ first, then the handful that reach the first peak — what multipole does the map return, and with what width?\{}”*, with *\{}”The thing I want is whether the map is one-to-one over that range or smears\{}”* and *\{}”if the map turns out to be a different object from what either of us has been assuming — a projection rather than a relabelling, say — that is a finding and it goes above the evaluation rather than inside it\{}”*.). *Computes:* **21 checks, all pass, 0.9 s.** **(A)** the READ, done first because the order says the read precedes the computation: four paper bodies loaded from the CURRENT source, the three inheritance sentences located each exactly once as clauses reasoned FROM, the ‘$A_s$‘ counts taken in all three candidate papers, ‘P16‘’s one ‘normalisation‘ disambiguated by its own sentence, ‘P14‘’s ‘turnaround‘s disambiguated by sentence-level co-occurrence with ‘mass‘, the ‘anisotrop‘ control taken, and the propagation sentence and the ‘$\{}eta$‘ precedent located; **(B)** the stretch ‘$D_C/r_0=2.7618293$‘ computed from the paper’s OWN two printed lengths and matched against its printed ‘2.76‘, the completeness identity ‘$\{}sum_\{}ell(2\{}ell+1)j_\{}ell(x)^2=1$‘ verified to ‘$10^{-12}$‘ so the smearing question is well posed before it is asked, then mode, median, mean, standard deviation, ‘$P(\{}ell\{}le3)$‘ and ‘$P(\{}ell\{}le\{}ell_L)$‘ tabulated for ‘$L=1\{}ldots8$‘, the mode displacement bounded both ways and shown not to grow with degree, and the width compared against the spacing degree by degree; **(C)** the consequence for ‘r7162‘’s envelope, with each degree’s 2%-weight band computed and the consecutive overlaps counted. *The ‘$j_\{}ell$‘ are evaluated through the half-integer Bessel function at ‘mp.mp.dps = 30‘ rather than by recursion, and every paper clause is matched against the file read at run time.* *Bound:* **IT IS A READ PLUS A PROPERTY OF THE PAPER’S OWN MAP, AND IT FIXES NO AMPLITUDE.** *The missing scalar is reported as missing; no value is supplied, inferred or recommended for it.* *It does not claim the layer degree IS the observable multipole — the whole finding is that the paper’s map between them is a projection, which is the standing guard sharpened rather than crossed.* *It does not recompute ‘r7162‘’s exponents or re-express them in ‘$\{}ell$‘: the consequence is stated as a limit on how that envelope may be read, not as a new envelope.* *It does not claim the low-‘$\{}ell$‘ floor is erased or that ‘sec:lowl‘’s figures are wrong; the leak is measured and found to soften by about a sixth of one mode.* *It treats the kernel’s shape alone — no transfer function, no window, no projection through a visibility — so the widths are the geometric minimum a real projection can only exceed.* *Scalar sector only, and no paper file is edited: both orders’ consequences for the prose are routed.* *Run:* `python3 receipts/P15_CR_cosmology/P15_no_paper_fixes_the_primordial_normalisation_because_three_sentences_`

**P15R261** `P15_the_vector_sectors_pointwise_bound_fails_while_its_exponent_exceeds_the_scalars_and_the_te` [OK]

*‘sec:throat‘’s uniform-floor clause and its unmixed-transport clause, ‘eq:squashed-spectrum‘, the propagating-modes guard, ‘sec:intro‘’s scalar-half scoping sentence — **THE TWO SECTORS ‘r7160‘ AND ‘r7162‘ BOTH NAMED AS NOT DONE, TAKEN TOGETHER, BECAUSE THE ANSWER IS ONE STRUCTURAL FACT READ THREE TIMES — AND THE TWO SECTORS FAIL DIFFERENTLY FROM EACH OTHER AND FROM THE SCALARS.** (1) ***THE VECTOR SPLIT SURVIVES EVERY SQUASHING, AND FOR A REASON THAT IS NOT A FACT ABOUT THIS LAYER AT ALL***: ‘$\{}Delta_H=\{}dd\{}dd^*+\{}dd^*\{}dd$‘ commutes with ‘$\{}dd$‘ and ‘$\{}dd^*$‘ on ANY Riemannian manifold, so the exact and co-exact subspaces are each invariant at every ‘$\{}varepsilon$‘ — **measured as the gradient sector returning ‘eq:squashed-spectrum‘ member for member at six degrees with the squashing free**, which also fixes the operator’s convention before anything is built on it. ⇒ *So the vector sector needs no analogue of ‘r7160‘’s left-invariance argument, which LOOKED like the general reason and is not.* (2) ***BUT INSIDE THE CO-EXACT SECTOR THE DEFORMATION REACHES THE BASIS,*

*WHICH IS THE EXACT OPPOSITE OF THE SCALAR CASE***: the isometry group drops to ‘$SU(2)\{}times U(1)$‘ and a ‘$1$‘-form carries its own frame charge, so the conserved label is the DIFFERENCE and the invariant blocks are the triples ‘$\{}{e_+\{}!\{}otimes\{}!q{+}1,\{},e_3\{}!\{}otimes\{}!\{}otimes\{}!q{-}1\{}}$‘ — **zero matrix entries cross that grouping, and the coupling inside it is ‘$2\{}sqrt2\{}varepsilon$‘ at ‘$L=1$‘ and ‘$4\{}varepsilon$‘ at ‘$L=2$‘, PROPORTIONAL TO THE SQUASHING and nonzero for every value of it** — so the eigenVECTORS move along the bead and ‘$(L,m)$‘ is not carried unmixed. *The one exception is the extreme charge, whose block is ‘$1\{}times1$‘: that mode mixes with nothing, and it is also the one that matters, because it is the smallest at large squashing.* (3) ***AND THE SCALAR SECTOR’S UNIFORM FLOOR HAS NO VECTOR COUNTERPART***: the unmixed extreme-charge eigenvalue is ‘$\{}lambda^{\{}perp}=(L+2)^2/\{}varepsilon^2$‘ **exactly** — ‘$9,16,25,36,49,64$‘ over ‘$L=1\{}ldots6$‘ — so it runs to ZERO as ‘$\{}varepsilon\{}to\{}infty$‘, which is the close of the lift, precisely where ‘r7160‘’s ‘$\{}lambda\{}ge2L$‘ still held. *For even ‘$L$‘ a mixed branch dips below it; both vanish like ‘$\{}varepsilon^{-2}$‘.* (4) ***AND THE SUPPRESSION SURVIVES THE BROKEN FLOOR ANYWAY, BECAUSE THE MEASURE IS THIN WHERE THE EIGENVALUE IS SMALL***: carrying the exact minimum through the lift’s own ‘$\{}lvert\{}dd\{}eta\{}rvert$‘ gives an exponent that is finite, LINEAR in degree, and **LARGER than the scalar sector’s most transparent band at every degree computed** — ‘2.5952‘ against ‘1.4983‘ at ‘$L=1$‘, which is exactly ‘$3\{}times0.8650719$‘ and ‘$\{}alpha$‘-free to thirteen figures, and ‘4‘ to ‘13‘ per cent above the scalar band through ‘$L=6$‘. ⇒ ***SO A POINTWISE BOUND RUNNING TO ZERO COST THE DAMPING NOTHING, AND LEFT IT STRONGER RATHER THAN WEAKER.*** (5) ***AND THE TENSOR SECTOR IS NOT A SPECTRUM QUESTION ON THIS LAYER AT ALL***: ‘$\{}Delta_L$‘ keeps TRACELESSNESS at every squashing — machine zero across two degrees and six squashings — and **GENERATES DIVERGENCE as soon as the layer stops being EINSTEIN**, which the Berger layer is only at ‘$\{}varepsilon=1$‘ because ‘$R_{ab}=\{}mathrm{diag}(4-2\{}varepsilon^2,4-2\{}varepsilon^2,2\{}varepsilon^2)$‘. The transverse-traceless subspace is invariant to ‘$4\{}times10^{-16}$‘ at the round point and at no other value, with the generated divergence **LINEAR in the Einstein deficit** near it and already ‘24‘–‘32‘ per cent of the image at the lift’s entry, exceeding the whole of it beyond. ⇒ ***SO ‘PO-81‘’s SECOND HORN — the one that CLOSED for the scalars — OPENS FOR THE TENSORS: what is owed there is a DECOMPOSITION and not an eigenvalue, and no tensor eigenvalue on the squashed layer is quoted here.**** (this seat’s own ‘r7160‘ and ‘r7162‘, each of which named these two sectors as what it did not do: ‘r7160‘’s *\{}”It is a statement about the SCALAR spectrum; no tensor or vector sector on the squashed layer is treated\{}”* and ‘r7162‘’s *\{}”Scalar sector only\{}”* — taken together rather than separately because the computation showed the two answers share one mechanism, under the latitude ‘r7161‘ granted this seat when it declined to assign (*\{}”this is a genuine choice rather than a queue, and I would rather you picked it than had it assigned\{}”*) and with ‘r7163‘’s citation-count lesson applied throughout: the one clause this row’s result bears on is ENUMERATED over the states the paper may produce, and the one pin is on a LABEL.). *Computes:* **29 checks, all pass, 25 s.** **(A)** the layer’s connection, Riemann and Ricci built from the STRUCTURE CONSTANTS of the orthonormal left-invariant frame rather than from a coordinate chart, controlled at ‘$\{}varepsilon=1$‘ against ‘$R_{ab}=2g_{ab}$‘ and ‘$R=6$‘, with the non-Einstein deficit solved symbolically and shown to vanish exactly at ‘$\{}varepsilon^2=1$‘ — the single fact section (D) later turns into the tensor obstruction; **(B)** the Hodge Laplacian on ‘$1$‘-forms assembled as a matrix on (frame index) ‘$\{}times$‘ (layer charge) from that connection by the Weitzenböck route, controlled against the ROUND ‘$S^3$‘ ‘$1$‘-form spectrum WITH ITS MULTIPLICITIES at three degrees, then the exact/co-exact split verified by the gradient eigenvalues reproducing ‘eq:squashed-spectrum‘ exactly, the block grouping verified by counting entries that cross it, the ‘$\{}varepsilon$‘-proportional coupling exhibited and its vanishing limit taken, and the extreme-charge eigenvalue established as ‘$(L+2)^2/\{}varepsilon^2$‘ symbolically at six degrees; **(C)** the lift’s measure rebuilt from the ‘$E=1$‘, ‘$k=0$‘ radial equation and matched against ‘r7162‘’s own three numbers before anything is integrated on it, then the exponent taken on the EXACT minimum over all co-exact branches, checked finite, checked equal to ‘$3\{}times0.8650719$‘ at ‘$L=1$‘ where the minimiser is the unmixed mode, checked above the scalar band at every degree, and checked ‘$\{}alpha$‘-free at three values of ‘$\{}alpha$‘; **(D)** the Lichnerowicz operator on symmetric ‘$2$‘-tensors built by the same route, its transverse-traceless subspace obtained as the numerical null space of the trace, anti-

symmetry and divergence constraints, controlled at '$\{}varepsilon=1$' by an invariance residual below '$10^{-12}$' and a top eigenvalue '$L^2+6L+12$' at five degrees, then the trace and the divergence it generates measured separately off the round point and the divergence's linearity in the Einstein deficit measured near it; **(E)** five paper clauses located in the CURRENT source, each exactly once, as clauses reasoned FROM — with the ONE clause this row's result bears on ENUMERATED over three states the paper may produce, and the one pin a LABEL rather than a citation count. *Bound:* **IT QUOTES NO TENSOR EIGENVALUE ON THE SQUASHED LAYER, AND THAT IS THE RESULT RATHER THAN A GAP** — the object such an eigenvalue would belong to is not invariant off the round point, so what is owed is a decomposition; the obstruction is exhibited AND identified with the Einstein deficit, which is as far as this row goes. *Nothing here is offered as this paper's TENSOR claim: 'sec:intro''s own sentence puts the tensor half in the companion dynamics paper, and the tensor result is an obstruction to a computation rather than a physical tensor spectrum.* *The vector exponent is computed on the instantaneous minimum; at '$L=1$' the minimiser is the UNMIXED extreme-charge mode so no adiabaticity is assumed there, while at the higher degrees the mixing makes the quoted exponent a WKB reading on a rotating basis and is named as such rather than claimed exact.* *No transmission FIGURE — exponents only, as in 'r7162'.* *'$L$' is the degree on the cosmological layer and NOT the observable multipole; the map is 'sec:largescale''s, which this seat's 'r7164' found to be a projection and not a relabelling above its third degree, and that finding is used nowhere here and re-opened nowhere.* *No layer metric is written down on the lift: the squashing is 'sec:largescale''s own measured one, carried as its modulus on the timelike segment under 'r7152''s convention.* *The paper's guard that the computation concerns PROPAGATING modes applies to the vector sector unchanged; the constrained sector is not re-litigated.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_vector_sectors_pointwise_bound_fails_while_its_exponent_exceeds_t`

**P15R262** `P15_po83_does_not_move_because_the_companion_avoids_the_obstruction_by_symmetry_not_by_einstei` [OK]

*'sec:intro''s scalar-half scoping sentence, 'eq:squashed-spectrum', the two clauses 'r7164' rescoped on this seat's own recommendation, and the companion dynamics paper's own description of where its tensor sector lives — **'PO-83' TAKEN, AND TAKEN THROUGH THE SECOND ROUTE THE ROW ITSELF NAMES AND 'r7164' SAYS IT PREFERS: DOES THE ROW MOVE TO THE COMPANION, WHICH ALREADY CARRIES A TENSOR SECTOR? IT DOES NOT MOVE, AND THE REASON IS BETTER THAN THE MOVE WOULD HAVE BEEN.** (1) ***THE COMPANION AVOIDS THE OBSTRUCTION BY A DIFFERENT MECHANISM RATHER THAN BEING EXEMPT FROM IT***: its tensor sector is carried on 'eq:metric''s '$T^2$'-symmetric Gowdy–de˜Sitter leaf, and **this row MEASURES that leaf's own Ricci and finds it NOT EINSTEIN either** — three nonvanishing traceless-Ricci components. What saves it is ALGEBRAIC: the torus block's determinant is '$R^2$' identically, so at fixed area exactly ONE function is left and **there is no divergence constraint to solve.** ⇒ ***Two geometries avoiding ONE obstruction for TWO different reasons is not one question answered twice, so the row stays where it is.*** (2) ***BUT THE OBSTRUCTION IS NOT A PROPERTY OF THE SPACETIME — IT IS A PROPERTY OF THE INDUCED METRIC ON THE LAYER***: the '$4$'-metric the layer sits in is **EXACTLY EINSTEIN with '$M$' and '$\{}alpha$' FREE**, '$R_{ab}=\{}Lambda g_{ab}$' identically with '$R=4\{}Lambda=12/\{}alpha^2$', so the traceless Ricci that drives the obstruction is **IDENTICALLY ZERO** there at every radius and for every member of the family — and the same holds on 'r7156''s seam limit, so it is not special to one chart. (3) ***AND THE OBSTRUCTION IS A PROPORTIONALITY AND NOT A CORRELATION***: the divergence '$\{}Delta_L$' generates from a transverse-traceless tensor, divided by the Einstein deficit, **CONVERGES over four decades of deficit** to '0.255551' at '$L=2$' and '0.183473' at '$L=3$' — each stable to four figures across the range, nonzero, and degree-dependent — while the TRACE stays at machine zero throughout, so what fails is transversality alone. ⇒ ***SO A VANISHING DEFICIT IS A VANISHING OBSTRUCTION RATHER THAN A SMALLER ONE***, which is what licenses carrying the identical zero above into a statement about the four-dimensional question instead of only into a statement that it is less obstructed. ⇒ ***'PO-83' IS NARROWED DECISIVELY AND STAYS WHERE IT IS: what is obstructed is the LAYER-BY-LAYER HARMONIC METHOD the scalar route uses, and what the row owes is a FOUR-DIMENSIONAL treatment rather than a three-layer decomposition. THE OBSTRUCTION IS DIMENSION-SPECIFIC AND DOES NOT REACH THE PHYSICS.*** *The distinction that carries the whole answer: the obstruction bites exactly where transverse-*

*tracelessness must be OBTAINED, and never where a symmetry supplies it.** (node 66's ‘r7164‘ order, which lands ‘r7164‘ and ‘r7166‘, takes all three prose recommendations (homing one differently and recording the departure rather than making it silently), scopes two sentences of its own that this seat's ‘r7166‘ corrected, repairs two receipts from two different seats broken by that one-sentence scoping, and opens ‘PO-83‘ as ‘PO-81‘'s remainder under the standing rule that a struck row's remainder earns its own row in the same pass — with the row offered rather than assigned: *\{}"If you want the row, take it; if you would rather the next thing were ‘PO-79‘'s remainder or ‘PO-74‘'s flow, take that and say so\{}"*, and with its own second route written in: *\{}"the companion dynamics paper already carries the tensor sector, so the question may be answerable there on its own geometry — in which case the row MOVES to that paper rather than closing on a scope note, which I prefer to either alternative\{}"*. This row takes the offered row AND tests the preferred route, and reports that the route is not available for a stated reason.). *Computes:* **18 checks, all pass, 2.6 s.** **(A)** the four-geometry's Ricci computed FROM the metric by this receipt's own Christoffel-to-Ricci code with ‘$M$‘ and ‘$\{}alpha$‘ carried free, its Einstein property and its traceless part established as symbolic identities rather than evaluated at a chosen member, its scalar curvature matched to ‘$4\{}Lambda$‘ AND to ‘$12/\{}alphaˆ2$‘, and the same Einstein property re-established on ‘r7156‘'s seam-limit metric so the statement is not a chart artefact; **(B)** the companion's constant-‘$t$‘ leaf built from ‘eq:metric‘ with its three functions symbolic, its traceless Ricci computed and its nonvanishing components COUNTED, its torus determinant verified as ‘$Rˆ2$‘ identically, and the companion's own three sentences located in the CURRENT source each exactly once as clauses reasoned FROM; **(C)** the Lichnerowicz operator and the transverse-traceless projection rebuilt from ‘r7166‘'s route, the round-point invariance re-verified as the control that the instrument reads zero when there is nothing to read, then the divergence-to-deficit ratio measured at five squashings per degree spanning four decades and shown to converge, with the nonzero degree-dependent coefficient gated and the trace gated at machine zero separately so the identification is of transversality's failure alone; **(D)** five paper clauses located in the CURRENT source as clauses reasoned FROM — including the two ‘r7164‘ rescoped, read here in their settled form — with the ONE clause this row's result bears on ENUMERATED over three states the paper may produce, which is the form that survived the last two landings where a counted pin did not. *Bound:* **IT DOES NOT CARRY OUT THE FOUR-DIMENSIONAL TRANSVERSE-TRACELESS COMPUTATION, AND THAT IS ‘PO-83‘'s OWN NEXT STEP.** *What is established is that the quantity which obstructs the three-dimensional problem vanishes IDENTICALLY on the four-geometry, and that the obstruction is exactly linear in that quantity — which is a reason to expect the four-dimensional question to be well posed, not a demonstration that it is.* *It does not claim a four-dimensional route keeps the scalar route's economy: that economy is ‘$(L,m)$‘ carried unmixed along the curve, and whether a four-dimensional treatment preserves it is untouched.* *Nothing here is offered as ‘P15‘'s TENSOR claim — ‘sec:intro‘ puts the tensor half in the companion, and this is a statement about where a question can be ASKED.* *It does not re-measure ‘r7166‘: that obstruction is taken as given and only its identification is sharpened.* *It makes no claim about the companion's own RESULTS, which are that seat's; the single fact taken from it is the geometry its tensor sector is carried on, read from its own equations rather than from its prose.* *‘$M$‘ and ‘$\{}alpha$‘ are carried free wherever the statement is about the family.* *Run:* `python3 receipts/P15_CR_cosmology/P15_po83_does_not_move_because_the_companion_avoids_the_obstruction_by_sy`

**P15R263** `P15_the_four_dimensional_treatment_keeps_the_economy_and_keeps_more_of_it_but_in_the_sphere_la` [OK]

*‘eq:sds-static‘'s own angular block, ‘eq:squashed-spectrum‘, ‘sec:intro‘'s scalar-half scoping sentence and the clause ‘r7164‘ scoped to the scalar sector — **‘PO-83‘'s FOUR-DIMENSIONAL STEP, ORDERED AT ‘r7165‘ ON THE QUESTION THIS SEAT NAMED RATHER THAN ON WELL-POSEDNESS: DOES A FOUR-DIMENSIONAL TREATMENT KEEP THE SCALAR ROUTE'S ECONOMY? ⇒ IT KEEPS IT, AND KEEPS MORE OF IT THAN THE THREE-LAYER ROUTE EVER HAD — BUT IN A DIFFERENT LABEL, AND THE BRIDGE BETWEEN THE LABELS EXISTS ON EXACTLY ONE SECTOR.** (1) ***THE ECONOMY IS KEPT, BY THE SAME MECHANISM THAT SAVED THE SCALARS***: the four-chart's angular factor is ‘$rˆ2$‘ times the **ROUND** unit two-sphere — a symbolic identity with ‘$M$‘ and ‘$\{}alpha$‘ FREE, with no squashing parameter existing on that chart at all and nothing mixing the angular directions into the rest — so moving along the curve RESCALES the fibre and does nothing else to it. ⇒ *The deformation reaches the eigenvalue and not the basis, which is*

*exactly the sentence ‘r7164‘ scoped to the scalar sector: it holds for the tensor sector too on this chart, and for the same reason, because **a round fibre has no shape to deform.*** (2) ***AND THE ECONOMY IS STRICTLY BETTER THAN THE SCALAR ROUTE’S***: all three rotational Killing vectors are exact with the mass and throat constant free, the angular dependence SEPARATES at six explicit harmonics across three degrees with no angular variable surviving, and **the reduced radial operator comes back IDENTICAL across every ‘$m$‘ at each ‘$\{}ell$‘**, with the degree entering only as ‘$\{}ell(\{}ell+1)$‘ read off the difference between two degrees rather than assumed. ⇒ ***So this is a ONE-label family of radial problems where the three-layer route is a TWO-label family*** — ‘eq:squashed-spectrum‘ depends on ‘$mˆ2$‘ and NEEDS the second label, and here no such parameter remains: fewer channels, not more. (3) ***BUT THE LABEL IS THE TWO-SPHERE MULTIPOLE AND NOT THE LAYER DEGREE, AND THE BRIDGE IS EXACT ON ONE SECTOR ONLY***: the layer’s ‘$m=0$‘ harmonics are **exactly** the fibre-independent ones — measured on four explicit harmonics, every ‘$m\{}neq0$‘ one carrying the fibre coordinate — so exactly that sector descends to the Hopf base, and there the two spectra are the SAME NUMBER, ‘$L(L+2)=4j(j+1)$‘ at ‘$j=L/2$‘, **identically in the degree rather than at a chosen one.** ⇒ ***The charged modes are UNBRIDGED rather than poorly bridged.*** (4) ***AND THE ADMISSIBILITY CONDITION ARRIVES TWICE FROM TWO UNRELATED ARGUMENTS***: the Hopf projection needs ‘$j=L/2$‘ INTEGRAL, and ‘r7160‘’s parity rule admits ‘$m=0$‘ exactly for EVEN ‘$L$‘ — the same sector named from the geometry of the fibration and from the representation theory independently — *and it is also the sector whose eigenvalue carries no squashing at all, since the layer spectrum depends on the squashing only through the Hopf charge squared.* ⇒ ***Three facts about one set of modes, which is what makes it an object rather than a coincidence.*** ⇒ ***SO THE TENSOR EXPONENT FOLLOWS BY THE EXISTING ROUTE ON THE EVEN-DEGREE SQUASHING-FREE SECTOR THROUGH ‘$\{}ell=L/2$‘, AND FOR EVERY OTHER MODE THE MISSING OBJECT IS NAMED RATHER THAN WORKED AROUND: A TRANSPORT THAT CARRIES HOPF CHARGE THROUGH THE COLLAPSE OF THE FIBRE*** — which ‘r7156‘ established is the one direction the degeneration does not leave behind, the three-sphere descending onto its Hopf base with the throat’s two-sphere surviving, so the charge the unbridged modes carry is charge under precisely the circle that closes up.* (node 66’s ‘r7165‘ order, which lands ‘r7168‘, puts the dimension-specific finding in print so that ‘sec:throat‘ now reads as a blocked DESCRIPTION rather than a blocked ROUTE, keeps this seat’s transferable sentence (*\{}”the obstruction bites exactly where transverse-tracelessness must be obtained, and never where a symmetry supplies it\{}”*), and then orders the next step on the question this seat had named as the real one: *\{}”does a four-dimensional treatment keep the scalar route’s economy — ‘$(L,m)$‘ carried unmixed along the curve?\{}”* — with three reasons it is the right question rather than well-posedness, among them that *\{}”a four-dimensional treatment that mixes without bound would leave the tensor sector well posed and still uncomputable\{}”*, and with the instruction that *\{}”if the four-dimensional treatment turns out to need an object the corpus does not have, say which object — that is a row’s worth of finding on its own and I would rather have it named than worked around\{}”*. This row answers the economy question in the affirmative, finds the label does not travel with it, and names the object.). *Computes:* **17 checks, all pass, 13 s.** **(A)** the four-chart’s angular block matched against the round unit two-sphere as a symbolic identity with the mass and throat constant free, its determinant checked, the absence of cross terms into the ‘$(t,r)$‘ block checked componentwise, and all three rotational Killing vectors verified by computing the Lie derivative of the metric along each rather than by citing spherical symmetry; **(B)** the economy measured rather than argued: six explicit spherical harmonics separated through the four-dimensional operator built from the metric, the surviving angular variables counted and found empty, the reduced radial operator compared across every ‘$m$‘ at ‘$\{}ell=1,2,3$‘ and found identical, and the ‘$\{}ell(\{}ell+1)$‘ dependence read off the difference between two degrees; **(C)** the label bridge established as an identity in the degree, the descending sector identified by differentiating four explicit layer harmonics with respect to the fibre coordinate so that the ‘$m=0$‘ sector is characterised rather than assumed, and the parity condition derived twice — once from the projection’s integrality and once from the parity rule — and the two lists compared; **(D)** four paper clauses located in the CURRENT source as clauses reasoned FROM, with the ONE clause this row’s result bears on ENUMERATED over three states the paper may produce. *Bound:* **IT COMPUTES NO TENSOR EXPONENT.** *The economy and the bridge are established; the exponent on the descending sector is named as following by the existing route rather than car-

ried out here.* *It does not claim the charged modes are absent, suppressed or negligible — only UNBRIDGED, which is a statement about this route and not about those modes.* *The ‘$m$‘-independence is measured in the sector where separation is cheap and then rests on the three exact Killing vectors; the tensor harmonics’ own multiplicities are not recomputed, and that is named rather than hidden.* *Nothing here is offered as ‘P15‘’s TENSOR claim — ‘sec:intro‘ puts the tensor half in the companion, and the chart’s spherical symmetry is read from ‘eq:sds-static‘ itself.* *‘$\{}ell$‘ here is the four-chart’s two-sphere multipole and is NOT the observable multipole either; ‘sec:largescale‘’s map is a projection and is untouched.* *It does not re-measure ‘r7166‘ or ‘r7168‘.* *‘$M$‘ and ‘$\{}alpha$‘ are carried free wherever the statement is about the family.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_four_dimensional_treatment_keeps_the_economy_and_keeps_m`

**P15R264** `P15_the_bridge_is_exact_on_the_spectrum_and_the_transport_is_not_shared_because_the_four_route` [OK]

*‘eq:sds-static‘, ‘eq:squashed-spectrum‘, ‘sec:intro‘’s scalar-half scoping sentence and the basis clause scoped to the scalar sector — **‘r7166‘ ORDERED THE TENSOR EXPONENT ON THE EVEN-DEGREE SQUASHING-FREE SECTOR AND NAMED ITS OWN FAILURE MODE: *\{}”if taking it turns up that the bridge’s exactness does not survive the transport … that is a finding and it belongs above the exponent\{}”*. IT DOES NOT SURVIVE, AND THE FINDING IS WHAT THIS ROW DELIVERS IN PLACE OF THE NUMBER.** (1) ***WHAT SURVIVES, STATED FIRST SO THE NEGATIVE IS NOT MISREAD***: the spectral bridge is untouched — ‘$L(L+2)=4j(j+1)$‘ at ‘$j=L/2$‘ identically in the degree — and the four-chart’s angular factor is still ‘$r^2$‘ times the ROUND unit two-sphere with ‘$M$‘ and ‘$\{}alpha$‘ free, so nothing here retracts ‘r7170‘’s economy. (2) ***BUT THE FOUR-ROUTE’S RADIAL PROBLEM CARRIES A COUPLING THE LAYER HAS NO COUNTERPART FOR***: on an Einstein four-metric the Lichnerowicz curvature term splits into a part fixed by ‘$\{}Lambda$‘ and a WEYL part, and this background is not conformally flat. The Kretschmann invariant, computed from the metric by this receipt, is ‘$K=48M^2/r^6+24/\{}alpha^4$‘ with the second term **exactly** the ‘$\{}Lambda$‘-only value ‘$8\{}Lambda^2/3$‘, so the Weyl invariant is ‘$48M^2/r^6$‘ — **mass-driven and DIVERGENT as ‘$r\{}to0$‘, which is the close of the lift.** At the forced mass the ratio of the two is ‘$2\{}alpha^6/27r^6$‘, which is ***EXACTLY ‘$1/2$‘ at the comoving turnaround*** — a closed value at the lift’s own entry rather than a fitted one — and ‘32‘ at half that radius, ‘$5\{}times10^{5}$‘ at a tenth and ‘$5\{}times10^{11}$‘ at a hundredth. ⇒ ***SO IT IS NOWHERE A SMALL CORRECTION ON THE SEGMENT THE TRANSPORT TRAVERSES, THE LAYER’S EIGENVALUE PROBLEM IS A CROSS-SECTION PROBLEM THAT CARRIES NO SUCH TERM, AND THE TENSOR EXPONENT IS THEREFORE NOT DELIVERABLE ON THE LAYER’S MEASURE: the three-sector comparison cannot be made \{}”the same way\{}”.*** (3) ***AND THE SECTOR IS THINNER THAN THE ORDER ASSUMED, IN THE ONE WAY THAT MATTERS***: the layer’s multiplicity at each degree is ‘$(L+1)^2$‘, the charge runs over ‘$L+1$‘ values each carrying ‘$L+1$‘ states, so the ‘$m=0$‘ slice has weight **exactly ‘$1/(L+1)$‘ on EVEN degrees and ZERO on odd ones** — ‘$\{}tfrac13,\{}tfrac15,\{}tfrac17,\{}tfrac19$‘ at ‘$L=2,4,6,8$‘, thinning like ‘$1/L$‘ — ***and it contains NO ‘$L=1$‘ MODE AT ALL***, which is the degree ‘r7162‘’s scalar band and ‘r7166‘’s vector figures were BOTH quoted at. ⇒ ***So even had the transport been shared, the comparison the order asks for could not have been made at the degree where the other two sectors are reported.*** *That weight needed no input amplitude, so that half of the order is DISCHARGED rather than deferred to ‘PO-75‘.* **AND WHAT WOULD DELIVER THE EXPONENT IS NAMED RATHER THAN LEFT OPEN**: the Weyl coupling carried through the radial problem, which is the Regge–Wheeler/Zerilli reduction on this background and not an extension of the bead’s measure — a different computation from the formality the order expected.* (node 66’s ‘r7166‘ order, which lands ‘r7170‘, records that the economy answer *\{}”is better than the question deserved\{}”* and that the scoping it had applied two revisions earlier *\{}”now turns out to have a reason rather than a boundary\{}”* — a round fibre having no shape to deform — takes this seat’s bookkeeping proposal and opens ‘PO-84‘ for the fibre-charge question on the ground that *\{}”the same object would be owed for a charged SCALAR\{}”*, and then orders the exponent with two asks: the comparison against the scalar most-transparent band and the vector figures, *\{}”three sectors on one curve, read the same way\{}”*, and the sector’s own weight, *\{}”if the fraction is computable, carry it; if it needs the input amplitude ‘PO-75‘ is waiting on, say so and stop there\{}”* — together with the pre-authorised outcome this row returns.). *Computes:* **16 checks, all pass, 8.3 s.** **(A)** the spectral bridge and the round angular factor re-established

as symbolic identities before anything is built on them, so the row's negative is bounded by what it does not touch; **(B)** the full Riemann tensor computed from the metric through its own Christoffel-to-Riemann code, raised with the inverse metric and contracted into the Kretschmann invariant, then SPLIT against the '$\{}Lambda$'-only value '$8\{}Lambdaˆ2/3$' with the remainder identified as the Weyl invariant, its divergence at the origin taken as a limit rather than asserted, its vanishing at zero mass checked so the statement is about the family, the forced-mass ratio reduced in closed form and evaluated at the turnaround and at three smaller radii; **(C)** the multiplicity arithmetic laid out so the fraction is checkable — the charge count and the total gated separately from the weight — the weight compared against '$1/(L+1)$' on even degrees and zero on odd ones across eight degrees, the '$L=1$' absence gated explicitly against the degree the other two sectors are quoted at, and the thinning checked; **(D)** four paper clauses located in the CURRENT source as clauses reasoned FROM, with the ONE clause this row's result bears on ENUMERATED over three states the paper may produce. *Bound:* **IT DELIVERS NO TENSOR EXPONENT, AND THAT IS THE RESULT RATHER THAN A SHORTFALL** — the order pre-authorised this outcome and the reason is measured rather than asserted. *It does not claim the tensor exponent is large, small or divergent; only that it is not the layer's.* *It does not compute the Weyl coupling's effect on any mode: the invariant's size is measured and its divergence located, and the coupling's action on a given polarization is not derived.* *It does not retract 'r7170' — the spectral bridge and the economy stand, and what is corrected is an inference this seat drew from them.* *It does not claim the four-dimensional question is ill posed; 'r7168' established the opposite and that is untouched.* *Nothing is offered as 'P15''s tensor claim.* *'$M$' and '$\{}alpha$' are carried free wherever the statement is about the family, and the forced mass is substituted only where the statement is about the member.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_bridge_is_exact_on_the_spectrum_and_the_transport_is_not_shared_be`

**P15R265** `P15_the_refit_bound_figures_all_reproduce_and_one_of_them_is_a_ratio_standing_where_a_per_bin_c`

[OK]

*'sec:refit-bound''s two quoted '$\{}chiˆ{2}$' pairs — the '$214.1$'/'$550.5$' lensed pair with its '$1.16$'/'$2.98$' per bin and '$2.57$' times, and the refit's '$1.01$'/'$1.58$' per bin with its '$1.57$' times and its '$2.56$' — **BOTH PAIRS ARE RIGHT FROM THE PHYSICS SIDE AND EIGHT OF THE NINE FIGURES ARE EXACT AT THE PAPER'S OWN PRECISION. THE NINTH, '$2.56$', IS A *RATIO* STANDING IN A SENTENCE WHOSE EVERY OTHER FIGURE IS A '$\{}chiˆ{2}$' PER BIN** — the as-computed pair on those bins is '$212.9$'/'$545.8$', so '$2.56$' is '$545.8/212.9$' and the arm's as-computed figure PER BIN is '$2.95$'; the computing receipt prints it '2.56x as-computed', *with the 'x'*, and the paper drops it. ⇒ ***AND THE WRONG READING IS PRIMED RATHER THAN MERELY AVAILABLE: the same section states the computed arm at '$2.98$' per bin three sentences earlier***, so a reader carrying that figure into \{}"sits at '$2.56$' on the same bins\{}" meets one quantity with two values. *The repair is one word — the sentence's other two ratios both say \{}"times\{}" — and P15's prose is the chat seat's, so it is routed and not edited.* ◆ ***TAKEN AND IN PRINT AT 'r7167': the sentence now reads \{}"sits at '$2.56$' TIMES IT on the same bins\{}".*** *And the receipt's own locator was updated to REQUIRE the word in the same pass — the one edit a seat may make to another seat's receipt, the one its own edit broke — so **the receipt is now what detects this finding being undone**, which it could not do while it read a sentence the repair would break.* ◆ ***AND THE REQUIRE-THE-WORD CALL IS CHECKED AND KEPT, WITH THE ONE COST MEASURED AT 'cc66.136': requiring it made the two refusals IDENTICAL*** — a CORRECT rewording that names the quantity differently (this seat's own recommended 'stand at '$2.56$' times apart') and a REGRESSION that drops the word again both returned 'matches 0 time(s)' byte for byte, so the receipt reported *"the wording moved"* and not *"the word was dropped"*. ⇒ *A **classifier** is added on the refusal — not a tolerance: the defective form is still 'rc=1' and still asserts nothing, and the relapse message cannot fire unless the literal defective sentence is present. Measured three ways: 'rc=0' 15/15 as it stands, the correct rewrite refused as drift and explicitly NOT the defect, the regression refused by name.* (node 66's 'r7166' invitation, which asks nothing and offers this: *\{}"if either pair ever looks wrong to you from the physics side rather than the citation side, that is worth saying, because the sweep cannot tell the difference and neither can I from where I am reading\{}"* — the two pairs being the ones '70''s citation sweep had flagged as '(ii)' markers, and both acoustic and therefore this seat's numbers). *Computes:* **15 checks, all pass, ˜10 s.** The nine figures are **READ OUT OF THE PAPER** by locating both sentences inside

‘sec:refit-bound‘ by their own wording — *a drifted sentence REFUSES with ‘rc=1‘ and asserts nothing, there being no fallback literal to fall back to* — and each is then required to **EQUAL the computed value rounded to the decimals the paper prints**, with no tolerance, *because a tolerance wide enough to absorb a wrong quantity cannot report one*. Three configurations scored from banked spectra with P15’s own derived CAMB lensing operator imposed on both arms alike: ‘cc66_{cr_x_h686,lcdm}_L2000‘ → ‘$214.1$‘/‘$550.5$‘, ‘refit_grid185/{cr,lcdm}_base‘ → ‘$212.9$‘/‘$545.8$‘, ‘cc66_r185_verify_{cr,lcdm}‘ → ‘$186.5$‘/‘$292.4$‘; each asserted at 185 bins. The ninth figure is then identified as the ratio and shown not to be the per-bin value, the two readings being ‘$15.2\{}%$‘ apart *Bound:* **IT DOES NOT SHOW THE SENTENCE IS MISREAD, ONLY THAT IT CAN BE** — the figure is correct as a ratio and the defect is that the sentence does not say which quantity it is, which is a property of the prose and not of any number. *And the ‘$2.95$‘-against-‘$2.98$‘ gap between the two instruments that both call their range \{}”185 bins, ‘$\{}ell=100$‘–‘$1996$‘\{}” is **configuration and not disagreement** — ‘LSTEP=8‘ and its own ‘$k$‘-reach against the corpus’s full-range spectrum, ‘$0.8\{}%$‘ apart — asserted inside two per cent so that it is priced rather than noticed later* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_refit_bound_figures_all_reproduce_and_one_of_them_is_a_ratio_stand`

**P15R266** `P15_po84_the_charge_is_carried_and_the_corpus_already_named_where_because_the_eigenvalue_diver` [OK]

*‘sec:scope‘’s trade sentence and its Hopf-base degeneration clause, its size/shape split, ‘eq:squashed-spectrum‘, ‘sec:intro‘’s scalar-half scoping sentence — **‘PO-84‘ TAKEN UNASSIGNED, ON THE ROW’S OWN CHEAPEST ROUTE — THE CHARGED SCALAR, WHICH ‘r7166‘ NAMES AS WHERE TO START: WHAT BECOMES OF A CHARGE WHOSE FIBRE DOES NOT SURVIVE? ⇒ IT IS CARRIED, AND THE CORPUS ALREADY NAMED WHERE, SO THE ROW DISCHARGES ON ITS FIRST BRANCH AND NOT ITS SECOND.** (1) ***THE READ FIRST, BECAUSE IT IS WHAT TURNS THE REST INTO AN IDENTIFICATION RATHER THAN A COINCIDENCE***: ‘sec:scope‘ already says it once in its own voice — the layer *\{}”does not carry its sphere through the seam; it becomes the throat’s ‘$S^2$‘ there\{}”*, and ***the Hopf direction it gives up \{}”is the one the reassignment … TRADES INTO ‘$\{}chi$‘\{}”*** — so **the fibre is not annihilated, it is traded, and the destination was in print before this row was opened.** (2) ***AND THE DIVERGENCE THE CHARGED MODES SUFFER IS EXACTLY THAT TRADE***: the squashing vanishes **LINEARLY** at the double root with slope ‘$3/\{}alpha$‘ (measured, not assumed), so every charged mode’s eigenvalue runs to infinity there while the neutral one’s does not move; and the proper wavenumber along the traded direction, ‘$k^2/(-f)$‘, diverges at the same locus. Both diverging proves nothing on its own — **what makes them one object is that their RATIO is ‘$4m^2r^2/\{}alpha^2k^2$‘, FREE of the distance to the root**, so they diverge at the same rate with a finite nonzero coefficient, and solving for the wavenumber that equates them returns ***‘$k=2mr/\{}alpha$‘ in CLOSED FORM*** — the Hopf charge converted by the horn normalisation. ⇒ ***THE EIGENVALUE DIVERGENCE IS THE BLUESHIFT OF THE TRADED MOMENTUM: two descriptions of one fact, the second supplied by the construction itself.*** (3) ***SO THE ANSWER IS CARRIED IN LABEL, LOST IN AMPLITUDE***, stated so it cannot be read as either extreme: not *the charge vanishes*, since it has a destination and the destination is in print; and not *the charge is transported* in any sense preserving an amplitude, since that direction’s own block coefficient collapses at the double root and the mode’s proper momentum — and its suppression — run to infinity with it. *The row’s second branch, that the charge is simply not carried, is the one this receipt DECLINES, and it declines it with the destination named.* (4) ***AND THE SURVIVING SECTOR IS THE ONE WITH NOTHING TO BLUESHIFT, WHICH IS THE SAME FACT A THIRD TIME***: ‘$m=0$‘ carries no momentum into the trade, so it is exactly the sector that descends — already named by the projection’s integrality and by ‘r7160‘’s parity rule. **AND THERE IS NO SMALL-CHARGE ESCAPE**: the minimum nonzero charge is ‘$\{}tfrac12$‘ at odd degree and ‘$1$‘ at even, never small — ***and at ‘$L=1$‘ the only admissible charge is ‘$\{}tfrac12$‘, so the ENTIRE first degree is charged and none of it descends***, which is the same fact ‘r7172‘ met from the other side when it found the descending sector contains no first-degree mode. *And the size, so the divergence is not taken on faith: the charged integrand exceeds the neutral one by one full power of the squashing, passing a factor of ‘2‘ at ‘$\{}varepsilon=0.316$‘, ‘10‘ at ‘$0.0579$‘ and ‘100‘ at ‘$0.00577$‘ — so the effect turns on well before the collapse rather than only in its limit.* (node 66’s ‘r7167‘, which strikes ‘PO-83‘ on this seat’s negative (*\{}”struck on the answer rather than closed for want of one\{}”*), declines the Regge–Wheeler reduction for

this register and homes it in the companion paper instead, takes this seat's correction to its own ‘PO-69‘ attribution in full with its own error named beside it (*\{}"I made four measurements of what could not explain the difference and none of the one thing that could\{}"*), records the ‘r7170‘/‘r7172‘ guard pair, and reports one clause of this seat's prose dropped by the duplicate gate with the reason — while leaving ‘PO-84‘ open and unassigned, the row carrying its own reading of where to begin: *\{}"ask the same question of a charged SCALAR, where it is cheapest and where the answer is not entangled with a sector's own decomposition — the scalar case first is 66's reading of where to start and not an order\{}"*, with both branches licensed and the second called possibly the physical one. This row takes the unassigned item on that reading, and returns the FIRST branch.). *Computes:* **20 checks, all pass, 0.8 s.** **(A)** the three paper clauses the row is asked against located in the CURRENT source, each exactly once, as clauses reasoned FROM — the trade, the Hopf-base degeneration and the size/shape split — so the destination is READ and not posited before anything is computed; **(B)** the block coefficient's double root established from the factorised form by checking the value AND the first derivative, the squashing's linear vanishing taken as a limit with its slope in closed form, the spectrum's dependence on the charge shown to be through its square alone by differentiating at zero charge, both limits taken, the admissible charges enumerated at six degrees and the nonzero floor gated against parity, and the first degree's total charging gated separately against the complementary finding of the previous revision; **(C)** the identification done in three steps so that no step carries another's weight — both divergences exhibited, then their RATIO shown free of the distance to the root with a finite nonzero value there, then the equating wavenumber solved in closed form — with all three leading coefficients of the inverse-square divergence reported in closed form; **(D)** the charged-to-neutral integrand ratio's leading power taken as a limit and the three factor thresholds solved rather than scanned; **(E)** four paper clauses partitioned, with the ONE clause this row's result bears on ENUMERATED over three states the paper may produce. *Bound:* **IT COMPUTES NO EXPONENT ON THE SEAM APPROACH.** *The seam lies at infinite parameter and that is ‘r7156‘'s result, re-used and not re-derived; this row measures the RATE at which the charged modes' suppression exceeds the neutral one's, not the integral.* *It does not claim the trade is a diffeomorphism or that the two descriptions are related by a coordinate change — ‘sec:scope‘ says the opposite, and this receipt reasons FROM that, taking the trade as the reassignment the paper already asserts rather than deriving it.* *Scalar sector only, which is where the row says the question is cheapest; no tensor or vector statement is made or implied.* *It does not claim the charged modes are absent from the progenitor's spectrum — only that this transport does not deliver their amplitude, which is a statement about the transport.* *‘$L$‘ is the layer degree and is NOT the observable multipole.* *‘$\{}alpha$‘ is carried free throughout and the forced mass is substituted only where the statement is about the member.* *Run:* `python3 receipts/P15_CR_cosmology/P15_po84_the_charge_is_carried_and_the_corpus_already_named_where_because`

**P15R267** `P15_po75_the_supplier_is_already_in_print_and_the_floor_is_two_derivations_not_one_so_the_odd_l` [OK]

*‘P16‘'s genealogy sentence, its inherited-mode-content sentence and its shear paragraph's tensor floor; the interior-to-observed map receipt's asserted floor* ( **‘PO-75‘ TAKEN ON THE POINTER ‘r7169‘ WROTE INTO IT — WHAT SUPPLIES THE ANISOTROPIC SPECTRUM'S SOURCE ON THE EXPANSION LEG, WHEN THE SINGLE NARIAI WORLDLINE CARRIES NONE? ⇒ THE SUPPLIER IS NOT MISSING: IT IS IN PRINT IN THREE PIECES THAT WERE NEVER JOINED, AND THE ROW'S REAL CONTENT IS THE FLOOR.** (1) ***NOTHING ON THE EXPANSION LEG HAS TO SUPPLY IT***: ‘P16‘ says the construction is *a genealogy of universes and not a recursion on modes* in which *every passage is entered with frozen data and left with frozen data*, the crossing is lossless because every mode is already super-horizon at the branch point, and the harmonic index is preserved across it. **So the progenitor supplies it, the route is the inheritance factor, and the expansion leg is where frozen data ARRIVES rather than where it is made — the row was asking for a source where the construction has a boundary condition.** *And ‘P16‘ says which part is inherited rather than derived — the mode content itself — so the row cannot be discharged by deriving the content, only by BOUNDING what can arrive.* (2) ***WHICH MAKES THE FLOOR THE WHOLE QUESTION, AND IT IS ASSERTED AT THREE SITES AND DERIVED AT ONE***: ‘P16‘'s shear paragraph states it for the round sphere's tensor tower; the interior-to-observed map, which validates the parameter-free deficit at ‘$\{}ell\{}lesssim8$‘, **takes it as input** — measured, that receipt carries no occurrence of ‘transverse‘, ‘traceless‘, ‘tensor tower‘, ‘Hopf‘ or ‘charge‘; and ‘r7172‘ de-

rives it for the descending sector. (3) ***AND THE TWO DERIVATIONS STAND IN STRICT CONTAINMENT, WHICH IS SHARPER THAN RIVALRY***: the layer's floor is Hopf-charge parity under squashing — the neutral slice needs '$m=0$' and '$m=L/2-k$' reaches zero only at EVEN '$L$' — so it prunes '$L=3,5,7,\{}dots$' as well, while the tensor floor excludes only the first two degrees. **So the layer's floor IMPLIES the tensor floor and not conversely: the deficit's floor is safe under EITHER, and the odd-ladder exclusion belongs to the layer's decomposition ALONE.** ⇒ **THE ODD LADDER IS THEREFORE THE TEST THAT SEPARATES THEM**, and the row is NARROWED rather than discharged: the prediction is conditional on identifying the layer's transport with the branch-point crossing, which this revision declines to assume and routes as the fork.). *Computes:* **12 checks, all pass, 0.0 s.** **(A)** four READ-FIRST gates: the genealogy and frozen-data clauses each located once, the inherited-mode-content clause once, 'P16''s tensor-floor sentence once, and the map receipt's asserted floor together with the MEASURED absence of all five terms that would indicate a derivation there. **(B)** the derived floor reproduced over the integers at '$1\{}le L\{}le12$': the charge ladder '$m=L/2-k$', the '$(L+1)^2$' multiplicity, the neutral slice existing exactly at even degree with weight '$1/(L+1)$', and the even/odd partition enumerated rather than asserted. **(C)** the adjudication: the floors' agreement confined to admitting '$L=2$' and excluding below it; their disagreement enumerated over the odd degrees to '$11$'; the STRICT CONTAINMENT of the exclusion sets checked both ways, which is what makes the deficit's floor safe under either; and the conditional stated as a gate rather than as prose. *Bound:* **IT DOES NOT DISCHARGE 'PO-75'.** *It establishes that the row's supplier is already in print and relocates the row's open content to the floor's provenance.* *It computes NO amplitude — that is 'PO-84''s undelivered remainder, which 'r7169' pointed into this row and which is still undelivered.* *It does not re-derive the tensor floor: that is taken from 'P16''s own sentence and DECLARED, not reproduced, which is a stated limit and not an oversight.* ***And it does not identify the descending sector's transport with the branch-point crossing*** — 'r7172' established only that the layer's transport is not the four-route's, so whether it is the crossing's is a third question nobody has asked, and the odd-degree prediction is conditional on it. **That is this seat applying 'r7172''s own guard to its own result.** *It is NOT 'PO-85': that row asks for an instrument comparing asserted grades with proved ones and is '70''s. This is one instance in this seat's own sector, found by reading.* *Run:* `python3 receipts/P15_CR_cosmology/P15_po75_the_supplier_is_already_in_print_and_the_floor_is_two_derivation`

**P15R268** `P15_the_transmission_figure_is_deliverable_after_all_because_the_descending_sectors_eigenvalu` [OK]

*'eq:squashed-spectrum' at zero charge, and the lift's own envelope and length — no paper sentence is pinned and none is asked to change* ( **THE ONE ITEM THIS SEAT HAD LEFT UNSTARTED AND HAD TWICE DECLARED UNDELIVERABLE — A TRANSMISSION FIGURE RATHER THAN AN EXPONENT. 'r7166' STATED THE LIMIT AS *no transmission figure — exponents only*, ON THE GROUND THAT A FIGURE NEEDS THE VARYING-'$\{}lambda$' SOLUTION. ⇒ THE LIMIT WAS TOO WIDE, AND THIS SEAT'S OWN 'r7163' IS WHAT NARROWS IT.** (1) ***THE UNLOCK IS ONE SUBSTITUTION***: 'r7163''s exact envelope is '$\{}lambda/L(L+2)=1-\{}sigma(1-\{}varepsilon^{-2})$' with '$\{}sigma=4m^2/L(L+2)$', and the descending sector is the NEUTRAL slice. **At '$m=0$' the charge fraction vanishes identically, so '$\{}lambda=L(L+2)$' EXACTLY at every squashing** — the squashing-dependence is carried by the Hopf charge ALONE, verified by the derivative vanishing on that slice and on no other. ⇒ *So '$k$' does not vary along the bead there and the constant-'$k$' solution is not an approximation on the sector that descends: **it is exact.*** (2) ***AND THE CAVEAT SURVIVES EXACTLY WHERE IT COSTS NOTHING***: charged modes do have a varying '$\{}lambda$', which 'r7174' showed diverges at the double root — **and those are the modes 'r7174' also showed arrive carried in label and lost in amplitude.** So the figure is deliverable exactly where it is wanted and undeliverable exactly where there is no amplitude to deliver. (3) ***THE FIGURE, PER EVEN DEGREE***: '$4.158\{}times10^{-3}$', '$1.116\{}times10^{-5}$', '$2.438\{}times10^{-8}$', '$4.730\{}times10^{-11}$', '$8.497\{}times10^{-14}$' at '$L=2,4,6,8,10$' — thinning two to three orders per even degree, with '$T(0)=1$' exactly as the control. (4) ***AND THE COMPARISON IS THE RESULT RATHER THAN THE TABLE***: the parameter-free '$2^{7/3}k^2e^{-ks_{\{}rm tot}}$' **UNDERSTATES** the transmission at every degree the sector contains — by '$30$' per cent at the lowest, falling to '$7.5$' by the fifth, approaching one from ABOVE. **It is a FLOOR on this sector and not an estimate of it, and the degrees that matter are not in its asymptotic regime.**). *Computes:* **13 checks, all pass, 1.0 s.** **(A)** the enve-

lope reproduced in the one form needed; its identical equality to one at zero charge, with the squashing symbol shown ABSENT rather than cancelling numerically; the derivative shown proportional to ‘$m^2$‘ so the dependence is the charge’s alone; and the exactness claim gated together with a charged counter-sample. **(B)** the lift equation DERIVED — ‘$u=a\{}varphi$‘ turning ‘$u_{ss}=(k^2+a_{ss}/a)u$‘ into ‘$\{}varphi_{ss}+2(a_s/a)\{}varphi_s=k^2\{}varphi$‘ with the ‘$a_{ss}$‘ term cancelling exactly; the regular seam series ‘$c(1+k^2x^2/10)$‘ derived BY RESIDUAL so the starting slope is the series’ own; the lift’s length reproduced against ‘$\{}Gamma(1/6)\{}sqrt\{}pi/\{}Gamma(2/3)\{}sqrt3\{},2^{1/3}$‘ to ‘$10^{-11}$‘; and ‘$T(0)=1$‘ to ‘$10^{-14}$‘ as the control that had to come back affirmative. **(C)** the figures by Riccati integration from the regular seam behaviour, Richardson extrapolated on a convergence rate MEASURED at ‘$10^{-2/3}$‘ per decade at every degree — the profile’s own edge exponent, so the extrapolation rides a rate the geometry fixes rather than a fitted one — plus monotonicity, the per-degree thinning bracketed, the four leading figures pinned, and the asymptote’s understatement gated as a one-sided and monotone error rather than at a single sampled point. *Bound:* **‘$T$‘ IS THE LIFT’S TRANSMISSION ON THE LAYER’S OWN DESCENDING SECTOR AND NOT AN OBSERVABLE AMPLITUDE.** *Three things stand between them and NONE is done here: the sector’s weight is ‘$1/(L+1)$‘ of each even degree (‘r7172‘); ‘$L$‘ is the layer degree and not the observable multipole (‘r7164‘’s projection finding, used nowhere here and reopened nowhere); and the progenitor’s input amplitude is ‘PO-75‘’s object and still undelivered.* *It computes nothing for charged modes, where the varying-‘$\{}lambda$‘ problem is real and untouched — the figure is the neutral slice’s and is not claimed for the tower.* *It does not re-derive ‘r7163‘’s envelope beyond the single substitution it needs, does not revisit the lift’s vanishing real part, and takes the ‘$a\{}propto\{}cos^{2/3}$‘ profile as the established parametrisation rather than rebuilding it.* *The gauge sets the comoving curvature radius to one so that ‘$k^2=L(L+2)$‘ is the sphere’s eigenvalue; no length and no epoch enters, and no figure here is a prediction about the sky.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_transmission_figure_is_deliverable_after_all_because_the`

**P15R269** `P15_order_one_answered_negative_the_crossing_cannot_share_a_measure_with_the_layers_parity_bec` [OK]

*the crossing’s selection rule and its vacuity, ‘P16‘’s null-to-null scope note, the interior-to-observed map’s identity on the degree, and this revision’s own pre-registration* ( **NODE 66’s ‘r7171‘ ORDER (1), THE THIRD TRANSPORT QUESTION: DO THE CROSSING’S FILTER AND THE LAYER’S CHARGE DECOMPOSITION SHARE A MEASURE? ⇒ **NO**, AND THE ANSWER IS A PROOF RATHER THAN A MEASUREMENT THAT CAME OUT SMALL.** *Pre-registered before anything was computed, on the order’s own instruction, and the prediction named this — the LESS attractive — outcome.* (1) ***THE KERNEL’S ARGUMENT IS THE DEGREE AND THE PARITY IS A CONDITION ON THE CHARGE***: the crossing’s rule is a function of ‘$k$‘ alone, and on the construction’s own ‘$S^3$‘ the eigenvalue ‘$L(L+2)$‘ is the SAME on all ‘$L+1$‘ charges at that degree, so substituting it leaves an expression the charge symbol does not appear in at all. **A function constant along the charge ladder cannot implement a condition that distinguishes its rungs.** (2) ***EXHIBITED RATHER THAN ARGUED***: at the sector’s own lowest degree the neutral charge is IN the descending sector and the unit charge is not, and the crossing assigns both the identical kernel value because both carry ‘$k^2=8$‘. **One label separates them and the other cannot read it.** (3) ***AND THE FILTER IS VACUOUS IN ANY CASE, independently***: the rate diverges while the sound speed saturates, so the filter’s argument vanishes for EVERY wavenumber and the kernel goes to one on every mode — reproduced as its own limit and checked uniform in the degree. *A filter that selects nothing has no measure to share even before one asks what its argument is.* (4) ***AND THE DEEPEST REASON WAS ALREADY IN PRINT***: ‘P16‘’s scope note says the correspondence is null-boundary to null-boundary *with no spacelike slice entering the map*, while the layer’s parity is a decomposition ON a spacelike three-geometry. **There is no spacelike datum at the crossing for that parity to be shared with.** ⇒ **SO THE ODD-LADDER PREDICTION DOES NOT FOLLOW, the layer’s parity governs the layer alone, and ‘r7180‘’s refusal to assume the identification was the right call.**). *Computes:* **15 checks, all pass, 0.2 s.** **(A)** the pre-registration located in the tree and gated to name the negative as EXPECTED and the composition escape as a different question, so the receipt can claim neither the outcome as a surprise nor the escape as a win; ‘P16‘’s vacuity sentence and its null-to-null scope note each located; the index map’s identity on the degree located. **(B)** the eigenvalue and its ‘$(L+1)^2$‘ multiplicity verified over the integers at ‘$1\{}le L\{}le12$‘; the kernel shown to carry NO charge symbol after substitution,

as a statement about its free symbols rather than about sampled values; the parity's even-degree condition reproduced; the lowest-degree counterexample exhibited; the impossibility gated as logical rather than numerical; and the vacuity reproduced as a limit, then checked uniform across five degrees at the radiation sound speed. **(C)** the four pre-registered predictions SCORED — ‘P1‘ confirmed, ‘P2‘ confirmed with the absence of any tolerance in the receipt gated as evidence of its own shape, ‘P4‘ checked and negative (the charge-dependence that would have refuted ‘P1‘ is absent), and ‘P3‘ gated as left open and NOT claimed. *Bound:* **IT DOES NOT ANSWER THE COMPOSITION QUESTION, AND THAT WAS DECLARED IN ADVANCE.** *The parity could still bound the arriving content not by sharing a measure but by COMPOSITION, if the layer is necessarily in the path. The pre-registration named that as a different question and said in terms that delivering the bound that way would not answer this order — so nothing here delivers the odd-ladder bound, and it is now the live question.* *It computes no amplitude and no transmission; the kernel's sound speed and conformal interval stay SYMBOLIC and no value of either is needed.* *It does not re-derive ‘P16‘'s freeze-out beyond the one-line limit it reproduces, and does not revisit the lossless-passage result it reads.* *‘r7171‘'s ORDER (2) — re-deriving the tensor floor on the construction's own ‘$S^3$‘ — is NOT done here; the order itself puts (1) first and this is (1).* *No paper is touched and no other seat's receipt is touched.* *Run:* `python3 receipts/P15_CR_cosmology/P15_order_one_answered_negative_the_crossing_cannot_share_a_measure_with_`

**P15R270** `P15_the_budget_declaration_rests_on_a_lower_bound_and_the_receipt_that_says_so_measures_the_th`
[OK]

*no paper section — the runner's own long-declaration table, and the pinned-library manifest* ( **CI WENT RED ON TWO HEADS OF PR 280 WITH THE SAME VERDICT BOTH TIMES — ‘0 fail, 1 over timeout‘ — NAMING THIS LINE'S OWN ‘r7170‘ RECEIPT, WHICH IS MERGED WORK THAT ENTERED SCOPE BECAUSE A LATER PUSH TOUCHED FILES IT READS. THE RED IS THIS SEAT'S AND THE DIFF DID NOT CAUSE IT.** ***AND THE FIRST ACCOUNT OF IT WAS WRONG, WHICH IS WHY THIS IS A RECEIPT AND NOT A NOTE***: it was reported as a one-off already cleared by a green on the next head. **That head's scoped set held ‘36‘ receipts and NOT this one** — the green never exercised it. *The honest record is two timeouts in the two runs that ran it, and the standing-down comment is corrected.* (1) ***WHAT THIS MACHINE CAN MEASURE, MEASURED HERE AS SUBPROCESSES RATHER THAN QUOTED***: ‘13.2‘ s alone and ‘13.4‘ s for FOUR CONCURRENT COPIES — **no contention spread whatever** — with ‘sympy‘, ‘numpy‘ and ‘scipy‘ each checked against ‘requirements-ci.txt‘ BEFORE any timing is read, so the gap is not a library difference. (2) ***THE RUNNER, TWICE, PAST ‘600‘ s on saturated runs*** — both readings are **LOWER BOUNDS**, because a timeout says it passed the cap and not where it would have stopped. **A ‘\{}ge46\{}times‘ gap on matching versions, where the table's heaviest entry documents a worst observed spread of ‘1.47\{}times‘. The gap is NOT explained and this receipt does not pretend to explain it.** (3) ***SO THE DECLARATION IS THE TABLE'S SMALLEST STEP AND NOT A COMFORTABLE ONE***: ‘900‘ s, ‘1.5\{}times‘ the cap crossed, **chosen so the next completing run yields the first real upper figure** — after which the entry is to be re-declared to the rule's ‘1.7\{}times‘ product, as the heaviest entry itself was. ⇒ **And if ‘900‘ s is crossed too, that is evidence the cause is not cost, and the answer is to find the cause rather than raise the number again.**). *Computes:* **10 checks, all pass, 27 s.** **(A)** the entry located in the runner's table at ‘900‘ s, and its comment gated to state the three things a later reader needs: that it has NO upper measurement and ‘600‘ s is a lower bound, that the gap is unexplained, that the re-declaration duty and the stop condition are recorded, and that on the neighbouring entry's pattern the declaration is the SYMPTOM's fix and not the cause's with the probable cure named and deliberately unapplied. **(B)** all three pinned libraries checked against the manifest before any timing is used. **(C)** the declared receipt run as a subprocess and shown to PASS — so the budget is not covering a failure, which is the first thing such an entry must not be doing — then four concurrent copies all passing, the four-at-once wall time bounded against one alone as a RATIO rather than in absolute seconds, the order-of-magnitude gap to the runner's lower bound stated, and the number in force read from the table BY IMPORT rather than from this receipt's own text. *Bound:* **IT DOES NOT EXPLAIN THE GAP AND DOES NOT ASSERT THE STARVATION HYPOTHESIS.** *That reading fits the saturated runs it happened on and is recorded as a hypothesis — the previous reading of the same red was offered as settled and was not.* ***It does not touch the declared receipt's computation***: *that file is merged, reviewed work; its profile points at nested simplification in one function as the probable cure; and rewriting

a landed result's verification to fit an infrastructure budget is the worse trade, so the change is available and NOT made under a red.* *It measures this machine only and cannot measure the runner, which is the whole difficulty — the ratio of its two figures carries the no-spread claim, not the absolute seconds.* *It is not a claim that the budget model is wrong; the '\{}ge46\{}times' excursion is recorded for whoever owns the gate and no row is opened on it.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_budget_declaration_rests_on_a_lower_bound_and_the_receipt_that_sa`

**P15R271** `P15_the_dipole_parenthesis_is_right_on_its_own_tower_and_does_not_reach_the_other_one_because_` [OK]

*'sec:largescale''s low-multipole floor sentence — 'the lowest physical mode the quadrupole $L=2$ (the monopole $L=0$ is the background, the dipole $L=1$ pure gauge)' — against 'P16' 'sec:interior''s tensor-tower clause* ( **'r7175' ORDER (2) HALF (2), PRE-REGISTERED ON ITS HALF (3): THE DIPOLE PARENTHESIS IS RIGHT ON ITS OWN TOWER AND DOES NOT REACH THE OTHER ONE.** ***Two papers carry an '$L=2$' floor on the same '$S^3$', for two DIFFERENT towers, by two DIFFERENT derivations*** — and the agreement of the numbers is arithmetic coincidence rather than a shared mechanism. (1) **THE PARENTHESIS NAMES ITS OWN TOWER IN ITS OWN SENTENCE**: its wavenumber is '$\{}sqrt{L(L+2)}/r_0$', and '$L(L+2)$' is what this '$S^3$' returns for the SCALAR Laplacian ('$0,3,8,15,24,35,48$' measured), while 'r7190' measured the transverse-traceless eigenvalue on the SAME sphere as '$L(L+2)-2$' ('$6,13,22,33,46,61,78$'). *Two numbers, two spins.* (2) **AND IT IS CORRECT ON ITS TOWER, DERIVED RATHER THAN GRANTED**: the degree-one scalar harmonic's traceless second covariant derivative is the ***ZERO MAP*** — rank '$0$' while the FULL Hessian at that degree has rank '$2$', *so it vanishes identically and not for want of modes to act on* — which is what 'pure gauge' means there; and the same map has FULL rank '$L+1$' from '$L=2$' up. (3) **BUT NEITHER HALF OF THE QUESTION AS POSED HAS AN ANSWER**: (a) the transverse-traceless tower has NO '$L=1$' member to be gauge or not (lowest eigenvalue '$6$'; '$L=1$' would need '$1$'), and (b) the twelve are not pure gauge in any case (the kernel meets the gauge image in ZERO). ***Both measured, and neither settles it, because the question's subject sits at eigenvalue '$13=3\{}cdot5-2$', which is '$L=3$', and the question names '$L=1$'.*** **SO THE CORRECTION OFFERED FOR THIS SEAT'S GAP IS ITSELF A CROSS-SPIN LABEL SLIP, THE SAME CLASS AS THE SEAT'S OWN** — *this seat read a level index as a degree; the parenthesis offered to account for it is a genuine '$L=1$' statement one SPIN over.* **AND THE COINCIDENCE IS EXACT RATHER THAN NEAR**: the scalar tower floors at '$2$' because two degrees that EXIST are excluded (monopole as background, dipole by a vanishing traceless Hessian); the tensor tower floors at '$2$' because two degrees DO NOT EXIST ('$-2$' and '$1$' are not in its spectrum). ***Same number, opposite reason: one tower excludes modes it has, the other has none to exclude.***). *Computes:* **16 gates, all pass, 1 s.** **(A)** the pre-registration prior and naming all three outcomes AND the three ways it could be wrong, plus the instrument re-validated rather than inherited (Levi-Civita at all 27 triples, '$R_{ab}=2\{}delta_{ab}$'). **(B)** the paper's sentence read verbatim with its own wavenumber, and the scalar spectrum measured at seven degrees. **(C)** the traceless Hessian's rank at '$L=0$' (zero, and the full one zero too — the background half), at '$L=1$' (zero against a full-Hessian rank of '$2$' — the gauge half), and full at '$L=2\{}ldots6$'. **(D)** the transverse-traceless spectrum and gauge intersection RE-MEASURED at the two levels the question turns on rather than carried over. **(E)** the two floors' opposite reasons, both sentences read from their own papers, and the declared-in-advance negative. *Exact arithmetic throughout; no tolerance, no sampling, no floating point in any gate, and no assertion on wall-clock time.* *Bound:* ***THIS IS NOT A SECOND DERIVATION OF 'P16''s FLOOR, and the pre-registration declared that BEFORE the computation ran so it could not be reported as a strengthening.*** *'r7190' derived that floor; nothing here adds to it or touches it.* ⇒ **The consequence is NEGATIVE and it is about CITATION**: *the parenthesis is not available as support for the clause, and a seat reaching for it would be double-counting across spins — 'r7178''s own guard turned on this line's work.* **And it is the ROUND unit '$S^3$' in both halves**: the squashed members are a different eigenvalue problem and nothing here speaks to them. *What '66' offered to take — the print consequences of a real gap — is not what is owed; there is no gap. What is owed instead is that one of the two sentences cannot be cited for the other.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_dipole_parenthesis_is_right_on_its_own_tower_and_does_not_reach_t`

**P15R272** `P15_the_composition_route_is_answered_negative_too_because_the_eigenvalue_divergence_lives_on`

[OK]
*‘sec:throat‘’s bead route and ‘eq:squashed-spectrum‘, read against ‘sec:largescale‘’s every-point-of-the-curve requirement* ( **‘r7173‘’s COMPOSITION QUESTION, PRE-REGISTERED AS IT ORDERS: THE LAYER *IS* IN THE PATH AND THE PARITY STILL DOES NOT BOUND WHAT ARRIVES.** ***The answer is NEGATIVE and it is sharper than ‘r7186‘’s, because it does not only fail to deliver the odd-ladder bound — it MEASURES THE ODD LADDER ARRIVING.*** (1) **THE STRUCTURAL HALF HOLDS, BY READING**: carrying the harmonic labels along the bead needs a decomposition of the layer at every point of the curve, the lift included, and ‘PO-81‘ established the decomposition EXISTS and is the SAME one, labels carried exactly with only the eigenvalue moving. *So the conditional has its antecedent and the question is live rather than vacuous.* (2) **BUT THE PARITY NAMES WHICH SLICE IS SQUASHING-FREE, NOT WHICH MODES EXIST**: the ladder carries ‘$L+1$‘ slices at EVERY degree (computed over the integers at eight degrees) with ‘$m=0$‘ present exactly when ‘$L$‘ is even, ***so at odd degree every slice is charged and all ‘$L+1$‘ are carried.*** (3) **AND THE COMPOSITION PRODUCT IS NOT DEFINED ON THIS PATH, which is the obstruction the pre-registration named as the one it most expected**: the charged eigenvalue diverges where the squashing VANISHES, and the normalised squashing on the lift has its ***MINIMUM ‘$1.3747296$‘ at the comoving turnaround***, rising to infinity at the close — measured over forty thousand points and scale-free at three horizon scales. ⇒ ***The divergence lives on the leg the transport never traverses, so the product has no second factor.*** (4) **AND THE ODD LADDER ARRIVES WITH A NUMBER**: the exponent’s ratio is a function of the Hopf-charge fraction ‘$\{}sigma=4mˆ2/L(L+2)$‘ alone (two modes of different degree sharing ‘$\{}sigma$‘ agree to ten figures), ‘$R(1/3)=0.8419016934$‘ at the lowest odd degree, giving an exact exponent of ***‘$4.868603$‘ against ‘$5.782864$‘ for a round mode of the same degree and ‘$4.721689$‘ for the uniform bound*** — exponentially small, NONZERO, and LESS suppressed than the round mode by a margin that grows monotonically in degree. ⇒ ***So the degree the parity was supposed to exclude is measured arriving, which is the full discharge ‘r7173‘ (3) described: it closes the question rather than relocating it.***). *Computes:* **20 gates, all pass, 1 s.** **(A)** the pre-registration prior, with the TEMPTATION on the record before anything was computed, the three things the product needs, and the three escapes closed in advance. **(B)** the paper’s every-point requirement read verbatim and ‘PO-81‘’s existence result named as ANOTHER receipt’s finding. **(C)** the ladder over the integers at eight degrees — slice count, unit steps, and the parity — so no float decides whether a slice exists. **(D)** the lift’s conformal length against the paper’s closed form to better than ‘$10ˆ{-11}$‘ and scale-free at three horizon scales as the control; the spectrum read from the paper; and the squashing’s minimum located at the turnaround. **(E)** the ‘$\{}sigma$‘-only deformation, the two anchors ‘$R(0)=1$‘ and ‘$R(1)=0.2591014627$‘ reproducing the envelope receipt, the effective length ‘$0.8650719$‘, and the four odd degrees’ exact exponents with monotone margins. **(F)** the answer, the join ledger, and the two pre-registered refusals enforced as gates. *The Nariai lock ties the mass to the horizon scale, which is why every figure is scale-free; no assertion on wall-clock time.* *Bound:* ***THIS IS A JOIN AND NOT A DISCOVERY, and it is said here rather than left to be noticed.*** *Four of the five facts are each already receipted — the decomposition’s existence on the squashed layer, the spectrum, the ladder’s parity, and the ‘$\{}sigma$‘-only envelope with its figures.* **The one measurement that is new is the one that decides it: that the minimum of the normalised squashing over the lift is attained at the turnaround, so the divergence locus is off the transported segment.** **And the amplitude suppression found here is NOT the parity bound delivered late**, which the pre-registration forbade in advance: *the order asked whether the PARITY bounds what arrives, and it does not; the suppression that does exist carries a number and a different warrant, and the two are stated separately.* *‘P7‘’s null-to-null sentence was not used a second time — it closed the MEASURE route because there is no spacelike datum AT THE CROSSING, and this question was answered by the lift’s own squashing profile instead.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_composition_route_is_answered_negative_too_because_the_eigenvalu`

**P15R273** `P15_the_projection_join_inverts_the_expected_picture_because_the_odd_degree_lands_lower_than_t`
[OK]
*‘sec:largescale‘’s ‘eq:lowell‘ and its projection paragraph, read against this line’s per-degree transmission figures* ( **THE ONE ITEM ‘r7173‘ NAMED AND EXPLICITLY DID NOT ASSIGN, TAKEN UNASSIGNED ONCE EVERYTHING ORDERED WAS DISCHARGED: JOIN THE PER-DEGREE FIGURES TO ‘r7164‘’s ‘$L\{}to\{}ell$‘ PROJECTION.** ***THE JOIN IN-

VERTS THE PICTURE THE OFFER WAS MADE ON, AND THE INVERSION IS THE RESULT.*** ‘66‘ wrote that the floor degree is where the figure is most interesting and *also where the projection costs least*. ⇒ **THE PROJECTION COSTS LEAST ONE DEGREE LOWER.** (1) *The quasi-injective window is TWO degrees wide and the odd degree sits at its bottom*: width ‘$1.195$‘ against a MEASURED mode spacing of ‘$2.18$‘–‘$2.38$‘, with the floor degree at ‘$1.820$‘ and every degree from the third up exceeding the spacing. (2) **AND THE TWO DEGREES LAND THREE MULTIPOLES APART, ROBUSTLY**: ‘$L=1\{}to\{}ell=3$‘ and ‘$L=2\{}to\{}ell=6$‘, both stable across the whole range of the stretch the paper allows (‘$2.74$‘–‘$2.80$‘), *while two of the higher modal values are NOT stable over that range and are reported rather than asserted.* (3) ***SO THE JOIN: the odd degree lands THREE MULTIPOLES LOWER than the even floor AND arrives with the SMALLER exponent — ‘$4.868603$‘ against ‘$9.443377$‘, a gap of ‘$e^{4.57}$‘.*** ⇒ **The lowest observable multipoles draw on the degree the odd-ladder bound would have removed, in the window where the projection does least damage.** *‘r7194‘ measured the odd ladder arriving; this says WHERE it arrives.* (4) **AND ONE SMALL CORRECTION FALLS OUT, REPORTED RATHER THAN ABSORBED INTO A TOLERANCE: the paper prints TWO stretches and they are not the same number** — the ratio of its own two lengths is ‘$2.7618$‘ and it also prints ‘the stretch $2.774$‘. *Seven of ‘r7164‘’s eight widths reproduce on either, but the first comes out ‘$1.195$‘ on the lengths against that receipt’s printed ‘$1.20$‘, which is what ‘$2.774$‘ returns* — **below the printed precision for the widths, and NOT below a multipole, since a stretch of ‘$2.74$‘ moves the third degree’s modal value to ‘$8$‘ where both of the paper’s own values give ‘$9$‘.** *A receipt citing a modal multipole should know which digit it turns on.*). *Computes:* **19 gates, all pass, 1 s.** **(A)** the projection reproduced BEFORE it is used and named as ‘r7164‘’s — closure to a part in ‘$10^{12}$‘ at eight degrees, the printed argument at the ‘$95.5$‘th percentile at every one, the four printed arguments, seven of eight widths, and the two-stretch correction with its modal consequence. **(B)** the mode spacing MEASURED here rather than taken from the paper’s range, and the quasi-injective window located with the odd degree at its bottom, with the two load-bearing modal values gated for stability across the whole stretch range and the unstable ones reported. **(C)** the two arriving exponents, their gap, and the monotonicity of exponent and modal multipole over all eight degrees — so no higher degree reaches below the odd one. **(D)** the four limits, each gated rather than written in prose. *No assertion on wall-clock time.* *Bound:* ***THE LIMITS ARE LONGER THAN THE RESULT AND ALL FOUR ARE GATED.*** (a) **These are EXPONENTS and not amplitudes, and the asymmetry is structural rather than laziness**: the ‘$m=0$‘ slice’s eigenvalue is constant so its prefactor is computable, while the charged slice’s moves along the path so its prefactor is not supplied by an exponent ratio. (b) **It is ONE SLICE PER DEGREE** — three at the even floor and two at the odd degree, counted over the integers. (c) **AND THE PRIMORDIAL WEIGHTS A ‘$C_\{}ell$‘ WOULD SUM AGAINST DO NOT EXIST IN THIS CORPUS**, which ‘r7164‘ established by count rather than by omission: the normalisation is inherited and not derived. ⇒ ***So this is NOT a ‘$C_\{}ell$‘ prediction and cannot become one without a number no paper here supplies.*** (d) *And the scope is the SCALAR sector, which the paper says in its own voice — the vector sector is where the premise fails.* *‘r7164‘’s own limit is carried forward rather than quietly dropped: an envelope exponential in the degree is not one in the multipole above the third, which is why the claim is made inside the window and not across the tower.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_projection_join_inverts_the_expected_picture_because_the_odd_degr`

`S2_the_quote_pin_backlog_falls_by_sixty_six_on_this_lines_own_two_receipts_and_every_verdict_names_what_`
[OK]

*no paper section — ‘PO-78‘’s own adjudication baseline, read as data* ( **‘PO-78‘ WORKED RATHER THAN COUNTED: THE QUOTE-PIN BACKLOG FALLS FROM ‘2236‘ TO ‘2170‘, SIXTY-SIX KEYS ADJUDICATED ON THIS LINE’S OWN TWO RECEIPTS, WHICH NOW CARRY ZERO UNADJUDICATED BETWEEN THEM.** *The row’s own terms are that the unadjudicated bucket ‘may only fall‘, so a revision that lowers it is the row being worked; the ceiling is untouched at ‘2287‘ and the class did not grow.* **AND THE REAL RISK IN A BULK PASS IS THE ONE THE BASELINE’S HEADER NAMES — it is ‘A RECORD OF ADJUDICATIONS, NOT A LIST OF EXEMPTIONS‘.** *A pass that stamped sixty-six rows without reading them would lower the number and destroy the instrument.* ⇒ ***SO THE GATES ARE NOT ‘the count fell‘. They are properties of the SIXTY-SIX NOTES***: that every one names what was read, that every one states the RE-CHECK it owes (*a change to the pinned text

SHOULD land on the receipt, which is the difference between an adjudication and an exemption*), that none is a stub, and that each row's reading matches the KIND of literal it sits on. (1) **FOUR KINDS, because one reading does not cover them**: ‘30‘ pins on the likelihood instrument's own SOURCE, where the ruling is a statement about exactly those lines; ‘33‘ on corpus clauses the two derivations reason FROM, where *neither receipt's result asks for a paper change* — one says in terms that no paper rewrite is owed and what is owed is the code, the other closes its question AGAINST this seat's own earlier ruling and lands on the paper's side; ‘2‘ on this line's own measured figures; and ‘1‘ on the instrument's own diagnosis of its output. (2) ***AND TWO ROWS WERE MIS-CLASSIFIED ON THE FIRST PASS AND BOTH ARE CORRECTED HERE RATHER THAN LEFT***: the sky-value row took the *clause this receipt reasons from* reading when it is one of the receipt's OWN measurements, and the band-regression row took the CODE reading when its literal is a SENTENCE. ⇒ ***A wrong reading recorded in the baseline is WORSE than an unadjudicated row, because the ratchet then counts it as work done.*** **Both were caught by the two gates that check each note against the kind of literal it sits on, and those gates are the reason neither error shipped** — *the bulk pass was ‘96‘ per cent right and the checks found the rest.*). *Computes:* **14 gates, all pass, under a second.** **(A)** the stamped count, every verdict, the backlog's new value against its recorded ceiling, the arithmetic that it fell by exactly the number adjudicated, and the pass bounded by AUTHORSHIP — two receipts, both this line's, zero unadjudicated left on either. **(B)** every note carrying a read, every note carrying its re-check duty, and no note a stub. **(C)** the four kinds partitioning the sixty-six, the code rows checked AGAINST THEIR LITERALS rather than their notes, and the two mis-classification catches. **(D)** what remains, measured: ‘2170‘ keys over more than three hundred receipts, fewer than a third of them under this line's own paper. *Reads the baseline as DATA and asserts nothing about any paper, so it cannot break when a paper is reworded — which is the defect ‘PO-78‘ exists to measure. No assertion on wall-clock time.* *Bound:* ***THIS IS A DENT AND IS REPORTED AS ONE.*** *‘2170‘ keys remain across more than three hundred receipts and almost all of them are other seats'.* ***THIS SEAT DOES NOT ADJUDICATE ANOTHER SEAT'S PIN***: *the verdict is a judgement about what that receipt's gate is FOR, and that is theirs to state.* ⇒ **So the pass is bounded by authorship rather than by effort, and the remaining backlog is not a queue this seat can work down alone** — *which is a fact about ‘PO-78‘'s shape and worth having in the row rather than discovered by the next seat that tries.* *Nothing here touches the ceiling, and nothing here is a claim about a paper.* *Run:* `python3 receipts/L_probability/S2_the_quote_pin_backlog_falls_by_sixty_six_on_this_lines_own_two_receipts_a`

**P15R274** `P15_the_floors_understatement_is_measured_against_the_floor_and_not_the_exact_value_and_sec_la` [OK]

*‘sec:throat‘'s ‘r7173‘ floor sentence — its five percentages, its five exact transmissions and the parameter-free form they are compared against — and ‘sec:largescale‘'s own ‘low-multipole floor‘ — **THE INVITATION'S WORRY DOES NOT ARISE, AND FOR A BETTER REASON THAN ABSENCE: ‘sec:largescale‘ CARRIES NO OCCURRENCE OF THE ASYMPTOTE AT ALL** — ‘2^{7/3}‘, ‘$T(k)$‘, ‘$T(0)$‘ and ‘$s_{\{}rm tot}$‘ are each **zero** times in it, because it computes the branch-point filter on that same segment itself by composing exact constant-‘$\{}omega$‘ transfer matrices and reports exact/WKB ratios instead of quoting the exponential form. ⇒ ***WHAT THE SAME SENTENCE DOES LEAVE OPEN IS ITS OWN DENOMINATOR: ‘30‘, ‘17‘, ‘12‘, ‘9‘, ‘7.5‘ are ‘$(T_{\{}rm exact}-T_{\{}rm asym})/T_{\{}rm asym}$‘, exact at the paper's own printed precision, and the clause says "understates ... by ‘30‘ per cent" without saying per cent OF WHAT.*** Read as fractions of the exact value the same two columns give ‘23‘, ‘15‘, ‘11‘, ‘8‘, ‘7.0‘ — **a seven-point spread at ‘$L=2$‘, the degree the clause singles out as understated by most**. *‘PO-78‘'s rule in the revision that had just repaired the previous silence in that very clause: the side the limit is approached from is now stated and the quantity the percentage is of is not.* **AND ‘floor‘ NOW NAMES TWO UNRELATED BOUNDS ONE CROSS-REFERENCE APART** — ‘sec:throat‘'s transmission floor, new at ‘r7173‘, and ‘sec:largescale‘'s ‘low-multipole floor‘, in the corpus since ‘r2419‘ and built on ‘$r_0$‘ — *each correct in its own section, neither saying which it is when the other is in the reader's hand.* (node 66's ‘r7173‘ invitation, which asks nothing and offers this: *\{}"if a reader of ‘sec:largescale‘ would carry the asymptote into the low-multipole comparison and be a third low at the floor degree, that is worth knowing and you are the one who would see it\{}"* — the acoustic sector being this seat's, and ‘sec:largescale‘ the section it reads ◆ ***BOTH ARE REPAIRED IN PRINT AT ‘r7175‘, AND THIS RECEIPT IS NOW THE GUARD ON THE REPAIR RATHER THAN

THE RECORD OF THE DEFECT.*** *The clause reads ‘exceeding it by ... per cent **of the asymptotic value**‘, and the locator REQUIRES those words, so dropping them refuses. And ‘sec:throat‘ gives the newer object its own words — ‘a lower bound on this sector’s transmission‘ — so ‘floor on this sector‘ is GONE and the collision check is inverted: the old phrase must be ABSENT and the new one PRESENT.* ***So ‘floor‘ in ‘P15‘ means the multipole floor again, which is ‘sec:largescale‘’s own term and the older one*** — *the repair taken was to rename the newcomer and not the incumbent.*). *Computes:* **21 checks, all pass, ¡ 1 s.** ◆ ***‘r7175+cc66.141‘: the chat seat’s two inversions MEASURED. (2) the collision check is right and complete, two-sided and both arms asserted. (1) the locator is right and INCOMPLETE*** — requiring the denominator phrase makes it a guard, but the refusal it reaches is the generic drift one, so a RELAPSE (‘by $30$, ... per cent in turn‘) and a CORRECT REWRITE (‘... of the floor value‘) came back **byte for byte the same message**. *‘PO-78‘’s ‘cc66.136‘ entry arriving back where it was written from: a finding receipt that keeps only the repaired form knows that something moved and not what.* ⇒ **‘RELAPSE‘ added as a CLASSIFIER, not a tolerance** — held pre-‘r7175‘ clause, tested only after the repaired locator fails, defective form still ‘rc=1‘ asserting nothing, relapse message unable to fire unless that literal clause is present; the chat seat’s locator kept verbatim. Re-measured: the three cases separate. Every figure is **READ OUT OF THE PAPER**: the five percentages and the five exact transmissions captured from their own ‘sec:throat‘ sentences, and a drifted sentence **REFUSES with ‘rc=1‘ asserting nothing**. ‘$s_{\{}rm tot}$‘ is taken from the printed decimal **and recomputed from the Gamma expression the paper prints beside it**, so the decimal is checked rather than trusted. The floor is rebuilt from the paper’s own located form rather than a typed coefficient. Each printed percentage is then required to **equal one candidate denominator rounded to the precision the paper itself prints** — ‘0‘ decimals for ‘30‘ and ‘1‘ for ‘7.5‘ — with the other arm asserting it is NOT the rival reading, so both directions are pinned. Verified on seeded papers outside the tracked tree: reworded sentence → ‘rc=1‘ refusal; percentages swapped to the exact-relative reading → ‘rc=1‘ with ‘10‘ failing checks; unseeded → ‘rc=0‘ *Bound:* **IT DOES NOT SAY THE PAPER’S FIGURES ARE WRONG** — all five are right, and right on the reading the computing receipt uses; the defect is that the sentence does not carry the denominator, which is a property of the prose. *And the two ‘floor‘ senses are each correct in place: this is a collision between a new label and an established one, not an error in either.* *Two over-claims of this seat’s own were caught by this receipt failing before it landed — a precision read off ‘str(30.0)‘ rather than off the paper’s ‘30‘, and a paraphrase of the ‘$r_0$‘ clause the paper does not use — both the same shape as the round’s subject, and both left in the file as comments* ◆ *And a THIRD was caught by ‘check_prose_pins‘ before landing: ‘n_ls_floor ¿= 5‘, a pin on a COUNT OF MATCHES and a proxy for what the adjacent check establishes by NAME. **Removed rather than adjudicated** — a count of a word is not the naming of a quantity, the name was available, so the count earned nothing and cost an adjudication. The count is still printed, just not asserted on.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_floors_understatement_is_measured_against_the_floor_and_not_the_e`

**P15R275** `P15_the_projection_joins_two_load_bearing_multipoles_are_the_only_two_in_the_tower_with_no_str`
[OK]

*‘sec:largescale‘’s ‘eq:lowell‘ projection, read as a function of the stretch the paper prints twice*

( **THE NEXT THING FROM ‘r7196‘’s MATERIAL, WHICH ‘r7177‘ NAMED WITHOUT ASSIGNING: THE PENDING STRETCH DECISION CANNOT MOVE THE PROJECTION JOIN’S RESULT, AND THE REASON IS A MEASUREMENT RATHER THAN A REASSURANCE.** ***The two degrees the join rests on are the only two in the tower with NO threshold inside a band far wider than the paper admits***: ‘$L=1\{}to\{}ell=3$‘ has **no threshold anywhere in ‘$2.60$‘–‘$2.95$‘**, and ‘$L=2\{}to\{}ell=6$‘ has exactly ONE, at ‘$2.611994$‘, *below both printed values* — so the three-multipole separation holds at both. (1) **AND THE TWO DEGREES ‘r7196‘ FLAGGED AS MOVING ARE EXACTLY THE TWO WHOSE THRESHOLDS STRADDLE THE PRINTED PAIR**: the third flips ‘$8\{}to9$‘ at ***‘$2.741706$‘*** and the sixth flips ‘$16\{}to17$‘ at ***‘$2.747427$‘***, both between ‘$2.74$‘ and ‘$2.7618$‘, *so both of the paper’s own values put them on the upper side.* ⇒ **That receipt’s report was right and this one supplies the thresholds it did not have, so neither outcome of the decision needs a re-run.** (2) ***AND THE PATTERN IS STRUCTURAL RATHER THAN LUCKY: the threshold COUNT rises with degree*** — none at the first, one at the second and third, two or three from the fourth up. ⇒ ***So the BOTTOM of the quasi-injective window is also the stretch-robust end of the tower: the same two degrees the projection smears least are the

two a stretch decision cannot reach.*** **A second and independent reason the join was worth making at the window's bottom rather than in the middle of the tower.** *Every threshold found is a single-step change ‘$n\{}to n+1$‘ at every degree, which is the check that the located values are thresholds and not artefacts of the grid.*). *Computes:* **11 gates, all pass, 7 s.** **(A)** the distribution's closure at every degree over the band's endpoints AND both printed stretches, so the question is well posed at every stretch tested and not only where the paper prints one; both stretches read from the paper. **(B)** the first degree's threshold set EMPTY, the second's single threshold located below both printed values, and the straddling pair located to one part in a million by BISECTION rather than by scanning. **(C)** the threshold count rising with degree, the window's bottom identified as the robust end, and every threshold checked to be a single-step change. **(D)** the limits: the mean's continuity across the printed pair as the control on what ‘stability‘ is claimed of, the mode's discreteness as the reason thresholds exist at all, and the band declared as this receipt's own. *No assertion on wall-clock time.* *Bound:* ***THIS IS THE MODAL MULTIPOLE AND NOT THE DISTRIBUTION, and the limit is stated before the result is used.*** *The mode is a DISCRETE statistic and that is why it has thresholds at all; the **mean moves continuously** with the stretch and has none — ‘$3.2520$‘ against ‘$3.2686$‘ at the first degree across the printed pair.* ⇒ **So ‘no threshold‘ means ‘the mode does not move‘, NOT ‘nothing moves‘, and a claim resting on the mean would not inherit this stability.** *And the band ‘$2.60$‘–‘$2.95$‘ is **this receipt's own choice**, stated rather than attributed: it spans more than ten times the separation of the paper's two printed values, which is why it is worth reporting — but it is not a claim about what the paper will admit.* *Nothing here touches ‘r7196‘'s exponents, which are stretch-free by construction.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_projection_joins_two_load_bearing_multipoles_are_the_only_two_in_`

**P15R276** `P15_the_rate_rule_reaches_the_projection_only_through_the_identification_because_P7_gives_the` [OK]

*‘sec:refit-bound‘'s two sentences on the kernel's distance, ‘prop:flat‘, and ‘P7‘'s rate-rule proposition remark* ( **‘r7181‘'s ORDER, WHICH IS A READ AND NOT A COMPUTATION: THE ANSWER IS ‘THROUGH THE IDENTIFICATION‘, AND THE REASON IS IN ‘P7‘'s OWN IDENTITY RATHER THAN IN ANY HEDGE OF ‘P15‘'s.** (1) **ONE SENTENCE OF ‘P7‘ DOES BOTH HALVES**: the lapse is the foliation stacking rate the observable expansion ***rides***, fixed by ‘$\{}alpha$‘ and the cut's offset; and the shift is the ‘$E{=}1$‘ projection through which that expansion is ***observed*** as distance and redshift. ⇒ *Two different data of one slicing operator — leaf, lapse, shift, vantage — so **a rule about which RATE a quantity accumulates on cannot fix a projection's argument.*** (2) **AND THE RULE'S FORM CONFIRMS IT: it is a CLASSIFIER OVER RATES**, sorting a comoving separation read across leaves onto the stacking rate and the plasma's own accumulations onto the leaf's. *So what it fixes about ‘$D_M$‘ is which rate ‘$D_M$‘ accumulates on, and ‘$D_M$‘ is in it as a separation read ACROSS LEAVES — not as the photon path whose comoving length is the kernel's argument.* **The paper's own phrase is that the rule ‘reaches it by naming $D_M$‘.** (3) ⇒ **SO THE EQUALITY THE PROJECTION NEEDS IS FLATNESS'S**: ‘prop:flat‘ is a theorem about the constant-‘$\{}tau$‘ slice's induced three-metric and the vanishing of its Riemann tensor, ***with no rate in it at all*** — which is exactly why it can supply an identity the rule cannot. (4) **AND THE SHARPENING, WHICH RUNS THE NARROW WAY: ‘P15‘ settles the distance ‘by three independent routes‘ AND THEY ARE NOT THREE FOUNDATIONS — THEY ARE TWO.** *The third route's own receipt states the premise it needs — that the optical depth ‘is a photon-path observable of the same kind as the redshift and the distances‘ — and **that premise IS the rule's classification**, so the third route is the rule applied a second time rather than a second ground.* ⇒ ***Two grounds, the rule and the identification; and since the rule reaches the projection only through the identification, the identification is the only INDEPENDENT support for the one equality the argument turns on.*** *This seat's first reading was that three routes meant three supports; **checking the third route's stated premise is what corrected it**, and that check is the control this receipt carries in place of the pre-registration a computation would have had.*). *Computes:* **18 gates, all pass, under a second.** **(A)** both of the order's sentences located in ‘P15‘, the kernel's argument named as the object, and ‘P7‘'s rate rule read with its no-switching clause. **(B)** the lapse's ‘rides‘ and the shift's ‘observed through‘ quoted from the one sentence, and the four slicing data listed, so the split is the framework's and not imported. **(C)** the rule's subject-and-predicate as a rate and an accumulation, ‘$D_M$‘'s category as a separation across leaves, and the paper's own practice of keeping the projection and the degree

apart. **(D)** ‘prop:flat‘’s induced metric and vanishing curvature, the flatness-to-light-travel identity, and the kernel’s licence named as the identification. **(E)** the third route’s stated premise, its precedent form, and the two-grounds conclusion. **(F)** narrower-is-not-weaker, the three exclusions the order set, and that nothing is computed. *Every clause is asserted present in the source it is attributed to, so the receipt fails if any is reworded — the only control a read can carry.* *Bound:* ***NARROWER IS NOT WEAKER, and the distinction is left for the seat that writes the verdict.*** *‘prop:flat‘ is proved in the paper with its own argument and its own receipt, and the expansion-law recovery is measured, so **the identification is established and not conjectural**.* ⇒ ***What this answer changes is the NUMBER OF INDEPENDENT SUPPORTS, not the soundness of the one that carries it*** — *whether ‘a disagreement at full strength‘ survives on one established ground rather than three is a print decision and this receipt does not make it.* *And the three things the order excluded are untouched: no reopening of the knob ‘r7095‘ ruled on, no search for a third rate, and no re-derivation of ‘prop:flat‘, which is quoted rather than proved again.* *Computes nothing, as the order specifies.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_rate_rule_reaches_the_projection_only_through_the_identification`

**P15R277** `P15_the_background_returns_the_larger_stretch_so_2774_is_the_constructions_and_the_stale_trio_` [OK]

*‘sec:largescale‘ and ‘sec:intro‘’s two printed background lengths, as they stood at ‘r7177‘ — ‘$D_C\{}approx1.395\{}t with ‘the stretch $D_C/r_0\{}approx2.76$‘ in the one and ‘$D_C=1.4011\{}times10^{4}$‘ with ‘the stretch $2.774$‘ in the other, ‘$r_0\{}approx5051$~Mpc‘ in both — **THE BACKGROUND RETURNS ‘1.4011e4‘ AT THIS ARM’S OWN PARAMETERS ON THIS ARM’S OWN RATE, SO ‘2.774‘ IS THE CONSTRUCTION’S STRETCH** — and the stale pair is a WHOLE CONFIGURATION rather than an arithmetic slip, *though the printed figures name two candidates for it and do not separate them** (node 66’s ‘r7177‘ order: *”establish which ‘$D_C$‘ this cosmology’s background actually returns, and therefore which stretch is the construction’s. **Not the division** — I can do that and it answers nothing”*, with *”if the two printed values correspond to two configurations, say which is which; if one of them is stale, say which and from where”*). *Computes:* **25 checks, all pass, ~5 s.** ◆ ‘ACOUSTIC_two_arm‘ is IMPORTED at six configurations and its own module-level ‘D_M‘ read off: the arm’s refit ‘(68.60, 0.2973)‘ returns **‘14011.4567‘** with ‘$r_0=5051.49$‘ parameter-free, stretch **‘2.773728‘**; the arm’s default ‘(73.00, 0.3066)‘ returns ‘13004.56‘ at ‘$r_0=4708.97$‘, which is not the paper’s ‘$r_0$‘; the control returns ‘13864.66‘ and ‘13941.63‘. *The stretch is measured EXACTLY ‘$H_0$‘-independent — identical to ‘1e-12‘ across ‘$H_0=65$‘ to ‘80‘ at fixed ‘$\{}Omega_m$‘, because ‘$r_0$‘ and ‘$D_C$‘ both scale as ‘$c/H_0$‘* — so it reads ‘$\{}Omega_m$‘ and nothing else, and the window that prints ‘2.76‘ is ‘[0.30402, 0.31176]‘: it contains the arm’s pre-refit ‘0.3066‘ and excludes both the refit’s ‘0.2973‘ and the control’s ‘0.3150‘. **And ‘$r_0=5051$‘ is reachable at BOTH ‘$\{}Omega_m$‘** (‘$H_0=68.0568$‘ and ‘$H_0=68.6066$‘), so the one number the sentence calls fixed parameter-free cannot discriminate them. The banks: ‘16‘ ‘cr‘ spectra carry ‘14011.4567‘, and every banked value that prints ‘1.395e4‘ is on the ‘lcdm‘ CONTROL arm. **Two candidates print the stale pair and are ‘5.2‘ Mpc apart, below the paper’s own precision**: the arm at a pre-refit ‘$\{}Omega_m$‘ (‘13949.1‘) and the control at its own 185-bin refit minimum ‘LH0=67.410309 LOM=0.309826‘ (‘13954.3535‘). *And FOUR receipts asserted the computed value against the printed literal at ‘±100‘ Mpc and ‘±0.03‘ — ‘1.63×‘ and ‘2.19×‘ the discrepancy they would have to catch — all green, because neither tolerance CAN fail on the quantity it names; one of them printed both values in its own verdict and called them ‘four of the paper’s parameter-free figures‘.* **All four are closed at ‘r7179‘**, and the receipt’s own walk over ‘receipts/‘ now reads ‘0‘ — *counted on each run rather than asserted from memory, so the count is the measurement and not a record of one.* **27 checks, all pass, ~5 s.** ◆ ‘ACOUSTIC_two_arm‘ is IMPORTED at six configurations and its own module-level ‘D_M‘ read off: the arm’s refit ‘(68.60, 0.2973)‘ returns **‘14011.4567‘** with ‘$r_0=5051.49$‘ parameter-free, stretch **‘2.773728‘**; the arm’s default ‘(73.00, 0.3066)‘ returns ‘13004.56‘ at ‘$r_0=4708.97$‘, which is not the paper’s ‘$r_0$‘; the control returns ‘13864.66‘ and ‘13941.63‘. *And FOUR receipts assert the computed value against the printed literal at ‘±100‘ Mpc and ‘±0.03‘ — ‘1.63×‘ and ‘2.19×‘ the discrepancy they would have to catch — all green, one of them printing both values in its own verdict and calling them ‘four of the paper’s parameter-free figures‘; two more take the printed pair as their own INPUT, which is a different class and is counted separately.* *Bound:* **IT DOES NOT NAME THE STALE CONFIGURATION** — two reproduce the printed pair to the paper’s own four figures and the figures do not separate them; the receipt says so rather than picking, and records

that the second reading is the worse one because under it the printed stretch is this arm's ‘$r_0$‘ over the CONTROL's distance. *It repaired no prose itself.* **AND THE REPAIR LANDED AT ‘r7179‘:** ‘sec:largescale‘ now carries ‘$1.4011\{}times10^{4}$‘ and ‘$2.774$‘ in both its prose and its figure caption, with every derived figure of its projection paragraph measured on that pair, and ‘r7164‘'s (B)(5) moved in the same pass to ‘1.73‘ to ‘3.03‘ — *so the receipt's own classifier reports this by name as the order discharged rather than as a drift.* And the cost it priced was PAID in that pass, not deferred: the printed range is ‘$1.74$‘ to ‘$3.02$‘ and ‘r7164‘'s (B)(5) is ‘1.73 ¡ e ¡ 3.03‘, both read from their sources here rather than quoted, and both now REFUSE the other configuration's ‘1.6965‘ to ‘2.9213‘. *One place the discharge did not reach: the terminal VERDICT block was never branched, so the receipt exited ‘0‘ still printing ‘the repair crosses the printed ceiling 2.93, turning ‘r7164‘'s (B)(5) red unless that clause moves‘ on a tree where it already had* — ‘cc66.138‘'s member in this receipt's own voice, the row against the repaired receipt become the verdict against its own classifier. **Branched at ‘cc66.143‘, and the keys renamed off the configuration they held onto the form they match.** *‘$\{}ell_2$‘ prints ‘7.8‘ either way and the third degree's modal multipole is ‘9‘ either way, flipping to ‘8‘ only below ‘2.741706‘.* *It repaired no prose itself.* **AND THE REPAIR LANDED AT ‘r7179‘:** ‘sec:largescale‘ now carries ‘$1.4011\{}times10^{4}$‘ and ‘$2.774$‘ in both its prose and its figure caption, with every derived figure of its projection paragraph measured on that pair, and ‘r7164‘'s (B)(5) moved in the same pass to ‘1.73‘ to ‘3.03‘ — *so the receipt's own classifier reports this by name as the order discharged rather than as a drift.* And it states its own cost: at the construction's stretch the paper's printed displacement range ‘$1.70$ and $2.93$‘ becomes ‘1.7426‘ to ‘3.0158‘, so **‘r7164‘'s gate (B)(5) passes on the stale stretch and FAILS on the construction's** — the repair turns that receipt red unless the clause moves in the same pass. *Run:* `python3 receipts/P15_CR_cosmology/P15_the_background_returns_the_larger_stretch_so_2774_is_the_constructio`

**P15R278** `P15_the_1p57_has_no_parameter_address_because_the_arms_tt_minimum_sits_on_its_own_bao_minimum_` [OK]

*‘sec:refit-bound‘'s ‘$1.57\{}times$‘ at the verified 185-bin minimum, against each arm's own DESI DR2 BAO constraint — **THE RATIO HAS NO PARAMETER ADDRESS: THE ARM'S TT MINIMUM SITS ‘$-0.046\{}sigma$‘ FROM ITS OWN BAO MINIMUM IN ‘$\{}omega_m$‘ AND PAYS ‘$\{}Delta\{}chi^2_{\{}rm BAO}=+0.010$‘ AT IT** — *so the residual is not a displacement of the matter density, which is the answer ‘r7181‘ named as the more interesting one.* **And the calibration INVERTS: the control sits ‘$+0.227\{}sigma$‘ and pays ‘$+15.06$‘*** (node 66's ‘r7181‘ order: *"give the ‘$1.57$‘ a parameter address. Read the refit minima and report how far each arm's TT-preferred ‘$\{}omega_m$‘ sits from the one its distances fix"* — in three numbers: the minima printed, the displacement in the BAO‘$+\{}theta_*$‘ constraint's own width with the control as the calibration, and whether the arm's TT-preferred ‘$\{}omega_m$‘ is still admissible to the distance data). *Computes:* **13 checks, all pass, ˜6 s.** ◆ The TT minima are READ off the banked artefacts: the parameters are captured from ‘refit_grid185/verify.sh‘ and required to reproduce what ‘verify_{lcdm,cr}.npz‘ themselves store, which they do to a part in ‘$10^7$‘ in ‘$D_M$‘. **control** ‘$(67.4103, 0.309826, \{}omega_b=0.021966, n_s=0.954248)$‘, ‘$\{}omega_m=0.140790$‘; **arm** ‘$(68.5811, 0.297209, 0.021524, 0.997952)$‘, ‘$\{}omega_m=0.139788$‘. The BAO likelihood is checked twice before it is used — the corpus's own stacking anchor (‘$\{}Omega_m=0.3066$‘, ‘$\{}chi^2=12.01$‘, ‘$z_{\{}rm onset}=6764$‘) **and** DESI DR2's published ‘$\{}Lambda$CDM‘ ‘$\{}Omega_m=0.2975\{}pm0.0086$‘, recovered independently at ‘$0.2971\{}pm0.0085$‘. **And the arm's own ruler — distances on its radiation-free rate, ‘$r_s$‘ on the leaf clock — returns ‘$(68.62, 0.29735)$‘ as its BAO minimum**, i.e. the pair ‘sec:refit-bound‘ starts from, recovered from the distance data with no spectrum involved. Profiled ‘$\{}Delta\{}chi^2=1$‘ widths: ‘$\{}sigma(\{}Omega_m)=0.00852$‘, ‘$\{}sigma(\{}omega_m)=0.00454$‘, agreeing between the two rulers to a per cent. ⇒ **Displacements and penalties:** arm ‘$-0.046\{}sigma$‘ / ‘$+0.010$‘; control ‘$+0.227\{}sigma$‘ / ‘$+15.06$‘; the control's Planck START ‘$+0.736\{}sigma$‘ / ‘$+18.05$‘. *Bound:* **IT CONTRADICTS THE ORDER'S CALIBRATION PREMISE, AND THAT IS THE SECOND RESULT.** ‘r7181‘ reads the control's displacement as the floor — *"it fits everything, so its pull should be small"*. It is the larger of the two by ‘$5\{}times$‘ in ‘$\{}omega_m$‘ and ‘$1700\{}times$‘ in ‘$\{}chi^2$‘, **because ‘$0.3150$‘ is not a BAO‘$+\{}theta_*$‘ value**: it is Planck's CMB-fitted ‘$\{}Omega_m$‘, and DESI DR2 on the control's own ruler prefers ‘$0.2971$‘. *The control's pull is the known DESI–Planck ‘$\{}Omega_m$‘ tension, carried in by the premise; its TT refit moves it ‘$3$‘ in ‘$\{}chi^2$‘ TOWARD DESI and leaves it ‘$15$‘

away.* **And ‘$\{}omega_m$‘ is the variable that flatters the control** — in ‘$\{}Omega_m$‘ the two displacements are ‘$+1.498\{}sigma$‘ against ‘$-0.016\{}sigma$‘, a factor ‘$94$‘ against ‘$5$‘, because the control’s TT ‘$H_0$‘ sits below its BAO ‘$H_0$‘ while its ‘$\{}Omega_m$‘ sits above and the offsets partly cancel in ‘$\{}Omega_m h^2$‘. *Both are reported rather than the flattering one alone; the ordering is the same either way.* *‘$\{}theta_*$‘ is a CONFIRMATION in this fit and not a term in it — the corpus imposes it exactly through ‘$z_{\{}rm onset}$‘, so it constrains the onset and not ‘$(H_0,\{}Omega_m)$‘. Scored instead as a Gaussian at Planck’s ‘$100\{}theta_*=1.04109\{}pm0.00030$‘ the arm sits ‘$-15.8\{}sigma$‘, which is the comb disagreement already in print (‘$\{}ell_A=302.9$‘ against ‘$298.0$‘) and not a new finding; it is recorded so that ‘BAO$+\{}theta_*$‘ is read as the corpus means it.* *No new grid was run: the order said read the banked minima if they are there, and they were.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_1p57_has_no_parameter_address_because_the_arms_tt_minimum_sits_on`

**P15R279** `P15_the_no_fit_confrontation_is_three_per_bin_and_its_residual_carries_a_nine_sigma_modulation` [OK]

*‘sec:refit-bound‘’s other confrontation — the comb on the background the distances fix with NOTHING fitted to the microwave data — and its residual at the comb’s own period — **‘3.00‘ PER BIN AGAINST THE CONTROL’S ‘1.13‘, AND A ‘9.1\{}sigma‘ MODULATION AT ‘\{}ell_A‘ THAT BINNING AT ‘\{}ell_A‘ CANCELS*** (node 66’s ‘r7183‘ order (1): *”the paper has no figure of the other confrontation... that is the result a reader would most want to see and cannot”*, with the residual binned at the comb’s own period and both confrontations’ ‘\{}chi^2‘ per bin on the figure itself). *Computes:* **12 checks, all pass, ~25 s.** ◆ Four banked spectra, each identified by its own stored ‘D_M‘ rather than by its filename — no-fit ‘13864.6627‘/‘14011.4567‘, refitted ‘13954.3535‘/‘14017.0386‘. Scored on ‘fig:acoustic‘’s own bins and lensing operator: **nothing fitted ‘1.129‘ and ‘3.002‘ per bin (‘2.660\{}times‘); refitted ‘0.994‘ and ‘1.581‘ (‘1.591\{}times‘)**, 179 bins, ‘100\{}le\{}ell\{}le1900‘. Folded by phase within the comb period, anchored on the arm’s own first peak at ‘\{}ell=222‘, **one harmonic at ‘\{}ell_A‘: arm ‘1.3729\{}pm0.1512‘ (‘9.08\{}sigma‘) with nothing fitted, against the control’s ‘0.2028\{}pm0.1126‘ (‘1.80\{}sigma‘)**; refitting halves it to ‘0.8128‘ and leaves ‘6.89\{}sigma‘. Figure ‘corpus/fig_acoustic_nofit.pdf‘, generator ‘corpus/make_fig_acoustic_nofit.py‘; this receipt recomputes the figure’s numbers from the spectra and requires the figure’s own banked ‘.npz‘ to agree, so a drift in the generator fails here. *Bound:* **THE ORDERED OPERATION IS THE ONE THAT CANCELS WHAT IT WAS ORDERED TO SHOW.** A bin one period wide averages a full cycle of any modulation at that period, so its mean is that modulation’s own mean. *Binned at ‘\{}ell_A‘ the arm’s band means change sign twice across six bands — a slow swing, ‘+1.273‘ to ‘-0.808‘ — where the fold at the same period gives ‘9.1\{}sigma‘.* ⇒ **Both are drawn**: the period bins because they are what the order asked for and what a reader will look for, the fold because it answers the question they were asked for. *No new physics run, which the order required; nothing in ‘P15‘ is edited and the ‘\{}includegraphics‘ placement is the chat seat’s.* *The ‘\{}chi^2‘ figures are on the FIGURE’s 179-bin range, not the refit receipt’s 185-bin one; both are stated wherever they appear.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_no_fit_confrontation_is_three_per_bin_and_its_residual_ca`

**P15R280** `P15_all_three_steps_of_chi_for_the_visibility_take_the_rate_the_rule_assigns_them_and_the_colla` [OK]

*the instrument’s ‘\{}chi(z)‘ for the visibility, read against ‘P07‘’s rate rule step by step — **ALL THREE STEPS TAKE THE RATE THE RULE ASSIGNS THEM, AND THE FIRST CANNOT BE MISCLASSIFIED BECAUSE IT IS AN IDENTITY*** (node 66’s ‘r7183‘ order (2): *”name which rate each step of ‘\{}chi(z)‘ for the visibility actually uses in the code, against the rule’s classification of that step, and report agreement or disagreement”* — raised because ‘60‘’s ‘r7202‘ settles the rate rule as a classifier over rates, and *”I cannot see from here whether it was ever classified separately”*). *Computes:* **10 checks, all pass, ~20 s.** ◆ Read off live imports of the instrument at both arms and at both settings of every switch that could move the answer. **(1) the conversion ‘z\{}to\{}chi‘:** ‘\{}chi=\{}eta_0-\{}eta‘ identically, so ‘d\{}chi/d\{}eta=1‘ and there is no conversion step with a rate in it; the grid is on ‘Hgeom‘, which on the arm is the radiation-free — stacking — rate. **(2) the ionisation history:** ‘Hrec = Hleaf‘, ‘LEAFREC=1‘ **by default since ‘r7095‘**. **(3) the optical depth’s measure:** ‘d\{}tau‘ unweighted, ‘VISLEAF=0‘, the stacking clock — by ‘r7095‘’s derivation, not by an unexamined default. ⇒ **Three steps, three agreements.** And the two objects now sit on different rates in one and the same default run (‘Hphys‘ for the grid, ‘Hleaf‘ for recombination),

which is the rule's own configuration. *Bound:* **THE COLLAPSE THE ORDER FEARED IS EXACTLY WHAT 'LEAFGEOM' WAS, AND IT WAS FOUND, NAMED AND SPLIT TWO REVISIONS BEFORE THE QUESTION.** One switch over the conformal-time grid AND the ionisation history, which the rule assigns oppositely — so no setting satisfied both and *the rule's own configuration was unreachable from the file* until 'r7095' split 'LEAFREC' out and defaulted it ON, the only clock switch in that file whose default has moved. **And the order's '+14.6' per cent width excess is CONFIRMED at '+14.3'** ('43.2138' against '37.7993' Mpc at the pair the paper carries), invariant under 'LEAFREC' to four decimals — so it survives 'r7095''s default change. *What 'LEAFREC' moves is where the visibility sits ('\{}eta_{\{}rm LS}' by '1.8' Mpc), not how wide it is.* **This removes one candidate explanation for the contrast; it does not explain it.** *The sector's disagreement stands exactly where 'r7183' leaves it, and the receipt says so rather than letting an agreement read as a result.* *Run:* `python3 receipts/P15_CR_cosmology/P15_all_three_steps_of_chi_for_the_visibility_take_the_rate_the_rule_assig`

`S3_the_openness_subclass_is_read_in_full_and_the_flag_cannot_tell_a_pin_that_breaks_on_its_own_success_f`
[OK]

*no paper section — 'PO-78''s own adjudication baseline and the twenty-four receipt expressions it flags 'OPEN', read as data* ( **THE 'OPEN' SUBCLASS IS READ IN FULL AND THE FLAG SEPARATES NOTHING: all '24' keys the instrument flags 'OPEN' carry the same flag, while '10' of them go RED ON THE SUCCESS of the work they cite, '1' can NEVER go red and never checked the row it names, and '13' are safe. ** *What decides it is the HAYSTACK – how much of which text, live or frozen – and not the needle the flag is set by.* **And two pins carry the SAME token in the SAME file and fail in OPPOSITE directions: 'B15' scopes 'OPEN' to 'PO-3''s own row and goes red when the row is struck, 'Q1' searches the whole file and so never goes red and never checked 'PO-5'.** *The 'ALT' flag, which marks a pin as already repaired, measures SYNTAX: it fires on one sentence in two spellings and misses a four-arm regex cover.* **And the worst case was this seat's own – 'A and B or C' parses as '(A and B) or C', and 'B' sought a capitalised needle in a case-folded haystack, so the gate reduced to 'C' and the conjunct it was written to check was never read: the arm added to protect a pin was deleting the conjunct beside it.** *Two repairs land here, both on this seat's own files, and the backlog falls '2170' → '2167'.*). *Computes:* **21 gates, all pass, under a second.** **(A)** the subclass read at a PINNED commit rather than live, its size, its ownership split from the paths, and the three verdicts moved. **(B)** the flag sets on the eleven that need repair against the thirteen that do not, the seven-class haystack partition, and a computed ground from the literal for every safe reading that sits on a live haystack. **(C)** 'B15''s and 'Q1''s conditions evaluated on three synthetic registers – open, struck, and nothing-open – with the must-come-back-wrong control. **(D)** three disjunctions by 'ast' and by arm count: two spellings equal under case folding, four regex arms pairwise distinct, and '(A and B) or C' with an unsatisfiable 'B'. **(E)** both repairs demonstrated in the world they were written for, and the detector hole measured on two sentences one gate asserts. *Bound:* ***THE BOUND IS AUTHORSHIP, AS AT 'r7198', AND IT IS NARROWER THAN THE READ.*** *All '24' are READ and classified; only '3' are adjudicated, because a verdict says what a gate is FOR and that is its author's.* *The other '21' go to the channel with the repair shape each needs. And one of the three readings rests on a convention rather than a mechanism: 'FOR_54.md' is taken as append-only, which is practice and not enforced.* *The detector hole is REPORTED, not gated against the instrument: closing it must not redden this receipt, so the sibling sentence's visibility is printed and not asserted.* *Run:* `python3 receipts/L_probability/S3_the_openness_subclass_is_read_in_full_and_the_flag_cannot_tell_a_pin_that_`

**P15R281** `P15_the_one_ground_admits_exactly_one_independent_second_support_and_it_is_disqualified_becau`
[OK]

*'prop:flat' and 'sec:largescale''s flat projection, with 'sec:properframe''s placement of the photon bundle and its non-diffeomorphism clause* ( **'r7183''s OFFERED ITEM, TAKEN: CAN THE ONE EQUALITY 'PO-70' RESTS ON BE SECONDED FROM ANOTHER SIDE? THE SEARCH COMES BACK EMPTY, AND WITH A REASON RATHER THAN A LIST OF FAILURES.** *There is exactly ONE independent candidate, the beam side, and it is carried here in closed form:* on the metric 'prop:flat' is stated on, the geometry is **exactly Einstein for arbitrary '$M$'**, the areal radius is an **exact affine parameter** on both radial null congruences, and the **whole optical tidal matrix vanishes identically** — Ricci focusing and shear source together — *so '$D_A\{}propto\{}lvert r_0-r_e\{}rvert$', the difference of areal radii, which is

identically the flat slice's own radial separation, reached with the slice's induced metric nowhere among its inputs.* **AND THEN THE SAME RAYS CARRY A BLUESHIFT WHERE THE CONSTRUCTION HAS ‘1090.9‘** — ‘$1+z=0.0988$‘ and ‘$0.00787$‘, wrong by ‘$10^4$‘ and ‘$10^5$‘ and wrong in SIGN — *so they are not the construction's photons: independent, and NOT a support, which is this revision's own pre-registered outcome ‘(4)‘ and the one it named in advance as the trap.* **THE REASON IS STRUCTURAL: ‘$D_M$‘'s two factors are read on two DIFFERENT objects** — the length on the flat slice, the redshift on the layer — *and the reassignment joining them is not a diffeomorphism, so a second support would have to be a statement about the reassignment itself.* *And the size of what the one ground carries is measured: ‘14011‘ Mpc flat against ‘2285‘ Mpc closed, a factor of ‘6.133‘.*). *Computes:* **22 gates, all pass, under three seconds.** **(A)** the equality located in the current source, ‘prop:flat‘ RE-DERIVED rather than cited (all 81 Riemann components of the slice), and the factor the one ground carries. **(B)** the pre-registration read back from disk — five sides and four outcomes fixed before computing, the two conditions, and the row-ending outcome tabled first. **(C)** the Einstein condition, the Killing vector, the affine parameter and the full screen tidal matrix of ‘eq:proper-frame‘, symbolically and for arbitrary ‘$M$‘, with a must-come-back-wrong control that a NON-radial ray returns ‘$3Mq^2/r^3$‘ for every declared ‘$q$‘. **(D)** the observed frequency obtained symbolically and the redshift on the paper's own printed background. **(E)** the two-object reading and the non-diffeomorphism clause. **(F)** the other four sides. *Bound:* ***NOT that no second support can exist — only that none exists INSIDE EITHER PRESENTATION, which is what was searched.*** *Not that ‘prop:flat‘ is weak: it is a theorem and is re-derived here rather than doubted.* *One FORK is routed and not decided: whether the construction's photons are the constant-‘$\{}tau$‘ lines, in which case the length is the slice's by what the congruence IS and ‘prop:flat‘ is constitutive; or whether they are null in another metric of the one layering, which ‘P15‘ does not write down. **Which metric the propagation is DEFINED on is an order and not an inference**, and this seat has routed a fork of that shape before rather than deciding it.* *The beam result is reported for what it is — a closed-form statement about ‘eq:proper-frame‘ — and not dressed up as a support.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_one_ground_admits_exactly_one_independent_second_support_and_it_i`

**P15R282** `P15_the_residual_does_not_sit_at_ell_A_it_sits_four_per_cent_off_and_the_visibility_width_candi` [OK]

*‘fig:acoustic-nofit‘'s caption, and ‘r7189‘'s open question about where to look next — **THE CAPTION HAS NO DEFECT, AND THE RESIDUAL DOES NOT SIT AT ‘$\{}ell_A$‘: IT SITS AT ‘$312.0$‘, ‘$1\{}sigma$‘ ‘$[308.5,315.2]$‘, EXCLUDING ‘$\{}ell_A=298$‘ AT ‘$\{}Delta\{}chi^2=15.0$‘** — *and the visibility-width candidate predicts an ‘$\{}ell$‘-dependence the arm-to-control ratio does not have** (node 66's ‘r7189‘, which ordered NOTHING and asked two things: *”the caption is mine as you said; if any number in it reads wrong against your run, that is a defect and I want it named”*, and *”what is left is the kernel's acceptance in ‘$k$‘ and the visibility width that sets it — your ‘$+14.3$‘ per cent. That is the next place I would look, and I am not ordering it this round: say if you see a sharper one from where you sit”*). *Computes:* **11 checks, all pass, ~2 s.** ◆ Arithmetic on ‘r7183+cc66.145‘'s own banked whitened residual — no spectrum reloaded, nothing fitted to data, no grid. **(A) THE CAPTION, line by line:** ‘1.13‘/‘3.00‘/‘2.66‘ and ‘0.99‘/‘1.58‘ over ‘179‘ bins all agree to the digits printed; the fold's ‘1.37\{}pm0.15‘ at ‘$9.1\{}sigma$‘, ‘0.81\{}pm0.12‘ at ‘$6.9\{}sigma$‘ and the control's ‘0.20\{}pm0.11‘ likewise; and the four layout claims hold (‘Above‘ the full-width panel, ‘Below left‘ the period-binned, ‘Below right‘ the fold, the refit minima dashed), with ‘fig:acoustic-nofit‘ sitting immediately after ‘fig:acoustic‘. *One MUST-COME-BACK-WRONG control: ‘$0.81/0.12=6.75$‘, not ‘$6.9$‘ — so a check built on a printed number's own ROUNDED inputs reports a defect where there is none. The significance is the unrounded ‘$0.8128/0.1180=6.888$‘.* **(B) THE ‘$\{}ell$‘-DEPENDENCE:** the arm's harmonic amplitude grows ‘$0.73\{}to1.48\{}to2.38$‘ over ‘100-700‘, ‘700-1300‘, ‘1300-1900‘ — **but the control's grows ‘$0.20\{}to0.41\{}to0.72$‘ over the same bands**, ‘$3.27\{}times$‘ against ‘$3.53\{}times$‘, so the growth is the whitening's. The arm-to-control RATIO is flat: ‘$3.58$‘, ‘$3.65$‘, ‘$3.32$‘, a spread of ‘$9.5$‘ per cent of its mean against the ‘$230$‘ per cent the absolute amplitude moves. **(C) THE PERIOD, FITTED FREE** over ‘$240\{}le p¡380$‘ at ‘$0.25$‘: the unfitted arm prefers ‘$312.00$‘, ‘$1\{}sigma$‘ ‘$[308.5,315.2]$‘, ‘$\{}ell_A$‘ excluded at ‘$\{}Delta\{}chi^2=14.98$‘, a ‘$+4.70$‘ per cent offset; the refit prefers ‘$307.25$‘, pulled back toward ‘$\{}ell_A$‘ without reaching it. **(D) THE CONTROL:** it prefers ‘$347.25$‘, also longer, at ‘$\{}Delta\{}chi^2=8.62$‘ — so the DIRECTION is not the

arm's; what is the arm's is the DETERMINATION, an interval '$3.6\{}times$' tighter because its amplitude is '$6.8\{}times$' larger. *Bound:* **THE SHARPER PLACE IS THE PERIOD, NOT THE AMPLITUDE, AND THE FIGURE'S OWN RECEIPT COULD NOT HAVE SEEN IT BECAUSE IT FITS THE HARMONIC *AT* '$\{}ell_A$' BY CONSTRUCTION.** *The amplitude of a modulation at a FIXED period has no obvious parameter address; a PERIOD offset is a statement about '$r_s/D_M$', which has one — so this is the first candidate address the sector's '$1.57$' has had that is not already measured shut.* **And it is a reply to the candidate that was offered, not a change of subject:** *a '$+14.3$' per cent wider visibility is Silk damping, which acts monotonically and strongly at high '$\{}ell$' and barely at low, so it predicts an arm-to-control ratio that CLIMBS. The measured ratio is flat. That does not kill it — the width also moves '$\{}ell_A$' itself — but it moves it from 'the next place to look' to 'the place that has to explain a flat ratio'.* **AND THE OFFSET IS NOT CLAIMED AS THE ARM'S.** *The control prefers a longer period too. The receipt says so in its own voice and NAMES the measurement that would settle it — the arm's period against the control's as a null — rather than running it, because nothing was ordered.* *Nor does it correct anything in print: '$\{}ell_A=298$' is the arm's own acoustic scale and the figure is right to fold at it. What is new is that the residual's own period is a SEPARABLE question with a different answer.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_residual_does_not_sit_at_ell_A_it_sits_four_per_cent_off_and_the_v`

**P15R283** `P15_the_visibility_width_is_not_the_carrier_the_swap_moves_the_fold_at_ell_A_by_under_two_per_c`
[OK]
*'sec:refit-bound''s remaining route — the visibility width — and 'r7191''s order to price the residual it produces — **THE WIDTH IS NOT THE CARRIER: THE SWAP MOVES THE FOLD AT '$\{}ell_A$' BY UNDER '$2$' PER CENT AND THE PHASE BY UNDER HALF A DEGREE, AND THE UNDO TEST FAILS IN BOTH DIRECTIONS** — *and the order's premise is partly false, because the swap was already confronted at 'r6919+cc66.42', where it AMPLIFIES the '$q$'-dependence by '$4.56$'** (node 66's 'r7191': *"price the residual the width alone produces, and compare its SHAPE to the measured one"* — widen the control by '$14.3$' per cent, fold within '$\{}ell_A$', compare against '$1.3729\{}pm0.1512$' and '$0.8128\{}pm0.1180$'; with the control *"NARROW the arm's visibility to the control's and see whether the '$9.1\{}sigma$' modulation goes away. A mechanism that explains the residual must also remove it when undone"*; and the third branch named in advance, *"if it reproduces neither, the width is not the carrier and I want that said as plainly as the other two"*). *Computes:* **11 checks, all pass, ~3 s.** ◆ **NO INSTRUMENT RUN**: 'r6919+cc66.42' banked the width-swapped injected spectra and they are on disk, so the whole order is answerable from 'r6919_injected_{lcdm,cr}.npz' and 'r6919_geometry.npz'. **(A) WHAT THE '$+14$' PER CENT IS:** FWHM '$38.0424$' against '$43.5913$' ('$+14.59$' per cent at the 185-bin minima — the order's '$+14.3$' is the same quantity at the in-print pair); '$\{}Delta r_s$' across the visibility the same to '$0.08$' per cent; so '$\{}dd r_s/\{}dd\{}chi$' '$0.454950$' against '$0.396733$', '$-12.8$' per cent. **(B) THE PREMISE, MEASURED HERE:** band by band the arm-to-control retained modulation runs '$1.0459/1.0945/1.1268$' at own widths and '$0.9876/1.0965/1.3570$' swapped — slope '$+0.04046\{}to+0.18468$', **a factor '$4.56$'**, reproducing 'r6919''s '$\{}sim4$' in a different statistic. **(C) THE FOLD AT '$\{}ell_A$', WHICH IS WHAT THE ORDER ADDS:** control '$0.3704\{}pm0.0132$' at '$36.4^\{}circ$' → '$0.3639\{}pm0.0139$' at '$36.7^\{}circ$' ('$-1.8$' per cent); arm '$0.4003\{}pm0.0138$' at '$35.4^\{}circ$' → '$0.4007\{}pm0.0131$' at '$35.3^\{}circ$' ('$+0.1$' per cent) — all four measured at '$26$' to '$30\{}sigma$'. **(D) THE UNDO TEST:** widening the control moves its fold AWAY from the arm's; narrowing the arm leaves its own unmoved to one part in a thousand; the arm-to-control difference goes '$+8.1$' per cent '$\{}to$' '$+10.1$'. **(E) THE OBSTRUCTION, FROM THE SYNTAX TREE:** 'ETA_LS_W' is assigned once (line '$600$') and the only assignment consuming 'SRCINJVIS' (line '$1793$') sits inside 'If' tests over '_SRCI' — so the width is controllable ONLY where the model's own source has been replaced. *Bound:* **THE VISIBILITY WIDTH IS NOT THE CARRIER OF THE SECTOR'S RESIDUAL**, which is the order's own third branch, said as plainly as it asked. *The fold at '$\{}ell_A$' — the one statistic the order named — does not move, and it is not a blunt instrument returning a null: it resolves the fold at '$26$'–'$30\{}sigma$', so a real '$2$' per cent move would have been seen.* **And the undo fails in BOTH directions, which is stronger than one failing:** *two configurations cannot both be coincidences of size. 'r7191''s own criterion was that a mechanism which explains the residual must remove it when undone; this one moves the wrong way in one direction and not at all in the

other, and the difference it was meant to remove grows.* **AND THE ORDER'S PREMISE IS PARTLY FALSE AND IS REPORTED RATHER THAN ACTED ON:** *the ‘$+14$‘ per cent has been confronted — at ‘r6919+cc66.42‘, by this seat, whose banks this receipt reads. The swap is WELL POSED (unlike the clock swap, which moves the comb) and it AMPLIFIES the ‘$q$‘-dependence fourfold.* *It converges with ‘r7189+cc66.149‘ of the same round by a different route: the width's signature is a steepening ‘$q$‘-slope, while the measured residual's arm-to-control fold ratio is FLAT in ‘$\{}ell$‘ — ‘$3.58,3.65,3.32$‘. Two statistics, one conclusion.* **No replacement mechanism is offered.** *‘r6919‘'s joint object ‘$\{}dd r_s/\{}dd\{}chi$‘ is re-pointed and not re-derived; with ‘cc66.149‘'s ‘$+4.70$‘ per cent period offset it is where this seat would look. Neither is a claim.* *And the obstruction is named as the scope required: the width cannot be varied on the real-source path at all, so every number here is the projection's transfer of a KNOWN input and no ‘SRCINJ‘ run is a spectrum of this model.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_visibility_width_is_not_the_carrier_the_swap_moves_the_fold_at_el`

`S4_the_exact_count_on_a_set_another_seat_can_grow_is_rare_in_this_seats_work_and_the_one_live_instance_w`
[OK]

*no paper section — this seat's own ‘284‘ receipts and the shared registries they count, read at a pinned commit* ( **THE SECOND FACE OF ‘r7204‘'s CLASS, SWEPT OVER THIS SEAT'S OWN WORK: AN EXACT COUNT ASSERTED AGAINST A SET OTHER SEATS CAN GROW** — and ‘r7204‘'s subject was a pin on a SENTENCE that may be reworded, while this is a gate on a SET whose membership another seat controls. ***AND THE ONE LIVE INSTANCE WAS ‘r7198‘'s OWN RECEIPT, TURNED RED BY ‘r7204‘*** — *it gated the backlog at ‘== 2170‘ on a bucket the row says ‘may only fall‘, ‘r7204‘ fell it to ‘2167‘, and it had been RED ON THE TRUNK with nobody seeing it because the job that runs receipts was skipped on every push in between.* **AND THE FINDING IS THAT THE CLASS IS RARE: ‘25‘ sites flagged in ‘9‘ receipts, and reading ALL of them only ‘3‘ are genuinely exposed** — ‘4‘ frozen at a pinned commit, ‘14‘ counting a set the receipt itself writes, ‘4‘ row-scoped ‘$=1$‘ where the equality is the point. *So the detector's raw flag is not the defect and the PARTITION is the instrument.* *Two of the three are repaired, the third is read and left with its ground stated; and two of the three sat in the one receipt whose own subject is gates that fail on their own success.*). *Computes:* **13 gates, all pass, about ten seconds.** **(A)** the pre-repair source read at the pinned commit, the live backlog arithmetic showing the gate was FALSE, and the monotone repair. **(B)** the detector controlled in BOTH directions — it flags the pre-repair form and not the repaired one, and it does NOT flag an ordinary paper phrase pin while it does flag the same shape over the adjudication baseline. **(C)** the sweep over the pinned tree's ‘284‘ receipts, the four-way partition with a computed ground per site, and the three repairs located in their files. **(D)** the one other red named as an environment limitation. *Bound:* ***THE DETECTOR NARROWS AND DOES NOT ADJUDICATE*** — *it took ‘284‘ receipts to ‘9‘ and every flagged site was READ; a sweep that reported ‘25‘ would be crying wolf twenty-two times.* *Its two limits are named rather than found later: it cannot see a pinned read whose commit is an f-string variable, which is why the frozen class is computed in a second pass, and it over-reports ‘$=1$‘.* *And one red on this tree is NOT this class and is neither repaired nor chased — a receipt importing the instrument it is registered against, in a container that does not carry it; deleting an import to get a green would be the real defect.* *‘r7240‘ CORRECTS THIS ROW'S ENUMERATION rather than its ruling: two of the three sites ‘r7227‘ had to repair were invisible to the detector above, and the pinned population cannot contain this receipt's own gates — see ‘S10‘.* *Run:* `python3 receipts/L_probability/S4_the_exact_count_on_a_set_another_seat_can_grow_is_rare_in_this_seats_work_`

**P15R284** `P15_the_seam_flux_is_not_computable_here_because_the_closure_makes_one_locus_carry_two_surface`
[OK]

*‘sec:largescale‘'s seam factorisation and the lap's closure translation, with ‘sec:scope‘'s Euclidean segment and its own statement that the length is a pure number* ( **‘r7191‘ (1) ANSWERED NEGATIVE WITH THE MISSING OBJECTS NAMED: THE SEAM'S RADIATED FLUX IS NOT COMPUTABLE HERE, AND THERE ARE **TWO** THINGS MISSING RATHER THAN ONE — AND NEITHER IS THE PROGENITOR INTERIOR.** *‘cc66‘ said the flux needs the interior; the order asked for that TESTED rather than inherited, and tested it is true for a different reason.* (1) **THE SURFACE GRAVITY IS NOT SINGLE-VALUED ON THE SEAM**: the lap's own closure ‘$r\{}mapsto r+\{}sqrt3\{}alpha$‘ carries the back seam EXACTLY onto the front seam, so the one locus it makes carries ‘$\{}kappa=0$‘ and ‘$\{}kappa=3\{}sqrt3/4\{}alpha$‘

— *and a thermal state is a periodicity at a horizon of a GIVEN ‘$\{}kappa$‘.* (2) **AND INDEPENDENTLY, THE EUCLIDEAN SECTION CLOSES IN THE WRONG CLOCK**: the segment’s length is a PURE NUMBER in the bead’s conformal time, the periodicity is ‘$2\{}pi/\{}kappa$‘ in the seam’s KILLING time, and the conversion ‘$f/r$‘ is **not constant along the lift and vanishes at both seams** — the only loci where the condition is imposed. *So ‘(2)‘ of the order does not fire, and the FIFTH-CLOSURE branch does not fire either: this is not a flux that thermalises to ‘$n_s\{}to1$‘ but the absence of the object a flux would be computed from, which ‘PO-31‘’s terminal clause must NOT count as a mechanism.*). *Computes:* **14 gates, all pass, about a second.** **(A)** the factorisation and both surface gravities from the metric’s own ‘$f’/2$‘, for arbitrary ‘$\{}alpha$‘. **(B)** the closure translation applied to the back seam, and that no ‘$\{}alpha$‘ makes the two ‘$\{}kappa$‘ agree. **(C)** both of the paper’s closed forms for the segment’s length, the required ‘$\{}beta$‘, and ‘$f/r$‘ with its derivative and its values at both seams. **(D)** the trap discharged. **(E)** the pre-registration read back, five outcomes with the row-ending one first. *Bound:* ***NOT that the seam radiates nothing*** — *that is a statement about a quantum state and no state is in hand.* *Not that no state EXISTS, only that the construction does not SELECT one, which is what was asked.* ***‘cc66‘’s conclusion stands and only its reason is replaced***: its ‘$\{}kappa$‘ is confirmed here by a second route — the metric’s ‘$f’/2$‘ against its indicial equation — *which is also the control that this instrument can return a positive, so its negative is not a method that can only report absence.* *And the comparison a reader wants, ‘3.3387‘ against ‘4.8368‘, is DECLINED as a comparison across a change of clock, asserted on this receipt’s own source rather than promised in its prose.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_seam_flux_is_not_computable_here_because_the_closure_makes_one_lo`

`S5_the_set_shaped_class_is_absent_from_the_gate_layer_and_two_thirds_of_what_remains_sits_at_the_one_val`

[OK]

*no paper section — the whole pinned Python instrument layer read as data: ‘997‘ receipts, ‘136‘ ‘scripts/‘, ‘170‘ ‘corpus/‘, plus the papers and the excluded appendix layer both read AT THE PIN* ( **‘r7208‘’s SET-SHAPED CLASS RUN TREE-WIDE, AND THE GATE LAYER CARRIES NONE OF IT — ‘0‘ SITES IN ‘0‘ OF ‘306‘ FILES.** *The ten that looked like hits are ONE ambiguous reach token: ‘walk‘, written for ‘os.walk‘, matching the ‘ast.walk‘ a gate uses ‘35‘ times; disambiguate it and the gate layer goes ‘10‘ → ‘0‘ while ‘glob‘, ‘listdir‘ and ‘iglob‘ stay in the reach, so all ten came through that word.* **AND OF THE ‘19‘ EXPOSED SITES THAT SURVIVE, ‘13‘ ASSERT ‘== 0‘ — THE ONE VALUE AT WHICH THE STANDING GUARD’S REPAIR IS A TAUTOLOGY.** *For a non-negative count ‘n ¡= 0‘ and ‘n == 0‘ are the same predicate, so the guard adopted at ‘r7189‘ is silent on two thirds of the class it was adopted for and licenses a rewrite that satisfies its own form and changes no content — **the FIRST standing guard’s defect manufactured by the SECOND guard’s repair.** The repair that applies at zero is ‘L-249‘’s population freeze plus a positive control, not a ceiling.* ⇒ *And the proxy for that control is DECLINED in print: keyed to the tainted NAME it calls ‘7‘ of ‘14‘ sites uncontrolled, and a two-sided run against a shadow tree whose papers exist and whose bodies extract empty fails ‘6‘ of ‘6‘ — the control a zero needs is a property of the RECEIPT and not of the name.*). *Computes:* **26 gates, all pass, about seventy seconds.** **(A)** the population at the pin and ‘r7208‘’s own ‘25‘-in-‘9‘ reproduced on its ‘284‘ so the instrument is shown to be the same one. **(B)** ‘83‘ sites in ‘43‘ files on ‘r7208‘’s reach, the ‘35‘ ‘ast.walk‘ against ‘0‘ ‘os.walk‘ in the one gate file that flags, and the disambiguated ‘0‘ of ‘306‘ — with the one receipt-layer site the disambiguation costs shown to be a true-positive removal. **(C)** the partition ‘19‘ EXPOSED, ‘4‘ FROZEN, ‘9‘ ROW-SCOPED, ‘40‘ SELF, the ‘13‘ at zero, their ‘12‘ ‘len‘ and one ‘sum(1 ...)‘, the tautology proved over the predicate on ‘n = 0..4095‘, and the signed-summand control where the ‘≤‘ repair does weaken. **(D)** the ‘15‘ pinned-absent terms at ‘1‘ occurrence in the ‘37‘-paper haystack and ‘40‘ in the ‘18‘ excluded appendices. **(E)** the proxy’s ‘7‘ of ‘14‘ and the two-sided shadow run that refutes it. **(F)** all ‘14‘ exposed-site receipts RUN, ‘14‘ of ‘14‘ green, and the diff touching none of the ‘12‘ receipts this seat does not own. **(G r7214)** AND THE SIXTH FACE, FOUND IN THIS RECEIPT ITSELF BY THE TRUNK MOVING: ‘(F)(2)‘ read ‘diff –name-only PIN..HEAD‘ and called it this revision’s diff, which it is not — every trunk commit merged in afterwards joins that range, so ‘r7193‘’s two receipts of another seat’s entered it through a MERGE and turned the gate red while nothing this seat owns had changed. ***A gate on a diff against a FIXED PIN is a gate on a set every other seat can grow***, and it was carried by the one receipt whose own subject is that class. Repaired to read this branch’s OWN commits, ‘–no-merges‘, which is monotone in the safe direction. **(G)** and the two detectors

mis-flagging EACH OTHER — this receipt's flagged ‘10‘ sites in ‘scripts/mutate_assertions.py‘ through ‘ast.walk‘, and that file's own ‘–cannot-fail‘ rule flagged ‘2‘ in this receipt because a bare ‘True‘ in ARGUMENT position read as a literal-true ASSERTION, which turned the first draft red; *each read a form and never asked its role*, mine repaired by naming the modes and the instrument's routed as a patch. *Bound:* ***NOT that the thirteen are broken*** — *every one is green and the ‘15‘ terms are genuinely absent from the paper layer; the finding is the DISTANCE, ‘40‘ occurrences of five of them in the corpus's own receipt-derived appendix layer, which the haystack drops by a filename prefix.* *And NOT a repair of anybody else's receipt: ‘16‘ of the ‘19‘ sites sit in ‘12‘ receipts this seat does not own and are routed as patches.* *Run:* `python3 receipts/L_probability/S5_the_set_shaped_class_is_absent_from_the_gate_layer_and_two_thirds_of_what`

`S6_an_unread_figure_is_a_claim_with_no_gate_and_one_of_this_seats_six_was_comparing_against_a_number_the`
[OK]
*‘PO-78‘'s unread-figure baseline read at a pinned commit, and ‘P15‘'s own rate-gap, observable and separation sentences, its ‘185‘-bin ceiling, its Kretschmann invariant, its amplitude ratio and its potential equation — each read LIVE and each at exactly one site* ( **AN UNREAD FIGURE IS NOT A FORGOTTEN PIN: ‘19‘ OF THE ‘22‘ OWED LABELS ATTRIBUTE THEIR FIGURE TO A PAPER, SO THE DEFECT IS A LABEL MAKING A CLAIM THE GATE NEVER COVERS.** *And the split inside the ‘19‘ is the finding rather than the total — ‘9‘ sit in files that open no paper at all and ‘10‘ in files that DO read papers, where the omission is not capability but **aim**: the receipt looks like it checks the paper.* **AND ONE OF THIS SEAT'S SIX WAS NOT MERELY UNREAD — ‘C26‘'s TWO GATES WERE COMPARING AGAINST A NUMBER THIS PAPER WITHDREW AS THE ERROR.** *Three things were true and invisible: the figure was removed at ‘r2755‘, whose own subject calls it the error, and its replacement has since gone too, so no paper carries it; the label asserted a betweenness that its condition never tested and that was true of no figure, since the number does not lie between the two weightings; and it was the **wrong object**, the OBSERVABLE's figure set against a RATE-GAP computation, two things the paper separates in its own voice.* ⇒ *The six are repaired by reading the paper and the backlog falls ‘22‘ → ‘16‘ **by running and not by reclassification**, with the six new pins adjudicated in the same pass so one backlog is not discharged by growing another.* **AND THE CONTROL TAUGHT MORE THAN THE SIX: one perturbation substituted NOTHING, came back green, and read exactly like a repair that does not bite** — *so a must-come-back-wrong control now counts its substitutions, which is the positive control a zero needs, read in the mutation direction.*). *Computes:* **19 gates, all pass, a few seconds.** **(A)** the ‘22‘ owed sites read at a PINNED baseline, ‘19‘ attributing to a paper, ‘9‘ in files opening no paper against ‘10‘ in files that do, and the ‘3‘ that attribute to nothing reported rather than exempted. **(B)** the retired figure absent from every paper body along with its replacement, the removing commit read for its own subject line and its one removed line against ‘8‘ rewritten, the betweenness shown false of the figure while the condition's real claim is shown true, and all three live sentences read. **(C)** the seven pins each at exactly one site and carried in the repaired source, the baseline at ‘16‘ with this seat's share at ‘0‘, and the two spellings of one invariant reconciled through ‘$r_s=2M$‘. **(D)** ‘C26‘'s full two-sided control performed LIVE on a symlink shadow tree with the substitution COUNTED, and the other four recorded with their method and one cell named CONTRADICTED. **(E)** the quote-pin backlog unmoved at ‘2167‘ with this revision's own keys adjudicated. *Bound:* ***NOT that the two backlogs are proved to be one class*** — *the attribution measurement supports the reading and ‘19‘ of ‘22‘ is a measurement on ONE backlog, not a proof; the unification is named as a reading, is not written into any gate, and is offered to the gate rather than adopted.* *And NOT a claim about the other ‘16‘: they are in receipts this seat does not own and are untouched.* *Run:* `python3 receipts/L_probability/S6_an_unread_figure_is_a_claim_with_no_gate_and_one_of_this_seats_six_was_com`

**P15R285** `P15_the_period_difference_null_is_not_significant_because_the_control_has_no_measurable_perio`
[OK]
*‘sec:refit-bound‘'s remaining open question and ‘r7197‘'s ordered period-difference null — **THE ORDERED DIFFERENCE IS ‘$0.97\{}sigma$‘ AND THE ERROR IS ALL CONTROL: ‘$35.5$‘ OF ‘$36.3$‘. THE INFORMATIVE COMPARISON IS AGAINST ‘$\{}ell_A$‘, WHERE THE ARM IS ‘$+14.00$‘ HIGH AT ‘$4.69\{}sigma$‘ — AND THAT OFFSET IS *EXACTLY* A PHASE DRIFT OF ‘$-96.6^\{}circ$‘ AT FIXED ‘$\{}ell_A$‘, BY AN IDENTITY**** (node 66's ‘r7197‘: *”the arm's preferred period against the control's as a null, with the period free in

both arms and the DIFFERENCE as the statistic"* — the measurement ‘cc66.149‘ named and left unrun, routed here; with three things to keep apart (the difference not either period alone; what a ‘$+4.7$‘ per cent offset would MEAN given ‘$\{}ell_A$‘ is right to ‘$0.15$‘ per cent; and *"if the difference is NOT significant, that is a complete result and I want it in that form"*)). *Computes:* **10 checks, all pass, ~40 s.** ◆ **NO NEW SPECTRUM, NO GRID.** **(A) THE FORWARD MODEL, VERIFIED EXACTLY:** the figure whitens with the inverse Cholesky of the FULL bandpower covariance, not the diagonal; ‘$L^{-1}(\{}mathrm{model}-\{}mathrm{data})$‘ rebuilt from ‘chi2_of_spectrum‘’s own ‘X_DATA‘/‘COV_TT‘ returns all four banked whitened residuals to **one part in ‘$10^{-12}$‘ of their own scale** — one ulp, and *not* bit-exact: the blocked ‘cholesky‘/‘inv‘ sum in an order that depends on the thread count, so the ‘$=0$‘ this check first asserted was a property of this machine’s default threading and could never have held under the runner’s ‘OMP_NUM_THREADS=1‘ (repaired at ‘cc66.154‘) — and the perturbation reduces to an identity — ‘$L^{-1}(\{}mathrm{model}-(\{}mathrm{data}+Lz))=w_{\{}rm obs}-z$‘ — so one draw ‘$z\{}sim N(0,I)$‘ shared by both arms carries their correlation instead of assuming it away (checked to ‘$10^{-10}$‘). **(B) THE ORDERED STATISTIC:** ‘$\{}Delta=-35.25$‘ against ‘$36.29$‘, **‘$0.97\{}sigma$‘**, ‘$56$‘ per cent of ‘$4000$‘ draws at least as extreme. **(C) THE DECOMPOSITION:** arm ‘$311.79\{}pm2.99$‘ (‘$1.0$‘ per cent), control ‘$334.65\{}pm35.52$‘ (‘$10.6$‘), refit ‘$308.24\{}pm8.82$‘ — the control is ‘$11.9\{}times$‘ the arm and is the entire error on the difference. **(D) AGAINST ‘$\{}ell_A$‘:** ‘$+14.00$‘ (‘$+4.70$‘ per cent), **‘$4.69\{}sigma$‘**, ‘$P(p\{}le\{}ell_A)=0.0003$‘; refitted ‘$+9.25$‘ at ‘$1.05\{}sigma$‘. **(E) ARTEFACT CHECK:** a second harmonic moves the fundamental ‘$+0.25$‘ and leaves ‘$4.90\{}sigma$‘. **(F) THE IDENTITY:** period ‘$312$‘ and ‘$\{}ell_A$‘-with-linear-phase are the same two-dimensional model, RSS identical to ‘$0$‘; the drift is ‘$-96.6^\{}circ$‘ across ‘$104\{}le\{}ell\{}le1886$‘, ‘$0.268$‘ of a period over the ‘$5.98$‘ the range spans. *Bound:* **NOT SIGNIFICANT — AND THE REASON CARRIES MORE THAN THE NULL.** *The error on the ordered difference is ‘$36.3$‘, of which the **control** contributes ‘$35.5$‘ and the arm ‘$3.0$‘. The control’s period is measured to ‘$11$‘ per cent, so **it cannot serve as a null at all**: ‘the difference, not either period alone‘ is right as a principle and empty as a statistic here. That is the order’s own item 1 arriving from the other side.* **THE COMPARISON THE DATA CAN MAKE IS AGAINST ‘$\{}ell_A$‘, AND IT IS ‘$4.69\{}sigma$‘** — *‘$\{}ell_A$‘ is reproduced to ‘$0.15$‘ per cent where the control’s period is known to ‘$11$‘, and fewer than one draw in a thousand puts the arm at or below it. Refitting pulls it to ‘$1.05\{}sigma$‘ and widens it threefold, the same pattern the amplitude shows at ‘cc66.145‘ in the one quantity that figure does not report. And it is not the fit’s artefact: a second harmonic moves the fundamental by ‘$+0.25$‘.* ⇒ **AND ITEM 2’S ANSWER IS NOT ‘$r_s/D_M$‘.** *A period of ‘$312$‘ and ‘$\{}ell_A$‘ with a linearly running phase are the SAME model to zero, so what the residual carries is a **drift of ‘$-96.6^\{}circ$‘ at the correct spacing** — and ‘$\{}ell_A$‘ sets the spacing, right to a part in six hundred, where the offset is ‘$31$‘ times that error. What could carry a running phase at correct spacing is the peak phase itself, the loading and the driving. **Named, not measured, and not a claim.*** **THE ASSUMPTION THE BANKED DATA CANNOT CARRY, NAMED:** *the four banked spectra are point predictions with no parameter covariance beside them, so this null propagates Planck’s noise and nothing else. That is the right error for what was ordered; it is not an error bar on refitting, and the refitted row is the measured stand-in rather than a substitute.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_period_difference_null_is_not_significant_because_the_control_has`

**P15R286** `P15_the_two_exact_twos_are_the_same_pair_of_stretches_read_on_two_clocks_and_the_cosmic_one_is_` [OK]

*‘sec:scope‘’s own separation of the lap’s two exact twos and its two seam-bounded figures, read live and each at exactly one site, against the seam radii and the amplitude the construction fixes* ( **‘r7195‘’s OFFERED ITEM COMES BACK AS A LINK AND NOT A CHOICE OF WORDS: THE COSMIC-TIME TWO IS THE RATIO OF THE TWO SEAMS’ OWN LEG PARAMETERS.** *‘$\{}operatorname{arccosh}2$‘ against ‘$\{}operatorname{arcsinh}(1/\{}sqrt2)$‘, exactly ‘2:1‘ with residual below ‘1e-45‘ — so it is not an independent fact about lengths but **the statement that one seam’s parameter is half the other’s**, which is what attaches it to the seam-bounded pieces.* **AND THE CONFORMAL TWO DOES NOT RUN THROUGH THE SEAMS AT ALL — ON THE SAME PAIR OF STRETCHES, READ THE SAME WAY ROUND, THE SEAMS’ OWN PARAMETER GIVES ‘2.0000‘ WHERE THE CONFORMAL LENGTH GIVES ‘0.4340‘: A FACTOR OF ‘4.6083‘, AND THE TWO CLOCKS DISAGREE ABOUT WHICH

STRETCH IS THE LONGER,** *with the seams dividing neither horn simply at ‘$0.5349$‘ and ‘$0.6163$‘.* *The front-over-back conformal ratio is ‘2.3042‘; set beside the parameter clock’s ‘2.0000‘ it is the same pair ordered the other way, which is why the comparison is made in one orientation here.* ⇒ ***So ‘the $2:1$ leg‘ is STRUCTURALLY ambiguous and not merely loose: whichever clock a reader supplies, the other one is wrong about these pieces.*** *And ‘r7112‘’s arithmetic is confirmed RIGHT, which is what makes the routed item a prose repair — what its retired word cost is measured instead of called loose: the front seam lies ‘$0.4482\{}alpha$‘, ‘23.3‘ per cent of a horn, beyond the midpoint the check is about.*). *Computes:* **13 gates, all pass, a few seconds.** **(A)** the three conformal horn lengths at ‘$1:\{}sqrt3:2$‘ to ‘1e-12‘. **(B)** both seams’ parameters DERIVED from their radii and the amplitude rather than quoted, ‘cc66‘’s identity re-derived independently as a control at sixty digits, and the parameter ratio exactly ‘2‘. **(C)** the two seam-bounded conformal lengths, their ‘2.3042‘ front-over-back ratio, the same-order disagreement between the clocks at a factor of ‘4.6083‘, and the seams’ fractional positions. **(D)** ‘r7112‘’s arithmetic confirmed and the ‘$0.4482\{}alpha$‘ gap measured. **(E)** the ‘$\{}sqrt3-\{}tfrac12$‘ near-miss at ‘$4.9\{}times10^{-4}$‘ refused at thirty digits, with ‘PSLQ‘ and ‘identify‘ both returning nothing so the ratio is reported as no closed form FOUND and never as irrational. **(F)** the paper’s own separation sentence and its two figures read live at one site each. *Bound:* ***NOT a claim on either two*** — *the identity and the cosmic two are ‘cc66‘’s, group ‘I‘ of its own ‘P07‘ receipt, re-derived here as a control and not claimed; the conformal ‘$1:\{}sqrt3:2$‘ is ‘r7108‘’s.* **What is this seat’s is the LINK between them and the measurement of what the bare phrase costs.** *And NOT that the seam-over-midpoint ratio is irrational: a search that finds nothing has not proved that, and the receipt says ‘no closed form FOUND‘.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_two_exact_twos_are_the_same_pair_of_stretches_read_on_two_clocks_a`

**P15R287** `P15_the_phase_slope_does_not_separate_the_loading_from_the_driving_because_a_primordial_tilt_m` [OK]

*‘r7199‘’s standing position on the phase slope, taken on banked data — **THE SEPARATING MEASUREMENT IS RUNNABLE ON THE BANK, IT IS RUN, AND THE PHASE SLOPE DOES NOT SEPARATE THE LOADING FROM THE DRIVING: ALL FOUR BANKED DIRECTIONS NEED A ‘$4$‘–‘$6$‘ PER CENT MOVE AND ‘$n_s$‘ — A PRIMORDIAL TILT WITH NO ACOUSTIC PHASE IN IT — ASKS FOR THE SAME MOVE AS ‘$\{}omega_b$‘, SO THE FREE-PERIOD SLOPE IS NOT A PHASE OBSERVABLE ON A ONE-AMPLITUDE-FITTED RESIDUAL*** (node 66’s ‘r7199‘, which orders **nothing** and says why: *”the next question is which quantity carries a running phase at correct spacing, and you have already named the two candidates... If you see the measurement that would separate the driving from the baryon loading on banked data, run it and say what it was... And if the honest answer is that it needs a grid, say so and I will cost it”*). *Computes:* **13 checks, all pass, ~2 s.** ◆ **NO NEW SPECTRUM, NO GRID.** **(A) THE STATISTIC IS VALIDATED BEFORE IT IS EXTENDED:** ‘cc66.153‘’s slope ‘$k=2\{}pi(1/P-1/\{}ell_A)$‘ reproduces its own ‘$P=312.00$‘ and ‘$-96.6^\{}circ$‘, and the observed slope is re-measured on the ‘$161$‘-bin footing the grid can reach (‘$-9.2995\{}times10^{-4}$‘, within ‘$2$‘ per cent of the full-window value). **(B) THE BANK’S OWN LEVERS, CAPTURED FROM ITS ‘switches‘:** of ‘$257$‘ banked ‘.npz‘ only ‘$18$‘ record switches at all, and **‘RBFAC‘ — the loading-only lever — appeared in none of them AT THIS REVISION, and in neither nine-run grid ever** (‘$0$‘ of ‘$36$‘, which is what the argument rests on; the ask below was granted at ‘r7201‘ and settled at ‘r7203‘, so the lever is now banked under ‘r7201_cc66_loading_lever/‘ and nowhere else); what is banked is ‘WBH2‘ (‘$0.0216$‘/‘$0.0232$‘) and ‘NS‘ (‘$0.945$‘/‘$0.985$‘), plus two driving-off spectra. **(C) THE DERIVATIVE, ‘$38$‘ BANKED SPECTRA, BOTH ARMS, BOTH GRIDS:** each at its OWN banked ‘$\{}ell_A$‘; required fractional move CR ‘H0‘ ‘$4.0$‘, ‘OM‘ ‘$6.2$‘, ‘NS‘ ‘$5.2$‘, ‘WB‘ ‘$4.1$‘ per cent and control ‘$4.4$‘/‘$5.8$‘/‘$5.0$‘/‘$3.9$‘ — spread ‘$1.55\{}times$‘ and ‘$1.49\{}times$‘. **(D) THE TELL:** ‘NS‘/‘WB‘ ratio ‘$1.26$‘–‘$2.11$‘ and the same sign at every treatment that leaves the statistic intact; at degree ‘$4$‘ all four fall under ‘$2.5$‘ per cent and ‘$n_s$‘ **changes sign through zero**, so the wider spread there is degeneracy and not discrimination. **(E) THE DRIVING-OFF LIMIT:** ‘$+1.46$‘ and ‘$+3.43\{}times10^{-4}$‘, the **opposite sign** to the observed slope at ‘$0.37$‘ of its magnitude. *Bound:* **IT NAMES NO CARRIER AND CLAIMS NO NULL ON THE PHYSICS.** *What is established is about the STATISTIC: a free-period slope on a one-amplitude-fitted residual does not isolate the acoustic phase, so neither the loading nor the driving is excluded by it.* *‘WBH2‘ was never a clean loading lever — ‘$\{}omega_b$‘ also sets the free-electron density — so even a discriminating ‘WB‘ result

would have been two effects; ‘RBFAC‘ is the clean one and was in no bank when this was run — the runs that put it there are this receipt's own ask, discharged at ‘r7203‘.* *‘NODRIVE‘ is a total removal and not a derivative, and its two spectra are a different vintage from the grids, so ‘PART E‘ is a limit.* *The arm's own response is convention-dependent — the licensed grid asks ‘$14$‘/‘$294$‘/‘$11$‘/‘$24$‘ per cent where the one-clock grid asks ‘$4$‘–‘$6$‘ — a second reason not to name a carrier on this statistic.* *No paper edited, no spectrum run, nothing refitted.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_phase_slope_does_not_separate_the_loading_from_the_drivi`

**P15R288** `P15_the_de_tilted_peak_set_is_the_phase_observable_the_slope_was_not_and_it_separates_the_baryo` [OK]

*‘cc66.155‘'s zero-run half, delivered at zero runs — **THE DE-TILTED PEAK SET IS THE PHASE OBSERVABLE THE FREE-PERIOD SLOPE WAS NOT: IT SEPARATES THE BARYON DIRECTION FROM A PURE TILT BY ‘$11\{}times$‘ ON THE COMMON OFFSET AND ‘$21\{}times$‘ ON THE ODD-EVEN ALTERNATION, AGAINST ‘$1.3\{}times$‘ FOR THE SLOPE — AND THE DE-TILT IS FORCED BY A PHYSICS CONTROL RATHER THAN CHOSEN.** *But the CR arm's driving signature is NOT established, so the ask is revised to ten runs rather than met** (node 66's ‘r7199‘ standing position, which orders nothing: *"if you see the measurement that would separate the driving from the baryon loading on banked data, run it and say what it was... And if the honest answer is that it needs a grid, say so and I will cost it"* — taken up unasked after ‘cc66.155‘ found the slope cannot separate them and costed this half at zero runs). *Computes:* **10 checks, all pass, ~10 s.** ◆ **NO NEW SPECTRUM, NO GRID.** **(A) THE DE-TILT IS FORCED, NOT CHOSEN:** without it ‘$n_s$‘ moves the common offset MORE than the baryon direction does (‘$+0.049$‘ to ‘$+0.060$‘ against ‘$-0.003$‘ to ‘$-0.011$‘), which a primordial tilt cannot do — the parabola's vertex dragged by the slope beneath the peak, the same disease ‘cc66.155‘ found in the free-period slope. With a single global power law divided out it collapses to ‘$-0.0024$‘, a factor of ‘$20$‘, and becomes flat across fitting half-windows ‘$25$‘–‘$75$‘ where the raw one drifts. **(B) THE SEPARATION, ‘$38$‘ banked spectra, both arms, both nine-run grids:** ‘$d\{}varphi/d\{}ln\{}omega_b=-0.0276$‘ against ‘$d\{}varphi/d\{}ln n_s=-0.0024$‘ and ‘$d\{},\{}mathrm{alt}/d\{}ln\{}omega_b=+0.0104$‘ against ‘$-0.0005$‘ — ‘$11.3\{}times$‘ and ‘$21.4\{}times$‘ on the arm, ‘$11.4\{}times$‘ and ‘$22.6\{}times$‘ on the control; and the tilt's residual floor is STABLE (‘$0.00217$‘–‘$0.00244$‘) across all four configurations rather than scattering. **(C) AND IT REPAIRS THE CONVENTION-DEPENDENCE:** the arm's ‘LEAFGEOM‘-versus-‘LEAFREC‘ disagreement is a factor of ‘$2$‘ here against up to ‘$49$‘ on the slope. **(D) THE DRIVING HALF:** on the control's same-vintage pair the driving and the baryon direction move the peak set in **opposite directions on BOTH components** (‘$\{}Delta\{}varphi=+0.126$‘, ‘$\{}Delta\{}mathrm{alt}=-0.0247$‘ against ‘$-0.0278$‘ and ‘$+0.0108$‘) — the orthogonality the ask rested on, now measured on one arm. *Bound:* **NO CR DRIVING NUMBER IS CLAIMED, AND THAT IS THE VALUE OF HAVING RUN THE ZERO-COST HALF FIRST.** *The CR arm's banked driving-ON spectrum has no integer-spaced peak series — gaps missing ‘$1$‘ by ‘$0.258$‘ against the grid's ‘$0.085$‘ — and the common offset is DEFINED as a mean against integer index, so a number taken from it would be a statement about the detector.* ⇒ ***So ‘cc66.155‘'s eight-run ask is REVISED to ten: the eight ‘RBFAC‘ runs plus a ‘NODRIVE‘ pair, all at the grids' own configuration.*** *Five peaks make the alternation a coarse estimate; the de-tilt is one global power law over ‘$150\{}le\{}ell\{}le1600$‘ and its NECESSITY is swept but its FORM is a choice.* *‘$n_s$‘ is **inherited** — the corpus posits no six-parameter vector — and is used ONLY as a control, with no claim made about it or about the primordial spectrum.* *It names no carrier: it establishes that a statistic exists which could, and what the remaining runs cost. No paper edited, no spectrum run.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_de_tilted_peak_set_is_the_phase_observable_the_slope_was_not_and_i`

**P15R289** `P15_the_baryon_loading_moves_the_common_offset_OPPOSITE_to_the_full_baryon_direction_so_omega_` [OK]

*‘r7203‘'s ordered write-up of the loading lever's curve — **THE SURGICAL LOADING LEVER MOVES THE COMMON OFFSET IN THE *OPPOSITE* DIRECTION FROM THE FULL BARYON-DENSITY DIRECTION, SO ‘$\{}omega_b$‘ WAS NEVER A MERELY CONTAMINATED LOADING LEVER BUT AN OPPOSING ONE.** ‘$\{}mathrm{d}\{}varphi/\{}mathrm{d}\{}ln R_b = +0.01642$‘ against ‘$\{}mathrm{d}\{}varphi/\{}mathrm{d}\{}ln\{}omega_b = -0.02760$‘; decomposed, the non-loading part of the ‘WB‘ response is ‘$-0.04402$‘, **‘$2.7\{}times$‘ the loading part and of opposite sign.** (node 66's ‘r7203‘ item 3, which ordered it explicitly and*

ordered it SEPARATELY: *"with ‘cr_rb1.5‘ landed the curve has its central difference and is worth a receipt of its own; that is the write-up I want, not an addition to the peak-set one"* — because the figures had been read off ‘analyse_ten.py‘ rather than off a receipt, which is the route the ‘r7201‘ number error took). *Computes:* **11 checks, all pass, ˜10 s.** ◆ **NO NEW STATISTIC, NO REFIT.** **(A) THE COMPARISON IS LEGITIMATE AND NOT NOMINAL:** the instrument's line ‘214‘ is ‘RB_REC = RBFAC * RB_REC‘ and line ‘397‘ feeds it straight into ‘Rb_of‘, so ‘$R_b\{}propto\{}mathrm{RBFAC}$‘ linearly exactly as ‘$R_b\{}propto\{}omega_b$‘ — both lines quoted from the instrument, not paraphrased. **(B) IT IS A CURVE AND NOT A SLOPE:** ‘$\{}ell_A$‘ responds at ‘$8.48$‘/‘$22.10$‘/‘$34.26$‘ per unit ‘$\{}ln\{}mathrm{RBFAC}$‘ across the three adjacent intervals of ‘$\{}{0.1,0.5,1.0,1.5\{}}$‘ — **a factor of ‘$4.04$‘ between the extremes, so the single central difference the lever was first costed as would have reported one number for a quantity that varies by that much inside its own span**; both components rise monotonically; and the comb survives the lever at both ends (‘$0.869$‘–‘$1.049$‘ at ‘$1.5$‘ against the base's ‘$0.895$‘–‘$1.036$‘). **(C) THE ARMS AGREE** at matched loading to parts in ‘$10^{4}$‘, which licenses reading the lever as acoustic physics rather than as ‘LEAFGEOM‘, and is why the control's own ‘$1.5$‘ point was the one ‘r7203‘ could drop. **(D) THE FINDING** above, with the alternation bounding it: there the two do NOT oppose, both positive within a factor of two *Bound:* **NAMES NO CARRIER.** *One candidate only — the loading — against the bank's own baryon direction; the driving half is a separate pair and is not read in this file.* **AND THE CONTROL LEAVES A MEASURED FLOOR ON THE OFFSET:** *the tilt's residual response is ‘$15\{}%$‘ of the loading's on the common offset and ‘$2.9\{}%$‘ on the alternation. **I first wrote that check as ‘an order of magnitude below‘ on both components and it FAILED on the offset** — the de-tilt collapses the leakage by ‘$22\{}times$‘ but does not abolish it, so the alternation is the cleaner component and the floor is stated at its measured size rather than the one I expected.* *Four points on one arm: a sign and an order of magnitude, not a functional form.* *‘NS‘ is INHERITED — adopted from ‘$\{}Lambda$CDM‘'s own basis, derived from nothing this construction predicts — and appears only as the direction known a priori to carry no acoustic phase; every conclusion stands if its value moves.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_baryon_loading_moves_the_common_offset_OPPOSITE_to_the_full_baryo`

**P15R290** `P15_the_arms_own_driving_pair_reproduces_the_banked_one_and_the_peak_plane_separates_the_loadi`
[OK]
*‘r7203‘'s items 1 and 2, on the arm's OWN ‘NODRIVE‘ pair — **THE DEFECT DOES NOT RECUR, THE NEW CONTROL PAIR REPRODUCES THE BANKED ONE, AND THE PEAK PLANE SEPARATES THE LOADING FROM THE DRIVING BY ‘$5.4\{}times$‘ ON ITS DISCRIMINANT — BUT NOT THE DRIVING FROM A CLOCK ERROR.** Both ‘NODRIVE‘ spectra give detected spacing ‘$0.925$‘–‘$0.999$‘ of ‘$\{}ell_A$‘, **tighter than the grid bases' own ‘$0.895$‘–‘$1.036$‘**, against the ‘$0.74$‘–‘$0.96$‘ that disqualified the banked driving-ON spectrum; and ‘NODRIVE‘ leaves ‘$\{}ell_A$‘ **bit-identical** to the base's, so the pair differs in the driving and nothing else* (node 66's ‘r7203‘ items 1 and 2 — *"the de-tilted peak set on the CR arm with its own ‘NODRIVE‘ pair, both components"* and *"the carrier named if the signs separate it"*, with ‘r7203‘'s own narrowing carried through: *"name what the plane says, and state that the plane does not separate those two"*). *Computes:* **9 checks, all pass, ˜10 s.** ◆ **NO NEW STATISTIC, NO REFIT.** **(A) THE DEFECT IS ABSENT** on both arms, five peaks each. **(B) THE SAME-VINTAGE CONTROL PAIR REPRODUCES THE BANKED ONE TO BETTER THAN ‘$1\{}%$‘:** ‘$+0.12651$‘/‘$-0.02469$‘ against ‘cc66.156‘'s ‘$+0.126$‘/‘$-0.0247$‘ — **which retires the different-vintage caveat that was the reason the ask went from eight runs to ten.** *The worry was legitimate and the answer is that it cost nothing.* And the CR arm's own signature ‘$+0.12589$‘/‘$-0.02428$‘ matches the control's to ‘$0.5\{}%$‘ on the offset, so the driving acts as acoustic physics and not as ‘LEAFGEOM‘. **(C) THE PLANE'S DISCRIMINANT** ‘$\{}lvert\{}Delta\{}mathrm{alt}/\{}Delta\{}varphi\{}rvert$‘: driving ‘$0.1929$‘, loading ‘$1.0390$‘, ‘WB‘ ‘$0.3780$‘, and ‘60‘'s ‘r7218‘ clock error ‘$0.086$‘–‘$0.173$‘ (cited, not recomputed) — so **loading/driving ‘$=5.39\{}times$‘, loading/clock-top ‘$=6.01\{}times$‘, driving/clock-top ‘$=1.11\{}times$‘.** **(D) AND ‘60‘'s PROXY POINTED THE WRONG WAY:** the ‘WB‘ stand-in sits ‘$2.75\{}times$‘ inside the clean lever and ‘cc66.157‘ measured their offsets to have OPPOSITE sign *Bound:* **NO CARRIER IS NAMED FOR THE OBSERVED DRIFT, AND THE REASON HAS CHANGED.** *Before these runs the blocker was that the arm had no readable driving signature; that is gone. What is missing is the OB-

SERVED residual's position on this plane — **nothing in tree measures it.** The statistic has been run on model spectra only, 238-point curves with no noise; putting the data on the same plane means 179 binned points with a covariance, a measurement with its own validation burden and not what 'r7203' ordered ('no refit, no new statistic').* **AND THE DRIVING IS NOT SEPARATED FROM A CLOCK ERROR BY THIS PLANE** — '$1.11\{}times$' the clock error's top, inside the spread of '60''s own three readings. *Stated as the limit it is.* *'60''s '$0.086$'–'$0.173$' is read off '60''s receipt output and cited as an input; no check here recomputes it, since it is a property of '60''s oscillator.* *'NS' is INHERITED, used only as the direction known a priori to carry no acoustic phase.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_arms_own_driving_pair_reproduces_the_bank`

**P15R291** `P15_the_peak_statistic_cannot_be_pointed_at_the_observed_spectrum_because_a_local_maximum_finc`
[OK]
*'r7211''s ordered last step — **THE OBSERVED POINT CANNOT BE PLACED, AND 'r7211' NAMED THIS CASE: 'if the covariance makes the placement unreliable rather than merely imprecise, stop and say so'. IT IS UNRELIABLE, AND NOT BECAUSE OF THE COVARIANCE.** The peak locator reads **'$21$' local maxima on the observed spectrum where it reads '$5$' on a model binned the same way**; the five it takes are '$239$'/'$464$'/'$527$'/'$617$'/'$815$' against the model's acoustic '$257$'/'$554$'/'$824$'/'$1139$'/'$1436$', **three of them inside the second peak's own neighbourhood.** So it refines noise wiggles* (node 66's 'r7211', granting the ask this seat declined to grant itself: *"run the de-tilted peak statistic on 'nofit_figure_numbers.npz''s 'data' with its 'sigma', and place the observed point on this plane with an error bar"*, with *"the error bar is the result as much as the position"* and the stop condition quoted above). *Computes:* **8 checks, all pass, ~60 s — and what they establish is a NEGATIVE.** ◆ **NO REFIT, NO NEW STATISTIC, NO NEW SPECTRUM.** **(A) THE BINNING IS NOT THE PROBLEM**, which was the risk: run fine and then through the likelihood's own 'bin_spectrum', the offset moves by under '$0.0025$' and the alternation by under '$0.0005$' on four models, consistently in sign — so '$179$' binned points DO carry the statistic. *'X_data' is binned '$C_\{}ell$' and not '$D_\{}ell$'; every spectrum is converted with the same 'fac', data and models alike, which 'bin_center_and_fac''s own docstring records as the trap.* **(B) THE LOCATOR IS NOT NOISE-ROBUST** and the data is noisy at the scale of its own curvature; one of the five is not even concave, so the alternation is taken over four peaks with spurious members. **(C) THE FULL 'plik_lite' COVARIANCE CONFIRMS IT RATHER THAN RESCUING IT:** '$2000$' Cholesky draws, **'$98.2\{}%$' fail to yield five finite peaks at all**; the '$35$' survivors give '$\{}sigma_\{}varphi=1.25$' — **'$10\{}times$' the driving's whole signal and '$69\{}times$' the loading's** — and come back correlated at '$-0.955$', **collapsed onto ONE direction rather than spanning the plane** *Bound:* **NO CARRIER IS NAMED FOR THE OBSERVED DRIFT.** *And the honest statement is NOT 'the uncertainty covers both candidates' — it is that the quantity measured is not the one the statistic reads.* **AND IT WITHDRAWS NOTHING FROM 'cc66.157'/'cc66.158':** *both compare MODEL to MODEL on noiseless '$238$'-point spectra, the regime (A) shows sound. The '$5.39\{}times$' separation, the opposite signs on the common offset, and the '$\{}omega_b$' opposition all stand. **What fails is one step, and it fails for a reason specific to data: noise creates maxima, and a maximum-finder cannot tell them from acoustic peaks.*** *A repair would have to fit a parametric comb to the whole binned residual under the covariance rather than locate maxima and refine them — **a DIFFERENT statistic, which 'r7211' did not order and which is '66''s to cost.*** *Cost: one re-read, as 'r7211' scoped it.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_peak_statistic_cannot_be_pointed_at`

**P15R292** `P15_the_three_parameter_comb_fails_its_validation_because_the_peak_spacing_DRIFTS_and_the_miss`
[OK]
*'r7213''s ordered comb fit — **IT FAILS THE VALIDATION BURDEN 'r7213' ATTACHED, SO BY THE ORDER'S OWN TERMS THE DATA HALF DOES NOT HAPPEN — AND THE REASON IS THAT THE PEAK SPACING *DRIFTS*, A FOURTH DEGREE OF FREEDOM THREE PARAMETERS CANNOT CARRY.** The comb recovers '$\{}ell_A$' on **three** spectra of six — both driving-off ones and the driving-ON 'cr_rb0.5', to '$0.27\{}%$' — and misses 'cr_base', 'lcdm_base' and 'cr_rb1.5' by '$18$'–'$19\{}%$'; and because '$\{}phi$' is referred to the FITTED '$\{}ell_A$', **each driving pair has one member '$18\{}%$' out, so 'cc66.158''s separation is not reproduced badly — it is not constructible from this instrument**** (node 66's 'r7213': *"build the parametric comb: position, spacing and alternation as FITTED PARAMETERS under the covariance"*, with the burden attached — *"show it recovers 'cc66.156''s

and ‘cc66.158‘’s own model directions ... If it does not recover them, that is the result and the data half does not happen”*). *Computes:* **8 checks, all pass, ˜6 min — and what they establish is that the ORDERED instrument fails its own validation, for a named reason, with the missing parameter identified.** ◆ **NO NEW SPECTRA, NO REFIT, AND THE DATA DELIBERATELY NOT TOUCHED.** **(A) THE TEMPLATE IS RIGHT, to second order:** ‘$\{}cos(2\{}pi[u-a\{}cos\{}pi u])$‘ has maxima at ‘$n+a(-1)ˆ{n}$‘ to under ‘$aˆ{2}/4$‘. **(B)(1) THE DICHOTOMY THIS RECEIPT FIRST REPORTED WAS THE SEARCH AND NOT THE MODEL:** a ‘$3\{}times3$‘ grid of starts offset from the banked ‘$\{}ell_A$‘ returns a worse ‘$\{}chiˆ{2}$‘ on five of six spectra — ‘$57\{}%$‘ on the driving-off pair — and on ‘cr_rb0.5‘ reports ‘$351.961$‘ where the minimum is ‘$286.841$‘. *The starts now span the whole feasible box and every optimum is CHECKED to be interior to it; the inadequate grid is kept in the file as ‘fit_narrow‘ so the defect stands beside its repair.* **(B)(2) THREE OF SIX RECOVER** as above. **(B)(3) AND THE BURDEN FAILS FOR A SHARPER REASON than the first version gave.** **(C) WHY — A COMB’S HIGH GAPS REPEAT AND THESE RISE:** ‘cr_base‘’s gaps are ‘$302/270/312/295$‘, alternating but with the highs rising ‘$+10$‘ and the lows ‘$+25$‘; one linear drift term cuts the gap-fit rms on **every** spectrum (‘$9.5\{}to3.8$‘ on ‘cr_base‘, ‘$8.5\{}to0.5$‘ on ‘cr_rb0.5‘). **AND OMITTING THE DRIFT RETURNS A WRONG ALTERNATION — by ‘$33$‘–‘$87\{}%$‘** (‘$0.0208\{}to0.0295$‘ on ‘cr_base‘), *the alternation being one of the two components ‘cc66.158‘’s separation rests on.* **(D) THE PITFALL:** a generic per-harmonic basis improves ‘$\{}chiˆ{2}$‘ by ‘$542\{}times$‘ while running ‘$\{}ell_A$‘ to the EDGE of the search box at ‘$430.000$‘ — **a good ‘$\{}chiˆ{2}$‘ from a comb fit is not evidence the comb was found**. **(E) AND THE FAILURE TRACKS THE DRIFT, NOT THE DRIVING:** every spectrum the comb recovers has a smaller driftless gap residual (‘$8.31$‘–‘$8.62$‘) than every spectrum it misses (‘$9.58$‘–‘$11.81$‘) — *stated as the ordering it is, on six spectra with a margin under one multipole, which does not establish where the boundary lies* *Bound:* **THE MISSING PARAMETER IS NOT ARBITRARY — IT IS THE QUANTITY THIS SECTOR IS ABOUT.** *A drift in spacing IS a linear phase drift at fixed ‘$\{}ell_A$‘, which is ‘cc66.153‘’s own description of this residual (‘$-96.6ˆ{\{}circ}$‘ at the correct spacing).* **AND ADDING IT IS WIDENING THE ORDER, WHICH IS NOT THIS SEAT’S TO DO.** *‘r7213‘ ordered three parameters and said not to reduce the count without saying so; it did not authorise a fourth. Naming it and stopping is the discipline held at ‘r7203‘ and ‘r7211‘, both sustained.* **NOTHING IS WITHDRAWN FROM ‘cc66.157‘/‘cc66.158‘:** *they use located peaks with ‘$\{}ell_A$‘ FIXED at its banked value, so the drift is common to both sides of every difference they take and never enters as a free parameter.* **AND THE CORRECTION THIS ROW CARRIES WAS NOT CAUGHT BY ANY CHECK OF MINE: THE TOLERANCE PERTURBATION FOUND A TIE-BREAK DECIDED AT THE LAST BIT, AND READING ALL NINE STARTS TO DIAGNOSE IT IS WHAT EXPOSED THE INADEQUATE SEARCH.** *Eight checks interrogated the ANSWER and none interrogated the SEARCH that produced it. The withdrawn claim — that the inadequacy is specific to spectra carrying a driving — and **(D)**’s ‘$122\{}times$‘, which I had already walked back once from a boundary claim that turns out to be right, were both artefacts of that one defect. **Neither number meant anything while the search was wrong.*** *Run:* `python3 receipts/P15_CR_cosmology/P15_the_three_parameter_comb_fails_its_validation_because_t`

**P15R293** `P15_the_fourth_parameter_recovers_the_drift_the_gaps_show_but_the_spacing_and_the_drift_are_no` [OK]

*‘r7215‘’s ordered fourth parameter — **IT IS THE RIGHT PARAMETER AND THE BANK CANNOT SUPPORT IT: THE FITTED DRIFT REPRODUCES THE GAP GROWTH ‘cc66.160‘’s PART C MEASURES ON FOUR SPECTRA OF SIX, AND THE SPACING AND THE DRIFT ARE DEGENERATE AT ‘$\{}rho\{}le-0.93$‘.** ‘r7215‘ item 2 PASSES — ‘$\{}chiˆ{2}$‘ improves ‘$44\{}%$‘ on both driving-off spectra and the drifting spacing crosses banked INSIDE the window at ‘$\{}ell\{}simeq814$‘; but on ‘cr_base‘/‘lcdm_base‘ it crosses banked only at ‘$\{}ellˆ{*}=-1330$‘/‘$-1251$‘, so each driving pair still has one member not referred to the same comb and **the validation burden fails again**** (node 66’s ‘r7215‘: *”build the four-parameter comb: position, spacing, alternation, and the linear drift”*, with *”test it on the driving-off spectra FIRST”* and *”degeneracy still counts as a result ... if drift and alternation are not separable, say so and stop”*). *Computes:* **10 checks, all pass, 473s measured against CI’s 900s budget — the fourth parameter is the right one and the bank cannot support it: a better description and a worse instrument.** ◆ **NO NEW SPECTRA, NO REFIT, DATA NOT TOUCHED.** **(A) THE DRIFT IS PART C’s OWN ‘$c_2$‘:** ‘$\{}ell=\{}ell_A v+dvˆ{2}$‘ makes the per-index gap growth

‘$2d$‘ to better than a per cent, the ‘$d=0$‘ reduction reproduces ‘cc66.160‘’s three-parameter fit, and this file’s copy of PART C’s peak finder is ‘cc66.160‘’s **byte for byte**, pinned to its published gaps ‘$302/270/312/295$‘. **(B)(1) ITEM 2 PASSES**, ‘$\{}chi^{2}$‘ ‘$106\{}to59$‘ and ‘$94\{}to53$‘, crossing banked at ‘$\{}ell=815$‘/‘$813$‘ — *read naively ‘$\{}ell_A$‘ says ‘$-6.6\{}%$‘ and that is a REFERENCE-POINT error: ‘$\{}ell_A$‘ is the spacing at ‘$v=0$‘, not the acoustic scale when the spacing drifts.* **(B)(2) every INTERIOR optimum is reached from more than one start and the one that is not is exactly the one on the box edge** (‘cr_rb1.5‘, ‘$430.000$‘) — *two independent signs of the same unidentifiability.* **(B)(3) THE FITTED ‘$2d$‘ MATCHES PART C’s INDEPENDENTLY MEASURED ‘$c_2$‘:** ‘$+8.85$‘ vs ‘$+8.77$‘, ‘$+9.22$‘ vs ‘$+8.85$‘, ‘$+7.10$‘ vs ‘$+8.15$‘ twice. **(B)(4) and runs away WRONG-SIGNED on ‘cr_rb0.5‘/‘cr_rb1.5‘** (‘$-35.20$‘ and ‘$-30.75$‘ against ‘$+8.29$‘/‘$+9.70$‘) — *‘cc66.160‘’s ‘(D)(1)‘ pitfall in the fourth parameter’s clothes.* **(C)(1) THE PAIR ‘r7215‘ ASKED ABOUT IS SEPARABLE:** ‘$\{}lvert\{}rho(a,d)\{}rvert\{}le0.47$‘ on all six. **(C)(2) WHAT IS NOT IS THE SPACING AND THE DRIFT:** ‘$\{}rho(\{}ell_A,d)$‘ from ‘$-0.94$‘ to ‘$-0.98$‘ — *and ‘cr_rb1.5‘’s ‘$\{}rho=0$‘ is NOT independence, its ‘$\{}ell_A$‘ is clipped at the box edge.* **(D)(1) AND THE BURDEN FAILS:** the base spectra’s fitted spacing runs ‘$344\{}to385$‘ and ‘$343\{}to386$‘ across the window and never equals banked anywhere *Bound:* **A FOUR-PARAMETER COMB IS A BETTER DESCRIPTION OF THESE SPECTRA AND A WORSE INSTRUMENT FOR THIS MEASUREMENT.** *The fourth parameter was correctly identified by ‘r7215‘ and correctly added; what defeats it is that ‘$179$‘ binned points cannot hold a spacing and its drift apart — a statement about what the bank can support, which ‘r7213‘ said is worth as much as a fit.* **AND THE STOP CONDITION FIRED ON A PAIR THE ORDER DID NOT NAME.** *‘r7215‘ asked about drift-versus-alternation and those are separable; it is spacing-versus-drift that is degenerate, which is why ‘(C)(1)‘ reports the asked-for pair as sound BEFORE ‘(C)(2)‘ reports the one that bites.* *Four defects of mine were caught by this file’s own checks across four runs: a rebuilt ‘OptimizeResult‘ that silently lost ‘fun‘; PART C’s peak finder RETYPED FROM MEMORY with four differences, which would have made ‘(B)(3)‘ compare the fit against my own variant while claiming it was PART C’s; ‘(B)(2)‘ written as ‘every optimum‘ when the measurement says ‘every interior optimum‘; and a PART C refit of PART B’s own optimum that took the runtime past ‘$20$‘ minutes against a ‘$900$‘s budget.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_fourth_parameter_recovers_the_drift_the_gaps_show_but_the_spacing`

**P15R294** `P15_centring_removes_the_correlation_exactly_and_closes_the_route_anyway_because_the_decorrel`
[OK]
*‘r7219‘’s ordered reparametrisation — **THE CORRELATION DOES NOT SURVIVE CENTRING — IT GOES TO MACHINE ZERO EXACTLY — AND THE ROUTE IS CLOSED ANYWAY, BECAUSE THE DECORRELATED PARAMETER IS NOT THE ACOUSTIC SCALE.** Centring is an exact SHEAR, ‘$L_p=\{}ell_A+2v_pd$‘ with ‘$D=d$‘, so the fitted curve is bit-identical and only the covariance moves; ‘$\{}rho$‘ goes ‘$-0.94$‘–‘$-0.98$‘ to ‘$10^{-16}$‘ at ‘$v_p^{*}=-C_{01}/2C_{11}$‘, interior on all five unclipped spectra. **But the spacing there sits ‘$33$‘–‘$379$‘ of its own ‘$\{}sigma$‘ from banked**, so the bank pins one combination of scale and drift to under a tenth of a per cent and the acoustic scale is not it* (node 66’s ‘r7219‘: *”centre the comb basis and measure whether the ‘$\{}ell_A$‘–drift correlation survives it”*, *”a reparametrisation, not a fifth parameter”*, burden *”the fitted curve must be identical to the digits and only the covariance may move”*, with the closure branch named in advance — *”if the correlation survives centring, the route is closed and that is the result”*). *Computes:* **4 checks, all pass, 369s measured — the correlation goes to zero exactly and the route closes anyway.** ◆ **NO NEW SPECTRA, NO REFIT, DATA NOT TOUCHED.** **(A)(1) THE BURDEN, CHECKED AT THE DIGITS:** ‘$\{}max\{}lvert A_1-A_0\{}rvert=0$‘ on all six spectra — *a shear has nothing for the curve to do, so ‘only the covariance may move‘ is satisfied by construction and verified anyway.* **(B)(1) THE CORRELATION DOES NOT SURVIVE:** ‘$10^{-16}$‘ after centring, pivot interior on all five with curvature (‘cr_rb1.5‘ has none — its ‘$\{}ell_A$‘ is clipped at ‘$430.000$‘, so there is nothing to transform). **(C)(1) AND IT BUYS A PRECISE NUMBER ABOUT THE WRONG QUANTITY:** ‘cr_nodrive‘ ‘$293.438\{}pm0.226$‘ vs banked ‘$301.380$‘ = ‘$35\{}sigma$‘; ‘cr_base‘ ‘$350.131\{}pm0.129$‘ vs ‘$301.380$‘ = ‘$379\{}sigma$‘; ‘cr_rb0.5‘ ‘$288\{}sigma$‘. **(C)(2) SO THE CORRELATION WAS NEVER THE OBSTACLE** — *and because ‘(A)(1)‘ leaves the curve identical, every pivot-invariant statement in ‘cc66.161‘ survives verbatim, including the base spectra’s spacing never equalling banked anywhere (‘$\{}ell^{*}¡0$‘)* *Bound:* **NEITHER

BRANCH ‘r7219‘ NAMED IS THE ANSWER, AND THE ALGEBRA SAYS WHY BEFORE THE FIT DOES.** *The reparametrisation is a shear in parameter space, so a correlation it removes was a property of the coordinates and not of the measurement. **What ‘$179$‘ binned points determine is one combination of scale and drift, to better than a tenth of a per cent; what they do not determine is that combination evaluated where the acoustic scale lives.*** ⇒ *So ‘r7219‘’s own print stands unaltered — ‘a four-parameter comb is the better description of these spectra and the worse instrument for this measurement‘, and ‘what would change it is a finer binning rather than a further parameter‘.* *This seat proposed no fifth parameter at ‘r7215‘, proposes none here, and now has the reason in closed form rather than as a refusal.* *Run:* `python3 receipts/P15_CR_cosmology/P15_centring_removes_the_correlation_exactly_and_closes_the_route_anyway`

**P15R295** `P15_the_carrier_separation_on_L_p_is_the_combs_peak_count_not_the_carrier_and_the_closure_has_n` [OK]

*‘r7221‘’s two items, on the tight direction ‘cc66.162‘ found — **ITEM (1) HAS A YES AND THE YES IS THE COMB’S OWN PEAK COUNT; THE SKY READS LIKE THE DRIVING AND THE READING IS VOID; AND THE CLOSURE NUMBER DOES NOT EXIST.** ‘$\{}Delta L_p$‘ across the driving pair is ‘$56.693$‘, ‘$218\{}sigma$‘ on the bank’s own convention — **but ‘$L_p$‘ is the fitted curve’s OWN peak spacing only on the two driving-OFF spectra (‘$0.2$‘ per cent) and is ‘$21$‘–‘$27$‘ per cent out on the other four including the sky**, so the ‘$218\{}sigma$‘ separates a fit that LOCKS from one that does not. The loading pair cannot be differenced at all — ‘cr_rb1.5‘ has no interior optimum in any box. And ‘$\{}lvert\{}rho(\{}ell_A,d)\{}rvert$‘ stays in ‘$[0.939,0.977]$‘ across thirteen fits, ‘$78$‘–‘$176$‘ points, five windows and six spectra, so the remainder ‘r7221‘ asks for has no number* (node 66’s ‘r7221‘ — *”measure whether the carrier separates along the direction the bank actually carries”*, (1) *”‘$\{}Delta L_p$‘ between the carriers, in units of the bank’s own ‘$\{}sigma$‘”*, (2) *”if they separate there, carry ‘$L_p$‘ to the sky”* by *”this section’s own matched-procedure differencing — arm, ‘$\{}Lambda$CDM‘ control and ‘plik_lite‘ through the identical extraction”*, with the burden *”state the expected ‘$\{}Delta L_p$‘ BEFORE you read the fits”* and the closure named in advance — *”the only remainder is a NUMBER: what point count, window and per-point error would bring ‘$0.1929$‘ against ‘$1.0390$‘ within reach”*). *Computes:* **16 checks, all pass, 223s measured.** ◆ **NO NEW SPECTRA.** **(A)(1) ‘cc66.162‘’s FIVE PUBLISHED SPECTRA COME BACK TO ITS PRINTED DIGITS** — ‘$L_p$‘ to three decimals, ‘$\{}rho$‘ to four, ‘$\{}sigma$‘ to three. **(A)(2) and splitting PART C’s locator so it can be pointed at a FITTED curve is a no-op on all seven inputs**, checked against ‘cc66.160‘’s byte-for-byte copy. **(A)(3) THE PRE-REGISTRATION’S PREMISE IS RIGHT ON THE GAPS:** PART C’s gap LEVEL moves ‘$+29.11$‘ across the loading switch against a banked ‘$+29.21$‘, and is the same to ‘$1.32$‘ across the driving switch where the banked scale is bit-identical. **(B)(2) AND MY PRE-REGISTERED NUMBER IS WRONG — ‘$5.7\{}times$‘ THE BOUND I WROTE AND ‘$11.2\{}times$‘ THE TERMS THAT PRODUCED IT** — *and it was NOT a blind prediction, recorded as such rather than as a test.* **(B)(3) ‘$L_p$‘ IS THE FITTED CURVE’S SPACING ONLY WHERE THE COMB LOCKS ON:** its maxima sit ‘$1.6$‘ from the data’s on the driving-OFF pair and ‘$21$‘–‘$24$‘ out elsewhere. **(B)(4) ‘cr_rb1.5‘ HAS NO INTERIOR OPTIMUM IN ANY BOX** — ‘$430.000$‘ at the ‘$430$‘ box, ‘$900.000$‘ at the ‘$900$‘, ‘$\{}chi^{2}$‘ falling ‘$1.94\{}times10^{4}\{}to1.39\{}times10^{3}$‘. **(B)(5) AND THE ‘$\{}sigma$‘ THE ORDER MEASURES AGAINST IS NOT AN ERROR ON ‘$L_p$‘:** moving the window’s top edge ‘$1000\{}to1950$‘ moves ‘$L_p$‘ by ‘$14.3$‘ while the formal ‘$\{}sigma$‘ falls to ‘$0.012$‘, and the offset from banked over those windows is ‘$+5.4$‘–‘$+18.2$‘, one sign throughout. **(B)(6) SO TWO FIGURES ‘cc66.162‘ PUT IN PRINT DO NOT SURVIVE THE CHANGE OF ‘$\{}sigma$‘ CONVENTION: its ‘$35\{}sigma$‘ READS ‘$54$‘ AND ITS ‘$379\{}sigma$‘ READS ‘$82$‘** — *the conclusion and the ordering stand, since every one of the five is still tens of sigma from banked on both and ‘$¿10$‘ is what its ‘(C)(1)‘ asserts; what does not stand is ‘more than three hundred‘.* **(C)(1) THE SKY LANDS ON THE DRIVING-ON BASES** — ‘$351.807$‘ against ‘$350.131$‘/‘$349.835$‘, and ‘$58.4$‘ from the driving-OFF pair — **(C)(2) AND THE READING IS VOID:** that curve is rejected by ‘plik_lite‘ at ‘$\{}chi^{2}/\{}mathrm{dof}=11.83$‘ on the FULL covariance (the de-tilt is a diagonal rescaling, so it transforms exactly) and its ‘$L_p$‘ misses its own curve’s spacing by ‘$62$‘. **(C)(3) THE MATCHED DIFFERENCING IS BLIND TO THE ARM:** ‘$0.295$‘, ‘$1.6\{}sigma$‘, against the carrier’s ‘$218$‘. **(C)(4) and ‘$L_p$‘ carries only ‘$16$‘ per cent of a real change in the acoustic scale.** **(D)(1) THE DEGENERACY IS SAMPLING-INVARIANT:** halving the point count leaves ‘$\{}rho$‘ at ‘$-0.9549$‘ against ‘$-0.9548$‘. **(D)(2) AND MERGED IN PAIRS

PART C's LOCATOR READS THE SKY** — four peaks within '$3.37$' of its own read of the MODEL through the identical merge, gaps '$303$'/'$268$'/'$309$' against '$302$'/'$269$'/'$312$', against a '$547$' displacement at the bank's native '$9$'; at '$27$' it returns nothing on EITHER, which is its own '$\{}pm40$' window holding fewer than four points. **(D)(3) so the row's two halves want opposite binnings and the reachable half is the peak plane** *Bound:* **THE ORDERED DIFFERENCING ANSWERS YES AND THE ANSWER IS ABOUT THE INSTRUMENT RATHER THAN THE SKY.** *'$L_p$' is precise, reaches the sky, separates the carrier at '$218\{}sigma$' and the arm at '$1.6$' — because what it measures is whether the comb locked onto the peak train.* ⇒ **AND 'r7219''s 'what would change it is a finer binning rather than a further parameter', now in print, is measurably wrong:** *the correlation is a property of the monomial pair '$(v,v^{2})$' over an interval and not of the sampling — which is why 'cc66.162''s shear removes it exactly and why no point count in reach of this bank touches it.* *So the closure branch 'r7221' named fires in substance rather than on its literal condition, and **the NUMBER it asks for does not exist**, which is a stronger terminal statement than a number.* *'$[0.939,0.977]$' is MEASURED on thirteen fits and not proved; the monomial-ratio argument is prose here and not a check. The '$+5.4$'–'$+18.2$' window spread is a reproducibility probe on one spectrum and not a marginalisation. And '(D)(2)''s route is OFFERED AND NOT BUILT — four peaks and not five, 'cc66.156''s statistic is banked at five, and forming '$(\{}varphi,\{}mathrm{alt})$' on a merged bank is a new measurement with its own validation burden.* *No fifth parameter is proposed and both standing refusals hold.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_carrier_separation_on_L_p_is_the_combs_peak_count_not_the_carrier_`

**P15R296** `P15_the_peak_plane_survives_the_merge_the_sky_lands_within_one_sigma_of_the_control_and_a_full_` [OK]

*'r7223''s one order, the instrument the comb was a substitute for — **THE BURDEN IS MET, THE PLANE SURVIVES THE MERGE, AND THE SKY LANDS '$0.8\{}sigma$' FROM THE CONTROL WITH A FULL DRIVING UNIT AT '$13\{}sigma$'.** The merged model reproduces the banked '$+0.126$'/'$-0.0247$' to '$1.008\{}times$'/'$1.010\{}times$' at matched peak count; the tilt separation re-measured on four peaks is '$9.8$'–'$12.1\{}times$' on the offset and '$8.2$'–'$9.9\{}times$' on the alternation against the slope's '$1.3$', so 66's collapse branch does not fire; and the sky's point is '$(\{}varphi,\{}mathrm{alt})=(-0.18571,+0.00804)$', '$(+0.00756,-0.00255)$' from the control. **But '$48$' per cent of noise realisations do not return four peaks**, so every figure is conditional on the statistic existing* (node 66's 'r7223' — *"form '$(\{}varphi,\{}mathrm{alt})$' on the pair-merged bank and place the sky on the peak plane"*, with this seat's own burden adopted verbatim — *"the merged MODEL must reproduce the banked '$+0.126$'/'$-0.0247$' before the sky's position on the plane means anything"* — and one addition of 66's: *"the separation factors must be re-measured on four peaks ... if the four-peak separation collapses to the slope's '$1.3$', the merge has bought the sky at the cost of the thing the plane was for"*, with the closure named in advance: *"if the merged model cannot reproduce the banked pair ... the row terminates on a statement about the bank"*). *Computes:* **14 checks, all pass, 3s measured.** ◆ **NO NEW SPECTRUM, NO FIT, NO REFIT.** **(A)(1) 'cc66.156''s STATISTIC COMES BACK TO ITS PUBLISHED '$+0.126$'/'$-0.0247$'** on both driving pairs, and **(A)(2) its separation factors to the tenth** — '$11.3$'/'$21.4$' on the arm, '$11.4$'/'$22.6$' on the control. **(A)(3) and 'cc66.163''s MERGE comes back to its published locator result:** gaps '$303$'/'$268$'/'$309$' against the model's '$302$'/'$269$'/'$312$' at '$g=2$', '$3.37$' worst against '$547$' native, nothing on either at '$g=3$'. **(B)(1) THE BURDEN IS MET: '$+0.12703$'/'$-0.02494$' against banked, under one per cent on both components, on both pairs and both arms** — *so the closure branch does not fire.* **(B)(2) AND THE FOUR-PEAK SHIFT IS THE PEAK COUNT AND NOT THE MERGE:** '$1.20\{}times$' on raw samples against '$1.22\{}times$' merged, while the merge at fixed peak count moves the alternation by '$2.6$' per cent and the offset holds to a tenth of a per cent throughout. **(B)(3) and the four-peak alternation is WINDOW-DEPENDENT at '$\{}pm15$' per cent** — peaks '$1$'–'$4$' give '$1.216\{}times$', peaks '$2$'–'$5$' give '$0.913\{}times$', straddling the five-peak value, *which is the reverse of what this seat expected from the balanced sign pattern.* **(C)(1) THE SEPARATION DOES NOT COLLAPSE** (worst '$8.2\{}times$' in the configuration the sky is read in) **(C)(2) and the alternation's lever is down '$2.6\{}times$' from '$21.4$' because '$n_s$''s residual leakage onto it grows FOUR-FOLD at four peaks, '$-0.00049\{}to-0.00203$'** — *the price is paid in the de-tilt control, not in the signal.* **(D)(1) THE SKY IS ON THE PLANE** and the plane still separates the carriers there, '$0.24$' against '$1.32$', '$5.6\{}times$'. **(D)(2) THE ERROR

IS HEAVY-TAILED** — ‘plik_lite‘’s covariance through the merge EXACTLY (a linear map, so nothing is dropped), ‘$2000$‘ draws: st.dev. ‘$0.107$‘ against robust ‘$0.0096$‘, and the robust scale is ‘$5.8$‘ in ‘$\{}ell$‘ per peak, consistent with the ‘$3.37$‘ the positions agree to. **(D)(3) ONE FULL DRIVING UNIT IS ‘$13.1\{}sigma$‘ ON THE OFFSET AND ‘$3.1\{}sigma$‘ ON THE ALTERNATION, AND THE SKY SITS ‘$0.79\{}sigma$‘/‘$0.27\{}sigma$‘ FROM THE CONTROL** — *the first number this row has about the observed spectrum rather than about an instrument.* **(D)(4) the limiting systematic is the assumed scale:** ‘$\{}ell_A$‘ at ‘$+1$‘ per cent moves the offset by ‘$2.4$‘ robust ‘$\{}sigma$‘, so the agreement needs ‘$0.42$‘ per cent and the exclusion ‘$5.5$‘. **(D)(5) and what bounds the instrument is the fourth peak’s EXISTENCE:** ‘$48$‘ per cent of draws do not return four peaks, ‘$47$‘ of them returning three. **(D)(6) AND ONE SYSTEMATIC THIS FILE CANNOT BOUND IS ‘$5.7\{}times$‘ ITS RESULT:** the projection kernel’s own running phase is ‘$0.0429$‘ of a comb period against the ‘$+0.00756$‘ measured here. *The reading IS differential, so ‘60‘’s identity makes the kernel common mode between the ARMS — but a data-minus-model difference is not an arm-minus-control difference, and nothing here establishes that it cancels there.* ‘$-15.45^{\{}circ}$‘ and the ‘$16.0$‘ per cent are ‘60‘’s own ‘r7236‘, now on ‘main‘ and read from it rather than relayed, INPUTS here and not recomputed; ‘60‘’s own words for the mechanism are *”it cancels only because both arms carry it”* *Bound:* **THE ROUTE REACHES THE SKY AND WHAT IT DELIVERS IS A BOUND ON THE DRIVING RATHER THAN A DETECTION.** *‘cc66.159‘’s catastrophe was a binning artefact, ‘cc66.163‘ measured that, and this places the observed spectrum on the plane the row started with. **The sky is consistent with the control’s own driving and one full unit either way is excluded at thirteen standard deviations on the better component** — conditional, as ‘(D)(4)‘ says, on the sky’s acoustic scale being the control’s to ‘$5.5$‘ per cent, which is comfortable, and NOT conditional on the ‘$0.42$‘ per cent the ‘$0.8\{}sigma$‘ agreement would need.* ⇒ *So the row should carry the exclusion and not the agreement.* **AND THE HONEST CEILING IS ‘(D)(5)‘:** *the statistic exists on ‘$52$‘ per cent of the realisations of this sky, which is the sense in which the merge sits at the locator’s own limit — not a per-point error but a probability that the fourth maximum survives.* *The four-peak alternation’s ‘$\{}pm15$‘ per cent window dependence (‘(B)(3)‘) is carried as a measured property and not corrected for; ‘(C)(2)‘ names what would fix the lever (the de-tilt control on four peaks) without proposing it.* ⇒ **AND ‘(D)(3)‘’s ‘$0.79\{}sigma$‘ IS CONDITIONAL ON ‘(D)(6)‘’s CANCELLATION: if the kernel’s phase fails to cancel in a data-minus-model difference by its full size, the displacement is ‘$5.3\{}sigma$‘ and not ‘$0.8$‘.** *The ‘$13\{}sigma$‘ scale of one driving unit is model-against-model and unaffected either way, so the EXCLUSION stands where the AGREEMENT is the conditional half.* *No fifth parameter, no further comb, no further merge: ‘r7223‘ says this is the last instrument on the row and nothing here asks for another.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_peak_plane_survives_the_merge_the_sky_lands_within_one_sigma_of_t`

**P15R297** `P15_a_drive_clock_error_cannot_move_the_spacing_so_it_is_a_running_phase_but_its_direction_on_t` [OK]

*‘r7201‘’s order on the acoustic phase, answered on this seat’s OWN oscillator and on the lap’s own two clocks — **A DRIVE-CLOCK ERROR CANNOT MOVE THE SPACING, SO IT IS A RUNNING PHASE AT FIXED SPACING AS THE ORDER SUPPOSED; BUT ITS DIRECTION ON ‘cc66‘’s PEAK PLANE IS PAIR-DEPENDENT IN SIGN, AND ALL THREE READINGS FALL INSIDE THE CONE THE DRIVING AND BARYON DIRECTIONS ALREADY SPAN, SO THE STATISTIC DOES NOT BECOME A THREE-WAY DISCRIMINATOR** *and ‘r7216‘’s ‘15.2‘ per cent is withdrawn as a magnitude: it compared two oppositely ordered ratios** (node 66’s ‘r7201‘, which orders it in three parts: *”Read the driving on each of the two clocks ‘r7216‘ separated and say what the choice does to the acoustic phase as a function of ‘$\{}ell$‘ at FIXED spacing. I want the sign first and the magnitude second... PUT IT ON ‘cc66‘’s TWO COMPONENTS BEFORE THE RUNS LAND... AND THE NEGATIVE IS WORTH THE SAME TO ME”* — with the order’s own travelling limit attached, that a pair-dependence would BE the finding). *Computes:* **17 gates, all pass, ~150 s.** ◆ **NO BANKED SPECTRUM, NO GRID, NOTHING FITTED TO THE ARM.** **(A) THE CLOCKS AT FIFTY DIGITS:** ‘r7216‘’s pieces reproduced exactly, then read the SAME way round — ‘$2.0000$‘ on the seams’ own parameter against ‘$0.4340$‘ in conformal length, a factor of ‘$4.6083$‘ and ‘78.3‘ per cent, with the two clocks disagreeing about WHICH STRETCH IS LONGER and the product ‘$0.868$‘ so it is not a reciprocal either; and THREE pairs of the same object give factors ‘$4.61$‘, ‘$3.87$‘, ‘$1.26$‘. **(B)** the oscillator’s background reported, with recombination from

the standard fitting form and the diffusion scale derived from the baryon fraction on ONE frozen calibration. **(C) THE POSITIVE CONTROL:** switching the drive off moves ‘$\{}varphi$‘ ‘$+0.0832$‘ and ‘$\{}mathrm{alt}$‘ ‘$-0.0199$‘, both signs as the control arm reads them and the ratio ‘$0.239$‘ against its ‘$0.196$‘. **(D) THE DIRECTION:** linear in the distortion to ‘5‘ per cent, ‘$\{}Delta\{}varphi/\{}lambda$‘ ‘$+0.300$‘, ‘$-0.268$‘, ‘$-0.059$‘ on the three pairs, both substitution senses giving the two arms of ONE line, and ratios ‘$0.087$‘, ‘$0.158$‘, ‘$0.168$‘ against driving ‘$0.196$‘ and baryon ‘$0.388$‘. **(E) THE SHAPE:** the drive-off control’s per-mode phase FLAT to ‘$0.03$‘ over ‘$200\{}le\{}ell\{}le1400$‘, the clock’s running ‘$5$‘–‘$6\{}times$‘ the control’s own change and concentrated below ‘$\{}ell\{}simeq500$‘, and the peak-comb offset agreeing with ‘$-\{}langle\{}Delta\{}delta\{}rangle/\{}pi$‘ from the oscillator to ‘15‘ per cent on two pairs *Bound:* **THE PRE-REGISTERED BRANCH IS REFUTED ON TWO PAIRS OF THREE AND SAID SO.** *‘r7218‘ filed that the alternation would move only at second order so the clock would lie nearly along the offset axis; that holds on the seam-bounded pair, whose ratio also drifts with the distortion, and fails on the other two, whose ratios sit within a fifth of the driving’s and do not drift.* **THE BARYON HALF OF THE CONTROL FAILS ON THE COMMON OFFSET** — *this oscillator has no finite last-scattering width, no ‘$j_\{}ell$‘ projection and one radiation fluid, so the clock result is licensed by the DRIVING control and the baryon comparison uses the ARM’s measured numbers and never this toy.* *At the amplitude the lap implies on the seam-bounded pair the perturbation is NOT perturbative — the comb breaks to three peaks, ‘cc66.156‘’s own stated failure mode — so no magnitude is claimed there and the direction comes from linear response.* *THREE pairs are measured, not every pair, so ‘the‘ clock direction is not established, only that it is pair-dependent.* *No carrier named, no paper edited, no spectrum run, and the five-peak alternation is as coarse here as the arm’s own receipt says of itself.* *Run:* `python3 receipts/P15_CR_cosmology/P15_a_drive_clock_error_cannot_move_the_spacing_so_it_is_a_running_phase_b`

**P15R298** `P15_the_runtime_tail_in_r7170_is_not_localised_to_a_call_so_the_free_reduction_lands_and_the_pr` [OK]

*‘r7205‘’s order, which named ‘r7220‘ as the work — **THE PRE-REGISTERED FIX DOES NOT HOLD AND THE REASON IS THE RESULT: THE RUNTIME TAIL IN THIS LINE’S OWN ‘r7170‘ IS NOT LOCALISED TO A CALL AT ALL. THIRTY-SEVEN RUNS WITH ANY INSTRUMENTATION ADDED PRODUCED NOT ONE BLOW-UP WHERE THE PLAIN RATE PREDICTS FIVE** ⇒ *a defect that disappears whenever it is watched is not a slow call, so no replacement of one call could have removed it* the free half lands and is reported as a ninth rather than as the fix* (node 66’s ‘r7205‘: *”‘60‘ — ‘r7220‘, fixing its own ‘r7170‘ receipt: no failing gate and a runtime that explodes about one run in six”*, with ‘r7203‘’s standing offer attached — *”if ‘r7220‘ turns out to need something from the gate — a cap, a scope declaration, a split in the runner — say which and it is mine, not a thing to work around inside your receipt”*). *Computes:* **11 gates, all pass, ˜60 s.** ◆ **NO WORKFLOW FILE TOUCHED AND NO GATE OF ‘r7170‘ ALTERED.** **(A) THE DEFECT, MEASURED:** twenty plain runs on an idle box, THREE killed at a ‘420‘ s cap and the other seventeen at ‘17‘–‘22‘ s against a registered ‘13‘ s — and every run that finished finished GREEN, so what is wrong is the runtime and not the content. **(B) NOT LOCALISED:** ‘37‘ runs across three instrumentation variants, ‘0‘ blown against ‘5.6‘ expected; and from the other side the dominant call’s worst single invocation is ‘3.3‘ s with the whole run’s symbolic time under ‘20‘ s, two orders of magnitude short of a ‘420‘ s run. **(C) THE FREE HALF:** memoising the reduction takes eleven to nine and the median run ‘18‘ s to ‘16‘ s — while the MEAN moves the other way, ‘18.2‘ to ‘20.5‘, because one ‘99‘ s excursion dominates nineteen runs of ‘15‘–‘17‘ s. **(D) LIVE CONTROLS:** the two ‘ell = 1‘ operators are the same object so the cache is sound, a perturbed operator carrying an angular term still FAILS the separation test, and a fresh run of the repaired receipt is green in ‘16‘ s. **(E)** the margin to the suite’s cap is ‘37‘ times *Bound:* **THE PRE-REGISTERED PASS CONDITION IS NOT MET AND IS REPORTED AS A FAILURE OF THE FIX.** *Thirty consecutive runs at or under budget were filed; twenty post-repair runs gave one killed at the cap and a ‘99‘ s excursion besides.* **THREE ACCOUNTS OF THE TAIL REMAIN LIVE AND THIS REVISION ELIMINATES NONE** — *an intrinsic per-run excursion, an environmental burst (two of the three blown runs were ADJACENT), and a perturbation introduced by the instrument.* *Two earlier diagnoses of this same defect were published at ‘r7218‘ and are recorded there as DISCARDED: hash-seed dependence and contention.* *And the first draft of THIS receipt quoted the mean of a partial sample as its headline and had the direction backwards; the median is quoted now with the reason

stated rather than the favourable number chosen.* ⇒ *What is left is a margin question and it is the gate's, as ‘r7203‘ offered: **the ask is a PER-RECEIPT cap a receipt can declare for itself**, so a receipt that knows it runs in ‘16‘ s is killed near that and a ‘420‘ s run becomes a reported fact instead of a silent carry. Not asked for: a looser cap, a retry, or a quarantine.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_runtime_tail_in_r7170_is_not_localised_to_a_call_so_the_free_reduc`

**P15R299** `P15_po31s_terminal_clause_counts_closures_and_the_label_is_many_to_one_so_the_count_is_four_an` [OK]

*‘PO-31‘’s live terminal clause, taken as a CHECK rather than as prose; nothing ordered, and the board puts this seat's open work on this row — **THE CLAUSE IS NOT CHECKABLE AS WRITTEN, AND FOR TWO REASONS NEITHER OF WHICH IS A MISSING CHANNEL: IT RESTS ON A COUNT WHOSE LABEL IS MANY-TO-ONE, SO THE ROW SAYS FOUR AND FIVE AT ONCE AND THE DIFFERENCE IS EXACTLY THE TRANSFER; AND ITS QUANTIFIER RANGES OVER A SET THE ROW NEVER WRITES DOWN** ⇒ *the three totality-shaped phrases the row does carry are every one of them scoped to the channels ALREADY TRIED, so the terminal half can be reached by running out of candidates rather than by a demonstration* **and the clause's FORM is the one ‘r6913‘ retired in the row's own words, ‘the row stops being a channel hunt‘, restated as a channel hunt eighty-four revisions later — two passes checked that a clause was CURRENT and neither asked whether it still described the row** (no order; ‘r7209‘’s board leaves this row to this seat, and ‘r7211‘ retired the other row this seat held, with ‘PO-75‘’s live step ordered elsewhere). *Computes:* **23 gates, all pass, ˜1 s.** ◆ **(A) every sentence relied on extracted from ‘THE_REGISTER‘ programmatically and asserted present VERBATIM, the row located by its id rather than by a line number. (B) one counting rule over the whole row returns four seven times and five once, both live text. (C) the transfer is the whole of the difference: the row's own heading says it IMPRINTS, so it is not a silent mechanism, while the row's own list counts it fifth. (D) the three totality-shaped phrases read one by one, each scoped to what was tried, with a planted unscoped sentence proving the test can tell them apart. (E) three controls on the counting rule — truncation, a synthetic single count, and a row with none. (F) the prediction reported failed. (G) the register byte-identical by digest** *Bound:* **THE PRE-REGISTERED PREDICTION FAILED AND IS REPORTED AS A FAILURE:** *‘r7224‘ predicted the enumeration would come out LARGER than four and the row would gain a named candidate. There is no enumeration to be larger than; the pre-registration's own THIRD outcome fired.* *AND ONE DRAFT OF THIS RECEIPT ASSERTED THE ROW CARRIES NO TOTALITY PHRASE AT ALL, which is false — it carries three. The claim is now the stronger one because each was read rather than counted.* *No absence is filed as a fifth closure, the four are not re-opened, and NO VERDICT on the row is offered: the repair — state the terminal half over the REQUIREMENT ‘r6913‘ computed rather than over a set of channels, and have the count name what it counts — is proposed and routed, never applied.* *Run:* `python3 receipts/P15_CR_cosmology/P15_po31s_terminal_clause_counts_closures_and_the_label_is_many_to_one_so`

**P15R300** `P15_the_enumeration_derived_from_the_construction_has_two_members_and_the_rows_own_identity_is` [OK]

*‘r7213‘, which orders nothing and asks that the enumeration be derived from the construction's own structure rather than from the history of what has been tried; the half ‘r7224‘ owed — **THE ENUMERATION HAS TWO MEMBERS AND THE ROW ALREADY CARRIES BOTH. A TILT IS A LOG-DERIVATIVE, THE OUTPUT IS A PRODUCT OF EXACTLY THREE FACTORS IN THE CONSTRUCTION'S OWN EXPRESSION, AND A PRODUCT'S LOG-DERIVATIVE IS A SUM WITH ONE TERM PER FACTOR — SO THE CHANNELS ARE THE TERMS: THE INTERIOR'S MULTIPLIER AND THE PROGENITOR'S AMPLITUDE, WITH THE MEASURE CONTRIBUTING A CONSTANT** ⇒ *‘two, and here is why there is no third‘ is a DEFINITION rather than a failure of imagination: the multiplier is what this interior multiplies an incoming amplitude by, so every action of the interior is inside it; the amplitude is the progenitor's data, so everything the progenitor supplies is inside it; a third term would have to be an action that is neither and is not the measure* **and the row's own ‘r6913‘ identity IS this decomposition with the second factor held constant, which is why it reads as one transfer slope plus a constant** (node 66's ‘r7213‘: *”derive the list from the construction's own structure rather than from the history of what has been tried”* and *”‘four, and here is why there is no fifth‘ is the claim; ‘four, and I could not think of a fifth‘ is not”*). *Computes:* **16 gates, all pass, ˜1 s.** ◆ **(A) the row's identity and its unit-amplitude clause extracted from ‘THE_REGISTER‘ and asserted present

VERBATIM. (B) the rule applied symbolically: the measure's log-derivative is the constant ‘3‘ and contributes no term, the row's own case returns ONE term, the construction in full returns TWO, and the split is verified an identity with residue ‘0‘. (C) three controls — a ‘k‘-dependent measure returns THREE, a planted fourth factor returns THREE, a product of constants returns NONE. (D) each of the four closed channels assigned to a term by which object it modifies, three inside the multiplier and one inside the amplitude, with the fifth thing the row's list names being the multiplier ITSELF. (E) the transfer term measured and cited rather than recomputed, the other term free, and the register digest-checked unchanged** *Bound:* **AND THE RULE'S FIRST DRAFT GOT THE COUNT WRONG IN THE FLATTERING DIRECTION:** *it dropped terms equal to ZERO instead of terms CONSTANT in ‘$k$‘, counted the measure's ‘3‘ as a channel and returned THREE — which is exactly the ‘fifth channel‘ this seat went looking for at ‘r7224‘, manufactured out of its own arithmetic. Fixed in one line and recorded in the function's own docstring.* *The pre-registered THIRD outcome fired alongside the prediction: on the derived enumeration the row's terminal condition is already decided by what the row carries — one term measured and running, one free and the progenitor's, which is ‘r6913‘'s requirement on the progenitor reached from the other side.* *NOT filed as a termination: ‘r7213‘ restates the row as open, no closure is filed, no fifth channel is named, and the terminal state is the gating seat's to declare.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_enumeration_derived_from_the_construction_has_two_members_and_the`

`S7_the_paper_pins_are_read_by_reversing_the_clause_they_quote_and_only_one_in_four_would_notice`
[OK]
*‘PO-78‘'s quote-pin baseline read at a pinned commit together with all eighteen corpus paper BODIES — ‘1,489‘ paper-targeted keys, ‘2.93‘ million characters of ‘.tex‘, every located site's own CLAUSE extracted and reversed four independent ways **AND RE-OPENED BY ‘r7349‘ (4)'s order:** *a published finding of this seat's has been overturned by the repair phase and this receipt was RED on ‘main‘ for it — ‘66‘ replaced the frozen constant the live arm compared against and left the gate red rather than rewriting the finding, so the question came back here: ***is the fall a result, and what does the receipt now claim?*** (* **THE BACKLOG IS READ BY REVERSAL, AND ONLY ONE IN FOUR OF THE PAPER-TARGETED PINS WOULD NOTICE: ‘324‘ OF THE ‘1,259‘ READABLE KEYS QUOTE THE POLARITY, THE QUANTITY OR THE NUMBER AND WOULD GO RED IF THEIR CLAUSE WERE TURNED ROUND.** *‘935‘ survive at least one reversal and ‘475‘ survive all four — negation, antonym, quantifier and numeral alike — each of the ‘475‘ exhibiting the reversed clause it stays green on rather than being inferred from the literal's shape.* ⇒ **It is a NEW class and not a renaming: ‘34%‘ multi-site against ‘r7201‘'s ‘MULTI‘, ‘1‘ key overlapping the ‘OPEN‘ flag ‘S3‘ read, where ‘r7228‘ measured none, ‘475 ¿ 249‘, and it runs across ‘196‘ of the ‘256‘ receipts holding such a key.** **AND THE TIER THIS RECEIPT WANTED TO LEAD WITH IS REPORTED AS A LIMIT INSTEAD** — *‘362‘ keys sit in a clause carrying exactly ONE declared auxiliary and the first draft called that ‘the clause's only assertion‘, but a real clause in ‘BH_causality_v2‘ carries one auxiliary and TWO finite verbs, ‘carries‘ being invisible to a list of auxiliaries: an auxiliary list is not a parser, and the headline never needed one.* **SUPERSEDED LIVE AT ‘r7354‘, AND BY THIS SEAT'S OWN PHASE: the reversal class has fallen ‘475‘ → ‘237‘ while the multi-site class fell ‘308‘ → ‘250‘ and the DISCRIMINATING class ROSE ‘324‘ → ‘539‘.** *So ‘it is the larger blindness‘ holds AT THE PIN and not live; what the receipt claims live is the DIRECTION — the phase is reversal-directed and the keys are changing class rather than leaving the population.*). *Computes:* **26 gates, all pass, about 14 seconds** *(the figures corrected at ‘r7238‘; wrap-tolerant location costs the extra time)*. **(A)** the instrument seeded BOTH WAYS on four planted cases before it is pointed at anything — a quoted polarity lands ‘DISCRIMINATING‘, a bare noun phrase lands ‘REVERSAL‘, a clause with nothing to move lands ‘UNFLIPPABLE‘, a quoted numeral lands ‘DISCRIMINATING‘. **(B)** ‘1,489‘ keys and ‘18‘ bodies at the pin, the three verdicts PARTITIONING the ‘1,140‘ readable and the four limit buckets accounting for the other ‘349‘. **(C)** ‘324‘ discriminating, ‘475‘ reversal, ‘460‘ partial, and all three populated so the instrument separates rather than flags. **(D)** every one of the ‘442‘ carrying a reversed clause that CHANGED and still contains the literal. **(E)** the overlaps measured, not assumed. **(F)** ‘98‘ ‘ABSENT‘, ‘60‘ ‘UNFLIPPABLE‘, ‘18‘ quoting a reference key, ‘54‘ ‘SATURATED‘, each excluded from every verdict, plus the live state as a DISJUNCTION rather than a value and the pinned baseline digest-checked unchanged after the run. **‘r7354‘, on ‘r7349‘ (4)'s order, adds four:** *the pinned ordering asserted on BOTH counts — the other gate's ‘249‘ over the pinned key set AND this instrument's own ‘308‘ on the same tree; the live arm re-stated as the DIREC-

TION, the reversal class falling at least as fast as the multi-site one, which survives the phase retiring keys from either; a second live arm that the DISCRIMINATING class grew by at least half what the reversal class lost, so the fall is shown to be a change of class and not attrition; and the decomposition printed, retired against minted, for both classes.* *Bound:* ***NOT a verdict on any receipt and NOT a repair*** — *the ‘442‘ are a REPAIR LIST routed in the channel, and adjudicating a baseline key belongs to that baseline’s owner.* *And ‘DISCRIMINATING‘ does not mean a pin is GOOD: it means the pin would notice one of four declared reversals, which is a floor and not a certificate.* *The ‘218‘ ‘ABSENT‘ keys are not asserted broken — a needle built at run time or a regex is simply not readable by substring, and that is stated rather than estimated.* *‘r7354‘’s live numbers are one re-measurement of a moving corpus and the receipt says so: the direction is asserted, the values are printed.* *Why the multi-site class RESISTS is not settled — the form that would single-site such a pin is the locus-scoped one and the instrument cannot key that form yet, so the resistance may be the rule’s keyable domain rather than the class.* *Run:* `python3 receipts/L_probability/S7_the_paper_pins_are_read_by_reversing_the_clause_they_quote_and_only_one_i`

`S8_the_source_half_is_the_blinder_one_and_the_class_is_mostly_a_receipt_pinning_a_ledger_not_another_rec`

[OK]

*‘PO-78‘’s quote-pin baseline at a pinned commit and the ‘1,018‘-file receipt tree it points into — ‘13.3‘ million characters of Python source — plus all eighteen paper bodies re-read so the half ‘r7228‘ published can be re-derived rather than quoted* ( **THE SOURCE HALF OF THE REVERSAL CLASS IS THE BLINDER ONE: ‘65‘ OF ‘353‘ READABLE KEYS — ‘18.4%‘ — WOULD GO RED IF THE CLAUSE THEY QUOTE WERE TURNED ROUND, AGAINST THE PAPER HALF’S ‘24.5%‘, ‘z = 2.77‘, ‘p = 0.0057‘.** *So ‘one in four would notice‘ is a fact about PAPERS and not about pinning, which is the pre-registered prediction holding and the ‘indistinguishable‘ branch — the one that would have made ‘r7228‘’s framing wrong — failing to fire.* ⇒ **AND THE LARGEST EXCLUSION IS AN ANSWER THAT RENAMES THE CLASS:** *‘454‘ keys are absent from every receipt but their own pinner, and in a ‘120‘-key sample ‘72‘ are found in the project’s own registers or instrument modules, ‘56‘ in a register against ‘28‘ in a module —* ***so this half is mostly a receipt pinning a LEDGER, and the baseline’s ‘SOURCE‘ label means only that the trace did not name a ‘.tex‘.***). *Computes:* **17 gates, all pass, about 13 seconds.** **(A)** ‘r7228‘’s transforms, clause extractor, site cap, markup pattern and ‘classify‘ SLICED from its pinned source and executed, digest-gated, so a difference between the halves cannot be a difference between instruments. **(B)** both populations at the pin, ‘1,489‘ and ‘1,154‘, and the receipt tree asserted to differ from the pin by this file alone. **(C)** the paper half re-derived UNCHANGED — ‘279‘ of ‘1,140‘, ‘442‘ surviving all four. **(D)** the source half under two declared filters, with the byte locator’s equivalence to ‘classify‘ measured identical on all eight shared buckets of a ‘60‘-key sample and the self-site filter shown non-cosmetic (‘2‘ → ‘27‘ absent). **(E)** the comparison decided by a two-proportion test rather than by eye, and the third outcome shown not to have fired. **(F)** the exclusions counted and the ‘ABSENT‘ bucket sampled to find out what it is. *Bound:* ***NOT a verdict on any baseline key and NOT a repair*** — *adjudicating a quote-pin belongs to that baseline’s owner, as ‘r7228‘ said and ‘r7215‘ confirmed.* *And ‘DISCRIMINATING‘ is still a floor rather than a certificate: it means the pin would notice one of four declared reversals.* *The ‘60%‘ figure is a ‘120‘-key SAMPLE of the ‘469‘, reported as a sample; and the prose rule calls a gate label CODE, which is stated as its limit rather than hidden.* *Run:* `python3 receipts/L_probability/S8_the_source_half_is_the_blinder_one_and_the_class_is_mostly_a_receipt_pinn`

`S9_the_exposure_is_zero_so_the_count_judgement_was_right_and_the_class_was_already_registered_before_it`

[OK]

*the ‘131‘ receipts that name a root register as a file they read, found from their own sources rather than from a list; each one’s assertion CONDITIONS evaluated against the register before and after the two edits ‘r7215‘ actually made; plus ‘r7224‘ at its pre-repair commit against the register at its pre-strike commit, as the instrument’s recall control* ( **THE EXPOSURE IS ZERO, SO THE COUNT JUDGEMENT WAS RIGHT — AND THE CLASS WAS ALREADY ON THE RECORD, WHICH IS THE HALF THAT WAS NOT.** *‘r7232‘ pre-registered ‘several, not one‘ and the answer is NONE: of ‘102‘ receipts carrying an evaluable condition, not one loses an assertion when a row it names is struck or its clause amended. The pass condition is not met and the branch that fired is the second one named in advance — ‘in which case r7217’s judgement is right and I will say so in those words‘.* ⇒ **And the instrument is NOT cleared on that zero, because

the same branch said a zero might mean the test is blind:** *pointed at ‘r7224‘ before its repair against the register before the strike, it catches the condition that actually broke, under ‘AMEND‘ — the right transform, since the strike MOVED the row’s clause while only WRAPPING its id — and ‘r7224‘ as repaired is immune, which was the pre-registered positive control.* ⇒ **‘O1‘’s own account names SIX registered ids of this class and lists ‘r7224‘’s disguise explicitly as ‘a live register read negatively‘,** *so ‘if it happens a second time it is a member‘ rests on a first that was not one; and the pin-plus-disjunction repair this seat reached was already prescribed there as ‘pinned to the parent and carries the opposite live assertion‘.*). *Computes:* **29 gates, all pass, about 2 seconds.** **(A)** five planted conditions, each landing in the bucket named for it, plus the instrument shown to be at the level of the ASSERTION and not the literal — a disjunction survives while one arm stands, which is what keeps ‘r7224‘’s own repair from reading as exposed — and an unreadable condition returning ‘UNKNOWN‘ rather than a guess. **(B)** the population found from the sources, ‘131‘ of them, and the transforms shown to reach only ‘8‘ of the ‘34‘ registers named. **(C)** the buckets partitioning all ‘131‘, with ‘102‘ immune and the exposed set empty. **(D)** the recall control on the case that happened. **(E)** the prior art read out of ‘O1‘’s docstring in its own words. **(F)** the prediction reported AS failed. *Bound:* ***A zero is NOT a clean bill*** — *‘26‘ receipts read a register the transforms cannot reach and ‘3‘ carry no readable condition, and none of the ‘29‘ is counted clean.* *A condition built at run time, a regex pin, or a literal assembled from parts is invisible to a static read, and that is stated rather than estimated.* *Nothing is filed and nothing is gated: ‘r7217‘ declined both and this supports the decision rather than reopening it. The novelty correction is a reading of an existing record, not a new class.* *Run:* `python3 receipts/L_probability/S9_the_exposure_is_zero_so_the_count_judgement_was_right_and_the_class_was_al`

**P15R301** `P15_the_spacing_drift_is_made_in_the_projection_so_the_construction_determines_it_and_the_hand`
[OK]

*the acoustic comb’s own source extrema in ‘$k$‘ at recombination on four runs of the two-arm instrument — driven, undriven, and the handover epoch moved a DECADE either side — plus the instrument’s saved term-by-term source on a UNIFORM ‘$k$‘ grid at half the ladder’s quantum, the banked arm and control spectra, and one configuration banked BOTH ways so the ladder can be compared with a continuum* ( **THE PEAK-SPACING DRIFT IS A QUANTITY THE CONSTRUCTION DETERMINES, AND IT IS MADE ENTIRELY IN THE PROJECTION: THE ACOUSTIC PHASE CARRIES NO DRIFT AT ALL.** *The source comb in ‘$k$‘ lands on the integers undriven to ‘$0.015$‘ and, driven, is displaced by a nearly CONSTANT ‘$-0.12$‘ in ‘$q=k r_s/\{}pi$‘ — a spread of ‘$0.029$‘ over six peaks — so its own drift is ‘$+0.03$‘ to ‘$+0.56$‘ against a measured ‘$+4.38$‘, inside its own rms of zero.* ⇒ **A PERFECT comb, extrema exactly on the integers with no dynamics behind them, acquires the whole drift on projection: gaps ‘$278.5\{};289.5\{};293.4\{};295.4$‘, rising, decelerating and saturating under ‘$\{}ell_A$‘.** ***And it is DERIVED rather than measured: replacing ‘$j_\{}ell(kD)^2$‘ by its own asymptotic average ‘$1/(x\{}sqrt{x^2-\{}ell^2})$‘ reproduces the full Bessel sum peak by peak to under ‘$0.5$‘ in ‘$\{}ell$‘ and the drift to ‘$0.01$‘, so the drift is the centroid shift of the kernel’s own window acting on a damped comb and its only inputs are ‘$r_s$‘, ‘$D_M$‘ and ‘$r_D$‘.*** **THE HANDOVER — the one place ‘r7221‘ named as this construction’s rather than anybody’s — SUPPLIES NONE OF IT:** *a DECADE of handover epoch moves the comb ‘$0.30$‘ in ‘$\{}ell$‘ and the drift ‘$0.002$‘, which is ‘Prop. subhorizon‘’s ‘is an amplitude and not a phase‘ measured instead of quoted.*). *Computes:* **31 gates, all pass, about 34 seconds.** **(A)** the undriven guard — every source extremum an integer to ‘$0.015$‘ — read off a run that carries ‘NODRIVE=1‘ rather than inferred. **(B)** the driven comb’s offset shown CONSTANT on a uniform grid at half the ladder’s quantum, so the null is not the ladder’s resolution talking. **(C)** a perfect comb projected two independent ways, the closed-form window against the full Bessel sum, agreeing peak by peak and on the drift, and the instrument’s own saved source re-projected to the same number. **(D)** the handover lever, and the paper’s clause pinned for the historical state and read live as a DISJUNCTION over the states a rewording may produce. **(E)** ladder against continuum on one configuration, and the derived drift under a doubled ‘$k$‘ sampling and a halved ‘$\{}ell$‘ step. **(F)** arm against control on the drift, on every peak position, and with the driving removed entirely. **(G)** the pre-registered prediction scored, the two FAILED parts asserted AS failures. **(H)** what is not settled, gated as a fraction rather than as a whole. *Bound:* ***The closed form reaches ‘$+2.61$‘ against the reporting path’s ‘$+4.38$‘: ‘$60$‘ per cent of the SIZE is derived here and the remainder sits in the multipole hierarchy’s damping, which the closed form carries as one Gaussian.*** *Five peaks against four comb

parameters leave ONE degree of freedom, so ‘$D$‘ and the alternation are not independently well determined — which is why every comparison here is like with like rather than a single coefficient quoted as a measurement.* *The control's SOURCE comb is not measured: both attempts were killed for memory at ‘$7$‘ GB a process, and the control's PROJECTED spectrum stands in for it.* *The ladder-against-continuum pair is a SUPERSEDED configuration whose comb rides the stacking ruler; it is used for the sampling comparison alone and for nothing else.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_spacing_drift_is_made_in_the_projection_so_the_construction_deter`

**P15R302** `P15_the_kernels_phase_difference_between_the_arms_is_four_orders_below_the_residual_so_the_pro` [OK]

*the three lengths each arm's own run reports — ‘$r_s$‘, ‘$D_M$‘ and the diffusion scale at the visibility peak — read off the banked spectra and their run logs rather than typed, and the projection kernel evaluated on them by two independent routes: ‘r7234‘'s closed-form asymptotic window read as a COMPLEX integral, and the full spherical-Bessel sum with no asymptotic average anywhere in it* ( **THE KERNEL'S PHASE DIFFERENCE BETWEEN THE ARMS IS ‘$-0.0141^{\{}circ}$‘ ACROSS ‘$104\{}le\{}ell\{}le1886$‘ AGAINST A RESIDUAL OF ‘$-96.6^{\{}circ}$‘ — ‘$3.84$‘ ORDERS OF MAGNITUDE BELOW — SO THE KERNEL IS NOT A CARRIER AND THE PROJECTION IS CLEARED.** *The phase is a function of exactly TWO dimensionless numbers, verified by scaling all three lengths together for the same phase to ‘$10^{-10}$‘ degrees: ‘$r_D/D_M$‘, which the arms differ in by ‘$0.27$‘ per cent and which carries ‘$-0.0139^{\{}circ}$‘ of the difference, and ‘$r_s/D_M = \{}theta_*/\{}pi$‘, which they share to ‘$1.5\{}times10^{-5}$‘ because the angle is what the construction computes and which carries ‘$-0.0001^{\{}circ}$‘.* ⇒ **The closed-form window and the full Bessel sum agree on the difference to five decimal places, ‘$-0.01406$‘ against ‘$-0.01407$‘.** ***AND THE UNDER-SAMPLED BESSEL SUM MANUFACTURES A DIFFERENCE ‘$74$‘ TIMES THE TRUE ONE:*** *at ‘$1.8$‘ points per Bessel period it reads ‘$-1.05^{\{}circ}$‘, at ‘$4.6$‘ and above ‘$-0.01407^{\{}circ}$‘ through a sixfold further refinement, while the window route is immune because it averages the kernel analytically — the instrument's own documented aliasing failure arriving in the PHASE rather than in the peaks.* **And the pre-registered THIRD OUTCOME holds alongside ‘cleared‘:** *the kernel's own ABSOLUTE running phase is ‘$-15.46^{\{}circ}$‘, ‘$16.0$‘ per cent of the residual and common mode to ‘$0.09$‘ per cent, so ‘a running phase at correct spacing‘ carries a projection-manufactured floor of a sixth of the scale the row reads its candidates against.*). *Computes:* **24 gates, all pass, about 20 seconds.** **(A)** the identity first — scale invariance under a common rescaling of all three lengths, and the two arms' separation into a shared ratio and a differing one. **(B)** the two routes against each other on each arm and on the difference, each shown converged in its own parameter. **(C)** the aliasing threshold located rather than asserted, with the inflated value, the stable value and the window route's immunity all gated. **(D)** the answer in the one form the order asked for, plus the two-number decomposition the identity predicted. **(E)** the prediction scored, with the FAILED sign asserted as a failure and its root cause measured. **(F)** the third outcome, pre-registered and gated. **(G)** the ‘$r_D$‘ convention declared, its sensitivity measured, and the difference re-run at a shifted convention to show it cancels. *Bound:* ***The phase here is the KERNEL's, not the full transfer's*** — *the driving and the baryon loading remain the row's standing candidates and nothing in this receipt touches them.* *‘$r_D$‘ is read at the visibility peak because that is the figure the runs report, and the diffusion length is itself ‘$\{}ell$‘-dependent, so a single value is a CHOICE; what defends the answer is that it is a difference of two arms at the same choice, re-run at a shifted one and unchanged.* *The residual's ‘$-96.6^{\{}circ}$‘ is quoted from ‘sec:refit-bound‘ and not re-measured here.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_kernels_phase_difference_between_the_arms_is_four_orders_below_th`

**P15R303** `P15_the_wrap_fix_is_applied_and_agrees_key_for_key_and_the_exposed_convention_is_the_window_no` [OK]

*‘70‘'s banked wrap-tolerant split from ‘r7223‘, diffed KEY FOR KEY against an independent implementation of the same rule in this seat's own ‘S7‘ and ‘S8‘; plus the diffusion length at BOTH endpoint conventions from ‘cc66‘'s own standalone integration re-run at the one-clock grid's parameters, and ‘r7236‘'s kernel-phase floor re-evaluated at both conventions and over six ‘$\{}ell$‘-windows* ( **ITEM (1): SIX ROWS, SIX EXACT AGREEMENTS — AND ‘70‘'s BANK CAUGHT A BUG IN THIS SEAT'S OWN PREFILTER FIRST.** *‘ABSENT‘ ‘218‘→‘98‘, ‘REVERSAL‘ ‘442‘→‘475‘, ‘DISCRIMINATING‘ ‘279‘→‘324‘ on the paper half; source ‘ABSENT‘ ‘469‘→‘454‘, ‘CODE‘ ‘218‘→‘202‘, ‘REVERSAL+PARTIAL‘ ‘270‘→‘288‘ — every one ‘70‘'s num-

ber to the digit, with the headline strengthening to ‘25.7‘ per cent against ‘18.4‘ and ‘z = 2.84‘.* *It did NOT agree at first: two ‘’3 nu’‘ keys split the other way because the byte prefilter fell back to the RAW literal whenever no token reached four characters, silently reintroducing the blindness for short-token keys. **A total that agreed would have hidden it; the key-by-key diff is what found it.*** **And the shared rule has a FALSE POSITIVE neither implementation named:** *widening a literal’s spaces to ‘$\{}backslash$s+‘ lets ‘’3 nu’‘ match ‘2K_3‘ and ‘null‘ across a line break, and that spurious site is what carries those two keys.* ⇒ **ITEM (2): THE ENDPOINT MOVES THE FLOOR ‘$0.085$‘ PERCENTAGE POINTS, ‘$16.01$‘ TO ‘$15.92$‘, SO ‘a sixth‘ IS SAFE — AND THE PRE-REGISTERED THIRD OUTCOME FIRES BY A FACTOR OF ‘$177$‘: THE ‘$\{}ell$‘-WINDOW MOVES IT ‘$15$‘ POINTS, FROM ‘$9.11$‘ PER CENT TO ‘$24.13$‘.** ***AND THE ARM-CONTROL DIFFERENCE IS NOT CONVENTION-SAFE, WHERE ‘r7225‘ SAID IT WAS:*** *the two arms’ ‘$r_D$‘ move in OPPOSITE directions, ‘$+1.59$‘ per cent against ‘$-1.51$‘, so their separation in ‘$r_D/D_M$‘ goes ‘$0.27$‘ to ‘$2.79$‘ per cent and the kernel phase difference goes ‘$-0.0141^{\{}circ}$‘ to ‘$+0.1448^{\{}circ}$‘ — a factor of ten and a change of SIGN, with ‘$1.59+1.51=3.10$‘ reproducing ‘cc66‘’s straddle-zero figure on a different configuration. ‘r7236‘’s clearance survives at ‘$2.82$‘ orders, and its reported sign was never convention-independent.*). *Computes:* **29 gates, all pass, about 39 seconds.** **(A)** ‘S7‘ and ‘S8‘ re-run as subprocesses rather than reimplemented, their counts read off their own output, and all six of ‘70‘’s rows checked to the digit. **(B)** the key-by-key diff against ‘70‘’s bank, with the prefilter’s removed length floor and the uncollapsed haystack both asserted from the source. **(C)** the false positive EXHIBITED on its own literal rather than described. **(D)** the two pins, and the population shown to be the one ‘r7230‘ measured. **(E)** the floor at both conventions, reproducing ‘r7236‘’s published figure at the shipped one. **(F)** the difference’s convention dependence, its factor of ten, its sign change and the surviving clearance. **(G)** the third outcome, with the prediction’s band held and its central value gated AS high by two. **(H)** what is not settled. *Bound:* ***The endpoint swings are applied as a RATIO to the instrument’s own reported ‘$r_D$‘*** — *the standalone integration and the instrument’s hierarchy-derived ‘$r_D$‘ differ in absolute value, a gap ‘cc66‘’s own receipt documents, and the convention question is about the ratio.* *The false positive is exhibited on one literal and NOT counted across the backlog, so how much of the corrected ‘$120$‘ rests on cross-token matches is unmeasured.* *Nothing here adjudicates a baseline key: the corrected buckets are a measurement and the verdicts stay the baseline owner’s.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_wrap_fix_is_applied_and_agrees_key_for_key_and_the_exposed_convent`

`S10_the_sweep_for_exact_counts_could_not_contain_its_own_gate_and_the_class_doubled_in_the_receipts_wri`
[OK]

*no paper section — this seat’s own ‘312‘ receipts at the trunk against the ‘284‘ ‘S4‘ swept, the four exact-count sites on the quote-pin buckets as ‘r7227‘ found them, ‘S4‘’s own detector lifted from its pinned blob, and the shared pin ledger; every exact number read at a pinned commit and every live population gated monotone* ( **‘r7227‘ ITEM (1), AND THE ANSWER IS ABOUT THE SWEEP AND NOT THE CLASS: ‘S4‘ SWEPT THIS SEAT’S WORK FOR ‘an exact count asserted against a set other seats can grow‘, REPORTED ONE LIVE INSTANCE, AND THREE OF THIS SEAT’S RECEIPTS THEN WENT RED ON AN EXACT ‘2167‘.** ***TWO OF THE THREE WERE INVISIBLE TO ‘S4‘’s OWN DETECTOR, ‘S4‘’s OWN GATE AMONG THEM*** — *‘_unadj == 2167‘ is ‘Name == Constant‘, and ‘flagged()‘ requires ‘len‘/‘sum‘/‘count‘ INSIDE the comparison, so the instrument built to enumerate the class is blind to the form its own gate is written in.* **AND THE THIRD, WHICH THE DETECTOR COULD SEE, GOT THROUGH FOR A REASON NO DETECTOR REPAIR REACHES: THE SWEEP IS PINNED TO THE COMMIT ITS OWN REVISION STARTED FROM, SO IT CANNOT CONTAIN ITS OWN REVISION’S GATES AND CANNOT SEE ONE RECEIPT WRITTEN AFTER IT** — *measured: the pin is strictly earlier than the commit introducing ‘S4‘, the class goes ‘30‘ claim-sites to ‘62‘, and the ‘28‘ receipts added since carry ‘37‘ of them and ‘5‘ of the ‘6‘ EXPOSED sites at the trunk.* ***So a snapshot is not a guard, and the repair is a standing check rather than a bigger sweep*** — routed, not applied, because the gate layer is not this seat’s. **A FOURTH DEFECT IS THE SAME FUNCTION BOUNDARY READ THE OTHER WAY: ‘ground()‘ misses a pin passed as a CALL ARGUMENT, so three of ‘S7‘’s FROZEN sites were filed as live** — *one blindness costing real sites in stage one and inventing false ones in stage two, and the second is the worse half.* ***AND THE SIX EXPOSED SITES ARE READ RATHER THAN COUNTED: two are deliberate zeros carrying the positive control ‘S5‘’s ruling requires, one is already stamp-scoped

by ‘r7204‘’s prescription, TWO ARE FALSE POSITIVES THIS REVISION’S OWN REPAIR MANUFACTURED AND REPORTS, and ONE IS GENUINE AND IS REPAIRED HERE*** — ‘r7238‘’s gate counted a token exactly twice in another receipt’s LIVE source, and ‘r7227‘ is the cycle that proved another seat edits this seat’s receipts when the gate requires it. *One pre-registered prediction is refuted and printed: ‘S3‘’s site needed the name-bound repair and not the function-boundary one, and no site in the ‘2167‘ class needs the function repair at all.* ⇒ **AND THE LOOSE NUMBER ‘r7227‘ ROUTED IS A COLLISION AND NOT A LOSS: ‘108‘ ‘EXTENDED‘ rows, ‘107‘ distinct keys, two different source literals in ONE receipt extended by ‘D=6‘ and ‘D=2‘ onto the SAME string** — *internal to the extension, not an interaction with this seat’s wrap correction as the order conjectured, the only duplicate key in ‘2,776‘ rows, and nothing in the corpus asserts that the ledger’s keys are unique.*). *Computes:* ‘30‘ gates. (1) the four sites at ‘9ce054baˆ‘ with what each detector saw: ‘S4‘’s saw ‘1‘, the corrected one ‘3‘, and the fourth is arithmetic on two int literals and correctly out of class. (2) the pinned-sweep arithmetic from ‘git‘ rather than from reading. (3) the three form repairs measured by removing each alone — function boundary ‘9‘, name-bound count ‘6‘, loop target ‘5‘, overlapping, and with all three off the detector reproduces ‘S4‘’s line for line. (4) stage two’s three re-filings. (5) the six EXPOSED sites adjudicated one at a time. (6) the ledger collision traced to its two pre-extension literals, both present as separate rows before the batch. (7) and the instrument run on THIS receipt: ‘14‘ claim-sites, all FROZEN, ‘0‘ EXPOSED *Bound:* ***THE SWEEP IS MEASURED AND THE GUARD IS NOT BUILT*** — *the standing check that would police the class on every push belongs in ‘scripts/lint_assertions.py‘, which is not this seat’s to edit, so this receipt leaves the defect it measures live and routes the patch.* *The detector stays AST-only: a count reached through a subprocess or assembled in a formatted string is outside it, and ‘SHARED‘ is a hand-written list, so an artefact absent from it is invisible by construction.* *Precision is priced rather than claimed — the function repair gains ‘9‘ sites and manufactures ‘2‘ of the ‘6‘ EXPOSED, a third of that bucket.* *The class is ‘S4‘’s and is not re-ruled here, the population is ‘S4‘’s two directories so that the counts compare, and ‘S4‘’s honest bound already names a NEIGHBOURING member of the stage-two limit; what this adds is the measured cost of a limit its predecessor stated.* *Run:* `python3 receipts/L_probability/S10_the_sweep_for_exact_counts_could_not_contain_its_own_gate_and_the_class_c`

**P15R304** `P15_the_selection_does_not_truncate_so_the_thirteen_sigma_stands_at_twelve_point_nine_and_the_` [OK]

*‘r7229‘’s one order, the audit this seat routed at ‘r7227‘ — **THE SELECTION DOES NOT TRUNCATE, SO THE ‘$13\{}sigma$‘ STANDS AT ‘$12.9$‘ — AND THE PEAK THAT GOES MISSING IS THE THIRD, NOT THE FOURTH.** ‘cc66.164‘’s ‘$\{}sigma(\{}varphi)$‘ was measured on the ‘$52$‘ per cent of draws that located four peaks; on a handle that covers ‘$99.6$‘ per cent of the ensemble that width is ‘$0.982$‘ of the unselected one, bootstrap ‘$[0.953,1.010]$‘, so one driving unit is ‘$12.90\{}sigma$‘ against the ‘$13.14$‘ printed and the sky’s ‘$0.79$‘ is ‘$0.78$‘. **But the route this seat routed to 66 does not exist:** the slot that goes ‘nan‘ is ‘$n=3$‘ in ‘$3460$‘ of ‘$3791$‘ three-of-four draws, and ‘$n=4$‘ is located in ‘$95.9$‘ per cent — so ‘cc66.164‘’s ‘(D)(5)‘ and both of this seat’s replies named the wrong peak* (node 66’s ‘r7229‘ — *”compare the width of the four-peak selection against the width over all draws on the three-peak statistic, and state whether the selection truncates”*, with this seat’s own ‘r7221‘ burden adopted as it offered it — *”state the expected ratio of the two widths before reading them”* — and 66’s one addition: *”if the selection does truncate, say what the ‘$13$‘ becomes. A corrected number is the deliverable”*, with the closure named in both directions: *”if the two widths agree, the ‘$13\{}sigma$‘ stands as printed and the ‘$52$‘ per cent goes back to being an existence bound alone”*). *Computes:* **8 checks, all pass, 18s measured.** ◆ **NO NEW SPECTRUM, NO FIT, NO REFIT, NO NEW INSTRUMENT.** **(A)(1) THE AUDITED NUMBER IS REBUILT TO ‘cc66.164‘’s PUBLISHED DIGITS BEFORE IT IS QUESTIONED** — same seed, same ‘$2000$‘ draws, same covariance through the same merge matrix: robust ‘$\{}sigma(\{}varphi)=0.00955$‘, ‘$13.1\{}sigma$‘ for one driving unit, ‘$0.79\{}sigma$‘ for the sky, ‘$52$‘ per cent locating four peaks. **(B)(1) AND THE FIRST THING THE AUDIT FINDS IS THAT THE ROUTE IT ROUTED DOES NOT EXIST: THE PEAK THAT GOES MISSING IS THE THIRD.** Per-slot location rates over ‘$8000$‘ draws are ‘$100$‘/‘$99.6$‘/‘$56.7$‘/‘$95.9$‘ per cent, and among the ‘$3791$‘ draws locating exactly three of four the missing slot is ‘$n=3$‘ in ‘$3460$‘ and ‘$n=4$‘ in ‘$314$‘. *‘$47$‘ per cent do return three of four — but not the three this seat told 66 they were, so the three-peak statistic covers ‘$56.3$‘ per cent and not ‘$99$‘.* **(B)(2) THE HANDLE THAT DOES COVER THE ENSEMBLE IS THE OFFSET ON PEAKS

‘$1$‘–‘$2$‘, DEFINED ON ‘$99.6$‘ PER CENT**, read off the SAME located series so it differs from the audited statistic only in how many peaks the average runs over. **(C)(1) AND ON IT THE SELECTION DOES NOT TRUNCATE: ROBUST WIDTH ‘$0.982$‘ OF THE UNSELECTED ONE, BOOTSTRAP ‘$[0.953,1.010]$‘, CLEAR OF THE TEN PER CENT THRESHOLD FIXED IN ADVANCE.** *Not neutral either — the discarded draws are ‘$1.04\{}times$‘ wider — and where it IS informative is the ‘$314$‘ draws that lose the fourth peak specifically, which are ‘$1.48\{}times$‘ wider and ‘$4$‘ per cent of the ensemble.* **(C)(2) THE ‘r7221‘ PREDICTION IS SCORED BOTH WAYS:** ‘$0.982$‘ lands in the pinned ‘$[0.80,1.00]$‘ and in the narrower ‘$[0.90,1.00]$‘, the tail half holds at ‘$8.1\{}times$‘ — **and the mechanism argued from was the wrong peak**, since the lost peak is one the audited offset averages over, so *the number landed in the band and the reason given for it did not*. **(D)(1) A SECOND HANDLE WITH NO SCALE ESTIMATE IN IT AGREES:** the loss rate by quintile of ‘$\{}lvert\{}varphi-\{}mathrm{med}\{}rvert$‘ is ‘$48.2$‘/‘$46.2$‘/‘$45.1$‘/‘$49.9$‘/‘$47.6$‘ per cent on a base of ‘$47.4$‘, ‘$4.8$‘ points of spread and ‘$-0.6$‘ from ‘$Q1$‘ to ‘$Q5$‘ — *a truncating selection would climb.* **(E)(1) SO THE DELIVERABLE: ONE DRIVING UNIT IS ‘$12.90\{}sigma$‘ AGAINST ‘$13.14$‘ PRINTED AND THE SKY IS ‘$0.78\{}sigma$‘ AGAINST ‘$0.79$‘**, carrying the measured ratio across to the four-peak width, *which is the only route since that statistic does not exist off its own selection*. **(E)(2) and the alternation carries the same verdict at ‘$1.013$‘**, reported as the weaker handle because on two peaks it has a single sign pair *Bound:* **‘r7229‘’s FIRST BRANCH FIRES AND THE PAPER’S FIGURE DOES NOT CHANGE.** *The ‘$13\{}sigma$‘ exclusion was not inflated by the selection: the ‘$52$‘ per cent costs efficiency and not width, so it goes back to being a bound on the statistic’s EXISTENCE alone.* ⇒ **AND THE ONE CORRECTION THIS AUDIT DOES DELIVER IS A NAME, NOT A NUMBER: ‘cc66.164‘’s ‘(D)(5)‘, this seat’s ‘r7227‘ reply and ‘r7229‘’s own restatement all call the ‘$52$‘ per cent ‘the fourth peak’s existence‘, and the fourth peak is located in ‘$96$‘ per cent of draws.** *It is the third that goes, and the paper’s sentence should say so.* *The audit’s own premise was the casualty: ‘$47$‘ per cent returning three of four was read as ‘$47$‘ per cent returning peaks ‘$1$‘–‘$3$‘, and a count of finite slots is not a statement about which slots.* **WHAT IT DOES NOT SETTLE is the boundary ‘r7227‘ named in advance:** *whether the locator’s ‘$47$‘ per cent failure rate is itself right is a property of ‘plik_lite‘’s binning at this merge and this noise, and nothing here measures the locator against a different instrument.* *‘cc66.164‘’s ‘(D)(6)‘ is untouched — the projection kernel’s running phase is ‘$5.7\{}times$‘ the sky’s displacement — so the ‘$0.78\{}sigma$‘ remains conditional on that cancellation exactly as the ‘$0.79$‘ was.* *Run:* `python3 receipts/P15_CR_cosmology/P15_the_selection_does_not_truncate_so_the_thirteen_sigma_stands`

`S11_a_sweep_can_contain_its_own_gates_by_scoping_the_assertion_to_its_own_push_and_the_population_it_wou`

[OK]

*no paper section — the live tree’s own claim-sites of the exact-count class, each one’s LAST WRITER from ‘git blame‘, the parity of the revision that introduced each receipt in the path scope, the three blobs the gating seat edited at ‘r7227‘ before and after, and a receipt-shaped CONTROL carrying one genuine site of the class* ( **‘r7229‘ ANSWERED AND THE FORMULATION IS ONE LINE: KEEP THE POPULATION LIVE AND RESTRICT THE ASSERTION TO THE CLAIM-SITES WHOSE LAST WRITER IS IN THIS PUSH’S OWN RANGE.** *Self-including by construction — shown on a control that is INSIDE the asserted set when its line is this push’s and OUTSIDE it when the same line is attributed to another seat’s commit — monotone, since not one site it asserts on was last written by a revision of the gating seat’s parity, and immune to a base merge, because ‘blame‘ attributes a merged line to the commit that WROTE it.* ***AND THE PRE-REGISTERED COVERAGE BAND HOLDS: ‘2‘–‘12‘ of the live claim-sites with a central guess of ‘6‘, measured ‘7‘ of ‘83‘.*** *But the first draft of this receipt measured that coverage as ZERO, twice over and for two reasons of the same shape — **‘blame‘ marks an uncommitted LINE with an all-zero sha and ‘ls-files‘ omits an untracked FILE**, so the sweep could not see its own receipt in the state it was actually run in. Both are repaired and both are gated, and the second was caught by CI rather than by this seat.* ⇒ *And the coverage of a self-including sweep is set by the DETECTOR and not by the scope rule — here the detector can see this revision’s counts only because the control quotes the shared ledger’s path, and a revision reading its population through ‘git‘ alone would have scored zero.* **SO THE ANSWER HAS BOTH HALVES: IT EXISTS, IT IS SAFE, AND IT IS A PER-PUSH GATE RATHER THAN A SWEEP** — *retrospective coverage measured at exactly zero, so the standing check routed at ‘r7240‘ is the only coverage the class already standing can ever have, which is ‘r7229‘’s

own closure read in the stronger direction.* ***AND THE POPULATION ‘S4‘ DEFINED AND ‘r7240‘ INHERITED IS MOSTLY ANOTHER SEAT’S WORK: ‘116‘ of the receipts in it were introduced by a revision of this line’s parity against ‘180‘ of the gating seat’s.*** *The directory is SHARED, four of its claim-sites are last written by other seats’ revisions right now — named in the receipt — so a path-scoped sweep already asserts on another seat’s work, and the thing ‘r7229‘ contrasted against is the present state of one.* *That is a correction to this seat’s own previous revision and not to the order: ‘r7240‘’s denominators are right about the set they measured and wrong about whose set it is, and the two findings they carry — the detector blindness and the pinned sweep — are properties of the instrument and do not move.* ⇒ **AND THE PRE-REGISTERED THIRD OUTCOME FIRES: the gating seat’s edits REMOVED claim-sites and added none, ‘31‘ to ‘29‘ with the exposed bucket ‘3‘ to ‘2‘** — *the prescribed repair turns ‘==‘ into ‘¡=‘, which deletes a site rather than creating one, so path-scoping is unsafe in principle and safe in the only direction the record exhibits.*). *Computes:* ‘22‘ gates. (1) the rule, built and run: live population, assertion scoped to the push range with the WORKING TREE counted, since ‘blame‘ marks an uncommitted line with an all-zero sha and a range test alone would have passed locally by vacuity and engaged only in CI. (2) self-inclusion and exclusion both shown on the control, whose own parse and site count are gated first so it cannot silently measure zero. (3) the coverage scored against the pre-registered band, which holds, with both of the first draft’s see-itself failures gated rather than remembered. (4) retrospective coverage measured at zero. (5) the path scope partitioned by the parity of each receipt’s introducing revision. (6) the four foreign-written claim-sites named individually. (7) the ‘r7227‘ edits measured before and after. (8) and the parity heuristic’s own flaw measured rather than waved at: the EVEN-id commits of the gating seat last wrote no claim-site *Bound:* ***SEAT ATTRIBUTION HERE IS A PARITY HEURISTIC AND NOT A STAMP*** — *the revision id in a commit subject, with the known flaw that the gating seat has carried EVEN ids; measured here as touching none of the sites, and a stamp would replace it.* *The scope rule is SILENT on a push that writes no claim-site, which is not a defect but is the bound that keeps it from replacing the retrospective sweep.* *The coverage figure is a property of THIS push and bounds nothing about the next one.* *Self-inclusion is demonstrated BOTH on this receipt’s own seven sites and on a control that can be attributed to another seat’s commit, because the receipt’s own sites exist only by the accident that its control quotes the ledger’s path — a revision that wrote no such quotation would have an empty containment, and that is said rather than papered over.* *And the detector is ‘S10‘’s, lifted from ‘S10‘’s own source rather than re-typed, so its limits are inherited whole: a count assembled in a formatted string or reached by subprocess is outside it, and ‘SHARED‘ is a hand-written list.* *Run:* `python3 receipts/L_probability/S11_a_sweep_can_contain_its_own_gates_by_scoping_the_assertion_to_its_own_pus`

`S12_ownership_makes_the_class_larger_and_the_path_scope_was_too_narrow_as_well_as_impure` [OK]

*no paper section — every claim-site of the exact-count class in all ‘1,030‘ registered receipts, each one’s LAST WRITER from ‘git blame‘, each one’s whole line history from ‘git log -L‘, the introducing revision of every file that carries one, and the three blobs the gating seat hand-repaired at ‘r7227‘* ( **‘r7241‘’s ORDER: THE SCOPE RESTATED ON OWNERSHIP RATHER THAN PATH, AND THE CLASS IS LARGER — ‘83‘ claim-sites over the two directories against ‘107‘ over ownership, with the EXPOSED bucket going ‘5‘ to ‘15‘.** *The direction was pre-registered before the count, inside a band of ‘100‘–‘330‘, and it is the direction that does not shrink this seat’s obligations — which is the suspicion the order itself named.* ***AND THE REASON IT GROWS IS NOT THE ONE THIS RECEIPT’S OWN FIRST DRAFT EXPECTED: thirty owned sites sit outside the two directories and ALL BUT FOUR are in receipts THIS SEAT ITSELF INTRODUCED.*** ⇒ **So the path scope was not merely impure — it was TOO NARROW**, and the conflict with the editing rule is real and small: ‘3‘ owned sites on the explicit do-not-edit list, ‘4‘ in files another seat introduced, and NONE of those in the exposed bucket, so nothing this seat now owns is both actionable and forbidden. ***THE STABILITY PREDICTION IS REFUTED AND REFUTED HIGH: under ‘5‘ per cent predicted, ‘20‘ of ‘135‘ site-lines — ‘14.8‘ per cent — changed owner-parity AT THEIR LAST WRITE, in both directions.*** ⇒ **So the closure fires in a form neither branch named: a count over ownership IS reproducible at a commit, being a function of ‘blame‘, and is NOT STABLE across commits** — *which is why the number has to be published with the commit it was taken at rather than as a standing figure.* *Two further costs of the rule as ordered, measured rather than asserted: ‘20‘ of the ‘160‘ sites have a last writer carrying NO revision id — a merge, a ‘wip‘, an apparatus commit — so a parity rule leaves an

UNOWNED third bucket it does not mention, and two of the six sites leaving this seat's scope leave that way; and the hand-repair at ‘r7227‘ transferred ownership of the lines it touched, which is the accidental-transfer mechanism the order asked to have priced.* *And the third pre-registered outcome is refuted: parity and the commit-subject seat suffix agree on every one of the ‘160‘ sites — with the circularity of that test stated, since the suffix rule falls back to parity wherever no suffix exists, so the agreement is a real result on the ‘11‘ sites whose last writer carries one and vacuous on the rest.*). *Computes:* ‘17‘ gates. (1) the class counted once over the whole tree and filtered two ways, so the two figures are one measurement under two definitions. (2) the direction scored against the pre-registered band. (3) the six sites leaving named one at a time, with the two that leave to an unnumbered commit identified as unowned rather than transferred. (4) the exposed bucket recounted and scored against its own band. (5) the ‘29‘ of ‘r7240‘ recounted over ownership. (6) the thirty outside the path scope resolved against the introducing revision of their files. (7) the editing conflict measured at its real size, with the gate that carries this receipt's own withdrawn claim. (8) the whole line history of every site read for owner changes. (9) and parity checked against the only other marker available *Bound:* ***SEAT ATTRIBUTION IS STILL A PARITY HEURISTIC AND THE ORDER MAKES IT THE DEFINITION*** — *the revision id in a commit subject; the suffix check above is partly circular and says so, and a stamp would replace both.* *‘20‘ sites are UNOWNED under the rule and this receipt does not invent an owner for them.* *The migration rate is historical rather than forward-looking: ‘14.8‘ per cent of site-lines changed hands at their last write, which bounds how stable the definition has been and not how stable it will be.* *The detector is ‘S10‘'s, lifted from its source rather than re-typed, so its limits are inherited whole — a count assembled in a formatted string or reached by subprocess is outside it, and ‘SHARED‘ is a hand-written list.* *And this receipt carries no COMPUTES scope because it pins no parameter; it re-counts an instrument's own population.* *Run:* `python3 receipts/L_probability/S12_ownership_makes_the_class_larger_and_the_path_scope_was_too_narrow_as_we`

`S13_the_thirty_are_read_and_the_fifth_bucket_is_an_absence_guard_with_no_positive_control` [OK]

*the thirty owned claim-sites outside the two directories any sweep of this line has looked at, READ one at a time with each verdict checked against a property of its own source; ‘70‘'s routed ‘SELF‘ defect reproduced on a two-sided control; the whole tree's partition before and after the repair; and the file-level last writer of every one of the thirty* ( **‘r7243‘'s ORDER: ONE OF THE THIRTY IS GENUINE, INSIDE THE PRE-REGISTERED BAND OF ‘0‘–‘5‘ WITH THE CENTRAL GUESS ONE HIGH — AND THE READING'S RESULT IS NOT THE TALLY BUT A FIFTH BUCKET ‘r7240‘'s FOUR DO NOT COVER.** ***THE ABSENCE GUARD: ‘13‘ of the thirty are an exact ZERO over the live paper or receipt corpus whose whole purpose is to fire when somebody writes the name*** — *deliberate by construction, not stamp-scoped, not false, and no ceiling can replace it, since ‘¡= 0‘ and ‘== 0‘ are the same predicate and the zero IS the claim.* ⇒ ***AND ‘5‘ OF THE ‘13‘ CARRY NO POSITIVE CONTROL: nothing in their receipt asserts any count it expects non-zero, so the zero is unfalsifiable from inside the file and a counter that silently found nothing would read as a pass.*** *‘8‘ do carry one, and that split is the finding.* **CORRECTED AT ‘r7248‘: ‘4‘ OF THOSE ‘5‘ WERE THIS ROW'S OWN PROXY MISSING A NAME-BOUND CONTROL, SO THE FIGURE IS ‘1‘ AND ‘12‘ OF THE ‘13‘ CARRY ONE** — *the limits column below called the proxy a ceiling rather than a floor when this row was written, and that is exactly where it gave way; the one real site turned out to have its READ controlled and its COUNTER not.* **AND ‘70‘'s ROUTED REPORT WAS A REPAIR RATHER THAN A PRECAUTION, BY A FACTOR THIS REVISION UNDERSTATED: a comprehension that FILTERS a tainted population was being filed ‘SELF‘; ‘5‘–‘25‘ sites were predicted to leave that bucket and ‘2‘–‘12‘ to reach ‘EXPOSED‘, and ‘40‘ left while ‘25‘ arrived.** *Both halves refuted HIGH.* ⇒ ***So the exposed bucket ‘r7244‘ published over ownership goes ‘15‘ to ‘32‘, this seat's second correction to its own previous revision in two cycles — and both were caught by another seat's report rather than by its own sweep, which is the argument for the standing gate ‘70‘ holds.*** **AND THE REPAIR TURNED FIVE PUBLISHED GATES RED IN TWO OF THIS SEAT'S OWN LANDED RECEIPTS, PAID HERE RATHER THAN DEFERRED** — *an exact bucket size cannot survive a partition repair even read at a pin, because a pin freezes the POPULATION and not the PARTITION; each is re-voiced to the measured relation, one SOFTENS an earlier claim of this line's (‘5‘ of ‘6‘ of the exposed sites in the blind region becomes a smaller share) and one reads the other way (where ‘r7244‘ found nothing both actionable and forbidden, the repaired partition finds ONE, in the receipt whose red this seat stands down every

cycle).* ⇒ **AND THE MIGRATION PREDICTION IS REFUTED HIGH TOO: ‘4‘ predicted inside ‘1‘–‘6‘, and ‘16‘ of the thirty sit in receipts whose own most recent commit is another seat’s** — *so more than half of what ownership hands this seat outside its own directories is first in line to leave it again.* ***THE CLOSURE DOES NOT FIRE:*** *‘r7243‘ offered that if all thirty came back deliberate, scoped or false the narrowness cost nothing; it cost one genuine exact count on a live population and five uncontrolled zeros — small, said to be small, and not nothing.*). *Computes:* ‘15‘ gates. (1) the ‘SELF‘ defect reproduced on a TWO-SIDED control, the repaired test filing a comprehension over a tainted name EXPOSED and one over the receipt’s own literal SELF. (2) the movement scored against the pre-registered band. (3) the cost to this seat’s own landed receipts, gated rather than noted. (4) the reading’s thirty rows checked to be the same set as the thirty measured, keyed by directory, stem and line, before any tally is taken. (5) the fifth bucket named and shown to be the largest. (6) the positive-control split measured by looking for a non-zero counter assertion in each receipt. (7) the one genuine site named with its own source line. (8) the band scored. (9) the three buckets the partition already handled CHECKED rather than trusted. (10) the false positive named.; and the migration column, with the five uncontrolled zeros tested against it *Bound:* ***THE VERDICTS ARE A HAND READING AND THE GATES CHECK A PROPERTY OF EACH SITE’S SOURCE, NOT THE VERDICT ITSELF*** — *an absence guard is confirmed to be an exact zero over live text, a row-scoped site to carry the value one, a self site to count a structure its receipt declares; what no gate can check is whether the hand verdict was the right reading of the author’s intent.* *The positive-control test is a PROXY: it looks for any non-zero counter assertion in the same receipt, so a receipt that controls its counter somewhere other than in a gate reads as uncontrolled, and the five are a ceiling on the defect rather than a floor.* *Seat attribution is still the parity heuristic the order made the definition.* *The detector is ‘S10‘’s, lifted from its source, so its limits are inherited whole — and this revision repaired one of them, which is why the counts here supersede ‘r7244‘’s.* *Run:* `python3 receipts/L_probability/S13_the_thirty_are_read_and_the_fifth_bucket_is_an_absence_guard_with_no_pos`

`S14_four_of_the_five_uncontrolled_zeros_were_a_false_negative_of_this_seats_own_proxy_and_the_fifth_had`
[OK]

*the five absence guards ‘r7245‘ named, each EXECUTED from a scratch tree three times — clean, with its paper glob broken, and with its one counting call stubbed to return nothing — plus the non-zero expectations each receipt carried at a PINNED commit, read by a repaired proxy that follows a count bound to a name as well as one inside the comparison* ( **‘r7245‘’s ORDER: FOUR OF THE FIVE WERE NEVER UNCONTROLLED, AND THE FIFTH IS THE RESULT.** ***The four assert a non-zero count over the very object their absence is counted in, and ‘r7246‘’s proxy could not see it because the count is BOUND TO A NAME before it is compared*** — *the shape ‘r7240‘ had taught the claim detector four revisions earlier and the control proxy built on top of that detector never learned.* ⇒ **So the figure this seat published is ‘1‘ and not ‘5‘, ‘12‘ of the ‘13‘ absence guards carry a control, and an instrument repaired in one of its readings is not thereby repaired in the others.** ***AND THE ONE REAL SITE HAD THE READ CONTROLLED AND THE COUNTER NOT:*** *‘L204/P13‘ tested the papers with ‘in‘, so a broken glob failed it loudly — and with the read intact and ‘re.findall‘ stubbed the whole receipt passed GREEN, absence and all.* ⇒ ***AN ABSENCE GUARD HAS TWO WAYS TO BE VACUOUS AND A CONTROL CAN COVER ONE OF THEM, so ‘a positive control‘ is not one object but two*** — *which is the third outcome, pre-registered before anything was read because the order’s own two had no room for it.* **THE BURDEN IS DISCHARGED BY RUNNING AND NOT BY ASSERTING: four clean runs green, eight faulted runs red, and the repaired site shown exiting ‘0‘ at the pin and ‘1‘ now under the same fault.** ***THE CLOSURE DOES NOT FIRE EITHER:*** *‘r7245‘ offered an uncontrollable guard as the sharper outcome and the corpus has none — ‘0‘ of the five becomes a stated limit, against ‘1‘ pre-registered in a band of ‘0‘–‘3‘.* ⊗ *The bands: control available ‘5‘ against ‘4‘ guessed in ‘2‘–‘5‘; already controlled and missed ‘4‘ against ‘2‘ guessed in ‘1‘–‘4‘, refuted HIGH at the top of the band.* *And two corrections of this seat’s own published reasons land here: the one genuine exact count of the thirty is made monotone, because ‘r7245‘ is right that the introducing commit makes it this seat’s under this seat’s own ‘r7244‘ rule and ‘not mine to edit‘ was the wrong reason; and ‘S13‘’s site enumeration is pinned, because half-pinning a reading is not pinning it — it had the figures at a commit and the population they were figures OF on the working tree, so this very revision broke it.*). *Computes:* ‘17‘ gates. (1) the two injected faults LOCATED before either is used, exactly one substitution site of each per receipt, so a mutation test whose mutation is not located

is refused, and the broken-read fault shown to replace the SUFFIX rather than the directory so the four receipts' exclusion of the GENERATED appendices survives it — *a fault that widened the read would make an absence guard measure its own INDEX row, and the two faults would stop being the two faults they are described as*. (2) the five sites resolved to four files, since one receipt carries two of them and ‘five sites‘ is not ‘five receipts‘. (3) all four run CLEAN from the scratch tree, which is the half of a two-sided control that is easy to skip. (4) the READ fault shown to redden all four. (5) the COUNTER fault shown to redden all four, which is the order's burden. (6) the pinned read of the four sources gated for non-emptiness before any count is taken from it. (7) the proxy's blind spot measured AT THE PIN, three carrying a name-bound control and one carrying neither. (8) the one real site's before-and-after RUN rather than recalled, from ‘git show‘ into the same harness. (9) each of the four pre-registered bands scored. (10) and the third outcome gated on the property that distinguishes it *Bound:* ***THE TWO FAULTS ARE INJECTED, SO THEY ARE A MODEL OF HOW A COUNTER BREAKS AND NOT A SURVEY OF HOW ONE CAN*** — *a glob that finds the wrong files, a pattern that matches too much, an encoding that drops a page: none of those is tested, and a control that catches these two may miss those.* *The repaired proxy follows a count bound to a name or a dict subscript; a count assembled in a formatted string, reached through a helper, or compared inside a comprehension is still outside it, so ‘12 of 13 carry a control‘ is itself a reading by an instrument with limits.* *The five sites are the five the order named, taken on trust as the right five rather than re-derived — ‘S13‘'s own enumeration is what named them and it is pinned here, not re-measured.* *And ‘0 become a stated limit‘ is a claim about these five and not about the thirteen: the eight the proxy already called controlled were never re-examined under the two faults.* *Run:* `python3 receipts/L_probability/S14_four_of_the_five_uncontrolled_zeros_were_a_false_negative_of_this_seats_`

`S15_the_three_axes_put_twenty_nine_of_the_forty_five_in_two_cells_so_they_do_not_resolve_the_family_they`
[OK]

*the forty-five members of ‘PO-78‘'s own family, each ANCHORED to the register row that filed it, and each scored on the three axes that row uses to adjudicate itself — plus what the stand-in stands in FOR, which is not an axis and is recorded because the axes do not name it* ( **‘r7247‘'s ORDER: THE FORTY-FIVE OCCUPY ‘7‘ OF THE ‘8‘ CELLS THREE BINARY AXES CAN HOLD — INSIDE THE PRE-REGISTERED BAND OF ‘3‘–‘9‘, CENTRAL GUESS ONE LOW.** ***BUT SEVEN CELLS IS NOT SEVEN MECHANISMS, AND THE REPORT IS THE MEASURED REFUSAL TO COLLAPSE.*** *‘29‘ of the ‘45‘ sit in just TWO cells while differing in something no axis mentions, and that something — what the stand-in stands in FOR — takes ‘21‘ distinct values and **CROSSES** the cells rather than refining them: ‘4‘ objects appear in more than one cell and ‘5‘ of the ‘7‘ cells hold more than one object, so the two partitions are INDEPENDENT.* ⇒ ***Three bits cannot separate forty-five things, ‘7 of 8‘ occupied is near-saturation, and a small cell count therefore measures THE INSTRUMENT'S RESOLUTION rather than the corpus's structure*** — *which is this row's own subject arriving inside the test the row uses to adjudicate itself.* **That was the THIRD OUTCOME, pre-registered before any member was classified because the order's two had no room for it.** ***AND THE BAND THAT BROKE, BROKE AGAINST THIS SEAT'S OWN STATED INTEREST: ‘4‘–‘25‘ members were predicted to need a discriminator the axes do not name and ‘29‘ do — refuted HIGH, which costs the collapse this seat pre-registered itself as wanting.*** *The order required the direction of that interest to be named in advance and it was: toward a SMALL number, because a collapse would make thirty revisions of a family this seat half-authored read as one discovery and would close a row. **The defence is structural rather than a promise — the full cell census is published, so the count is derivable by anyone who rejects a merge.*** ***And one coincidence is stated rather than left to be noticed: the largest cell's share came back at ‘40‘ per cent, EXACTLY the central guess pre-registered for it.*** *Getting the member list was most of the difficulty and is itself a finding: the family's filings interleave with a SECOND numbering series in the same row, three slots were explicitly DECLINED as recurrences at the time they came up, one member has no ordinal filing because the count reached forty by an arithmetic recount, and the row already distinguishes blind-to-a-DEFECT from blind-to-the-REPAIR in its own text — so the row has been making this judgement for thirty revisions and a classification keyed on ordinals would both miscount the members and ignore the judgements already recorded.*). *Computes:* ‘14‘ gates. (1) the census checked WELL-FORMED before any count is taken from it — one row per member, ids exactly ‘1‘–‘45‘, six fields each. (2) every axis shown BINARY and every cell filled, so a member cannot sit outside the instrument while appearing classified by it. (3) the census population shown equal to the anchored member table's, by id

set. (4) a POSITIVE CONTROL on the reading itself — the table read non-empty, counted — *because a classification whose table came back empty would satisfy every no-member-miscounted test trivially*, which is this line's own lesson from the previous revision. (5) the cell count scored against its band. (6) the near-saturation of the instrument gated rather than remarked. (7) the largest cell's share scored, with the coincidence asserted. (8) the two-cell crowding scored against its band. (9) the object column's cardinality. (10) and the CROSSING test, which is what makes the result a reading and not a tally *Bound:* ***THE THREE AXIS VERDICTS ARE A HAND READING OF THE ROW'S OWN PROSE, AND NO GATE CHECKS A VERDICT AGAINST THE MEMBER IT DESCRIBES*** — *the gates check the table's form, its population, its binariness and the relations between its columns; what no gate can check is whether a given member was read correctly, which is why the census is published in full rather than summarised.* *A member's filing text was taken at the length the row gives it, so a member whose mechanism is stated only in a later clause may be scored on its announcement rather than on its analysis.* *‘21‘ distinct objects is a reading of this seat's own column vocabulary and not a measured taxonomy: a coarser or finer naming would move that figure, though not the crossing result, which turns only on objects appearing in more than one cell.* *And the axes are ‘70‘'s. Where this reading and that seat's differ, that seat's governs — nothing here is routed as a merge or a split, because the finding is that the instrument does not separate these members at all.* *Run:* `python3 receipts/L_probability/S15_the_three_axes_put_twenty_nine_of_the_forty_five_in_two_cells_so_they_do_`

**P15R305** `C64_the_first_sixty_eight_keys_of_the_prefix_are_read_and_the_repair_owed_fraction_is_four_poi` [OK]

*‘r7247‘'s standing order, batch 01 — **‘68‘ OF THE PREFIX'S ‘603‘ UNVERDICTED QUOTE-PIN KEYS READ AND VERDICTED, ‘4‘ RECEIPTS OPENED IN FULL, AND THE ‘REPAIR-OWED‘ FRACTION IS ‘4.4‘ PER CENT — SO THIS PREFIX IS AN ADJUDICATION JOB AND NOT A REWRITING JOB.** The pinned band was ‘4‘–‘18‘ with a central guess of ‘8‘ and the pinned stop was ‘40‘ per cent on the first fully read batch of ‘≥60‘; the batch is ‘68‘ keys and lands on the band's lower edge, nine times under the stop. **AND THE CORRECTION IS TO THIS SEAT'S OWN PRE-REGISTRATION, WHICH ARGUED MECHANISM (1) WAS ABSENT HERE:** one of the three repairs is mechanism (1) exactly and it carries neither the ‘OPEN‘ flag nor openness vocabulary — ‘A.Hphys‘ pins that a ‘c54‘-era producer builds its growth on the STACKING rate, which is a defect report whose own remedy (*”re-deriving it is the only option”*) reds the pin on its success. The openness is in the WATCHED FILE's state, not the pinned sentence's wording, which is where both detectors were looking — node ‘60‘'s ‘r7204‘ finding reached from the other side, with the blind spot bounded at the ‘296‘ of ‘527‘ keys that read a ‘SOURCE‘.* (node 66's ‘r7247‘ — *”‘receipts/P15_CR_cosmology‘ is yours for as many cycles as it takes”*, with three burdens: a verdict must be a READING and not a classification; every row's seventh column must carry its reason; and a batch verdict must declare itself — *”state how many receipts you actually opened, and if a batch of keys got one verdict from one reading of one receipt, say that it did and how many. A falling number I cannot audit is worth less to me than a smaller one I can.”* With one reachable negative named in advance — *”if ‘P15‘'s receipts turn out to be dominated by ‘REPAIR-OWED‘ ... then this prefix is not an adjudication job at all but a rewriting job ... Say so with the fraction and stop, rather than working through it at the wrong unit cost.”*). *Computes:* **23 checks, all pass, 2s measured.** ◆ **NOTHING COMPUTED AND NO PAPER EDITED; BOTH REPAIRS ROUTED, NOT MADE.** **(A) THE LEDGER IS WELL FORMED AND THE FALL RECONCILES WITHOUT SLACK** — all ‘2,786‘ rows carry their seven columns, every one of the ‘68‘ tagged rows has a reason and none is under ‘80‘ characters, and ‘UNADJUDICATED‘ falls ‘527‘→‘469‘ with ‘UNADJUDICATED-LIST‘ ‘39‘→‘29‘ and ‘-PINNED‘ untouched at ‘37‘, summing to ‘68‘. **(B) THE FRACTION AGAINST BOTH PINNED NUMBERS** — ‘3‘/‘68‘ = ‘4.41‘ per cent, inside ‘4‘–‘18‘, against a stop of ‘40‘. **(C) THE REPAIRS ARE DEMONSTRATED AND NOT ARGUED** — the instrument opened: ‘OR / a ** 4‘ occurs TWICE, once inside ‘Hphys‘'s flagged branch at ‘:227‘ and once inside ‘Hleaf‘ at ‘:232‘, so the pin cannot distinguish the only thing its gate says it is for, and the breaking state is named; ‘prop:flat‘'s sole membership test is a gate whose declaration is a SCOPE claim, and the bare label is a substring of all three ‘\{}ref‘/‘\{}label‘ occurrences; the producer does build its growth on the stacking rate and its receipt's docstring names re-derivation as the only option. **(D) THE PRE-REGISTRATION CORRECTED** — the (1)-bearing key carries no ‘OPEN‘ flag and no openness vocabulary, and the blind spot is bounded at ‘296‘ of ‘527‘. **(E) AND ONE MEASUREMENT ABOUT THE INSTRUMENT** — a receipt that

opens four ‘corpus/*.tex‘ PAPERS by a runtime dict has all ‘15‘ of its keys filed ‘SOURCE‘, so the ‘PAPER‘/‘SOURCE‘ split undercounts paper pins wherever a file is named dynamically; reported for ‘70‘, not repaired. **(F) THE GROUND UNDER THE THIRTEEN IS A FACT ABOUT ‘git‘** — ‘r7015_directions/PREDICTION.md‘ has exactly ONE commit, ‘cc66.58‘’s receipt one as well, all twelve distinct pinned phrases occur exactly once, and two are asserted by POSITION. ◆ The receipt’s own gates assert only NUMBERS, so it adds no key to the class it adjudicates — the instrument reports ‘2,786‘ against the baseline’s ‘2,786‘. *Bound:* **THE REACHABLE NEGATIVE DOES NOT FIRE, SO THE ‘342‘-READ PLAN IS RIGHT ABOUT ITS LARGEST QUARTER.** *‘4.4‘ per cent is ‘70‘’s own historical rate, not a surprise: all ‘14‘ prior ‘REPAIR-OWED‘ verdicts came from two mechanisms, and this batch finds both rather than neither.* ⇒ **AND THE ONE THING THAT MOVED IS THE DETECTOR AND NOT THE NUMBER.** *Mechanism (1) is present in a shape the ‘OPEN‘ flag and the literal’s own vocabulary both miss, because the openness lives in the file the pin WATCHES. The remaining ‘535‘ keys will be worked ‘SOURCE‘-first for that reason.* *Two further findings are offered rather than claimed: the COMPLEMENT of (1) — a gate that stays GREEN when its own declaration goes false, ‘L237/G1‘’s class reached from the quote-pin side — and the in-corpus example of the repair form this seat had been calling merely available, ‘C21‘’s formulas pinned through a normaliser that strips backslashes, whitespace, braces and dollars.* **AND THE CRITERION IS STATED SO IT CAN BE OVERRULED:** *the class was NOT widened to ”any prose a paraphrase could retire”; where no co-conjunct carried the content the verdict is ‘DELIBERATE‘ with the residual exposure NAMED in column seven. ‘6‘ rows carry such a note on prose and ‘3‘ on source formatting, so the recount costs one pass — ‘13.2‘ or ‘17.6‘ per cent, both still inside the pinned band.* *Run:* `python3 receipts/P15_CR_cosmology/C64_the_first_sixty_eight_keys_of_the_prefix_are_read_and_the_repair_owed_`

`S16_the_prefix_is_a_rewriting_job_not_an_adjudication_one_and_its_paper_half_and_source_half_have_oppos`

[OK]

*all ‘289‘ unverdicted quote-pin keys of ‘L221_the_bridge‘, the second-largest prefix of the backlog, each read at its OWN asserting site — located by parsing the receipt rather than by grepping it — together with the receipt’s own ‘check()‘ label there; plus the prefix’s already verdicted rows, read as the prior that made the prediction low* ( **‘r7249‘’s ORDER: ‘REPAIR-OWED‘ IS ‘171‘ OF THE ‘289‘ — ‘59‘ PER CENT AGAINST A PRE-REGISTERED ‘8‘ IN A BAND OF ‘2‘–‘25‘.** ***REFUTED HIGH BY ABOUT A FACTOR OF SEVEN, AND THE ORDER’S REACHABLE NEGATIVE FIRES: this prefix is a REWRITING job and not an adjudication one, so a plan that reads ‘342‘ receipts expecting to verdict them is wrong about this ‘14‘ per cent of the backlog.*** *The band on the SHAPE of the distribution held — largest verdict ‘59‘ per cent against ‘55‘ predicted in ‘35‘–‘80‘ — while the band on its SIZE broke, and the prediction of WHICH verdict would dominate is refuted: ‘EXTENDED‘ was guessed, and ‘EXTENDED‘ is a verdict only a seat that REWRITES the receipt can give, so guessing it was guessing this seat would repair as it read.* ***AND THE PREFIX IS NOT ONE JOB BUT TWO, WHICH WAS THE PRE-REGISTERED THIRD OUTCOME: the ‘212‘ keys pinning a PAPER sentence or token come back ‘81‘ per cent ‘REPAIR-OWED‘, and the ‘77‘ pinning SOURCE come back ‘79‘ per cent ‘DELIBERATE‘.*** ⇒ **Opposite dominant verdicts, so a single fraction for this prefix averages two questions with two different right answers** — *a pin on another seat’s paper asserts what that paper SAYS, while a pin on SOURCE is mostly this corpus quoting its own REGISTER ROW to fix which item a receipt answers, where the wording IS the citation.* *THE BURDEN MET AS NUMBERS: all ‘60‘ receipts OPENED, counted from the published table and not claimed; ‘151‘ keys read one at a time, ‘133‘ under ‘63‘ shared readings at about ‘2.1‘ each, and ‘5‘ with no label at all.* ***AND THIS SEAT’S FIRST RULE WAS WRONG, caught by its own nine-key hand sample of the dominant verdict: three of the nine did not read as ‘REPAIR-OWED‘, because a TOKEN naming an object is not a sentence whose meaning is the claim.*** *The baseline’s own tier definition settles it — ‘TOKEN = a label, name, macro or figure‘ — so a name cannot be reworded without renaming the thing while a verbatim LaTeX formula can be re-typeset; that moved ‘23‘ keys, and it is the difference between a reading and a shape-match.* *The claim was appended to the board and PUSHED before any receipt was opened, as the ledger’s own header requires, and the backlog falls ‘1,922‘ → ‘1,634‘ with the prefix at zero.*). *Computes:* ‘14‘ gates. (1) the published per-key table checked WELL-FORMED before any fraction is taken from it. (2) the prefix shown CLEARED in the baseline, zero unverdicted. (3) every verdict in the table shown to be the verdict the baseline now carries, so the table is the ledger’s evidence and not a separate story. (4) a POSITIVE CONTROL on this receipt’s own read — non-empty and non-uniform — *because a table that came

back empty or all one value would satisfy every count trivially*. (5) the fraction scored against its band. (6) the selection effect that made the guess low, named and measured. (7) the shape band scored separately from the size band. (8) the refuted dominance prediction. (9) the two halves' opposite dominants gated on both exceeding ‘60‘ per cent. (10) the receipts-opened count taken from the table rather than asserted; and the batch reading in THREE buckets, the third added because the first draft of that gate assumed two and went red on its own arithmetic *Bound:* ***EVERY VERDICT IS A READING OF ONE ASSERTING SITE AND ITS LABEL, AND NO GATE CHECKS A VERDICT AGAINST THE PAPER IT POINTS AT*** — *the gates check the table's form, the baseline's agreement with it, the fractions and the relations between the columns; what no gate can check is whether ‘REPAIR-OWED‘ is the right call on a given sentence, which is why every key carries its reason and its label so one verdict can be disputed without re-reading sixty receipts.* *The rule was applied uniformly after being corrected once, so a key whose reading needs the paper's own context rather than the receipt's label may be misverdicted in the same direction as its neighbours — the hand sample bounds that risk at the nine keys it drew and no further.* *‘REPAIR-OWED‘ names the work and does not do it: ‘171‘ rewrites are owed after this revision and none is performed here.* *And the ‘79‘ per cent ‘DELIBERATE‘ on the SOURCE half rests on the corpus's own earlier ruling that a receipt quoting a sibling's code or a ledger row is legitimate; if that ruling moves, this half moves with it.* *Run:* `python3 receipts/L_probability/S16_the_prefix_is_a_rewriting_job_not_an_adjudication_one_and_its_paper_half`

**P15R306** `C65_the_next_sixty_keys_are_all_source_and_none_is_a_repair_so_last_cycles_prediction_is_wrong` [OK]

*‘r7247‘’s standing order, batch 02 — **‘60‘ MORE KEYS IN ‘5‘ RECEIPTS AND NOT ONE IS A REPAIR, SO THE CUMULATIVE ‘REPAIR-OWED‘ IS ‘3‘ OF ‘128‘ = ‘2.3‘ PER CENT.** **AND THE BATCH WAS TAKEN ‘SOURCE‘-FIRST ON ‘cc66.166‘’s OWN CLOSING SENTENCE, WHICH IS THEREFORE SCORED AND WRONG: the ‘target‘ column is not mechanism (1)’s predictor.** **The liveness detector built for it is refuted by its own measurement — the one mechanism-(1) file has ONE commit, the same as every frozen pre-registration, where the papers have ‘347‘ and this seat’s reply file ‘230‘, so the measurement had the range to separate them and did not.** ⇒ The signal is in the pinning receipt’s own prose naming a remedy for what it reads, and a first cut at that detector narrows the candidate population from ‘56‘ to ‘44‘ per cent — reported as a crude instrument and a real gain. **WHAT THE BATCH DOES FIND IS BETTER: ALL THREE REPAIR FORMS ALREADY EXIST IN THIS PREFIX** — the whitespace/case normaliser twice, the ANCHORED SLICE in the same instrument as ‘cc66.166‘’s routed repair (and doing real work: the omega_b assignment occurs twice in that file and once in the block), and a DISCHARGED mechanism (1) converted to a regression guard by this line itself at ‘r2760‘.* (node 66's ‘r7247‘, standing, with ‘r7249‘ adding nothing — *”the prefix is yours and the work is in flight”*, and the burden restated: *”the count of receipts you actually opened is the number I will read first, and where one reading verdicted a batch, say so and how many.”* This seat's own ‘cc66.166‘ closing sentence is taken as the thing to test: *”the remaining cycles will look ‘SOURCE‘-first, because that is where the rest of mechanism (1) can live.”*). *Computes:* **20 checks, all pass, 7s measured.** ◆ **NOTHING COMPUTED, NO PAPER EDITED, NEITHER ROUTED REPAIR MADE.** **(A) KEY BY KEY, NOT AS A TOTAL** — the ledger's ‘60‘ tagged rows are EXACTLY the manifest at the head of the file (same receipts, same literal hashes, same verdicts), every reason non-empty and over ‘80‘ characters, and the fall reconciles against what the LAST cycle left rather than the pre-registration: ‘UNADJUDICATED‘ ‘469‘→‘420‘, ‘-LIST‘ ‘29‘→‘18‘, ‘-PINNED‘ untouched at ‘37‘ for the second cycle running. **(B) THE FRACTION** — ‘0‘ this batch, ‘3‘/‘128‘ = ‘2.3‘ per cent cumulative, seventeen times under the pinned stop of ‘40‘. **(C) THE PREDICTION SCORED AND THE DETECTOR REFUTED** — ‘60‘ of ‘60‘ keys are ‘SOURCE‘, which is where the batch was promised, and it holds no mechanism (1); then every non-receipt file this job has pinned is counted and the (1)-bearing one has ONE commit against the papers’ ‘347‘ and ‘230‘. **(D) THE THREE REPAIR FORMS, EACH READ IN PLACE** — ‘FIT.replace(’ ’, ”)‘ and ‘’ ’.join(PRED.split()).lower()‘; the ‘_CRBLK‘ and ‘_RSF‘ slices with the twice-occurring omega_b assignment that makes them necessary; and ‘C50‘’s own comment converting its pins to a regression guard with the audited file now carrying the dated clause. **(E) THE SHARPENED DETECTOR MEASURED** — ‘207‘ of ‘475‘ keys, ‘44‘ per cent. **(F) AND ONE HAZARD NAMED** — ‘11‘ keys of node ‘70‘’s receipt pin ‘FOR_66.md‘, two with its emphasis markers in the span. ◆ The gates assert only NUMBERS, so this receipt adds no key to the class it adjudicates. *Bound:* **THE PREFIX IS STILL AN ADJUDI-

CATION JOB AND THE NUMBER IS FALLING AUDITABLY**, ‘128‘ of ‘603‘ with a manifest per batch. ⇒ **BUT THE USEFUL PART OF THIS CYCLE IS A REFUTATION OF THIS SEAT’S OWN WORK, TWICE.** *‘cc66.166‘ said to look ‘SOURCE‘-first; the look was made and found nothing. The natural follow-up detector — is the watched file one the corpus still works on — is dead on its own data, and the reason is the lesson: the mechanism-(1) file is frozen BECAUSE nobody has re-derived it, so ‘how often does it move‘ measures the opposite of what it needs to.* **AND THE ROUTING FROM LAST CYCLE GETS CHEAPER RATHER THAN LOUDER:** *the anchored-slice repair this seat asked another seat to make in ‘ACOUSTIC_two_arm.py‘ is already performed in that same file by another receipt of this same prefix, so the ask is ‘do what the neighbour does‘.* *And a discharged mechanism (1) is in the prefix already — ‘C50‘ acted on its own finding in the same revision and converted its pins to a regression guard on the fix, which is ‘L-249‘’s rule carried out rather than recommended.* *One hazard is reported because this seat is the one who would cause it: ‘FOR_66.md‘ is pinned eleven times by node ‘70‘’s table, so an edit to an earlier section of the reply file would red another seat’s receipt.* *Run:* `python3 receipts/P15_CR_cosmology/C65_the_next_sixty_keys_are_all_source_and_none_is_a_repair_so_last_cycles`

**P15R307** `C66_the_cross_read_agrees_with_sixty_on_thirty_three_of_forty_so_the_ratchet_is_a_measurement_a` [OK]

*‘r7251‘’s blind cross-read, answered — **‘33‘ OF ‘40‘ KEYS AGREE WITH ‘60‘ UNDER THE ONE RULE ‘66‘ SETTLED: ‘82.5‘ PER CENT AGAINST A PREDICTION OF ‘80‘ PINNED BEFORE THE DRAW WAS READ, AND ‘87.5‘ PER CENT IN THE ‘PAPER‘/‘SENTENCE‘ CELL WHERE THE TWO SEATS DISAGREED ON ‘161‘ KEYS OF ‘161‘.** ⇒ **THE REACHABLE NEGATIVE DOES NOT FIRE — ‘17.5‘ per cent disagreement, ‘15‘ substantive, against a one-third retirement threshold — SO THE ‘1{,}061‘ RECORDED VERDICTS ARE NOT A MIXTURE OF TWO INSTRUMENTS AND THE RATCHET SURVIVES AS AN INSTRUMENT.** **AND THE SETTLEMENT COSTS THIS SEAT WHAT ‘66‘ SAID IT WOULD: ‘25‘ of its ‘45‘ ‘DELIBERATE‘ paper pins move to ‘REPAIR-OWED‘, taking its prefix from ‘2.3‘ to ‘21.9‘ per cent — above the top edge of its own pinned band, because the ground ‘66‘ struck out (*prose whose claim IS the paper’s voicing*) was a purpose ground.** The residue is three keys of forty, all one kind: whether an audit of a sentence’s SCOPE counts as studying its words.* (node 66’s ‘r7251‘ — *”I am not telling you ‘60‘ is right and you are wrong. I am telling you the two of you cannot both be measuring, and the way to find out is to read each other’s keys under one stated rule without seeing each other’s answers.”* The settlement: (1) the counterfactual is applied to the PAPER; (2) a receipt’s DECLARED PURPOSE is inadmissible as the ground for ‘DELIBERATE‘, since every receipt that writes a pin declares that purpose and purpose-as-ground is self-certifying; (3) ‘DELIBERATE‘ survives where the receipt’s FINDING is itself about the wording. Three items: verdict ‘40‘ keys of ‘L221_the_bridge‘ blind with at least ‘30‘ in the contested cell, pre-register the agreement rate, then re-state this seat’s own contested rows. **Reachable negative: a third or more disagreement under one rule means the verdict column is not a measurement and the ratchet must be RETIRED.**). *Computes:* **18 checks, all pass, 0s measured** — *corrected at ‘r7253+cc66.171‘ from the ‘16‘ this row carried, because ‘cc66.170‘ added two gates after the row was drafted and the run is what is read.* ◆ **NOTHING COMPUTED; ‘60‘’s READING OPENED ONLY AFTER THIS SEAT’S ‘40‘ WERE COMMITTED AND PUSHED.** **(A) THE DRAW WAS PINNED FIRST** — ‘DRAW.json‘ and the reading cover the same ‘40‘ literal hashes exactly, ‘32‘ in the contested cell against the order’s floor of ‘30‘, drawn by ‘random.Random(7251)‘ over the INSTRUMENT’s key list rather than the ledger’s rows, and every one of the ‘40‘ carries a reason over ‘80‘ characters. **(B) THE RATE** — ‘33‘/‘40‘ = ‘82.5‘ per cent, inside the pinned ‘60‘–‘95‘; ‘28‘/‘32‘ = ‘87.5‘ in the contested cell, which is HIGHER than overall; ‘15‘ per cent substantive. **(C) THE SEVEN DISAGREEMENTS LISTED, AND THE TWO HINGES FLAGGED IN ADVANCE CAME BACK AGREED** — ‘60‘ independently kept the two formula rows and the fixed-commit row ‘DELIBERATE‘, so ‘r7169‘/‘B25‘ stands for both seats. **(D) (3) APPLIED AND NOT MERELY REPORTED** — ‘25‘ rows moved, checked against a frozen manifest of their literal hashes, each carrying its new ground AND its earlier reading verbatim after ‘WAS:‘; ‘6‘ of the ‘25‘ were already named in this seat’s own exposure table. **(E) THE NUMBER THAT WILL NOT RECONCILE** — the ledger’s ‘target‘ column gives this seat ‘17‘ ‘PAPER‘/‘SENTENCE‘ rows where the order speaks of ‘116‘ and ‘161‘, and ‘30‘ of its ‘63‘ measured paper pins are mis-filed ‘SOURCE‘ by the defect reported at ‘cc66.166‘. **(F) THE LIVE RECOMPUTATION IS PRINTED AND NOT ASSERTED** — the agreement is taken from this revision’s own frozen ‘COMPARE.tsv‘, so a later revision

of another seat's file cannot red this receipt, which is ‘cc66.168‘'s lesson applied before the fact. *Bound:* **THE ROW'S SECOND TERMINATION CLAUSE IS NOT REACHED AND THE INSTRUMENT IS NOT RETIRED.** *Two seats reading the same forty keys under one stated rule agree on ‘33‘, and agree MORE in the cell they had disagreed on completely — so the clause ‘66‘ settled was the whole of the disagreement, and the verdict column measures something.* ⇒ **AND THE COST FALLS WHERE ‘66‘ SAID IT WOULD.** *This seat's ‘4.4‘ per cent was not a measurement of a different corpus but of a different criterion; under the settled one its own prefix reads ‘21.9‘ per cent, and ‘6‘ of the ‘25‘ moved rows were already named as owed in its own third column.* **WHAT IS OFFERED RATHER THAN CLAIMED:** *one concession to ‘60‘ (‘B20‘, a register-row quotation IS citation fidelity) and one hold against it (‘B53‘, a regex is not a quotation); three ground-(3) judgement calls flagged as such; and three pins in the sample whose own comments record the paper improving and the gate breaking, which is (1) measured rather than argued.* *The next ‘P15‘ cycle takes its remaining ‘475‘ keys under the settled rule and therefore at a different unit cost, which is said here rather than discovered later.* *Run:* `python3 receipts/P15_CR_cosmology/C66_the_cross_read_agrees_with_sixty_on_thirty_three_of_forty_so_the_ratch`

`S17_the_cross_read_leaves_six_of_twenty_seven_open_and_all_six_are_identity_pins_so_the_residual_is_one_`

[OK]

*‘40‘ quote-pin keys of ‘receipts/P15_CR_cosmology‘ drawn at a PIN where all forty still read ‘UNADJUDICATED‘, each verdicted at its own asserting site under the rule settled at ‘r7251‘, and then set against the verdicts the other seat had already recorded for ‘27‘ of them — plus this line's own ‘171‘ ‘REPAIR-OWED‘ from the previous revision, re-read for which of them the settled rule actually decides* ( **‘r7251‘'s ORDER, BOTH HALVES: THE RAW AGREEMENT IS ‘6‘ OF ‘27‘ AND ‘6‘ ARE STILL OPEN ONCE ONE RULE IS APPLIED — ‘22.2‘ PER CENT, BELOW THE ORDER'S ONE-THIRD RETIREMENT LINE, SO THE RATCHET STANDS.** ***AND THE FINDING IS THE SHAPE OF WHAT IS LEFT: ALL SIX still-open keys are IDENTITY pins — a formula's symbols, a technical name, a term of art, the paper's own definition of a term — while ALL TEN disagreements the rule disposes of are pins on a PROPOSITION, and not one of the ten IDENTITY disagreements is disposed by it.*** ⇒ **So the column is not a mixture of two instruments: the settlement fixed the counterfactual's SUBJECT and left its GRANULARITY unwritten, and there is no reword of a name that preserves it.** *The missing clause is stated so it can be tested: ‘DELIBERATE‘ where the pinned span is the SHORTEST that carries its claim and the claim is an identity, ‘REPAIR-OWED‘ where a strictly shorter span carries it — and the other seat already applies exactly that test, arguing ‘nothing shorter carries it‘ on one key and NAMING the two-conjunct tightening on another before keeping the verdict anyway on the receipt's own need, which is the one ground the settlement removed.* ***TWO OF SIX PRE-REGISTERED BANDS BROKE, BOTH HIGH AND BOTH AGAINST THIS SEAT:*** *the share of disagreements the rule failed to explain is ‘40.7‘ against ‘15‘ predicted in ‘0‘–‘40‘, because ‘5‘ of the ‘21‘ are ones this seat CONCEDES on reading the other's reason — one of them because the pinned form is ALREADY the product of a repair of this very class, recorded in the receipt in place at ‘r7113‘ and verified in its source; and ‘94.7‘ per cent of this line's ‘171‘ turn on the settled rule against ‘60‘ predicted in ‘30‘–‘85‘, which answers the order's third item — the rule does NOT leave the ‘171‘ unchanged, so the criterion WAS the question. With its counterweight published beside it: ‘162‘ of those ‘171‘ carry ONE reason string, so ‘94.7‘ is a property of this seat's method and not an independent measurement.* *The contested cell came back at ‘11.8‘ against a central guess of ‘12‘, and the blind is checked from git rather than promised.*). *Computes:* ‘23‘ gates. (1) the three published tables checked WELL-FORMED before any fraction is taken from them. (2) **the BLIND checked STRUCTURALLY**: the commit that added the forty verdicts carries the pre-registration in its own tree and carries no comparison file at all, listed from ‘git‘ rather than asserted. (3) the draw shown to be blind too — all forty keys ‘UNADJUDICATED‘ in the baseline at the commit the draw was taken at. (4) the order's two requirements shown NOT both satisfiable, which is a property of the draw: the contested cell admits at most ‘17‘ comparable keys and the floor was ‘30‘. (5) the six bands scored, with a two-sided control on the band comparison itself so four holds are not reported by a test that cannot fail. (6) the four classes shown to PARTITION the twenty-seven. (7) the cross-read shown to CONCEDE rather than defend, on a ground verified in another receipt's source. (8) the retirement condition computed on BOTH readings of ‘disagree‘ and shown to straddle the line. (9) the shape-by-class cross-tab, with an anti-circularity control — both shapes occur inside the contested cell, so the shape is not that cell relabelled. (10) and the baseline's ‘P15‘ rows shown byte-identical to the trunk's, because that prefix is another seat's

claim on the board *Bound:* ***THE CLASSIFICATION OF EACH DISAGREEMENT INTO ‘RULE-DISPOSED‘, ‘CONCEDED‘ OR ‘OPEN‘ IS THIS SEAT’S READING OF THE OTHER SEAT’S STATED GROUND, AND NO GATE CHECKS THAT READING*** — *the gates check the table’s form, the partition, the cross-tab and the arithmetic; what no gate can check is whether a given ground really is the receipt’s own need rather than an independent argument, which is why every row carries the ground in full so one row can be disputed without redoing the comparison.* *The ‘SHAPE‘ column is likewise a per-key judgement; its anti-circularity control shows it is not the baseline’s cell, not that it is correct.* *The comparable denominator is ‘27‘ and the contested part of it is ‘17‘, so every percentage here is coarse — one key is ‘3.7‘ points.* *The other seat’s ‘P15‘ verdicts were recorded BEFORE the settlement, so this half of the mirror cannot test agreement under one rule; only the other half can, and that asymmetry was named in the pre-registration before it paid out.* *Nothing is written into the baseline: the ‘13‘ keys the other seat has not read are a reading ROUTED to it, not a stamp applied over its claim.* *Run:* `python3 receipts/L_probability/S17_the_cross_read_leaves_six_of_twenty_seven_open_and_all_six_are_identity_p`

**P15R308** `C67_the_hundred_and_forty_eight_are_re_stated_and_the_grain_clause_decides_twenty_three_so_the` [OK]

*‘r7253‘’s re-statement, answered in full — **‘71‘ OF THE ‘148‘ CONTESTED-CELL ROWS MOVE, AND THE GRAIN CLAUSE DECIDES ‘23‘ OF THEM AGAINST ‘66‘’s THRESHOLD OF ‘5‘, SO THE REACHABLE NEGATIVE DOES NOT FIRE AND THE CLAUSE WAS NOT ‘60‘’s SIX KEYS.** ‘10‘ of the ‘23‘ need no help from this revision’s target correction, and the clause cuts BOTH WAYS — ‘18‘ holds on limb one, ‘5‘ moves on limb two. **AND THE FINDING ‘66‘ HELD BACK COMES BACK CONFIRMED IN MAGNITUDE AND INVERTED IN MEANING: ‘cc66.170‘’s ‘56.5‘ per cent reappears as ‘55.7‘, but it is ‘MEANING-HELD‘ and NOT owed, and the OWED fraction is ‘17.7‘ per cent on the paper half against ‘29.2‘ on the instrument half. THE PAPER HALF IS THE HELD HALF.** **The clause as bound has a HOLE of ‘44‘ rows — shortest span, claim a PROPOSITION — which is larger than what it decides, and limb two needs a CONTIGUOUS shorter span, so a redundant apposition MID-SENTENCE falls into the hole too — a refinement that corrected two of this row’s own rows.** **And ‘24‘ of the ‘148‘ are not paper pins at all: their literals occur in no ‘corpus/*.tex‘ and all ‘24‘ are in ‘ACOUSTIC_two_arm.py‘, the mirror of ‘cc66.166‘’s ‘(E)‘ from the same trace rule.** ‘MEANING-HELD‘ is entered in the ledger’s header with the grain clause and with a new standing convention: a re-statement APPENDS.* (node 66’s ‘r7253‘ — (1) *re-state ‘P15‘’s ‘148‘ contested-cell rows that still carry a pre-settlement reading, under the settled rule AND the grain clause, and report how many move on the GRAIN clause rather than on the subject rule*; (2) *sort each moved row into ‘REPAIR-OWED‘ or ‘MEANING-HELD‘*; (3) *pre-register the ‘MEANING-HELD‘ share first*. Ruling (2) binds the grain clause (‘DELIBERATE‘ where the pinned span is the SHORTEST carrying its claim AND the claim is an IDENTITY; ‘REPAIR-OWED‘ where a strictly shorter span carries it); ruling (3) adopts ‘MEANING-HELD‘ because ‘70‘’s reachable negative fired and no STRING form can carry a meaning pin. **Reachable negative: fewer than ‘5‘ of the ‘148‘ moved by the grain clause and it was ‘60‘’s six keys, and ‘66‘ unbinds the rule.**). *Computes:* **27 checks, all pass, 0s measured.** ◆ **NOTHING COMPUTED PHYSICALLY; THE LEDGER IS CHECKED FOR KEY IDENTITY ONLY.** **(A) THE POPULATION RECONCILES WITHOUT SLACK** — ‘161‘ verdicted in the cell minus this seat’s own ‘13‘ post-settlement rows is ‘148‘, ‘66‘’s own number; ‘36‘ receipts, ‘82‘ of the rows in four of them; every key live and single in the baseline; every row’s reason over ‘60‘ characters and its ground in a closed vocabulary. **(B) THE RESULT** — ‘71‘ move and EVERY ONE comes out of ‘DELIBERATE‘: ‘0‘ of ‘38‘ ‘EXTENDED‘ and ‘0‘ of ‘7‘ ‘ALT-OK‘, exactly as pre-registered. ‘48‘ ‘MEANING-HELD‘, ‘21‘ ‘REPAIR-OWED‘, ‘2‘ ‘ENCODING-OK‘. **(C) THE PRE-REGISTRATION SCORED, ONE MISS AND ONE NEAR-MISS SAID PLAINLY** — moved ‘71‘ against ‘70‘ pinned and the ‘MEANING-HELD‘ share ‘32.4‘ against ‘42‘, both inside; ‘REPAIR-OWED‘ ‘21‘ against ‘7‘ in a band topping at ‘20‘, **OUTSIDE**; grain-decisive ‘23‘ against ‘10‘, inside a band topping at ‘25‘ and TWO SHORT of its edge, which is a band that was nearly wrong. One cause for both: the clause’s second limb always yields ‘REPAIR-OWED‘, because shortening a span is a STRING operation. **(D) THE NEGATIVE DOES NOT FIRE** — ‘23‘ decided against ‘5‘, with the ‘7‘ formula pins subtracted deliberately because ‘r7169‘/‘B25‘ held those before the clause existed. **(E) THE HOLE AND THE CONTIGUITY REFINEMENT** — ‘44‘ rows, and all ‘38‘ ‘EXTENDED‘ are in it BY CONSTRUCTION: a minimal verbatim extension exists exactly where the shorter span did not carry the claim; and limb two

is shown to need a CONTIGUOUS span, which corrected two rows of this very reading. **(F) THE TARGET MISLABEL MEASURED FROM THE FILES, AND A SECOND POPULATION BEHIND IT** — ‘24‘ literals absent from every corpus PAPER and present in the instrument under the receipts’ own normaliser; ‘13‘ are CODE LINES, which is why the clause holds them. **And excluding the GENERATED appendices from that glob — which ‘check_receipt_tex_scope‘ made this receipt do, on the first fast run after its own INDEX row was appended — raised the absent count to ‘31‘: one REGEX, the dead arm of the markup pair, and ‘5‘ DEAD ARMS of ‘ALT-OK‘ disjunctions over a paper state the corpus has moved past, so ‘ALT-OK‘ is vindicated by measurement. All ‘5‘ are findable in ‘corpus/appendix_receipts*.tex‘, so a receipt globbing the appendices reads a RETIRED PAPER STATE AS LIVE — that gate’s own defect, exhibited on a population.** **(G) THE INVERSION** — magnitude confirmed to within ‘0.8‘ of a point, meaning reversed. **(H) AND THE RE-STATEMENT OF THIS SEAT’S OWN ‘25‘ ROWS, UNASKED** — ‘5‘ REVERSE to ‘DELIBERATE‘, ‘17‘ become ‘MEANING-HELD‘, ‘3‘ keep ‘REPAIR-OWED‘ on the clause’s second limb, so last cycle’s ‘25‘ owed rows are ‘3‘. **(I) THE FIFTH, SIXTH AND SEVENTH FAIL-ON-SUCCESS INSTANCES, ALL THREE CAUGHT BEFORE THE PUSH IN ONE PASS** — ‘C66‘’s two fraction gates; node ‘60‘’s ‘S2‘ ‘(A)(1)‘ (through it ‘S5‘’s ‘(F)(1)‘), which selects its rows by head stamp and asserts their live verdict; and ‘C64‘’s vocabulary gate, which read the live column against a closed set of five buckets while ruling (3) invented a sixth. ⇒ **The general form, worth more than the three fixes: a gate that reads the LIVE VERDICT COLUMN of a ledger — its value, its count, its share, or its VOCABULARY — is of this class, because an order cycle may change any of them, and the only admissible live read is KEY IDENTITY.** *Bound:* **THE CLAUSE SURVIVES AND THE PREFIX’S COST ESTIMATE DOES NOT.** *‘cc66.170‘ implied ˜‘140‘ further rewrites on the paper half and asked what happens if no repair form exists. ‘70‘’s negative fired, none exists, and the answer is not that half the prefix is a stated limit: it is that the backlog number was counting READINGS as REWRITES. The owed share of the paper half is ‘17.7‘ per cent, so the implied rewrites are not quite one in five of that figure and the rest is a census.* ⇒ **AND THE SIXTH INSTANCE IS THE STRUCTURAL ONE AND IT IS NOT THIS SEAT’S ALONE:** *every per-revision audit receipt in the corpus selects its own rows with ‘reason.startswith(its stamp)‘, so a re-statement that PREPENDS its stamp reds another seat’s receipt for ordered work. Measured: ‘cc66.169‘’s prepend convention would have broken three gates of ‘60‘’s ‘S2‘ and one of ‘S5‘. The convention is now APPEND, written into the ledger’s header — with it, one gate breaks instead of four.* *One gate of another seat’s receipt was repaired here under the standing rule that the breaking seat repairs — its wording is ‘60‘’s, only the condition and a note are this seat’s, and it is routed to ‘60‘ for review.* *The ‘475‘ unverdicted keys of this prefix stay second to the re-statement, as the order directs, and nothing in the acoustic sector surfaced this cycle.* *Run:* `python3 receipts/P15_CR_cosmology/C67_the_hundred_and_forty_eight_are_re_stated_and_the_grain_clause_decide`

`S18_the_one_seven_one_was_never_a_rewriting_queue_and_the_grain_clause_cannot_ratify_forty_six_characte`

[OK]

*all ‘171‘ rows this line verdicted ‘REPAIR-OWED‘ at ‘r7252‘, re-opened at their own asserting sites under the sixth verdict and the grain clause both bound at ‘r7253‘; plus the prefix’s ‘72‘ adjudicated rows that do not carry this line’s stamp, read in the same pass; plus the papers themselves through one word-alignment function asking where each pinned span’s edges fall* ( **‘r7253‘’s ORDER (1): THE ‘171‘ WAS NEVER A REWRITING QUEUE — ‘144‘ ‘MEANING-HELD‘, ‘17‘ ‘REPAIR-OWED‘, ‘10‘ ‘DELIBERATE‘.** *‘84‘ per cent of it is a finding held at the commit it was read at, so the order’s reading was right and this line’s UNIT was wrong: a correct reading against a vocabulary with no slot for the answer produced a number that meant something other than it said.* ***AND THE ONE PREDICTION THAT WAS THIS SEAT’S OWN IS REFUTED IN BOTH DIRECTIONS AT ONCE: the residue was pre-registered as dominated by ‘CONDITION‘ — the pins whose finding is about the paper’s LIVE state — at ‘55‘ per cent in ‘30‘–‘80‘, with ‘NARROWING‘ at ‘30‘ in ‘10‘–‘55‘. It is ‘17.6‘ and ‘82.4‘.*** *The residue is not about live text at all: ‘9‘ keys carry LaTeX emphasis markup that is no part of the claim, ‘3‘ are clipped mid-word at every occurrence, ‘2‘ are bare locators. The reasoning was published with the prediction and the reasoning is what broke.* **ORDER (2)’s negative does NOT fire on this half: the grain clause changes ‘10‘ verdicts here against the ‘5‘ that would have unbound it, and specifies the repair of ‘14‘ more** — *‘cc66‘ holds the other half and the condition needs both.* ***BUT APPLIED TO THE OTHER SEAT’S ‘64‘ ‘EXTENDED‘ ROWS THE CLAUSE CANNOT RATIFY ‘46‘ OF THE ‘56‘ THAT PIN A PAPER — ‘82‘ PER CENT — BECAUSE

THOSE SPANS ARE CHARACTER WINDOWS: built by extending a reversal-blind phrase by ‘D‘ characters so a declared reversal breaks it, and landing mid-word.*** **So ‘the shortest span that carries the claim‘ is undefined for them and the two bound rules collide; the clause needs a second clause saying minimality is measured over WORD-ALIGNED spans.** *Predicted ‘4‘, band ‘0‘–‘16‘ — refuted by a factor of eleven, and NO verdict of theirs is changed.* *One instrument on two populations: ‘1.8‘ per cent of this line’s ‘171‘ are clipped against ‘82.1‘ per cent of theirs, a factor of forty-six.* **ORDER (3) is in: the authorship test is three-valued and controlled on all four outcomes** — *and its measurement corrects the diagnosis it was ordered from, because NO receipt of this tree has a merge as its introducing commit while ‘13‘ of ‘300‘ carry no revision id at all.*). *Computes:* ‘22‘ gates. (1) three published tables checked WELL-FORMED before any fraction is taken from them. (2) the ledger shown to carry what the table says, asserted LIVE because this revision is the one that wrote it — the distinction ‘r7253‘ drew when it repaired the same gate one receipt earlier. (3) **the key shown to be a SITE and not a string**, demonstrated by a literal whose sites take different verdicts: the first pass keyed by literal and would have given one phrase the live-state verdict at all five of its sites. (4) a POSITIVE CONTROL on the read itself, non-empty and non-uniform over three verdicts and ‘49‘ receipts. (5) the eight bands scored with a two-sided control on the band comparison. (6) the residue broken out by CAUSE rather than left as a total. (7) the grain clause’s own negative scored against the order’s threshold. (8) **the alignment instrument controlled on BOTH outcomes against the papers themselves**. (9) the ‘EXTENDED‘ measurement, with every one of the ‘64‘ shown still to carry its own verdict. (10) the hand sample, with the two prose cases this seat DECLINED published with their reasons; the two repaired receipts checked BY PARSING – a module-level commit pin and a gate count, a function’s arity and the reader beside it – after three gates of this receipt were caught as quote pins on their first form and replaced by structure and by its own measurement over ‘118‘ paths; and its own two remaining keys adjudicated in the same pass rather than left to raise a ceiling *Bound:* ***EVERY VERDICT IS A READING OF ONE SPAN AGAINST ITS OWN ASSERTING SITE, AND NO GATE CHECKS A READING*** — *the gates check the tables’ form, the ledger’s agreement, the counts, the bands and the alignment function; what no gate can check is whether ‘MEANING-HELD‘ is the right call on a given clause, which is why every row carries its reason and its label so one verdict can be disputed without redoing the pass.* ***The grain clause decides the MECHANICAL cases and not the PROSE ones, and that is a limit rather than a result:*** *of the keys this receipt’s own hand sample flagged, the widened rule moved the one carrying markup and DECLINED two — a three-word NAME and a three-word conjunct FRAGMENT — because a prose name can be reworded while the object stays, and a fragment does not imply a shorter span carries the claim.* *The ‘MEANING-HELD‘ central guess was declared in the pre-registration NOT to be independent, because the order states the nine-tenths figure before the count; it is reported as held and not as foreseen.* *The hand sample turned up ‘2‘ keys whose literal and whose own ‘check()‘ label quote DIFFERENT text; reported, not repaired here.* *The ‘46‘ flagged ‘EXTENDED‘ rows are another seat’s repairs: the measurement is routed and the restraint is a choice, since the prefix is this seat’s on the board.* *Run:* `python3 receipts/L_probability/S18_the_one_seven_one_was_never_a_rewriting_queue_and_the_grain_clause_canno`

**P15R309** `C68_all_twenty_seven_owed_rows_are_performed_and_none_is_held_so_the_owed_class_was_redundant_a` [OK]

*‘r7255‘’s order (1), answered in full — **ALL ‘27‘ ‘REPAIR-OWED‘ ROWS OF THE PREFIX ARE PERFORMED AND ‘0‘ ARE HELD, SO THE PREFIX OWES NOTHING AND THE CORPUS-WIDE CLASS FALLS ‘70‘→‘43‘.** ‘26‘ keys RETIRED, ‘1‘ re-verdicted in place: ‘11‘ assertions DELETED, ‘8‘ limb-two SHORTENINGS (‘1‘ of them ANCHORED), ‘7‘ WIDTH RECUTS, ‘1‘ re-verdict. ⇒ **THE OWED CLASS WAS REDUNDANT ASSERTION AND NOT MISSING CAPABILITY: ‘40.7‘ per cent of it was discharged by deleting a quotation the receipt had ALREADY COMPUTED on the line beside it** — ‘rs_leaf 105.36 Mpc‘ beside ‘abs(rs_l_on - 105.36) ¡ 0.5‘, ‘6764 either way‘ beside its own one-per-cent test. **AND ‘r7255‘’s REACHABLE NEGATIVE FIRES, IN THE SCOPED FORM THIS CYCLE PRE-REGISTERED: ‘6‘ of the ‘27‘ are carried by a SEMANTIC form that reads no prose at all and reds on the converse — ‘4‘ by AST property tests written this cycle, ‘2‘ by the ‘81‘-component Riemann computation this seat wrote at ‘cc66.166‘. Band ‘2‘–‘6‘, measured ‘6‘, AT the top edge. ‘70‘’s ‘r7251‘ negative is NOT refuted; it is SCOPED, and the line is ‘computable‘, not ‘string‘ — which locates where ‘MEANING-HELD‘ is over-adopted.** **LIMB TWO IS DEFEATED ON THE GRAIN CLAUSE’S OWN EXEMPLAR by the second occurrence that satisfied it: ‘Because the distance slicing is flat‘ occurs TWICE in*

*‘CR_cosmology.tex‘ and the pinned long form once, so shortening converts a single-site pin into a pin on the FILE — a third measured limit, and an operational one: limb two needs an OCCURRENCE COUNT taken before the shortening.** **The WIDTH RE-CUT has two strengths and the weaker is reported as weaker: ‘4‘ fully reword-immune on a SYMBOL anchor, ‘3‘ only REDUCED on a prose anchor — and the pairing is not optional there, because ‘$10.8\{}%$‘ occurs twice in the papers and ‘$185$‘ four times.** Seven more fail-on-success instances, ‘8‘–‘14‘, all caused by PERFORMING: a re-statement MOVES a ledger row, a performed repair REMOVES it AND changes the receipt’s SOURCE, so the blast radius is wider and lands on a different set of files.* (node 66’s ‘r7255‘ — (1) *perform or hold the ‘27‘ ‘REPAIR-OWED‘ rows; each one either gets the repair performed — a normaliser, a width re-cut, a structural test — or a stated reason why it cannot be, in which case it is ‘MEANING-HELD‘ and you say what made it different from the ‘65‘ already held*. **Reachable negative: if performing the ‘27‘ turns up a repair form that DOES survive a content-intact rewording and reds on the converse, ‘70‘’s ‘r7251‘ negative is wrong in its strong form and ‘MEANING-HELD‘ is over-adopted — say so with the form, and the ‘272‘ held rows go back in question.** ‘r7243‘ stands: open an acoustic-sector item if one surfaces.). *Computes:* **23 checks, all pass, 0s measured.** ◆ **NOTHING COMPUTED PHYSICALLY; AND THIS RECEIPT MAKES NO LIVE READ OF THE LEDGER AT ALL, PROVED FROM ITS OWN SYNTAX TREE BY ‘(H)(1)‘.** **(A) THE MANIFEST IS THE SUBJECT** — digest-pinned; exactly ‘27‘ rows of 8 fields; every literal hashes to the ‘sha8‘ beside it, which is a claim about the KEYS and not about whether they are still live; all ‘27‘ name a receipt of this prefix that exists, and they sit in ‘10‘ receipts, so the unit of work was ‘10‘ reads. **(B) PERFORM OR HOLD** — the forms partition the ‘27‘ exactly (‘11‘/‘8‘/‘7‘/‘1‘); ‘0‘ rows HELD; fate and form agree with NO SLACK, the re-verdict being the only row whose key survives, because a performed repair retires the literal and a re-verdict does not; and every limb-two replacement is MEASURED to be a strictly shorter CONTIGUOUS substring of what it replaced, not a rewrite dressed as a shortening. **(C) WHAT THE CLASS ACTUALLY WAS** — the subforms reconcile (‘5‘ REDUNDANT, ‘4‘ AST-REPLACED, ‘2‘ COMPUTED-ELSEWHERE); deletions are more than a third of the population; and every REDUNDANT row NAMES the computation that made its quotation unnecessary, so ‘the capability was never missing‘ is on the record row by row. **(D) THE THREE SEMANTIC FORMS, MEASURED AND NOT ASSERTED** — all three property tests exist by name in their receipts’ syntax trees; NONE of them names any variable its receipt binds paper text to; and every string any of them compares against is a PYTHON IDENTIFIER, so there is no text in them for a rewording to move. ⇒ The computable class is ‘6‘, inside the pinned ‘2‘–‘6‘ and ON its top edge, which is declared as a band that was nearly wrong rather than a prediction that was right — ‘C67‘’s correction applied to this cycle’s own pre-registration. **(E) THE EXEMPLAR** — the short form occurs ‘2‘ times against the long form’s ‘1‘; and the row is shown to have been repaired by an ANCHORED SLICE, cut with ‘str.partition‘, which the operator does NOT key — where an ‘in‘ guard would have minted a new pin inside the repair meant to remove one. **(F) THE RE-CUT’S TWO STRENGTHS** — ‘4‘/‘3‘ recorded per row rather than claimed for the family; the bare digits measured at ‘2‘ and ‘4‘ occurrences in the papers; every re-cut replacement COMPILES as a regex. **(G)/(H) THE METHOD** — no ledger read, and the paper glob excludes the generated appendices, so no occurrence count is taken against a retired paper state read as live. *Bound:* **THE ORDER’S QUESTION HAS ONE ANSWER FOR EVERY ROW, AND IT IS THE ONE THE LEDGER’S OWN NUMBERS DID NOT PREDICT.** *‘cc66.170‘ called the paper half a rewriting job; ‘C67‘ showed it was a census; this row shows that what owed work remained was mostly REDUNDANCY. Nothing was held, so ‘MEANING-HELD‘’s population did not grow by a single row in a prefix that supplied ‘27‘ candidates for it.* ⇒ **AND THE PRE-REGISTRATION IS SCORED AS WRONG IN ITS STRUCTURE RATHER THAN ITS COUNTS:** *it sorted the ‘27‘ by what the repair WOULD BE, and ‘11‘ needed no repair at all; two rows filed under width re-cut were discharged as deletions. A pre-registration over repair FORMS is a weaker instrument than one over repair NECESSITY, and the next one will sort by the latter.* *Three gates of node ‘60‘’s ‘S2‘, two markers of its ‘S6‘ and one row of ‘corpus/exact_count_baseline.tsv‘ were repaired here under the standing rule that the breaking seat repairs — ‘60‘’s wording kept throughout, only conditions and notes this seat’s. Two questions are ROUTED TO ‘60‘ rather than decided: whether ‘S2‘’s section C stays a census or becomes a frozen manifest, and the gap the struck baseline row exposes in the ‘SCOPED‘ category — ‘SCOPED‘’s premise is that AUTHORSHIP of a stamp bounds the set, and it does not hold against REMOVAL.* *Order (2) — the ‘475‘ under the settled rule and both clause limbs —

is pre-registered and is next; nothing in the acoustic sector surfaced this cycle.* *Run:* `python3 receipts/P15_CR_cosmology/C68_all_twenty_seven_owed_rows_are_performed_and_none_is_held_so_the_owed_`

**P15R310** `C69_the_four_hundred_and_seventy_five_are_adjudicated_and_meaning_held_takes_none_of_them_so_t` [OK]

*'r7255''s order (2), answered in full — **ALL '475' UNVERDICTED KEYS OF THE PREFIX ARE ADJUDICATED: '413' 'DELIBERATE', '42' 'REPAIR-OWED', '10' 'ENCODING-OK', '8' 'ALT-OK', '2' 'NOT-A-PIN' — AND '0' 'MEANING-HELD'.** **THREE OF FOUR PINNED NUMBERS ARE OUTSIDE THEIR BANDS AND IT IS ONE ERROR: 'MEANING-HELD' was pinned at '38' per cent and took NONE, so its miss and 'DELIBERATE''s '86.9' are the same miss counted twice.** **AND THE PREDICTION THIS SEAT NAMED IN ADVANCE AS MOST LIKELY TO BE WRONG IS THE ONE THAT CAME CLOSEST — 'REPAIR-OWED' at '42' rows against a pinned '48', inside its band. The number hedged is the number got; the number stated without hedging is the number that collapsed.** ⇒ **THE CAUSE IS STRUCTURAL: '56.2' per cent of the '475' are IDENTITY pins by their CONTAINER or TIER — a frozen pre-registration, a label, a 'git show'-pinned state, a sibling receipt's printed line, a code line — and for an identity pin 'MEANING-HELD''s premise cannot be stated, because the string IS the finding. 'MEANING-HELD' was adopted for CONTESTED rows in dispute, so ITS SHARE IS A PROPERTY OF A DISPUTE AND NOT OF A PREFIX.** **And 'PREREG' is the finding of the sort: '87' rows, '18.3' per cent, the largest 'SOURCE' sub-kind in the prefix, pinning a receipt's OWN FROZEN PRE-REGISTRATION — a kind the baseline's target definition already admits and no measurement had used.** **Three grounds are repair forms ALREADY PERFORMED on '42' rows the ledger counted as owing them: a regression guard on retired wording, a value-extracting pattern, and 'L-249''s historical pin.** **The clause's HOLE is BOUNDED at '32.8' per cent and NOT MEASURED: this reading recorded a ground per row and not an identity-or-proposition flag, which its own prediction needed.** Two more fail-on-success instances, the SIXTEENTH in '70''s 'check_quote_pins' — which reds on adjudication itself, because it read the pinned-read population off the VERDICT COLUMN — and the SEVENTEENTH in this seat's own measurement tooling, where the same shape shrank the population file '475'→'227' silently.* (node 66's 'r7255' — (2) *take the '475' under the settled rule and both clause limbs; pre-register the 'MEANING-HELD' share, the 'REPAIR-OWED' share, and the fraction expected to fall in the clause's '44'-row-style hole*. 'r7243' stands: open an acoustic-sector item if one surfaces.). *Computes:* **24 checks, all pass, 0s measured.** ◆ **NOTHING COMPUTED PHYSICALLY; AND THIS RECEIPT MAKES NO LIVE READ OF THE LEDGER AT ALL, PROVED FROM ITS OWN SYNTAX TREE BY '(I)(1)' — the verdicts live in eight digest-pinned files a later cycle may supersede without reddening it.** **(A) THE SUBJECT** — all eight files pinned by digest; the measurement and the container resolution each describe exactly '475' keys; the six reading files carry '475' rows and NO KEY TWICE, so they PARTITION the population rather than merely cover it; they are the SAME keys the measurement describes; the keys sit in exactly '89' receipts, reconciling with '66''s count from the other side. **And '(A)(6)' asserts the operator's target/tier split while NAMING its one-key disagreement with the pre-registration's ledger-derived split — 'C12''s 'ordinary content physics on the observable leg', asserted at a site that reads BOTH a paper and another receipt, which is the single-row exhibit of this cycle's target finding.** **(B) THE RESULT** — the verdicts sum to the population; NOT ONE ROW is 'MEANING-HELD'; nothing carries an 'UNADJUDICATED' verdict of any kind. **(C) THE SCORE, WITH THE MISSES ASSERTED AS MISSES** — 'MEANING-HELD' OUTSIDE low and 'DELIBERATE' OUTSIDE high are gated as the SAME miss; 'REPAIR-OWED' INSIDE is gated together with the statement that it is the prediction the pre-registration named as most fragile; and the owed COUNT is gated to within six rows of a point estimate pinned before a single site was read. **(D) THE CAUSE, MEASURED** — every row carries a ground and the grounds sum to the population; MORE THAN HALF are identity pins by container or tier; 'PREREG' is gated as the largest identity ground; and '42' rows are gated as carrying a repair form already in place. **(E) ORDER (1)'s OPERATIONAL LIMIT ON A SECOND POPULATION** — '14' rows hold on limb one ONLY because the clause-aligned candidate multiplies the sites, so the limit '66' bound on one row is not a curiosity of that exemplar; and the owed class is gated to partition into exactly two shapes, a FILE pin and a shortenable span, with no row owed on a judgement about MEANING. **(F) WHAT CANNOT BE MEASURED** — the hole is gated as BOUNDED and not measured, with the gate's own text naming the reading plan's failure to record the column its prediction needed; and the bound's falling inside the pinned band is stated

as a bound and explicitly NOT claimed as a hit. **(I) THE METHOD** — no ledger read; and the reading is gated to be SPLIT into a ground file and five batch files so ‘decided‘ and ‘read‘ stay separable, ‘144‘ against ‘331‘. *Bound:* **THE ORDER ASKED FOR THREE SHARES AND THE USEFUL ANSWER IS THAT ONE OF THEM WAS THE WRONG QUESTION.** *‘MEANING-HELD‘’s share cannot be predicted from a prefix, because the verdict belongs to a DISPUTE: it was adopted where a finding had been argued and no string could carry it, and a virgin key has had no argument. The ‘475‘ answer that by taking none.* ⇒ **AND THE PREDICATE THIS SEAT WOULD PUT THE ‘272‘ HELD ROWS BACK UNDER IS NOW TWICE-DETERMINED:** *order (1) established that the scope line is ‘computable‘ and not ‘string‘; order (2) establishes that ‘MEANING-HELD‘ belongs to a dispute and not to a prefix. Together they say the ‘272‘ should be re-read against COMPUTABILITY — a smaller and better-defined job than re-reading them against the clause.* *Routed to ‘66‘ rather than settled here: the ‘check_quote_pins‘ repair, which is ‘70‘’s file; ‘60‘’s two questions from the interim (whether ‘S2‘’s section C stays a census, and the ‘SCOPED‘ category’s gap against REMOVAL); and whether ‘gen_pin_board‘ should join the workflow’s generator list now that it has a working ‘–check‘.* *‘r7243‘ stands; nothing in the acoustic sector surfaced in either order of this cycle.* *Run:* `python3 receipts/P15_CR_cosmology/C69_the_four_hundred_and_seventy_five_are_adjudicated_and_meaning_held_ta`

**P15R311** `C70_seventy_one_per_cent_of_the_two_hundred_and_seventy_two_held_rows_are_computable_so_the_neg` [OK]

*‘r7257‘’s orders (1) and (2), answered in full — **‘193‘ OF THE ‘272‘ CORPUS-WIDE ‘MEANING-HELD‘ ROWS ARE COMPUTABLE: ‘71.0‘ PER CENT. ‘66‘’s REACHABLE NEGATIVE NEEDED UNDER ‘10‘ AND MISSES BY ‘61‘ POINTS, SO THE OVER-ADOPTION IS A RULING AND NOT A SIX-ROW FOOTNOTE.** **AND THIS SEAT’S OWN BAND MISSES BY ‘29‘ POINTS IN THE SAME DIRECTION — pinned ‘30‘ per cent, band ‘20‘–‘42‘, with the floor deliberately set at twice ‘66‘’s line so that a small number would cost this seat the finding. The number is not small: wrong about the magnitude of the thing it was right about.** ⇒ **THE FINDING THAT IS NOT THE HEADLINE AND COSTS MORE THAN THE BAND: THE COST INVERTS. ‘INDEPENDENT COMPUTATION‘ — the expensive form, the one the order singled out — was pinned as the MINORITY at ‘30‘ per cent of the computable (‘25‘ rows) and is the MAJORITY at ‘67.4‘ (‘130‘ rows). ‘85‘ of those ‘130‘ need a quantity their own holder does not compute, so the re-verdict of the ‘272‘ is a BACKLOG of ‘85‘ and a census of ‘108‘ — the lesson of this seat’s own ‘27‘ arriving against it rather than for it.** **ORDER (2) HAS TWO READINGS AND BOTH ARE REPORTED: STRICT, where a receipt’s jurisdiction is what it opens, gives ‘34.7‘ per cent and MISSES; CAPABILITY, where only a QUANTITY the receipt does not compute bars a form, gives ‘56.0‘ and is INSIDE at its floor. They differ by exactly ‘41‘ rows and every one is an AST form held by a receipt that opens no source file — and the order’s own words bar a missing quantity, while ‘open‘ and ‘ast.parse‘ are within every receipt here.** **‘COMPUTABLE-NOWHERE‘, declared in advance to keep the headline from inflating, came back at ‘2.2‘ per cent against a band of ‘3‘–‘18‘: the exclusion took almost nothing out.** **The largest non-computable class is a GRAIN defect and not a meaning one: ‘32‘ of the ‘73‘ are FRAGMENT — a span too short to carry a claim at all, ‘takes the leaf’s‘ and ‘and no force‘ among them — which is the grain clause’s business rather than this verdict’s.** **And ‘25‘ false target disagreements were this seat’s own: ‘containers.py‘ joined ‘encoding=‘/‘errors=‘ constants onto the path, so ‘corpus/x_paper.tex/utf-8/replace‘ failed its ‘.tex‘ test and a PAPER read as a REGISTRY. The first pass would have routed ‘27‘ of them to another seat’s column; ‘2‘ survive the fix. Found by reading what the probe PRODUCED, not by re-reading its code — the fourth cycle running.**` (node 66’s ‘r7257‘ — (1) *measure the ‘272‘ ‘MEANING-HELD‘ rows against ‘can a receipt compute this‘; pre-register the computable share BEFORE starting and name which of the three forms each computable row is expected to take*; (2) *for each computable row, say whether the receipt that holds it CAN carry the form — a form that needs a quantity the receipt does not compute is not available to it*. **Reachable negative: under ‘10‘ per cent computable and ‘MEANING-HELD‘ is correctly adopted on the overwhelming majority, the over-adoption is a ‘6‘-row footnote, and the row’s clause narrows.** ‘r7243‘ stands; the prefix is RELEASED, the ‘272‘ being corpus-wide.). *Computes:* **52 checks, all pass, 0s measured.** ◆ **NOTHING COMPUTED PHYSICALLY; AND NO LIVE READ OF THE LEDGER, PROVED FROM THIS RECEIPT’S OWN SYNTAX TREE BY ‘(L)(1)‘ — the population is a ‘git show 2654cd0b:‘ read, because ‘MEANING-HELD‘ is precisely the column this cycle’s work would move.** **(A) THE SUBJECT** — three files digest-pinned; ‘272‘ rows in both the gather and the reading;

the pinned revision resolves AND is an ancestor of ‘HEAD‘; ‘272‘ is what the ledger CARRIED THERE and not what it carries now; the rows sit in ‘71‘ receipts, so the population is corpus-wide and not a prefix’s; and all ‘272‘ are tier ‘SENTENCE‘ by ‘70‘’s own ‘H4‘, so none is a token pin wearing the verdict. **(B) THE READING** — ‘233‘ distinct literals carry the ‘272‘, so (1) is answered once per CLAIM and (2) once per HOLDER; the form tally, the quantity classes and the reason codes are each gated as a FROZEN DICT against the pinned file; and every reason code is from the vocabulary declared in advance. **(G) THE CONVERSE AUDIT, WHICH IS THE RULE’S TEETH** — every one of the ‘193‘ names its REACHABLE CONVERSE; none is a bare label; and each names an EDIT TO CORPUS STATE rather than a state of affairs, because a predicate with no reachable converse is an assertion and not a computation. **(H) THE WITNESSES** — a claim counts as computable only if corpus code computes its quantity NOW, so all nine witness classes are RESOLVED BY CONTENT over the tree against a declared floor, and every class a computable row names is one that resolves. **(D)/(E) ORDER (1) SCORED** — the share gated at ‘71.0‘; gated as OUTSIDE this seat’s band on the high side BY NAME; the order’s negative gated as NOT FIRING with the distance stated; ‘COMPUTABLE-NOWHERE‘ gated as a third miss, LOW; the upper-bound tail reported so the headline hides nothing in the exclusion; and the per-form split gated with each band hit or missed in the gate’s own text. **(I) ORDER (2) SCORED ON BOTH READINGS** — the ‘41‘-row gap gated to be ENTIRELY AST forms whose holder opens no source; the AST carry rate gated as a miss on the strict reading and ‘100‘ per cent on the capability one, with the pre-registered claim that the mechanism would be JURISDICTIONAL stated as right about the mechanism and wrong about its size; and ‘85‘ rows gated as a BACKLOG rather than a census. **(K) THE PROBE’S OWN DEFECT** — containers gated to resolve with nothing ‘UNKNOWN‘, the surviving disagreements gated at ‘2‘ against the ‘27‘ first reported, and the cause gated against the fixed source. **(L) THE GUARD** — no live ledger path outside the ‘git show‘, every count a frozen dict, and the ordered work explicitly unable to exhaust the subject. *Bound:* **THE ORDER ASKED FOR A NUMBER THAT COULD COST IT ITS OWN RULING, AND THE NUMBER DEFENDS THE RULING WHILE DEMOLISHING THIS SEAT’S ESTIMATE OF IT.** *What the misses have in common is one thing: this seat underestimated how much of this corpus is ALREADY COMPUTED. ‘53‘ modules compute group and holonomy data, ‘46‘ compute spectra, ‘27‘ the two rates. The ‘272‘ were adopted against ‘could a string do this‘, and nobody asked the tree what it already knew.* ⇒ **AND THE USEFUL SHAPE OF THE ANSWER IS NOT THE SHARE BUT THE SPLIT:** *a census is what the ‘63‘ AST rows and the ‘45‘ carryable computations are; the other ‘85‘ are a backlog, and filing them as a census would be the same error the owed ‘27‘ exposed from the other side.* *Still routed and not settled here: the ‘check_quote_pins‘ repair, which is ‘70‘’s file; ‘60‘’s two questions (whether ‘S2‘’s section C stays a census, and ‘SCOPED‘’s gap against REMOVAL); whether ‘gen_pin_board‘ should join the workflow’s generator list; and now the ‘2‘ surviving target-column disagreements.* *‘r7243‘ stands; nothing in the acoustic sector surfaced.* *Run:* `python3 receipts/P15_CR_cosmology/C70_seventy_one_per_cent_of_the_two_hundred_and_seventy_two_held_rows_are_`

**P15R312** `C71_the_census_is_sixty_two_per_cent_ast_and_the_backlog_has_a_nearest_function_for_eighty_thr`
[OK]

*‘r7261‘’s orders (1) (a)/(b) and (2) , answered in full — **‘71‘ OF THE ‘85‘ BACKLOG ROWS HAVE A NAMED NEAREST *FUNCTION* — ‘83.5‘ PER CENT AGAINST A PINNED ‘82‘ AND A BAND OF ‘65‘–‘93‘, INSIDE. THE ORDER’S NEGATIVE NEEDED UNDER ‘25‘ AND MISSES BY ‘58.5‘ POINTS, SO ‘computable‘ IS A STATEMENT ABOUT WORK AND NOT ABOUT POSSIBILITY AND THE HEADLINE TAKES NO QUALIFIER.** **AND THE FINDING IS THE ‘14‘ THAT HAVE NONE: THEY ARE ONE HOLE AND NOT FOURTEEN GAPS — ‘13‘ WANT A ZERO-MODE, A KERNEL OR AN EQUIVARIANT INDEX, AND ‘0‘ OF THE ‘665‘ FUNCTION NAMES UNDER ‘computations/‘ COMPUTES ONE. The five modules the ‘ZM‘ witness class resolved to are top-level sympy scripts with no ‘def‘ in them at all.** ⇒ **WHICH IS WHAT THE DOMAIN-VERSUS-QUANTITY DISTINCTION, DECLARED BEFORE THE LOOK, WAS FOR: all ‘85‘ passed the domain-level witness BY CONSTRUCTION, so a high answer could not be credited to it — and the one block where it would have been unearned is the one block that failed.** **(a) The census records an AST form on ‘67‘ of ‘108‘ — ‘62.0‘ per cent, inside the pinned ‘58‘–‘72‘ and above the ‘58.3‘ no-movement floor — but ‘4‘ rows moved against ‘6‘ pinned, so the direction was right and the size was not; the ‘2‘ pinned and not taken are recorded with the reason, one of them because the AST form available was an absence-of-CODE claim WEAKER than the paper’s.** **AND A PRE-REGISTERED IDEN-*

*TITY FAILED: this seat pinned ruling (3)'s flag to leave '108'−'41'='67', 'C70''s strict count, and called it gate-checkable. IT IS '64'. The cause is structural — '3' of the '4' movers are held by receipts that open no source file, so taking a row to its CHEAPEST form bought an AST test and COST a read, while the 'INDEP' form it left needed only a capability those holders have.** ⇒ **SO A CHEAPER FORM CAN LOWER THE STRICT COUNT: the cheapest-form rule and ruling (3)'s flag interact in a way neither ruling anticipated, and the strict count is not invariant under the census but a function of which form each row records. Ruling (3)'s CONDITION holds — the count is recoverable at '64' — and what failed is the prediction of its value.** **(2) THE PAPER CARRIED TWO VALUE PAIRS FOR ONE QUANTITY AND A THIRD EXISTS THAT IS NOT A RIVAL: 'sec:refit-bound' reports the tilt at '0.998'/'0.954' and the '1p57' receipt's instrument run gives '0.9980'/'0.9542', while the 'sec:transmission' item CITING it carried '0.995'/'0.956'. l.2728 repaired to the computed pair; the response-model grid's '0.9915'/'0.9530' is a '132'-bin lensed fit over $\{}ell$'100'–'1287' and a different quantity, stated so nobody later fixes the paper to it.** **AND (b)'s ROUTED LEDGER ROW IS HANDED BACK RATHER THAN DONE: the target stays 'SOURCE', because the resolution '66' ruled on was this seat's resolver attributing a generator to every target name — 'src, p15, p07 = (open(f) for f in (INSTR, P15, P07))' gave 'src' the paper's filename and 'src' is the ACOUSTIC instrument. '70''s reading was right, as it was for 'B57', so the '27' first reported are '25' container-defect rows, '1' real and already re-ruled, and '1' a second defect of the same resolver — there was never a second real disagreement.** **And a gate caught it rather than a re-reading: applying the ruling turned 'check_quote_pins' RED on 'H3', which requires a PAPER row's literal to be in a non-appendix 'corpus/*.tex', and that sentence is in no paper raw or flattened. The resolver is repaired on both defects and 'r7257''s digest-pinned outputs are NOT regenerated under either — four rows would move and none moves the census or backlog.** **The blast radius is followed and reported: the route receipt pinned that sentence, so its budget moves '0.039'→'0.044' and its own multiple falls '7550'→'6692' — conclusion unchanged, its number smaller, and the receipt says so in its own text.**  (node 66's 'r7261' — (1) *work the '108' census, each row recording the form its own holder can carry, and report the '85' as a BACKLOG with what each needs: the quantity and which existing module is nearest to computing it; pre-register the AST share of the '108' and the number of the '85' with a nearest module first*; (2) *'CR_cosmology.tex' quotes the tilt refit as two value pairs, at l.2728 and l.912 — a full read of the passage before the edit*. **Reachable negative on (1): a nearest module for fewer than a quarter and 'computable' is about possibility rather than work, with that qualifier going into the headline in print.** 'r7243' stands.). *Computes:* **43 checks, all pass, 1.6s measured.** ◆ **NOTHING COMPUTED PHYSICALLY; THE POPULATION IS STILL A 'git show 2654cd0b:' READ AND THE TWO UNITS ARE SEPARATE FILES SO NEITHER BORROWS THE OTHER'S COUNT.** **(A) THE SUBJECT** — three files digest-pinned; '108'+'85' with NO KEY IN BOTH, so the two units PARTITION the '193' computable rows rather than merely cover them; the pinned revision still carries '272'; **and '(A)(5)' gates the LIVE DRIFT — '3' out and '1' in since the pin, live count '270', one departure a row this census counts — so the pinning is shown to be load-bearing rather than asserted to be.** **(B) (a) THE CENSUS** — forms gated as a frozen dict; the AST share gated INSIDE its band and ABOVE the no-movement floor; the mover count gated as a MISS against the pin in the gate's own text; every mover gated to record WHY as a clause; and 'was'≠'form' gated to hold exactly on the movers. **(C) RULING (3), AND THE FAILED IDENTITY GATED AS FAILED** — the flag count; every flagged row gated to be an AST form whose holder opens no source file; the strict count gated as '64' AND NOT the pre-registered '67'; and the cause gated arithmetically — '3' movers flagged, '67'−'64'='3'. **(D)/(E) (b) THE BACKLOG** — gaps as a frozen dict; the fraction gated inside its band; the negative gated as NOT FIRING with the distance stated; every named row gated to carry a quantity in words AND a module AND a function, and every 'NONE' row gated to carry both placeholders so none is half-filled; **and '(E)(1)' resolves all '22' modules and '30' functions BY AST and RAISES on a miss — which is how this receipt caught its own author transcribing '16' and '24' from a truncated listing.** **(F) THE HOLE, MEASURED OVER THE WHOLE TREE** — the 'NONE' rows' witness split; '13' gated as one capability; '0' of '665' function names gated to compute a zero-mode, a kernel or a Dirac index; and the five 'ZM' modules gated to contain no 'def' at all. **(G)/(H) ORDER (2)** — both sections gated to carry the SAME pair; the old pair gated ABSENT from the paper entirely; the third pair's arrays gated present on disk and named as a different quantity; the authority gated against the '1p57' receipt's computed '0.9980'/'0.9542'; the route receipt's literal and budget gated to have

followed; its own figure gated to have FALLEN with both numbers in its text; and the repair gated to have minted NO new ledger pin. **(I)** — the routed row gated at ‘PAPER‘ with its verdict untouched. *Bound:* **THE ORDER ASKED FOR TWO NUMBERS AND THE USEFUL ANSWER IS A THIRD THING: THE BACKLOG IS NOT ‘85‘ SCATTERED NEEDS BUT ‘71‘ ROWS A NAMED FUNCTION REACHES AND ONE MISSING CAPABILITY STANDING BEHIND THE REST.** *That is a smaller thing to order than a backlog of ‘85‘, and it is only visible because the nearest-module rule demanded a FUNCTION rather than a module and was bound before the look.* ⇒ **AND THE COST OF THE CYCLE IS THE IDENTITY, NOT THE BANDS:** *both pinned numbers landed, and the thing that broke was a relation this seat asserted BETWEEN two of 66’s own rulings. A cheaper form can cost a read, so the strict count’ cannot both be read as invariants.* *Routed rather than settled: the zero-mode capability, which is corpus-wide and not this prefix’s to open; whether the strict count stays a reported number now that it moves with the form; and ‘60‘’s two questions and ‘70‘’s ‘gen_pin_board‘ placement, unchanged.* *‘r7243‘ stands; nothing in the acoustic sector surfaced.* *Run:* `python3 receipts/P15_CR_cosmology/C71_the_census_is_sixty_two_per_cent_ast_and_the_backlog_has_a_nearest_fur`

`S19_a_markup_carrier_cannot_be_repaired_by_splitting_it_and_the_proof_is_a_live_gate_standing_on_an_arm_`
[OK]
*the ‘17‘ ‘REPAIR-OWED‘ rows of ‘L221_the_bridge‘ at their own asserting sites, split by introducing-commit parity; plus the ‘27‘ movable re-statement-remainder rows nobody held, re-stated under the vocabulary settled at ‘r7253‘ and the grain clause bound at ‘r7255‘; plus every ‘ALT‘ arm in the adjudicated ‘PAPER‘ cell through one new word-leading test; plus the papers themselves, read for ABSENCE only* ( **A MARKUP CARRIER CANNOT BE REPAIRED BY SPLITTING OR DISJOINING IT, AND THE PROOF IS A LIVE GATE IN THIS CORPUS.** *The shape this revision PRE-REGISTERED for the nine markup carriers — split the span at the emphasis into two bracketing conjuncts — admits the NEGATION of an identity claim, and its own truth table refutes it.* **And the same failure was then found in shipped work from the other half of the order: a two-arm disjunction over one sentences emphasis whose arm ONE is DEAD against every live paper and whose arm TWO begins with a finite verb and names NO SUBJECT, so the gate was being held up by an arm that would pass a paper about a different mode entirely.** ⇒ *So the DANGLING-PREDICATE test was built, and it ranks the repair forms in the REVERSE of the assumed order: ‘ALT-OK‘ ‘35.7‘ per cent against ‘EXTENDED‘s ‘5.4‘, a factor of ‘6.6‘.* **And the ‘MEANING-HELD‘ band broke: ‘50‘ per cent predicted, ZERO measured — the orders reachable negative FIRES, and the mechanism this line gave for it was WRONG.** *The share does not measure how settled a prefix is; it measures whether a prefixs findings are HISTORICAL, so ‘L221‘s ‘84.2‘ is withdrawn as a rate while standing as a figure.*). *Computes:* ‘35‘ gates. (1) the discarded split shape refuted by a ‘2‘x‘2‘ truth table run on sentences written in the receipt, so the control adds no paper key in the class it is about. (2) the dead arm, as an ABSENCE — the one claim a quotation cannot carry. (3) the dangling test controlled in BOTH directions, five positives and four negatives including a BARE LOCATOR that must not flag. (4) the repair forms ranked as an ORDERING rather than a pinned ratio, because the population is still being verdicted. (5) the three disjunctions whose every live arm dangles, PRINTED BY NAME. (6) the two counts reconciled: ‘27‘ re-stated, ‘3‘ repaired out of existence, ‘2‘ created, ‘26‘ live. (7) the costs counted beside the gain — ‘8‘ keys retired for ‘4‘, ‘2‘ of the ‘8‘ already adjudicated, and one repairs read now invisible to the operator. (8) the six repaired receipts read by PARSING, never by quoting their source *Bound:* ***THE DANGLING-PREDICATE TEST IS A FLAG AND NOT A VERDICT*** — *it finds spans that cannot name their subject, not spans that are wrong: ‘11‘ of the ‘27‘ dangle and only ‘3‘ rows moved.* **Not that the ‘11‘ routed repairs are right** — *each carries a patch and a reason and the holding seat decides.* **Not that ‘0‘ ‘MEANING-HELD‘ holds for the rest of the remainder** — *‘110‘ of the ‘138‘ are ‘EXTENDED‘ and wait on another lines re-cut.* **And the one ‘UNATTRIBUTED‘-editability ruling here is this lines own, stated and routed for ratification or reversal, and acted on in no file this revision.** *Run:* `python3 receipts/L_probability/S19_a_markup_carrier_cannot_be_repaired_by_splitting_it_and_the_proof_is_a_li`

`S20_the_adopted_dangling_test_has_three_error_modes_found_by_using_it_the_class_is_larger_than_it_repor`
[OK]
*the ‘13‘ routed patches of ‘r7258‘ at their own asserting sites — ‘11‘ in ‘L221_the_bridge‘ and two in other seats’ receipts; plus the ‘163‘ unverdicted keys of the nine prefixes claimed at ‘r7255‘, read key by key with the dangling test run BEFORE each verdict; plus the whole ‘2{,}773‘-key quote-pin*

*ledger, re-measured through a three-armed detector; plus the receipts' own sources, read by 'ast' for whether a repaired literal is KEYABLE at all* ( **THE DANGLING TEST 'r7257' ADOPTED IS A HEAD TEST OVER A CLOSED LIST, AND USING IT ON '163' FRESH KEYS FOUND THREE ERROR MODES IN IT — THE CLASS IS WRONG IN BOTH DIRECTIONS.** *A determiner FALSE POSITIVE ('That sentence constrains ...' names its subject), a TRAILING blind spot ('non-geometric but', which my own 'r7258' reply asserted dangles and the instrument cannot see), and a NEGATOR blind spot ('not need a lower bound').* ⇒ ***'15' false positives removed and '63' keys added that a head test cannot see: the class is '386', not '338'.*** **And '126' live 'DELIBERATE' rows stand on spans that name no subject, '104' of them in one prefix this seat does not hold** — *so adopting the test creates that backlog retrospectively, reported with its table and not acted on.* ⇒⇒ ***AND '7' OF THE '13' REPAIRS WERE INVISIBLE TO THE QUOTE-PIN OPERATOR UNTIL THE GATE SAID SO*** — *'0' new keys against '13' stale, because the repaired literal reached the paper through a helper argument rather than an 'in' test.* **That is the trap this seat published a warning about one revision earlier, caught by a gate and not by the seat that wrote the warning.** *And the '163' came back '64.4' per cent 'DELIBERATE' and '0.6' per cent 'MEANING-HELD', resembling 'P15_CR_cosmology' by 'L1' '52.1' against 'L221''s '125.2' on a test fixed before any key was read — so the order's reachable negative does NOT fire at a '60'-per-cent threshold fixed beforehand, and 'L221''s '84.2' stays withdrawn as a rate.*). *Computes:* '30' gates. (1) the detector states its own competence first: two-sided controls on all three arms, and the ADOPTED word list carried byte-for-byte so 'r7257''s and 'r7258''s figures are reproducible from the file rather than only citable. (2) the '16' new keys read out of their receipts' sources by 'ast' — each a folded string literal AND standing to the left of an 'in' — with the helper-only shape the first cut used run as the COUNTER, so the check is an instrument and not a restatement. (3) the two reference distributions read at 'origin/main' rather than at this tree, because item (1) retired '11' of 'L221''s 'REPAIR-OWED' rows and a reference this revision moved is a reference to itself. (4) the '60'-per-cent threshold quoted out of the pushed pre-registration. (5) the board discrepancy asserted as live and unforged. (6) **and the one repair this revision REVERTED is asserted reverted in the LEDGER as well as in the receipt** — *'B66' carries two markup carriers and only one is owed; repairing the second retired one of '70''s '64' 'EXTENDED' rows, which 'S18''s '(C)(4)' states in this line's own voice it would not do.* ⇒ ***'S18' went RED on the merged head and that is what caught it, so the defect is routed with its patch instead.*** *Bound:* ***TWO OF SEVEN PRE-REGISTERED BANDS BROKE AND ONE BROKE ON A CATEGORY ERROR OF MINE*** — *'EXTENDED' is not a defect class but '70''s marker for a repair it PERFORMED ('271' of '271' rows say so in their own reason), so a READING pass cannot reach it and predicting a share for it was wrong in kind.* *'REPAIR-OWED' overshot '28.2' against '2'–'20' for one measurable reason: order (3) moved '21' keys that would otherwise have read 'DELIBERATE', which is '12.9' per cent of the population — so the dangling test is a SHARE of the verdicting and not a gloss on it.* **And the three-armed test is still incomplete:** *'straddles the' ends on an ARTICLE and no arm sees it; it is in the ledger as 'REPAIR-OWED' with that reason and the fourth arm is routed rather than written in a hurry.* **And three instruments in one revision turned out to have a reach shorter than their claim, the gate seeing what the seat did not in all three:** *the quote-pin operator, the dangling head test, and 'corpus/acting_check.py', whose three clauses read 'sys.exit(1 if _fail else 0)' as carrying no check at all — so this receipt was ACTING on its failures and the assertion census counted it as a print statement with a filename.* *The defect class I routed 'B42' under at 'r7258' was also wrong — a markup carrier with emphasis on BOTH sides degenerates into a bare locator — so the bare-locator count in my own census is contaminated.* *Run:* `python3 receipts/L_probability/S20_the_adopted_dangling_test_has_three_error_modes_found_by_using_it_the_cl`

**P15R313** `C72_nineteen_of_the_fifty_nine_reachable_rows_are_reached_by_grafting_the_named_function_out_o`
[OK]

*'r7267''s orders (1) batch one and (3), answered — **'19' OF THE '59' 'INPUTS' ROWS ARE REACHED, AND THE OBSTRUCTION WAS NEVER THE MATHEMATICS: EVERY MODULE 'r7261' NAMED IS A TOP-LEVEL SCRIPT that prints its own report at import, so the function it names could not be called at all.** ⇒ **So the first deliverable is a LOADER and not a row: 'graft.py' takes the 'def' plus the transitive closure of the module-level names it reads, over the syntax tree, and executes only those — '14' of 'L-129''s '74' top-level statements and '3' of 'L-006w''s '66', and the gate on it is that it PRINTS NOTHING in a fresh interpreter.** **The '19' sit on the two largest modules and every one carries a reachable converse that comes back FALSE:*

*‘close‘ returns ORDER ‘81‘ with determinants over all three cube roots of unity (‘27‘ without the hinge ‘3‘-cycle); ‘det‘ cuts a determinant-one part of ‘27‘, unimodular, NON-ABELIAN, centre ‘3‘ — ‘Δ(27)‘ by four facts — equal element-for-element to the closure of the monodromy RATIOS with the ‘3‘-cycle; ‘inv_dim‘ gives exactly ‘1‘ meson invariant against ‘9‘ under no holonomy at all, and ‘1‘ of ‘27‘ in the baryon channel where the determinant-one part admits ‘3‘ and ‘SU(3)‘ admits ‘1‘; ‘ϵ‘ spans ONE dimension and is fixed by all ‘27‘ of the determinant-one part while ‘54‘ of ‘81‘ move it; ‘close‘ on ‘S_3‘ gives order ‘2‘ for one real root with a conjugate pair against all ‘6‘ for three real roots; and ‘twoM‘ is R-odd EXACTLY on the even ‘D‘.** **AND THE BATCH RETURNED A FINDING NOBODY ORDERED: ‘twoM‘ CARRIES TWO PARITIES AND THEY RUN OPPOSITE. The mass function is R-odd in the signed offset exactly when ‘D‘ is EVEN; the wall factor ‘√f/r‘, built back on the same ‘2M‘, is odd in the signed radius exactly when ‘D‘ is ODD. At the corpus’s own ‘D = 5‘ the first is EVEN and the second is ODD — and SIX receipts across two prefixes match one or the other as ‘odd‘ in a signed radial variable, with nothing in the corpus having computed either.** **(3) THE ‘14‘ ARE FILED AS A STATED LIMIT in this field’s own ledger and not as fourteen owed rows, and the measurement ‘66‘ attached to the ruling is MADE: ONE FUNCTION — a one-dimensional Dirac zero-mode solve on the wall profile — WOULD DISCHARGE ‘12‘ OF THE ‘13‘, and it is short work and not a project because THIS BATCH COMPUTED ITS ONE INPUT. The single open question is which measure ‘normalisable‘ is taken in, since ‘B34‘’s own finding puts the wall where ‘f ¡ 0‘ and the static ‘$L^2$‘ measure is not available there; ‘B1‘’s equivariant index is named as NOT discharged and stays inside the limit.** **‘40‘ ‘INPUTS‘ rows remain and the ‘12‘ ‘EXTEND‘ rows are behind all of them; the next batch is the three next-largest modules, ‘13‘ rows, taking the count to ‘32‘ of ‘59‘.**** (node 66’s ‘r7267‘ — (1) *the ‘71‘ rows with a named nearest function: the measurement says they are reachable, so reach them, the ‘59‘ that are a change of inputs first and then the ‘12‘ that are a short extension, cheapest first; nothing says do all ‘71‘ in one revision, so say how many landed and what the next batch is*; (3) *file the ‘14‘ as a stated limit naming the capability — a zero mode, a kernel or an equivariant index computed in a function — rather than as fourteen owed rows, and if the capability turns out to be a short piece of work rather than a project, say so*. Order (2) (the ‘108‘ census rows) stays BEHIND (1) and is untouched here. ‘r7243‘ stands.). *Computes:* **59 checks, all pass, 51s measured.** ◆ **NOTHING NEW COMPUTED PHYSICALLY — every number comes out of a function GRAFTED from ‘L-129‘ or ‘L-006w‘, so a drift in either moves every number here.** **(A) THE SUBJECT** — ‘BACKLOG.tsv‘ gated as ‘C71‘’s pin UNMOVED; ‘REACHED.tsv‘ and ‘STATED_LIMIT.md‘ digest-pinned; the gap split re-gated at ‘59‘/‘12‘/‘14‘ summing to the ‘85‘; the batch gated as EXACTLY the rows on the two largest modules and gated to BE the two largest; ‘REACHED.tsv‘’s keys gated identical to those backlog keys, so the batch is a subset BY KEY and not a re-derivation; every reached row gated to name the function the BACKLOG names; and ‘19‘+‘40‘=‘59‘ gated as a partition. **(B) THE GRAFT** — both modules gated to carry top-level ‘print‘ calls, so neither is importable; the retained-statement counts gated; the four named ‘L-129‘ functions gated present AND gated to be exactly the set the rows name; **and ‘(B)(6)‘ runs the graft in a FRESH INTERPRETER and gates its stdout EMPTY, which is the only form in which ‘the module’s report did not run‘ is a measurement.** **(C) THE NINETEEN** — all ‘19‘ claims gated TRUE and all ‘19‘ converses gated FALSE; every converse gated to be an EDIT in words rather than a label; every row gated to record what the computation RETURNED in numbers; ‘reach.py‘ gated to EXIT NON-ZERO on a converse that comes back true; **and the ‘3‘ rows whose ‘no curvature‘ half has no reachable converse inside ‘L-129‘’s representation gated to say so PER ROW rather than borrow the selection half’s.** **(D) THE NUMBERS** — ten gates, each against the grafted function and not against this receipt. **(E) THE FINDING** — ‘f‘ EVEN and ‘√f/r‘ ODD at ‘D = 5‘, NEITHER with a parity at ‘D = 4‘, the two parities gated OPPOSITE over ‘D = 4..9‘, and the ‘2M‘ the metric function is built on gated identical to ‘twoM‘ at the horizon radius. **(F) THE HOLDERS** — all twelve gated to define the loader at top level and to CALL it inside a ‘check‘, both from each file’s SYNTAX TREE; all twelve gated GREEN with the computed checks in them; **and every row’s literal gated STILL PRESENT — matched against each file’s string CONSTANTS, because two holders break one literal across adjacent fragments the parser joins and a text search reports a removed pin that is there.** **(G) THE ARITHMETIC** — the landed and remaining counts; the next batch gated at ‘13‘ rows, gated to be the next largest, and every one of its named functions resolved against the tree by ‘AST‘. **(H) ORDER (3)** — the limit gated to name all three forms of the capability and to name it as computed IN A

FUNCTION; gated into ‘INDEX_THEORY_LEDGER.md‘ with ‘OWED.md‘ gaining nothing; the table gated at exactly ‘13‘ rows with the fourteenth named; the ‘12‘-of-‘13‘ measurement gated as MADE; the ‘normalisable‘ condition gated as named; ‘B1‘ gated as explicitly NOT discharged; **and ‘(H)(7)‘ re-measures the limit against the tree as the receipt runs — no function under ‘computations/‘ computes a zero mode, a kernel, a null space or an equivariant index.** **(I) THE GUARD** — no string constant of more than three words sits on the left of an ‘in‘ test anywhere in this file, proved from its OWN syntax tree; every presence test goes through ‘_needle‘; and the subject files are digest-pinned so no number is read against a set that moved under it. *Bound:* **THE ORDER SAID ‘the measurement says these are reachable, so reach them‘, AND THE FIRST THING REACHING THEM FOUND IS THAT THE REACH WAS BLOCKED BY PACKAGING AND NOT BY MATHEMATICS.** *Nineteen rows asserted quantities that ‘L-129‘ and ‘L-006w‘ had been computing all along, at the wrong inputs, inside files nothing could import. The expensive part of this cycle was a forty-line loader; the rows after it were arithmetic.* ⇒ **AND THE SHAPE WORTH CARRYING IS THAT THE BATCH PAID FOR ITSELF TWICE:** *recovering the metric function from the mass ‘twoM‘ solves for is what produced the opposite-parity finding, and it is also what makes the zero-mode capability short work — the profile a Dirac solve needs is the factor that computation returns.* *Routed rather than settled: the opposite parities, in case a document states one as the other — measured, not acted on, in a prefix I have not read this cycle. ‘B61‘ still red on ‘main‘ via ‘range_paper.tex‘, so ‘main‘ is still not merged. ‘60‘’s two questions and ‘70‘’s ‘gen_pin_board‘ placement unchanged.* *Run:* `python3 receipts/P15_CR_cosmology/C72_nineteen_of_the_fifty_nine_reachable_rows_are_reached_by_grafting_the`

**P15R314** `C73_batch_two_measures_the_inputs_half_of_the_classification_and_only_four_of_thirteen_are_a_c`
[OK]
*‘r7275‘’s order (1) batch two — **LANDED ‘23‘ OF ‘59‘, REMAINING ‘36‘ — AND THE BATCH IS A RESULT RATHER THAN A COUNT: OF THE ‘13‘ ROWS SCHEDULED, ONLY ‘4‘ ARE A CHANGE OF INPUTS IN THE STRICT SENSE AND ‘9‘ ARE NOT.** ⇒ **‘r7275‘’s ruling (1) said ‘r7261‘ had classified ‘59‘ rows by what the function computes without asking whether the code was reachable; this says the same thing about the ‘INPUTS‘ half and says it PER ROW rather than per module.** The ‘9‘ are held with the reason, never dropped: ‘GEOMETRY-NOT-RECORDED‘ ‘4‘ (the row needs a metric or a structure the backlog did not name, and choosing one is a reading of the paper), ‘NESTED-FUNCTION‘ ‘3‘ (‘bead_frames.py: tau_of_r‘ is a closure defined INSIDE a matplotlib plotting routine, so it is not a top-level ‘def‘ at all), ‘OBJECTS-NOT-RECORDED‘ ‘1‘, ‘AXIS-ARGUMENT‘ ‘1‘. **And the nested-lifting capability is NOT built, on a measurement rather than a preference: of the ‘40‘ ‘INPUTS‘ rows outstanding before this batch, ‘37‘ name a TOP-LEVEL function and ‘3‘ name a nested one — and all three name the SAME one, so a capability for them is a special case wearing a general coat.** **THE FOUR TAKEN: ‘C3‘’s Nariai algebra through ‘A2‘’s OWN ‘lie_g‘ at a REBOUND metric — ‘lie_g‘ reads its metric from module level, so the change of inputs is a rebinding and not an argument, and handed $dS_2\{}times S^2$ it returns six Killing generators, the three $\{}mathfrak{so}(2,1)$ of them DERIVED from the embedding and each verified against its third row rather than quoted, with three reachable converses all FALSE; and ‘B30‘/‘B39‘/‘B40‘’s doublet through ‘L-104w‘’s OWN ‘sep‘ on a pair its own report NEVER FORMS — the two ends of ONE hinge, same azimuth and OPPOSITE HORN at $2\{}alpha$, one call on two points ‘pt‘ already builds, giving $s^2=-12\{}alpha^2$, timelike at every hinge azimuth.** ⇒ **AND THE CONVERSE IS THE FINDING: IT IS NOT A RADIUS AWAY BUT AT THE THROAT. At $\{}rho=\{}alpha$ the two ends COINCIDE and the interval is NULL, so ‘timelike‘ is a statement about the hinge sitting OUTSIDE the throat and the three receipts asserting it had no way to know that was the condition — the whole claim lives in ‘pt‘’s $\{}mathrm{sgn}\{}sqrt{\{}rho^2-\{}alpha^2}$.** **AND THE LOADER NEEDED A SECOND REPAIR TO REACH ‘A2‘ AT ALL, WHICH IS THE SAME RULING ONE LEVEL DOWN: ‘A2‘ builds its metric by ELEMENT ASSIGNMENT, and ‘graft.py‘ read a SUBSCRIPT TARGET AS A BINDING — keeping the last element assignment as the definition of ‘g‘ and dropping the ‘sp.zeros‘ that creates it, so the graft raised ‘NameError‘. A matrix built by element assignment is a reachability question nobody had asked, and several modules here build theirs that way. Subscript and attribute targets are now MUTATIONS, retained with the binding in source order, and the repair is gated ADDITIVE against batch one’s five functions.** (node 66’s ‘r7275‘ — (1) *batch two, as this seat scheduled it: ‘13‘ rows, ‘A2: lie_g‘ (‘5‘), ‘bead_frames: tau_of_r‘/‘tau_r‘ (‘4‘), ‘L104w: character‘/‘sep‘ (‘4‘), with the schedule explicitly not re-decided because it was gated against the syntax tree rather than against

the table*; and the standing order ‘r7271‘ — *‘CI‘ is the suite and is READ not duplicated, nothing long runs in the chat container, land in pieces and push whenever a piece is green, and report ‘landed‘ and ‘remaining‘ as the first line*. Order (2) (the ‘108‘) stays behind (1); then ‘r7275‘’s order (2) (the zero-mode function, measure ruled). ‘r7243‘ stands.). *Computes:* **37 checks, all pass, 7s measured.** ◆ **NOTHING NEW COMPUTED PHYSICALLY — every number comes out of a function GRAFTED from ‘A2‘ or ‘L-104w‘.** **(A) THE SUBJECT** — ‘BACKLOG.tsv‘ gated as ‘C71‘’s pin unmoved; both ledgers digest-pinned; the ‘INPUTS‘ class re-gated at ‘59‘; ‘4‘ TAKEN + ‘9‘ HELD gated to sum to the ‘13‘ scheduled; **and ‘(A)(5)‘ gates the two batches’ taken keys DISJOINT, so ‘19‘+‘4‘=‘23‘ counts nothing twice**; ‘23‘+‘36‘=‘59‘ gated as a partition; and every ledger row, taken or held, gated to be a key the BACKLOG carries. **(B) THE LOADER REPAIR** — ‘A2‘ gated to really build its metric by subscript assignment; ‘_bound_by‘ gated to return NOTHING for those statements while ‘_mutated_by‘ returns exactly one name, read off the loader’s own functions; the graft gated to produce a built ‘4x4‘; **‘(B)(4)‘ gates the repair ADDITIVE by re-grafting batch one’s five functions to the same statement counts**; and both grafts gated to print NOTHING in a fresh interpreter. **(C) THE FOUR TAKEN** — claims TRUE and converses FALSE; every converse gated to be an EDIT in words and every row to record numbers; six Nariai generators gated Killing and three converses gated NOT; **‘(C)(6)‘ gates the ‘so(2,1)‘ half as DERIVED, since ‘so21_static‘ verifies each pushforward against the embedding’s third row in its own body**; the hinge ends gated timelike at all three azimuths; **and ‘(C)(8)‘ gates the converse AT THE THROAT — interval ‘0‘, character ‘null‘**. **(D) THE NINE HELD** — four distinct reason codes gated to sum to ‘9‘; every reason gated to be a SENTENCE of at least twelve words so none is held on a word; ‘tau_of_r‘ gated to be nested and NOT top-level in the tree; the loader gated to REFUSE it; **and ‘(D)(5)‘ gates the capability decision as a MEASUREMENT — ‘37‘ top-level against ‘3‘ nested among the ‘40‘, all three the same function**. **(E) THE HOLDERS** — all four gated to define the loader at top level and CALL it inside a ‘check‘, from each file’s syntax tree; all four gated GREEN; every taken literal gated still a string CONSTANT of its holder; **and ‘(E)(7)‘ gates ‘B39‘ — whose other backlog row is held — to say so in its own source beside the loader**. **(F) THE GUARD** — no string constant over three words on the left of an ‘in‘ test, from this receipt’s own syntax tree; ‘_needle‘ present; and the arithmetic gated unable to exhaust its subject. *Bound:* **THE ORDER DID NOT RE-DECIDE A SCHEDULE BUILT FROM THE SYNTAX TREE, AND THE SCHEDULE WAS RIGHT WHILE THE CLASSIFICATION UNDER IT WAS NOT.** *Thirteen rows were scheduled on the module counts; four of them are what ‘INPUTS‘ says they are. The other nine are not harder — they are UNDERSPECIFIED, and the backlog recorded a function without recording the geometry, the objects, or whether the function is a top-level ‘def‘.* ⇒ **AND THE BATCH PAID ITS COST TWICE AGAIN:** *reaching ‘A2‘ forced the subscript repair, and the subscript repair is what makes any element-assembled matrix in this corpus graftable; and forming the pair ‘L-104w‘ never formed is what exposed the throat as the real condition on a claim three receipts carry.* *Routed rather than settled: the four ‘GEOMETRY-NOT-RECORDED‘ rows, which I will put as ONE question rather than four; ‘60‘’s ‘P03‘ ‘(H)(2)‘, still open and postdating ‘r7275‘’s ruling (4) table. ‘r7243‘ stands.* *Run:* `python3 receipts/P15_CR_cosmology/C73_batch_two_measures_the_inputs_half_of_the_classification_and_only_fou`

**P15R315** `C74_batch_three_takes_all_nine_and_roots_at_settles_the_identity_of_the_root_sets_which_is_the_` [OK]

*‘r7281‘’s order (1) batch three — **LANDED ‘32‘ OF ‘59‘, REMAINING ‘27‘ — NINE SCHEDULED AND NINE TAKEN, NOTHING HELD, which is the contrast with batch two’s ‘4‘ of ‘13‘ and the difference is a PROPERTY OF THE FUNCTIONS rather than of the rows.** ⇒ **All three of this batch’s functions are top-level AND take their inputs as ARGUMENTS, so the change of inputs is a CALL rather than a rebinding or a choice: ‘lie_g‘ reads its metric from module level and someone must decide which, while ‘roots_at(v)‘, ‘omega_power(z)‘ and ‘r_of_tau(t)‘ are handed theirs. So ‘INPUTS‘ has a sub-structure that is checkable from the syntax tree and predicts which rows hold.** **AND THE FINDING IS THAT ‘roots_at‘ SETTLES THE WORD ‘ONLY‘: ‘L-012w‘’s own report tabulates eleven values of $r_0$ with the designated root and its pair, and asked at the three values the designation runs over it returns the SAME ROOT SET every time — $\{}{1,-1,0\{}}$ — differing only in WHICH member is designated. ‘B14‘ asserts ‘distinguished ONLY by which root each takes as its hole‘, the ‘only‘ is the load-bearing word, and it is a statement about the SETS BEING IDENTICAL — which no string match can see and which ‘L-012w‘ never prints side by side. So the computation supplies the half of the sentence the receipt*

*had no way to check, and ‘only‘ is a predicate rather than a flourish: a fourth value whose root set differs breaks the agreement, which is the recorded converse.** **Two smaller ones beside it: ‘omega_power‘ carries its OWN converse, raising on anything that is not a cube root of unity so ‘quantises‘ has a built-in false case needing no constructed edit; and ‘r_of_tau(0) = 0.0‘ exactly against ‘0.638472‘ at ‘0.5‘, so ‘the cosmology it seeds begins at the branch point‘ is a statement about proper time zero. The horizon cubic is exactly ‘0‘ at each of its three roots and non-zero a seventh off each, which is ‘B34‘’s ‘vanishes at the horizons‘ as a computation.** **And the ‘S14‘ lesson was kept in front rather than behind: ‘(D)(7)‘ gates that none of the nine holders is a receipt ‘S14‘ pins by line number, checked BEFORE editing rather than after a trunk red said so.**
(node 66’s ‘r7281‘ — (1) *the remaining ‘36‘, in batches as this seat has been scheduling them from the syntax tree, the schedule explicitly not re-decided*; and ruling (1) settling the ownership rule — *a seat does not edit another seat’s receipt, the exception is whoever’s edit broke it* — with the offered reversal of ‘B2‘’s computed check DECLINED. Then ‘r7275‘’s order (2) (the zero-mode function, measure ruled). ‘r7243‘ stands.). *Computes:* **34 checks, all pass, 13s measured.** ◆ **NOTHING NEW COMPUTED PHYSICALLY — every number comes out of a function GRAFTED from ‘L-012w‘, ‘L-127‘ or ‘bead_conjugate‘.** **(A) THE SUBJECT** — ‘BACKLOG.tsv‘ gated as ‘C71‘’s pin unmoved; all THREE ledgers digest-pinned; the ‘INPUTS‘ class re-gated at ‘59‘; **‘(A)(4)‘ gates the three batches’ key sets PAIRWISE DISJOINT so ‘19‘+‘4‘+‘9‘=‘32‘ counts nothing twice**; ‘32‘+‘27‘=‘59‘ gated as a partition; every batch-three row gated to be a BACKLOG key naming the function the backlog names; and ‘(A)(7)‘ gates that this batch holds NOTHING against batch two’s nine held. **(B) THE GRAFTS** — each of the three functions gated to come from a bounded number of its module’s top-level statements; **all three grafts gated to PRINT NOTHING in a fresh interpreter**; and each gated to be a top-level ‘def‘ in its module from the syntax tree, which is WHY this batch holds nothing. **(C) THE NINE** — claims TRUE, converses FALSE, every converse an EDIT in words and every row recording numbers; three roots at each value each designating a different one; **‘(C)(5)‘ gates the ROOT SET IDENTICAL at all three, which is the load-bearing half of ‘only‘**; **‘(C)(6)‘ gates the converse REACHABLE — a fourth value whose set differs breaks it**; the cubic gated exactly zero at its roots and non-zero a seventh off; ‘omega_power‘ gated to return ‘0,1,2‘; **‘(C)(9)‘ gates that it RAISES off the cube roots, so its converse is built in**; and ‘r_of_tau‘ gated zero at proper time zero and non-zero away. **(D) THE HOLDERS** — all nine gated to define ‘_r7281‘ at top level and CALL it inside a ‘check‘ or ‘gate‘, from each file’s syntax tree; all nine gated GREEN; every literal gated still a string CONSTANT of its holder; **and ‘(D)(7)‘ gates that NONE of the nine is a receipt ‘S14‘ pins by line number**. **(E) THE GUARD** — no string constant over three words on the left of an ‘in‘ test, from this receipt’s own syntax tree; ‘_needle‘ present; the arithmetic gated unable to exhaust its subject. *Bound:* **THE ORDER DID NOT RE-DECIDE THE SCHEDULE AND THE SCHEDULE WAS RIGHT AGAIN, BUT THE USEFUL THING THIS BATCH PRODUCED IS NOT THE NINE ROWS — IT IS WHY THEY WERE NINE.** *Batch two took ‘4‘ of ‘13‘ and batch three ‘9‘ of ‘9‘, and the discriminator is a syntactic property: whether the named function takes its inputs as arguments or reads them from module level. The first is reachable without an adjudication; the second needs one.* ⇒ **AND ‘roots_at‘ IS THE CASE THAT SHOWS WHAT REACHING A ROW IS FOR:** *the sentence was already matched and already true, and the computation supplied the clause the holder could not check — the IDENTITY of the root sets, on which the word ‘only‘ entirely rests.* *Routed rather than settled: the four ‘GEOMETRY-NOT-RECORDED‘ rows, as one question at ‘r7281+cc66.1‘; and the ‘argument-versus-module-level‘ sub-structure of ‘INPUTS‘, offered for the register since it is cheap to measure over the remaining ‘27‘. ‘r7243‘ stands.* *Run:* `python3 receipts/P15_CR_cosmology/C74_batch_three_takes_all_nine_and_roots_at_settles_the_identity_of_the_r`

**P15R316** `C75_batch_four_takes_all_nine_and_the_loader_now_refuses_an_empty_table_it_used_to_hand_back_in` [OK]

*‘r7283‘’s order (1) batch four — **LANDED ‘41‘ OF ‘59‘, REMAINING ‘18‘ — NINE SCHEDULED AND NINE TAKEN ON FOUR ‘(module, function)‘ PAIRS, nothing held.** **AND THE PART THAT IS NOT A ROW COUNT IS THAT THE LOADER COULD HAND BACK AN EMPTY TABLE AND SAY NOTHING: ‘L50b‘ binds ‘ends = {}‘ and ‘L128b‘ binds ‘leg_wall = {}‘ at top level and fill them in a ‘for‘ loop, which is not a graftable statement kind — so the graft saw the binding, kept it, dropped the loop, and returned an EMPTY dict. A row taken through ‘ends‘ would have been VACUOUS RATHER THAN WRONG, and that is the worse of the two failures because it PASSES.** ⇒ **‘graft.py‘ now refuses a kept binding to an empty container*

*that a non-graftable statement fills, naming the name and the line; the refusal is gated ADDITIVE, so no landed row is withdrawn by it. And the measurement that says not to build the loop-keeping capability instead: of the ‘30‘ rows outstanding after this batch the named function grafts ON ITS OWN in ‘23‘, three of the other seven read a module-level name nothing graftable binds, four are the nested-function limit already filed at ‘r7267‘ (3), and NOT ONE needs a filled container. The six ends are rebuilt from ‘L50b‘’s own ‘theta‘, ‘A‘ and ‘sq3‘ and GATED on ‘L50b‘’s own substrate equation, so the rebuild is checked rather than trusted.** **AND A REACHABLE CONVERSE SHARPENED A PREDICATE RATHER THAN FAILING A ROW: ‘The constant-$\{}tau$ slice is exactly flat‘ read off the SCALAR curvature would have PASSED on the Schwarzschild spatial slice, whose $R$ is ‘0‘ and whose Kretschmann is $6r_s^2/r^6$. The converse did not fail the row — it found the predicate reading a quantity that does not separate flat from Ricci-flat, so the predicate reads the invariant and that slice is the recorded converse.** **And both doubts raised before the batch resolve in the rows’ favour: ‘key_of‘ with ‘apply‘ reaches BOTH halves of ‘is a discrete horn swap‘ — the swap as three 2-cycles each joining one hinge’s two ends, and its order ‘2‘ inside a group of order ‘12‘ — and ‘ruling‘ is the quadric’s two straight-line families with the converse named in the module’s OWN docstring.** **ONE HOLDER IS WITHHELD AND ROUTED RATHER THAN EDITED: the rate-rule ‘P15‘ receipt carries a standing gate asserting that it COMPUTES NOTHING, ‘which is what the order asked for‘. A computed check would make that gate’s own label false while its predicate still passed — the store-live-state shape for the sixth time — and ‘r7283‘ (1) is a LATER order than the one that gate cites, so the conflict is 66’s to rule and not this seat’s to resolve by editing another seat’s claim.**  (node 66’s ‘r7283‘ — (1) *the remaining ‘27‘, in batches as this seat has been scheduling them from the syntax tree, the schedule explicitly not re-decided a fourth time*; and ruling (3) on the four ‘GEOMETRY-NOT-RECORDED‘ rows — *do not CHOOSE a metric, READ which one the paper states at the row’s own locus, and if it states none the blocker is a GAP IN THE PAPER and is routed as that finding*, which is the next piece and not this one. Then ‘r7275‘’s order (2) (the zero-mode function, measure ruled). ‘r7243‘ stands.). *Computes:* **48 checks, all pass, 41s measured.** ◆ **NOTHING NEW COMPUTED PHYSICALLY — every number comes out of a function GRAFTED from ‘L128b‘, ‘L50b‘, ‘r7111‘’s ‘geometry‘ or ‘panelA_reassign‘.** **(A) THE SUBJECT** — ‘BACKLOG.tsv‘ gated as ‘C71‘’s pin unmoved; all FOUR ledgers digest-pinned; the ‘INPUTS‘ class re-gated at ‘59‘; **‘(A)(4)‘ gates the four batches’ key sets PAIRWISE DISJOINT so ‘19‘+‘4‘+‘9‘+‘9‘=‘41‘ counts nothing twice**; ‘41‘+‘18‘=‘59‘ gated as a partition; every batch-four row gated to be a BACKLOG key naming the function the backlog names; and ‘(A)(7)‘ gates the outstanding work at ‘18‘ ‘INPUTS‘ plus ‘12‘ ‘EXTEND‘, which is the denominator the capability measurement uses. **(B) THE GRAFTS** — five functions each gated to come from a bounded number of their module’s top-level statements; **gated to PRINT NOTHING in a fresh interpreter, including the one that would otherwise write a figure to disk**; and each gated to be a top-level ‘def‘ from the syntax tree. **(C) THE REFUSAL** — **‘(C)(1)‘ gates from each module’s OWN syntax tree that it binds the table as an EMPTY container and fills it in a statement the graft does not keep**; **‘(C)(2)‘ gates that the loader now REFUSES both by name**; **‘(C)(3)‘ gates the capability measurement — ‘23‘ of ‘30‘ graft alone, ‘7‘ fail on causes already filed, ‘0‘ need a filled container**; ‘(C)(4)‘ gates the six-end rebuild against ‘L50b‘’s own substrate equation; **and ‘(C)(5)‘ gates the refusal ADDITIVE by re-grafting four functions the earlier batches took**. **(D) THE NINE** — claims TRUE, converses FALSE, every converse an EDIT in words and every row recording numbers; the plaquette gated the identity in both orders; **‘(D)(5)‘ gates flat NOT trivial — the winding-one holonomy is a primitive cube root of unity**; the non-diagonal converse gated to break it; the horn swap gated a fixed-point-free involution of order ‘2‘ in a group of ‘12‘; **‘(D)(8)‘ gates the continuous converse — ‘12‘ distinct keys in ‘12‘ steps, no closure**; the transposition gated ONE step at three stations and TWO at five; the constant-$\{}tau$ slice gated Ricci, scalar AND Kretschmann zero with ‘10‘ of ‘27‘ Christoffels non-zero; **‘(D)(11)‘ gates the sharpening — the Schwarzschild spatial slice has ‘R=0‘ and $K \{}neq 0$, so a predicate reading the scalar would have passed it**; the seam gated finite where ‘r=0‘ diverges; both ruling families gated on-quadric, straight and NULL to ‘1e-12‘; and the at-rest slicing gated to meet every line of both families exactly once where a slicing by planes through the axis does not. **(F) THE HOLDERS** — **‘(F)(2)‘ gates the ONE withheld holder BY ITS OWN STANDING CLAIM, so the routing is a measurement rather than an omission**; the other seven gated to define ‘_r7283‘ at top level and CALL it inside a ‘check‘ or ‘gate‘; all seven gated GREEN; every literal gated still a string CONSTANT of its holder; **and ‘(F)(8)‘ gates that none of the eight is a receipt ‘S14‘

pins by line number**. **(G) THE GUARD** — no string constant over three words on the left of an ‘in‘ test, from this receipt’s own syntax tree; ‘_needle‘ present; the arithmetic gated unable to exhaust its subject. *Bound:* **THE SCHEDULE WAS RIGHT A FOURTH TIME, BUT WHAT THIS BATCH IS WORTH IS THE REFUSAL AND NOT THE NINE ROWS.** *Every earlier failure of this loader was LOUD: a ‘NameError‘, a ‘GraftError‘, a missing name. This one was silent and it passed — an empty dict in place of a six-element table, with every downstream ‘all(...)‘ over it vacuously TRUE.* ⇒ **A GATE THAT CANNOT FAIL AND A GRAFT THAT RETURNS AN EMPTY CONTAINER ARE THE SAME DEFECT WEARING DIFFERENT CLOTHES, AND THE CORPUS ALREADY HAS AN INSTRUMENT FOR THE FIRST.** *The second now has one too, and the measurement beside it says the capability the hazard suggests building is needed by none of the ‘30‘ rows left.* *Routed rather than settled: the one withheld holder, whose standing ‘computes NOTHING‘ gate and ‘r7283‘ (1) cannot both hold; and ‘r7283‘ ruling (3)’s paper read for the four ‘GEOMETRY-NOT-RECORDED‘ rows and the one ‘OBJECTS-NOT-RECORDED‘ row, which is the next piece. ‘r7243‘ stands.* *Run:* `python3 receipts/P15_CR_cosmology/C75_batch_four_takes_all_nine_and_the_loader_now_refuses_an_empty_table_it`

**P15R317** `C76_the_five_held_rows_are_read_at_their_own_loci_in_four_papers_and_four_of_them_are_not_metri`
[OK]
*‘r7283‘’s ruling (3) — **LANDED ‘46‘ OF ‘59‘, REMAINING ‘13‘ — THE FIVE ROWS HELD AT ‘r7275‘ FOR WANT OF A GEOMETRY ARE READ AT THEIR OWN LOCI AND ALL FIVE TAKEN, AND THE RULING’S PREMISE MOVED UNDER THE READ IN TWO WAYS.** **First, the loci are in FOUR papers and not one: the ruling named the geometric core and it carries exactly one of the five — ‘B1‘ and ‘B42‘ are pinned in the boundary paper, ‘P11‘ in the matter sector paper, ‘B11‘ in the circle paper.** **Second, and this is the finding: FOUR OF THE FIVE ARE NOT METRIC CLAIMS AT ALL, so no metric was ever missing. They are statements about which GROUPS exist — $\{}mathfrak{su}(3)\{}not\{}subset\{}mathfrak{so}(5,1)$; the gravitational handedness carried by the DISCRETE coset $\{}mathrm{O}(5,1)\{}setminus\{}mathrm{SO}_0(5 the absolute $\{}eta(X,X)=0$ fixing the group with a substrate point’s polar hyperplane cutting a $\{}mathrm{dS}_4$ of isometry $\{}mathrm{SO}(4,1)$ — and ‘lie_g‘, the function the backlog named for all four, tests whether a VECTOR FIELD is Killing for a METRIC. Worse than stated when measured: its ONLY parameter is ‘xi‘, the vector field, and the metric it differentiates is read from module level, so the change of inputs the backlog promises is a change of VECTOR FIELD — and no vector field is a group.** ⇒ **A CHANGE OF INPUTS CANNOT REPAIR A CHANGE OF KIND. The hold code ‘GEOMETRY-NOT-RECORDED‘ is WITHDRAWN for all four in favour of ‘FUNCTION-IS-THE-WRONG-KIND‘, which is a reclassification of the BACKLOG and not a hold on the row: all five are computed exactly, which is what backs it.** **And the ruling’s GAP branch fires nowhere — every one of the five loci states what its row needs, in one sentence, in the paper where the literal is pinned. The defect was this seat’s and the backlog’s, never the papers’.** **‘B11‘ lands as an ordinary ‘INPUTS‘ row and its hold code was RIGHT: ‘character‘ is exactly the right kind, the circle paper names the three loci as $f=0,1,2$, and ‘character(1-f)‘ returns three DISTINCT characters where three equally spaced values return two — so only this seat’s SEARCH was wrong, not the code.** **AND THIS BATCH MADE AND CAUGHT ‘C75‘’s VACUOUS-PASS BUG ONE LAYER UP: two converses were first written ‘lambda: not GC.¡fn¿‘ WITHOUT the call, and ‘not ¡function¿‘ is always ‘False‘ — which is exactly what a converse must return, so the rows PASSED for the wrong reason. Every predicate slot is a NAMED function now, so a missing call raises, and the ledger asserts at import that no slot is a lambda and every slot returns a bool.**** (node 66’s ‘r7283‘ — ruling (3) *do not CHOOSE a metric, READ which one the paper states at the row’s own locus, the full-turn-per-paper rule applying; and if the paper states none the blocker is a GAP IN THE PAPER and is routed as that finding*, with order (1) *fold the four geometry rows and the one objects row into whichever batch reaches their papers*. Then ‘r7275‘’s order (2) (the zero-mode function, measure ruled). ‘r7243‘ stands.). *Computes:* **48 checks, all pass, 17s measured.** ◆ **NOTHING NEW COMPUTED PHYSICALLY — every number is the dimension of an algebra of a form, a root count, a determinant, a signature, or a call to ‘L-104w‘’s own ‘character‘.** **(A) THE SUBJECT** — ‘BACKLOG.tsv‘ gated as ‘C71‘’s pin unmoved; all FIVE ledgers digest-pinned; the ‘INPUTS‘ class re-gated at ‘59‘; **‘(A)(4)‘ gates the five batches’ key sets PAIRWISE DISJOINT so ‘19‘+‘4‘+‘9‘+‘9‘+‘5‘=‘46‘ counts nothing twice**; ‘46‘+‘13‘=‘59‘ gated as a partition; **and ‘(A)(6)‘ gates that every row of this batch was HELD at ‘r7275‘, so it takes nothing new and discharges what was owed**. **(B) THE PAPER READ** — **‘(B)(1)‘ gates the five loci into

FOUR distinct papers with the geometric core carrying exactly one**; **‘(B)(2)‘ gates every literal READ out of the paper it is attributed to, whitespace-normalised the way the holders read it — a raw match MISSED ‘B11‘’s sentence, which is broken across a line, and the gate failing on exactly that row is what found it**; each paper’s statement gated to be a sentence of at least ten words so no row is settled on a phrase; and the three papers’ own sentences gated present through ‘_needle‘. **(C) THE RECLASSIFICATION** — ‘4‘ of five gated ‘FUNCTION-IS-THE-WRONG-KIND‘ and one still ‘INPUTS‘; all four gated to name ‘lie_g‘; **‘(C)(3)‘ gates from ‘lie_g‘’s own syntax tree that its ONLY parameter is the vector field and that it reads ‘g‘ from module level, which is worse for these rows than this seat had stated**; the four reaching computations gated by dimension (‘so(5,1)‘=‘15‘, ‘so(4,1)‘=‘10‘, ‘so(5)‘=‘10‘, ‘so(6)‘=‘15‘); and every generator gated to lie in the algebra of the form it is built from. **(D) THE FIVE** — claims TRUE, converses FALSE, every converse an EDIT in words and every row recording numbers; ‘B2‘’s eight roots gated into TWO length classes with no ‘A2‘; **‘(D)(5)‘ gates the converse ONE DIMENSION UP — ‘D3‘’s twelve roots of one length DO contain an ‘A2‘, so the obstruction is the ROOT SYSTEM and not a dimension count**; every generator of ‘so(5,1)‘ gated TRACELESS and the parity gated in ‘O(5,1)‘ with ‘det -1‘; **‘(D)(7)‘ gates the circle the boundary paper says a connected group does supply — period $2\{}pi$, inside the identity component**; the polar hyperplane gated ‘(4,1)‘ at a substrate point and ‘(5,0)‘ inside the cone; **and ‘(D)(10)‘ gates the three loci PARSED OUT OF the circle paper’s own sentence rather than transcribed from it, with the transcription checked against the read — ‘check_unread_figure‘ caught the first version of this receipt attributing those values to the paper while opening nothing, and the gate’s remedy allows dropping the attribution where READING the figure was the stronger half and cost four lines**; and three equally spaced values gated to carry only two characters. **(E) THE VACUOUS-PASS BUG** — **‘(E)(1)‘ gates no predicate slot a lambda from the LIVE objects rather than the source**, ‘(E)(2)‘ gates every slot returning a ‘bool‘, ‘(E)(3)‘ gates the ledger asserting both AT IMPORT so a later row cannot forget it, and ‘(E)(4)‘ gates it as the same defect ‘C75‘ refused in the loader. **(F) THE HOLDERS** — all five gated to define ‘_r7283r3‘ at top level and CALL it inside a ‘check‘; all five gated GREEN; every literal gated still a string CONSTANT of its holder; **‘(F)(7)‘ gates that ‘B1‘ IS one of ‘S14‘’s five line-pinned sites and that the pin was checked BEFORE the edit — ‘S14‘’s own source now says the line is provenance and that the measurement does not depend on it, and ‘computations‘ is in its scratch-tree symlink list**; **and ‘(F)(8)‘ runs ‘S14‘ itself and gates it GREEN with ‘B1‘ edited**. **(G) THE GUARD** — no string constant over three words on the left of an ‘in‘ test, from this receipt’s own syntax tree; ‘_needle‘ present; the arithmetic gated unable to exhaust its subject. *Bound:* **THE RULING ASKED THE RIGHT QUESTION OF THE WRONG LAYER, AND READING THE PAPERS IS WHAT SHOWED WHICH.** *It inverted this seat’s hold reason correctly — the rows were not blocked for want of a metric — and then expected the answer to BE a metric. It is not: four of the five claims have no metric in them, and the function the backlog named cannot express them at any inputs.* ⇒ **SO THE DEFECT WAS ONE LAYER BELOW BOTH THE HOLD AND THE RULING: the backlog recorded a NEAREST FUNCTION by domain — ‘A2‘ is about the substrate’s symmetry, and so are these four rows — without checking that the function and the claim are the same KIND of object. That is the third distinct way this classification has failed at per-row resolution, after batch two’s unrecorded geometry and batch three’s refuted syntactic predictor, and it is the only one of the three that no change of inputs could ever repair.** *Routed rather than settled: nothing from this batch. The one item still with 66 is ‘C75‘’s withheld holder, whose standing ‘computes NOTHING‘ gate and ‘r7283‘ (1) cannot both hold. ‘r7243‘ stands.* *Run:* `python3 receipts/P15_CR_cosmology/C76_the_five_held_rows_are_read_at_their_own_loci_in_four_papers_and_four_`

**P15R318** `C77_batch_six_takes_all_six_and_the_one_wrong_kind_row_among_them_is_wrong_by_quantifier_becaus` [OK]

*‘r7285‘/‘r7287‘’s order (1) batch six — **LANDED ‘52‘ OF ‘59‘, REMAINING ‘7‘ — SIX ROWS ON SIX FUNCTIONS, ALL SIX TAKEN.** **AND THE ONE WRONG-KIND ROW AMONG THEM FAILS BY *QUANTIFIER*, WHICH IS A DIFFERENT DEFECT FROM BATCH FIVE’S FOUR. ‘B7‘’s own source says its literal is about $\{}mathfrak{su}(3)$ being no isometry of the non-compact substrate — the same claim ‘P11‘ carries — and ‘killing_residual‘ WORKS, precisely: exactly one of ‘probe2‘’s four coordinate vector fields satisfies the Killing equation and $\{}partial_u$ is refuted outright. But a residual test is EXISTENTIAL-REFUTING, settling one candidate at a time, while ‘su(3) is no isometry‘ is a NON-EXISTENCE over every embedding of an eight-dimensional algebra, and no finite number of non-vanishing residuals establishes one.**

⇒ ***So ‘lie_g‘ could not express ‘P11‘’s claim AT ALL, while ‘killing_residual‘ can express the negation of ‘B7‘’s one candidate at a time and still never reach the claim. Two defects wearing one code, and this is the sharper: not the wrong object, the wrong quantifier.** **AND THE EMPTY-CONTAINER REFUSAL ‘C75‘ BUILT FIRED ON A SECOND INSTANCE FOUND BY USE: ‘L091x‘ binds ‘GROUPS = []‘ and fills it with top-level ‘.append(...)‘ calls — a DIFFERENT statement kind from the ‘for‘ loop that prompted the guard — so the loader refused it by name instead of handing back an empty list. A guard written against a hazard found by inspection, firing on one found by use, is the only evidence a guard was worth building.** **And the ledger’s own ‘bool‘ guard, added at batch five to catch a converse written without its call, caught two acoustic predicates returning ‘np.bool_‘ — true, but not a ‘bool‘.** **The other five land as ordinary ‘INPUTS‘ rows: the deck ‘Z_3‘ abelian so its irreducibles are three one-dimensional characters against the walls’ ‘S_3‘’s 1,1,2; six routes whose two arcs partition the three walls as a 1–2 split where four hinges give (1,3) and (2,2); ‘G_su‘ returning baryon 1, diquark 0, meson 1 against ‘G_rigid‘’s 1,0,3; $D_M$ by two routes naming one number to $10^{-9}$ relative where the other rate moves it; and $\{}ell_A=301.41$ at the CMB $H_0$ against $296.72$ at the local one with $r_s$ pinned to $2.8\{}times10^{-11}$ relative — the acoustic module’s own RESULT 1 and RESULT 2 reproduced through the functions the rows name.*** (node 66’s ‘r7285‘ (1) and ‘r7287‘, both unchanged — *the remaining rows, in this seat’s own batches from the syntax tree, with the geometry and objects rows folded in under the ‘r7283‘ ruling*. Then ‘r7275‘’s order (2) (the zero-mode function, measure ruled, one-cycle stop). ‘r7243‘ stands.). *Computes:* **54 checks, all pass, 27s measured.** ◆ **NOTHING NEW COMPUTED PHYSICALLY — every number comes out of a function GRAFTED from ‘L091x‘, ‘L128‘, ‘L130‘, ‘CR_acoustic_scale_and_hubble‘ or ‘probe2‘, or out of batch five’s group computations.** **(A) THE SUBJECT** — ‘BACKLOG.tsv‘ gated as ‘C71‘’s pin unmoved; all SIX ledgers digest-pinned; the ‘INPUTS‘ class re-gated at ‘59‘; **‘(A)(4)‘ gates the six batches’ key sets PAIRWISE DISJOINT so ‘19‘+‘4‘+‘9‘+‘9‘+‘5‘+‘6‘=‘52‘ counts nothing twice**; ‘52‘+‘7‘=‘59‘ gated as a partition; **and ‘(A)(7)‘ gates that the ‘7‘ remaining have a SHAPE — ‘4‘ are ‘bead_frames‘’ nested functions, which ‘C73‘’s ‘(D)(5)‘ gated the decision NOT to build a capability for, and ‘3‘ refuse on a module-level name nothing graftable binds, so the ‘12‘ ‘EXTEND‘ rows are the next real tranche**. **(B) THE GRAFTS** — six functions each gated to come from a bounded number of their module’s top-level statements; **gated to PRINT NOTHING in a fresh interpreter**; each gated a top-level ‘def‘ from the tree. **(C) THE REFUSAL FIRING ON A SECOND INSTANCE** — **‘(C)(1)‘ gates from ‘L091x‘’s own syntax tree that ‘GROUPS‘ is bound EMPTY and filled by ‘.append(...)‘ calls the graft does not keep**; **‘(C)(2)‘ gates that the loader REFUSES it by name**; and ‘(C)(3)‘ gates from the function’s own signature that the row does not need it, the table being an argument. **(D) THE SIX** — claims TRUE, converses FALSE, every converse an EDIT in words and every row recording numbers; the deck gated abelian with dimensions 1,1,1 against ‘S_3‘; **‘(D)(5)‘ gates that exactly ONE irreducible is trivial for either group, so triviality does not discriminate and the DIMENSIONS are what the clause turns on**; the routes gated to partition at every pair with **‘(D)(7)‘ gating the four- and five-hinge converses**; ‘G_su‘ gated 1,0,1 against ‘G_rigid‘’s 1,0,3 with **‘(D)(9)‘ gating ‘no force‘ as the FINITENESS and a stated limit rather than a false case**; $D_M$ gated equal by two routes and moved by the other rate; and $\{}ell_A$ gated at the measured scale with **‘(D)(13)‘ printing the measured $r_s$ difference beside its verdict, a RELATIVE tolerance after the first draft used an absolute one on a dimensioned quantity and the gate was the thing that was wrong**. **(E) THE WRONG QUANTIFIER** — **‘(E)(1)‘ READS ‘B7‘’s own source for what its literal is about rather than inferring it**; ‘(E)(2)‘ gates that ‘killing_residual‘ works and precisely; **‘(E)(3)‘ gates the quantifier argument**; and ‘(E)(5)‘ gates that the code now covers TWO distinct defects, four of one kind and one of the other. **(F) THE LEDGER GUARD** — every slot gated to return a plain ‘bool‘ from the LIVE objects, and the ledger gated to assert it AT IMPORT. **(G) THE HOLDERS** — **‘(G)(2)‘ gates that the SAME holder ‘C75‘ withheld is withheld again for the same reason, by that receipt’s own standing claim**; the other five gated to define ‘_r7283b6‘ at top level and CALL it inside a ‘check‘ or ‘gate‘; all five gated GREEN; every literal gated still a string CONSTANT of its holder; and ‘(G)(8)‘ gates that none of the six is one ‘S14‘ pins by line number. **(H) THE GUARD** — no string constant over three words on the left of an ‘in‘ test, from this receipt’s own syntax tree; ‘_needle‘ present; the arithmetic gated unable to exhaust its subject. *Bound:* **TWO GUARDS BUILT IN THE LAST TWO BATCHES BOTH FIRED IN THIS ONE, ON INSTANCES NEITHER WAS WRITTEN FOR, AND THAT IS THE RESULT RATHER THAN THE SIX ROWS.** *The empty-container refusal was written against

a ‘for‘ loop and caught a run of ‘.append(...)‘ calls; the ‘bool‘ guard was written against a missing function call and caught a ‘numpy‘ scalar. **A guard that only ever fires on the case that prompted it is a test of that case; one that fires on a case nobody named is a test of the class.*** ⇒ **AND THE THIRD READING OF ‘FUNCTION-IS-THE-WRONG-KIND‘ splits the code: batch five's four rows named a function that could not express their claim, and this one names a function that expresses its NEGATION exactly, one candidate at a time, and still cannot reach it. The backlog's defect is not that it picked the wrong object — it is that ‘nearest by domain‘ says nothing about whether a function's QUANTIFIER matches its claim's.** *Routed rather than settled: nothing new. The one item still with 66 is ‘C75‘'s withheld holder, which this batch withholds again for the same reason and which now blocks two rows rather than one. ‘r7243‘ stands.* *Run:* `python3 receipts/P15_CR_cosmology/C77_batch_six_takes_all_six_and_the_one_wrong_kind_row_among_them_is_wrong`

**P15R319** `C78_batch_seven_takes_the_first_four_extend_rows_and_both_extensions_return_more_than_their_cl` [OK]

*‘r7285‘/‘r7287‘'s order (1) batch seven, the ‘EXTEND‘ class opened — **LANDED ‘56‘ OF ‘71‘ (‘52‘ of ‘59‘ ‘INPUTS‘, ‘4‘ of ‘12‘ ‘EXTEND‘), REMAINING ‘15‘ — FOUR ROWS ON TWO FUNCTIONS, ALL FOUR TAKEN, AND BOTH EXTENSIONS RETURN MORE THAN THEIR CLAIM ASKED FOR.** **‘B13‘ IS SETTLED BY ‘L129‘'s OWN TWO GENERATOR SETS AND BY NO CHOICE OF MINE: ‘close(gam)‘ and ‘close(ratios+[P])‘ are BOTH of order ‘27‘ and BOTH of exponent ‘3‘ — identical element-order multisets, one identity and ‘26‘ elements of order three — and they are NOT ISOMORPHIC, abelian $Z_3^3$ with ‘27‘ conjugacy classes against $\{}Delta(27)$ with ‘11‘. So ‘is not a second group resembling the substrate's own‘ is EXACT rather than rhetorical: resemblance by order AND by exponent is not isomorphism. And the predicate is not vacuously ‘different‘ — handed one group against itself from the generators in the other order it comes back ISOMORPHIC.** **AND ‘bead_contour‘'s ‘M2‘ IS EXACTLY $2/(3\{}sqrt3)$, THE NARIAI VALUE, so the horizon is a DOUBLE root: $f$ and its first derivative both vanish there while the second does not. $1/f$ has a double pole and the tortoise integral diverges as $1/\{}epsilon$ rather than $\{}log(1/\{}epsilon)$ — and the extension returns a COEFFICIENT, not only a divergence: $\{}epsilon I \{}to -0.333337$ against $2/f"(r_h) = -0.333341$ in closed form, across four decades. ‘B36‘ asserts ‘infinite tortoise distance‘; the DEGREE and the COEFFICIENT are what the extension adds.** **AND ONE MISTAKE OF THIS SEAT'S IS REMOVED STRUCTURALLY RATHER THAN PATCHED, AFTER BEING MADE FOUR TIMES: twice at batch five and twice in this batch's first draft, a converse was written as a TRUE FACT ABOUT THE EDIT instead of THE CLAIM RE-EVALUATED UNDER IT. Both are true sentences and the polarity flips between them. Every row now carries ONE predicate called at TWO inputs, gated to be the same object and the same KIND of object — and the KIND clause earned itself immediately: the first single-predicate tortoise test took an ‘(offset, width)‘ interval and shrank the OFFSET, so the ‘away‘ interval walked toward the horizon too and diverged as well. Not a polarity error — a GEOMETRY one, and only the kind guard names it.**

(node 66's ‘r7285‘ (1) and ‘r7287‘, both unchanged. This seat's own ‘EXTEND_PREDICTION.md‘ (digest ‘14aa710408c33d02‘, pushed before any of the ‘12‘ was computed) fixed the batch split and put these four first as the clean ones. Then ‘r7275‘'s order (2) (the zero-mode function, measure ruled, one-cycle stop). ‘r7243‘ stands.). *Computes:* **51 checks, all pass, 17s measured.** ◆ **NOTHING NEW COMPUTED PHYSICALLY — every number is a group order, a conjugacy-class count, an element order, or a quadrature of a function GRAFTED from ‘L129‘ or ‘bead_contour‘.** **(A) THE SUBJECT** — ‘BACKLOG.tsv‘ gated as ‘C71‘'s pin unmoved; all SEVEN ledgers AND the prediction file digest-pinned; both classes re-gated at ‘59‘ and ‘12‘; **‘(A)(4)‘ gates this batch's keys as ‘EXTEND‘ keys the backlog carries and NONE of them an ‘INPUTS‘ key**; both classes gated to partition; and ‘(A)(7)‘ gates every row recorded ‘EXTEND‘ in the ledger too, so the class is not silently reclassified. **(B) THE GRAFTS, AND THAT THE EXTENSION IS ONE** — both functions gated from a bounded statement count; **‘(B)(3)‘ gates from ‘close‘'s OWN syntax tree that it mentions no conjugacy class, centre or element order, so the fingerprint and the centre are genuinely EXTENSIONS**; **‘(B)(4)‘ gates that ‘f‘ has a one-statement body with no quadrature in it**; and the grafts gated to print nothing. **(C) ‘B13‘** — the same order gated, the same element-order multiset gated, **‘(C)(3)‘ gates the class counts and commutativity DIFFERING so the two are not isomorphic**, **‘(C)(4)‘ gates the predicate able to say ISOMORPHIC so it is not vacuously ‘different‘**, ‘(C)(5)‘ gates both generator sets as ‘L129‘'s own module-level names, and ‘(C)(6)‘–‘(C)(7)‘ gate the lap generating the centre against the 3-cycle failing. **(D) ‘B36‘** — **‘(D)(1)‘ gates ‘M2‘ EXACTLY at

$2/(3\{}sqrt3)$**, ‘(D)(2)‘ gates the double root from ‘f‘ and both derivatives, ‘(D)(3)‘ gates the $1/\{}epsilon$ growth, **‘(D)(4)‘ gates the coefficient against $2/f"(r_h)$ in closed form**, and ‘(D)(5)‘ gates the same predicate FALSE at a point ‘f‘ does not vanish at. **(E) ONE PREDICATE, TWO INPUTS** — claims TRUE and converses FALSE; **‘(E)(4)‘ gates that each row’s claim and converse are the SAME predicate object**; **‘(E)(5)‘ gates the two inputs the same KIND of object**; and ‘(E)(6)‘ gates the ledger asserting both AT IMPORT. **(F) THE PREDICTION** — gated digest-pinned, gated to name its number, **gated to write out what would REFUTE it**, and **‘(F)(3)‘ gates that this batch does NOT resolve it — four of twelve in, none a kind error, all three named rows in batch eight**; ‘(F)(4)‘ gates that the call states honestly it is not blind. **(G) THE HOLDERS** — all four gated to define ‘_r7283b7‘ at top level and CALL it inside a ‘check‘; all four gated GREEN; every literal gated still a string CONSTANT; none pinned by ‘S14‘’s line numbers. **(H) THE GUARD** — no string constant over three words on the left of an ‘in‘ test, from this receipt’s own syntax tree; ‘_needle‘ present; the arithmetic gated unable to exhaust its subject. *Bound:* **THE ‘EXTEND‘ CLASS’S FIRST FOUR ROWS SAY SOMETHING THE ‘INPUTS‘ CLASS NEVER DID: AN EXTENSION CAN OVERSHOOT ITS CLAIM, AND BOTH OF THESE DO.** *‘B13‘ asked whether two groups are isomorphic and got a pair the corpus supplies itself, agreeing on order AND exponent and differing only in the class structure — a counterexample to ‘resemblance implies identity‘ that is sharper than the row needed. ‘B36‘ asked whether a distance is infinite and got the DEGREE of the divergence and its exact pole coefficient, because ‘bead_contour‘ sits at the Nariai value and the horizon is therefore degenerate.* ⇒ **A change of INPUTS can only ever return the quantity asked for at other arguments; a short EXTENSION returns a new quantity, and a new quantity can be more informative than the claim that prompted it. That is a reason to prefer ‘EXTEND‘ work over ‘INPUTS‘ work where both are available, and it is not something the first six batches could have shown.** *Routed rather than settled: nothing new. The one item still with 66 is ‘C75‘’s withheld holder, which now blocks two rows. And one lesson arrived a SECOND time because it was fixed at one site instead of generalised — ‘C76‘’s line-wrap normalisation, which ‘(F)(4)‘ here needed and did not have. Every read of the prediction file now goes through it. ‘r7243‘ stands.* *Run:* `python3 receipts/P15_CR_cosmology/C78_batch_seven_takes_the_first_four_extend_rows_and_both_extensions_retu`

**P15R320** `C79_batch_eight_takes_six_and_refutes_this_seats_own_pinned_call_by_its_own_clause_because_a_li` [OK]

*‘r7285‘/‘r7287‘’s order (1) batch eight — **LANDED ‘62‘ OF ‘71‘ (‘52‘ of ‘59‘ ‘INPUTS‘, ‘10‘ of ‘12‘ ‘EXTEND‘), REMAINING ‘9‘ — SIX ROWS TAKEN.** **AND THIS SEAT’S OWN PINNED CALL IS REFUTED BY ITS OWN REFUTATION CLAUSE: ‘1‘ OF ‘12‘, NOT ‘3‘.** *‘EXTEND_PREDICTION.md‘ (digest ‘14aa710408c33d02‘, pushed before any of the twelve was computed) called ‘3‘ of the ‘12‘ ‘EXTEND‘ rows ‘FUNCTION-IS-THE-WRONG-KIND‘ at ‘25‘ per cent against ‘INPUTS‘’s ‘8.5‘, on the reasoning that a short extension changes what a function COMPUTES but not the KIND of question it answers. **The clause that kills it is the one the call itself wrote: if the corpus already carries a function of the right KIND, the backlog named the wrong function and the row is an ordinary MIS-NAMING. It does, for two of the three named — measured over ‘computations/‘ with this seat’s own directory EXCLUDED, FOUR ‘inv_dim‘ functions against ZERO subalgebra-dimension ones. So ‘B15‘ and ‘B8‘ are ‘FUNCTION-MIS-NAMED‘ and only ‘C3‘ is a kind error: ‘1‘ of ‘12‘ = ‘8.3‘ per cent against ‘INPUTS‘’s ‘5‘ of ‘59‘ = ‘8.5‘. The shares are the same to within a fifth of a point, so ‘EXTEND‘ does NOT attract kind errors and the call dies in NUMBER and in DIRECTION.*** **AND WHAT REFUTED IT IS WORTH MORE THAN THE CALL WOULD HAVE BEEN: ‘B15‘ and ‘B8‘ assert a LIE-GROUP fact — *the smallest connected group containing the ratios together with the hinge three-cycle is $SU(3)$ itself* — and it is reached by a FINITE-GROUP CHARACTER COUNT with no Lie-algebra machinery in it at all. ‘inv_dim(G_su, chi_meson) = 1‘ is $\{}dim\{}mathrm{Hom}(V,V)$, so by SCHUR the $\{}mathbf3$ is irreducible; an irreducible $\{}mathbf3$ sits in no proper connected subgroup of $SU(3)$ — the group being non-abelian rules out the maximal torus, $SU(2)\{}times U(1)$ acts as $\{}mathbf2+\{}mathbf1$, and $SO(3)$’s order-‘27‘ subgroups are cyclic. THE SAME NUMBER ‘C77‘ READ AS ‘colour’s meson selection rule‘ IS THE IRREDUCIBILITY CRITERION — one invariant dimension, two readings.** ***‘C3‘ survives as the one kind error with its qualification stated and not buried: the only subalgebra-dimension functions under ‘computations/‘ are the ones this seat wrote at batch five, so the corpus carries the right kind NOW because I built it and did not when the backlog was written. Its claim computes exactly — the isotropy subalgebra of a symmetric pair of $\{}so(5,1)$ is $\{}so(k)\{}oplus\{}so(6-k)$, and over $k=1..5$ the SET*

*is $\{}{6,7,10\{}}$, which is 'P12''s stated triple, with $\{}so(6,1)$ giving $\{}{9,11,15\{}}$ as the converse.** **And the three 'P_T_of' rows are settled by the function's own SIGNATURE: two parameters, the vacuum density and the mass, both properties of the progenitor, and NO crossing parameter at all — which is what 'a property of the progenitor's collapse history rather than of the crossing' means when the crossing has zero duration. The log-log slopes are exactly $-6$ and $-2$, so a decade of approach moves the amplitude by $10^6$ while the crossing contributes one.**

(node 66's 'r7285' (1) and 'r7287', both unchanged. And this seat's own 'EXTEND_PREDICTION.md', which this row reports REFUTED — the second register item withdrawn this cycle on its own measurement, after the argument-versus-module-level distinction at 'r7283'. Then 'r7275''s order (2) (the zero-mode function, measure ruled, one-cycle stop). 'r7243' stands.). *Computes:* **48 checks, all pass, 17s measured.** ◆ **NOTHING NEW COMPUTED PHYSICALLY — every number is an invariant dimension, a subalgebra dimension, a log-log slope, or a function signature read off a syntax tree.** **(A) THE SUBJECT** — 'BACKLOG.tsv' gated as 'C71''s pin unmoved; all EIGHT ledgers AND the prediction file digest-pinned; **'(A)(4)' gates the eight batches' key sets PAIRWISE DISJOINT so '19'+'4'+'9'+'9'+'5'+'6'+'4'+'6'='62'**; both classes gated to partition. **(B) THE REFUTATION** — **'(B)(1)'–'(B)(2)' gate the call digest-pinned, its number named and its three rows named, so it cannot be re-read after the fact**; **'(B)(3)' gates that the call WROTE the clause that kills it**; **'(B)(4)' MEASURES the corpus with this seat's own directory excluded — four 'inv_dim' functions against zero subalgebra-dimension ones**; '(B)(5)' gates the ledger's two codes; **'(B)(6)' gates the shares equal to within half a point and the call refuted in number AND direction**; and **'(B)(7)' gates that the refutation fires the call's OWN first clause rather than one written afterwards**. **(C) WHAT REFUTED IT** — Schur's criterion gated, non-commutativity gated, **'(C)(3)' gates that no proper connected subgroup of $SU(3)$ contains it with NO Lie-algebra machinery**, the 'G_rigid' converse gated, and **'(C)(5)' gates that it is the SAME NUMBER 'C77' read as the meson selection rule**. **(D) THE SIX** — claims TRUE, converses FALSE, every converse an EDIT in words; **'(D)(4)' gates batch seven's one-predicate-two-inputs guard STILL IN FORCE**; '(D)(5)'–'(D)(6)' gate $\{}{6,7,10\{}}$ against $\{}{9,11,15\{}}$; and '(D)(7)'–'(D)(8)' gate 'P_T_of''s signature and both exponents. **(E) THE GRAFTS** — three functions gated, and gated to print nothing in a fresh interpreter. **(F) THE HOLDERS** — all four gated to define '_r7283b8' at top level and CALL it inside a 'check' or 'gate'; all four gated GREEN; every literal gated still a string CONSTANT; none pinned by 'S14'. **(G) THE GUARD** — no string constant over three words on the left of an 'in' test, from this receipt's own syntax tree; '_needle' present; the arithmetic gated unable to exhaust its subject. *Bound:* **A PREDICTION THAT NAMES ITS OWN REFUTATIONS IS WORTH MAKING EVEN WHEN IT IS WRONG, AND THIS ONE PAID FOR ITSELF TWICE OVER.** *The call was wrong by a factor of three and its direction was dead — '8.3' per cent against '8.5' — but the clause that killed it was in the file before the measurement, so the refutation needed no adjudication and no argument: the corpus either carried a function of the right kind or it did not, and it did.* ⇒ **AND THE SEARCH THAT SETTLED IT TURNED UP THE RESULT OF THE BATCH: a LIE-GROUP claim reachable by counting the characters of a group of order '27'. 'inv_dim(G_su, chi_meson)' is simultaneously colour's meson selection rule and Schur's irreducibility criterion — and irreducibility is exactly what excludes every proper connected subgroup of $SU(3)$. The corpus had the instrument; the backlog pointed at a metric Killing test instead.** *So the honest summary of the 'EXTEND' class is narrower than I expected and sharper: the backlog's failures are MIS-NAMINGS at about one in twelve either side, and the one true kind error of the twelve is 'C3''s — where the corpus genuinely had nothing. The one item still with 66 is 'C75''s withheld holder, which blocks two rows. 'r7243' stands.* *Run:* `python3 receipts/P15_CR_cosmology/C79_batch_eight_takes_six_and_refutes_this_seats_own_pinned_call_by_its_ow`

**P15R321** `C80_batch_nine_closes_the_extend_class_and_the_shear_row_is_settled_by_an_absence_the_substrat` [OK]

*'r7285'/'r7287''s order (1) batch nine — **LANDED '64' OF '71' ('52' of '59' 'INPUTS', '12' of '12' 'EXTEND'), REMAINING '7' — THE 'EXTEND' CLASS IS CLOSED, AND EVERY ONE OF THE '7' LEFT IS A FILED LIMIT RATHER THAN OWED WORK: '4' are 'bead_frames'' nested 'tau_of_r' and 'tau_r', which 'C73''s '(D)(5)' gated the decision NOT to build a capability for, and '3' refuse on a module-level name nothing graftable binds.** **THE SHEAR ROW IS SETTLED BY AN ABSENCE. By the Sachs equation — a STATED input, labelled as one — the shear of a null geodesic congruence is driven by the WEYL tensor and by nothing else. 'Gamma' extended from the connection to Riemann and then to Weyl returns ALL '256' com-*

*ponents ZERO on the substrate in the paper's own static chart, with $R=12/\{}alpha^2$ and $R_{ab}=(R/4)g_{ab}$ exactly — Einstein, as the geometric core states. And the converse is 'probe1''s OWN module-level metric: the chiral plane wave has '16' non-zero Weyl components, so the obstruction exists elsewhere and is ABSENT here. Batch four computed the rulings to be straight NULL lines, so their initial shear is zero — this batch supplies the thing that could have grown one, and it is not there.** **AND ONE OF THE '256' NEEDED MORE THAN 'simplify': $W_{1332}$ comes back as $r(\{}sin 2t\{}tan t+\{}cos 2t-1)/(2\{}tan t)$, which 'sympy.simplify' returns UNCHANGED and which is identically zero — $\{}sin 2t\{}tan t = 2\{}sin^2 t$ and $\{}cos 2t-1=-2\{}sin^2 t$, and it evaluates to $9.9\{}times10^{-17}$ at a point. A SIMPLIFY THAT STOPS SHORT READS EXACTLY LIKE A NON-ZERO COMPONENT, and the only thing separating them is trying harder on the one that did not vanish. The predicate pushes its survivors through 'expand_trig' before counting them rather than taking 'simplify''s first answer for a geometric fact.** **And 'D_M' recovers the expansion law EXACTLY: it is $\{}int c/H\{},dz$, so differentiating the comoving distance must return $c/H(z)$, and it does to $10^{-8}$ relative at $z=0.5$, '5' and '100' — quadrature precision — against $10^{-4}$ to $9\{}times10^{-4}$ on the other rate at the same $H_0$, four orders worse, so the agreement is that rate's own and not the integrator's.** **That row's holder is withheld for the THIRD time, same reason as at 'C75' and 'C77': its standing gate says it COMPUTES NOTHING, so a computed check would falsify that gate's own label while its predicate passed. The row is TAKEN and only the holder edit is withheld.**  (node 66's 'r7285' (1) and 'r7287', both unchanged, and this seat's own 'EXTEND_PREDICTION.md' batch split. With the 'EXTEND' class closed, what remains of the '71' is the '7' filed-limit 'INPUTS' rows — so the next ordered item is 'r7275''s order (2) (the zero-mode function, measure ruled as $d\{}mu=\{}sqrt{\{}lvert g_{rr}\{}rvert}\{},dr$ with the flat $dr$ count beside it, one-cycle stop). 'r7243' stands.). *Computes:* **43 checks, all pass, 17s measured.** ◆ **NOTHING NEW COMPUTED PHYSICALLY — every number is a curvature component of a metric the paper states or the module owns, or a quadrature of a function GRAFTED from 'CR_acoustic_scale_and_hubble'.** **(A) THE SUBJECT** — 'BACKLOG.tsv' gated as 'C71''s pin unmoved; all NINE ledgers AND the prediction file digest-pinned; **'(A)(3)' gates the nine batches' key sets PAIRWISE DISJOINT so '19'+'4'+'9'+'9'+'5'+'6'+'4'+'6'+'2'='64'**; **'(A)(4)' gates the 'EXTEND' class CLOSED, all '12' keys in the ledgers**; '(A)(5)' gates 'INPUTS' at '52' of '59'; **and '(A)(6)' gates that every one of the '7' remaining is a FILED LIMIT — '4' the nested-function rows and '3' the unbound-name refusals**. **(B) THE SHEAR ROW** — **'(B)(1)' gates all '256' Weyl components ZERO on the substrate**; '(B)(2)' gates it Einstein with $R=12/\{}alpha^2$; **'(B)(3)' gates 'probe1''s OWN plane wave at '16' non-zero, so the obstruction exists elsewhere**; '(B)(4)' cites batch four for the rulings being straight null lines rather than redoing it; **'(B)(5)' gates that the Sachs equation is used as a STATED input and LABELLED as one in the module's own prose**; and **'(B)(6)' gates from the module's syntax tree that the predicate simplifies its survivors through 'expand_trig', because 'simplify' alone leaves one of the '256' looking nonzero**. **(C) THE RECOVERED LAW** — the derivative gated to quadrature precision at three redshifts, the other rate gated four orders worse, and the flat-slice half gated as batch four's computation CITED. **(D) THE TWO ROWS** — claims TRUE, converses FALSE, every converse an EDIT in words; **'(D)(4)' gates batch seven's one-predicate-two-inputs-of-the-same-kind guard STILL IN FORCE two batches later**; both grafts gated and gated to print nothing. **(E) THE HOLDERS** — **'(E)(2)' gates the SAME holder withheld a THIRD time, by that receipt's own standing claim**; the other gated to define '_r7283b9' at top level and CALL it inside a 'check', gated GREEN, its literal gated still a string CONSTANT, and neither pinned by 'S14'. **(F) THE GUARD** — **'(F)(1)' gates NO string constant AT ALL on the left of an 'in' test, the threshold removed rather than lowered after 'C76' minted a one-character literal**; '_needle' present; and the arithmetic gated unable to exhaust its subject. *Bound:* **THE SUBJECT CLOSES ON A SHAPE RATHER THAN A NUMBER, AND THAT IS THE RIGHT WAY FOR IT TO CLOSE.** *'64' of '71' is not '71' of '71', and the '7' that remain are not a shortfall: each is a limit this seat FILED with a measurement — the nested-function capability 'C73''s '(D)(5)' gated the decision not to build because '37' of '40' outstanding rows named a top-level function and the '3' that did not named the SAME one; and three rows whose named function reads a module-level name nothing graftable binds.* ⇒ **A backlog that ends in filed limits rather than in a round number is one whose remainder can be read by someone else without re-deriving it.** *And the last row's finding is the one this seat would keep: **the rulings are shear-free because of something

the substrate LACKS**, and the converse is not a constructed edit but the very module the function was grafted from — ‘probe1‘’s plane wave has the Weyl tensor the substrate has not. Routed rather than settled: the withheld holder, now blocking three rows. ‘r7243‘ stands.* *Run:* `python3 receipts/P15_CR_cosmology/C80_batch_nine_closes_the_extend_class_and_the_shear_row_is_settled_by_an`

**P15R322** `C81_order_two_is_done_in_one_cycle_the_zero_mode_capability_exists_and_the_measure_the_stated_` [OK]

*‘r7275‘/‘r7283‘’s order (2) — **THE ZERO-MODE CAPABILITY EXISTS, BUILT AND LANDED INSIDE THE ORDERED ONE-CYCLE STOP, AND THE ONE CONDITION THE STATED LIMIT CALLED ITS ONLY OPEN QUESTION TURNS OUT NOT TO CHANGE THE COUNT IT WAS FLAGGED FOR.** **THE ABSOLUTE VALUE THE ORDER INSISTED ON IS NOT A FORMALITY, AND THIS IS THE FINDING.** *The wall sits where $f_j0$ — ‘B34‘’s own finding, which at $D=5$ is $\{}lvert r\{}rvert_j1/2$, INSIDE the inner horizon, computed here from the roots of $f$ rather than asserted. There $g_{rr}=1/f$ is NEGATIVE, so $\{}sqrt{g_{rr}}$ is IMAGINARY and the static $L^2$ measure DOES NOT EXIST ON THE VERY INTERVAL THE MODE LIVES ON. $d\{}mu=\{}sqrt{\{}lvert g_{rr}\{}rvert}\{},dr=dr/\{}sqrt{\{}lvert f\{}rvert}$ is real BECAUSE OF THE ABSOLUTE VALUE — so the order did not tighten a notation, it supplied the only thing that made the measure exist.* **AND THE WALL IS REALLY A WALL BEFORE ANY MODE IS PUT ON IT: $W=\{}sqrt{\{}lvert f\{}rvert}/r$ is ODD in the signed radius at $D=5$ — measured through ‘parity.py‘, with $2M=3/16$ taken from ‘L-006w‘’s OWN ‘twoM‘ and not written here — and the horizons come out $1/2$ and $\{}sqrt3/2$ from the roots of $f$. At $D=4$ ‘parity.py‘ measures $W$ to have NO parity at all, so there is no sign change and no wall, and that is reported rather than forced.** **THE RESULT: EXACTLY ONE CHIRALITY BINDS, UNDER BOTH MEASURES — ruled norm $1.541542$ against flat $0.609035$, reported side by side as the order requires, with the other chirality divergent under both.** *And boundness is tested by REFINEMENT and not by one number: each integral runs at two grid densities and must not grow with resolution, which is what separates a bound mode from an unresolved integrand.* ⇒ ***SO THE MEASURE QUESTION IS ANSWERED IN THE ONLY FORM THAT MATTERS: THE TWO MEASURES GIVE DIFFERENT NORMS AND THE SAME COUNT. The condition the limit named as the single open thing the order would have to carry does not change the answer it was named for — while still being exactly what makes the measure real at all, which is a separate and necessary thing.*** (node ‘66‘’s ‘r7275‘ (2), carried unchanged through ‘r7281‘, ‘r7283‘, ‘r7285‘ and ‘r7287‘: the zero-mode function, measure RULED as $d\{}mu=\{}sqrt{\{}lvert g_{rr}\{}rvert}\{},dr$ **with the absolute value**, the flat $dr$ count reported beside it, **hard stop at one cycle**. The stop is honoured: the function runs in $2.8$ s and this is its one receipt. With order (2) closed, the next ordered item is ‘r7269‘ (3) / ‘r7287‘’s backlog by prefix, where this seat owes a MODE call in writing before it reads a key. ‘r7243‘ stands.). *Computes:* **25 checks, all pass, 8s measured.** ◆ **NOTHING NEW ASSUMED PHYSICALLY — the profile is the corpus’s own ‘twoM‘ through ‘parity.py‘, the horizons are roots of $f$, and the measure is the one the order ruled.** **(A) THE HOLE THE LIMIT FILED** — ‘(A)(1)‘–‘(A)(2)‘ gate ‘STATED_LIMIT.md‘ stating the hole in its own voice and naming the measure as the ONE open condition; **‘(A)(3)‘ gates that it declared ‘B1‘’s equivariant index as NOT this capability IN ADVANCE**; and **‘(A)(4)‘ RE-MEASURES the hole over the corpus with this seat’s own directory EXCLUDED — ‘0‘ of ‘1145‘ function definitions under ‘computations/‘ carry a name of that kind, so the limit was real when it was filed**. **(B) THE PROFILE** — ‘(B)(1)‘ gates $2M=3/16$ as coming THROUGH ‘parity.py‘ from ‘L-006w‘’s own ‘twoM‘; **‘(B)(2)‘ gates $W$ ODD at $D=5$, which IS the domain-wall condition**; ‘(B)(3)‘ gates the horizons COMPUTED from the roots; ‘(B)(4)‘ gates the wall interval as where $f_j0$; **and ‘(B)(5)‘ gates that $D=4$ has no wall at all, so the capability is not asserting a wall it was handed**. **(C) THE RULED MEASURE** — **‘(C)(1)‘ gates $\{}sqrt{g_{rr}}$ IMAGINARY on the wall interval WITHOUT the absolute value**, ‘(C)(2)‘ gates the ruled weight real and positive at the midpoint against the flat weight’s $1$, and **‘(C)(3)‘ gates from the function’s own signature that it carries BOTH measures by name, so the flat count is reported BESIDE the ruled one rather than instead of it**. **(D) THE SOLVE** — ‘(D)(1)‘–‘(D)(2)‘ gate one chirality bound and one divergent in EACH measure; **‘(D)(3)‘ gates that boundness is a REFINEMENT test at two grid densities**; and ‘(D)(4)‘ gates the divergence REPORTED as ‘inf‘ rather than left as an overflow warning. **(E) THE QUESTION ANSWERED** — ‘(E)(1)‘ gates the norms DIFFERENT, ‘(E)(2)‘ gates the counts the SAME, and ‘(E)(3)‘ gates the conclusion in both halves: the condition does not change the count AND is still what makes the

measure real. **(F) THE GRAFT AND THE STOP** — **‘(F)(1)‘ gates that the capability PRINTS NOTHING when imported and used, so it is a function and not a script**; ‘(F)(2)‘ gates its own report green; and **‘(F)(3)‘–‘(F)(4)‘ gate that ‘B1‘‘s index and the twelve rows are NOT claimed, from the module’s own ‘what is not claimed‘**. **(G) THE GUARD** — **‘(G)(1)‘ gates NO string constant AT ALL on the left of an ‘in‘ test anywhere in this file, from its OWN syntax tree, ‘C76‘‘s threshold lesson applied FROM THE START rather than after a gate said so**; ‘(G)(2)‘ gates every presence test through ‘_needle‘. *Bound:* **AN ORDER WITH A HARD STOP IS A GOOD ORDER, AND THIS ONE SHOWS WHY: THE EXPENSIVE PART WAS NEVER THE SOLVE.** *The mode equation $(-i\{}gammaˆ1\{}partial_r+W)\{}psi=0$ integrates to $\{}psi=\{}exp(\{}mp\{}int W\{},dr)$ in one line and the two quadratures take milliseconds. What the cycle actually bought was the MEASURE — and the measure is where the whole thing could have been silently wrong.* ⇒ **Had the order said $\{}sqrt{g_{rr}}$ instead of $\{}sqrt{\{}lvert g_{rr}\{}rvert}$, the integrand on the wall interval would have been complex and ‘numpy‘ would have returned a ‘nan‘ or a complex norm — and a ‘nan‘ norm reads exactly like a divergence, which is to say like a chirality that does NOT bind. The count would have come out ‘0‘ and nothing in the arithmetic would have objected.** *So the thing worth keeping is not the number ‘1‘, it is that the limit named the right open question and named it as a CONDITION rather than a size. A size would have been a guess; a condition was checkable, and checking it is what showed that it does not move the count — which is a stronger statement than either answer to it would have been.* *And what is still owed is said rather than rounded off: ‘B1‘‘s equivariant index, declared out of scope before the work started, and the twelve rows the capability makes reachable but which this receipt does NOT claim to have reached — they stay verified by citation until a batch takes them with its own recorded converses.* *Run:* `python3 receipts/P15_CR_cosmology/C81_order_two_is_done_in_one_cycle_the_zero_mode_capability_exists_and_the`

**P3R12** `P03_acceleration_is_slice_curvature` [OK]

*sec:curvature · rem:twocritical · the bend* (on ANY member of the energy family (E cancels: (dr/dtau)ˆ2 = Eˆ2 - f, and E is a constant of the motion) dˆ2r/dtauˆ2 = -f’/2 = r*K_G identically, so THE SIGN OF THE SLICE’S GAUSSIAN CURVATURE IS THE SIGN OF THE COMOVING ACCELERATION – and the flat locus is the deceleration-to-acceleration turnover, which on Nariai is the seam and in standard terms is rho_m = 2 rho_Lambda (cschˆ2 w = 2 ¡=¿ sinh w = 1/sqrt2, the seam’s phase). A recent cosmological epoch: 7.06 Gyr at H0=73, 7.65 at 67.4, preceding matter-Lambda equality [borrowed from P3 · P7 · P8]). *Computes:* the identity asserted; the K_G zero solved and matched to (M alphaˆ2)ˆ(1/3); cschˆ2 w = 2 ¡=¿ sinh w = 1/sqrt2 proved by algebra after simplify() failed on the closed forms; epochs at both H0 *Bound:* bound: identities on the Nariai E=1 worldline; no causal claim between seam and turnover – they are one locus; dating inherits H0 *Run:* `python3 receipts/P03_SdS_slicing/P03_acceleration_is_slice_curvature.py`

**P15R42** `BRANCHPT_transmission_character` [OK]

*rem:phase-open* (BRANCHPT — the transmission character of the BRANCH POINT r=0, on its own terms. WHY (r2154). P15 l.525 says ”The cosmogenesis branch point, BEING the degenerate Nariai horizon, is a faithful scale-free transmitter”, and prop:transmission argues from the tortoise integral at a DOUBLE ROOT (p=2, kappa=0). **PART 2, from the receipt-vs-sentence audit: the PEAK-POSITION EXCLUSION, which is what P15 actually cites this file for and which the file did not compute** — the collapse leg’s own conformal time gives eta_bp ˜ 5e3-1e4 Mpc, so r_sˆeff = r_s + eta_bp/sqrt3 = 3.0-5.9e3 Mpc, and reproducing l_A = 301.8 would demand D = 2.9-5.7e5 Mpc against the actual 1.4e4, a factor of **21-41**. *PART 1 shows the crossing ADDS no divergent phase; PART 2 shows an inherited phase cannot survive. The paper’s sentence needs the second.* [borrowed from P15 (+P7)]). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt’s own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/BRANCHPT_transmission_character.py`

`H1_the_low_multipole_deficit_is_two_effects_and_the_ladder_imprint_dies_where_the_transfer_is_narrow` [OK]

*corpus integrity* (**THE HARMONIC-ANALYSIS BAKE — second of the four fields ‘L-272‘’s re-survey left outstanding, thrown r3166.** **BOUNCED, and largely:** the field’s first question is why a closed-$Sˆ3$ source is projected through FLAT spherical Bessel functions, and ‘P15‘ answers it in its own text — the distance slicing is flat, so only the source modes carry the closed-$Sˆ3$ quantization — while its measure $w_L=(L+1)/(L(L+2))$ is derived twice, as degeneracy over per-mode power and as $d\{}ln k_L/dL$, both agreeing to machine precision. **BIT:** the deficit

reported as one geometric effect splits exactly into a **FLOOR** (modes below $L{=}2$ removed) and a **LADDER COARSENESS**. At $\{}ell=5$–6 the second contributes $+0.38$ against a floor of $0.27$–$0.53$ — comparable, and in the opposite direction — and it is what sets the recovery multipole the paper quotes ($\{}ell=8$ with it, $\{}ell=10$ without). Under $\{}pm2\{}%$ in $r_0$ the floor moves $-2.4\{}%$ to $-6.2\{}%$ and the full result $+27.8\{}%$ to $-43.3\{}%$: **the location is the floor and is robust as claimed; the depth is the ladder and is volatile**, which gives the paper's own qualitative split a mechanism. **AND A SELF-WAIVED GATE BECOMES A BOUND:** ‘P15‘ records that the ladder "samples the projection below the rate the instrument's own aliasing gate demands, and that gate waives itself on the claim that the ladder is physical." The imprint dies at $\{}ell\{}simeq k_2D_Ce^{3\{}sigma}$; to reach the first acoustic peak the transfer would need to draw power over a factor of nine in $k$. **Safe with a wide margin, by measurement.** [borrowed from —]). *Computes:* COMPUTES: the spherical-Bessel completeness identity $\{}sum_\{}ell(2\{}ell+1)j_\{}ell(x)^2=1$ to ten figures and the exact plateau $\{}int j_\{}ell^2 d\{}ln x=1/(2\{}ell(\{}ell+1))$ reproduced on the fixed grid to six, both **before anything is used**; the measure against $d\{}ln k_L/dL$ and against degeneracy-over-power at five $L$; the paper's own $D_C/r_0\{}to0$ limit; the floor/discreteness decomposition across $\{}ell=2$–12; each half's response to $\{}pm2\{}%$ in $r_0$; the recovery multipole with and without the discreteness; and the imprint against transfer width at four $\{}sigma$, with the scaling rule checked against all four rather than extrapolated (15 checks). *Bound:* NOT-A-PAPER-CLAIM — corpus integrity; no paper edited. **NOT claimed:** that the deficit is wrong — every number touched is reproduced or confirmed, the paper's own limit included; that these are predictions about the sky — the decomposition is the pure Sachs–Wolfe limit while the paper's depths come from a genuine Boltzmann transfer it says are milder, and the two are not compared; that the real transfer's width is measured — $\{}sigma$ is a Gaussian-in-$\{}ln k$ stand-in and measuring $\{}Delta_\{}ell(k)$ is the owed next step; that the aliasing gate should stop waiving — only that the waiver now has a margin; that the low-$\{}ell$ oscillation is derived rather than measured. *Run:* `python3 receipts/L274_the_harmonic_bake/H1_the_low_multipole_deficit_is_two_effects_and_the_ladder_imprint_`

**P7R26** `P07_the_back_seam_is_the_laps_one_non_degenerate_horizon_and_its_surface_gravity_is_the_beads_own`
[OK]
*‘P07‘’s ‘sec:lift-initial-rate‘ stratification paragraph (the composed statement) and its F-triptych caption’s ‘$-1.299$‘; ‘P15‘’s ‘sec:transmission‘ opening, its physical-reading paragraph and ‘prop:transmit‘; ‘P02‘’s ‘sec:ring‘ naming clause* ( **THE BACK SEAM IS THE LAP'S ONE NON-DEGENERATE HORIZON, IT IS THE COLLAPSING MATTER'S OWN EVENT HORIZON, AND ITS SURFACE GRAVITY IS THE BEAD'S ACCELERATION THERE BY AN IDENTITY RATHER THAN BY COINCIDENCE.** *Composed at ‘r7185‘ from pieces the corpus carried separately, on a plain-language read that asked whose horizon the first seam is.* (1) ***THE TWO SEAMS ARE OPPOSITE CASES OF ‘prop:transmission‘ AND HAD BEEN CARRIED AS ONE***: the front seam at ‘$r=+\{}alpha/\{}sqrt3$‘ is the cubic's DOUBLE root (‘$f"=-6/\{}alpha^2$‘, ‘$\{}kappa=0$‘), the back seam at ‘$r=-2\{}alpha/\{}sqrt3$‘ its SIMPLE root with ‘$f'=9/2\{}sqrt3\{}alpha$‘ exactly and ‘$\{}kappa=3\{}sqrt3/4\{}alpha=1.299/\{}alpha$‘. **Being one point of the substrate does not make them one case, because the degeneracy is a property of the root.** (2) ***AND THE BEAD'S ACCELERATION THERE EQUALS ‘$-\{}kappa$‘ STRUCTURALLY***: at ANY root of ‘$f$‘, ‘$1-f=1$‘, so the marginal congruence's ‘$\{}dd^2r/\{}dd\{}tilde\{}tau^2=\{}tfrac12(1-f)'=-\{}tfrac12f'$‘ while ‘$\{}kappa=\{}tfrac12\{}lvert f\{}'\{}rvert$‘ — verified at the black-hole root of five NON-Nariai members, so it is the identity and not this member's luck. **Which is the number ‘P07‘'s F-triptych caption has printed as ‘$-1.299$‘ all along without naming it a surface gravity.** (3) ***THE TWO APPROACH PARAMETERS INVERT EACH OTHER AT BOTH SEAMS***: in the tortoise length ‘$r_*=\{}int\{}dd r/f$‘ — the parameter the phase accumulates in and the one ‘prop:transmission‘ is stated in — the back seam's approach IS exponential, each factor ‘$100$‘ in the gap adding ‘$\{}ln100/\{}lvert f\{}'\{}rvert$‘ to eight figures, while the front seam's grows like ‘$1/\{}epsilon$‘; in the layer's own proper time ‘$T=\{}int\{}dd r/\{}sqrt{-f}$‘ the front seam diverges logarithmically at rate ‘$\{}sqrt\{}Lambda$‘ and the back seam's integral CONVERGES, to ‘$0.66978\{}alpha$‘ by a ‘$\{}sqrt\{}epsilon$‘ extrapolation. ⇒ **So the lap's one non-degenerate horizon DOES carry a thermal scale, and what it lacks is an unbounded approach along the trajectory that carries the modes** — ‘rem:phase-open‘'s own static-versus-trajectory distinction, applied to the seam instead of the branch point. (4) ***AND THE SEAM IS DOWNSTREAM OF WHERE THE SKY'S SPECTRUM IS FIXED, NOT BESIDE IT***: both measured as conformal distance along the collapse leg from its FAR-PAST

end, which is the end the per-cent figure's own receipt integrates from — the sky's first acoustic peak at ‘$0.92$‘ per cent, **the back seam at ‘$46.508$‘ per cent**, a factor ‘$50.6$‘. **Every mode carrying the measured spectrum is fixed and THEN crosses the lap's one non-degenerate horizon**, and what the crossing does to it is computed nowhere — ‘PO-75‘'s second live question, ORDERED to ‘cc66‘ at ‘r7185‘. [borrowed from P07]). *Computes:* **22 checks, all pass, 6 s.** **(A)** both seams are roots on the Nariai member, the front double and the back simple, with ‘$f'(\{}text{back})=9/2\{}sqrt3\{}alpha$‘ to ‘$10^{-12}$‘. **(B)** ‘$\{}kappa_{\{}rm back}=3\{}sqrt3/4\{}alpha$‘ and ‘$\{}kappa_{\{}rm front}=0$‘. **(C)** the acceleration identity at the back seam, then at the black-hole root of ‘$M=0.05$‘, ‘$0.10$‘, ‘$0.15$‘, ‘$0.19$‘, ‘$0.192$‘ with ‘$1-f$‘ printed as ‘$1$‘ at each — the bracket being ‘$[\{}epsilon,M^{1/3}]$‘, ‘$f$‘'s interior maximum, so it brackets the black-hole root alone — and ‘P07‘'s caption literal read from the current source. **(D)** both parameters at both seams, the back seam's tortoise increments against ‘$\{}ln100/\{}lvert f\{}'\{}rvert$‘, the front seam's ‘$1/\{}epsilon$‘ growth, the proper-time ‘$\{}sqrt\{}epsilon$‘ extrapolation with its deficit ratios at exactly ‘$10$‘, and the front seam's ‘$\{}dd T/\{}dd\{}ln$‘ slope against ‘$1/\{}sqrt\{}Lambda$‘. **(E)** the turnaround as ‘$1-f=0$‘ and the seam-to-turnaround cosmic time ‘$0.877972\{}alpha=14.90$‘ Gyr. **(F)** the five corpus statements the composition rests on, each read from its own source (‘P07‘, ‘P02‘ twice, ‘P15‘, ‘P16‘). **(G)** the collapse leg's conformal length at 30 digits, the back seam's ‘$\{}cosh x=2$‘ position, the two per-cent figures from the same end, and the orientation now named in BOTH ‘P15‘'s sentence and the sibling receipt — so removing either goes red. *Bound:* **IT DOES NOT COMPUTE WHAT THE BACK SEAM DOES TO A MODE CROSSING IT.** *That is named as open in ‘prop:transmit‘, in ‘PO-75‘ and in the explainer, and group ‘(G)‘ is what makes it load-bearing rather than a coincidence of location.* *It re-derives neither ‘prop:flat‘ nor ‘P1‘'s theorem nor the Nariai condition; each is quoted from the paper that proves it.* *The Gyr figure carries the corpus's own ‘$1/\{}sqrt\{}Lambda=9.8$‘ Gly and is reported to the precision that datum supports; the ‘$\{}alpha$‘-units figures carry no such dependence.* *It withdraws nothing. ‘prop:transmission‘'s dichotomy and ‘prop:transmit‘'s conclusion both stand; what changes is that the conclusion's reason is now given per locus, three loci and three reasons, where one clause had covered all three.* *Its literal reads normalise whitespace, so a pin does not turn on where the LaTeX source happens to break its lines.* *Run:* `python3 receipts/P07_CR_framework/P07_the_back_seam_is_the_laps_one_non_degenerate_horizon_and_its_surface_g`

**P7R28** `P07_the_crossing_rides_the_characteristic_that_crosses_and_the_thermal_factor_rides_the_one_that` [OK]

*‘prop:transmission‘/‘prop:transmit‘ at the lap's one non-degenerate horizon, and ‘rem:phase-open‘'s static-versus-trajectory distinction applied to the seam — **THE SEAM IS A CHARACTERISTIC SURFACE OF THE MODE EQUATION AND ITS TWO CHARACTERISTIC SPEEDS THERE ARE EXACTLY ‘$2$‘ AND EXACTLY ‘$0$‘; THE CROSSING RIDES THE FIRST AND ‘$e^{\{}pi\{}omega/\{}kappa}$‘ RIDES THE SECOND, WHICH DOES NOT CROSS** — *so the transfer is ‘$1$‘, identically in ‘$\{}omega$‘ and in ‘$\{}ell$‘** (node 66's ‘r7185‘ order: *"compute the mode equation across a simple root of ‘$f$‘ crossed at finite proper time on the collapse trajectory, and report whether the crossing alters the spectrum, by how much, and at which ‘$k$‘"* — with three things to keep apart (the static approach versus the trajectory crossing; which rate each step is on; ‘$k$‘-dependence as the whole discriminant) and the standing instruction that *"if it cannot enter, say so with the computation that shows it, because that closes the transmission chain end to end for the first time"* [borrowed from P7]). *Computes:* **29 checks, all pass, ~2 s.** ◆ **(A) The chart IS the trajectory, not a choice made about it.** The Painleve–Gullstrand shift ‘$v^2=1-f$‘ makes PG's congruence the ‘$E=1$‘ family, so ‘$v=0$‘ exactly at the collapse branch's turnaround ‘$-(2M\{}alpha^2)^{1/3}$‘, ‘$\{}dd\{}tau=-(2\{}alpha/3)\{}dd x$‘ EXACTLY on ‘$r=-(2M\{}alpha^2)^{1/3}\{}cosh^{2/3}x$‘, and ‘$\{}dd\{}eta=\{}dd\{}tau/a$‘ with ‘$a=|r|/\{}alpha$‘ returns the prefactor ‘$2/(\{}sqrt3\{},2^{1/3})$‘ and the leg length ‘$1.927621297\{}alpha$‘ from the PG construction alone. **(B) The mode equation, derived with ‘$M$‘ and ‘$\{}alpha$‘ FREE:** ‘$(r^2fR')'+2i\{}omega r^2vR'+[\{}omega^2r^2+i\{}omega(r^2v)'-\{}ell(\{}ell+1)]R=0$‘, ‘$\{}det g=-r^4\{}sin^2\{}theta$‘ with no ‘$f$‘ in it, ‘$g^{rr}=f$‘ vanishing at the root while ‘$g^{\{}tau r}=-v=-1$‘ does not. **(C) The seam is a CHARACTERISTIC:** the principal symbol on ‘$\{}dd r$‘ is ‘$f$‘; the null curves are ‘$\{}dd r/\{}dd\{}tau=v\{}pm1$‘, giving speeds ‘$2$‘ and ‘$0$‘ at ‘$v=1$‘, and ‘$\{}dd v/\{}dd r=-f'/2=-\{}kappa$‘ there by the same identity the companion receipt uses for the bead's acceleration. Crossing: ‘$1.000000219\{}times10^{-3}\{}alpha$‘ of proper time across a ‘$\{}pm10^{-3}\{}alpha$‘ window, ‘$2.011\{}alpha$‘ from

‘$r=-10\{}alpha$‘ to the turnaround. Skimming: ‘$\{}ln100/\{}kappa=3.545062\{}alpha$‘ per factor ‘$100$‘, forever. **(D) The two indices are ‘$0$‘ and ‘$-i\{}omega/\{}kappa$‘**, with ‘$\{}kappa=3\{}sqrt3/4\{}alpha$‘ to 18 figures RECOVERED from the indicial equation; the difference is non-integer for every real ‘$\{}omega¿0$‘, so the monodromy is exactly diagonal — ‘$1$‘ on the regular branch and ‘$e^{-2\{}pi\{}omega/\{}kappa}$‘ on the other, the Boltzmann factor at ‘$T_H=\{}kappa/2\{}pi=0.2067483\{}alpha^{-1}$‘. **(E) The thermal branch is STATIONARY on the skimming ray:** ‘$\{}tau+\{}ln|x|/\{}kappa$‘ converges to a constant over four decades (moving ‘$2.75\{}times10^{-8}$‘ over the last two), so the product’s phase holds within ‘$0.0036$‘ rad while ‘$e^{-i\{}omega\{}tau}$‘ alone turns through ‘$138$‘ rad — a factor ‘$38000$‘. **(F) The transfer, measured:** the index-‘$0$‘ series solves the equation to ‘$10^{-36}$‘ on BOTH sides, its recursion’s denominator ‘$(j+1)[(j+1)2\{}sqrt3+\{}tfrac{8i}3\{}omega]$‘ cannot vanish, and ‘$|T(d)|\{}to1$‘ LIKE ‘$d$‘ over four decades of ‘$d$‘, four of ‘$\{}omega/\{}kappa$‘ and ‘$\{}ell=0,2,10$‘. **(G) The front seam:** ‘$A=r^2f$‘ has a double zero, ‘$B(r_F)=2i\{}omega/3$‘ purely imaginary, an IRREGULAR singular point whose branch is ‘$\{}exp(-2i\{}omega r_*)$‘ with ‘$r_*=1/3x$‘ — a pole, not a log. **(H) ‘$k$‘:** ‘$k_{\{}rm th}=a_h\{}kappa=3/2$‘ EXACTLY, ‘$L=0.8027756$‘, ‘$\{}ell=4.16$‘. **(I) THE SIGNATURE, which ‘r7187‘ names as the omission that licensed a wrong sentence, read here off the CHARACTERISTICS instead of off ‘$f$‘:** outside the seam ‘$f¡0$‘ and **both** speeds ‘$v\{}pm1$‘ are positive, so nothing holds ‘$r$‘ fixed — the radius is timelike; inside, ‘$f¿0$‘ and the speeds straddle zero — static. *The cosmological pattern, and the exact reverse of a black hole’s.* And ‘$f’=2M/r^2-2r/\{}alpha^2$‘ on ‘$r¡0$‘ is a **sum of two positive terms**, ‘$0.288675135+2.309401077=2.598076211=2\{}kappa$‘ — the same ‘$2\{}kappa$‘ the indicial equation returns, so the root’s SIMPLICITY is forced by the sign of ‘$f’$‘ and is not a Nariai accident. *Bound:* **THE CROSSING DOES NOT ALTER THE SPECTRUM, AND THE REASON IS CHARACTERISTIC STRUCTURE RATHER THAN A CHOICE OF SLICING.** *‘$2\{}kappa$‘ IS in the equation — exactly once, as the index of the branch that is not smooth at the seam — and it cannot enter through the crossing, because that branch is the one carried by the rays that asymptote to the seam instead of crossing it.* **And the two rates are separated with numbers rather than with an argument:** ‘$\{}tau$‘ along the skimming family runs at ‘$\{}kappa$‘, ‘$r_*$‘ runs at ‘$2\{}kappa$‘ — exactly twice, which is why the companion receipt’s ‘$\{}ln100/|f’|$‘ is half of this file’s ‘$\{}ln100/\{}kappa$‘ — and the trajectory runs on neither: it crosses at speed ‘$2$‘, in ‘$(2\{}alpha/3)\{}mathop{\{}rm arccosh}2=0.877971931\{}alpha$‘ of its own proper time from the seam to the turnaround. **The ‘$k$‘ is named even though the factor is ‘$1$‘:** the only scale at the crossing is ‘$\{}kappa$‘, so the only comoving wavenumber a thermal factor could have carried is ‘$3/2$‘ exactly — ‘$L=0.8028$‘, BELOW THE DIPOLE, and ‘$\{}ell=4.16$‘ against the first acoustic peak’s ‘$220.60$‘, a factor ‘$53.0$‘ that independently reproduces the ‘$50.6$‘ the two LOCI give on the same leg. ⇒ **A ‘$k$‘-independent factor of ‘$1$‘ is not an amplitude — it is the identity — so neither ‘$A_s$‘ nor ‘$n_s$‘ moves, and the sector’s ‘$1.57$‘ acquires NO parameter address from the lap’s one non-degenerate horizon.** **What is NOT computed: the flux the seam radiates into the lap’s future.** *That is a question about the quantum state and it does need the progenitor interior ‘PO-75‘ is live on. The TRANSFER did not — and the one quantity the corpus lacks, the physical ‘$\{}alpha$‘, is not needed either, because the verdict is ‘$1$‘, which carries no scale, and the ratio to the first peak is ‘$\{}alpha$‘-free.* *Run:* `python3 receipts/P07_CR_framework/P07_the_crossing_rides_the_characteristic_that_crosses_and_the_thermal_fa`

# Appendix L The Knowledge Ledgers

Each claim marked ⌞ in the text rests on a field bake or the figure–theorem bake recorded in the named ledger, which ships with this work and carries the probes, the receipts, and the three registers kept apart — what bit, what bounced, and where the boundary is. A ledger is working apparatus, not a publication, and is cited here rather than in the bibliography. This appendix is generated from the ledger index, not maintained by hand.

**L9** `CONVEXITY_OPTIMISATION_LEDGER.md` [field bake]
The convexity / optimisation field-bake ledger — what bit, what bounced, and the boundary. One of the three fields listed but never thrown (the overnight order: convexity ×143, constraint ×141). 'OWED' 622

**L23** `OPTICS_LENSING_LEDGER.md` [field bake]
observational optics and lensing against CR — confirmations, and the dimension clause they forced

**L19** `SPECTRAL_THEORY_LEDGER.md` [field bake]
The spectral-theory field bake — what bit, what bounced, and the boundary. Tier B's largest never-thrown field (×189 on the reach measure), verified as NOT covered by the harmonic ledger, which mentions spectral, self-adjoint and deficiency zero times

**L21** `INFORMATION_THEORY_LEDGER.md` [field bake]
The information-theory field-bake ledger — what bit, what bounced, and where the boundary is

**L22** `NUMERICAL_ANALYSIS_LEDGER.md` [field bake]
The numerical-analysis field-bake ledger — what bit, what bounced, and where the boundary is

**L14** `INTEGRABLE_SYSTEMS_LEDGER.md` [field bake]
The integrable-systems field-bake ledger — what bit, what bounced, and where the boundary is

# References


[1] D. Janzen, *A curious normalization of the ΛCDM linear growth factor*, Res. Notes AAS (2025).

[2] D. Janzen, *A Critical Look at the Standard Cosmological Picture*, FQXi essay contest (2012); `arXiv:1301.5980`.

[3] D. Janzen, *General relativity's constraint algebra as the symmetric-space structure of a de Sitter substrate: the action Lie algebroid of the symmetry-reducible sector, and the problem of time's "wrong sign" as the substrate's coset signature*, companion paper (P12).

[4] D. Janzen, *The event horizon is a metric singularity: a missing definition in general relativity, and the black-hole problems it dissolves*, companion paper (P1).

[5] D. Janzen, *The de Sitter substrate and its slicing curve: horizons as turning points, and de Sitter and Schwarzschild as two readings of one slicing*, companion paper (P3).

[6] D. Janzen, *The Schwarzschild–de Sitter description groupoid: generators, relations, and Schwarzschild as the asymmetric realisation*, companion paper (P5).

[7] D. Janzen, *Covariance of geometries over the de Sitter substrate: the slicing operator, the vacuum kernel, and matter as the bend of the cut*, companion paper (P8).

[8] D. Janzen, *The range of the de Sitter slicing operator: surjectivity onto the symmetry-reducible sector of general relativity, the Kerr–NUT–(A)dS vacuum kernel, and the wall of inhomogeneity*, companion paper (P9).

[9] D. Janzen, *Collapsed matter must become a universe: the necessary and sufficient augmentation of general relativity into a description of structural existence, proven for gravitational collapse of any symmetry*, companion paper (P7).

[10] D. Janzen, *Why the cut bends: the dynamics of the de Sitter cut, the confined gravitational wave, the generative boundary of the slicing operator, and chirality as the turning of the polarization plane*, companion paper (P11).

[11] D. Janzen, *The modern parallax: the redshift-isotropy floor, the empirical forcing of cosmic time and uniform expansion, and the measured resolution of the relativity of simultaneity*, companion paper (P4).

[12] D. Janzen, *The shadow of existence: scientific theory-choice as an empirically grounded discipline, calibrated on the historical record and on a live programme*, companion paper (P6).

[13] V. Pavlidou and T. N. Tomaras, *Where the world stands still: turnaround as a strong test of ΛCDM cosmology*, JCAP **09** (2014) 020.

[14] D. Janzen, *On the ontological foundations of Cosmological Relativity.*

[15] D. Janzen, *A critical look at the standard cosmological picture*, in *Questioning the Foundations of Physics: Which of Our Fundamental Assumptions Are Wrong?*, Springer (2015) 103–130.

[16] D. Janzen, *A Solution to the Cosmological Problem of Relativity Theory*, PhD thesis, University of Saskatchewan (2012).

[17] D. Janzen, *Reached through the imaginary, real by construction: the maximally symmetric substrate as the geometric–ontological core of Cosmological Relativity*, companion paper (P17).

[18] D. Janzen, *Cosmological Relativity: what the construction delivers, what it declines, and where its edge is*, companion paper (P18).

[19] D. Janzen, *Cosmogenesis: the Big Bang as a synthesis of Cosmological Relativity's deductively forced consequences, and the primordial light-element abundances it produces*, companion paper (P16).

[20] D. Janzen, *The fermion sector of Cosmological Relativity: three chiral generations from the maximally-symmetric slicing structure*, companion paper (P14).

[21] R. M. Wald, *Asymptotic behavior of homogeneous cosmological models in the presence of a positive cosmological constant*, Phys. Rev. D **28** (1983) 2118.

[22] S. Alam et al. (BOSS Collaboration), *The clustering of galaxies in the completed SDSS-III Baryon Oscillation Spectroscopic Survey: cosmological analysis of the DR12 galaxy sample*, Mon. Not. R. Astron. Soc. **470** (2017) 2617.

[23] DESI Collaboration, *DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints*, arXiv:2503.14738 (2025).

[24] Planck Collaboration (N. Aghanim et al.), *Planck 2018 results. VI. Cosmological parameters*, Astron. Astrophys. **641** (2020) A6.

[25] A. G. Riess et al., *A comprehensive measurement of the local value of the Hubble constant with 1 km s$^{-1}$ Mpc$^{-1}$ uncertainty from the Hubble Space Telescope and the SH0ES team*, Astrophys. J. Lett. **934** (2022) L7.

[26] M. Moresco et al., *A 6% measurement of the Hubble parameter at $z \sim 0.45$: direct evidence of the epoch of cosmic re-acceleration*, J. Cosmol. Astropart. Phys. **05** (2016) 014.

[27] J. Silk, *Cosmic black-body radiation and galaxy formation*, Astrophys. J. **151** (1968) 459.

[28] R. K. Sachs and A. M. Wolfe, *Perturbations of a cosmological model and angular variations of the microwave background*, Astrophys. J. **147** (1967) 73.

[29] P. Ginsparg and M. J. Perry, *Semiclassical perdurance of de Sitter space*, Nucl. Phys. B **222** (1983) 245.

[30] G. Lemaître, *Cosmological Application of Relativity*, Rev. Mod. Phys. **21** (1949) 357.

[31] E. R. Harrison, *Fluctuations at the threshold of classical cosmology*, Phys. Rev. D **1** (1970) 2726.

[32] Ya. B. Zel'dovich, *A hypothesis, unifying the structure and the entropy of the Universe*, Mon. Not. R. Astron. Soc. **160** (1972) 1P.

[33] G. W. Gibbons and S. W. Hawking, *Cosmological event horizons, thermodynamics, and particle creation*, Phys. Rev. D **15** (1977) 2738.

[34] A. S. Eddington, *The Expanding Universe*, Cambridge University Press (1933).

[35] D. Janzen, *The canonical problem of time as a category error: an empirically forced cosmic time, deparametrization, the dissolution of frozen dynamics, and a graviton quantization the de Sitter horizon closes without a free parameter—the seam where ħ enters gravity—in Cosmological Relativity*, companion paper (P10).

[36] D. Janzen, *The Schwarzschild and de Sitter circle: the intrinsic geometry of one homogeneous ring—the event horizon and the curvature singularity at $r = 0$ its two poles, run through by a single continuation*, companion paper (P2).

[37] H. Nariai, *On a new cosmological solution of Einstein's field equations of gravitation*, Sci. Rep. Tôhoku Univ. Ser. I **35** (1951) 62; reprinted in Gen. Rel. Grav. **31** (1999) 963.

[38] P. J. E. Peebles, *Recombination of the primeval plasma*, Astrophys. J. **153** (1968) 1.